JEE MAIN & ADVANCED ULTIMATE FORMULA & REVISION HANDBOOK

DIAMOND HAND BOOK

Complete All-in-One Master Formula, Reaction & Concept Encyclopedia

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PHYSICS CHEMISTRY MATHEMATICS

PART I : PHYSICS

29 High-Yield Chapters (Pages 7 - 92)

- Mechanics & Gravitation
- Fluids, Waves & Thermodynamics
- Electrodynamics & EMI
- Optics, Modern Physics & Semiconductor Devices

PART II : CHEMISTRY

34 High-Yield Chapters (Pages 94 - 230)

- Physical Chemistry & Thermodynamics
- Inorganic Metallurgy, Coordination & Blocks
- Organic Chemistry (GOC, Reactions & Mechanisms)
- Comprehensive Organic Reagent Atlas & Salt Analysis

PART III : MATHEMATICS

19 High-Yield Chapters (Pages 232 - 289)

- Calculus (Differential & Integral)

EXCLUSIVELY CRAFTED FOR RAPID FORMULA & SHORTCUT REVISION

Complete Exhaustive Edition 289 Master Pages Target: JEE Main & Advanced

Aditya Soni

Curator, Compiler & Author | Diamond Hand Book

A dedicated JEE handbook engineered for aspirants aiming for top ranks in JEE Main and JEE Advanced. Built to deliver maximum formula retention in minimum time.

2025-2026 Comprehensive Master Revision Edition **289 Complete Pages**

THE OBJECTIVE & GOAL OF THIS BOOK

The primary goal of the Diamond Hand Book is purely and exclusively for rapid formula revision. During high-pressure JEE preparation, students spend countless hours sifting through bulky 500-page notes just to review standard identities, key physical constants, reaction mechanisms, and coordinate geometry results. This book consolidates all essential, exam-tested formulas across Physics, Chemistry, and Mathematics into a single, ultra-clean, highly dense revision manual. Use it for nightly 20-minute reviews and last-minute pre-exam revision.

CONTENT SOURCES & AI SYNTHESIS ACKNOWLEDGEMENTS

Foundational Curriculum Source: Physics Wallah (PW)

All foundational educational materials, formulas, short notes, and reaction compilations in this handbook were originally sourced from Physics Wallah (PW). Sincere appreciation is extended to Alakh Pandey sir and the extraordinary PW academic faculty for creating invaluable educational resources for Indian students.

AI Enhancement & Computational Synthesis:

The content was curated, restructured, cross-referenced, LaTeX-formatted, and augmented using cutting-edge AI models: Google Gemini, Anthropic Claude, OpenAI ChatGPT, and Google Antigravity to ensure flawless academic accuracy, high typographical readability, and seamless navigation.

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Diamond Hand Book **The Gold Standard in JEE Formula Revision**

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DIAMOND HAND BOOK CORE REVISION STRATEGY

Daily 30-Minute Targeted Drill: Review one subject module every day before starting problem-solving.

Boundary & Dimensional Checks: Verify physical units, asymptotic behavior (x -> 0, x -> inf), and limiting conditions.

Organic Reaction Pathways: Use the Organic Reagent Atlas (p. 227) for fast multi-step synthesis and named reactions.

Speed Shortcuts & Results: Memorize standard area shortcuts, standard integrals, and conic properties for rapid speed.

PART I

PHYSICS

Mechanics | Waves & Heat | Electrodynamics | Optics | Modern Physics

29 Chapters | Pages 7 to 92

THERMAL PHYSICS & THERMODYNAMICS

JEE Main & Advanced Complete Formula & Reference Sheet

JEE REVISION SHEET

1. TEMPERATURE SCALES & THERMAL EXPANSION

Solids, Liquids & Cavities

Temperature Scale Conversion

$$\frac{\text{Reading} - \text{LFP}}{\text{UFP} - \text{LFP}} = \text{Constant}$$

$$\frac{C}{5} = \frac{F - 32}{9} = \frac{K - 273.15}{5} = \frac{R}{4}$$

LFP = Lower Fixed Point; UFP = Upper Fixed Point.

Coefficients of Expansion & Relation

$$\alpha = \frac{\Delta L}{L_0 \, \Delta T}, \quad \beta = \frac{\Delta A}{A_0 \, \Delta T}, \quad \gamma = \frac{\Delta V}{V_0 \, \Delta T}$$

$$\alpha : \beta : \gamma = 1 : 2 : 3 \quad \Rightarrow \quad \beta = 2\alpha, \quad \gamma = 3\alpha$$

Units: $^\circ C^{-1}$ or K^{-1}. Dimensions: $[K^{-1}]$.

Thermal Stress & Expansion Applications

- **Thermal Strain:** $\frac{\Delta L}{L} = \alpha \, \Delta T$
- **Thermal Stress:** $\frac{F}{A} = Y \, \alpha \, \Delta T$ *(Rod fixed rigidly)*
- **Thermal Force / Tension:** $F = Y \, A \, \alpha \, \Delta T$
- **Constant Difference in Length:** $L_1 \alpha_1 = L_2 \alpha_2$
- **Bimetallic Strip:** Strip with larger α expands more, forming the *convex side*.
- **Cavity Expansion:** $\Delta r = r_0 \, \alpha \, \Delta T$ (Cavity expands same as solid body).

Pendulum Clock Time Loss / Gain

$$T = 2\pi \frac{\sqrt{L}}{\sqrt{g}} \quad \Rightarrow \quad \frac{\Delta T}{T} = \frac{1}{2} \alpha \, \Delta\theta$$

- **Fractional Time Loss:** $\Delta t = \frac{1}{2} \alpha \, \Delta\theta \, t$
- **Time Lost per Day (86400 s):** $\Delta t_{\text{day}} = 43200 \, \alpha \, \Delta\theta \, \text{seconds}$

Temp ↑: T ↑, clock runs slow. Temp ↓: clock runs fast.

Density Variation & Liquid Expansion

$$\rho' = \frac{\rho}{1 + \gamma \, \Delta T} \approx \rho \, (1 - \gamma \, \Delta T)$$

- **Real vs Apparent:** $\gamma_{\text{real}} = \gamma_{\text{apparent}} + 3\alpha_{\text{vessel}}$
- **Apparent Volume Change:** $\Delta V_{\text{app}} = V_0 \, (\gamma_{\text{real}} - 3\alpha_{\text{solid}}) \, \Delta T$
- **Anomalous Expansion:** Water has maximum density ($1000 \, \text{kg/m}^3$) at **4°C**.

Scale Measurement Error

$$\text{True Length} = \text{Scale Reading} \times (1 + \alpha_{\text{scale}} \, \Delta T)$$

- If $T > T_{\text{calibrated}}$: Scale expands, reading is smaller (True > Reading).
- If $T < T_{\text{calibrated}}$: Scale contracts, reading is larger (True < Reading).

2. CALORIMETRY & PHASE CHANGES

Heat Capacity, Phase Transformations & Mechanical Equivalent

Specific & Molar Heat Capacity

$$\Delta Q = m \, c \, \Delta T \quad \Rightarrow \quad c = \frac{\Delta Q}{m \, \Delta T}$$

$$C = \frac{\Delta Q}{n \, \Delta T} = M \, c$$

Variable c: $Q = \int m \, c(T) \, dT$.

Water Equivalent & Capacity

$$\text{Heat Capacity} = m \, c = n \, C$$

$$W = m \, c \, \text{grams}$$

Water equivalent W absorbs/releases same heat for same ΔT.

Latent Heat (Phase Change)

$$\Delta Q = m \, L$$

- $L_{\text{fusion (ice)}} = 80 \, \text{cal/g}$
- $L_{\text{vap (steam)}} = 540 \, \text{cal/g}$

Temp remains constant during phase change.

Principle of Calorimetry & Mixture Temperature

$$\text{Heat Lost} = \text{Heat Gained}$$

$$T_{\text{mix}} = \frac{\sum m_i \, c_i \, T_i}{\sum m_i \, c_i} = \frac{m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3}{m_1 c_1 + m_2 c_2 + m_3 c_3}$$

Bounds: $T_{min} \le T_{mix} \le T_{max}$ *(no phase change).*

Mechanical Energy Conversion & Ice-Water Systems

- **Mechanical Equivalent:** $W = J \, Q$ where $J \approx 4.2 \, \text{J/cal}$.
- **Potential Energy:** $m \, g \, h = m' \, L + m' \, c \, \Delta T$
- **Kinetic Energy:** $\frac{1}{2} m \, v^2 = m' \, L + m' \, c \, \Delta T$
- **Ice-Water Mixing:** Balance heat needed to bring ice to 0°C & melt it against heat available from cooling water to 0°C.

3. HEAT TRANSFER: CONDUCTION

Fourier's Law, Thermal Resistance & Composite Slabs

Fourier's Law of Heat Conduction

$$H = \frac{dQ}{dt} = -K A \frac{dT}{dx} = \frac{K A (T_1 - T_2)}{L}$$

Steady state ($dT/dt = 0$). K = Thermal conductivity.

Thermal Electrical Analogy

$$H = \frac{\Delta T}{R_{th}} \quad \text{where} \quad R_{th} = \frac{L}{K A}$$

Heat current $H \equiv I$, Temp diff $\Delta T \equiv \Delta V$, Thermal Resistance $R_{th} \equiv R$.

Combination of Conductors & Junction Temperature

- **Series:** $R_{eq} = R_1 + R_2 = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A}$
- **Series Equivalent Conductivity:** $K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}}$
- **Series Junction Temp (T_j):** $T_j = \frac{K_1 L_2 T_1 + K_2 L_1 T_2}{K_1 L_2 + K_2 L_1}$
- **Parallel:** $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow K_{eq} = \frac{K_1 A_1 + K_2 A_2}{A_1 + A_2}$

Radial & Spherical Conduction [Added JEE Adv]

- **Cylindrical Shell (r_1, r_2, Length L):**

$$R_{th} = \frac{\ln(r_2/r_1)}{2\pi K L} \Rightarrow H = \frac{2\pi K L (T_1 - T_2)}{\ln(r_2/r_1)}$$

- **Spherical Shell (r_1, r_2):**

$$R_{th} = \frac{r_2 - r_1}{4\pi K r_1 r_2} \Rightarrow H = \frac{4\pi K r_1 r_2 (T_1 - T_2)}{r_2 - r_1}$$

- **Ice Growth on Pond:** $t = \frac{\rho L_f}{2 K \theta} (y_2^2 - y_1^2)$

4. THERMAL RADIATION LAWS

Stefan-Boltzmann, Wien's, Kirchhoff's & Newton's Law of Cooling

Absorptance, Reflectance, Transmittance

$$a = \frac{Q_a}{Q}, \quad r = \frac{Q_r}{Q}, \quad t = \frac{Q_t}{Q}$$

$$a + r + t = 1$$

Black body: $a = 1, r = 0, t = 0$.

Emissive Power & Emissivity

$$E = \frac{\Delta Q}{A \Delta t} \text{ } \text{(W/m}^2\text{)}$$

$$e = \frac{E_{body}}{E_{blackbody}} \quad (0 \le e \le 1)$$

Total emissive power: $E = \int E_\lambda \, d\lambda$.

Kirchhoff's Law

$$\frac{E_\lambda}{a_\lambda} = E_{\lambda, \text{blackbody}}$$

$$\frac{e_\lambda}{a_\lambda} = 1 \Rightarrow e_\lambda = a_\lambda$$

"Good absorbers are good emitters."

Stefan-Boltzmann Law & Net Heat Loss

$$P_{emitted} = \frac{dQ}{dt} = e \sigma A T^4$$

$$P_{net} = \frac{dQ}{dt} = e \sigma A (T^4 - T_0^4)$$

$$\text{Rate of Cooling} = \frac{dT}{dt} = \frac{e \sigma A}{m c} (T^4 - T_0^4)$$

Stefan's constant $\sigma = 5.67 \times 10^{-8}$ text{ W m}$^{-2}$ text{K}$^{-4}$. T, T_0 in Kelvin!

Wien's Displacement Law

$$\lambda_m T = b = 2.89 \times 10^{-3} \text{ m·K}$$

- $\lambda_m \propto \frac{1}{T}$ (Peak wavelength shifts left as T increases).
- Area under $E_\lambda - \lambda$ curve = σT^4.
- Ratio of peak intensities: $\frac{E_{\lambda 1}}{E_{\lambda 2}} = (\frac{T_1}{T_2})^5$ [Added JEE Adv].

Newton's Law of Cooling

Valid for small temperature difference ($T - T_0 << T_0$).

$$\frac{dT}{dt} = -k (T - T_0) \quad \text{where} \quad k = \frac{4 e \sigma A T_0^3}{m c}$$

- **Exact Form:** $T(t) = T_0 + (T_i - T_0) e^{-kt}$
- **Practical Mean Form:**

$$\frac{T_1 - T_2}{\Delta t} = K [(\frac{T_1 + T_2}{2}) - T_0]$$

Solar Constant & Radiation Pressure [Added JEE Adv]

- **Solar Constant (S):** $S = (\frac{R_S}{R_{ES}})^2 \sigma T_S^4$
- **Radiation Pressure (P_{rad}):**
  - Absorbing Surface: $P_{rad} = \frac{I}{c}$
  - Reflecting Surface: $P_{rad} = \frac{2I}{c}$ (I = Intensity)

5. FIRST LAW OF THERMODYNAMICS & CORE CONCEPTS

Work Done, Internal Energy, Heat & Gas Laws

First Law of Thermodynamics (FLOT)

$$\Delta Q = \Delta U + W \quad \Rightarrow \quad dQ = dU + dW$$

- **Sign Conventions:**
  - $+\Delta Q$: Heat added to system.
  - $+W$: Work done *by* system.
  - $+\Delta U$: Internal energy increases.

Work Done in P-V Diagram

$$W = \int P\, dV = \text{Area under P-V curve}$$

- **Cyclic Process Work:** Net area enclosed by cycle.
  - Clockwise: $W > 0$ (+ve)
  - Anti-clockwise: $W < 0$ (-ve)

Internal Energy (State Function)

$$\Delta U = n\, C_v\, \Delta T = n\left(\frac{f}{2} R\right) \Delta T$$

U depends only on temperature for ideal gas ($\Delta U_{cycle} = 0$).

Degrees of Freedom (f) & Specific Heats

Gas Atomicity	f	C_v	C_p	$\gamma = \frac{C_p}{C_v}$
Monoatomic (He, Ar)	3	$\frac{3}{2} R$	$\frac{5}{2} R$	$\frac{5}{3} \approx 1.67$
Diatomic / Rigid (N_2, O_2)	5	$\frac{5}{2} R$	$\frac{7}{2} R$	$\frac{7}{5} = 1.40$
Non-Rigid Diatomic (High T)	7	$\frac{7}{2} R$	$\frac{9}{2} R$	$\frac{9}{7} \approx 1.29$
Polyatomic / Non-linear	6	$3 R$	$4 R$	$\frac{4}{3} \approx 1.33$

Mayer's Relation: $C_p - C_v = R$.

Gas Mixtures Relations [Added JEE Main/Adv]

- **Equivalent Molar Mass:** $M_{mix} = \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2}$
- **Mixture Specific Heat $C_{v,mix}$:**

$$C_{v,mix} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

- **Mixture Adiabatic Exponent (γ_{mix}):**

$$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

6. STANDARD THERMODYNAMIC PROCESSES

Isothermal, Isochoric, Isobaric, Adiabatic & Polytropic

Process	Governing Equation	Work Done (W)	Heat Exchange (ΔQ)	Molar Heat Capacity (C)
Isothermal ($T = \text{const}$, $\Delta T = 0$)	$P V = \text{const}$ $P_1 V_1 = P_2 V_2$	$n R T \ln\left(\frac{V_2}{V_1}\right) = 2.303\, n R T \log\left(\frac{P_1}{P_2}\right)$	$\Delta Q = W$ (since $\Delta U = 0$)	$C = \infty$
Isochoric ($V = \text{const}$, $\Delta V = 0$)	$\frac{P}{T} = \text{const}$ $\frac{P_1}{P_2} = \frac{T_1}{T_2}$	$W = 0$	$\Delta Q = \Delta U = n C_v \Delta T$	$C = C_v = \frac{R}{\gamma - 1}$
Isobaric ($P = \text{const}$, $\Delta P = 0$)	$\frac{V}{T} = \text{const}$ $\frac{V_1}{V_2} = \frac{T_1}{T_2}$	$P(V_2 - V_1) = n R (T_2 - T_1)$	$\Delta Q = n C_p \Delta T$	$C = C_p = \frac{\gamma R}{\gamma - 1}$
Adiabatic ($\Delta Q = 0$)	$P V^{\gamma} = \text{const}$ $T V^{\gamma - 1} = \text{const}$	$\frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{n R (T_1 - T_2)}{\gamma - 1}$	$\Delta Q = 0$	$C = 0$
Polytropic [Adv]	$P V^{n} = \text{const}$	$\frac{P_1 V_1 - P_2 V_2}{n - 1} = \frac{n R (T_1 - T_2)}{n - 1}$	$n C \Delta T$	$C_v + \frac{R}{1 - n}$

Slopes of P-V Curves & Work Comparisons

- **Isothermal Slope:** $\left(\frac{dP}{dV}\right)_{iso} = -\frac{P}{V}$
- **Adiabatic Slope:** $\left(\frac{dP}{dV}\right)_{dia} = -\gamma \frac{P}{V} = \gamma \times$ (Isothermal Slope)
- **Expansion Work Order ($V_1 \to V_2$):**

$$W_{isobaric} > W_{isothermal} > W_{adiabatic} > W_{isochoric}$$

Bulk Modulus of Gases & Sound Speed [Added JEE Main]

- **Isothermal Bulk Modulus:** $B_{iso} = P$
- **Adiabatic Bulk Modulus:** $B_{dia} = \gamma P \quad \Rightarrow \quad \frac{B_{dia}}{B_{iso}} = \gamma$
- **Free Expansion:** Expansion into vacuum ($P_{ext} = 0$). $W = 0$, $\Delta Q = 0 \Rightarrow \Delta U = 0 \Rightarrow T = \text{const}$.

7. HEAT ENGINES, CARNOT CYCLE & REFRIGERATORS

Efficiency, COP & Cascaded Engines

Heat Engine Efficiency

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

Q_1 = Heat from source (T_1); Q_2 = Heat to sink (T_2).

Carnot Engine Efficiency

$$\eta_{Carnot} = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$

Ideal reversible engine. T_1, T_2 in Kelvin.

Refrigerator & Heat Pump COP

$$\beta_{ref} = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$$

$$\beta_{HP} = \frac{Q_1}{W} = \frac{T_1}{T_1 - T_2}$$

Relation: $\beta_{HP} = \beta_{ref} + 1$.

Cascaded Heat Engines (Two Engines in Series)

- **Same Work Output ($W_1 = W_2$):**

$$T = \frac{T_1 + T_2}{2} \text{ (Arithmetic Mean)}$$

- **Same Efficiency ($\eta_1 = \eta_2$):**

$$T = \sqrt{T_1 T_2} \text{ (Geometric Mean)}$$

Entropy Change Formulas

[Added JEE Adv]

- **General Change:** $\Delta S = \int \frac{dQ_{rev}}{T}$
- **Isothermal:** $\Delta S = \frac{Q}{T} = n R \ln(\frac{V_2}{V_1})$
- **Isochoric:** $\Delta S = n C_v \ln(\frac{T_2}{T_1})$
- **Phase Change:** $\Delta S = \frac{m L}{T}$

8. IMPORTANT CONSTANTS & CONVERSIONS

Standard Values for Numerical Problem Solving

- **Gas Constant (R):** 8.314 J/(mol·K) $\approx 2.0 \text{ cal/(mol·K)}$
- **Boltzmann Const (k_B):** $1.38 \times 10^{-23} \text{ J/K}$
- **Stefan-Boltzmann (σ):** $5.67 \times 10^{-8} \text{ W/(m}^2\text{·K}^4\text{)}$
- **Wien's Const (b):** $2.89 \times 10^{-3} \text{ m·K}$
- **1 atm:** $1.013 \times 10^5 \text{ N/m}^2$
- **Calorie:** $1 \text{ cal} = 4.186 \text{ J}$

MECHANICAL PROPERTIES OF SOLIDS

Comprehensive JEE Main & Advanced Formula Revision Sheet

PHYSICS REVISION

1. Stress & Strain Fundamentals

Normal / Longitudinal Stress (σ)

$$\sigma = F_{res} / A$$

- **Tensile:** Increases length ($\Delta L > 0$)
- **Compressive:** Decreases length ($\Delta L < 0$)

SI Unit: N/m^2 or Pa $[ML^{-1}T^{-2}]$

Volumetric Stress (ΔP)

$$\sigma_v = F_{normal} / A = \Delta P$$

- Uniform perpendicular force per unit area over entire surface.
- Causes change in volume without shape change.

Tangential / Shear Stress (τ)

$$\tau = F_{tangential} / A$$

- Deforming force applied parallel to surface.
- Causes change in shape (angle of shear) at constant volume.

1. Longitudinal Strain

$$\varepsilon = \Delta L / L$$

Dimensionless ratio of change in length to original length.

2. Volumetric Strain

$$\varepsilon_v = \Delta V / V$$

Ratio of change in volume to initial volume.

3. Shearing Strain (φ)

$$\varphi = \tan \varphi = x / L$$

x = relative displacement of top face; L = height. (φ in rad)

2. Elastic Moduli & Hooke's Law

Hooke's Law: Within elastic limit, *Stress* $\propto$ *Strain* $\Rightarrow$ **Stress = E × Strain** (E = Modulus of Elasticity).

Young's Modulus (Y)

$$Y = \sigma / \varepsilon = (F / A) / (\Delta L / L) = F\,L / (A\,\Delta L)$$

For wire of radius r suspended with mass M:

$$Y = (M\,g\,L) / (\pi\,r^2\,l)$$

- Applies only to solids (wires/rods).

Bulk Modulus (B or K)

$$B = -\Delta P / (\Delta V / V) = -V\,(\Delta P / \Delta V)$$

- Negative sign indicates V decreases with ↑ pressure.

$$\text{Compressibility } (C) = 1 / B = -\Delta V / (V\,\Delta P)$$

- Valid for Solids, Liquids, and Gases.

Modulus of Rigidity (η)

$$\eta = \tau / \varphi = (F_{tangential} / A) / \varphi = F\,L / (A\,x)$$

- Applies only to solids. Fluids have η = 0.
- Rigid Body: E = ∞; Rubber < Steel in elasticity.

3. Poisson's Ratio (σ) & Inter-Elastic Constant Relations

Core JEE Advanced

Poisson's Ratio (σ)

$$\sigma = -\text{Lateral Strain} / \text{Longitudinal Strain} = -(\Delta d / d) / (\Delta L / L)$$

- **Theoretical Range:** $-1 < \sigma < 0.5$
- **Practical / Real Range:** $0 \le \sigma \le 0.5$
- **Incompressible Material ($\Delta V = 0$):** $\sigma = 0.5$
- **Relation for fractional volume change:** $\Delta V / V = (\Delta L / L) \times (1 - 2\sigma)$

4 Golden Inter-Elastic Relations (Y, B, η, σ)

Relation #	Formula
1. Y & B relation	$Y = 3\,B\,(1 - 2\sigma)$
2. Y & η relation	$Y = 2\,\eta\,(1 + \sigma)$
3. σ elimination	$9 / Y = 3 / \eta + 1 / B \Rightarrow Y = 9\,B\,\eta / (3B + \eta)$
4. σ calculation	$\sigma = (3B - 2\eta) / (6B + 2\eta)$

4. Elastic Strain Energy & Work Done

Total Elastic Potential Energy (U)

$$U = W = ½ \times F \times \Delta L = ½ \times (Y\,A / L) \times (\Delta L)^2$$

Work done by stretching force F against internal restoring force.

$$U = F^2\,L / (2\,Y\,A)$$

Energy Density (u = Energy per unit volume)

$$u = U / \text{Vol} = ½ \times \text{Stress} \times \text{Strain} = ½\,Y \times (\text{Strain})^2$$

$$u = (\text{Stress})^2 / (2\,Y)$$

Area under Stress-Strain curve = Strain Energy Density.

5. Deformations under Self-Weight & Thermal Effects

High Yield

Self-Weight Elongation & Potential Energy

$$\Delta L = M g L / (2 A Y) = \rho g L^2 / (2 Y)$$

• Equivalent to entire weight acting at C.G. (L/2).

$$U_{self\text{-}weight} = M^2 g^2 L / (6 A Y) = \tfrac{1}{3} F_{max} \Delta L$$ [Added]

Thermal Stress & Strain (Rigidly Fixed End)

$$\text{Thermal Strain} = \alpha \Delta\theta$$

$$\text{Thermal Stress} = Y \alpha \Delta\theta$$

$$\text{Thermal Tension / Force} = Y A \alpha \Delta\theta$$

α = Coeff. of linear expansion; Δθ = temperature change.

6. Breaking Stress, Equivalent Springs & Advanced Shortcuts

Breaking Stress & Max Length

- **Breaking Force:** $F_{breaking}$ = Breaking Stress × Area ($A \propto d^2$). Independent of length L!
- **Max Length under own weight:** L_{max} = Breaking Stress / (ρ g)

Equivalent Spring Constant of Wire (k)

$$k = Y A / L \Rightarrow \text{Wire behaves as spring of constant } k$$

• Series Combination: $1/k_{eq} = 1/k_1 + 1/k_2$

• Parallel Combination: $k_{eq} = k_1 + k_2$

JEE Advanced Special Results

- **Composite Rod Elongation:**

$$\Delta L_{net} = F \sum (L_i / A_i Y_i)$$

- **Tension at distance x from free end (Massless/Hanging):**

$$T(x) = (M / L) g x \Rightarrow \Delta L = \int_0^L [T(x) / (A Y)] dx = M g L / (2 A Y)$$

- **Rotational Extension of Uniform Rod (Omega ω):**

$$\Delta L = \rho \omega^2 L^3 / (3 Y) = M \omega^2 L^2 / (3 A Y)$$ [Added]

JEE FORMULA SHEET

Centre of Mass, Collisions & Circular Motion

JEE MAIN & ADVANCED

1. Position of Centre of Mass (COM) — Fundamentals

Discrete System of n Particles:

$$r_{CM} = (\sum m_i r_i) / (\sum m_i) = (\sum m_i r_i) / M$$

$$x_{CM} = (\sum m_i x_i)/M,\ y_{CM} = (\sum m_i y_i)/M,\ z_{CM} = (\sum m_i z_i)/M$$

COM is a unique hypothetical point where total mass can be assumed concentrated for external forces.

Continuous Mass Distribution:

$$r_{CM} = (\int r\, dm) / (\int dm) = (1/M) \int r\, dm$$

$$x_{CM} = (\int x\, dm)/M,\ y_{CM} = (\int y\, dm)/M,\ z_{CM} = (\int z\, dm)/M$$

Two Particle System:

$$r_1 = [\, m_2 / (m_1 + m_2)\,]\, L,\ r_2 = [\, m_1 / (m_1 + m_2)\,]\, L$$

$$m_1 r_1 = m_2 r_2 \Rightarrow r \propto 1/m$$

r_1, r_2 are distances of m_1, m_2 from COM; L is separation. Total linear moment about COM is zero: $\sum m_i r_{i/CM} = 0$.

Shift in COM (Removal of Mass): [Added]

$$r_{rem_CM} = (M r_{total} - m r_{removed}) / (M - m)$$

Used for cut-out cavities (e.g. disc with circular hole).

2. COM of Standard Rigid Bodies — Reference

Object / Body Shape	COM Position (y_c from base/center)
Uniform Rod (Length L)	$y_c = L / 2$
Triangular Plate / Triangular Lamina	$y_c = h / 3$ (from base on median)
Semi-circular Ring (Radius R)	$y_c = 2R / \pi$, $x_c = 0$
Semi-circular Disc (Radius R)	$y_c = 4R / (3\pi)$, $x_c = 0$
Hemispherical Shell / Hollow Hemisphere	$y_c = R / 2$, $x_c = 0$
Solid Hemisphere (Radius R)	$y_c = 3R / 8$, $x_c = 0$
Hollow Circular Cone (Height h)	$y_c = h / 3$ (from base)
Solid Circular Cone (Height h)	$y_c = h / 4$ (from base)
Circular Sector Wire (Angle 2α) [Added]	$y_c = (R \sin\alpha) / \alpha$
Circular Sector Disc (Angle 2α) [Added]	$y_c = (2R \sin\alpha) / (3\alpha)$

3. Dynamics & Motion of COM — Kinematics

Velocity & Momentum of COM:

$$v_{CM} = (\sum m_i v_i) / M = P_{sys} / M$$

$$P_{sys} = M v_{CM}$$

Acceleration & External Force:

$$a_{CM} = (\sum m_i a_i) / M = F_{ext,net} / M$$

$$F_{ext} = M\, (dv_{CM} / dt) = dP_{sys} / dt$$

Internal forces cancel out in pairs (Newton's 3rd Law): $F_{int,net} = 0$.

Conservation of Linear Momentum:

$$\text{If } F_{ext,net} = 0 \Rightarrow a_{CM} = 0$$

$$P_{sys} = \text{constant} \Rightarrow v_{CM} = \text{constant}$$

If initial system is at rest and $F_{ext} = 0$, COM remains stationary ($\Delta r_{CM} = 0$).

Kinetic Energy of System:

$$K_{sys} = \tfrac{1}{2} M v_{CM}^2 + K_{w.r.t.\ COM}$$

$K_{sys} \neq \frac{1}{2} M v_{CM}^2$ in general. It equals $\frac{1}{2} M v_{CM}^2$ only when all internal motion relative to COM is zero.

4. Variable Mass System & Rocket — Advanced

General Equation of Motion:

$$F_{ext} + F_{thrust} = M\, (dv / dt)$$

$$F_{thrust} = v_{rel}\, (dm / dt)$$

v_{rel} = velocity of mass addition/ejection w.r.t system. If mass decreases, $dm/dt < 0$.

Rocket Propulsion (Under Gravity):

$$\text{Velocity at time t: } v(t) = v_0 - gt + u \ln(M_0 / M)$$

$$\text{Upward Thrust Force: } F_{th} = u\, (-dM/dt)$$

$$\text{Burnt-out Speed (g=0): } v_{max} = u \ln(M_0 / M_f)$$

u = exhaust speed w.r.t. rocket, M_0 = initial total mass, M = mass at time t.

5. Linear Impulse & Collisions — Impact

Impulse-Momentum Theorem:

$$J = \int F\, dt = \Delta P = P_f - P_i$$

Area under Force vs Time (F–t) graph gives impulse.

Coefficient of Restitution (e):

$$e = -(v_{2f} - v_{1f}) / (v_{2i} - v_{1i}) = (v_{sep}) / (v_{app})$$

$$e = (\text{Impulse of Reformation}) / (\text{Impulse of Deformation})$$

- *e = 1 : Perfectly Elastic (KE conserved)*
- *0 < e < 1 : Inelastic (KE lost)*
- *e = 0 : Perfectly Inelastic (bodies stick together, $v_{sep} = 0$)*

1D Collision Velocities (General e):

$$v_{1f} = [(m_1 - e\, m_2)/(m_1 + m_2)]\, v_{1i} + [(1 + e) m_2/(m_1 + m_2)]\, v_{2i}$$

$$v_{2f} = [(1 + e) m_1/(m_1 + m_2)]\, v_{1i} + [(m_2 - e\, m_1)/(m_1 + m_2)]\, v_{2i}$$

Special Head-On Elastic Cases (e = 1):
- $m_1 = m_2$: Velocities interchange ($v_{1f} = v_{2i}$, $v_{2f} = v_{1i}$).
- $m_1 \gg m_2$, $v_{2i} = 0$: Heavy body unaffected ($v_{1f} = v_{1i}$), light body gets double speed ($v_{2f} = 2\, v_{1i}$).
- $m_1 \ll m_2$, $v_{2i} = 0$: Light body rebounds ($v_{1f} = -v_{1i}$), heavy body remains at rest ($v_{2f} = 0$).

Kinetic Energy Loss in Collision:

$$\Delta K = [(m_1\, m_2) / 2(m_1 + m_2)]\, (1 - e^2)\, (v_{1i} - v_{2i})^2$$

Fractional Loss (Perfectly Inelastic, $v_{2i}=0$): $\Delta K / K_i = M / (m + M)$

Ballistic Pendulum:

$$v_{combined} = [m / (m + M)]\, u$$

Bullet Speed: $u = [(m + M) / m]\, \sqrt{(2gy)}$

m = bullet mass, M = block mass, y = vertical height raised by pendulum.

Oblique / 2D Collisions: [Added]

1. Apply momentum conservation along common tangent (no impulse): $v_{1t} = u_{1t}$, $v_{2t} = u_{2t}$.
2. Apply restitution along line of impact (normal): $v_{2n} - v_{1n} = e\,(u_{1n} - u_{2n})$.
3. Apply momentum conservation along line of impact.

Rebound of Ball from Floor: [Added]

Speed after n^{th} bounce: $v_n = e^n\, u = e^n \sqrt{(2gh)}$

Height after n^{th} bounce: $h_n = e^{2n}\, h$

Total distance before stopping: $H_{total} = h\, [(1 + e^2) / (1 - e^2)]$

Total time before stopping: $T_{total} = \sqrt{(2h/g)}\, [(1 + e) / (1 - e)]$

6. Circular Motion Kinematics

Rotation

Angular Variables & Basics:

$\theta = s / r$ (rad), $\omega = d\theta/dt$ (rad/s), $\alpha = d\omega/dt = d^2\theta/dt^2$ (rad/s^2)

Time Period: $T = 2\pi / \omega = 1 / f$, Frequency: $f = n$ (r.p.s.)

Small displacement $d\boldsymbol{\theta}$ is a vector, but finite displacement $\Delta\boldsymbol{\theta}$ is scalar (doesn't commute: $\theta_1+\theta_2 \neq \theta_2+\theta_1$).

Vector Relations:

$v = \omega \times r \Rightarrow v = \omega\, r$

$a = (d\omega/dt) \times r + \omega \times (dr/dt) = a_t + a_c$

Tangential & Centripetal Acceleration:

Tangential Acceleration: $a_t = dv/dt = \alpha\, r$

Centripetal (Radial) Acceleration: $a_c = v^2 / r = \omega^2\, r = \omega\, v$

Net Acceleration: $a_{net} = \sqrt{(a_c^2 + a_t^2)}$

Angle with radius vector: $\tan \varphi = a_t / a_c$

- *Uniform Circular Motion (UCM): v = const $\Rightarrow a_t = 0$, $a_{net} = a_c$.*
- *Non-Uniform Circular Motion: v changes $\Rightarrow a_t \neq 0$, $a_c \neq 0$.*

7. Dynamics of Circular Motion

Vehicles & Roads

Centripetal Force:

$$F_c = m\, a_c = m\, v^2 / r = m\, \omega^2\, r$$

Real force directed towards center; provided by tension, friction, gravity, normal, etc.

Bending of a Cyclist:

$$\tan \theta = v^2 / (r\, g)$$

θ is angle of cycle with vertical to avoid skidding.

Vehicle on Flat Unbanked Level Road:

$$v_{max} = \sqrt{(\mu_s\, r\, g)}$$

Friction provides centripetal force. Skidding occurs if $v > v_{max}$.

Vehicle on Banked Road (With Friction):

$$v_{max} = \sqrt{[r\, g\, (\tan \theta + \mu_s) / (1 - \mu_s \tan \theta)]}$$

$$v_{min} = \sqrt{[r\, g\, (\tan \theta - \mu_s) / (1 + \mu_s \tan \theta)]}$$

Optimum Safe Speed ($\mu = 0$): $v_0 = \sqrt{(r\, g \tan \theta)}$

θ = angle of banking. If friction angle $\varphi = \tan^{-1}(\mu_s)$, $v_{max} = \sqrt{[r\, g \tan(\theta + \varphi)]}$.

Conical Pendulum: [Added]

Tension: $T = mg / \cos \theta$

Angular velocity: $\omega = \sqrt{(g / L \cos \theta)}$

Time Period: $T_{period} = 2\pi \sqrt{(L \cos \theta / g)}$

θ = semi-vertical angle, L = length of string.

8. Vertical Circular Motion & Torque

JEE Advanced

Vertical Circular Motion (String + Mass): [Added]

Min velocity at bottom for full loop: $u_{min} = \sqrt{(5gL)}$

Min velocity at top for full loop: $v_{top} = \sqrt{(gL)}$

Tension at angle θ with lowest point: $T(\theta) = m\, [(u^2/L) - g(2 - 3 \cos \theta)]$

Tension difference: $T_{bottom} - T_{top} = 6\, mg$

- *If $\sqrt{(2gL)} < u < \sqrt{(5gL)}$: String goes slack ($T=0$) before top and leaves circular path.*
- *If $u \leq \sqrt{(2gL)}$: Oscillates between $\theta = \pm 90°$.*

Vertical Loop in Light Rigid Rod: [Added]

Min bottom velocity for complete loop: $u_{min} = \sqrt{(4gL)} = 2\sqrt{(gL)}$

Min velocity at top: $v_{top} = 0$

Moment of Force / Torque (τ):

$\tau = r \times F \Rightarrow |\tau| = r\, F \sin \theta = F\, r_\perp$

Cartesian Form: $\tau = (y\, F_z - z\, F_y)\hat{i} + (z\, F_x - x\, F_z)\hat{j} + (x\, F_y - y\, F_x)\hat{k}$

SI Unit: N·m, Dimensions: $[M\, L^2\, T^{-2}]$. Direction via Right-Hand Thumb Rule. Anti-clockwise torque positive (+), clockwise negative (−).

Relative Angular Velocity:

$$\omega_{AB} = (v_{AB})_{\perp} / r_{AB}$$

$$\omega_{AB} = (v_A \sin\theta_1 + v_B \sin\theta_2) / r$$

$(v_{AB})_{\perp}$ is the component of relative velocity perpendicular to line joining A and B.

MOTION IN A STRAIGHT LINE

JEE Main & Advanced Comprehensive Formula Sheet

Kinematics 1D
Quick Revision Sheet

1. KINEMATICS FUNDAMENTALS & VECTOR DEFINITIONS

Position & Displacement Vectors

Position Vector: $\mathbf{r}_A = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

Displacement Vector: $\Delta\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A = (x_2-x_1)\hat{i} + (y_2-y_1)\hat{j} + (z_2-z_1)\hat{k}$

- Distance = Total path length (Scalar, ≥ 0).
- Displacement = Shortest straight-line distance (Vector).
- **Distance ≥ |Displacement|**
- If Distance > |Displacement| ⇒ Path is curved or motion reversed direction (velocity became zero at least once).

Motion on a Circular Arc

A particle moves from A to B along a circular path of radius R covering angle θ at speed v:

- Distance covered: $s = R\theta$
- Displacement magnitude: $|\Delta\mathbf{r}| = AB = 2R\sin(\theta/2)$
- Ratio: $|\text{Displacement}| / \text{Distance} = \sin(\theta/2) / (\theta/2)$
- Average Velocity magnitude: $|\mathbf{v}_{avg}| = [2v\sin(\theta/2)] / \theta$
- Average Acceleration magnitude: $|\mathbf{a}_{avg}| = [2v^2\sin(\theta/2)] / (R\theta)$ **[Added]**

2. SPEED, VELOCITY & ACCELERATION RELATIONS

Velocity Definitions & Forms

- Average Speed: $v_{avg} = \text{Total Distance} / \text{Total Time} = \Delta s / \Delta t$
- Average Velocity: $\mathbf{v}_{avg} = \Delta\mathbf{r} / \Delta t = (\mathbf{r}_f - \mathbf{r}_i) / (t_f - t_i)$
- Instantaneous Velocity: $\mathbf{v} = d\mathbf{r} / dt = (dx/dt)\hat{i} + (dy/dt)\hat{j} + (dz/dt)\hat{k}$
- Instantaneous Speed: $v = |\mathbf{v}| = |d\mathbf{r} / dt| = \sqrt{[(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2]}$

Note: $|v_{avg}| \le$ Average Speed, but Instantaneous Speed = |Instantaneous Velocity|

Acceleration Forms & Differential Relations

- Average Acceleration: $\mathbf{a}_{avg} = \Delta\mathbf{v} / \Delta t = (\mathbf{v}_2 - \mathbf{v}_1) / \Delta t$
- Instantaneous Acceleration: $\mathbf{a} = d\mathbf{v} / dt = d^2\mathbf{r} / dt^2$
- 1D Position-Dependent Acceleration: $a = v\,(dv / dx)$
- Time-Position Differential Form: If $t = f(x)$, then:

$$v = (dt/dx)^{-1} \quad | \quad a = -v^3\,(d^2t / dx^2) = -(d^2t / dx^2) / (dt/dx)^3$$

3. AVERAGE SPEED FORMULATIONS (SPECIAL CASES)

Case Condition	Formula / Mathematical Relation	Remarks
Equal Time Intervals $t_1 = t_2 = \ldots = t_n = t$	$v_{avg} = (v_1 + v_2 + \ldots + v_n) / n$	Arithmetic Mean of speeds.
Unequal Time Intervals	$v_{avg} = (v_1t_1 + v_2t_2 + \ldots + v_nt_n) / (t_1 + t_2 + \ldots + t_n)$	Weighted Time Average.
Equal Distance Parts Distance d split into n equal parts	$1 / v_{avg} = (1/n)\,[\,(1/v_1) + (1/v_2) + \ldots + (1/v_n)\,]$	Harmonic Mean of speeds.
Two Equal Distances (d/2 each)	$v_{avg} = (2\,v_1\,v_2) / (v_1 + v_2)$	Standard 2-part distance average.
Three Equal Distances (d/3 each) **[Added]**	$v_{avg} = (3\,v_1\,v_2\,v_3) / (v_1v_2 + v_2v_3 + v_3v_1)$	Standard 3-part distance average.

4. EQUATIONS OF MOTION (CONSTANT ACCELERATION: A = CONST)

Standard Equations (Scalar & Vector Form)

1. $\mathbf{v} = \mathbf{u} + \mathbf{a}t$
2. $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2 = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$
3. $\mathbf{s} = [\,(\mathbf{u} + \mathbf{v}) / 2\,]\,t$
4. $v^2 = u^2 + 2\,\mathbf{a}\cdot\mathbf{s}$ (or $\mathbf{v}\cdot\mathbf{v} = \mathbf{u}\cdot\mathbf{u} + 2\,\mathbf{a}\cdot\mathbf{s}$)
5. Displacement in n^{th} second:

$$\mathbf{S}_{n^{th}} = \mathbf{u} + \tfrac{1}{2}\,\mathbf{a}\,(2n - 1)$$

Strict Requirement: Acceleration a MUST be uniform/constant.

Galileo's Law of Odd Numbers & Ratios

For a body starting from rest ($u = 0$) with constant acceleration a:

- Ratio of displacements in equal successive time intervals (Δt):

$$S_1 : S_2 : S_3 : \ldots : S_n = 1 : 3 : 5 : \ldots : (2n - 1)$$

- Ratio of total displacements at time t, 2t, 3t, ...:

$$x_1 : x_2 : x_3 : \ldots = 1^2 : 2^2 : 3^2 : \ldots = 1 : 4 : 9 : \ldots$$

• Ratio of times taken to cover equal successive distances (s):
$t_1 : t_2 : t_3 : ... = 1 : (\sqrt{2} - 1) : (\sqrt{3} - \sqrt{2}) : ... : (\sqrt{n} - \sqrt{(n-1)})$

5. STOPPING DISTANCE, STOPPING TIME & BULLET PLANK RESULTS

Stopping Distance & Time

Given retardation a_0 and initial speed u_0:

• Stopping Distance: $s = u_0^2 / (2a_0) \Rightarrow s \propto u_0^2$

• Stopping Time: $t = u_0 / a_0 \Rightarrow t \propto u_0$

If speed is doubled, stopping distance becomes 4 times!

Mid-Point Velocity of Train

Train passing a point with front end speed u and rear end speed v (constant a):

$$v_{mid} = \sqrt{[\,(u^2 + v^2) / 2\,]}$$

Independent of train length.

Bullet-Plank Stopping Formula

A bullet loses $1/n$-th of its velocity in penetrating 1 plank. Minimum identical planks required to stop it:

$$N = n^2 / (2n - 1)$$

If velocity changes from u to v per plank: $N = u^2 / (u^2 - v^2)$

6. ACCELERATING-DECELERATING PROFILE & SEGMENTED MOTION

Accelerating & Retarding Motion (Rest to Rest)

A body starts from rest, accelerates at constant rate α for time t_1, then immediately decelerates at rate β for time t_2 to come to rest. Total time elapsed = $t = t_1 + t_2$.

• Peak Velocity: $v_{max} = [\,(\alpha\,\beta) / (\alpha + \beta)\,]\,t$

• Total Displacement: $s = ½\,[\,(\alpha\,\beta) / (\alpha + \beta)\,]\,t^2 = ½\,v_{max}\,t$

• Ratio of Time Intervals: $t_1 / t_2 = \beta / \alpha$

• Time Fractions: $t_1 = [\,\beta / (\alpha + \beta)\,]\,t$ | $t_2 = [\,\alpha / (\alpha + \beta)\,]\,t$

Unequal Distances in Equal Time Intervals

A particle travels distance S_1 in 1st time interval t and distance S_2 in 2nd equal time interval t with constant acceleration a:

• Initial Velocity: $u = (3S_1 - S_2) / (2t)$

• Acceleration: $a = (S_2 - S_1) / t^2$

7. MOTION UNDER GRAVITY (NO AIR RESISTANCE)

Vertical Projection from Ground (u Upwards)

Sign Convention: Upward = +ve, Downward = −ve, $a = -g$

• Maximum Height Attained: $H_{max} = u^2 / (2g)$

• Time of Ascent = Time of Descent: $t_a = t_d = u / g$

• Total Time of Flight: $T = 2u / g$

• Velocity at return point: $v = -u$ (same speed, downward)

• Speed Symmetry: Speed at any height h is identical during upward and downward journey: $v(h) = \sqrt{(u^2 - 2gh)}$

Body Crossing Same Height h at Times t_1 and t_2

A body thrown upwards crosses a point at height h at times t_1 (ascending) and t_2 (descending):

• Height of the point: $h = ½\,g\,t_1\,t_2$

• Initial Velocity of projection: $u = ½\,g\,(t_1 + t_2)$

• Maximum Height: $H_{max} = ⅛\,g\,(t_1 + t_2)^2$

• Total Time of Flight: $T = t_1 + t_2$

Projection from Height H (3-Body Problem)

Three identical particles projected from top of tower with same speed u:

1. Particle 1 thrown vertically UP (time t_1 to reach ground)
2. Particle 2 thrown vertically DOWN (time t_2 to reach ground)
3. Particle 3 dropped freely from REST / Horizontally (time t_3)

• Time Relation: $t_3 = \sqrt{(t_1\,t_2)}$

• Tower Height: $H = ½\,g\,t_1\,t_2$

• Speed of Projection: $u = ½\,g\,(t_1 - t_2)$ [Added]

Motion Under Air Drag / Resistance

Air drag causes constant retardation a opposing motion:

• **Ascending Motion:** Net retardation = $g + a$
$t_1 = \sqrt{[\,2h / (g + a)\,]}$ | $v_{projection} = \sqrt{[\,2(g + a)h\,]}$

• **Descending Motion:** Net acceleration = $g - a$
$t_2 = \sqrt{[\,2h / (g - a)\,]}$ | $v_{impact} = \sqrt{[\,2(g - a)h\,]}$

• **Result: $t_2 > t_1$ (Time of descent > Time of ascent) and $v_{impact} < v_{projection}$.**

8. RELATIVE MOTION IN 1-DIMENSION

Relative Position, Velocity & Acceleration

- Relative Velocity of A w.r.t. B: $\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B$
- Relative Velocity of B w.r.t. A: $\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = -\mathbf{v}_{A/B}$
- Relative Acceleration: $\mathbf{a}_{A/B} = \mathbf{a}_A - \mathbf{a}_B$
- Relative Displacement: $\mathbf{s}_{A/B} = \mathbf{s}_A - \mathbf{s}_B = \mathbf{u}_{A/B}\, t + \frac{1}{2}\, \mathbf{a}_{A/B}\, t^2$

Overtaking & Collision Conditions

Two bodies separated by distance d moving along same line:

- **Same direction:** Relative speed = $|v_A - v_B|$; Time to overtake = $d / |v_A - v_B|$
- **Opposite directions:** Relative speed = $v_A + v_B$; Time to collide = $d / (v_A + v_B)$
- **Minimum Distance to Avoid Collision:** If car A (speed u_A) behind car B (speed u_B, $u_A > u_B$) brakes with retardation a:

$$d_{min} = (u_A - u_B)^2 / (2a)$$ [Added]

9. COMPREHENSIVE GRAPHICAL ANALYSIS RULES

Graph Type	Slope Represents	Area Under Curve Represents
Position–Time ($x - t$)	$\tan\theta = dx/dt$ = **Instantaneous Velocity (v)** • Straight line ⇒ Uniform velocity • Parabola opening up ⇒ $a > 0$ • Parabola opening down ⇒ $a < 0$	*No physical significance.*
Velocity–Time ($v - t$)	$\tan\theta = dv/dt$ = **Instantaneous Acceleration (a)** • Slope > 0 ⇒ Acceleration • Slope < 0 ⇒ Retardation • Slope = 0 ⇒ Constant velocity	$\int v\, dt$ = **Displacement (Δx)** (taking sign into account) $\int \lvert v \rvert\, dt$ = **Total Distance** (area summed as positive values)
Acceleration–Time ($a - t$)	da/dt = **Jerk** [Added]	$\int a\, dt$ = **Change in Velocity ($\Delta v = v_2 - v_1$)** *(Note: Area does NOT give final velocity directly, but Δv).*
Velocity–Position ($v - x$) [Added]	dv/dx Acceleration: $a = v \cdot$ (Slope)	*No direct simple integral.*
$v^2 - x$ Graph [Added]	$d(v^2)/dx = 2a$ Acceleration: $a = \frac{1}{2} \times$ (Slope)	*No direct simple integral.*

10. VARIABLE ACCELERATION & CALCULUS INTEGRATION FRAMEWORK

Differentiation & Integration Flow

Displacement (x) ⇄ (d/dt | $\int dt$) Velocity (v) ⇄ (d/dt | $\int dt$) Acceleration (a)

- If $v = f(t)$: $x = \int v\, dt$ | $a = dv/dt$
- If $a = f(t)$: $v = v_0 + \int a\, dt$ | $x = x_0 + \int v\, dt$
- If $a = f(x)$: $\int v\, dv = \int a(x)\, dx \Rightarrow \frac{1}{2}(v^2 - u^2) = \int_{x_0}^{x} a(x)\, dx$
- If $a = f(v)$: $t = \int [1 / a(v)]\, dv$ | $x = \int [v / a(v)]\, dv$

Distance vs. Displacement via Integration

When velocity reverses sign in time interval $[t_1, t_2]$ (i.e. $v(t) = 0$ at $t = t_0$):

- **Displacement:** $\Delta x = \int_{t_1}^{t_2} v(t)\, dt$
- **Total Distance:** $s = \int_{t_1}^{t_2} |v(t)|\, dt = \left|\int_{t_1}^{t_0} v\, dt\right| + \left|\int_{t_0}^{t_2} v\, dt\right|$

Crucial Step for JEE Main/Advanced: Always check roots of $v(t) = 0$ before computing total distance!

11. JEE ADVANCED USEFUL SHORTCUT IDENTITIES

Quadratic Time-Distance Relation

If motion is governed by $t = \alpha x^2 + \beta x$ (where α, β are constants):

• Velocity: $v = (2\alpha x + \beta)^{-1}$

• Retardation: $a = -2\alpha v^3$

Direct application in JEE Advanced numericals.

Max Distance Between Two Cars [Added]

Car A (ahead, moving at constant v) and Car B (behind, accelerating from rest with a):

• Max separation occurs when velocity of B equals velocity of A ($v_B = v$).

• Maximum Separation: $x_{max} = d_{initial} + v^2 / (2a)$

12. IMPORTANT CONSTANTS & UNITS

• Acceleration due to gravity (g): $9.8\ m/s^2 \approx 9.81\ m/s^2 \approx \pi^2\ m/s^2 \approx 32.2\ ft/s^2$

• Standard SI Unit Conversion: $1\ km/h = 5/18\ m/s \approx 0.278\ m/s$ | $1\ m/s = 18/5\ km/h = 3.6\ km/h$

• Dimension Vectors: Displacement $[L]$, Velocity $[LT^{-1}]$, Acceleration $[LT^{-2}]$, Jerk $[LT^{-3}]$

SYSTEM OF PARTICLES & ROTATIONAL MOTION

JEE Main & Advanced Revision Formula Sheet • High-Density Master Reference

PHYSICS REFERENCE

1. CENTRE OF MASS (COM) — DISCRETE

Position Vector: $\mathbf{r_{cm}} = \frac{\sum m_i r_i}{\sum m_i} = \frac{m_1 r_1 + m_2 r_2 + \dots}{M}$

Components: $x_{cm} = \frac{\sum m_i x_i}{M}$, $y_{cm} = \frac{\sum m_i y_i}{M}$, $z_{cm} = \frac{\sum m_i z_i}{M}$

Two Particle System: $r_1 = \frac{m_2 r}{m_1 + m_2}$, $r_2 = \frac{m_1 r}{m_1 + m_2}$

Origin at COM; r_1, r_2 are distances from m_1, m_2 to COM. ($m_1 r_1 = m_2 r_2$)

Velocity of COM: $\mathbf{V_{cm}} = \frac{\sum m_i v_i}{M} \Rightarrow \mathbf{P_{sys}} = \mathbf{M\, V_{cm}}$

Acceleration of COM: $\mathbf{a_{cm}} = \frac{\sum m_i a_i}{M} = \frac{F_{ext,\,net}}{M}$

Internal forces sum to zero ($F_{int,\,net} = 0$).

2. COM — CONTINUOUS & CAVITY

Continuous Distribution: $\mathbf{r_{cm}} = \frac{\int r\, dm}{\int dm} = \frac{1}{M}\int r\, dm$

Mass Densities: $dm = \lambda\, dl$ *(1D)*, $dm = \sigma\, dA$ *(2D)*, $dm = \rho\, dV$ *(3D)*

Cavity / Mass Removal Formula:

$$\mathbf{r_{rem}} = \frac{M_{orig}\, r_{orig} - m_{rem}\, r_{cav}}{M_{orig} - m_{rem}} = \frac{-\, m_{rem}\, r_{cav}}{M_{remaining}}$$

Origin assumed at original COM ($r_{orig} = 0$). COM shifts opposite to cavity.

Conservation of Linear Momentum:

If $\mathbf{F_{ext}} = 0 \Rightarrow \mathbf{a_{cm}} = 0 \Rightarrow \mathbf{V_{cm}} = \text{const} \Rightarrow \mathbf{P_{sys}} = \text{const}$

3. STANDARD COM POSITIONS FOR RIGID BODIES

Body / Shape	Position of Centre of Mass (from base / center / axis)	Conditions / Symmetry
Uniform Rod / Rectangular Lamina / Sphere / Cube	At geometric center (L/2, Center, Center)	Uniform mass distribution
Semicircular Wire Ring	$y_{cm} = \frac{2R}{\pi} \approx 0.637\, R$	On symmetry axis ($x_{cm} = 0$)
Semicircular Disc	$y_{cm} = \frac{4R}{3\pi} \approx 0.424\, R$	On symmetry axis
Hemispherical Shell (Hollow)	$y_{cm} = \frac{R}{2}$	From center of base sphere
Solid Hemisphere	$y_{cm} = \frac{3R}{8}$	From flat base center
Hollow Cone (Open Base) [Added]	$h_{cm} = \frac{h}{3}$	Measured along axis from base (h = height)
Solid Cone	$h_{cm} = \frac{h}{4}$	Measured along axis from base

4. IMPULSE & LINEAR COLLISIONS

Impulse: $\mathbf{J} = \int F\, dt = \Delta P = M\,(V_f - V_i)$

Area under Force vs Time graph = Impulse

Coefficient of Restitution (e):

$$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{\text{Velocity of separation (along LOI)}}{\text{Velocity of approach (along LOI)}}$$

Impulsive Definition: $e = \frac{\int F_{reformation}\, dt}{\int F_{deformation}\, dt}$

Collision Types:

- **e = 1:** Perfectly Elastic (KE conserved)
- **0 < e < 1:** Inelastic (KE not conserved)
- **e = 0:** Perfectly Inelastic (Bodies stick, $v_1 = v_2$)

5. POST-COLLISION VELOCITIES

Direct 1D Collision Formulas:

$$v_1 = \frac{(m_1 - e\, m_2)\, u_1 + m_2\,(1 + e)\, u_2}{m_1 + m_2}$$

$$v_2 = \frac{(m_2 - e\, m_1)\, u_2 + m_1\,(1 + e)\, u_1}{m_1 + m_2}$$

Special Cases (e = 1):

- If $m_1 = m_2 \Rightarrow v_1 = u_2$, $v_2 = u_1$ (Velocities swap)
- If $m_1 \gg m_2$ and $u_2 = 0 \Rightarrow v_1 \approx u_1$, $v_2 \approx 2\, u_1$ [Added]

KE Loss in 1D Collision:

$$\Delta K = \frac{1}{2}\,\frac{m_1 m_2}{m_1 + m_2}\,(u_1 - u_2)^2\,(1 - e^2)$$

For e = 1, $\Delta K = 0$. Maximum loss when e = 0.

6. MOMENT OF INERTIA (I) & CORE THEOREMS

Discrete System: $I = \sum m_i r_i^2$

Continuous Body: $I = \int r^2 \, dm$

r = perpendicular distance of dm from axis.

Physical Nature & Unit: Tensor quantity (scalar for fixed axis). SI Unit: $kg \cdot m^2$. Dim: $[M L^2 T^0]$.

Radius of Gyration (k): $I = M k^2 \Rightarrow k = \sqrt{(I / M)}$

Reduced Mass (2 Particles): $I_{cm} = \mu r^2$ where $\mu = \frac{m_1 m_2}{m_1 + m_2}$

1. Parallel Axis Theorem (Any 3D Body):
$I = I_{cm} + M d^2$
CRITICAL: *One axis MUST pass through the Centre of Mass! d = distance between parallel axes.*

2. Perpendicular Axis Theorem (2D Plane Lamina Only):
$I_z = I_x + I_y$
Valid ONLY for flat laminar bodies in xy-plane. Axes x, y lie in plane, z is ⊥ to plane.

7. MASTER MOMENT OF INERTIA (I) TABLE FOR STANDARD RIGID BODIES

Body / Geometry	Axis of Rotation	Moment of Inertia (I)	Radius of Gyration (k)
Thin Circular Ring (Mass M, Radius R)	Through center, ⊥ to plane	$M R^2$	R
	About any diameter	$\frac{1}{2} M R^2$	$R / \sqrt{2}$
	About tangent in plane / ⊥ to plane	$\frac{3}{2} M R^2$ / $2 M R^2$	$\sqrt{(3/2)} R$ / $\sqrt{2} R$
Uniform Thin Disc (Mass M, Radius R)	Through center, ⊥ to plane	$\frac{1}{2} M R^2$	$R / \sqrt{2}$
	About any diameter	$\frac{1}{4} M R^2$	$R / 2$
	About tangent in plane / ⊥ to plane	$\frac{5}{4} M R^2$ / $\frac{3}{2} M R^2$	$\sqrt{(5)/2} R$ / $\sqrt{(3/2)} R$
Thin Uniform Rod (Length L)	Through COM, ⊥ to length	$\frac{1}{12} M L^2$	$L / \sqrt{12}$
	Through one end, ⊥ to length	$\frac{1}{3} M L^2$	$L / \sqrt{3}$
Solid Cylinder	Central longitudinal axis	$\frac{1}{2} M R^2$	$R / \sqrt{2}$
Hollow Thin Cylinder / Hoop	Central longitudinal axis	$M R^2$	R
Solid Sphere	About any diameter (through COM)	$\frac{2}{5} M R^2$	$\sqrt{(2/5)} R$
	About any tangent line	$\frac{7}{5} M R^2$	$\sqrt{(7/5)} R$
Spherical Shell (Hollow)	About any diameter	$\frac{2}{3} M R^2$	$\sqrt{(2/3)} R$
	About any tangent line	$\frac{5}{3} M R^2$	$\sqrt{(5/3)} R$
Rectangular Plate / Sheet (Sides a, b)	Through COM, ⊥ to plate plane	$M \frac{(a^2 + b^2)}{12}$	$\sqrt{[(a^2+b^2)/12]}$
Square Plate (Side L)	Through COM, parallel to one side	$\frac{1}{12} M L^2$	$L / \sqrt{12}$

8. TORQUE & ROTATIONAL DYNAMICS

Torque Vector: $\tau = r \times F$

Magnitude: $\tau = r F \sin\theta = F r_\perp = F_\perp r$

$r_\perp$ = moment arm (⊥ distance to line of action).

Law of Rotation (Rigid Body):
$\tau_{ext,\,net} = I \alpha$ *(Analogue of **F** = m **a**)*

Conditions for Static Equilibrium:
1. Translational: $\sum F_{ext} = 0$
2. Rotational: $\sum \tau_{ext} = 0$ (about any arbitrary point)

9. ANGULAR MOMENTUM & CONSERVATION

Particle Angular Momentum:
$L = r \times p = m (r \times v) \Rightarrow |L| = m v r \sin\theta = p r_\perp$

Rigid Body (Fixed Axis): $L = I \omega$

Torque & Angular Momentum: $\tau_{ext} = \frac{dL}{dt}$

Conservation of Angular Momentum:
If $\tau_{ext} = 0 \Rightarrow L = const \Rightarrow I_1 \omega_1 = I_2 \omega_2$
As Moment of Inertia (I) increases, Angular Speed (ω) decreases.

Angular Impulse: $J_\theta = \int \tau \, dt = \Delta L = I (\omega_f - \omega_i)$

Principle of Moments: $F_1 d_1 = F_2 d_2$ *(Load × Load Arm = Effort × Effort Arm)*

10. KINEMATICS & WORK-ENERGY ANALOGY TABLE

Linear Motion Quantity	Rotational Analogue	Rotational Formula / Equation	Conditions / Notes
Displacement s	Angular Displacement θ	$s = \theta r$	Tangential relation
Velocity v	Angular Velocity ω	$v = \omega \times r \Rightarrow v = \omega r$	Direction given by Right-Hand Rule
Acceleration a	Angular Acceleration α	$a_t = \alpha r$, $a_c = \omega^2 r = v^2/r$	$a_{total} = a_t + a_c$
Equation 1	Constant α Kinematics	$\omega = \omega_0 + \alpha t$	Constant angular acceleration only
Equation 2	Constant α Kinematics	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$	Constant angular acceleration only
Equation 3	Constant α Kinematics	$\omega^2 - \omega_0^2 = 2 \alpha \theta$	Constant angular acceleration only
Kinetic Energy	Rotational Kinetic Energy	$K.E._{rot} = \frac{1}{2} I \omega^2 = \frac{L^2}{2I}$	Fixed axis / pure rotation
Work Done	Rotational Work	$W = \int \tau \, d\theta$ *(Uniform $\tau \Rightarrow W = \tau \theta$)*	Work-Energy Theorem: $W_{net} = \Delta K$
Power	Rotational Power	$P = \tau \cdot \omega = \tau \omega \cos \varphi$	Instantaneous power

11. RIGID BODY KINEMATICS & PURE ROLLING MECHANICS

Velocity of Point on Rigid Body:

$V_B = V_A + \omega \times r_{B/A}$

Rigid Body Distance Constraint:

Velocity components of two points along the line joining them must be equal:

$V_A \cos \theta_1 = V_B \cos \theta_2$

> **Pure Rolling Condition (Stationary Surface):**
> Velocity of point of contact relative to ground is zero:
> $V_{cm} = R \omega \mid a_{cm} = R \alpha$

Slipping Conditions:

- Forward Slipping: $V_{cm} > R \omega$ (Friction f_k acts backward)
- Backward Slipping: $V_{cm} < R \omega$ (Friction f_k acts forward)

Velocity of General Point on Rolling Body:

$V_p = \sqrt{[V_{cm}^2 + (r \omega)^2 + 2 V_{cm} r \omega \cos \theta]}$

At distance d from contact point: $V_p = 2 V_{cm} \sin(\theta/2)$

Velocities at Specific Points (Pure Rolling):

- Topmost Point: $V_{top} = V_{cm} + R \omega = 2 V_{cm}$
- Center of Mass: $V_{mid} = V_{cm}$
- Bottommost Contact Point: $V_{bottom} = 0$
- Mid-Height Lateral Points: $V = \sqrt{2} V_{cm}$

12. TOTAL ENERGY & ANGULAR MOMENTUM IN ROLLING

Total Rolling Kinetic Energy:

$$K_{total} = K_{trans} + K_{rot} = \frac{1}{2} M V_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

> **Energy in terms of Radius of Gyration (k):**
> $$K_{total} = \frac{1}{2} M V_{cm}^2 \left(1 + \frac{k^2}{R^2}\right)$$
> *Ratio:* $\frac{K_{rot}}{K_{total}} = \frac{k^2/R^2}{1 + k^2/R^2}$, $\frac{K_{trans}}{K_{total}} = \frac{1}{1 + k^2/R^2}$

Angular Momentum about Origin O (Combined Motion):

$L_O = L_{cm} + r_{cm} \times M V_{cm} = I_{cm} \omega + r_{cm} \times M V_{cm}$

For pure rolling on horizontal ground: $L_O = I_{cm} \omega + M V_{cm} R = M V_{cm} R (1 + k^2/R^2)$

13. ROLLING DOWN AN INCLINED PLANE

Linear Acceleration down Incline (Angle θ):

$$a = \frac{g \sin \theta}{1 + k^2/R^2}$$

Velocity at Bottom (Height h):

$$v = \sqrt{\frac{2 g h}{1 + k^2/R^2}}$$

Time of Descent:

$$t = \frac{1}{\sin \theta} \sqrt{\frac{2 h}{g}} \left(1 + \frac{k^2}{R^2}\right)$$

> **Minimum Static Friction for Pure Rolling:**
> $$\mu_{min} = \frac{\tan \theta}{1 + R^2/k^2}$$
> *Static friction acts UP the incline. No energy dissipated by static friction.*

14. JEE ADVANCED KEY RESULTS, SHORTCUTS & SPECIAL COLLISION CASES

1. Collision / Combination of Rotating Discs:
Two coaxial discs (I_1, ω_1 and I_2, ω_2) brought into contact:

• Common Angular Velocity (Same direction): $\omega_f = \frac{I_1\,\omega_1 + I_2\,\omega_2}{I_1 + I_2}$

• Common Angular Velocity (Opposite direction): $\omega_f = \frac{I_1\,\omega_1 - I_2\,\omega_2}{I_1 + I_2}$

• Energy Loss during Coupling: $\Delta K = \frac{I_1\,I_2}{2\,(I_1 + I_2)}\,(\omega_1 \mp \omega_2)^2$

2. Instantaneous Axis of Rotation (ITAR / IAOR): [Added]
Point of zero instantaneous velocity. Distance from COM:
$r_{ITAR} = \frac{V_{cm}}{\omega}$. Body behaves as undergoing pure rotation about ITAR.

3. Angular Velocity on Release of Hanging Rod:
Rod of length L hinged at top, released from horizontal:
$\omega = \sqrt{\frac{3\,g}{L}}$ | Initial angular acceleration: $\alpha = \frac{3\,g}{2\,L}$

4. Toppling / Tipping Condition on Rough Surface: [Added]
A block of width b and height h pulled by horizontal force F at top:

• Toppling occurs before sliding if: $\mu > \frac{b}{h}$

• Critical Force for Toppling: $F_{crit} = \frac{M\,g\,b}{2\,h}$

5. Center of Percussion (Impulse without Hinge Force): [Added]
Distance from hinge to point of application of impulse such that hinge reaction force is zero:
$x = \frac{k_{hinge}^2}{r_{cm}}$ *(For uniform rod hinged at end: $x = \frac{2}{3}\,L$)*

6. Fast Comparison — Inclined Plane Motion:
For sphere, disc, hollow sphere, ring rolling down same incline:
• Acceleration: $a_{Solid\ Sphere} > a_{Disc} > a_{Hollow\ Sphere} > a_{Ring}$
• Time to reach bottom: $t_{Ring} > t_{Hollow\ Sphere} > t_{Disc} > t_{Solid\ Sphere}$

NEWTON'S LAWS OF MOTION & FRICTION

Complete JEE Main & Advanced Revision Sheet

Physics — Class 11

1. FUNDAMENTALS & NEWTON'S LAWS

First & Second Laws of Motion

1st Law (Inertia): Body stays at rest or uniform motion unless acted upon by external net force $F_{net} \neq 0$. Inertia depends purely on mass.

2nd Law (Momentum): Net force equals time rate of change of linear momentum:

$$F_{net} = dp/dt = d(mv)/dt$$

If mass m is constant: $F_{net} = ma$

Components: $F_x = m\,a_x$, $F_y = m\,a_y$, $F_z = m\,a_z$

3rd Law, Momentum & Impulse

3rd Law: To every action, equal & opposite reaction ($F_{AB} = -F_{BA}$). Forces occur in pairs on *different* bodies.

Linear Momentum: $p = mv$ [Unit: kg·m/s]

Impulse (J): Change in linear momentum produced by large force over small time Δt:

$$J = \int F\,dt = F_{avg}\,\Delta t = p_f - p_i = \Delta p$$

Impulse = Area under $F - t$ graph.

2. SPECIAL DYNAMIC SYSTEMS & FORCES

Variable Mass / Liquid Jets

Force exerted by continuous fluid jet striking vertical wall:

• Jet stopped at wall: $F_{wall} = \rho A u^2$

• Jet bouncing back: $F_{wall} = 2\rho A u^2$

• Oblique impact at angle θ to normal:

$$F_{jet} = 2\rho A u^2 \cos\theta$$

Rocket Propulsion [Added]:

$F_{thrust} = v_{rel}\,(-dm/dt)$

$v = v_0 + v_{rel}\,ln(m_0/m) - gt$

Frames & Pseudo Force

• **Inertial Frame:** Unaccelerated ($a_{frame} = 0$). NLM holds valid directly.

• **Non-Inertial Frame:** Accelerated ($a_{frame} \neq 0$). NLM requires **Pseudo Force**:

$$F_{pseudo} = -m\,a_{frame}$$

Magnitude = $m\,a_0$, Direction = opposite to frame acceleration a_0.

Apparent gravity in car accelerating horizontally at a_0: $g' = \sqrt{(g^2 + a_0^2)}$

Apparent Weight in Lift

Observer measures normal reaction N on weighing machine:

• Rest / Uniform Velocity ($a = 0$): $N = mg$

• Accelerating Upwards ($a \uparrow$): $N = m(g + a)$

• Accelerating Downwards ($a \downarrow$): $N = m(g - a)$

• Free Fall ($a = g \downarrow$): $N = 0$ (Weightlessness)

• Super Fall ($a > g \downarrow$): Block leaves floor, contacts ceiling with $N = m(a - g)$

3. SPRING SYSTEMS & CONSTRAINT RELATIONS

Spring Properties & Combinations

Hooke's Law: Restoring force $F_s = -k\,x$ (where x is deformation).

Length Rule: For a given spring, $k \times l = \text{constant}$.

• If cut in ratio $m : n$, new lengths $l_1 = m\,l/(m+n)$, $l_2 = n\,l/(m+n)$:

$$k_1 = [(m + n)/m]\,k \;|\; k_2 = [(m + n)/n]\,k$$

Combinations:

• Series: $1/k_{eq} = 1/k_1 + 1/k_2 + \ldots$

• Parallel: $k_{eq} = k_1 + k_2 + \ldots$

Constraint Equations (Wedge & Pulley)

1. String Constraint: Total length of inextensible string is constant:

$$\sum T \cdot v = 0 \Rightarrow \sum T \cdot a = 0$$

2. Movable Pulley: If pulley acceleration is a_p and blocks are a_1, a_2:

$$a_p = (a_1 + a_2)/2$$

3. Wedge Constraint: Perpendicular relative velocity/acceleration at contact surface = 0 (surfaces remain in contact).

4. STANDARD CONNECTED BLOCK SYSTEMS & PULLEYS

System Configuration	Acceleration (a) Formula	Tension / Normal Forces
Atwood Machine (Masses $m_1 > m_2$ over ideal pulley)	$a = [(m_1 - m_2)/(m_1 + m_2)]\, g$	$T = [2\, m_1\, m_2 / (m_1 + m_2)]\, g$
Horizontal Table Pulley (Mass M on smooth table connected to hanging m)	$a = [m / (M + m)]\, g$	$T = [M\, m / (M + m)]\, g$
Block on Smooth Wedge (Mass m on wedge M free to move)	$a_{wedge} = (M\, g \cos\theta \sin\theta) / (M + m \sin^2\theta)$	$N = (M\, m\, g \cos\theta) / (M + m \sin^2\theta)$
Blocks in Contact (Force F pushes m_1, m_2, m_3 horizontally)	$a = F / (m_1 + m_2 + m_3)$	Contact force $N_{12} = [(m_2 + m_3)/(m_1 + m_2 + m_3)]\, F$

FRICTION & ADVANCED JEE RELATIONS

Complete JEE Main & Advanced Revision Sheet

Physics — Class 11

5. LAWS OF FRICTION & CORE CONCEPTS

Static, Limiting & Kinetic Friction

Friction opposes relative motion or tendency of relative motion along contact surface. Independent of macroscopic contact area.

• **Static Friction (f_s):** Self-adjusting force. $0 \le f_s \le f_{s,max}$

• **Limiting Friction (f_{lim}):** Max static friction before slipping:

$$f_{lim} = \mu_s N$$

• **Kinetic Friction (f_k):** Acts during active slipping:

$$f_k = \mu_k N \quad \text{(Note: } \mu_k < \mu_s\text{)}$$

Contact Force, Angle of Friction & Repose

• **Total Contact Force (R):** Vector sum of normal reaction N and friction force f:

$$R = \sqrt{(N^2 + f^2)} \Rightarrow R_{max} = N\sqrt{(1 + \mu_s^2)}$$

• **Angle of Friction (λ):** Angle made by net contact force R with normal N at verge of slipping:

$$\tan\lambda = f_{lim} / N = \mu_s \Rightarrow \lambda = \tan^{-1}(\mu_s)$$

• **Angle of Repose (α):** Max angle of incline for block to remain stationary:

$$\tan\alpha = \mu_s \Rightarrow \alpha = \lambda = \tan^{-1}(\mu_s)$$

6. INCLINED PLANES & FORCE OPTIMIZATIONS

Pulling vs. Pushing on Rough Surface

Force applied at angle θ to horizontal:

• **Pulling Force:** Normal $N = mg - F\sin\theta$

$$F_{pull} = (\mu_s\, mg) / (\cos\theta + \mu_s \sin\theta)$$

• **Pushing Force:** Normal $N = mg + F\sin\theta$

$$F_{push} = (\mu_s\, mg) / (\cos\theta - \mu_s \sin\theta)$$

Key Result: $F_{pull} < F_{push}$ (Pulling is easier than pushing). Minimum force to move block: $F_{min} = (\mu_s\, mg) / \sqrt{(1 + \mu_s^2)}$ at optimal angle $\theta = \tan^{-1}(\mu_s) = \lambda$.

Block on Rough Inclined Plane ($\theta > \alpha$)

When incline angle θ exceeds angle of repose α:

• Acceleration sliding down: $a = g\,(\sin\theta - \mu_k \cos\theta)$

• Retardation moving up: $a = g\,(\sin\theta + \mu_k \cos\theta)$

• Force parallel to plane to prevent sliding (F_{min}):

$$F_{min} = mg\,(\sin\theta - \mu_s \cos\theta)$$

• Force parallel to plane to push block up (F_{max}):

$$F_{max} = mg\,(\sin\theta + \mu_s \cos\theta)$$

7. TWO-BLOCK SYSTEMS & RELATIVE SLIPPING

Two-Block System (m on top of M)

Rough interface (μ), ground smooth:

Case A: Force F applied on bottom mass M

• Max friction on top mass: $f_{max} = \mu\, m\, g$

• Max common acceleration without slipping: $a_{max} = f_{max} / m = \mu\, g$

• Max force without slipping:

Force Applied on Top Mass m

Case B: Force F applied on top mass m

• Max acceleration of bottom block M provided by friction: $a_{max} = (\mu\, m\, g) / M$

• Max force without slipping:

$$F_{max} = (M + m)\, a_{max} = (1 + m/M)\, \mu\, m\, g$$

If $F > F_{max}$ (Slipping occurs):

$$F_{max} = (M + m)\, a_{max} = (M + m)\, \mu\, g$$

If $F > F_{max}$ (Slipping occurs):

$a_{top} = \mu\, g$ | $a_{bottom} = (F - \mu\, m\, g) / M$

$a_{top} = (F - \mu\, m\, g) / m$ | $a_{bottom} = (\mu\, m\, g) / M$

8. JEE ADVANCED SPECIAL RESULTS & SHORTCUTS

Advanced Incline & Friction Results

• **Smooth vs Rough Incline Time Ratio:** If time taken to slide down rough incline of angle θ is n times that on smooth incline:

$$\mu_k = \tan\theta\,(1 - 1/n^2)$$

• **No Slipping on Accelerating Wedge:** Horizontal acceleration a of wedge to keep block stationary w.r.t wedge:

$a = g\tan\theta$ (for smooth wedge)

$a_{min} = g\tan(\theta - \lambda)$ | $a_{max} = g\tan(\theta + \lambda)$ (for rough wedge)

Equilibrium & Special Problems

• **Lami's Theorem (3-Force Equilibrium):** For three coplanar concurrent forces in equilibrium:

$$F_1 / \sin\alpha = F_2 / \sin\beta = F_3 / \sin\gamma$$

• **Man-Cage Problem:** Man of mass m in cage of mass M pulling rope to hold cage stationary:

Tension $T = [(M + m)/2]\, g$ | Normal force $N = [(m - M)/2]\, g$

Condition to stand inside: $m \geq M$.

MOTION IN A PLANE

JEE Main & Advanced Complete Revision Formula Sheet PHYSICS SHEET 01

1. Vectors & Position Vectors

Position Vector:
$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, $|\mathbf{r}| = \sqrt{(x^2 + y^2 + z^2)}$

Unit Vector: $\hat{\mathbf{r}} = \mathbf{r} / |\mathbf{r}| = (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) / \sqrt{(x^2 + y^2 + z^2)}$

Displacement Vector (P_1 to P_2):
$\Delta\mathbf{r} = (x_2 - x_1)\hat{\mathbf{i}} + (y_2 - y_1)\hat{\mathbf{j}} + (z_2 - z_1)\hat{\mathbf{k}}$
$|\Delta\mathbf{r}| = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$

Direction Cosines (in 3D):
$\cos\alpha = x / |\mathbf{R}|$, $\cos\beta = y / |\mathbf{R}|$, $\cos\gamma = z / |\mathbf{R}|$
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Parallel Vectors Condition:
$\mathbf{a} \parallel \mathbf{b} \Rightarrow a_1/b_1 = a_2/b_2 = a_3/b_3 = m$

Vector Operations:
- Dot Product: $\mathbf{A} \cdot \mathbf{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$
- Vector Product: $|\mathbf{A} \times \mathbf{B}| = AB\sin\theta$

$\mathbf{A} \times \mathbf{B} = (A_yB_z - A_zB_y)\hat{\mathbf{i}} - (A_xB_z - A_zB_x)\hat{\mathbf{j}} + (A_xB_y - A_yB_x)\hat{\mathbf{k}}$

2. Ground-to-Ground Projectile Motion

Projection angle θ with horizontal, initial speed u.

Components:
$u_x = u\cos\theta$, $a_x = 0$, $u_y = u\sin\theta$, $a_y = -g$

Instantaneous Velocity (time t):
$v_x = u\cos\theta$, $v_y = u\sin\theta - gt$
$v = \sqrt{(v_x^2 + v_y^2)} = \sqrt{[u^2 - 2ugt\sin\theta + g^2t^2]}$
Angle with horizontal (α): $\tan\alpha = v_y / v_x = \tan\theta - (gt / u\cos\theta)$

Core Quantities:
- Time of Flight: $T = (2u\sin\theta) / g = 2u_y / g$
- Max Height: $H = (u^2\sin^2\theta) / (2g) = u_y^2 / (2g) = gT^2 / 8$
- Range: $R = (u^2\sin 2\theta) / g = (2u_x u_y) / g$

Key Relation: $4H = R\tan\theta = (1/2)gT^2$

Complementary Angles:
Range is identical for angles θ and $(90° - \theta)$ or $(45° \pm \alpha)$.
- Max Range: $R_{max} = u^2 / g$ (at $\theta = 45°$)
- Height at R_{max}: $H = R_{max} / 4 = u^2 / (4g)$

Equation of Trajectory:
$y = x\tan\theta - (g x^2) / (2u^2\cos^2\theta) = x\tan\theta(1 - x / R)$

3. Projectile Special Cases & Shortcuts

Complementary Angles Ratios (θ and 90° − θ):
- Time of flights T_1, T_2: $T_1 / T_2 = \tan\theta$
- Max Heights H_1, H_2: $H_1 / H_2 = \tan^2\theta$
- Product of T_1, T_2: $T_1 \cdot T_2 = 2R / g$
- Product of H_1, H_2: $H_1 \cdot H_2 = R^2 / 16 \Rightarrow R = 4\sqrt{(H_1 H_2)}$
- Sum of Squares: $T_1^2 + T_2^2 = (4 R_{max}) / g$
- Sum of Heights: $H_1 + H_2 = R_{max} / 2$

Two Points at Same Height h (at t_1 and t_2):
$t_1 + t_2 = T = (2u\sin\theta)/g$
$t_1 \cdot t_2 = 2h / g \Rightarrow h = (1/2) g t_1 t_2$
Average Velocity between these points: $v_{avg} = u\cos\theta$

Energy & Momentum Relations:
- KE at Highest Point: $K_{top} = K_0\cos^2\theta$ ($K_0 = 1/2\ m\ u^2$)
- Momentum at Highest Point: $\mathbf{p}_{top} = m\ u\cos\theta\ \hat{\mathbf{i}}$
- Change in Momentum (Full Flight): $\Delta\mathbf{p} = -2\ m\ u\sin\theta\ \hat{\mathbf{j}} = -m\ g\ T\ \hat{\mathbf{j}}$

Time when Velocity ⊥ Initial Velocity:
Condition: $\mathbf{u} \cdot \mathbf{v} = 0 \Rightarrow t = u / (g\sin\theta)$

Radius of Curvature $\rho = v^2 / a_\perp$:
- At Point of Projection: $\rho = u^2 / (g\cos\theta)$
- At Highest Point: $\rho = (u^2\cos^2\theta) / g$

Max Height Throw vs Max Range:
If a person throws max horizontal distance X, max vertical height he can throw is $H_{max} = X / 2$.

4. Projection From a Height h

Case I: Horizontal Projection ($u_x = u$, $u_y = 0$):
- Time of Flight: $T = \sqrt{(2h / g)}$
- Horizontal Range: $R = u\ T = u\sqrt{(2h / g)}$
- Velocity at time t: $v = \sqrt{(u^2 + g^2t^2)}$, $\tan\theta = gt / u$
- Striking Velocity: $v_{strike} = \sqrt{(u^2 + 2gh)}$

Case II: Oblique Upward (at angle θ above horizontal):
Equation: $-h = (u\sin\theta)\ t_1 - (1/2)\ g\ t_1^2$
$t_1 = [T + \sqrt{(T^2 + 8h/g)}] / 2$ (where $T = 2u\sin\theta / g$)
Range $R = (u\cos\theta)\ t_1$

Case III: Oblique Downward (at angle θ below horizontal):
Equation: $h = (u\sin\theta)\ t_2 + (1/2)\ g\ t_2^2$
$t_2 = [\sqrt{(T^2 + 8h/g)} - T] / 2$
Range $R = (u\cos\theta)\ t_2$

5. Projectile Motion on an Inclined Plane

Incline angle = β (or α), Projection angle w.r.t incline = α (or θ).

Motion UP the Incline:

$a_x = -g \sin\beta$, $a_y = -g \cos\beta$

- Time of Flight: $T = (2u \sin\alpha) / (g \cos\beta)$
- Max Height from Plane: $H = (u^2 \sin^2\alpha) / (2g \cos\beta)$
- Range on Incline R:

$R = (2u^2 \sin\alpha \cos(\alpha + \beta)) / (g \cos^2\beta) = (u^2 / g \cos^2\beta) [\sin(2\alpha + \beta) - \sin\beta]$

- Maximum Range ($\alpha = 45^\circ - \beta/2$):

$R_{max,\,up} = u^2 / [g (1 + \sin\beta)]$

Motion DOWN the Incline:

$a_x = +g \sin\beta$, $a_y = -g \cos\beta$

- Time of Flight: $T = (2u \sin\alpha) / (g \cos\beta)$
- Range on Incline R:

$R = (2u^2 \sin\alpha \cos(\alpha - \beta)) / (g \cos^2\beta) = (u^2 / g \cos^2\beta) [\sin(2\alpha - \beta) + \sin\beta]$

- Maximum Range ($\alpha = 45^\circ + \beta/2$):

$R_{max,\,down} = u^2 / [g (1 - \sin\beta)]$

6. Relative Motion & Overtaking

Relative Velocity: $\mathbf{v}_{AB} = \mathbf{v}_A - \mathbf{v}_B$, $\mathbf{v}_{BA} = \mathbf{v}_B - \mathbf{v}_A$

1D Overtaking / Chasing (Separation d, Lengths L_1, L_2):

- Same direction: $t = (d + L_1 + L_2) / (v_1 - v_2)$
- Opposite direction: $t = (d + L_1 + L_2) / (v_1 + v_2)$
- Accelerated: $d + L_1 + L_2 = (u_1 - u_2)t + (1/2)(a_1 - a_2)t^2$

Minimum Distance / Collision Avoidance:

To avoid collision of two braking vehicles behind each other:

$d \geq (u_1 - u_2)^2 / (2\, a_{rel})$

Closest Approach of 2 Particles in 2D:

$d_{min} = |\mathbf{r}_0 \times \mathbf{v}_{rel}| / |\mathbf{v}_{rel}|$, Time: $t = -(\mathbf{r}_0 \cdot \mathbf{v}_{rel}) / |\mathbf{v}_{rel}|^2$

Escalator Problem:

Man walking time t_1, Moving escalator time t_2.

- Walking on moving escalator (same direction): $t = (t_1\, t_2) / (t_1 + t_2)$
- Walking against moving escalator: $t = (t_1\, t_2) / (t_1 - t_2)$

7. Rain-Man & Swimmer-River Problems

Rain-Man Problem:

$\mathbf{v}_{rm} = \mathbf{v}_r - \mathbf{v}_m$

If rain falls vertically (v_r) and man moves horizontally (v_m):

- Relative speed: $v_{rm} = \sqrt{(v_r^2 + v_m^2)}$
- Umbrella angle with vertical: $\tan\theta = v_m / v_r$

Swimmer-River Problem (River width d, speed u, Swimmer v):

Resultant velocity: $\mathbf{v}_{net} = \mathbf{v}_{swimmer/water} + \mathbf{v}_{river} = \mathbf{v} + \mathbf{u}$

- **Shortest Time to Cross:**

Head perpendicular to bank ($\theta = 0°$):

$t_{min} = d / v$, Drift: $x = u \cdot (d / v)$

- **Shortest Path / Direct Crossing (Zero Drift, if v > u):**

Head upstream at angle θ to normal:

$\sin\theta = u / v$, Time: $t = d / (v \cos\theta) = d / \sqrt{(v^2 - u^2)}$

- **Minimum Drift (if v < u):**

Head upstream at angle $\sin\theta = v / u$ w.r.t. normal.

Min Drift: $x_{min} = d \sqrt{(u^2 - v^2)} / v$, Time: $t = d / (v \cos\theta)$

8. Kinematics of Circular Motion

Angular Variables:

$\omega = d\theta / dt = 2\pi / T = 2\pi f$, $\alpha = d\omega / dt = d^2\theta / dt^2$

Vector relation: $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, $v = \omega\, r$

Equations of Angular Motion (Constant α):

$\omega = \omega_0 + \alpha t$

$\Delta\theta = \omega_0 t + (1/2)\alpha t^2$

$\omega^2 = \omega_0^2 + 2\alpha(\Delta\theta)$

Accelerations in Circular Motion:

- Centripetal / Radial (dir: center): $a_c = v^2 / r = \omega^2 r = v\,\omega$
- Tangential (dir: tangent): $a_t = dv / dt = \alpha\, r$
- Total Acceleration: $\mathbf{a} = \mathbf{a}_c + \mathbf{a}_t \Rightarrow a = \sqrt{(a_c^2 + a_t^2)}$
- Angle with radius: $\tan\varphi = a_t / a_c$

Uniform Circular Motion (UCM): $v = \text{const}$, $a_t = 0$, $a = a_c = \omega^2 r$

9. Dynamics of Circular Motion & Banking

Centripetal Force: $F_c = m\, v^2 / r = m\, \omega^2\, r$

Level Circular Road (Friction μ):

Max safe speed to avoid skidding: $v_{max} = \sqrt{(\mu\, r\, g)}$

Banked Road (Angle θ, Friction μ):

- Optimum Speed (No friction needed): $v_0 = \sqrt{(r\, g \tan\theta)}$
- Max Safe Speed (Skidding outward):

$v_{max} = \sqrt{[\, r\, g\, (\tan\theta + \mu) / (1 - \mu \tan\theta)\,]}$

- Min Safe Speed (Skidding inward):

$v_{min} = \sqrt{[\, r\, g\, (\tan\theta - \mu) / (1 + \mu \tan\theta)\,]}$

Bending of a Cyclist:

Angle with vertical: $\tan\theta = v^2 / (r\, g)$

Conical Pendulum [Added]:

String length L, angle θ with vertical.

$T \cos\theta = mg$, $T \sin\theta = m\, \omega^2\, (L \sin\theta)$

Time period: $T_p = 2\pi \sqrt{(L \cos\theta / g)}$

10. Vertical Circular Motion (VCM)

Mass m tied to string of length R.

Tension at Angle θ from Bottom:

$T = (m\, v^2 / R) + m\, g \cos\theta$

Critical Conditions for Full Loop (String):

- Min Speed at Bottom: $v_{bottom,\,min} = \sqrt{(5\, g\, R)}$
- Min Speed at Top: $v_{top,\,min} = \sqrt{(g\, R)}$
- Min Speed at Horizontal: $v_{mid,\,min} = \sqrt{(3\, g\, R)}$
- Tension at Bottom (min speed): $T_{bottom} = 6\, mg$
- Tension at Top (min speed): $T_{top} = 0$
- Difference in Tension: $T_{bottom} - T_{top} = 6\, mg$

Condition for Light Rod (Rigid):

Min speed at bottom to complete loop: $v_{bottom} = \sqrt{(4\, g\, R)}$

Oscillation vs Leaving Circle:

- Oscillates in lower hemisphere ($\theta \leq 90°$): $v_0 \leq \sqrt{(2\, g\, R)}$
- Leaves circular path ($90° < \theta < 180°$): $\sqrt{(2\, g\, R)} < v_0 < \sqrt{(5\, g\, R)}$

SIMPLE HARMONIC MOTION (SHM)

JEE MAIN & ADVANCED HIGH-DENSITY FORMULA REVISION SHEET

PHYSICS REVISION

1. FUNDAMENTAL CONDITION & DIFFERENTIAL EQUATION

Linear S.H.M.

Restoring Force is directly proportional to displacement towards mean position:

$$F = -k\,x \quad \Rightarrow \quad a = -(k/m)\,x = -\omega^2 x$$

Differential Equation:

$$(d^2x / dt^2) + \omega^2 x = 0$$

where angular frequency $\omega = \sqrt{(k/m)} = 2\pi / T = 2\pi f$

Angular S.H.M.

Restoring Torque is proportional to angular displacement:

$$\tau = -C\,\theta \quad \Rightarrow \quad \alpha = -(C/I)\,\theta = -\omega^2\,\theta$$

Differential Equation:

$$(d^2\theta / dt^2) + \omega^2\,\theta = 0$$

where $\omega = \sqrt{(C/I)}$, C = torsional constant, I = moment of inertia.

2. KINEMATICS OF SIMPLE HARMONIC MOTION

Quantity	Formula / Expression	Extreme & Mean Values
Displacement (x)	$x(t) = A\sin(\omega t + \varphi_0)$ or $x(t) = A\cos(\omega t + \varphi_0)$ Phase angle = $(\omega t + \varphi_0)$; Initial phase = φ_0	At Mean (x = 0): $x = 0$ At Extreme (x = ±A): $x = \pm A$
Velocity (v)	$v = dx/dt = A\omega\cos(\omega t + \varphi_0) = \pm\omega\sqrt{(A^2 - x^2)}$	At Mean (x = 0): $v_{max} = \pm A\omega$ At Extreme (x = ±A): $v = 0$
Acceleration (a)	$a = dv/dt = -A\omega^2\sin(\omega t + \varphi_0) = -\omega^2 x$	At Mean (x = 0): $a = 0$ At Extreme (x = ±A): $a_{max} = \pm\omega^2 A$
Relations from Data	$A = \sqrt{[\,(v_1^2x_2^2 - v_2^2x_1^2) / (v_1^2 - v_2^2)\,]}$ $\omega = \sqrt{[\,(v_1^2 - v_2^2) / (x_2^2 - x_1^2)\,]}$	$\omega = a_{max} / v_{max}$ $A = v_{max}^2 / a_{max} \Rightarrow T = 2\pi\,(v_{max} / a_{max})$

Phase Differences (Δφ)

- Phase difference between **x** and **v**: **π / 2 (90°)** (v leads x)
- Phase difference between **v** and **a**: **π / 2 (90°)** (a leads v)
- Phase difference between **x** and **a**: **π (180°)** (opposite phase)

Standard Phase Values (φ_0 from Mean)

- At $x = 0$ moving +ve: $\varphi_0 = 0 \Rightarrow x = A\sin(\omega t)$
- At $x = A/2$ moving +ve: $\varphi_0 = \pi/6$ (30°)
- At $x = A/\sqrt{2}$ moving +ve: $\varphi_0 = \pi/4$ (45°)
- At $x = \sqrt{3}A/2$ moving +ve: $\varphi_0 = \pi/3$ (60°)

3. TIME INTERVAL CALCULATIONS

Standard Time Intervals (Starting from x = 0)

From $x = 0$ to $x = A/2$:	**t = T / 12**
From $x = A/2$ to $x = A$:	**t = T / 6**
From $x = 0$ to $x = A/\sqrt{2}$:	**t = T / 8**
From $x = 0$ to $x = \sqrt{3}A/2$:	**t = T / 6**
From $x = 0$ to $x = A$:	**t = T / 4**

General Time Formula

Time taken to travel between phases φ_1 and φ_2:

$$\Delta t = \Delta\varphi / \omega = [\,(\varphi_2 - \varphi_1) / 2\pi\,]\,T$$

Meeting Time of 2 Particles:

If two particles with same A, ω cross each other: $x_1 = x_2 \Rightarrow \Delta\varphi$ can be found using phasor circle projection.

4. ENERGY OF SIMPLE HARMONIC MOTION

Kinetic Energy (K.E.)

$K = \frac{1}{2} m v^2$

$K = \frac{1}{2} m \omega^2 (A^2 - x^2)$

$K = \frac{1}{2} m \omega^2 A^2 \cos^2(\omega t + \varphi)$

- $K_{max} = \frac{1}{2} m \omega^2 A^2$ (at $x = 0$)
- $K_{min} = 0$ (at $x = \pm A$)

Potential Energy (P.E.)

$U = \frac{1}{2} k x^2 = \frac{1}{2} m \omega^2 x^2$

$U = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t + \varphi)$

- $U_{max} = \frac{1}{2} m \omega^2 A^2$ (at $x = \pm A$)
- $U_{min} = 0$ (at $x = 0$, assuming $U_0 = 0$)

Total Energy (E)

$E = K + U = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} k A^2$

Constant for conservative system.

If $U_0 \neq 0$ at mean position Added:

$E = U_0 + \frac{1}{2} m \omega^2 A^2$

Important Energy Relations & Frequency Notes:

- At $x = \pm A / \sqrt{2} \Rightarrow$ K.E. = P.E. = E / 2 (Equal energy point)
- At $x = \pm A / 2 \Rightarrow$ P.E. = E / 4 and K.E. = 3E / 4
- **Frequency Note:** If particle oscillates with frequency f (or angular frequency ω), then K.E. and P.E. oscillate with frequency **2f** (or **2ω**). Total energy remains constant (frequency = 0).
- **Average Energies over a complete cycle:** $\langle K.E. \rangle_{time} = \frac{1}{4} m \omega^2 A^2 = E / 2$, $\langle P.E. \rangle_{time} = E / 2$ Added
- **Average Energies over space (x = −A to +A):** $\langle K.E. \rangle_{space} = (2/3) E$, $\langle P.E. \rangle_{space} = (1/3) E$ Added

SHM SYSTEMS & COMBINATIONS

SPRING SYSTEMS, PENDULUMS & LIQUID COLUMNS

PAGE 2

5. SPRING-MASS SYSTEMS & CUTTING OF SPRINGS

Basic Spring-Mass Systems

- **Horizontal / Vertical Block:** $T = 2\pi \sqrt{(m / k)}$
- **Two-Body System (Spring between 2 masses):** $T = 2\pi \sqrt{(\mu / k)}$ where reduced mass $\mu = (m_1 m_2) / (m_1 + m_2)$
- **Inertia & Force Factors:** $T = 2\pi \sqrt{(\text{Inertia Factor} / \text{Force Factor})}$

Cutting of Springs & Stiffness Law

- Stiffness is inversely proportional to length: $k \propto 1 / L \Rightarrow k L = \text{constant}$
- If spring of length L is cut into lengths L_1 & L_2: $k_1 = k (L_1 + L_2) / L_1$, $k_2 = k (L_1 + L_2) / L_2$
- Ratio of spring constants: $k_1 / k_2 = L_2 / L_1$
- **Ratio example (1:2:3):** $k_1 = 6k$, $k_2 = 3k$, $k_3 = 2k$

Combination	Equivalent Spring Constant (k_{eq})	Time Period (T)
Series Combination (Same force F, total $x = x_1 + x_2$)	$1 / k_{eq} = (1 / k_1) + (1 / k_2) \Rightarrow k_{eq} = (k_1 k_2) / (k_1 + k_2)$	$T = 2\pi \sqrt{[m (k_1 + k_2) / (k_1 k_2)]}$ $T_{series}^2 = T_1^2 + T_2^2$
Parallel Combination (Same elongation x, $F_{total} = F_1 + F_2$)	$k_{eq} = k_1 + k_2$	$T = 2\pi \sqrt{[m / (k_1 + k_2)]}$ $(1 / T_{parallel}^2) = (1 / T_1^2) + (1 / T_2^2)$
Inclined Springs (Angle θ)	$k_{eq} = k_1 + 2k_2 \cos^2\theta$ E.g. For $\theta = 60°$: $k_{eq} = k + 2k(1/4) = 3k/2$	$T = 2\pi \sqrt{(2m / 3k)}$

6. SIMPLE, PHYSICAL & TORSIONAL PENDULUMS

Simple Pendulum & Effective Gravity

Physical / Compound Pendulum

Rigid body pivoted at point O at distance d from C.M.:

$$T = 2\pi \sqrt{[I / (m g d)]}$$

where I = moment of inertia about pivot O = $I_{cm} + m d^2$.

Special Standard Cases:

- Standard Time Period (small θ): $T = 2\pi \sqrt{(l / g_{eff})}$
- **Second's Pendulum:** T = 2 seconds ⇒ length $l \approx 1$ meter (99.3 cm).
- **Lift accelerating upwards (a):** $g_{eff} = g + a \Rightarrow T = 2\pi \sqrt{[l / (g+a)]}$
- **Lift accelerating downwards (a):** $g_{eff} = g - a \Rightarrow T = 2\pi \sqrt{[l / (g-a)]}$
- **Free Fall / Weightlessness:** $g_{eff} = 0 \Rightarrow T =$ &infty; (does not oscillate).
- **Horizontally accelerating car (accel a):** $g_{eff} = \sqrt{(g^2 + a^2)} \Rightarrow T = 2\pi \sqrt{[l / (g^2 + a^2)^{1/4}]}$
- **Bob immersed in liquid (Density σ, liquid ρ):** $g_{eff} = g (1 - \rho/\sigma) \Rightarrow T = 2\pi \sqrt{[l / \{g(1 - \rho/\sigma)\}]}$
- **Electric field E down on charge +q:** $g_{eff} = g + (qE/m)$

- **Uniform Thin Rod (length L) pivoted at end:** $d = L/2$, $I = mL^2/3 \Rightarrow T = 2\pi \sqrt{(2L / 3g)}$
- **Uniform Circular Ring (radius R) pivoted at rim:** $d = R$, $I = 2mR^2 \Rightarrow T = 2\pi \sqrt{(2R / g)}$
- **Uniform Disc (radius R) pivoted at rim** Added: $d = R$, $I = (3/2)mR^2 \Rightarrow T = 2\pi \sqrt{(3R / 2g)}$

Torsional Pendulum

Restoring torque $\tau = -C\theta$

$$T = 2\pi \sqrt{(I / C)}$$

where C = torsional constant (torque per unit twist) and I = M.I. about suspension wire axis.

Very Long Pendulum (Comparable to Earth Radius R)

If length l is comparable to Earth's radius R:

$$T = 2\pi \sqrt{[1 / \{ g (1/l + 1/R) \}]}$$

• If $l >> R$ (Infinitely long pendulum, l → &infty;):

$$T = 2\pi \sqrt{(R / g)} \approx 84.6 \text{ minutes}$$

7. FLUID & ATMOSPHERIC SHM SYSTEMS

Oscillation of Liquid Column in U-Tube

U-tube with cross-sectional area A, filled with liquid of density ρ and total liquid length L:

$$T = 2\pi \sqrt{(L / 2g)} \text{ or } T = 2\pi \sqrt{[m / (2 \rho g A)]}$$

If tube is inclined at angles α and β Added: $T = 2\pi \sqrt{[L / \{g(\sin\alpha + \sin\beta)\}]}$

Floating Cylinder / Block

Body of mass m, density σ, cross-section A floating in liquid of density ρ (submerged depth h):

$$T = 2\pi \sqrt{[m / (\rho g A)]} = 2\pi \sqrt{(h / g)}$$

where $h = (\sigma / \rho) H$ is the equilibrium submerged height.

8. SUPERPOSITION OF TWO SIMPLE HARMONIC MOTIONS

Same Direction & Same Frequency

Superposition of $x_1 = A_1 \sin(\omega t)$ and $x_2 = A_2 \sin(\omega t + \theta)$:

Resultant: $x = A \sin(\omega t + \varphi)$

$A = \sqrt{[\, A_1^2 + A_2^2 + 2 A_1 A_2 \cos\theta \,]}$

$\tan\varphi = (A_2 \sin\theta) / (A_1 + A_2 \cos\theta)$

- Constructive ($\theta = 0$): $A_{max} = A_1 + A_2$
- Destructive ($\theta = \pi$): $A_{min} = |A_1 - A_2|$

Perpendicular SHMs (Lissajous Figures) Added

Superposition of $x = A \sin(\omega t)$ and $y = B \sin(\omega t + \delta)$:

$(x^2/A^2) + (y^2/B^2) - (2xy/AB) \cos\delta = \sin^2\delta$

- If $\delta = 0 \Rightarrow$ Straight Line: $y = (B/A)\, x$
- If $\delta = \pi/2 \Rightarrow$ Ellipse: $(x^2/A^2) + (y^2/B^2) = 1$ (Circle if A=B)
- If $\delta = \pi \Rightarrow$ Straight Line: $y = -(B/A)\, x$

9. METHOD OF SMALL OSCILLATIONS (POTENTIAL ENERGY METHOD)

For a system near stable equilibrium position x_0:

1. Find equilibrium position x_0 by setting $\mathbf{dU / dx\,|_{x=x_0} = 0}$
2. Check stability: $\mathbf{d^2U / dx^2\,|_{x=x_0} > 0}$ (Stable equilibrium condition)
3. Calculate angular frequency of small oscillation:

   $\mathbf{\omega = \sqrt{[\, U''(x_0) / m \,]} = \sqrt{[\, (d^2U / dx^2\,|_{x=x_0}) / m \,]}}$
4. Time Period: $\mathbf{T = 2\pi / \omega = 2\pi \sqrt{[\, m / U''(x_0) \,]}}$

10. DAMPED HARMONIC OSCILLATIONS

Equation of Motion & Amplitude

Damping force $F_d = -b\, v$ (where b = damping constant). Net force:

$m\,(d^2x / dt^2) + b\,(dx / dt) + k\,x = 0$

Displacement equation:

$x(t) = A_0\, e^{-bt / 2m} \cos(\omega' t + \varphi)$

Damped Amplitude: **$A(t) = A_0\, e^{-bt / 2m}$**

Frequency & Energy Decay

Damped Angular Frequency:

$\omega' = \sqrt{[\, (k/m) - (b^2 / 4m^2) \,]} = \sqrt{[\, \omega_0^2 - (b / 2m)^2 \,]}$

Mechanical Energy at time t:

$E(t) = \tfrac{1}{2}\, k\, A_0^2\, e^{-bt / m} = E_0\, e^{-bt / m}$

Relaxation time $\tau = m / b \Rightarrow E(t) = E_0\, e^{-t / \tau}$ Added

Quality factor $Q = \omega_0\, \tau = (m\, \omega_0) / b$ Added

11. FORCED OSCILLATIONS & RESONANCE

Driving Force & Amplitude

External Periodic Driving Force: $F(t) = F_0 \cos(\omega_d\, t)$

Steady State Displacement: $x(t) = A \sin(\omega_d\, t + \varphi)$

Amplitude of Forced Oscillation:

$A = (F_0 / m) / \sqrt{[\, (\omega_0^2 - \omega_d^2)^2 + (b\, \omega_d / m)^2 \,]}$

Resonance Conditions

- **Resonance:** When driving frequency matches natural frequency: $\omega_d = \omega_0 = \sqrt{(k/m)}$
- At resonance, Amplitude A and Velocity Amplitude v_m reach maximum.
- **Amplitude at Resonance (Small damping):** $A_{max} \approx F_0 / (b\, \omega_0)$
- **Sharpness of Resonance:** Higher Quality factor Q implies sharper resonance peak.

12. JEE ADVANCED SHORTCUT FORMULAS & SPECIAL CASES

Tunnel along Earth's Chord/Diameter:	Particle dropped in tunnel through Earth: $T = 2\pi\sqrt{(R / g)} \approx 84.6\ \text{min}$ (independent of chord length or position). Added
Piston in Gas Cylinder (Adiabatic):	Mass m on piston of area A in volume V at pressure P: $T = 2\pi\sqrt{[\,(m\,V) / (\gamma\,P\,A^2)\,]}$ Added
Rolling Cylinder + Spring:	Cylinder of mass m with spring k attached at center (pure rolling): $T = 2\pi\sqrt{[\,(3m) / (2k)\,]}$ Added
Conical Pendulum (for comparison):	Mass revolving in horizontal circle of radius r (string length L, angle θ with vertical): $T = 2\pi\sqrt{[\,(L\cos\theta) / g\,]}$ Added

WORK, POWER, ENERGY & COLLISIONS

Complete Ultimate JEE (Main & Advanced) Revision Sheet

Units & Dimensions: $[M\,L^2\,T^{-2}]$

1 eV = 1.6×10^{-19} J | 1 kWh = 3.6×10^{6} J

1. WORK DONE MECHANICS

Constant & Multiple Forces

Work by Constant Force:

$$W = F \cdot S = F\,S\cos\theta$$

- *Positive Work: $0° \le \theta < 90°$*
- *Negative Work: $90° < \theta \le 180°$*
- *Zero Work: $\theta = 90°$ OR $S = 0$ OR $F = 0$*

Work by Multiple Forces:

$$W_{net} = (\sum F) \cdot S = W_1 + W_2 + W_3 + \dots$$

Variable Force & Graphical Calculation

Work by Variable Force:

$$W = \int dW = \int F \cdot ds = \int (F_x dx + F_y dy + F_z dz)$$

- Graphical Method: W = Area under F_x vs x graph.
- 3D Vector Form: $\mathbf{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$, $d\mathbf{s} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

Work Done by Specific Systems

1. Inclined Plane: $W_g = m\,g\,h = m\,g\,l\sin\theta$

2. Staircase / Step Motion: $W_g = m\,g\,(h_1 + h_2 + \dots + h_n) = m\,g\,h_{total}$

3. Spring Force: $F_s = -k\,x$

$$W_s = -\tfrac{1}{2}\,k\,(x_f^2 - x_i^2)$$

W_{ext} against spring = $+\tfrac{1}{2}\,k\,(x_f^2 - x_i^2)$

Standard JEE Result: Pulling Hanging Chain

Uniform chain of Mass M, Length L, hanging over smooth table with (1/n)th portion hanging.

Work done in pulling hanging portion back onto table:

$$W = M\,g\,L / (2\,n^2)$$

Target Center of Mass raised by $h_{cm} = L / (2n)$

2. CONSERVATIVE FORCES & POTENTIAL ENERGY

Conservative vs Non-Conservative Forces

Conservative Force: Work done is path-independent; $W_{round\text{-}trip} = 0$. (e.g., Gravitational, Electrostatic, Spring).

Non-Conservative Force: Work is path-dependent; $W_{round\text{-}trip} \neq 0$. (e.g., Friction, Viscous drag).

Potential Energy (U) Definition & Differential Relations

$$\Delta U = U_f - U_i = -W_{conservative} = -\int_{r_i}^{r_f} F \cdot dr$$

1D Relation: $F(x) = -\,dU / dx$

3D Gradient: $F = -\,(\partial U/\partial x\,\hat{i} + \partial U/\partial y\,\hat{j} + \partial U/\partial z\,\hat{k})$

Equilibrium State	Force Condition	First Derivative	Second Derivative (d^2U/dx^2)	Potential Energy (U)
Stable Equilibrium	F = 0	dU/dx = 0	**Positive (> 0)**	Minimum
Unstable Equilibrium	F = 0	dU/dx = 0	**Negative (< 0)**	Maximum
Neutral Equilibrium	F = 0	dU/dx = 0	Zero (= 0)	Constant

3. WORK-ENERGY THEOREM & ENERGY CONSERVATION

Work-Energy Theorem (Universal)

$$W_{all\ forces} = \Delta K = K_f - K_i$$

$$W_c + W_{nc} + W_{ext} + W_{pseudo} = \tfrac{1}{2}\,m\,v_f^2 - \tfrac{1}{2}\,m\,v_i^2$$

Valid for point masses and rigid bodies in inertial & non-inertial frames.

Mechanical Energy Conservation

If $W_{nc} = 0$ and $W_{ext} = 0$:

$$E_{total} = K + U = \text{Constant}$$

$$K_i + U_i = K_f + U_f$$

If non-conservative forces act: $W_{nc} = \Delta E_{mech} = (K_f + U_f) - (K_i + U_i)$

Kinetic Energy & Linear Momentum

$$K = p^2 / (2m) \quad | \quad p = \sqrt{(2\,m\,K)}$$

- m constant: $K \propto p^2$
- p constant: $K \propto 1/m$
- K constant: $p \propto \sqrt{m}$

Variable Mass Systems (Rocket Propulsion) Added

$$F_{thrust} = v_{rel}\,(dm/dt)$$

$$v(t) = v_0 + v_{rel}\,\ln(m_0 / m) - g\,t$$

v_{rel} = exhaust velocity relative to rocket.

4. POWER MECHANICS

Core Definitions & Units

$$P_{avg} = W / t$$

$$P_{inst} = dW / dt = F \cdot v = F\,v\cos\theta$$

- Units: 1 Watt = 1 J/s = 10^7 erg/s
- Horsepower: 1 HP = 746 W
- Slope of Work-Time graph: $\tan\theta = dW/dt = P_{inst}$
- Area under P-t graph: $\int P\,dt$ = Work done (W)

Motion with Constant Power Delivery

If Engine delivers constant power P to mass m starting from rest:

1. Velocity: $v(t) = (2\,P\,t / m)^{1/2} \Rightarrow v \propto t^{1/2}$

2. Displacement: $S(t) = (8\,P / 9\,m)^{1/2}\,t^{3/2} \Rightarrow S \propto t^{3/2}$

3. Force: $F(t) = (m\,P / 2\,t)^{1/2} \Rightarrow F \propto t^{-1/2}$

Elevator Engine Power

$$T = (M + m)(g + a)$$

$$P_{inst} = T \cdot v = (M + m)(g + a)\,v$$

For uniform speed (a = 0): P = (M + m) g v

Water Pump Efficiency & Power

$$P_{out} = (dm/dt)\,[g\,h + \tfrac{1}{2}\,v^2]$$

If outlet velocity $v \approx 0$: $P_{out} = (dm/dt)\,g\,h = \rho\,A\,v_{in}\,g\,h$

$$\text{Efficiency } \eta = (P_{output} / P_{input}) \times 100\%$$

Instantaneous Power in Projectile Motion

$$P(t) = F \cdot v = (-m\,g\,\hat{j}) \cdot [u\cos\theta\,\hat{i} + (u\sin\theta - g\,t)\,\hat{j}] = -m\,g\,(u\sin\theta - g\,t) = m\,g^2\,t - m\,g\,u\sin\theta$$

- *Power at Highest Point: P = 0 (since **F** ⊥ **v**).*
- *Power over complete flight time: $P_{avg} = 0$.*

5. VERTICAL CIRCULAR MOTION (MASS ATTACHED TO STRING)

General Relations at Angle θ from Lowest Point

$$v(\theta) = \sqrt{(u^2 - 2\,g\,R\,(1 - \cos\theta))}$$

$$T(\theta) = (m / R)\,[u^2 - g\,R\,(2 - 3\cos\theta)]$$

Lowest Point A (θ = 0°): $T_A = m\,u^2/R + m\,g$

Horizontal Point B (θ = 90°): $T_B = m\,u^2/R - 2\,m\,g$

Highest Point C (θ = 180°): $T_C = m\,u^2/R - 5\,m\,g$

Critical Velocity Conditions (Light String)

1. Complete Vertical Loop: $u \geq \sqrt{(5\,g\,R)}$

At minimum critical speed $u = \sqrt{(5\,g\,R)}$:

$$v_A = \sqrt{(5gR)},\ v_B = \sqrt{(3gR)},\ v_C = \sqrt{(gR)}$$

$$T_A = 6\,mg,\ T_B = 3\,mg,\ T_C = 0$$

2. Oscillation Condition ($\theta_{max} \leq 90°$): $u \leq \sqrt{(2\,g\,R)}$

3. String Slackening (Leaves Circular Path): $\sqrt{(2gR)} < u < \sqrt{(5gR)}$

Light Rigid Rod System Added

Since rod can sustain compression (T tension can be negative):

$$v_{min\ at\ top} = 0$$

$$u_{min\ at\ bottom\ for\ full\ loop} = \sqrt{(4\,g\,R)}$$

Slackening Angle & Projectile Path Added

String slacks when T = 0 at angle θ (> 90°):

$$\cos\theta = (u^2 - 2gR) / (3gR)$$

Particle executes projectile motion from point of slackening.

6. ONE-DIMENSIONAL COLLISIONS (HEAD-ON)

Coefficient of Restitution (e)

e = (Rel. Velocity of Separation) / (Rel. Velocity of Approach)

$$e = (v_2 - v_1) / (u_1 - u_2)$$

• Elastic: e = 1 | Inelastic: 0 < e < 1 | Perfectly Inelastic: e = 0

General 1D Final Velocity Formulas

$$v_1 = [(m_1 - e\, m_2)\, u_1 + (1 + e)\, m_2\, u_2] / (m_1 + m_2)$$

$$v_2 = [(m_2 - e\, m_1)\, u_2 + (1 + e)\, m_1\, u_1] / (m_1 + m_2)$$

Special Cases for Perfectly Elastic 1D Collisions (e = 1)

Case 1: Equal Masses ($m_1 = m_2$)

$$v_1 = u_2 \,,\; v_2 = u_1$$

Velocities interchange completely.

Case 2: Heavy Projectile ($m_1 >> m_2$)

$$v_1 \approx u_1$$

$$v_2 \approx 2u_1 - u_2$$

Case 3: Heavy Target ($m_2 >> m_1$)

$$v_1 \approx -u_1 + 2u_2$$

$$v_2 \approx u_2$$

KINETIC ENERGY LOSS & INELASTIC COLLISIONS

Kinetic Energy Loss in 1D Collision

$$\Delta K = \tfrac{1}{2} [(m_1\, m_2) / (m_1 + m_2)] (1 - e^2) (u_1 - u_2)^2$$

For reduced mass $\mu = (m_1 m_2)/(m_1+m_2)$: $\Delta K = \tfrac{1}{2} \mu (1 - e^2) v_{rel}^2$

Perfectly Inelastic Collision (e = 0)

Bodies stick together and move with common velocity v_{com}:

$$v_{com} = (m_1 u_1 + m_2 u_2) / (m_1 + m_2)$$

$$\Delta K_{max} = \tfrac{1}{2} [(m_1\, m_2) / (m_1 + m_2)] (u_1 - u_2)^2$$

Fractional K.E. Transfer & Loss (Target at Rest $u_2 = 0$)

1. Fractional Loss in Inelastic Collision:

$$\Delta K / K_i = [m_2 (1 - e^2)] / (m_1 + m_2)$$

2. Fractional K.E. Transferred in Elastic Collision (e=1):

$$\Delta K / K_1 = (4\, m_1\, m_2) / (m_1 + m_2)^2$$

Max energy transferred when $m_1 = m_2$ (100% transfer).

Ratio of Final to Initial K.E. (e = 0, $u_2 = 0$)

$$K_f / K_i = m_1 / (m_1 + m_2)$$

$$\Delta K / K_i = m_2 / (m_1 + m_2)$$

7. REBOUNDING OF BALL FROM HORIZONTAL FLOOR

N-th Rebound Velocity & Height

Ball dropped from height H_0 with initial speed $v_0 = \sqrt{(2gH_0)}$:

1. Velocity after n-th rebound: $v_n = e^n v_0$

2. Height after n-th rebound: $H_n = e^{2n} H_0$

3. Time of flight for n-th bounce: $t_n = 2\, e^n \sqrt{(2H_0/g)}$

Total Distance & Total Time Until Motion Stops

Summing up infinite bounces ($n \to \infty$):

Total Distance $D = H_0 [(1 + e^2) / (1 - e^2)]$

Total Time $T_{total} = \sqrt{(2H_0/g)} [(1 + e) / (1 - e)]$

8. TWO-DIMENSIONAL (OBLIQUE) COLLISIONS

Line of Impact (LOI) & Velocity Components

Line of Impact: Common normal to contact surface.

• Momentum is conserved ONLY along the Line of Impact (if no ext force).

• Velocity components *perpendicular to LOI* remain UNCHANGED during collision.

$$e = (v_{2,\,LOI} - v_{1,\,LOI}) / (u_{1,\,LOI} - u_{2,\,LOI})$$

Standard JEE Result: Identical Masses ($m_1 = m_2$)

If $m_1 = m_2$ and target m_2 is initially at rest ($u_2 = 0$):

1. Perfectly Elastic Collision (e = 1):

Particles move at right angles ($\theta + \phi = 90°$) to each other post-collision.

2. Inelastic Collision (e < 1):

Angle between post-collision velocity vectors: $\theta + \phi < 90°$.

Oblique Bounce off Smooth Fixed Plane Added

Ball strikes smooth plane at angle α to normal with speed u:

$$v_x = u \sin \alpha \quad \text{(parallel to plane)}$$

$$v_y = e\, u \cos \alpha \quad \text{(normal to plane)}$$

$$v' = u \sqrt{(\sin^2 \alpha + e^2 \cos^2 \alpha)}$$

$$\tan \beta = (\tan \alpha) / e \quad \textit{(β = angle of rebound to normal)}$$

Center of Mass Frame Analysis Added

$$v_{1,\,com} = u_1 - v_{com} = [\, m_2 / (m_1 + m_2) \,]\, u_{rel}$$

$$v'_{1,\,com} = -e\, v_{1,\,com}$$

$$K_{com} = \tfrac{1}{2}\, \mu\, v_{rel}^{\,2}$$

In COM frame, total momentum is identically ZERO before and after collision.

JEE MAIN & ADVANCED ULTIMATE REVISION BOOKLET

Units & Measurements • Mathematical Tools • Vector Algebra

1. UNITS & DIMENSIONAL ANALYSIS

Base Units & System Systems

7 Base SI Units: Length (m), Mass (kg), Time (s), Current (A), Temp (K), Luminous Int (cd), Amount (mol).

Supplementary Units:

• Plane Angle: $d\theta = ds / r$ [rad] ($1^\circ = \pi/180$ rad $\approx 1.745\times10^{-2}$ rad, $1' = 2.91\times10^{-4}$ rad, $1'' = 4.85\times10^{-6}$ rad)

• Solid Angle: $d\Omega = dA / r^2$ [sr] (Sphere total solid angle = 4π sr)

Unit Conversion Rule: $n_1u_1 = n_2u_2$

$\Rightarrow n_2 = n_1 [M_1/M_2]^a [L_1/L_2]^b [T_1/T_2]^c$

Important Distance Units:

• 1 AU = 1.496×10^{11} m (Earth-Sun mean dist)

• 1 ly = 9.46×10^{15} m (Light year distance)

• 1 pc (parsec) = 3.08×10^{16} m = 3.26 ly

Homogeneity & Dimensionless Combinations

Principle of Homogeneity: Only quantities of the same dimensions can be added/subtracted.
If $Z = A \pm B \Rightarrow [Z] = [A] = [B]$.

Dimensionless Quantities: Strain, Refractive Index, Relative Density, Angle, Solid Angle, Susceptibility, Poisson's Ratio, Dispersive Power.

JEE Standard Time/Length Combinations:

• Time Period T: $T \propto \sqrt{(l/g)}$, $T \propto \sqrt{(m/k)}$, $T \propto \sqrt{(R/g)}$

• Electromagnetic Time Constants: $\tau = L/R = RC = \sqrt{(LC)}$ [T^1]

• Characteristic Impedance: $Z = \sqrt{(L/C)} = \sqrt{(\mu_o/\varepsilon_o)}$ [$M^1L^2T^{-3}A^{-2}$] (Ω)

• Speed of light: $c = 1 / \sqrt{(\mu_o\varepsilon_o)}$ [L^1T^{-1}]

2. ERRORS IN MEASUREMENT & EXPERIMENTAL INSTRUMENTS

Errors in Measurement

Mean Absolute Error: $\Delta a_{mean} = \frac{\sum |\Delta a_i|}{n}$ where $\Delta a_i = a_{mean} - a_i$

Relative & Percentage Error: Relative = $\frac{\Delta a_{mean}}{a_{mean}}$; % Error = Relative × 100%

Error Combination Rules:

• Addition/Subtraction: $Z = A \pm B \Rightarrow \Delta Z = \Delta A + \Delta B$ (Absolute errors add)

• Product/Division: $Z = A \cdot B$ or $A/B \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$

• Power Formula: $Z = \frac{A^p B^q}{C^r} \Rightarrow \frac{\Delta Z}{Z} = p\frac{\Delta A}{A} + q\frac{\Delta B}{B} + r\frac{\Delta C}{C}$

• Max % Error in $Z = (p\%A + q\%B + r\%C)$

Vernier Calipers & Screw Gauge

Vernier Calipers:

• Least Count (LC) = 1 MSD − 1 VSD

• If n VSD = $(n-1)$ MSD $\Rightarrow$ LC = $\frac{1\text{ MSD}}{n}$

• Total Reading = MSR + (Coinciding VSD × LC) − (Zero Error)

• Positive Zero Error: Vernier zero is right of Main Scale zero.

• Negative Zero Error: Vernier zero is left of Main Scale zero.

Screw Gauge (Micrometer):

• Pitch = $\frac{\text{Distance moved on linear scale}}{\text{Number of full rotations}}$

• LC = $\frac{\text{Pitch}}{\text{Total divisions on circular scale}}$

• Total Reading = LSR + (CSR × LC) − (Zero Error)

Significant Figures Rules

1. All non-zero digits are significant (e.g., 42.3 has 3 SF).
2. Zeros between non-zero digits are significant (5.03 has 3 SF).
3. Leading zeros are never significant (0.006 has 1 SF).
4. Trailing zeros with a decimal are significant (4.330 has 4 SF).
5. Exponential notation $A \times 10^x$: SF determined solely by A.

Rounding Off & Arithmetical Operations

Rounding Off 5: If dropped digit is 5 followed by zeros, preceding digit is left unchanged if even, raised by 1 if odd (e.g., 3.250 → 3.2; 3.750 → 3.8).

Addition/Subtraction: Result reported to the least number of *decimal places* in original numbers.

Multiplication/Division: Result reported to the least number of *significant figures* in original numbers.

3. MATHEMATICAL TOOLS: ALGEBRA & TRIGONOMETRY

Quadratic & Binomial

Quadratic Eq: $ax^2 + bx + c = 0$

Roots: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$

Sum $x_1+x_2 = -b/a$, Prod $x_1x_2 = c/a$

Binomial Approx ($x \ll 1$):

$(1 + x)^n \approx 1 + nx$

$(1 - x)^n \approx 1 - nx$

$(1 + x)^{-n} \approx 1 - nx$

Progressions & Series

AP: $S_n = \frac{n}{2}[2a + (n-1)d]$

GP: $S_n = \frac{a(1 - r^n)}{1 - r}$

Infinite GP ($|r| < 1$): $S_\infty = \frac{a}{1 - r}$

Logarithms & Triangle Laws

Log Rules:

$\log(mn) = \log m + \log n$

$\log(m/n) = \log m - \log n$

$\log(m^n) = n \log m$

$\ln m = 2.303 \log_{10} m$

Standard Summations:

- $\sum n = \frac{n(n+1)}{2}$
- $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$

Triangle Sine & Cosine Laws:

Sine Law: $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

Cosine Law: $\cos A = \frac{b^2+c^2-a^2}{2bc}$

4. DIFFERENTIAL & INTEGRAL CALCULUS

Differential Calculus & Kinematic Applications

Standard Derivatives:

- $\frac{d}{dx}(x^n) = n\,x^{n-1}$, $\frac{d}{dx}(\sin x) = \cos x$, $\frac{d}{dx}(\cos x) = -\sin x$, $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(\ln x) = \frac{1}{x}$

Product & Quotient Rules:

- Product: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
- Quotient: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v(du/dx) - u(dv/dx)}{v^2}$

Kinematic Relations: Velocity $v = \frac{dx}{dt}$, Acceleration $a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = v\frac{dv}{dx}$

Integral Calculus, Maxima/Minima & Averages

Standard Integrals:

- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C\ (n \neq -1)$; $\int \frac{1}{x}\,dx = \ln|x| + C$
- $\int e^x\,dx = e^x + C$; $\int \sin x\,dx = -\cos x + C$; $\int \cos x\,dx = \sin x + C$

Maxima and Minima of $y = f(x)$:

- For Maximum: $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$ (Negative)
- For Minimum: $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$ (Positive)

Average of a Continuous Quantity:

$\langle y \rangle = \frac{\int_{x1}^{x2} y\,dx}{\int_{x1}^{x2} dx} = \frac{1}{x_2 - x_1}\int_{x1}^{x2} y\,dx$

5. VECTOR ALGEBRA & RESOLUTION

Vector Addition & Components

Parallelogram Law of Addition:

Magnitude $R = |A + B| = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$

Direction angle α (with vector A): $\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$

Direction angle β (with vector B): $\tan\beta = \frac{A\sin\theta}{B + A\cos\theta}$

Special Cases of Resultant R:

- Parallel ($\theta = 0°$): $R_{max} = A + B$
- Anti-parallel ($\theta = 180°$): $R_{min} = |A - B|$
- Perpendicular ($\theta = 90°$): $R = \sqrt{(A^2 + B^2)}$
- Equal Magnitudes ($|A| = |B| = A$): $R = 2A\cos(\theta/2)$

Resolution & Unit Vectors

Unit Vector: $\hat{a} = \frac{A}{|A|}$ (Dimensionless, magnitude = 1).

3D Resolution of Vector:

$A = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$

Magnitude: $|A| = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$

Direction Cosines:

$\cos\alpha = A_x/|A|$, $\cos\beta = A_y/|A|$, $\cos\gamma = A_z/|A|$

Identity: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Or $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Vector Products (Dot & Cross)

Scalar (Dot) Product:

$A \cdot B = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$

Angle between vectors: $\cos\theta = \frac{A \cdot B}{|A||B|}$

Projection of A on $B = A \cdot \hat{B} = \frac{A \cdot B}{|B|}$

Vector (Cross) Product:

$A \times B = (AB\sin\theta)\,\hat{n} = -(B \times A)$

Matrix Form: $A \times B = det\,|\,\hat{\imath}\ \ \hat{\jmath}\ \ \hat{k}\,|$ with rows (A_x, A_y, A_z) and (B_x, B_y, B_z)

$= \hat{\imath}(A_yB_z - A_zB_y) - \hat{\jmath}(A_xB_z - A_zB_x) + \hat{k}(A_xB_y - A_yB_x)$

Geometrical Applications & Theorems

Area Calculations:

- Area of Parallelogram with sides A & $B = |A \times B|$
- Area of Parallelogram with diagonals d_1 & $d_2 = \frac{1}{2}|d_1 \times d_2|$

[Added]

- Area of Triangle with sides A & $B = \frac{1}{2}|A \times B|$

Lami's Theorem (Coplanar Equilibrium):

If $F_1 + F_2 + F_3 = 0$, then $\frac{F_1}{\sin\theta_1} = \frac{F_2}{\sin\theta_2} = \frac{F_3}{\sin\theta_3}$

Vector Calculus Derivatives:

$\frac{d}{dt}(A \cdot B) = \frac{dA}{dt} \cdot B + A \cdot \frac{dB}{dt}$; $\frac{d}{dt}(A \times B) = \frac{dA}{dt} \times B + A \times \frac{dB}{dt}$

6. JEE ADVANCED VECTOR & COORDINATE RESULTS

Scalar & Vector Triple Products [Added]

Scalar Triple Product (Volume of Parallelepiped):

$[A\ B\ C] = A \cdot (B \times C) = det\ |\ A_x\ A_y\ A_z\ ;\ B_x\ B_y\ B_z\ ;\ C_x\ C_y\ C_z\ |$

Properties: $[A\ B\ C] = [B\ C\ A] = [C\ A\ A]$; If coplanar, $[A\ B\ C] = 0$.

Vector Triple Product (BAC-CAB Rule):

$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$

$(A \times B) \times C = B(A \cdot C) - A(B \cdot C)$

Relative Velocity & Shortest Distance

Relative Position & Velocity:

$r_{B/A} = r_B - r_A$, $v_{B/A} = v_B - v_A$

Condition for Collision of Two Particles:

Relative velocity $v_{B/A}$ must be directed along relative position $r_{A/B}$:

$$\frac{r_B - r_A}{|r_B - r_A|} = \frac{v_A - v_B}{|v_A - v_B|}$$ [Added]

Shortest Distance between moving particles:

$$d_{min} = \frac{|r_{BA} \times v_{BA}|}{|v_{BA}|}$$

7. MASTER REFERENCE: 84 PHYSICAL QUANTITIES & DIMENSIONS

#	Physical Quantity	Defining Formula	Dimensional Formula	SI Unit
1	Length	Base Quantity	$[M^0\ L^1\ T^0]$	m
2	Mass	Base Quantity	$[M^1\ L^0\ T^0]$	kg
3	Time	Base Quantity	$[M^0\ L^0\ T^1]$	s
4	Electric Current	Base Quantity	$[A^1]$	A
5	Thermodynamic Temp	Base Quantity	$[K^1]$	K
6	Luminous Intensity	Base Quantity	$[cd^1]$	cd
7	Amount of Substance	Base Quantity	$[mol^1]$	mol
8	Plane Angle	s / r	$[M^0\ L^0\ T^0]$	rad
9	Solid Angle	A / r^2	$[M^0\ L^0\ T^0]$	sr
10	Area	Length × Breadth	$[M^0\ L^2\ T^0]$	m^2
11	Volume	Area × Height	$[M^0\ L^3\ T^0]$	m^3
12	Velocity / Speed	Displacement / Time	$[M^0\ L^1\ T^{-1}]$	m/s
13	Acceleration	Velocity / Time	$[M^0\ L^1\ T^{-2}]$	m/s^2
14	Linear Momentum / Impulse	Mass × Velocity	$[M^1\ L^1\ T^{-1}]$	kg·m/s (N·s)
15	Force / Weight	Mass × Acceleration	$[M^1\ L^1\ T^{-2}]$	N
16	Work / Energy / Torque	Force × Displacement	$[M^1\ L^2\ T^{-2}]$	J (N·m)
17	Power	Work / Time	$[M^1\ L^2\ T^{-3}]$	W (J/s)
18	Pressure / Stress / Modulus	Force / Area	$[M^1\ L^{-1}\ T^{-2}]$	Pa (N/m^2)
19	Density	Mass / Volume	$[M^1\ L^{-3}\ T^0]$	kg/m^3
20	Frequency	1 / Time Period	$[M^0\ L^0\ T^{-1}]$	Hz (s^{-1})
21	Angular Velocity	Angle / Time	$[M^0\ L^0\ T^{-1}]$	rad/s
22	Angular Acceleration	$\Delta\omega$ / Δt	$[M^0\ L^0\ T^{-2}]$	rad/s^2
23	Moment of Inertia	$m\ r^2$	$[M^1\ L^2\ T^0]$	$kg·m^2$
24	Angular Momentum / Planck's h	$I\omega$ or E/ν	$[M^1\ L^2\ T^{-1}]$	J·s
25	Surface Tension / Spring Const	Force / Length	$[M^1\ L^0\ T^{-2}]$	N/m
26	Velocity Gradient	dv / dx	$[M^0\ L^0\ T^{-1}]$	s^{-1}
27	Coefficient of Viscosity	$F / (A \cdot dv/dx)$	$[M^1\ L^{-1}\ T^{-1}]$	Pa·s (kg/(m·s))
28	Gravitational Constant (G)	$Fr^2 / (m_1 m_2)$	$[M^{-1}\ L^3\ T^{-2}]$	$N·m^2/kg^2$
29	Latent Heat	Heat / Mass	$[M^0\ L^2\ T^{-2}]$	J/kg
30	Specific Heat Capacity	Heat / (Mass × ΔT)	$[M^0\ L^2\ T^{-2}\ K^{-1}]$	J/(kg·K)
31	Thermal Conductivity	$Q·l / (A·\Delta T·t)$	$[M^1\ L^1\ T^{-3}\ K^{-1}]$	W/(m·K)
32	Stefan's Constant (σ)	$E / (A·t·T^4)$	$[M^1\ L^0\ T^{-3}\ K^{-4}]$	$W/(m^2·K^4)$
33	Wien's Constant (b)	$\lambda_m T$	$[M^0\ L^1\ T^0\ K^1]$	m·K
34	Boltzmann Constant (k)	*Energy / Temperature*	$[M^1\ L^2\ T^{-2}\ K^{-1}]$	J/K
35	Universal Gas Constant (R)	$PV / (n\ T)$	$[M^1\ L^2\ T^{-2}\ K^{-1}\ mol^{-1}]$	J/(mol·K)
36	Electric Charge	Current × Time	$[M^0\ L^0\ T^1\ A^1]$	C
37	Electric Current Density	Current / Area	$[M^0\ L^{-2}\ T^0\ A^1]$	A/m^2
38	Electric Potential / EMF	Work / Charge	$[M^1\ L^2\ T^{-3}\ A^{-1}]$	V (J/C)

#	Physical Quantity	Defining Formula	Dimensional Formula	SI Unit
39	Electric Field Intensity	Force / Charge	$[M^1 L^1 T^{-3} A^{-1}]$	N/C or V/m
40	Electric Dipole Moment	Charge × Distance	$[M^0 L^1 T^1 A^1]$	C·m
41	Electric Flux	Electric Field × Area	$[M^1 L^3 T^{-3} A^{-1}]$	V·m
42	Capacitance	Charge / Potential	$[M^{-1} L^{-2} T^4 A^2]$	F (C/V)
43	Resistance / Impedance	Potential / Current	$[M^1 L^2 T^{-3} A^{-2}]$	Ω (V/A)
44	Resistivity	$R \cdot A / l$	$[M^1 L^3 T^{-3} A^{-2}]$	Ω·m
45	Electrical Conductivity	1 / Resistivity	$[M^{-1} L^{-3} T^3 A^2]$	S/m ($\Omega^{-1}m^{-1}$)
46	Magnetic Field (B)	$F / (q\,v)$	$[M^1 L^0 T^{-2} A^{-1}]$	T (Wb/m^2)
47	Magnetic Flux (Φ)	Magnetic Field × Area	$[M^1 L^2 T^{-2} A^{-1}]$	Wb ($T \cdot m^2$)
48	Magnetic Dipole Moment	Current × Area	$[M^0 L^2 T^0 A^1]$	$A \cdot m^2$ or J/T
49	Self / Mutual Inductance	*Flux / Current*	$[M^1 L^2 T^{-2} A^{-2}]$	H (Wb/A)
50	Permeability of Free Space (μ_0)	$2\pi F d / (I_1 I_2 l)$	$[M^1 L^1 T^{-2} A^{-2}]$	H/m or N/A^2
51	Permittivity of Free Space (ε_0)	$q_1 q_2 / (4\pi F r^2)$	$[M^{-1} L^{-3} T^4 A^2]$	F/m or $C^2/(N \cdot m^2)$
52	Magnetization (M)	Magnetic Moment / Volume	$[M^0 L^{-1} T^0 A^1]$	A/m
53	Magnetic Intensity (H)	B / μ	$[M^0 L^{-1} T^0 A^1]$	A/m
54	Magnetic Susceptibility	M / H	$[M^0 L^0 T^0]$	Dimensionless
55	Refractive Index	Speed in vacuum / Speed in medium	$[M^0 L^0 T^0]$	Dimensionless
56	Dielectric Constant (ε_r)	Permittivity ratio	$[M^0 L^0 T^0]$	Dimensionless
57	Relative Permeability (μ_r)	Permeability ratio	$[M^0 L^0 T^0]$	Dimensionless
58	Strain	Change in dim / Original dim	$[M^0 L^0 T^0]$	Dimensionless
59	Poisson's Ratio	Lateral / Longitudinal strain	$[M^0 L^0 T^0]$	Dimensionless
60	Mechanical Eq of Heat (J)	Work / Heat	$[M^0 L^0 T^0]$	J/cal
61	Rydberg Constant (R)	Wave number formula	$[M^0 L^{-1} T^0]$	m^{-1}
62	Compton Wavelength	$h / (m\,c)$	$[M^0 L^1 T^0]$	m
63	Bohr Magneton	$e\,h / (4\pi\,m)$	$[M^0 L^2 T^0 A^1]$	$A \cdot m^2$
64	Electric Polarizability	Dipole moment / Electric Field	$[M^{-1} L^{-2} T^4 A^2]$	$C \cdot m^2/V$
65	Charge Mobility	Drift Velocity / Electric Field	$[M^{-1} T^2 A^1]$	$m^2/(V \cdot s)$
66	Decay Constant (λ)	Rate of decay per nucleus	$[M^0 L^0 T^{-1}]$	s^{-1}
67	Radioactive Activity	dN / dt	$[M^0 L^0 T^{-1}]$	Bq (disintegrations/s)
68	Solar Constant	Energy / (Area × Time)	$[M^1 L^0 T^{-3}]$	W/m^2
69	Force Constant (Spring)	Force / Extension	$[M^1 L^0 T^{-2}]$	N/m
70	Impulse	Force × Time	$[M^1 L^1 T^{-1}]$	N·s
71	Angular Frequency	2π × Frequency	$[M^0 L^0 T^{-1}]$	rad/s
72	Force Gradient	dF / dx	$[M^1 L^0 T^{-2}]$	N/m
73	Kinematic Viscosity	Viscosity / Density	$[M^0 L^2 T^{-1}]$	m^2/s
74	Radiant Energy	Energy of EM waves	$[M^1 L^2 T^{-2}]$	J
75	Radiant Flux / Luminosity	Energy / Time	$[M^1 L^2 T^{-3}]$	W
76	Wave Intensity	Power / Area	$[M^1 L^0 T^{-3}]$	W/m^2
77	Acoustic Impedance	Pressure / Volume Velocity	$[M^1 L^{-4} T^{-1}]$	$Pa \cdot s/m^3$
78	Verdet Constant	Angle / (Field × Length)	$[M^{-1} L^{-1} T^2 A^1]$	$rad/(T \cdot m)$
79	Volume Flow Rate	Volume / Time	$[M^0 L^3 T^{-1}]$	m^3/s
80	Photon Momentum	h / λ	$[M^1 L^1 T^{-1}]$	kg·m/s
81	Gravitational Potential	Work / Mass	$[M^0 L^2 T^{-2}]$	J/kg
82	Gravitational Field Strength	Force / Mass	$[M^0 L^1 T^{-2}]$	N/kg (m/s^2)
83	Magnetic Permeability (μ)	B / H	$[M^1 L^1 T^{-2} A^{-2}]$	T·m/A or H/m
84	Dispersive Power	$(\mu_v - \mu_r) / (\mu_y - 1)$	$[M^0 L^0 T^0]$	Dimensionless

GRAVITATION — JEE ADVANCED FORMULA BOOKLET

SHEET 1 OF 3

1. UNIVERSAL LAW OF GRAVITATION

Scalar Form: $F = G \cdot m_1 \cdot m_2 / r^2$

G = 6.67 × 10^{-11} N·m²/kg² | [G] = [$M^{-1} L^3 T^{-2}$]

Vector Form: $F_{21} = -(G \cdot m_1 \cdot m_2 / r_{12}^3)\, r_{12}$

Action-reaction pair: $F_{21} = -F_{12}$. Central & conservative.

Superposition: $F_{net} = \sum F_i$

Independent of intervening medium or external bodies.

2. SYMMETRICAL MASS SYSTEMS

3 Mass Polygon (Equilateral Triangle side l):

$v = \sqrt{(G \cdot M / (l \cdot \sqrt{3}))}$ |

$T = 2\pi \cdot \sqrt{(l^3 \cdot \sqrt{3} / (3G \cdot M))}$

Masses move in circle under mutual gravitational attraction.

4 Mass Polygon (Square side l):

$v = 1/2 \cdot \sqrt{(G \cdot M / l) \cdot (4 + \sqrt{2})}$

Binary Star System: [Added]

Radii: $r_1 = r \cdot m_2 / (m_1 + m_2)$,

$r_2 = r \cdot m_1 / (m_1 + m_2)$

Orbital Speeds: $v_1 = \sqrt{(G \cdot m_2^2 / ((m_1 + m_2) r))}$

Period: $T^2 = 4\pi^2 \cdot r^3 / (G(m_1 + m_2))$

3. ACCELERATION DUE TO GRAVITY (G)

Surface Value: $g = G \cdot M / R^2 = 4/3 \cdot \pi \cdot G \cdot \rho \cdot R$

For constant density ρ, g ∝ R. Earth g ≈ 9.81 m/s².

Height Variation (h):

- Exact: $g_h = g \cdot (1 + h/R)^{-2} = G \cdot M / (R+h)^2$
- Approx (h << R, h < 100km): $g_h \approx g \cdot (1 - 2h/R)$
- Fractional Decrease: $\Delta g / g = 2h / R$

Depth Variation (d):

- Exact: $g_d = g \cdot (1 - d/R) = g \cdot r / R$
- Fractional Decrease: $\Delta g / g = d / R$

Linear inside uniform sphere; zero at center (d = R).

4. SHAPE & ROTATION VARIATIONS OF G

Latitude Rotation (λ): $g' = g - \omega^2 \cdot R \cdot \cos^2\lambda$

- At Equator (λ = 0°): $g_{eq} = g - \omega^2 \cdot R$ (Min)
- At Poles (λ = 90°): $g_{pole} = g$ (Max, no rotation effect)
- Weight Difference: $m(g_{pole} - g_{eq}) = m \cdot \omega^2 \cdot R$

Weightlessness by Rotation:

Equator becomes weightless if

$\omega = \sqrt{(g / R)} \approx 1.23 \times 10^{-3}$ rad/s

Earth's day duration would drop from 24h to ~1.41 hours.

Elliptical Shape of Earth:

$R_{eq} - R_{pole} \approx 21$ km ⇒ $g_{pole} > g_{eq}$

5. GRAVITATIONAL FIELD INTENSITY (I OR E_G)

Definition: $I = F / m = -\nabla V$ (Unit: N/kg or m/s²)

Point Mass: $I = G \cdot M / r^2$ (Radially inward)

Uniform Spherical Shell (Mass M, Radius R):

- $r > R$ (Out): $I = G \cdot M / r^2$
- $r = R$ (Surf): $I = G \cdot M / R^2$
- $r < R$ (In): $I = 0$

Uniform Solid Sphere (Mass M, Radius R):

- $r > R$ (Out): $I = G \cdot M / r^2$
- $r = R$ (Surf): $I = G \cdot M / R^2$
- $r < R$ (In): $I = G \cdot M \cdot r / R^3 = 4/3 \cdot \pi \cdot G \cdot \rho \cdot r$

6. EXTENDED BODY FIELDS (ADVANCED RESULTS)

Ring of Mass M, Radius R (Axial distance x):

$I = G \cdot M \cdot x / (R^2 + x^2)^{3/2}$

- Center (x = 0): $I = 0$
- Max Field at: $x = R / \sqrt{2}$ ⇒

$I_{max} = 2G \cdot M / (3\sqrt{3} \cdot R^2)$

Uniform Disc (Radius R, Axial distance x):

$I = (2G \cdot M / R^2) \cdot [\, 1 - x / \sqrt{(x^2 + R^2)}\,]$

Uniform Infinite Rod / Wire: [Added]

$I = 2G \cdot \lambda / r$ (where λ = mass per unit length)

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7. GRAVITATIONAL POTENTIAL (V)

Definition: `V = W_ext / m = -∫ I□ · dr□` (SI Unit: J/kg)
Scalar quantity. Reference V = 0 at r = ∞.

Point Mass: `V = - G·M / r`

Uniform Spherical Shell:
- `r ≥ R` (Out/Surf): `V = - G·M / r`
- `r < R` (Inside): `V = - G·M / R` (Uniform constant)

Uniform Solid Sphere:
- `r ≥ R` (Out): `V = - G·M / r`
- `r < R` (In): `V = - (G·M / 2R³) · (3R² - r²)`
- Center (`r = 0`):

`V_center = - 1.5 · (G·M / R) = 1.5 V_surf`

Thin Ring Axis: `V = - G·M / √(R² + x²)`

8. POTENTIAL ENERGY (U) & WORK DONE

Two Mass System: `U = - G·m₁·m₂ / r`

N-Mass System:

`U_total = ∑_{all pairs i<j} (- G·m_i·m_j / r_ij)`

Work Done by External Force:

`W_ext = ΔU = U_final - U_initial = - W_grav`

Lifting Mass m to Height h:

`W = m·g·h / (1 + h/R)`

- Special cases: `h = R ⇒ W = 1/2 m·g·R` | `h = R/2 ⇒ W = 1/3 m·g·R`

9. ESCAPE VELOCITY (V_E)

From Surface:

`v_e = √(2G·M / R) = √(2g·R) = √(8/3 · π·G·ρ·R²)`

For Earth: v_e ≈ 11.2 km/s. Independent of mass/angle of projection.

From Height h: `v_e(h) = √(2G·M / (R+h))`

From Planet Body Center (Solid Sphere): [Added]

`v_e(center) = √(3G·M / R) = √3 · v_surface`

10. JEE PROJECTION & SPEED TRICKS

Case 1: Velocity given v > v_e (i.e. v = n·v_e, n > 1):
Final speed in interstellar space (`r → ∞`):

`v_final = √(v² - v_e²) = v_e · √(n² - 1)`

Case 2: Velocity given v < v_e (i.e. v = n·v_e, n < 1):
Maximum height reached above surface:

`h = R · [ n² / (1 - n²) ] = R · [ v² / (v_e² - v²) ]`

Tunneling Through Earth (SHM Motion): [Added]
Force inside tunnel:

`F = - (G·M·m / R³) r = - m·ω²·r`

Time period through tunnel:

`T = 2π · √(R / g) ≈ 84.6 min`

Max velocity at center: `v_max = √(g·R) ≈ 7.9 km/s`

11. CIRCULAR SATELLITE MECHANICS

Orbital Speed (v_o):

`v_o = √(G·M / r) = √(g·R² / (R+h))`

For near-Earth satellite (h ≈ 0): v_o = √(g·R) ≈ 7.92 km/s.

Relation to Escape Velocity: `v_e = √2 · v_o`

Time Period: `T = 2π·r / v_o = 2π·√(r³ / (G·M))`

`T² = (4π² / G·M) r³` ⇒ `T² ∝ r³`

Energy Spectrum of Satellite (Orbit r):
- Kinetic Energy: `K = G·M·m / (2r)`
- Potential Energy: `U = - G·M·m / r`
- Mechanical Energy: `E = K + U = - G·M·m / (2r)`
- Binding Energy: `B.E. = -E = + G·M·m / (2r)`

Key Inter-relations: `U = 2E = -2K` | `K = -E`

12. GEOSTATIONARY & POLAR SATELLITES

Geostationary Satellite Parameters:
- Orbit: Equatorial plane | Direction: West to East
- Time period: `T = 24 hours`
- Height above surface: `h ≈ 35,800 km` (`r ≈ 6.6 R_e`)
- Orbital speed: `v_o ≈ 3.08 km/s`

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13. KEPLER'S LAWS OF PLANETARY MOTION

First Law (Orbits): Planets move in elliptical orbits with the Sun at one focus.

Second Law (Areas): Radial line sweeps equal areas in equal times.

$dA / dt = L / (2m) = Constant$

Direct consequence of Angular Momentum Conservation (Torque = 0).

Third Law (Periods): $T^2 \propto a^3$ (a = Semi-major axis)

$T^2 = (4\pi^2 / G \cdot M_sun) \cdot a^3$

14. ELLIPTICAL ORBIT GEOMETRY & SPEEDS

Orbital Distances:

- Perihelion (Nearest): $r_min = r_p = a(1 - e)$
- Aphelion (Farthest): $r_max = r_a = a(1 + e)$
- Relation: $a = (r_p + r_a) / 2$ | $e = (r_a - r_p) / (r_a + r_p)$

Velocity Conservation (Angular Momentum & Energy):

$v_p \cdot r_p = v_a \cdot r_a \Rightarrow v_p / v_a = r_a / r_p = (1 + e) / (1 - e)$

Speeds at Extremes: [Added]

$v_p = \sqrt{((G \cdot M / a) \cdot ((1+e) / (1-e)))}$

$v_a = \sqrt{((G \cdot M / a) \cdot ((1-e) / (1+e)))}$

Total Mechanical Energy in Ellipse:

$E = - G \cdot M \cdot m / (2a)$ (Depends purely on semi-major axis a)

15. NEUTRAL POINT & ZERO POTENTIAL

Neutral Point (I_net = 0): Separation r between M_1 & M_2

Distance from M_1: $x = r \cdot [\sqrt{M_1} / (\sqrt{M_1} + \sqrt{M_2})]$

Potential at Neutral Point:

$V_N = - (G / r) \cdot (\sqrt{M_1} + \sqrt{M_2})^2$

16. ORBIT TRANSFER & MANOEUVRES

Work Done in Changing Orbits ($r_1 \rightarrow r_2$):

- Circular to Circular:

$W = E_2 - E_1 = G \cdot M \cdot m/2 \cdot (1/r_1 - 1/r_2)$

Hohmann Transfer Orbit: [Added]

Transfer semi-major axis: $a_trans = (r_1 + r_2) / 2$

Velocity boost needed at r_1:

$\Delta v_1 = \sqrt{(G \cdot M / r_1)} \cdot [\sqrt{(2r_2 / (r_1+r_2))} - 1]$

17. SUMMARY OF FIELD & POTENTIAL GRAPHS

Body	Region	Field (I)	Potential (V)
Shell	r < R	0	-GM/R (Const)
	r > R	$\propto 1/r^2$	$\propto -1/r$
Solid Sphere	r < R	$\propto$ r (Linear)	Parabolic (Min at r=0)
	r > R	$\propto 1/r^2$	$\propto -1/r$

18. CRITICAL JEE CONSTANTS & VALUES

Quantity	Symbol	Standard Value
Gravitational Constant	G	6.67×10^{-11} N·m²/kg²
Mass of Earth	M_e	5.98×10^{24} kg
Radius of Earth	R_e	6.37×10^6 m (≈ 6400 km)
Mean Sun-Earth Distance	1 AU	1.496×10^{11} m
1st Cosmic Velocity (Orbital)	v_o1	7.9 km/s
2nd Cosmic Velocity (Escape)	v_e2	11.2 km/s

1. DIFFERENTIAL EQUATION & CLASSIFICATION

Wave Motion Definition: Transfer of energy and momentum through a medium without net transport of matter. Medium must possess *Inertia* (for KE) and *Elasticity* (for PE).

Classifications:

- **Mechanical:** Requires medium (Sound, String, Water).
- **Non-Mechanical:** Travels in vacuum (EM Waves).
- **Transverse:** Particle vibration ⊥ wave direction (Strings, light, surface water waves). Transmits in solids & liquid surfaces.
- **Longitudinal:** Particle vibration || wave direction (Sound). Transmits via volume elasticity in solids, liquids, & gases.

General 1D Wave Differential Equation:

$$\partial^2 y / \partial t^2 = v^2 (\partial^2 y / \partial x^2)$$

Any physical state function y(x,t) representing a wave must be continuous and bounded for all x and t.

Functional Forms: $y(x,t) = f(t \pm x/v) = g(vt \pm x)$

- $f(t - x/v)$ or $f(kx - \omega t)$ → Moving towards **+x direction**
- $f(t + x/v)$ or $f(kx + \omega t)$ → Moving towards **−x direction**

2. PLANE PROGRESSIVE HARMONIC WAVES

Standard Wave Equation:

$$y(x,t) = A \sin(\omega t - kx + \phi_0) = A \sin 2\pi (t/T - x/\lambda + \phi_0/2\pi)$$

Parameters & Definitions:

- Amplitude (A): Maximum particle displacement.
- Wavelength (λ): Distance between consecutive identical phases.
- Period (T) & Frequency (f): $T = 1/f$.
- Angular Frequency: $\omega = 2\pi f = 2\pi/T$
- Wave Vector / Propagation Constant: $k = 2\pi/\lambda = \omega/v$
- Wave Velocity: $v = f\lambda = \omega/k$
- Total Phase: $\Phi(x,t) = \omega t - kx + \phi_0$

Phase & Path Difference Relations:

$$\Delta\Phi = (2\pi / \lambda)\, \Delta x = (2\pi / T)\, \Delta t = k\, \Delta x$$

3. KINEMATICS OF WAVE PARTICLES

Particle Velocity (u):

$$u = \partial y/\partial t = A\omega \cos(\omega t - kx + \phi_0) \Rightarrow u_{max} = A\omega$$

Wavefront Slope: $\partial y/\partial x = -k A \cos(\omega t - kx + \phi_0)$

Fundamental Particle-Wave Velocity Relation:

$$\text{Particle Velocity } (u) = -v \times (\text{Slope of Waveform } \partial y/\partial x)$$

Particle Acceleration (a_p):

$$a_p = \partial^2 y/\partial t^2 = -\omega^2 A \sin(\omega t - kx + \phi_0) = -\omega^2 y$$

$a_{p,max} = \omega^2 A$. Note that $a_p = v^2 (\partial^2 y/\partial x^2)$.

Maximum Particle Speed Condition:

If $u_{max} = n \times v$, then $A\omega = n(\omega/k) \Rightarrow A = n/k = n\lambda / (2\pi)$.

4. VELOCITY OF TRANSVERSE WAVES ON STRINGS

Fundamental Formula:

$$v = \sqrt{(T / \mu)} = \sqrt{(T / (\rho A))} = \sqrt{(\sigma / \rho)}$$

T = Tension, $\mu = m/L$ (linear mass density), ρ = volume density, σ = stress.

Special Cases for Variable Tension:

- **Massless string with hanging block M:** $v = \sqrt{(Mg / \mu)}$
- **Submerged block in liquid (Density σ_b, Liquid ρ_l):** $T = Mg\,(1 - \rho_l/\sigma_b) \Rightarrow v = \sqrt{[Mg(1 - \rho_l/\sigma_b) / \mu]}$
- **Massive Rope (Mass M, Length L) hanging vertically:**
  Tension at distance x from free bottom end: $T(x) = (M/L)\,x\,g = \mu\,g\,x$
  Velocity at position x: $v(x) = \sqrt{(gx)}$
  Time taken by pulse to travel bottom to top: $t = 2\sqrt{(L / g)}$
  Acceleration of wave pulse: $a = dv/dt = g / 2$ (Constant)
- **Massive Rope hanging with attached block M_0:**
  $T(x) = (M_0 + \mu x)\,g \Rightarrow v(x) = \sqrt{([M_0 + \mu x]\,g / \mu)}$ [Added]
  $t = (2/g)\,[\sqrt{(M_0+M)g / \mu)} - \sqrt{(M_0 g / \mu)}]$
- **Rotating String (Mass M, Radius R, Angular Speed ω):** [Added]
  Tension at distance r from axis: $T(r) = ½\,\mu\,\omega^2\,(R^2 - r^2)$

5. SPEED OF LONGITUDINAL / SOUND WAVES

General Elastic Medium Formula:

> $v = \sqrt{(E / \rho)}$

- In Solid Rods: $v = \sqrt{(Y / \rho)}$ *(Y = Young's Modulus)*
- In Bulk Liquids & Fluids: $v = \sqrt{(K / \rho)}$ *(K = Bulk Modulus)*
- Ordering: $v_{solid} > v_{liquid} > v_{gas}$ (due to elasticity dominance).

Sound Speed in Gases (Thermodynamic Form):

- **Newton's Formula (Isothermal):** $K = P \Rightarrow v = \sqrt{(P / \rho)}$ (Gives 280 m/s for air at STP — Inaccurate).
- **Laplace Correction (Adiabatic):** $K = \gamma P \Rightarrow v = \sqrt{(\gamma P / \rho)} = \sqrt{(\gamma R T / M)}$
  ($\gamma = C_p/C_v$: Monoatomic = 5/3, Diatomic = 7/5, Polyatomic = 4/3).
- **Temperature Effect:** $v \propto \sqrt{T}$ (in Kelvin).
  For low temperatures: $v_t \approx v_0 + 0.61\,t$ (m/s, t in °C).
- **Humidity Effect:** Density of moist air $\rho_m < \rho_d$ (dry air) $\Rightarrow v_{moist} > v_{dry}$.
- **Pressure Effect:** No effect on v at constant temperature since $P/\rho = RT/M$.

6. POWER TRANSMITTED & ENERGY DENSITY

Energy Transport in String Waves:

- Kinetic Energy per unit length: $dk/dx = \frac{1}{2} \mu (\partial y/\partial t)^2$
- Potential Energy per unit length: $dU/dx = \frac{1}{2} T (\partial y/\partial x)^2$
- In a travelling wave, $dk/dx = dU/dx$ at all points!
- Total Energy per unit length: $dE/dx = \mu \omega^2 A^2 \cos^2(\omega t - kx)$
- Average Energy Density: $u = \frac{1}{2} \rho \omega^2 A^2 = \frac{1}{2} \mu \omega^2 A^2 / \text{Area}$

Power Transmitted Across a Section:

$$P(x,t) = -T (\partial y/\partial x)(\partial y/\partial t) = \mu v \omega^2 A^2 \cos^2(\omega t - kx)$$

Average Power:

$$\langle P \rangle = \frac{1}{2} \mu v \omega^2 A^2 = 2 \pi^2 f^2 A^2 \mu v$$

7. WAVE INTENSITY & SOUND LOUDNESS

Intensity Definition: Energy passing per unit time per unit normal area: $I = \langle P \rangle / S$.

$$I = \frac{1}{2} \rho v \omega^2 A^2 = 2 \pi^2 f^2 A^2 \rho v$$

Geometric Spreading & Source Geometries:

- **Isotropic Point Source (3D):** Power P_0 uniform in all directions.
  $I = P_0 / (4\pi r^2) \Rightarrow I \propto 1/r^2 \Rightarrow$ Amplitude $A \propto 1/r$
- **Line Source (2D Cylinder):** [Added]
  $I = P_0 / (2\pi r L) \Rightarrow I \propto 1/r \Rightarrow$ Amplitude $A \propto 1/\sqrt{r}$
- **1D Unattenuated Wave (String/Pipe):** $I =$ constant $\Rightarrow A =$ constant.

Loudness & Decibel Scale (β):

$$\beta \text{ (in dB)} = 10 \log_{10} (I / I_0)$$

$I_0 = 10^{-12}\ W/m^2$ *(Standard threshold of human hearing at 1 kHz).*

Loudness Difference Formula:

$$\Delta\beta = \beta_2 - \beta_1 = 10 \log_{10} (I_2 / I_1) = 20 \log_{10} (A_2 / A_1)$$

If distance changes from r_1 to r_2 for point source: $\Delta\beta = 20 \log_{10} (r_1 / r_2)$

8. PRESSURE & DENSITY WAVES IN SOUND

Sound in fluids can be expressed as a **Displacement Wave** $s(x,t)$ or a **Pressure Wave** $\Delta P(x,t)$.

Derivation & Relations:

- Displacement Wave: $s(x,t) = s_0 \sin(\omega t - kx)$
- Volumetric Strain: $dV/V = \partial s/\partial x$
- Excess Pressure: $\Delta P = -B (dV/V) = -B (\partial s/\partial x)$
- Substituting $s(x,t)$:
  $\Delta P = B k s_0 \cos(\omega t - kx) = \Delta P_0 \cos(\omega t - kx)$

Pressure Amplitude (ΔP_0):

$$\Delta P_0 = B k s_0 = \rho v \omega s_0 = 2\pi \rho v f s_0$$

Density Wave: $\Delta\rho(x,t) = \rho_0 k s_0 \cos(\omega t - kx) = \Delta\rho_0 \cos(\omega t - kx)$ where $\Delta\rho_0 = (\rho_0 / v) \omega s_0$.

Intensity in terms of Pressure Amplitude:

$$I = (\Delta P_0)^2 / (2 \rho v) = (\Delta P_0)^2 v / (2 B)$$

Phase Node/Antinode Phase Shift Rule: Pressure wave leads displacement wave by **90° (π/2)**. Displacement nodes are pressure antinodes (maximum pressure variation), and displacement antinodes are pressure nodes (constant atmospheric pressure).

9. PRINCIPLE OF SUPERPOSITION & INTERFERENCE

When two waves $y_1 = A_1 \sin(\omega t - kx)$ and $y_2 = A_2 \sin(\omega t - kx + \phi)$ overlap:

$$y(x,t) = A_R \sin(\omega t - kx + \theta), \quad A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos\phi}$$

$\tan\theta = (A_2 \sin\phi) / (A_1 + A_2 \cos\phi)$. Resultant Intensity: $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\phi$.

Condition	Phase Difference (ϕ)	Path Difference (Δx)	Resultant Amplitude (A_R)	Resultant Intensity (I)
Constructive (Max)	$2n\pi$ (n = 0,1,2...)	$n\lambda$	$A_{max} = A_1 + A_2$	$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$
Destructive (Min)	$(2n-1)\pi$ (n = 1,2...)	$(2n-1)\lambda/2$	$A_{min} = \lvert A_1 - A_2 \rvert$	$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Identical Sources ($I_1 = I_2 = I_0$): $I = 4 I_0 \cos^2(\phi/2)$, $I_{max} = 4 I_0$, $I_{min} = 0$.

Fringe Contrast / Visibility Ratio: $I_{max} / I_{min} = (A_1 + A_2)^2 / (A_1 - A_2)^2 = (\sqrt{I_1} + \sqrt{I_2})^2 / (\sqrt{I_1} - \sqrt{I_2})^2$

10. BOUNDARY REFLECTIONS & TRANSMISSION COEFFICIENTS

Boundary Phase Shift Rules:

- **Rigid / Fixed End / Denser Boundary:** Phase changes by π (180°).
  Incident: $y_i = A_i \sin(\omega t - k_1 x) \Rightarrow$ Reflected: $y_r = -A_r \sin(\omega t + k_1 x)$
- **Free / Open End / Rarer Boundary: No phase change (0°).**
  Reflected: $y_r = A_r \sin(\omega t + k_1 x)$
- Transmitted wave **NEVER** undergoes a phase change regardless of boundary!

Amplitude Coefficients (Medium 1 → Medium 2):

$$A_r = [(v_2 - v_1)/(v_1 + v_2)]\, A_i = [(k_1 - k_2)/(k_1 + k_2)]\, A_i$$
$$A_t = [2v_2/(v_1 + v_2)]\, A_i = [2k_1/(k_1 + k_2)]\, A_i$$

Power Conservation: $P_i = P_r + P_t \Rightarrow \mu_1 v_1 A_i^2 = \mu_1 v_1 A_r^2 + \mu_2 v_2 A_t^2$

Transmission/Reflection Power Fractions: $R = P_r/P_i = (v_2 - v_1)^2/(v_1 + v_2)^2$, $T = P_t/P_i = 4v_1v_2/(v_1 + v_2)^2$

11. STANDING / STATIONARY WAVES MATHEMATICS

Formed by superposition of two identical waves travelling in opposite directions: $y_1 = A \sin(\omega t - kx)$, $y_2 = A \sin(\omega t + kx + \phi_0)$.

$y(x,t) = (2A \cos kx) \sin \omega t$ (Fixed at x = λ/4, 3λ/4) or $y(x,t) = (2A \sin kx) \cos \omega t$ (Fixed at x = 0)

Nodes (Zero Displacement, Permanently at Rest):

- For $y = 2A \sin(kx) \cos(\omega t)$: $\sin(kx) = 0 \Rightarrow kx = n\pi$
- Positions: $x = 0, \lambda/2, \lambda, 3\lambda/2, \ldots, n(\lambda/2)$
- Strain & Pressure variation are **MAXIMUM** at nodes!

Antinodes (Maximum Amplitude = 2A):

- For $y = 2A \sin(kx) \cos(\omega t)$: $|\sin(kx)| = 1 \Rightarrow kx = (2n + 1)\pi/2$
- Positions: $x = \lambda/4, 3\lambda/4, 5\lambda/4, \ldots, (2n + 1)(\lambda/4)$
- Strain & Pressure variation are **ZERO** at antinodes!

Key Properties of Standing Waves:

- Distance between consecutive Nodes or Antinodes = $\lambda / 2$. Distance between adjacent Node and Antinode = $\lambda / 4$.
- **No net flow of energy** across nodes (energy remains localized in loops).
- Particles in the same loop vibrate in the **same phase**. Particles in adjacent loops vibrate in **opposite phase (180° shift)**.

12. VIBRATIONS OF STRETCHED STRINGS (FIXED & FREE ENDS)

Case A: String Fixed at Both Ends (Length L)

- Nodes at both ends: $L = n(\lambda / 2) \Rightarrow \lambda_n = 2L / n$
- Harmonic Frequencies:
  $f_n = n(v / 2L) = (n / 2L)\sqrt{(T / \mu)}$ (n = 1, 2, 3...)
- Fundamental Frequency (1st Harmonic): $f_1 = (1 / 2L)\sqrt{(T / \mu)}$
- Overtones: $f_2 = 2 f_1$ (1st overtone / 2nd harmonic), $f_3 = 3 f_1$ (2nd overtone / 3rd harmonic).
- **Ratio of Harmonics:** $f_1 : f_2 : f_3 : \ldots = 1 : 2 : 3 : 4\ldots$ (All harmonics present).

Case B: String Fixed at One End, Free at Other

- Node at fixed end, Antinode at free end: $L = (2n - 1)(\lambda / 4)$
- Harmonic Frequencies:
  $f_n = (2n - 1)(v / 4L) = [(2n - 1) / 4L]\sqrt{(T / \mu)}$ (n = 1, 2, 3...)
- Fundamental (1st Harmonic): $f_1 = (1 / 4L)\sqrt{(T / \mu)}$
- Overtones: $f_3 = 3 f_1$ (1st overtone / 3rd harmonic), $f_5 = 5 f_1$ (2nd overtone / 5th harmonic).
- **Ratio of Harmonics:** $f_1 : f_3 : f_5 : \ldots = 1 : 3 : 5\ldots$ (Only odd harmonics).

Sonometer Laws (Fixed String): $f \propto 1/L$ (Law of length), $f \propto \sqrt{T}$ (Law of tension), $f \propto 1/\sqrt{\mu}$ (Law of mass density).

13. ORGAN PIPES (SOUND WAVES IN TUBES)

Open Organ Pipe (Open at Both Ends)

- Antinodes at both open ends.
- Wavelengths: $\lambda_n = 2L / n$
- Frequencies: $f_n = n\,(v / 2L)$ (n = 1, 2, 3...)
- Ratio of Harmonics: $f_1 : f_2 : f_3 = 1 : 2 : 3$ (All harmonics).
- Pressure nodes at ends; displacement antinodes at ends.

Closed Organ Pipe (Closed at One End)

- Node at closed end, Antinode at open end.
- Wavelengths: $\lambda_n = 4L / (2n - 1)$
- Frequencies: $f_n = (2n - 1)\,(v / 4L)$ (n = 1, 2, 3...)
- Ratio of Harmonics: $f_1 : f_3 : f_5 = 1 : 3 : 5$ (Odd harmonics only).
- Fundamental Frequency ratio: $f_{open} = 2\,f_{closed}$ for same length!

End Correction (Lord Rayleigh Correction):

Displacement antinode extends slightly beyond the physical open boundary by end correction $e = 0.6\,r = 0.3\,D$ (r = tube radius, D = diameter).

- **Open Pipe (2 open ends):** $L_{eff} = L + 2e = L + 1.2\,r \Rightarrow f_1 = v / [2(L + 1.2r)]$
- **Closed Pipe (1 open end):** $L_{eff} = L + e = L + 0.6\,r \Rightarrow f_1 = v / [4(L + 0.6r)]$

14. RESONANCE TUBE EXPERIMENT

Used to determine speed of sound in air using tuning fork of frequency f:

- 1st Resonance (Length l_1): $l_1 + e = \lambda / 4$
- 2nd Resonance (Length l_2): $l_2 + e = 3\lambda / 4$
- 3rd Resonance (Length l_3): $l_3 + e = 5\lambda / 4$

Key Results & Formulas:

> **Wavelength:** $\lambda = 2\,(l_2 - l_1)$
> **Speed of Sound:** $v = 2\,f\,(l_2 - l_1)$
> **End Correction:** $e = (l_2 - 3l_1) / 2$

If diameter D is given: $v = 2\,f\,(l_2 - l_1)$ and $e = 0.3\,D$.

15. BEATS & TUNING FORK PHYSICS

Beat Phenomenon: Periodic variation in sound intensity due to superposition of two waves of slightly different frequencies ($f_1 \approx f_2$).

- Beat Frequency: $f_b = |f_1 - f_2|$
- Beat Period: $T_b = 1 / |f_1 - f_2|$

Tuning Fork Alteration Rules:

- **Filing a prong:** Mass decreases ⇒ Frequency **INCREASES** ($f \uparrow$).
- **Waxing / Loading:** Mass increases ⇒ Frequency **DECREASES** ($f \downarrow$).

n-Tuning Forks Series: [Added]

If n tuning forks are arranged in increasing order of frequency, each producing x beats/sec with the preceding one:

> $f_{last} = f_{first} + (n - 1)\,x$

16. DOPPLER EFFECT IN SOUND (EXHAUSTIVE MASTER FORMULAS)

Apparent change in frequency heard by observer due to relative motion between Source, Observer, and Medium.

$$f' = f\left[\frac{(v \pm v_m \pm v_O)}{(v \pm v_m \mp v_S)}\right]$$

v = speed of sound in still medium, v_m = wind velocity along source-observer line, v_O = observer speed, v_S = source speed.

Motion Case	Apparent Frequency (f')	Apparent Wavelength (λ')
Source moves towards Observer (Rest)	$f' = f\left[\frac{v}{(v - v_S)}\right] > f$	$\lambda' = (v - v_S)/f < \lambda$
Source moves away from Observer (Rest)	$f' = f\left[\frac{v}{(v + v_S)}\right] < f$	$\lambda' = (v + v_S)/f > \lambda$
Observer moves towards Source (Rest)	$f' = f\left[\frac{(v + v_O)}{v}\right] > f$	$\lambda' = \lambda$ (Unchanged in medium!)
Observer moves away from Source (Rest)	$f' = f\left[\frac{(v - v_O)}{v}\right] < f$	$\lambda' = \lambda$ (Unchanged in medium!)
Both approaching each other	$f' = f\left[\frac{(v + v_O)}{(v - v_S)}\right]$	$\lambda' = (v - v_S)/f$
Both receding from each other	$f' = f\left[\frac{(v - v_O)}{(v + v_S)}\right]$	$\lambda' = (v + v_S)/f$
Both moving in same direction (Source leading)	$f' = f\left[\frac{(v + v_O)}{(v + v_S)}\right]$	$\lambda' = (v + v_S)/f$

17. ADVANCED DOPPLER SPECIAL CASES

1. Oblique Motion (Angle Doppler):
When motion is not along line joining source and observer, resolve velocities along line of sight:

$$f' = f\left[\frac{(v - v_O \cos\theta_2)}{(v - v_S \cos\theta_1)}\right]$$

θ_1, θ_2 are angles made by v_S, v_O with the line of sight.

2. Echo & Moving Wall Reflections:

- **Source moving towards rigid wall at v_S:**
  Frequency incident on wall: $f_w = f\left[\frac{v}{(v - v_S)}\right]$
  Frequency of echo heard by driver: $f' = f\left[\frac{(v + v_S)}{(v - v_S)}\right]$
  Beat frequency heard by driver: $f_b = f' - f = f\left[\frac{2v_S}{(v - v_S)}\right]$
- **Moving Wall (Speed v_w towards stationary source):**
  Echo frequency: $f' = f\left[\frac{(v + v_w)}{(v - v_w)}\right]$ [Added]

3. Low Speed Approximations ($v_S, v_O \ll v$): [Added]

$$\text{Fractional Shift: } \frac{\Delta f}{f} \approx \frac{v_{rel}}{v}$$

18. OPTICAL DOPPLER EFFECT (LIGHT)

Unlike sound, light requires no medium. Depends purely on relative velocity v_r between source and observer (c = speed of light):

$$f' = f\sqrt{\frac{(1 + v/c)}{(1 - v/c)}} \approx f(1 + v_{rel}/c)$$

- **Redshift (Receding Source):** Wavelength increases, frequency decreases. $\Delta\lambda/\lambda \approx +v/c$.
- **Blueshift (Approaching Source):** Wavelength decreases, frequency increases. $\Delta\lambda/\lambda \approx -v/c$.

19. MUSICAL SCALES & KEY DEFINITIONS

- **Tone:** Pure sound of single frequency.
- **Harmonics:** Frequencies that are integral multiples of fundamental (f, $2f$, $3f$...).
- **Overtone:** Any resonant frequency above fundamental.
- **Octave:** If $f_2 = 2f_1$, f_2 is 1 octave higher than f_1.
  For n octaves: $f_2 = 2^n f_1$.
- **Unison:** Two vibrating bodies with equal frequencies ($f_1 = f_2$).

20. IMPORTANT JEE CONSTANTS & DIRECT SHORTCUT RESULTS

- Speed of sound in air at 0°C (STP): $v_0 \approx 332$ m/s.
- Threshold Intensity of hearing: $I_0 = 10^{-12}$ W/m^2.
- Threshold of Pain Intensity: $I_{pain} \approx 1$ W/m^2 = 120 dB.
- Persistence of Hearing for human ear: $\Delta t \approx 0.1$ s.
- Minimum distance of reflector for distinct Echo: $d = v\,\Delta t/2 \approx (332 \times 0.1)/2 = 16.6$ m (or ~17 m).
- **Organ Pipe Clamping/Cutting Shortcuts:**
  If open pipe of length L is cut into two equal halves: fundamental frequency doubles ($f' = 2f$).
  If open pipe is dipped half in water: acts as closed pipe of length L/2 ⇒ fundamental frequency remains **same** ($f' = f$)!
- **Two Strings Joined:** Reflected & Transmitted pulse speed relation: $v_1/v_2 = \sqrt{(\mu_2/\mu_1)}$.

1. IDEAL GAS LAWS & ASSUMPTIONS OF KTG

Core Assumptions of KTG

- Molecules are tiny, rigid, perfectly elastic spheres.
- Molecular volume is negligible vs total gas volume: $V_{mol} << V_{gas}$.
- No intermolecular forces of attraction or repulsion (PE = 0).
- Collisions with container walls produce gas pressure.
- Time of collision is negligible compared to time between collisions.
- Molecules obey Maxwell-Boltzmann speed distribution.

Fundamental Gas Laws

- **Boyle's Law:** $P \propto 1/V \Rightarrow P_1V_1 = P_2V_2$ *(n, T const)*
- **Charles's Law:** $V \propto T \Rightarrow V_1/T_1 = V_2/T_2$ *(n, P const)*
- **Gay-Lussac's Law:** $P \propto T \Rightarrow P_1/T_1 = P_2/T_2$ *(n, V const)*
- **Avogadro's Law:** $V \propto n \Rightarrow$ Equal P, V, T contain equal N
- **Dalton's Law:** $P_{total} = \sum P_i = (\sum n_i) R T / V$
- **Graham's Law of Diffusion:** Rate $r \propto 1/\sqrt{(\rho)} \propto 1/\sqrt{(M)}$

Equations of State & Universal Constants

$P V = n R T = N k_B T$ | $P = (\rho / M) R T$

- **n (Moles):** $n = m / M = N / N_A$
- **R (Gas Constant):** $8.314\ J \cdot mol^{-1}K^{-1} \approx 2.0\ cal \cdot mol^{-1}K^{-1} = 0.0821\ L \cdot atm \cdot mol^{-1}K^{-1}$
- **k_B (Boltzmann):** $R / N_A = 1.38 \times 10^{-23}$ J/K
- **N_A (Avogadro No.):** $6.022 \times 10^{23}\ mol^{-1}$
- **STP Conditions:** T = 273.15 K (0°C), P = 1 atm = 1.013×10^5 Pa, V_{molar} = 22.4 L
- **SATP Conditions:** T = 298.15 K (25°C), P = 1 bar, V_{molar} = 24.8 L

2. KINETIC INTERPRETATION & MOLECULAR SPEEDS

Pressure & Kinetic Energy

$P = (1/3) \rho v_{rms}^2 = (1/3) (m N / V) v_{rms}^2 = (2/3) E_v$

- **E_v:** Translational KE per unit volume = $(1/2) \rho v_{rms}^2 = (3/2) P$
- **E_k (per molecule):** $(1/2) m v_{rms}^2 = (3/2) k_B T$
- **E_{mole} (per mole):** $(3/2) R T = (3/2) P V$
- **Energy Density:** $u = (3/2) P \Rightarrow$ Pressure $P = (2/3) u$

Molecular Speed Statistics

Speed Type	Mathematical Expression	Ratio Value
v_{rms}	$\sqrt{(3RT/M)} = \sqrt{(3P/\rho)} = \sqrt{(3k_BT/m)}$	$\sqrt{(3)} \approx 1.732$ (1.225)
v_{avg}	$\sqrt{(8RT/\pi M)} = \sqrt{(8k_BT/\pi m)}$	$\sqrt{(8/\pi)} \approx 1.595$ (1.128)
v_{mp}	$\sqrt{(2RT/M)} = \sqrt{(2k_BT/m)}$	$\sqrt{(2)} \approx 1.414$ (1.000)

$v_{mp} : v_{avg} : v_{rms} = 1 : 1.128 : 1.225$ | $v_{rms} > v_{avg} > v_{mp}$

Maxwell-Boltzmann Speed Distribution Function [Added]

$dN_v = 4\pi N (m / (2\pi k_B T))^{3/2} v^2 \exp(-m v^2 / (2 k_B T)) dv$

• dN_v = Number of molecules with speeds between v and v + dv.

- Peak of curve corresponds to v_{mp}.
- As T increases: Curve flattens, peak shifts right (v_{mp} increases), area under curve remains N.
- As M increases (at same T): Peak height increases, peak shifts left.

3. DEGREES OF FREEDOM & EQUIPARTITION THEOREM

Gas Atomicity / Structure	Degrees of Freedom (f)	U (per mole)	$C_v = (f/2)R$	$C_p = (f/2 + 1)R$	$\gamma = C_p/C_v = 1 + 2/f$
Monoatomic (He, Ne, Ar)	f = 3 (3 Trans)	(3/2) R T	(3/2) R	(5/2) R	5/3 ≈ 1.67
Diatomic (Room T) (O_2, N_2, H_2)	f = 5 (3 Trans + 2 Rot)	(5/2) R T	(5/2) R	(7/2) R	7/5 = 1.40
Diatomic (High T)	f = 7 (3 Trans + 2 Rot + 2 Vib)	(7/2) R T	(7/2) R	(9/2) R	9/7 ≈ 1.29
Triatomic Non-linear (H_2O, SO_2)	f = 6 (3 Trans + 3 Rot)	3 R T	3 R	4 R	4/3 ≈ 1.33

Gas Atomicity / Structure	Degrees of Freedom (f)	U (per mole)	$C_v = (f/2)R$	$C_p = (f/2 + 1)R$	$\gamma = C_p/C_v = 1 + 2/f$
Triatomic Linear (CO_2, CS_2)	f = 5 (3 Trans + 2 Rot)	(5/2) R T	(5/2) R	(7/2) R	7/5 = 1.40

** Law of Equipartition: Each quadratic energy term contributes (1/2)k_BT per molecule. Each active vibrational mode has 2 degrees of freedom (KE + PE).*

GAS MIXTURES, REAL GASES & MEAN FREE PATH

Thermodynamics & Molecular Dynamics Dynamics

4. GASEOUS MIXTURES & REAL GAS BEHAVIOR

Properties of Gas Mixtures

- **Total Moles:** $n_{total} = n_1 + n_2 + ... + n_k$
- **Equivalent Molecular Mass:** $M_{mix} = (\sum n_i M_i) / (\sum n_i)$
- **Equivalent C_v:** $C_{v,mix} = (\sum n_i C_{vi}) / (\sum n_i)$
- **Equivalent C_p:** $C_{p,mix} = (\sum n_i C_{pi}) / (\sum n_i)$
- **Equivalent γ_{mix}:**

  $$\gamma_{mix} = C_{p,mix} / C_{v,mix} = 1 + [(n_1 + n_2) / ((n_1/(\gamma_1 - 1)) + (n_2/(\gamma_2 - 1)))]$$

- **Internal Energy of Mixture:** $U_{mix} = \sum U_i = (1/2) R T \sum (n_i f_i)$

van der Waals Equation of Real Gases [Added]

$$(P + a n^2 / V^2) (V - n b) = n R T$$

- **Correction Factor 'a':** Measures intermolecular attraction forces. Units: atm·L^2/mol^2 or N·m^4/mol^2.
- **Correction Factor 'b':** Co-volume / Excluded volume = 4 × (Actual volume of molecules) = 4 × (4/3 π r^3 N_A). Units: L/mol or m^3/mol.
- **Low Pressure Limit:** $(P + a/V_m^2)V_m = RT \Rightarrow Z = 1 - a/(R T V_m) < 1$
- **High Pressure Limit:** $P(V_m - b) = RT \Rightarrow Z = 1 + (P b)/(R T) > 1$

Compressibility Factor (Z) & Critical Constants [Added]

$$Z = V_{real} / V_{ideal} = (P V_m) / (R T)$$

- **Z = 1:** Ideal gas behavior.
- **Z < 1:** Attractive forces dominate (Easily liquefiable).
- **Z > 1:** Repulsive forces dominate.
- **Boyle Temperature (T_B):** Temperature at which real gas acts ideal over wide P range.

  $$T_B = a / (R b)$$

- **Inversion Temperature (T_i):** $T_i = 2 a / (R b) = 2 T_B$
- **Critical Temperature:** $T_c = 8 a / (27 R b)$
- **Critical Pressure:** $P_c = a / (27 b^2)$
- **Critical Volume:** $V_c = 3 b$
- **Critical Z:** $Z_c = P_c V_c / (R T_c) = 3/8 = 0.375$

5. MEAN FREE PATH & TRANSPORT PHENOMENA

Mean Free Path (λ) Derivation & Relations

$$\lambda = 1 / (\sqrt{2} \pi d^2 n_V) = (k_B T) / (\sqrt{2} \pi d^2 P)$$

- **d:** Diameter of gas molecule | **n_V:** Number density N/V
- **Dependence on Variables:**
  - At Constant P: $\lambda \propto T$
  - At Constant V: λ is constant (independent of T).
  - At Constant T: $\lambda \propto 1/P$
- **Dependence on Mass:** $\lambda \propto 1 / (m \rho)$

Collision Frequency & Mean Free Time

- **Collision Frequency (Z_1 per molecule):**

  $$Z_1 = v_{avg} / \lambda = \sqrt{2} \pi d^2 n_V v_{avg} = 4 d^2 n_V \sqrt{(\pi k_B T / m)}$$

- **Total Collisions per unit volume per sec (Z_{11}):**

  $$Z_{11} = (1/2) n_V Z_1 = (1/\sqrt{2}) \pi d^2 n_V^2 v_{avg}$$

- **Mean Free Time (Relaxation Time τ):** $\tau = \lambda / v_{avg} = 1 / Z_1$

6. FIRST LAW OF THERMODYNAMICS (FLOT) DEEP DIVE

First Law Statement & Sign Conventions

$$\Delta Q = \Delta U + W \Rightarrow dQ = dU + dW$$

Work Done Calculation Methods

- **ΔQ > 0:** Heat absorbed BY system | **ΔQ < 0:** Heat released
- **W > 0:** Work done BY system (Expansion) | **W < 0:** Work done ON system (Compression)
- **ΔU > 0:** Internal energy increases | **ΔU < 0:** Internal energy decreases
- **Path Functions:** Q and W depend on exact thermodynamic path taken.
- **State Functions:** U, P, V, T, S depend ONLY on initial and final states.

  For any closed cyclic process: $\oint dU = 0 \Rightarrow Q_{net} = W_{net}$

- **General Work Formula:** $W = \int_{V_i}^{V_f} P\, dV$
- **Indicator Diagram (P-V Graph):**

  W = Area under P-V curve bounded by V-axis

- **Cyclic Process Work:**
  - **Clockwise Loop:** $W_{net} > 0$ (Positive, Engine)
  - **Counter-Clockwise Loop:** $W_{net} < 0$ (Negative, Refrigerator)
  - W_{net} = Enclosed Area inside the P-V cycle loop.

THERMODYNAMIC PROCESSES & POLYTROPIC ANALYSIS

Detailed Process Equations & Property Transformations

7. COMPREHENSIVE ANALYSIS OF STANDARD PROCESSES

Process	Governing Condition	Work Done (W)	Internal Energy (ΔU)	Heat Exchanged (ΔQ)	Molar Heat Capacity (C)
Isobaric	P = Const (dP = 0) V / T = Const	$P(V_f - V_i) = n R \Delta T$	$n C_v \Delta T$	$n C_p \Delta T$	$C_p = (f/2 + 1)R = \gamma R / (\gamma - 1)$
Isochoric	V = Const (dV = 0) P / T = Const	**0**	$n C_v \Delta T$	$\Delta Q = \Delta U = n C_v \Delta T$	$C_v = (f/2)R = R / (\gamma - 1)$
Isothermal	T = Const (dT = 0) P V = Const	$n R T \ln(V_f / V_i)$ $= n R T \ln(P_i / P_f)$	**0**	ΔQ = W	**C = ∞**
Adiabatic	Q = Const (dQ = 0) $P V^{\gamma}$ = Const	$(P_i V_i - P_f V_f) / (\gamma - 1)$ $= n R(T_i - T_f) / (\gamma - 1)$	ΔU = −W $= n C_v \Delta T$	**0**	**C = 0**
Polytropic [Added]	$P V^x$ = Const ($x \neq 1, \gamma$)	$(P_i V_i - P_f V_f) / (x - 1)$ $= n R \Delta T / (1 - x)$	$n C_v \Delta T$	$n C \Delta T$	$C = C_v + R / (1 - x)$ $= R / (\gamma - 1) + R / (1 - x)$

8. ADVANCED POLYTROPIC PROCESS ANALYSIS ($PV^X = C$) [ADDED]

Polytropic Exponent (x) Classification

- **x = 0:** Isobaric Process (P = Const, $C = C_p$)
- **x = 1:** Isothermal Process (T = Const, C = ∞)
- **x = γ:** Adiabatic Process (Q = Const, C = 0)
- **x = ±∞:** Isochoric Process (V = Const, $C = C_v$)
- **Condition for Heat Absorption during Expansion:**

  If $1 < x < \gamma \Rightarrow$ Molar heat capacity C < 0! Gas absorbs heat yet its temperature drops.

Slope & Bulk Modulus Relations

- **P-V Curve Slope:** $dP / dV = -x (P / V)$
  - Isothermal Slope: $(dP/dV)_{iso} = -P/V$
  - Adiabatic Slope: $(dP/dV)_{adia} = -\gamma (P/V)$
  - $(Slope)_{adiabatic} = \gamma \times (Slope)_{isothermal}$
- **Bulk Modulus of Elasticity ($B = -V\, dP/dV$):**

  $B_{isothermal} = P$ | $B_{adiabatic} = \gamma P$ | $B_{polytropic} = x P$

9. SPECIAL PROCESSES & GRAPHICAL SHORTCUTS

Free Expansion & Variable Processes [Added]

Energy Distribution & Work Fractions

- **Free Expansion (Expansion into Vacuum):**

  $W = 0, Q = 0 \Rightarrow \Delta U = 0 \Rightarrow T_i = T_f$

  • Highly irreversible! Ideal gas temp stays constant; real gas temp drops due to internal work against attraction.
- **Linear P-V Process ($P = P_0 - \alpha V$):**
  - Work Done W = Area of Trapezoid = $(1/2)(P_1 + P_2)(V_2 - V_1)$
  - Max Temperature occurs at: $V = P_0 / (2\alpha)$
  - $T_{max} = P_0^2 / (4 \alpha n R)$

- **Isobaric Process Ratios:**

  $\Delta U : W : \Delta Q = 1 : (\gamma - 1) : \gamma = (1/\gamma) : (1 - 1/\gamma) : 1$
- Fraction of heat converted to internal energy: $\Delta U / Q = 1 / \gamma$
  - Monoatomic ($\gamma = 5/3$): 60% internal energy, 40% work
  - Diatomic ($\gamma = 7/5$): 71.4% internal energy, 28.6% work
- Fraction of heat converted to work: $W / Q = 1 - 1/\gamma = (\gamma - 1) / \gamma$

10. SECOND LAW, ENGINES & REFRIGERATORS

Heat Engines

$\eta = W / Q_1 = 1 - Q_2 / Q_1$

- Q_1: Heat absorbed from source (T_1).
- Q_2: Heat rejected to sink (T_2).
- **Kelvin-Planck:** $\eta < 100\%$ always.

Refrigerators & Heat Pumps

$\beta = Q_2 / W = Q_2 / (Q_1 - Q_2)$

- **β:** Coeff. of Performance (COP).
- **Heat Pump COP (α):** $\alpha = Q_1/W = \beta + 1$
- **Relation:** $\beta = (1 - \eta) / \eta$

Carnot Engine Cycle

$\eta_{Carnot} = 1 - T_2 / T_1 = 1 - T_L / T_H$

- Reversible ideal engine.
- $Q_1 / T_1 = Q_2 / T_2$
- $\beta_{Carnot} = T_2 / (T_1 - T_2)$

ADVANCED CYCLES, ENTROPY & HEAT TRANSFER

JEE Advanced Problem Solving & Thermal Physics Supplement

11. OTHER THERMODYNAMIC CYCLES [ADDED]

Otto Cycle (Ideal Petrol Engine)

- Consists of 2 Isochoric processes + 2 Adiabatic processes.
- **Compression Ratio (r):** $r = V_{max} / V_{min} = V_1 / V_2$
- **Thermal Efficiency:**

  $\eta_{Otto} = 1 - 1 / r^{\gamma - 1}$

Diesel Cycle & Stirling Cycle

- **Diesel Cycle:** 1 Isobaric heat addition + 1 Isochoric heat rejection + 2 Adiabatic lines.

  $\eta_{Diesel} = 1 - (1 / r^{\gamma-1}) [(r_c^{\gamma} - 1) / (\gamma (r_c - 1))]$

  (r_c = Cut-off ratio = V_3/V_2)
- **Stirling Cycle:** 2 Isothermal + 2 Isochoric lines. Efficient as Carnot.

12. ENTROPY & REVERSIBILITY [ADDED]

Entropy Definition & Calculation ($dS = dQ_{rev} / T$)

- **State Function:** ΔS depends only on initial and final thermodynamic states.
- **General Ideal Gas ΔS Formula:**

  $\Delta S = n C_v \ln(T_f / T_i) + n R \ln(V_f / V_i)$

  $\Delta S = n C_p \ln(T_f / T_i) - n R \ln(P_f / P_i)$
- **Phase Change Entropy:** $\Delta S_{fusion} = m L_f / T_m$ | $\Delta S_{vap} = m L_v / T_b$

Second Law Statements & Universe Entropy

- **Clausius Statement:** Heat cannot spontaneously flow from a colder to hotter body without external work.
- **Principle of Entropy Increase:** For any natural/irreversible process in an isolated system:

  $\Delta S_{univ} = \Delta S_{system} + \Delta S_{surrounding} > 0$
- **Reversible Process:** $\Delta S_{univ} = 0$

13. HEAT TRANSFER MECHANISMS (CONDUCTION & RADIATION) [ADDED]

Thermal Conduction & Thermal Resistance

Heat Current $H = dQ / dt = K A (T_1 - T_2) / L$

- **Thermal Resistance (R_{th}):** $R_{th} = L / (K A) \Rightarrow H = \Delta T / R_{th}$
- **Series Combination:** $R_{eq} = R_1 + R_2 \Rightarrow L_{eq} / K_{eq} = L_1 / K_1 + L_2 / K_2$
- **Parallel Combination:** $1 / R_{eq} = 1/R_1 + 1/R_2 \Rightarrow K_{eq} A_{eq} = K_1 A_1 + K_2 A_2$
- **Junction Temp (T_j in Series):** $T_j = (K_1 L_2 T_1 + K_2 L_1 T_2) / (K_1 L_2 + K_2 L_1)$

Thermal Radiation & Stefan-Boltzmann Law

$P_{emitted} = e \sigma A T^4$ | $P_{net} = e \sigma A (T^4 - T_0^4)$

- **σ (Stefan Constant):** $5.67 \times 10^{-8}\ W \cdot m^{-2} \cdot K^{-4}$
- **Blackbody Radiation ($e = 1$):** $P = \sigma A T^4$
- **Wien's Displacement Law:** $\lambda_{max} T = b = 2.898 \times 10^{-3}\ m \cdot K$
- **Newton's Law of Cooling (Small ΔT):** $dT/dt = -k (T - T_0)$

  $(T_1 - T_2) / t = K [(T_1 + T_2)/2 - T_0]$

14. STANDARD JEE SHORTCUT RESULTS & CONSTANTS SUMMARY

- **Work in Adiabatic Process:** $W = (P_1V_1 - P_2V_2)/(\gamma - 1) = n R (T_1 - T_2)/(\gamma - 1)$
- **Adiabatic Bulk Modulus:** $B = \gamma P$
- **Isothermal Bulk Modulus:** $B = P$
- **Slope Ratio:** $(dP/dV)_{Adiabatic} / (dP/dV)_{Isothermal} = \gamma$
- **Velocity Comparison:** $v_{rms} = 1.732 \sqrt{(RT/M)}$, $v_{avg} = 1.595 \sqrt{(RT/M)}$, $v_{mp} = 1.414 \sqrt{(RT/M)}$
- **Specific Heat Ratio:** $\gamma = C_p/C_v = 1 + 2/f$
- **Mean Free Path:** $\lambda \propto T/P$ (at constant volume, λ is constant)
- **Isobaric Work Fraction:** $W / Q = (\gamma - 1) / \gamma$

MODERN PHYSICS

JEE MAIN + ADVANCED · FORMULA REVISION SHEET

Dual nature · Photoelectric · X-rays · Atoms
Nuclei · Radioactivity · Nuclear reactions
★ = standard JEE result filled from syllabus gaps

Boxed gold bar = must-remember relation Navy bar = definition / SI form Conditions in amber Source PDF is primary; signs/ε_0/$\hbar$ typos in the source are corrected here

0 · CONSTANTS & CONVERSIONS

Quantity	JEE working value
Planck h	$6.626\times10^{-34}\,\text{J s} = 4.136\times10^{-15}\,\text{eV s}$
$\hbar = h/2\pi$	$1.055\times10^{-34}\,\text{J s}$
hc	$12400\,\text{eV}\cdot\text{Å} = 1240\,\text{eV}\cdot\text{nm}$
$c,\ e,\ m_e$	$3\times10^{8}\,\text{m/s};\ 1.6\times10^{-19}\,\text{C};\ 9.1\times10^{-31}\,\text{kg}$
Rydberg R_∞	$1.097373\times10^{7}\,\text{m}^{-1}$
Bohr radius a_0	$0.529\,\text{Å} = 5.29\times10^{-11}\,\text{m}$
$1\,\text{eV},\ 1\,\text{u}$	$1.6\times10^{-19}\,\text{J};\ 931.5\,\text{MeV}/c^2 = 1.6605\times10^{-27}\,\text{kg}$
Fine structure α ★	$\alpha = e^2/(4\pi\varepsilon_0\hbar c)\approx 1/137$
Bohr magneton μ_B ★	$e\hbar/2m_e = 9.27\times10^{-24}\,\text{J/T}$
Compton λ_C ★	$h/(m_e c) = 2.43\times10^{-12}\,\text{m} = 0.0243\,\text{Å}$
$k = 1/(4\pi\varepsilon_0)$	$9\times10^{9}\,\text{N m}^2\text{C}^{-2}$

Workhorse: photon energy $E(\text{eV}) = 12400/\lambda(\text{Å}) = 1240/\lambda(\text{nm})$.

1 · DUAL NATURE OF RADIATION & MATTER

1.1 Photons ★

$$E = h\nu = \frac{hc}{\lambda},\qquad p = \frac{h}{\lambda} = \frac{E}{c},\qquad m_{\text{rel}} = \frac{E}{c^2}$$

Number of photons in time t from power P: $N = Pt/(h\nu) = P\lambda t/(hc)$.
Intensity $I = (\text{energy/area/time})$.

A photon interacts with **one** electron (Einstein). Intensity $\propto$ number of photons, not energy of one photon.

1.2 Photoelectric effect

Emission of electrons from a metal by light of frequency $\geq \nu_0$ (or $\lambda \leq \lambda_0$). Instantaneous — no measurable time lag.

Einstein's photoelectric equation

$$h\nu = \phi_0 + K_{\max} = h\nu_0 + \tfrac{1}{2}mv_{\max}^2$$
$$K_{\max} = eV_s = h(\nu - \nu_0)$$
$$K_{\max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

$\phi_0 = h\nu_0 = hc/\lambda_0$ = work function. V_s = stopping potential (retarding voltage that makes photocurrent zero).

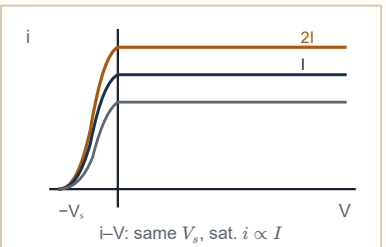
Emission only if $\nu \geq \nu_0$. Raising intensity cannot start emission below threshold.

Change	Current i	$K_{\max}$, V_s
Intensity ↑ (ν fixed $> \nu_0$)	sat. $i \propto I$	unchanged
Frequency ↑ (I fixed)	sat. i same	both ↑
Different metal	—	ν_0 changes
Point source distance ★	$I \propto 1/r^2$	—

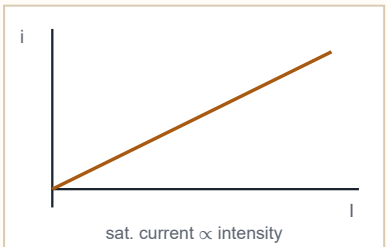
Graphs (JEE Main staples)

$$V_s = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

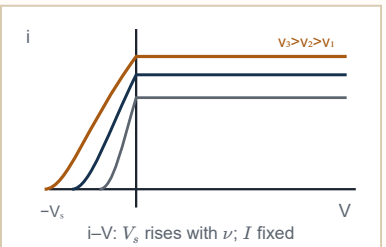
Slope h/e (universal). ν-intercept $= \nu_0$. $K_{\max}$–ν slope $= h$. Parallel lines for different metals.



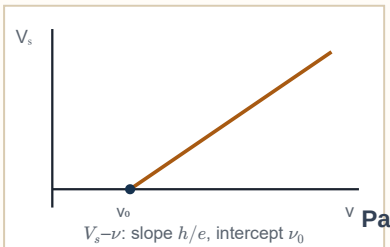
i–V: same V_s, sat. $i \propto I$



sat. current $\propto$ intensity



i–V: V_s rises with ν; I fixed



V_s–ν: slope h/e, intercept ν_0

1.3 Compton effect ★

Photon–free electron elastic scatter. Wavelength increases; photon loses energy.

$$\Delta\lambda = \lambda' - \lambda = \lambda_C(1-\cos\theta)$$
$$\lambda_C = \frac{h}{m_e c} = 0.0243\,\text{Å}$$

$\theta = 0 \Rightarrow \Delta\lambda = 0;\ \theta = 90^\circ \Rightarrow \Delta\lambda = \lambda_C;\ \theta = 180^\circ \Rightarrow \Delta\lambda = 2\lambda_C$ (max).

$$K_e = h\nu - h\nu' = \frac{h\nu\,\alpha(1-\cos\theta)}{1+\alpha(1-\cos\theta)}$$
$$\alpha = h\nu/(m_e c^2)$$

$\Delta\lambda$ independent of λ and of the scatterer. Recoil electron and photon: coplanar; ϕ (electron) vs θ linked by momentum conservation.

1.4 Pair production / annihilation ★

Photon $\rightarrow e^- + e^+$ in Coulomb field of a nucleus. Threshold $E_\gamma \geq 2m_e c^2 = 1.022\,\text{MeV}$. Reverse: annihilation yields two $511\,\text{keV}$ photons (to conserve p).

1.5 de Broglie (matter) waves

Every material particle has wavelength $\lambda = h/p$. Observable for light particles (electrons).

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}\qquad (K = \tfrac{1}{2}mv^2 \text{ non-rel.})$$

Charged particle accelerated by V: $K = qV$, so $\lambda = h/\sqrt{2mqV}$.

$$\lambda_e = \frac{12.27}{\sqrt{V}}\ \text{Å}\qquad (V \text{ in volts};\ m_e = 9.1\times10^{-31}\,\text{kg})$$

Source numerical: $h = 6.62\times10^{-34}$. Some keys use $12.25/\sqrt{V}$ or $12.26/\sqrt{V}$. Use the value given in the question.

★ Other species (non-rel., V in volts): $\lambda_p \approx 0.286/\sqrt{V}$ Å, $\lambda_\alpha \approx 0.101/\sqrt{V}$ Å.

Relativistic (when $K \gtrsim mc^2$) ★

$$\lambda = \frac{hc}{\sqrt{K(K+2mc^2)}}$$

From $E_{\text{tot}}^2 = p^2c^2 + m^2c^4$. Use when $K \gtrsim mc^2$.

★ Bohr orbit as a standing wave: $2\pi r_n = n\lambda \Leftrightarrow mvr = n\hbar$ (quantisation).

★ Group velocity $=$ particle velocity v; phase velocity $v_p = E/p = c^2/v > c$ (massive particle).

Davisson–Germer: electron diffraction from Ni; Bragg $2d\sin\theta = n\lambda$. Historic peak: $54\,\text{eV},\ \theta = 50^\circ$.

1.6 Heisenberg uncertainty ★

$$\Delta x\,\Delta p_x \geq \frac{\hbar}{2} = \frac{h}{4\pi},\qquad \Delta E\,\Delta t \geq \frac{\hbar}{2}$$

JEE keys also accept $\geq \hbar$ or $\geq h/4\pi$ if the problem states the form. Estimate ground-state energy of a particle confined to L: $\Delta p \sim h/(4\pi L)$, $K \sim (\Delta p)^2/2m$.

If a photon of λ locates an electron, $\Delta x \sim \lambda$ and $\Delta p \sim h/\lambda$.

2 · X-RAYS

EM waves, $\lambda \sim 0.01$–$1\,\text{nm}$. Coolidge tube: electrons of KE eV strike a metal target.

2.1 Continuous (bremsstrahlung)

Electron decelerated in nuclear field $\rightarrow$ photons of a continuum of energies up to eV.

$$\lambda_{\min} = \frac{hc}{eV} = \frac{12400}{V}\ \text{Å},\qquad \nu_{\max} = \frac{eV}{h}$$

Duane–Hunt. $\lambda_{\min}$ independent of target Z; decreases as tube voltage ↑.

★ Intensity of continuum $\propto ZV^2$ (empirical). Most electrons heat the target, not X-rays.

2.2 Characteristic X-rays

Incident electron knocks out a K (or L, …) electron. A higher-shell electron fills the vacancy $\rightarrow$ photon of discrete energy (K, L, M series).

Line	Jump	Photon energy
K_α	$2 \rightarrow 1$	$10.2(Z-1)^2\,\text{eV}$
K_β	$3 \rightarrow 1$	$13.6(Z-1)^2(1-1/9)\,\text{eV}$

L_α	$3 \to 2$	$13.6(Z-7.4)^2(1/4 - 1/9)$ eV ★

$\lambda_{K_\alpha} > \lambda_{K_\beta}$ and $E_{K_\beta} > E_{K_\alpha}$. Characteristic lines appear only if $eV \geq$ ionisation energy of that shell.

2.3 Moseley's law ★

$$\sqrt{\nu} = a(Z-b)$$
$$K_\alpha:\ \nu = \tfrac{3}{4}Rc(Z-1)^2,\ \ b=1$$

$b \approx 7.4$ for L series. $\sqrt{\nu}$ vs Z is linear. Explains ordering of Co/Ni etc.

2.4 Bragg diffraction ★

$$2d\sin\theta = n\lambda \qquad (n = 1, 2, \dots \text{ order})$$

X-ray spectrum: cutoff $\lambda_{\min}$ + characteristic spikes

3 · ATOMS — RUTHERFORD SCATTERING

α on thin Au foil, ZnS scintillations:

- Most α undeflected or deflected $\sim 1^\circ \Rightarrow$ atom mostly empty; +charge in a tiny nucleus.
- A few deflected by $\geq 90^\circ \Rightarrow$ nucleus massive, concentrated charge.

Distance of closest approach (head-on, KE K) ★

$$d = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{K} \qquad (\alpha \text{ on nucleus } Z)$$

At turning point $K = U(d)$. Estimates nuclear size (upper bound).

Impact parameter ★

$$b = \frac{k\,q_1 q_2}{2K}\cot\frac{\theta}{2} = \frac{k\,Ze^2}{K}\cot\frac{\theta}{2}\ (\alpha)$$

$b = 0 \Rightarrow \theta = 180^\circ$; large $b \Rightarrow$ small θ.

★ Scattered number: $N(\theta) \propto \dfrac{ntZ^2}{K^2\sin^4(\theta/2)}$ (Rutherford formula). t = foil thickness, n = number density.

4 · BOHR MODEL (H AND H-LIKE)

Postulates. (1) Electron in circular Coulomb orbit about nucleus; stationary states — no radiation. (2) Radiation only in a jump: $h\nu = |E_i - E_f|$. (3) Angular momentum quantised:

$$L = mvr = n\hbar = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$

Force balance + quantisation ⇒

$$r_n = \frac{4\pi\varepsilon_0 n^2\hbar^2}{mZe^2} = \frac{n^2 a_0}{Z}$$
$$a_0 = 0.529\ \text{Å}$$
$$v_n = \frac{Ze^2}{4\pi\varepsilon_0 n\hbar} = \frac{Z\alpha c}{n}$$
$$v_1(\text{H}) = 2.18\times10^6\ \text{m/s}$$
$$E_n = -\frac{13.6\,Z^2}{n^2}\ \text{eV}$$

SI energy: $E_n = -mZ^2e^4/(8\varepsilon_0^2 n^2\hbar^2)$. H: $Z=1$; He^+: $Z=2$; Li^{2+}: $Z=3$. Source PDF omitted Z and the minus sign on E_n.

Qty	Relation	$\propto$
KE	$K = -E_n = 13.6Z^2/n^2$ eV	Z^2/n^2
PE	$U = 2E_n = -27.2Z^2/n^2$ eV	Z^2/n^2
Virial	$\lvert U\rvert = 2K$, $E = K+U = -K$	—
r_n	n^2a_0/Z	n^2/Z
v_n	$\alpha c\,Z/n$	Z/n
T_n	$T_1 n^3/Z^2$; $T_1 = 1.52\times10^{-16}$ s	n^3/Z^2
$f = 1/T$	$f_1 Z^2/n^3$	Z^2/n^3
$I = ef$ ★	$ev/(2\pi r)$	Z^2/n^3
$\mu = IA$ ★	$n\mu_B = ne\hbar/(2m)$	n
B at nucleus ★	$\mu_0 I/(2r)$	Z^3/n^5

★ Ionisation energy from level n: $13.6Z^2/n^2$ eV. Excitation $1 \to n$: $13.6Z^2(1-1/n^2)$ eV.

★ Ground-state values (H): $r_1 = a_0$, $v_1 = c/137$, $E_1 = -13.6$ eV, $K = 13.6$ eV, $U = -27.2$ eV.

4.1 Spectra of H / H-like

Photon energy = difference of stationary-state energies (emission: $n_u \to n_l$; absorption: reverse).

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_l^2} - \frac{1}{n_u^2}\right)$$
$$R = 1.097\times10^7\ \text{m}^{-1}$$
$$\Delta E = 13.6\,Z^2\left(\frac{1}{n_l^2} - \frac{1}{n_u^2}\right)\text{eV}$$

Series	n_l	Region	H first line / limit
Lyman	1	UV	1216 Å / 911 Å
Balmer	2	visible	6563 Å / 3646 Å
Paschen	3	IR	18751 Å / 8204 Å
Brackett	4	IR	limit 14584 Å
Pfund	5	IR	limit 22790 Å

Series limit: $1/\lambda = RZ^2/n_l^2$. First line of a series: $n_u = n_l + 1$.

$$N_{\text{lines}}(n_2 \to n_1) = \frac{(n_2-n_1)(n_2-n_1+1)}{2}$$
$$n_2 \to 1:\quad \frac{n_2(n_2-1)}{2}$$

H energy ladder (eV): $n=1: -13.60;\ 2: -3.40;\ 3: -1.51;\ 4: -0.85;\ 5: -0.54;\ \infty: 0$.

Visible Balmer: Hα 6563 Å (red), Hβ 4861, Hγ 4340, Hδ 4102.

4.2 Reduced mass, isotopes, shortcuts ★

$$\mu = \frac{mM}{m+M}, \qquad R_H = R_\infty\frac{\mu}{m} = \frac{R_\infty}{1+m/M}$$

Replace $m \to \mu$ in r_n, E_n. Heavier nucleus ⇒ slightly larger $|E|$ and R. Deuterium lines slightly blue-shifted vs H.

★ For H-like in state n, $Z_{\text{eff}} = Z$. Muonic atom: $m_\mu \approx 207m_e$ ⇒ radii $\sim 1/207$, energies $\times 207$.

★ Photon wavelength from $n_2 \to n_1$: $\lambda = hc/|\Delta E|$. If λ in Å, $\lambda = 12400/\Delta E_{\text{eV}}$.

★ Resonance absorption: incident photon energy must match a Bohr difference (to within natural width).

★ If KE of electron in first Bohr orbit is K, then $E = -K$, $U = -2K$, $v = \sqrt{2K/m}$, $r = ke^2/(2K)$.

5 · ATOMIC NUCLEUS

Size $\sim$ femtometre ($1\,\text{fm} = 10^{-15}$ m). Mass number $A = Z + N$.

$$R = R_0 A^{1/3}, \qquad R_0 \approx 1.2\,\text{fm} = 1.2\times10^{-15}\,\text{m}$$

Some keys use $R_0 = 1.1$ or 1.3 fm — use the given value.

Volume $\propto A$. Nuclear density nearly constant:

$$\rho = \frac{3m_N}{4\pi R_0^3} \approx 2.3\times10^{17}\,\text{kg m}^{-3}$$

★ Surface area $\propto A^{2/3}$. Closest approach of two nuclei $\sim R_1 + R_2 = R_0(A_1^{1/3} + A_2^{1/3})$.

5.1 Mass defect & binding energy ★

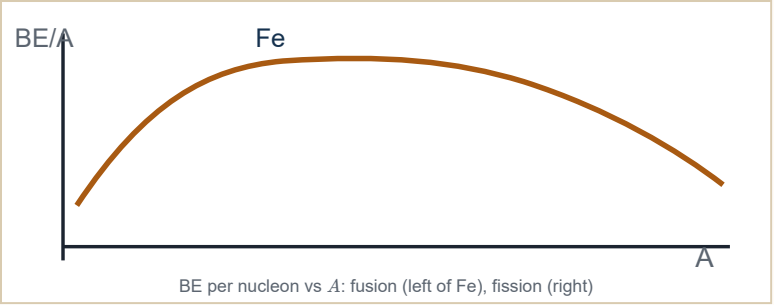
$$\Delta m = Zm_p + (A-Z)m_n - M_{\text{nuc}}$$
$$\text{atomic form: } \Delta m = Zm_H + (A-Z)m_n - M_{\text{atom}}$$
$$\text{BE} = \Delta m\,c^2 = \Delta m(\text{u}) \times 931.5\,\text{MeV}$$

BE per nucleon = BE/A. Peak near ^{56}Fe (≈ 8.8 MeV). Higher BE/A ⇒ more stable.

Packing fraction $f = (M-A)/A$. Negative $f \Leftrightarrow$ generally tightly bound.

Useful masses: $m_p = 1.007276$ u, $m_n = 1.008665$ u, $m_H = 1.007825$ u, $m_e = 0.0005486$ u, $\alpha = 4.002603$ u. BE(α) ≈ 28.3 MeV; deuteron ≈ 2.22 MeV.

★ Separation energy of a nucleon ≈ BE difference of parent and residual nucleus.



BE per nucleon vs A: fusion (left of Fe), fission (right)

Nuclear force: short-range ($\sim 2\,\text{fm}$), charge-independent, saturating, spin-dependent, strongest of the three common interactions. Magic numbers ★: $2, 8, 20, 28, 50, 82, 126$.

$N/Z \approx 1$ (light); $N/Z \to \sim 1.6$ (heavy) for β-stability.

5.2 Q-value & nuclear reactions ★

$a + X \to Y + b$. Conserved: A, Z, energy, momentum, lepton number (in β).

$$Q = (m_{\text{initial}} - m_{\text{final}})c^2$$

$Q > 0$ exoergic (possible at low KE); $Q < 0$ endoergic, threshold in lab (target at rest):

$$K_{\text{th}} = -Q\frac{m_a + m_X}{m_X} \approx -Q\left(1 + \frac{m_a}{m_X}\right)$$

Fission: heavy nucleus $\to$ two medium fragments + 2–3 n. $^{235}\text{U} + n \to$ fragments $+ \sim 200\,\text{MeV}$. Fusion: light nuclei combine; $4p \to {}^4\text{He} + 2e^+ + 2\nu + 26.7\,\text{MeV}$ (solar pp chain net).

6 · RADIOACTIVITY

- Statistical: only probabilities; a given nucleus may decay at once or "never".
- Independent of T, P, E/B fields (chemical state too, except tiny electron-capture effects).
- Energy from the nucleus itself. α/β change $Z \Rightarrow$ a new element.

Decay law

$$\frac{dN}{dt} = -\lambda N, \qquad N = N_0 e^{-\lambda t}, \qquad A = \lambda N = A_0 e^{-\lambda t}$$

λ = decay constant (probability of decay per unit time). A = activity.

Units ★: $1\,\text{Bq} = 1$ decay/s; $1\,\text{Ci} = 3.7 \times 10^{10}\,\text{Bq}$.

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}, \qquad \tau = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693} = 1.443\,T_{1/2}$$

$T_{1/2}$: time for $N \to N_0/2$. Mean (average) life τ: average survival time.

$$N = N_0\,2^{-t/T_{1/2}}$$
$$t = \frac{2.303}{\lambda}\log_{10}(N_0/N)$$

After n half-lives: remaining fraction $= 2^{-n}$; decayed $= 1 - 2^{-n}$.

Survival probability of one nucleus in time t: $e^{-\lambda t}$. Decay probability: $1 - e^{-\lambda t}$.

For $t \ll T_{1/2}$: $\approx \lambda t$.

★ Specific activity $= A/m = \lambda N_A/M$ for a pure sample (M = molar mass).

★ $\ln N$ (or $\ln A$) vs t: straight line, slope $= -\lambda$.

6.1 α, β, γ decay ★

Mode	$\Delta A, \Delta Z$	Key relation
α	$-4, -2$	${}^A_Z X \to {}^{A-4}_{Z-2}Y + \alpha$
β^-	$0, +1$	$n \to p + e^- + \bar{\nu}$; $Q = [M_X - M_Y]c^2$
β^+	$0, -1$	$Q = [M_X - M_Y - 2m_e]c^2$
EC	$0, -1$	$p + e^- \to n + \nu$; $Q = [M_X - M_Y]c^2 - B_e$
γ	$0, 0$	nuclear de-excitation

α energy sharing (recoil)

$$K_\alpha = \frac{A-4}{A}Q, \qquad K_{\text{recoil}} = \frac{4}{A}Q$$

β spectrum is continuous (0 to $K_{\max}$) because $\bar{\nu}/\nu$ shares energy. End-point $K_{\max} \approx Q$ (ignore recoil). Neutrino: Pauli, ≈ 0 rest mass in JEE.

★ β^+ needs $Q \geq 1.022\,\text{MeV}$ (two electron masses). EC can occur when β^+ is forbidden.

★ Geiger–Nuttall (advanced): $\log\lambda = a + b\log R_\alpha$ or $\ln\lambda = A - B/\sqrt{Q_\alpha}$ — shorter $T_{1/2} \Leftrightarrow$ higher Q_α.

6.2 Successive decay, mixture, dating ★

$A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C$ (stable), $N_B(0) = 0$:

$$N_B = \frac{\lambda_A N_{A0}}{\lambda_B - \lambda_A}(e^{-\lambda_A t} - e^{-\lambda_B t})$$
$$t_m = \frac{\ln(\lambda_B/\lambda_A)}{\lambda_B - \lambda_A}$$

Secular eq. ($\lambda_B \gg \lambda_A$, parent long-lived): $A_B = A_A$ i.e. $\lambda_B N_B = \lambda_A N_A$.

Transient eq. ($\lambda_B > \lambda_A$): $A_B/A_A = \lambda_B/(\lambda_B - \lambda_A)$.

Two independent species: $A(t) = A_1 e^{-\lambda_1 t} + A_2 e^{-\lambda_2 t}$.

Parallel (branching) modes: $\lambda = \lambda_1 + \lambda_2$, $T_{1/2} = 0.693/\lambda$. Branching ratio $= \lambda_i/\lambda$.

Carbon dating (^{14}C, $T_{1/2} = 5730\,\text{yr}$):

$$t = \frac{1}{\lambda}\ln\frac{A_0}{A} = \frac{T_{1/2}}{0.693}\ln\frac{A_0}{A}$$

A_0: activity of living sample (same species, same mass). Also works with N_0/N.

7 · HIGH-YIELD JEE SHORTCUTS ★

- If λ is halved at fixed energy-intensity I, photon energy $\times 2$, photon number $\times 1/2$, so sat. current halves and $K_{\max}$ changes via Einstein. Many JEE Main keys treat intensity as photon flux, so they keep i independent of ν.
- $eV_s = 12400(1/\lambda - 1/\lambda_0)$ with λ in Å, V_s in volts.
- de Broglie of e^- in Bohr n: $\lambda_n = 2\pi r_n/n = 2\pi a_0 n/Z$.
- Ratio $r_n : v_n : T_n : E_n = n^2 : 1/n : n^3 : 1/n^2$ (fixed Z).
- For H-like, $E \propto Z^2$, $\lambda \propto 1/Z^2$, $v \propto Z$, $r \propto 1/Z$.
- A photon from $n_2 \to n_1$ ionises ground-state H only if $\Delta E \geq 13.6\,\text{eV}$ (Lyman limit / continuum). Lyman α is $10.2\,\text{eV}$ — it excites, it does not ionise.
- To excite H from ground, minimum $\lambda = hc/10.2\,\text{eV} = 1216\,\text{Å}$ (not 911 — that's ionisation).
- X-ray: if V doubled, $\lambda_{\min}$ halves; characteristic λ unchanged (depends on Z).
- If K_α of an element matches Lyman α of H-like ion, use Moseley = Bohr with $Z - 1$.
- Activity in n half-lives: $A = A_0/2^n$. Mass of radioactive atoms $\propto N \propto A$.
- Mixture of α and β emitters: total $A = \sum \lambda_i N_i$; detected particles depend on branching.
- In α decay of rest parent, $p_\alpha = p_{\text{recoil}}$, so $K \propto 1/m$.
- 1 u mass defect $\Leftrightarrow 931.5\,\text{MeV}$. Never use 931 vs 931.5 inconsistently with the paper.
- Nuclear density independent of A; atomic density is not.
- Compton $\Delta\lambda/\lambda$ is large for X / γ, negligible for visible ($\lambda \gg \lambda_C$).
- If stopping potential is V_s at λ, at 2λ it is not $V_s/2$ — use Einstein, not proportion.
- Number of photoelectrons $\propto I \times (\text{area}) \times t/(h\nu)$ times quantum efficiency ≤ 1.

8 · GRAPH & TRAP CHECKLIST

- i–V photoelectric: current zero at $V = -V_s$, then rises to saturation; reversing anode–cathode beyond saturation does not raise $K_{\max}$.
- KE of photoelectrons is a distribution $0 \to K_{\max}$; $K_{\max}$ is for the least-bound (surface) electrons.
- Work function is of the **emitter**, not the collector (unless specified).
- Bohr E_n is **negative**; transitions use $|\Delta E|$. Do not write $E = +13.6/n^2$.
- Rydberg: n_l is the **lower** (smaller) quantum number for emission formula as written.
- X-ray $\lambda_{\min}$ from tube voltage; characteristic λ from target Z. Do not mix.
- Radioactive $A = \lambda N$, not N. A sample can have large N and small A if λ is tiny.
- β^+ Q formula with atomic masses must subtract $2m_e c^2$; β^- must not.
- Half-life of a given nuclide is independent of N_0; activity is not.
- de Broglie: use kinetic energy, not total/relativistic energy, unless the problem is relativistic.

Corrected vs source PDF: Bohr r_n includes ε_0 and $\hbar$; v_n uses $\hbar$ not h; $E_n = -13.6Z^2/n^2$ eV (minus sign, Z); Einstein $K_{\max} = eV_s$; half-life algebra is $N_0/2 = N_0 e^{-\lambda T_{1/2}}$.

COMMUNICATION SYSTEMS

JEE Main & Advanced Revision Formula Sheet

PHYSICS
CLASS 12 • CH-15

1. SIGNAL BANDWIDTH & SPECTRUM ALLOCATION

Core Fundamentals

Bandwidth (B): The range of frequencies over which a signal or medium operates: $B = f_{max} - f_{min}$.

Signal Type	Frequency Band / Range	Bandwidth
Speech / Voice [Added]	300 Hz – 3100 Hz	**2.8 kHz**
Music Signal [Added]	20 Hz – 20 kHz	**20 kHz**
Video Signal	DC – 4.2 MHz	**4.2 MHz**
TV Signal (Video + Audio) [Added]	Video + Voice band	**6.0 MHz**

Transmission Medium	Usable Bandwidth
Standard Wire Pair [Added]	~ 18 kHz
Coaxial Cable	~ **750 MHz** (operated up to 750 MHz)
Optical Fiber [Added]	> 100 GHz (1 THz to 1000 THz spectrum)
Free Space Radio Waves	Few hundred kHz to a few GHz

Standard Radio Frequency Spectrum Allocations [Added]:

AM Broadcast: 540 – 1600 kHz | **FM Broadcast:** 88 – 108 MHz | **TV Broadcast:** 54 – 890 MHz | **Cellular Mobile:** 840 – 935 MHz | **Satellite:** 3.7 – 6.4 GHz

2. PROPAGATION MODES OF EM WAVES

Wave Optics & Atmosphere

Ground (Surface) Wave

- Glides over Earth's surface.
- Useful up to **f ≤ 2 MHz**.
- Attenuation increases rapidly with frequency due to ground eddy current losses.
- Requires vertically polarized antenna.

Sky Wave (Ionospheric)

- Reflected by Ionosphere layers.
- Useful range: **2 MHz to 30 MHz**.
- Waves with $f > f_c$ penetrate ionosphere into space.
- Subject to fading & diurnal changes.

Space Wave (Line of Sight)

- Direct LOS propagation.
- Useful range: **f > 30 MHz** (VHF, UHF, Microwaves).
- Limited by Earth's curvature.
- Used in TV, FM, Satellite links.

A. Ionospheric Sky Wave Formulas [Added]

• **Critical Frequency (f_c):** Maximum frequency reflected back at normal incidence ($\theta = 0^\circ$):

$$f_c = 9\sqrt{N_{max}} = 9\,(N_{max})^{1/2}$$

Where N_{max} = maximum electron density of ionosphere in m^{-3}.

• **Maximum Usable Frequency (MUF):** For reflection at angle of incidence θ:

$$f_{MUF} = f_c \sec\theta = f_c\sqrt{1 + (D/2h)^2}$$

Where D = Skip Distance, h = Virtual Height of ionospheric layer.

• **Skip Distance (D_{skip})** [Added]: Minimum distance from transmitter on Earth's surface at which sky wave of frequency $f > f_c$ returns to earth: $D_{skip} = 2h\sqrt{(f/f_c)^2 - 1}$.

B. Space Wave / Line-of-Sight (LOS) Range Formulas

• **Range of Single Tower (d):** Distance to horizon from antenna height h:

$$d = \sqrt{2Rh}$$

where Earth radius $R \approx 6400\ km = 6.4 \times 10^6\ m$.

• **Max Line-of-Sight Distance (d_M):** Between transmitting antenna (h_T) & receiving antenna (h_R):

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

• **Maximum Coverage Area (A):** Area covered by TV broadcast tower of height h_T:

$$A = \pi d^2 = 2\pi R h_T$$

• **Population Covered (P_{pop})** [Added]: Given average population density ρ (people/m^2):

$$P_{pop} = \rho \times A = 2\pi R h_T \rho$$

3. ANTENNA LENGTH & RADIATED POWER

Transmitter / Receiver

Minimum Antenna Length (L):

To radiate signals effectively, antenna size must be comparable to signal wavelength λ:

$$L \geq \frac{\lambda}{4} = \frac{c}{4f}$$

(At least $\sim \lambda/4$; quarter-wave dipole antenna).

Power Radiated by Antenna (P) [Added]:

Power radiated by linear antenna of length L for wavelength λ:

$$P \propto \left(\frac{L}{\lambda}\right)^2 \propto f^2$$

Higher frequency $f \rightarrow$ smaller $\lambda \rightarrow$ shorter antenna & exponentially higher radiated power

4. AMPLITUDE MODULATION (AM) — WAVE EQUATIONS & SPECTRUM

Modulation

Definition: The amplitude of high-frequency carrier wave is varied in accordance with the message signal, keeping frequency and phase constant.

Primary Signal Equations:

- **Carrier Wave:** $c(t) = A_c \sin(\omega_c t)$
- **Modulating (Message) Signal:** $m(t) = A_m \sin(\omega_m t)$
- **Modulated AM Wave:**

$$C_m(t) = [A_c + A_m \sin(\omega_m t)] \sin(\omega_c t)$$

$$C_m(t) = A_c [1 + \mu \sin(\omega_m t)] \sin(\omega_c t)$$

Modulation Index / Modulation Depth (μ):

$$\mu = \frac{A_m}{A_c} = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

Where: $A_{max} = A_c + A_m$ and $A_{min} = A_c - A_m$.

• **Condition to avoid distortion:** $\mu \leq 1$ (or percentage modulation $\%\mu = \mu \times 100\%$).

• If $\mu > 1$: *Overmodulation* occurs, causing envelope distortion.

Spectral Expansion & Frequencies Present in AM Wave:

$$C_m(t) = A_c \sin(\omega_c t) + \frac{\mu A_c}{2} \cos(\omega_c - \omega_m)t - \frac{\mu A_c}{2} \cos(\omega_c + \omega_m)t$$

Component	Angular Frequency (ω)	Frequency (f)	Amplitude
Lower Sideband (LSB)	$\omega_c - \omega_m$	$f_c - f_m$	$\mu A_c / 2$
Carrier Component	ω_c	f_c	A_c
Upper Sideband (USB)	$\omega_c + \omega_m$	$f_c + f_m$	$\mu A_c / 2$

• **Bandwidth of AM Signal:** $Bandwidth = f_{USB} - f_{LSB} = (f_c + f_m) - (f_c - f_m) = 2 f_m = 2 \times (Modulating\ Frequency)$

5. POWER RELATIONS & MULTI-SIGNAL MODULATION

Power Analysis

AM Power Distribution Formulas:

• **Carrier Power (P_c):**

$$P_c = \frac{A_c^2}{2R}$$

• **Total Transmitted Power (P_t):**

$$P_t = P_c \left(1 + \frac{\mu^2}{2}\right)$$

• **Sideband Power (P_{sb}):**

$$P_{sb} = P_{LSB} + P_{USB} = \frac{\mu^2}{2} P_c$$

• **Power in Each Sideband ($P_{LSB} = P_{USB}$):**

$$P_{LSB} = P_{USB} = \frac{\mu^2}{4} P_c$$

• **Modulation Efficiency (η):** Fraction of total power contained in useful sidebands:

$$\eta = \frac{P_{sb}}{P_t} = \frac{\mu^2}{2 + \mu^2}$$

For 100% modulation ($\mu = 1$): $\eta_{max} = 1/3 = 33.3\%$ (Carrier carries 66.7% power).

• **Multiple Modulating Signals** [Added]: If carrier is modulated simultaneously by independent signals with indices $\mu_1, \mu_2, \mu_3 \ldots$:

$$\mu_{net} = \sqrt{\mu_1^2 + \mu_2^2 + \mu_3^2 + \ldots}$$

$$P_t = P_c \left(1 + \frac{\mu_{net}^2}{2}\right)$$

6. AM GENERATION & ENVELOPE DETECTION

Circuits & Filtering

A. Generation via Square Law Device:

• Input: $x(t) = m(t) + c(t) = A_m \sin(\omega_m t) + A_c \sin(\omega_c t)$

• Non-linear output: $y(t) = B\,x(t) + C\,x(t)^2$

• Frequencies generated: $0, \omega_m, 2\omega_m, \omega_c, 2\omega_c, (\omega_c - \omega_m), (\omega_c + \omega_m)$.

• Passed through **Band Pass Filter (BPF)** centered at ω_c to reject DC, ω_m, $2\omega_m$, $2\omega_c$, leaving standard AM wave.

B. Demodulation / Envelope Detector:

Process: AM Signal → Rectifier (Diode) → Envelope Detector (RC filter).

• **Time Constant Condition for Detector:**

$$\frac{1}{f_c} \ll RC \ll \frac{1}{f_m} \quad \Rightarrow \quad T_c \ll \tau \ll T_m$$

• *Condition to avoid Diagonal Clipping* [Added]:

$$RC \leq \frac{\sqrt{1 - \mu^2}}{\mu\, \omega_m}$$

7. FREQUENCY MODULATION (FM) & DIGITAL COMMUNICATION [ADDED]

Advanced JEE Concepts

FM Modulation Index (m_f):

$$m_f = \frac{\Delta f}{f_m}$$

FM Bandwidth (Carson's Rule):

$$B_{FM} \approx 2(\Delta f + f_m) = 2 f_m (m_f + 1)$$

FM bandwidth is much larger than AM, giving

Sampling Rate & Bit Rate (PCM):

$$f_s \geq 2 f_{max} \quad (Nyquist\ Rate)$$

$$Bit\ Rate\ R_b = n \times f_s = 2\,n\,f_{max}$$

ATOMS & NUCLEI

JEE Main & Advanced Complete Formula & Revision Sheet

PHYSICS
MODERN PHYSICS

1. RUTHERFORD MODEL & ALPHA SCATTERING

Nuclear Scattering Dynamics

Distance of Closest Approach (r_0)

At head-on collision ($\theta = 180°$), total initial Kinetic Energy of α-particle converts to Electrostatic Potential Energy at turning point:

$$K = \frac{1}{4\pi\varepsilon_0}\cdot\frac{(2e)(Ze)}{r_0} \Rightarrow r_0 = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K} = \frac{4kZe^2}{m\,v^2}$$

→ *$K = ½\ m\ v^2$ is initial KE of α-particle; Z = Target atomic number.*

Impact Parameter (b) & Scattering

Perpendicular distance of initial velocity vector from central nucleus:

$$b = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2\cot(\theta/2)}{K}$$ [Added]

Number of scattered particles at angle θ:

$$N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$$ [Added]

→ *$b = 0 \Rightarrow \theta = 180°$ (Head-on); Large $b \Rightarrow \theta \approx 0°$ (Undeviated).*

2. BOHR MODEL FOR HYDROGENIC ATOMS (Z = 1, 2, 3...)

Quantization & Orbit Dynamics

Fundamental Bohr Postulates

1. Centripetal Balance:

$$\frac{m\,v^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{Z\,e^2}{r^2}$$

2. Quantization of Angular Momentum:

$$L = m\,v\,r = \frac{n\,h}{2\pi}\ (n = 1,2,3...)$$

3. Transition Energy:

$$\Delta E = E_i - E_f = h\,\nu = \frac{h\,c}{\lambda}$$

Key Hydrogenic Orbit Parameters (Standard Formulas)

Physical Quantity	Exact & Analytical Expression	Proportionality & Values
Bohr Radius (a_0)	$a_0 = \frac{\varepsilon_0 h^2}{\pi m e^2}$ (Radius of 1st orbit of H-atom)	$a_0 = 0.529$ Å $= 0.0529$ nm
Orbit Radius (r_n)	$r_n = \frac{\varepsilon_0 h^2 n^2}{\pi m e^2 Z} = a_0\frac{n^2}{Z}$	$r_n = 0.529\frac{n^2}{Z}$ Å $\ (r_n \propto \frac{n^2}{Z})$
Orbital Velocity (v_n)	$v_n = \frac{e^2}{2\varepsilon_0 h}\frac{Z}{n} = v_0\frac{Z}{n}$ where $v_0 = \frac{c}{137}$	$v_n = 2.19\times10^6\frac{Z}{n}$ m/s $\ (v_n \propto \frac{Z}{n})$
Time Period (T_n) [Added]	$T_n = \frac{2\pi r_n}{v_n} = \frac{4\varepsilon_0^2 h^3 n^3}{m e^4 Z^2}$	$T_n \propto \frac{n^3}{Z^2}$ $\ (T_1 : T_2 : T_3 = 1 : 8 : 27)$
Revolution Frequency (f_n) [Added]	$f_n = \frac{1}{T_n} = \frac{m e^4 Z^2}{4\varepsilon_0^2 h^3 n^3}$	$f_n \propto \frac{Z^2}{n^3}$
Equivalent Current (I_n) [Added]	$I_n = \frac{e}{T_n} = e\,f_n = \frac{m e^5 Z^2}{4\varepsilon_0^2 h^3 n^3}$	$I_n \propto \frac{Z^2}{n^3}$
Magnetic Dipole Moment (M_n) [Added]	$M_n = I_n A_n = I_n(\pi r_n^2) = n\cdot\frac{e\hbar}{2m}$	$M_n \propto n\,Z$ (Bohr Magneton $\mu_B = \frac{e\hbar}{2m}$)
Magnetic Field at Nucleus (B_n) [Added]	$B_n = \frac{\mu_0 I_n}{2 r_n}$	$B_n \propto \frac{Z^3}{n^5}$

3. ENERGY SPECTRUM & DE BROGLIE WAVE MECHANICS

Energy Partitioning & Standing Waves

Energy State Formulas

- **Kinetic Energy (K_n):** $K_n = \frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r_n} = +13.6\frac{Z^2}{n^2}$ eV
- **Potential Energy (U_n):** $U_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n} = -27.2\frac{Z^2}{n^2}$ eV
- **Total Energy (E_n):** $E_n = -\frac{1}{8\pi\varepsilon_0}\frac{Ze^2}{r_n} = -13.6\frac{Z^2}{n^2}$ eV

Central Force Relation: $E_n = -K_n = \frac{U_n}{2} \Rightarrow U_n = 2$

De Broglie Standing Wave Condition

An electron in a stable orbit forms a stationary standing wave:

$$2\pi r_n = n\,\lambda_{dB} \ \Rightarrow\ \lambda_{dB} = \frac{h}{p} = \frac{h}{m\,v_n}$$ [Added]

Circumference of n-th orbit contains exactly n de Broglie wavelengths.

Ionization Energy (I.E.): $E_\infty - E_1 = +13.6\ Z^2\ eV$
Ionization Potential: $V_{ion} = +13.6\ Z^2$ Volts

4. ATOMIC TRANSITION SPECTRA & SPECTRAL SERIES

Rydberg Equation & Photon Emission

Rydberg Formula for Hydrogenic Atoms

Wavelength λ of photon emitted during transition from $n_2 \rightarrow n_1$ ($n_2 > n_1$):

$$\frac{1}{\lambda} = \bar{\nu} = R\,Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Rydberg Constant: $R = \frac{m e^4}{8 \varepsilon_0^2 h^3 c} \approx 1.097 \times 10^7 m^{-1}$

Energy equivalence: $R\,h\,c \approx 13.6\ eV \;|\; \frac{1}{R} \approx 912\ \text{Å}$

Total Number of Spectral Lines Emitted

- Transition from state n_2 to state n_1:

$$N = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} = \frac{\Delta n(\Delta n + 1)}{2}$$

- Transition from state n to Ground State ($n_1 = 1$):

$$N = \frac{n(n-1)}{2}$$

Hydrogen Spectral Series Summary Table

Series	Lower State (n_1)	Upper State (n_2)	Spectral Region	Shortest Wavelength (λ_{min})	Longest Wavelength (λ_{max})
Lyman	$n_1 = 1$	2, 3, 4... ∞	Ultraviolet (UV)	$\frac{1}{R} \approx 912$ Å ($n_2=\infty$)	$\frac{4}{3R} \approx 1216$ Å ($n_2=2$)
Balmer	$n_1 = 2$	3, 4, 5... ∞	Visible Region	$\frac{4}{R} \approx 3646$ Å ($n_2=\infty$)	$\frac{36}{5R} \approx 6563$ Å ($n_2=3$)
Paschen	$n_1 = 3$	4, 5, 6... ∞	Near Infrared	$\frac{9}{R} \approx 8206$ Å	$\frac{144}{7R} \approx 18752$ Å
Brackett	$n_1 = 4$	5, 6, 7... ∞	Infrared	$\frac{16}{R} \approx 14585$ Å	$\frac{400}{9R} \approx 40512$ Å
Pfund	$n_1 = 5$	6, 7, 8... ∞	Far Infrared	$\frac{25}{R} \approx 22790$ Å	$\frac{900}{11R} \approx 74588$ Å

5. FINITE NUCLEAR MASS EFFECT & X-RAY SPECTROSCOPY

Reduced Mass & Characteristic Lines

Motion of Nucleus (Reduced Mass μ)

When mass of nucleus M is finite, electron & nucleus revolve around common CM:

$$\mu = \frac{M\,m}{M + m}$$

- Modified Radius: $r_{n'} = r_n\left(\frac{m}{\mu}\right) = (0.529\ \text{Å})\frac{n^2}{Z}\left(\frac{m}{\mu}\right)$
- Modified Energy: $E_{n'} = E_n\left(\frac{\mu}{m}\right) = (-13.6\ \text{eV})\frac{Z^2}{n^2}\left(\frac{\mu}{m}\right)$
- Modified Rydberg Constant: $R' = R\left(\frac{\mu}{m}\right)$

Continuous X-Rays & Cutoff Wavelength

Deceleration of incident high-speed electrons (Bremsstrahlung):

$$eV = h\nu_{max} = \frac{hc}{\lambda_{min}} \Rightarrow \lambda_{min} = \frac{hc}{eV} = \frac{12400}{V\ (\text{in Volts})}\ \text{Å}$$

*$\rightarrow \lambda_{min}$ depends **ONLY** on accelerating potential V, completely independent of target material!*

Characteristic X-Rays & Moseley's Law

Inner-shell electron ejection followed by higher-level electronic decay transitions:

Line Wavelengths:

- K_α line ($L \rightarrow K$): $\lambda_{K_\alpha} = \frac{hc}{E_K - E_L}$
- K_β line ($M \rightarrow K$): $\lambda_{K_\beta} = \frac{hc}{E_K - E_M}$

Moseley's Law Equation:

$$\sqrt{\nu} = a(Z - b)$$

For K_α line: Screening constant $b = 1$

$$\frac{1}{\lambda} = R(Z-1)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = \frac{3}{4}R(Z-1)^2$$

6. NUCLEAR PROPERTIES, SIZE & BINDING ENERGY

Nuclear Structure & Stability

Nuclear Size & Density Invariance

- **Nuclear Radius:** $R = R_0 A^{1/3}$ where $R_0 \approx 1.1 \times 10^{-15}\ m = 1.1\ fm$
- **Nuclear Volume:** $V = \frac{4}{3}\pi R^3 \propto A$ (Directly proportional to mass no. A)

Mass Defect & Binding Energy (BE)

- **Mass Defect (Δm):** Difference in constituent and nucleus mass:

$$\Delta m = [Z\,m_p + (A - Z)\,m_n] - M_{nucleus}$$

- **Binding Energy (BE):** Energy required to break nucleus:

7. NUCLEAR REACTIONS, Q-VALUE & ENERGETICS

Reaction Energetics & Fission/Fusion

Q-Value of Nuclear Reaction: $A + B \rightarrow C + D + Q$

1. Mass Basis:	2. BE Basis:	3. KE Basis:
$Q = [(m_A + m_B) - (m_C + m_D)]\, c^2$	$Q = BE(C) + BE(D) - [BE(A) + BE(B)]$	$Q = K_C + K_D - [K_A + K_B]$

- **Exothermic ($Q > 0$):** Energy released; Mass decreases; Products more bound.
- **Endothermic ($Q < 0$):** Energy absorbed; Mass increases; Needs threshold KE.

[JEE Adv] **Threshold Kinetic Energy for Endothermic Reaction (Target B at rest):** $K_{threshold} = |Q| \left(1 + \frac{m_A}{m_B}\right)$

Nuclear Fission

Heavy nucleus ($A > 230$) splits into two medium fragments:

$^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + Q$

- Energy released per fission event ≈ 200 MeV (≈ 0.85 MeV/nucleon).

Nuclear Fusion

Light nuclei ($A \leq 4$) fuse to form a heavier stable nucleus:

$4\,({}^{1}_{1}H) \rightarrow {}^{4}_{2}He + 2\,{}^{0}_{+1}e + 2\,\nu + 26.7\ MeV$

- Energy released per unit mass in fusion is **much higher** than in fission!

8. LAW OF RADIOACTIVE DECAY & KINETICS

Disintegration Laws & Activity

Decay Rate & Exponential Law

Rate of disintegration is proportional to active nuclei present:

$$\frac{dN}{dt} = -\lambda N \quad \Rightarrow \quad N(t) = N_0\, e^{-\lambda t}$$

- **Activity (R or A):** $R = -\frac{dN}{dt} = \lambda N = R_0\, e^{-\lambda t}$
- **Survival Probability at time t:** $P(t) = \frac{N(t)}{N_0} = e^{-\lambda t}$
- **Units:** 1 Bq = 1 decay/s | 1 Ci = 3.7×10^{10} Bq | 1 Rd = 10^{6} Bq

Half-Life, Mean Life & Decay Fractions

- **Half-Life ($T_{1/2}$):** Time for half sample to decay:

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

- **Mean / Average Life (τ):** Sum of lives / total nuclei:

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} \approx 1.443\, T_{1/2}$$

- **Nuclei after n half-lives ($t = n\, T_{1/2}$):**

$$N = N_0 \left(\frac{1}{2}\right)^{n} = N_0 \left(\frac{1}{2}\right)^{t / T_{1/2}}$$

Parallel / Simultaneous Decay Modes

Nucleus decays simultaneously via two modes (λ_1, λ_2):

$$\lambda_{eff} = \lambda_1 + \lambda_2 \quad \Rightarrow \quad \frac{1}{T_{eff}} = \frac{1}{T_1} + \frac{1}{T_2}$$

Fraction of decay via mode 1: $f_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2}$ | Mean life $\tau_{eff} = \frac{\tau_1 \tau_2}{\tau_1 + \tau_2}$

Series / Successive Decay ($A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C$) [Adv]

Number of daughter nuclei B at time t (if $N_{B0} = 0$):

$$N_B(t) = \frac{\lambda_A N_{A0}}{\lambda_B - \lambda_A} \left(e^{-\lambda_A t} - e^{-\lambda_B t}\right)$$

Time for maximum N_B: $t_{max} = \frac{\ln(\lambda_B / \lambda_A)}{\lambda_B - \lambda_A}$

Radioactive Equilibrium: $\lambda_A N_A = \lambda_B N_B \Rightarrow \frac{N_A}{N_B} = \frac{T_A}{T_B}$

9. ALPHA, BETA & GAMMA DECAY ENERGETICS

Decay Equations & Recoil Kinetics

Decay Type	Nuclear Reaction Equation	Q-Value & Kinetic Energy Distribution
α-Decay	${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He + Q$	$Q = [m(X) - m(Y) - m(\alpha)]\, c^2$ KE of α-particle: $K_\alpha = \left(\frac{A-4}{A}\right) Q$ | Daughter Recoil: $K_Y = \left(\frac{4}{A}\right) Q$
β^--Decay (e^-)	${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu} + Q$ $(n \rightarrow p + e^- + \bar{\nu})$	$Q = [m(X) - m(Y)]\, c^2$ (Shared continuously between e^- and antineutrino $\bar{\nu}$)
β^+-Decay (e^+)	${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \nu + Q$ $(p \rightarrow n + e^+ + \nu)$	$Q = [m(X) - m(Y) - 2\, m_e]\, c^2$ (In terms of atomic masses)
K-Capture	${}^{A}_{Z}X + {}^{0}_{-1}e \rightarrow {}^{A}_{Z-1}Y + \nu + Q$	$Q = [m(X) - m(Y)]\, c^2$ (Emits characteristic X-rays)

DUAL NATURE OF RADIATION AND MATTER

JEE Main & Advanced Complete Formula Sheet

1. Types of Electron Emission

Minimum energy to escape a metal surface is the **Work Function (Φ or W_0)**.

- **Thermionic:** Due to thermal energy (heating).
- **Field/Cold:** Due to application of a very high electric field.
- **Photoelectric:** Due to suitable electromagnetic radiation (light).
- **Secondary:** Due to striking of fast moving primary electrons.

2. Photon Theory of Light

Light consists of quanta/packets of energy called photons. They travel at speed c, are neutral, and have zero rest mass ($m_0 = 0$).

- **Energy of a Photon:**

$$\boxed{E = h\nu = \frac{hc}{\lambda}}$$

- **Shortcut Form [Added]:**

$$\boxed{E(\text{in eV}) \approx \frac{12400}{\lambda(\text{in Å})} \approx \frac{1240}{\lambda(\text{in nm})}}$$

- **Momentum (p):** $p = \dfrac{h}{\lambda} = \dfrac{E}{c}$
- **Dynamic/Equivalent Mass:** $m = \dfrac{E}{c^2} = \dfrac{h}{\lambda c}$
- **Number of Photons & Intensity [Added]:** Let n = no. of photons emitted per second, P = Power, A = Area.

$$P = n \times E \quad \Longrightarrow \quad n = \frac{P\lambda}{hc}$$

$$\text{Intensity } (I) = \frac{P}{A} = \frac{nh\nu}{A}$$

3. Photoelectric Effect

Ejection of electrons when light of suitable frequency hits a metal.

- **Threshold Frequency (ν_0) / Wavelength (λ_0):** Minimum frequency (maximum wavelength) below which no emission occurs.

$$\Phi = h\nu_0 = \frac{hc}{\lambda_0}$$

- **Einstein's Photoelectric Equation:**

$$\boxed{K_{\text{max}} = E - \Phi = h\nu - h\nu_0 = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}}$$

- **Stopping Potential (V_0 or V_s):** Minimum retarding potential where photocurrent becomes zero.

$$\boxed{K_{\text{max}} = eV_0 = \frac{1}{2}mv_{\text{max}}^2}$$

$$V_0 = \left(\frac{h}{e}\right)\nu - \frac{\Phi}{e}$$

Key Laws of Photoelectric Emission:

- Photocurrent $\propto$ Intensity (for fixed $\nu > \nu_0$).
- K_{max} depends on frequency (ν) and is independent of Intensity.
- Photoelectric emission is an instantaneous process.

4. Important Graphical Relations

- **Current vs Intensity:** Straight line passing through the origin.
- **Current vs Retarding Potential (V):**
  - *Diff Intensity ($I_3 > I_2 > I_1$), Same ν:* Different saturation currents, but **same** stopping potential V_0.
  - *Diff ν ($\nu_3 > \nu_2 > \nu_1$), Same Intensity:* Same saturation current, but **different** V_0 (higher ν means more negative V_0).
- **Stopping Potential (V_0) vs Frequency (ν):**
  - Parallel straight lines for different metals.
  - **Slope** $= h/e$ (Universal constant).
  - **Intercept on ν-axis** $= \nu_0$ (Threshold).
  - **Intercept on V_0-axis** $= -\Phi/e$.

5. Radiation Pressure & Force

When radiation hits a surface, it exerts pressure and force. Let I = intensity, c = speed of light, r = reflectivity (0 to 1).

- **Radiation Pressure:**

$$\boxed{P_r = \frac{I}{c}(1 + r)}$$

- **Radiation Force [Added]:** $F = P_r \times A = \frac{\text{Power}}{c}(1 + r)$
- **Special Cases:**
  - Perfectly Absorbing ($r = 0$): $P_r = I/c$
  - Perfectly Reflecting ($r = 1$): $P_r = 2I/c$

6. de-Broglie Matter Waves

Every moving particle has an associated wave.

- **Fundamental Formula:**

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}}$$

- **For Charged Particle (accelerated by V volts):**

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

- **Direct Formula Shortcuts [Added for JEE Main]:**
  - **Electron:** $\lambda_e = \frac{12.27}{\sqrt{V}}$ Å $= \sqrt{\frac{150}{V}}$ Å
  - **Proton:** $\lambda_p \approx \frac{0.286}{\sqrt{V}}$ Å
  - **Deuteron:** $\lambda_d \approx \frac{0.202}{\sqrt{V}}$ Å
  - **Alpha Particle:** $\lambda_\alpha \approx \frac{0.101}{\sqrt{V}}$ Å
- **For Gas Molecule / Thermal Particle:**

$$\lambda = \frac{h}{\sqrt{3mkT}}$$

(k = Boltzmann constant, T = Temp in Kelvin)

7. Heisenberg's Uncertainty Principle

It is impossible to exactly measure position and momentum simultaneously.

$$\boxed{\Delta x \cdot \Delta p \geq \frac{h}{4\pi} = \frac{\overline{h}}{2}}$$

(Where $\overline{h} = h/2\pi$, Δx = position uncertainty, Δp = momentum uncertainty)

8. Motion of Electron in Fields

- **Electric Field:** Force $F_E = eE$. (Accelerates linearly or traces parabola if $v \perp E$).
- **Magnetic Field:** Force $F_B = Bev \sin\theta$.
  - If $v \parallel B$, $F_B = 0$.
  - If $v \perp B$, circular path radius $R = \frac{mv}{eB}$.

9. Specific Experiments (Advanced)

A. Compton Scattering: Increase in X-ray photon wavelength after striking an electron. Let m_0 = rest mass, ϕ = scattering angle.

$$\boxed{\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c}(1 - \cos\phi)}$$

B. J.J. Thomson's Method (for e/m): Crossed electric ($\vec{E}$) and magnetic ($\vec{B}$) fields. Undeflected beam condition: $F_E = F_B \implies eE = Bev \implies \mathbf{v = E/B}$. Specific charge (if deflected by shift x):

$$\frac{e}{m} = \frac{E}{B^2 R} = \frac{V_x}{B^2 l L d}$$

C. Millikan's Oil Drop Method (Quantization): Measures electron charge. Drop radius (r) and charge (q):

$$r = \sqrt{\frac{9\eta v_0}{2(\rho - \sigma)g}}$$

$$q = \frac{18\pi\eta(v_1 + v_0)}{E}\sqrt{\frac{\eta v_0}{2(\rho - \sigma)g}}$$

(v_0=terminal vel. under gravity, v_1=under field+gravity, η=viscosity, ρ=density of oil, σ=density of medium)

D. Davisson-Germer Experiment: Confirmed wave nature of electrons using Ni crystal. Maximum intensity observed at: **Voltage:** $V = 54$ V, **Scattering Angle:** $\theta = 50°$.

E. Electron Microscope: Uses accelerated electron beams to study minute objects (magnification $\sim 10^5$).

WAVE OPTICS

JEE Main & Advanced Revision Formula Sheet

PHYSICS FORMULA BOOKLET

1. ELECTROMAGNETIC NATURE OF LIGHT

- **Sinusoidal Wave Equations:**

$$E_y = E_0 \sin(kx \mp \omega t) \quad | \quad B_z = B_0 \sin(kx \mp \omega t)$$

Light is a transverse non-mechanical EM wave. E & B are in phase.

- **Speed of Light in Medium:**

$$v = \frac{1}{\sqrt{(\mu\, \varepsilon)}} = \frac{c}{\mu} \;\Rightarrow\; E_0 = v\, B_0$$

- **Refractive Index (μ):** $\mu = \frac{c}{v} = \frac{\lambda_{vac}}{\lambda_{med}} = \sqrt{(\mu_r\, \varepsilon_r)} \approx \sqrt{(\varepsilon_r)}$ [Added]
- **Intensity of EM Wave:** [Added]

$$I = \frac{1}{2}\, \varepsilon\, v\, E_0^2 = \frac{1}{2}\, c\, \varepsilon_0\, E_0^2 \text{ (in vacuum)} \;\Rightarrow\; I \propto A^2$$

2. WAVEFRONTS & HUYGENS' PRINCIPLE

- **Wavefront:** Locus of all medium particles oscillating in the same phase. Light rays are always *perpendicular* to wavefronts.
- **Shapes of Wavefronts:**

Source Type	Wavefront Shape	Intensity Relation
Point Source	Spherical	$I \propto 1/r^2$ $(A \propto 1/r)$
Line Source	Cylindrical [Added]	$I \propto 1/r$ $(A \propto 1/\sqrt{r})$
Distant Source	Plane	$I = \text{const}$ $(A = \text{const})$

- **Secondary Wavelets:** Every point on a wavefront acts as a secondary spherical wavelet source traveling at speed v.

3. PRINCIPLE OF INTERFERENCE

Superposition of two coherent waves: $y_1 = A_1 \sin(\omega t - kx)$, $y_2 = A_2 \sin(\omega t - kx - \phi)$

Resultant Amplitude & Phase:

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos\phi$$

$$\tan\alpha = \frac{A_2 \sin\phi}{A_1 + A_2 \cos\phi}$$

Resultant Intensity Equation:

$$I = I_1 + I_2 + 2\sqrt{(I_1 I_2)} \cos\phi$$

- **Phase Diff (ϕ) vs Path Diff (Δx):**

$$\phi = \frac{2\pi}{\lambda}\Delta x \quad | \quad \Delta x = \frac{\lambda}{2\pi}\phi$$

- **Coherent vs Incoherent Sources:**
  - *Coherent:* Constant ϕ with time. $I = I_1 + I_2 + 2\sqrt{(I_1 I_2)} \cos\phi$
  - *Incoherent:* Random $\phi(t)$. $I_{net} = I_1 + I_2$ (No stable fringes)
- **Slit Width & Intensity Relation:**

$$\frac{W_1}{W_2} = \frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$$

4. MAXIMA & MINIMA CONDITIONS

Constructive Interference (Maxima):

- $\phi = \pm 2n\pi$ (n = 0, 1, 2, ...)
- Path Diff: $\Delta x = \pm n\lambda$
- Max Amplitude: $A_{max} = A_1 + A_2$
- Max Intensity: $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$

Destructive Interference (Minima):

- $\phi = \pm (2n - 1)\pi$ (n = 1, 2, 3, ...)
- Path Diff: $\Delta x = \pm (2n - 1)\frac{\lambda}{2}$
- Min Amplitude: $A_{min} = |A_1 - A_2|$
- Min Intensity: $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Special Case (Identical Sources $I_1 = I_2 = I_0$):

$$I = 4I_0 \cos^2\frac{\phi}{2} \quad | \quad I_{max} = 4I_0,\; I_{min} = 0$$

Contrast Ratio & Visibility:

$$\frac{I_{max}}{I_{min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(\sqrt{r} + 1)^2}{(\sqrt{r} - 1)^2} \quad \text{where } r = \frac{I_1}{I_2}$$

Fringe Visibility: $V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$ [Added]

5. YOUNG'S DOUBLE SLIT SETUP (YDSE)

Setup: Slit separation d, Screen distance D ($D \gg d$).

- **Path Difference at Angle θ:**

$$\Delta x = d \sin\theta \approx d \tan\theta = \frac{yd}{D} \text{ (for small } \theta)$$

- **Position of n^{th} Bright Fringe (Central Max at $y = 0$):**

$$y_n = \frac{n\lambda D}{d} \quad (n = 0, \pm1, \pm2, ...)$$

- **Position of m^{th} Dark Fringe:**

$$y_m = (2m - 1)\frac{\lambda D}{2d} \quad (m = 1, 2, 3, ...)$$

- **Linear Fringe Width (β):**

$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$$

  - Fringe width is constant for all orders in YDSE.
  - Bright & dark fringes are equally spaced.
- **Angular Fringe Width (θ_0):**

$$\theta_0 = \frac{\beta}{D} = \frac{\lambda}{d} \text{ (in radians)}$$

- **YDSE in Medium of Refractive Index μ:** [Added]

$$\lambda' = \frac{\lambda}{\mu} \;\Rightarrow\; \beta' = \frac{\beta}{\mu}$$

6. OPTICAL PATH & GLASS SLAB SHIFT

- **Optical Path Length:** Distance in vacuum equivalent to path t in medium: $L_{opt} = \mu t$.
- **Path Difference Introduced by Thin Sheet:**

$$\Delta x_{extra} = (\mu - 1)t$$

- **Shift of Fringe Pattern (y_0):**

$$y_0 = \frac{D}{d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t$$

  - Shift occurs towards the side where the slab is placed.
  - Fringe width β remains UNCHANGED.
- **Number of Fringes Shifted (N):** [Added]

$$N = \frac{y_0}{\beta} = \frac{(\mu - 1)t}{\lambda}$$

7. OBLIQUE YDSE & THIN FILMS [ADDED]

- **Oblique Incidence in YDSE:**
  If light strikes slits at angle α:

$$\Delta x = d(\sin\theta \mp \sin\alpha)$$

  Central maximum shifts to $\theta = -\alpha$.
- **Thin Film Interference (Thickness t, Refractive Index μ):**
  Reflected Light (includes π phase flip at denser boundary):
  - Maxima (Bright): $2\mu t\cos r = (2n - 1)\frac{\lambda}{2}$
  - Minima (Dark): $2\mu t\cos r = n\lambda$

  Transmitted Light (No extra phase flip):
  - Maxima (Bright): $2\mu t\cos r = n\lambda$
  - Minima (Dark): $2\mu t\cos r = (2n - 1)\frac{\lambda}{2}$
- **Wedge-Shaped Thin Film Fringe Spacing:**

$$\beta = \frac{\lambda}{2\mu\theta}$$ (θ in radians)

8. SINGLE SLIT DIFFRACTION

Slit width = a (or d), Screen distance = D.

- **Intensity Formula:**

$$I(\theta) = I_0\frac{\sin^2\alpha}{\alpha^2} \text{ where } \alpha = \frac{\pi a\sin\theta}{\lambda}$$

- **Condition for Minima (Dark Points):**

$$a\sin\theta = \pm n\lambda$$ (n = 1, 2, 3, ... ; n ≠ 0)

- **Condition for Secondary Maxima:**

$$a\sin\theta \approx \pm(n + \frac{1}{2})\lambda$$ (n = 1, 2, 3, ...)

- **Width of Central Maximum:**
  - Angular Width: $W_\theta = 2\theta_1 = \frac{2\lambda}{a}$
  - Linear Width on Screen: $W = \frac{2\lambda D}{a}$
- **Angular Width of Secondary Maxima:**

$$\theta_s = \frac{\lambda}{a}$$ (half the central max width)

- **Intensity Ratio of Peak Maxima:** [Added]

$$I_0 : I_1 : I_2 : I_3 = 1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2} \ldots \approx 1 : 0.045 : 0.016$$

9. CIRCULAR APERTURE & RESOLUTION

- **Airy Disc (Circular Aperture of Diameter d):**
  Angular radius of first dark ring:

$$\theta_R = 1.22\frac{\lambda}{d}$$

  Linear radius on focal plane of lens (f):

$$r = 1.22\frac{\lambda f}{d}$$

- **Rayleigh's Criterion for Resolution:**
  Two point objects are just resolved when central maximum of one falls on first minimum of the other.
- **Telescope Resolution Limits:**
  - Limit of Resolution: $d\theta = 1.22\frac{\lambda}{d}$
  - Resolving Power: $RP = \frac{1}{d\theta} = \frac{d}{1.22\lambda}$
- **Microscope Resolving Power:** [Added]
  - Limit of Resolution: $\Delta x = \frac{\lambda}{2\mu\sin\theta}$
  - Resolving Power: $RP = \frac{2\mu\sin\theta}{\lambda} = \frac{2NA}{\lambda}$

$NA = \mu\sin\theta$ is Numerical Aperture.

10. DIFFRACTION GRATING [ADDED]

- **Grating Element ($d = a + b$):**

$$d\sin\theta = n\lambda$$ (Principal Maxima)

- **Dispersive Power:** $\frac{d\theta}{d\lambda} = \frac{n}{d\cos\theta}$
- **Resolving Power:** $RP = \frac{\lambda}{d\lambda} = nN$ (N = total lines)

11. POLARIZATION & PHOTOMETRY

- **Unpolarized Light through Polarizer:**

$$I_{out} = \frac{I_0}{2}$$ (Intensity halved always)

- **Malus' Law (Plane Polarized Light through Analyzer):**

$$I = I_0\cos^2\theta$$

θ = angle between transmission axes of polarizer & analyzer.

- **Two Polaroids in Series:** $I_2 = I_1\cos^2\theta'$
- **Brewster's Law (Polarization by Reflection):**

$$\tan\theta_p = \mu \Rightarrow \theta_p = \tan^{-1}(\mu)$$

  - At Brewster's angle $i = \theta_p$, reflected ray is 100% plane polarized.
  - Reflected & refracted rays are perpendicular: $\theta_p + r = 90°$.
- **Polarization Types** [Added]:
  • *Circular:* Superposition of 2 perpendicular equal-amplitude waves with $\Delta\phi = \pi/2$.
  • *Elliptical:* Superposition with unequal amplitudes or arbitrary $\Delta\phi$.
- **Optical Activity & Specific Rotation:**

$$[\alpha]_t^{\circ}C^{\lambda} = \frac{\theta}{L \times C}$$

θ = rotation angle (°), L = path length in decimeters (dm), C = concentration in g/cm³.

RAY OPTICS & OPTICAL INSTRUMENTS

JEE Complete Formula Sheet

1. Fundamental Properties & Reflection

Optical Path & Speed of Light

- Speed of light in vacuum: $c = 3 \times 10^8$ m/s.
- Speed in medium: $v = c/\mu$.
- **Optical Path Length**: $L = \mu d = c \times t$
- *Note*: Wavelength and velocity change in a medium ($\lambda_m = \lambda/\mu$), but **frequency and colour do not change**.

Laws of Reflection

- Angle of incidence = Angle of reflection: $\angle i = \angle r$
- **Vector Form**: $\hat{r} = \hat{e} - 2(\hat{e} \cdot \hat{n})\hat{n}$
  ($\hat{e}$ = incident, $\hat{r}$ = reflected, $\hat{n}$ = outward normal)
- **Deviation (δ)**: $\delta = 180^\circ - 2i$ (for single reflection)

Plane Mirrors

- **Image**: Virtual, erect, laterally inverted, same size, equidistant behind mirror.
- **Rotation**: If mirror rotates by θ (keeping incident ray fixed), reflected ray rotates by **2**θ.
- **Relative Velocity**: If object moves towards mirror at v, image approaches at v. Speed of image w.r.t object $= 2v$.
- **Field of View Minimums**:
  - To see own full height: $H_{mirror} \geq H_{person}/2$
  - To see full wall behind (standing in middle): $H_{mirror} \geq H_{wall}/3$
- **Images by Inclined Mirrors** (Angle θ): Let $m = 360^\circ/\theta$.
  - If m is even: $n = m - 1$ (always)
  - If m is odd: $n = m$ (asymmetric obj), $n = m - 1$ (obj on angle bisector)
  - If m is fraction: $n =$ integral part of m.

2. Spherical Mirrors

Basic Formulas (Use Sign Convention: Pole is origin, direction of incident ray is +ve)

- Focal length: $f = R/2$
- **Mirror Formula**: $\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}}$
- **Optical Power**: $P = -\frac{1}{f}$ (in m^{-1} or Dioptres)

Magnification

- **Transverse (m)**: $m = \frac{h_i}{h_o} = -\frac{v}{u}$
- Shortcut form: $\boxed{m = \frac{f}{f-u} = \frac{f-v}{f}}$ *[Added]*
- **Longitudinal (m_L)** (for small object along axis):

$$m_L = -\frac{dv}{du} = -\left(\frac{v}{u}\right)^2 = -m^2$$

Newton's Formula & Kinematics

- **Newton's Formula**: $x_1 x_2 = f^2$
  (x_1, x_2 are object and image distances from **Focus**)
- **Velocity of Image** (moving along axis):

$$\vec{v}_I = -m^2 \vec{v}_O \quad \text{(w.r.t mirror)}$$

3. Refraction of Light

Snell's Law

- Scalar: $\mu_1 \sin i = \mu_2 \sin r$
- Vector: $\mu_1(\hat{e} \times \hat{n}) = \mu_2(\hat{r} \times \hat{n})$

Apparent Depth & Shifts

- **Apparent Depth** (seen from rarer medium):

$$d_{Ap} = \frac{d_{Ac}}{\mu_{rel}} = \frac{d_{Ac}}{\mu}$$

- **Apparent Height** (seen from denser medium):

$$d_{Ap} = \mu \cdot d_{Ac}$$

- **Normal Shift** (Δx): Distance image appears to move.

$$\Delta x = t\left(1 - \frac{1}{\mu_{rel}}\right) \quad \text{(Shift is in dir. of incident ray)}$$

- **Multiple Media**: $d_{Ap} = \sum \frac{d_i}{\mu_i}$
- **Lateral Shift** (Glass Slab):

$$y = \frac{t \sin(i-r)}{\cos r}$$

Total Internal Reflection (TIR)

- **Conditions**: Ray must travel from Denser to Rarer medium. Angle of incidence $i > \theta_C$.
- **Critical Angle (θ_C)**: $\boxed{\sin\theta_C = \frac{\mu_{\text{Rarer}}}{\mu_{\text{Denser}}} = \frac{1}{\mu}}$

4. Prism & Dispersion

Prism Formulas

- Angle of Prism: $A = r_1 + r_2$
- Angle of Deviation: $\delta = i_1 + i_2 - A$
- **Minimum Deviation (δ_m)**: Occurs when $i_1 = i_2 = i$, and $r_1 = r_2 = A/2$. Ray passes symmetrically. $\boxed{\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin(A/2)}}$

- **Maximum Deviation**: Occurs at grazing incidence ($i_1 = 90°$). $\delta_{max} = 90° + i_2 - A$.
- **Small Angled Prism** ($A \leq 10°$): $\delta = (\mu - 1)A$
- **Condition for No Emergence**: TIR at second surface regardless of incidence angle if $A > 2\theta_C$. *[Added]*

Dispersion

- Angular Dispersion: $\theta = \delta_V - \delta_R$
- **Dispersive Power (ω)**:

$$\omega = \frac{\delta_V - \delta_R}{\delta_Y} = \frac{\mu_V - \mu_R}{\mu_Y - 1}$$

- **Cauchy's Equation**: $\mu = A + \frac{B}{\lambda^2}$ *[Added]*

5. Curved Surface & Lenses

Refraction at Single Spherical Surface

- **Main Formula**: $\boxed{\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}}$
- Transverse Magnification: $m = \frac{\mu_1 v}{\mu_2 u}$

Lens Formulas

- **Lens Maker's Formula**: $\boxed{\frac{1}{f} = (\mu_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$
- **Lens Formula**: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
- **Power of Lens**: $P = \frac{1}{f}$ (+ve for Convex, -ve for Concave).
- Transverse Mag: $m = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$ *[Added]*
- Longitudinal Mag: $m_L = \frac{dv}{du} = m^2$ *[Added]*
- Velocity of Image: $\vec{v}_I = m^2 \vec{v}_O$ *[Added]*

Combination of Thin Lenses

- In contact: $P_{eq} = P_1 + P_2 \implies \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$
- Separated by distance d:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \quad (P = P_1 + P_2 - dP_1P_2)$$

Displacement Method (Finding convex lens f)

- Requires object-screen distance $D > 4f$.
- If d is distance between two lens positions forming sharp images: $\boxed{f = \frac{D^2 - d^2}{4D}}$ *[Added]*
- Object size (O) from two image sizes (I_1, I_2): $O = \sqrt{I_1 I_2}$

Silvering of Lenses (Behaves as a Mirror)

- **Equivalent Power**: $P_{eq} = P_m + 2P_L$
- Equivalent focal length: $f_{eq} = -\frac{1}{P_{eq}}$
- Plano-convex (Plane surface silvered): $f = \frac{R}{2(\mu-1)}$
- Plano-convex (Convex surface silvered): $f = \frac{R}{2\mu}$

6. Optical Instruments

Note: D = Least distance of distinct vision (usually 25cm).

Simple Microscope (Magnifying Glass)

- Max Magnification (image at D): $M = 1 + \frac{D}{f}$
- Normal Adjustment (image at ∞): $M = \frac{D}{f}$

Compound Microscope

- Max Magnification (final image at D):

$$M = -\frac{v_o}{u_o}\left(1 + \frac{D}{f_e}\right), \quad \text{Length } L = v_o + |u_e|$$

- Normal Adjustment (final image at ∞):

$$M = -\frac{v_o}{u_o}\left(\frac{D}{f_e}\right), \quad \text{Length } L = v_o + f_e$$

Astronomical Telescope

- Final image at D (Max Strain):

$$M = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right), \quad \text{Length } L = f_o + |u_e|$$

- Normal Adjustment (final image at ∞):

$$M = -\frac{f_o}{f_e}, \quad \text{Length } L = f_o + f_e$$

Resolving Power & Limits

- Resolving Power $= 1/$Limit of Resolution
- **Microscope Limit of Resolution**: $\Delta d = \frac{1.22\lambda}{2\mu \sin\theta}$
- **Telescope Limit of Resolution**: $\Delta\theta = \frac{1.22\lambda}{a}$ (a = aperture diameter of objective)
- Eye Resolving Limit: $\Delta\theta \approx 1' = (1/60)°$

ELECTROSTATICS & CAPACITANCE

JEE Main & Advanced Complete Revision Formula Sheet

JEE TARGET

1. Electric Charge & Coulomb's Law

Properties of Charge & Induction

- **Quantization:** $Q = \pm ne$ ($n \in \mathbb{Z}$, $e = 1.6 \times 10^{-19}\ C$)
- **Relativistic Invariance:** Charge is independent of reference frame speed.
- **Induced Charge Formula:**

$$q' = -q\,(1 - 1/K)$$

Where K is dielectric constant. For conductors ($K = \infty$), $q' = -q$.

Coulomb's Law in Vacuum & Medium

- **Force magnitude:** $F = 1/4\pi\varepsilon_0\varepsilon_r \cdot |q_1 q_2|/r^2$
- **Vector Form:** $\boldsymbol{F}_{12} = 1/4\pi\varepsilon_0 \cdot q_1 q_2/|r_1 - r_2|^3\,(\boldsymbol{r}_1 - \boldsymbol{r}_2)$
- **Constants:** $k = 1/4\pi\varepsilon_0 = 9 \times 10^9\ N{\cdot}m^2/C^2$, $\varepsilon_0 = 8.85 \times 10^{-12}\ F/m$
- **Medium Effect:** $F_m = F_0/\varepsilon_r = F_0/K$

2. Electric Field Intensity (E) & Charge Density

Electric Field Definition

$\boldsymbol{E} = lim_{q_0 \to 0}\,(\boldsymbol{F}/q_0)$

Units: N/C or V/m

Dimension: $[M\ L\ T^{-3}\ A^{-1}]$

Superposition: $\boldsymbol{E}_{net} = \sum \boldsymbol{E}_i$

Charge Densities

Linear (λ): $\lambda = dq/dl$ (C/m)

Surface (σ): $\sigma = dq/dA$ (C/m^2)

Volume (ρ): $\rho = dq/dV$ (C/m^3)

Resultant of Two Fields

$E = \sqrt{(E_1^2 + E_2^2 + 2E_1E_2\cos\theta)}$

$\tan\alpha = (E_2 \sin\theta)/(E_1 + E_2\cos\theta)$

3. Standard Electric Field (E) Distributions

Charged Body Configuration	Electric Field Formula (E)	Key Conditions / Peak Values
Point Charge	$E = k\,q/r^2$	Radially outward ($q>0$) or inward ($q<0$)
Uniform Ring (Axis)	$E = (k\,Q\,x)/(x^2 + R^2)^{3/2}$	$E_{max} = (2\,k\,Q)/(3\sqrt{3}\,R^2)$ at $x = \pm R/\sqrt{2}$
Uniform Finite Rod (Perp. bisector)	$E = \lambda/(2\pi\varepsilon_0\,r) \cdot 1/\sqrt{(1 + 4r^2/L^2)} = (k\,q)/(r\sqrt{(L^2/4 + r^2)})$	Symmetric line; $\lambda = q/L$
Infinite Line Charge / Rod	$E = \lambda/(2\pi\varepsilon_0\,r) = 2\,k\,\lambda/r$	As length $L \to \infty$
Thin Spherical Shell / Hollow Sphere	• $r < R$: $E = 0$ • $r \geq R$: $E = k\,Q/r^2 = (\sigma\,R^2)/(\varepsilon_0\,r^2)$	Surface field: $E_{surface} = \sigma/\varepsilon_0$
Uniform Solid Non-Conducting Sphere	• $r < R$: $E = (k\,Q\,r)/R^3 = (\rho\,r)/(3\,\varepsilon_0)$ • $r \geq R$: $E = k\,Q/r^2$	Inside field is directly proportional to r ($E \propto r$)
Infinite Non-Conducting Sheet	$E = \sigma/(2\,\varepsilon_0)$	Uniform field, independent of distance r
Conducting Sheet / Near Conductor Surface	$E = \sigma/\varepsilon_0$	Perpendicular to surface (Local boundary)
Uniformly Charged Disc [Added]	$E = (\sigma/2\varepsilon_0)\,[\,1 - x/\sqrt{(x^2 + R^2)}\,]$	Axis at distance x from center

4. Gauss's Law & Electrostatic Energy Density

Gauss's Law Statement

$$\Phi_E = \oint E \cdot dA = q_{enclosed}/\varepsilon_0$$

- Flux Φ_E depends *only* on charges enclosed inside closed Gaussian surface.
- E on LHS is net field due to *all* charges (inside + outside).

Electrostatic Energy Density (u)

$u = \frac{1}{2}\,\varepsilon_0\,E^2$ (in vacuum) | $u = \frac{1}{2}\,\varepsilon_0\,\varepsilon_r\,E^2$ (in medium)

Stored electrostatic energy per unit volume in the electric field region.

5. Electric Dipole Mechanics & Interactions

Dipole Basics & Field Equations

- **Dipole Moment:** $\boldsymbol{p} = q\,\boldsymbol{d}$ (Directed from $-q$ to $+q$)
- **Axial Field ($\theta = 0°$):** $E_{axial} = (2\,k\,\boldsymbol{p}) / r^3$ (Parallel to $\boldsymbol{p}$)
- **Equatorial Field ($\theta = 90°$):** $E_{eq} = -(k\,\boldsymbol{p}) / r^3$ (Antiparallel)
- **General Point (r, θ) Field:**

$$E = (k\,p / r^3)\,\sqrt{(1 + 3\cos^2\theta)}$$

- **Field Angle (α with radius vector):** $\tan\alpha = \tfrac{1}{2}\tan\theta$

External Field Effects & Interactions

- **Torque:** $\boldsymbol{\tau} = \boldsymbol{p} \times \boldsymbol{E} \;\Rightarrow\; \tau = pE\sin\theta$
- **Potential Energy:** $U = -\boldsymbol{p}\cdot\boldsymbol{E} = -pE\cos\theta$
- **Rotation Work:** $W_{\theta 1 \to \theta 2} = pE\,(\cos\theta_1 - \cos\theta_2)$
- **Force in Non-Uniform Field:** $F = p\,(dE/dx)$
- **Axial Dipole-Dipole Interaction:**

$F = (1/4\pi\varepsilon_0)\cdot(6\,p_1\,p_2 / r^4)$ | $U = -(1/4\pi\varepsilon_0)\cdot(2\,p_1\,p_2 / r^3)$

- **Broadside Dipole-Dipole Interaction:**

$F = (1/4\pi\varepsilon_0)\cdot(3\,p_1\,p_2 / r^4)$ | $U = (1/4\pi\varepsilon_0)\cdot(p_1\,p_2 / r^3)$

6. Electrostatic Potential (V) & Potential Energy (U)

Electrostatic Potential Energy (U)

- **Two Point Charges:** $U = (k\,q_1\,q_2) / r$
- **N-Charge System:** $U = \sum_{i<j}(k\,q_i\,q_j) / r_{ij}$
- **Work-Energy Relation:** $W_{ext} = \Delta U = q\,(V_f - V_i)$ (Slow movement)

Potential (V) & Field (E) Relations

- **Definition:** $V(r) = -\int_{\infty}^{r} \boldsymbol{E}\cdot d\boldsymbol{r}$
- **Gradient Form:** $\boldsymbol{E} = -\nabla V = -(\partial V/\partial x\,\hat{\boldsymbol{i}} + \partial V/\partial y\,\hat{\boldsymbol{j}} + \partial V/\partial z\,\hat{\boldsymbol{k}})$
- **Potential Difference:** $V_B - V_A = -\int_A^B \boldsymbol{E}\cdot d\boldsymbol{r}$

7. Standard Electric Potential (V) Distributions

System	Potential Formula (V)	Key Characteristics & Notes
Point Charge	$V = k\,q / r$	Scalar; includes sign of charge q
Uniform Ring	• Axis: $V = (k\,Q) / \sqrt{(x^2 + R^2)}$ • Center ($x=0$): $V_{center} = k\,Q / R$	Max potential occurs at center ($x=0$)
Spherical Shell / Conducting Sphere	• $r \geq R$: $V = k\,Q / r$ • $r \leq R$: $V = k\,Q / R$ (Constant)	Entire conductor volume is equipotential
Solid Non-Conducting Sphere	• $r \geq R$: $V = k\,Q / r$ • $r \leq R$: $V = [k\,Q\,(3R^2 - r^2)] / (2\,R^3) = \rho(3R^2 - r^2) / (6\varepsilon_0)$	$V_{center} = 1.5\,V_{surface} = (3\,k\,Q) / (2\,R)$ Parabolic variation inside
Uniform Charged Disc	$V = (\sigma / 2\varepsilon_0)\,[\,\sqrt{(x^2 + R^2)} - x\,]$	Along central axis distance x
Electric Dipole	• General (r,θ): $V = (k\,p\cos\theta) / r^2$ • Axial ($\theta=0°$): $V = k\,p / r^2$ | Equatorial: $V = 0$	Valid for point dipoles ($r \gg d$)
Infinite Line Charge	$V_B - V_A = -2\,k\,\lambda\,\ln(r_B / r_A)$	Absolute potential is undefined at ∞
Thin Non-Conducting Sheet	$V_B - V_A = -[\,\sigma / (2\varepsilon_0)\,]\,(r_B - r_A)$	Linear change along field lines
Thin Conducting Sheet	$V_B - V_A = -(\sigma / \varepsilon_0)\,(r_B - r_A)$	Field is double non-conducting sheet

8. Equipotential Surfaces & Conductors

- Electric field lines are **always perpendicular** to equipotential surfaces ($\boldsymbol{E} \perp d\boldsymbol{r}$).
- Work done moving charge q between any two points on an equipotential surface is **zero** ($W = 0$).
- Conductor surfaces in electrostatic equilibrium are **equipotential regions** with zero internal field ($E_{in} = 0$).

9. Capacitance (C) Fundamentals & Geometries [Added]

Core Definition & Isolated Sphere

- **Definition:** $C = Q / V$ (Unit: Farad [F], 1 F = 1 C/V)
- **Isolated Conducting Sphere:**

$$C = 4 \pi \varepsilon_0 R$$

Parallel Plate Capacitor (PPC)

- **Vacuum / Air PPC:** $C_0 = (\varepsilon_0 A) / d$
- **PPC with Slab (thickness $t < d$):**

$$C = (\varepsilon_0 A) / [\, d - t\,(1 - 1/K)\,]$$

- **PPC with Metal Slab ($K = \infty$):** $C = (\varepsilon_0 A) / (d - t)$

Cylindrical Capacitor [Added]

$C = (2 \pi \varepsilon_0 L) / [\, \ln(b / a)\,]$

Coaxial cylinders of radii a (inner) and b (outer).

Spherical Capacitor [Added]

- **Outer Grounded:** $C = (4 \pi \varepsilon_0 a b) / (b - a)$
- **Inner Grounded:** $C = (4 \pi \varepsilon_0 b^2) / (b - a)$

10. Energy Stored, Forces & Combination of Capacitors [Added]

Energy Stored (U)

$$U = \tfrac{1}{2} C V^2 = \tfrac{1}{2} Q^2 / C = \tfrac{1}{2} Q V$$

Plate Force & Pressure

Force: $F = Q^2 / (2\varepsilon_0 A) = \tfrac{1}{2} \varepsilon_0 E^2 A$

Pressure: $P = \sigma^2 / (2\varepsilon_0) = \tfrac{1}{2} \varepsilon_0 E^2$

Series / Parallel Rules

Series: $1/C_{eq} = \sum 1/C_i$ (Same Q)

Parallel: $C_{eq} = \sum C_i$ (Same V)

11. Charge Redistribution & Dielectric Insertion [Added]

Connecting Two Charged Capacitors / Conductors

- **Common Potential (V_c):** $V_c = (C_1 V_1 + C_2 V_2) / (C_1 + C_2) = (Q_1 + Q_2) / (C_1 + C_2)$
- **Heat Loss during sharing (ΔH):**

$$\Delta H = \tfrac{1}{2} [\, (C_1 C_2) / (C_1 + C_2)\,] (V_1 - V_2)^2$$

Dielectric Insertion Effects (Dielectric Constant K)

Quantity	Battery Connected (V Const)	Battery Removed (Q Const)
Capacity (C)	$C' = K C$	$C' = K C$
Charge (Q)	$Q' = K Q$	$Q' = Q$
Voltage (V)	$V' = V$	$V' = V / K$
Field (E)	$E' = E$	$E' = E / K$
Energy (U)	$U' = K U$	$U' = U / K$

12. JEE Advanced Shortcuts: Shells, Self-Energy & RC Transients [Added]

Self-Energy of Spheres

- **Thin Conducting Spherical Shell:** $U_{self} = (k Q^2) / (2 R)$
- **Uniform Solid Non-Conducting Sphere:**

$$U_{self} = (3 k Q^2) / (5 R)$$

Earthing/Grounding Concentric Shells

- When a conductor is earthed, its net potential becomes zero ($V_{earth} = 0$).
- Charge flows to/from earth to satisfy $V_{shell} = \sum (k q_i / r_i) = 0$.

Transient RC Circuits (Charging & Discharging Dynamics) [Added]

1. Charging Circuit ($q(0) = 0$):

- Charge: $q(t) = Q_0 (1 - e^{-t/\tau})$ where $Q_0 = C \varepsilon$
- Current: $I(t) = I_0 e^{-t/\tau}$ where $I_0 = \varepsilon / R$

2. Discharging Circuit ($q(0) = Q_0$):

- Charge: $q(t) = Q_0 e^{-t/\tau}$
- Current: $I(t) = I_0 e^{-t/\tau}$ where $I_0 = Q_0 / (RC)$

Time Constant (τ): $\tau = R C$. At $t = \tau$, charging capacitor reaches ~63.2% of max charge, discharging reaches ~36.8%.

CURRENT ELECTRICITY

JEE Main & Advanced Formula Sheet

1. ELECTRIC CURRENT & DRIFT VELOCITY

- **Average Current:** $I_{av} = \frac{\Delta q}{\Delta t}$
- **Instantaneous Current:** $I = \lim_{\Delta t \to 0} \frac{\Delta q}{\Delta t} = \frac{dq}{dt}$
- **Current in Conductor:** $I = neAv_d$
  (n: charge carrier density, e: electron charge, A: cross-sectional area, v_d: drift velocity)
- **Drift Velocity:** $\vec{v}_d = -\frac{e\vec{E}}{m}\tau$ (magnitude $v_d = \frac{eE}{m}\tau$)
  ($\vec{E}$: applied electric field, m: mass of electron, τ: average relaxation time)
- **Current Density ($\vec{J}$):** A vector quantity with magnitude I/A normal to the charge flow. $\vec{J} = \frac{dI}{dA}\hat{n} \implies I = \int \vec{J} \cdot d\vec{A}$.
  Relation to drift velocity: $\vec{J} = -ne\vec{v}_d$
- **Mobility (μ):** $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$ (SI Unit: m^2/Vs)
- **Current in terms of Mobility:** $I = neA\mu E$

2. OHM'S LAW & RESISTANCE

- **Ohm's Law:** $V = IR$
- **Vector Form of Ohm's Law:** $\vec{J} = \sigma\vec{E}$
- **Resistance:** $R = \rho\frac{l}{A}$
- **Resistivity (ρ):** $\rho = \frac{m}{ne^2\tau}$
- **Conductivity (σ):** $\sigma = \frac{1}{\rho} = \frac{ne^2\tau}{m}$
- **Dependence on Temperature:**
  - Resistance: $R_T = R_0[1 + \alpha(T - T_0)]$
  - Resistivity: $\rho_T = \rho_0[1 + \alpha(T - T_0)]$

  (α: temperature coefficient; positive for conductors, negative for semiconductors and insulators)
- **[Added] Stretching of Wire:** If a wire is stretched to n times its original length (volume remains constant):
  $R \propto l^2 \propto \frac{1}{A^2} \propto \frac{1}{r^4} \implies R_{\text{new}} = n^2 R_{\text{old}}$.

3. GROUPING OF RESISTANCES

- **Series Combination:** $R_{eq} = R_1 + R_2 + \cdots + R_n$
  (Current remains the same through all resistors)
- **Parallel Combination:** $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$
  (Potential difference is the same across all resistors)
- **[Added] Voltage Divider Rule (Series):**
  Voltage across R_1: $V_1 = V\left(\frac{R_1}{R_1+R_2}\right)$
- **[Added] Current Divider Rule (Parallel):**
  Current through R_1: $I_1 = I\left(\frac{R_2}{R_1+R_2}\right)$

[Added] Carbon Resistor Color Code:
*Mnemonic: "**BB ROY** of **G**reat **B**ritain had **V**ery **G**ood **W**ife"*

Color	Digit	Mult.	Color	Digit	Mult.
Black	0	10^0	**G**reen	5	10^5
Brown	1	10^1	**B**lue	6	10^6
Red	2	10^2	**V**iolet	7	10^7
Orange	3	10^3	**G**rey	8	10^8
Yellow	4	10^4	**W**hite	9	10^9

Tolerance: Gold = $\pm 5\%$, Silver = $\pm 10\%$, None = $\pm 20\%$.

4. CELLS, EMF & GROUPING

- **Terminal Voltage (V):**
  - Discharging (supplying current): $V = E - Ir$
  - **[Added]** Charging (receiving current): $V = E + Ir$
  - Open Circuit ($I = 0$): $V = E$
- **Cells in Series:** $E_{eq} = E_1 + E_2 + \ldots$ (watch polarity!), $r_{eq} = r_1 + r_2 + \ldots$
  Current for n identical cells: $I = \frac{nE}{R+nr}$.
  If s cells are reversed: $E_{eq} = (n - 2s)E$.
- **Cells in Parallel:**
  $E_{eq} = \frac{\sum(E_i/r_i)}{\sum(1/r_i)}$, $\frac{1}{r_{eq}} = \sum\frac{1}{r_i}$
  Current for n identical cells: $I = \frac{E}{R+(r/n)}$.
- **Mixed Grouping:** n cells in series, m rows in parallel (Total cells $N = mn$).
  $E_{eq} = nE$, $r_{eq} = \frac{nr}{m}$, $I = \frac{nE}{R+\frac{nr}{m}}$.
  Maximum current is drawn when $R = \frac{nr}{m}$.
- **[Added] Maximum Power Transfer Theorem:**
  Power delivered to an external resistance R is maximum when $R = r_{eq}$.
  Maximum Power: $P_{max} = \frac{E_{eq}^2}{4r_{eq}}$.

5. KIRCHHOFF'S LAWS

- **KCL (Junction/Node Law):** $\sum I_{in} = \sum I_{out}$. (Based on the Law of Conservation of Charge).
- **KVL (Loop/Voltage Law):** $\sum IR + \sum E = 0$ around any closed loop. (Based on the Law of Conservation of Energy).

6. MEASURING INSTRUMENTS

- **Galvanometer to Ammeter:** Connect a small resistance (shunt S) in *parallel*.

$I_g R_g = (I - I_g)S \implies S = \frac{I_g R_g}{I - I_g}$.
Ideal ammeter has zero resistance ($R_A = 0$).

- **Galvanometer to Voltmeter:** Connect a high resistance (R_S) in *series*.
  $V = I_g(R_g + R_S) \implies R_S = \frac{V}{I_g} - R_g$.
  Ideal voltmeter has infinite resistance ($R_V = \infty$).
- **Wheatstone Bridge:** Balanced (null point, $I_g = 0$) condition: $\frac{P}{Q} = \frac{R}{S}$.
- **Meter Bridge:** Practical application of Wheatstone bridge. Used to find unknown resistance X.
  $X = R\left(\frac{100-l}{l}\right)$ (l is balance length from end A in cm).

7. POTENTIOMETER

Acts as an ideal voltmeter because it draws zero current from the test circuit at the balance point.

- **Potential Gradient (x):** Potential drop per unit length of potentiometer wire.
  $x = \frac{V_{wire}}{L} = \frac{I \cdot R_{wire}}{L} = \left(\frac{E_{driver}}{R_{wire}+r+R_h}\right)\frac{R_{wire}}{L}$.
- **Comparing EMFs of Two Cells:** $\frac{E_1}{E_2} = \frac{l_1}{l_2}$.
- **Internal Resistance of a Cell:**
  $r = R\left(\frac{l_1}{l_2} - 1\right) = R\left(\frac{l_1 - l_2}{l_2}\right)$.
  (l_1: balance length without shunt R [open circuit], l_2: balance length with shunt R [closed circuit]).

8. HEATING EFFECT & POWER

- **Electric Power:** $P = VI = I^2R = \frac{V^2}{R}$.
- **Joule's Law of Heating:**
  Heat produced $H = \int P dt = I^2Rt = \frac{V^2}{R}t = VIt$ (in Joules).
  In calories: $H = \frac{I^2Rt}{4.18}$ cal.
- **Bulbs in Series:** $\frac{1}{P_{eq}} = \frac{1}{P_1} + \frac{1}{P_2} + \dots$
  Brightness $\propto R \propto \frac{1}{P_{rated}}$.
  (The bulb with the lowest wattage glows brightest).
- **Bulbs in Parallel:** $P_{eq} = P_1 + P_2 + \dots$
  Brightness $\propto \frac{1}{R} \propto P_{rated}$.
  (The bulb with the highest wattage glows brightest).

9. THERMOELECTRIC EFFECTS

- **Seebeck Effect:** E.m.f developed due to a temperature difference at two junctions of dissimilar metals.
  $e = aT + bT^2$.
- **Neutral Temperature (T_n):** Temperature of the hot junction at which thermo-e.m.f is maximum ($de/dT = 0$).
  $T_n = -\frac{a}{2b}$.
- **Inversion Temperature (T_i):** Temperature at which e.m.f becomes zero. $T_i = -\frac{a}{b}$.
  Relation: $T_i - T_n = T_n - T_0 \implies T_i = 2T_n - T_0$
  (T_0: temperature of the cold junction).
- **Peltier Coefficient (π):** $\pi = \frac{\Delta H}{\Delta Q}$
  (Heat absorbed or evolved per unit charge at a junction).
- **Thomson Coefficient (σ):** $\sigma = \frac{\Delta H}{\Delta Q \Delta T}$.

10. CHEMICAL EFFECT OF CURRENT

- **Faraday's 1st Law of Electrolysis:** Mass deposited/liberated is directly proportional to charge passed.
  $m = Zq = ZIt$
  (Z: Electrochemical Equivalent, E.C.E).
- **Faraday's 2nd Law:** When the same charge is passed through different electrolytes, mass deposited is proportional to chemical equivalent.
  $\frac{m_1}{m_2} = \frac{E_1}{E_2} = \frac{Z_1}{Z_2}$.
  Chemical Equivalent $E = \frac{\text{Atomic Weight}}{\text{Valency}}$.
- **Faraday Constant (F):** $F = \frac{E}{Z} = 96500$ C/mol.
  (Charge required to liberate 1 gram equivalent of substance). 1 Faraday $= N_A \times e$.

SEMICONDUCTORS & DEVICES

JEE Main & Advanced Complete Revision Formula Booklet

PHYSICS REVISION SHEET

1. Band Theory & Intrinsic Carriers

Forbidden Energy Gap (E_g): Energy gap between Valence Band (VB) and Conduction Band (CB).

- **Conductors:** $E_g = 0$ (VB and CB overlap).
- **Semiconductors:** $E_g < 3\ eV$ (Si: $1.1\ eV$, Ge: $0.7\ eV$, GaAs: $1.43\ eV$). At $T = 0\ K$, behaves as insulator.
- **Insulators:** $E_g > 3\ eV$ (e.g., Diamond $E_g \approx 5.4\ eV$).

Intrinsic Semiconductor Carrier Concentration

$$n_e = n_h = n_i$$

n_e = electron density, n_h = hole density, n_i = intrinsic carrier density.

Mass Action Law (Thermal Equilibrium)

$$n_e \cdot n_h = n_i^2$$

Holds true for both intrinsic and extrinsic semiconductors at equilibrium.

Temperature Dependence ADDED

$$n_i^2 = A \cdot T^3 \cdot \exp\left(-\frac{E_g}{k_B T}\right)$$

2. Current & Electrical Conductivity

Total Current in Semiconductor

$$I = I_e + I_h = e\,A\,(n_e\,v_e + n_h\,v_h)$$

Current Density & Conductivity ADDED

$$J = \sigma\,E = e\,(n_e\,\mu_e + n_h\,\mu_h)\,E$$

$$\sigma = e\,(n_e\,\mu_e + n_h\,\mu_h), \quad \rho = \frac{1}{\sigma}$$

μ_e, μ_h = drift mobilities. Always $\mu_e > \mu_h$ because electrons move in CB while holes move in VB.

3. Extrinsic (Doped) Semiconductors

Property	N-Type	P-Type
Dopant	Pentavalent (P, As, Sb)	Trivalent (B, Al, In, Ga)
Majority	Electrons ($n_e \approx N_d$)	Holes ($n_h \approx N_a$)
Minority	Holes ($n_h \approx n_i^2 / N_d$)	Electrons ($n_e \approx n_i^2 / N_a$)
Impurity Level	Donor level E_d (just below CB)	Acceptor level E_a (just above VB)

Ionization Energy: Si ≈ 0.05 eV, Ge ≈ 0.01 eV. Semiconductor remains overall *electrically neutral*.

4. P-N Junction & Biasing

- **Depletion Layer:** Region devoid of mobile charge carriers, containing uncompensated immobile donor (N_d^+) and acceptor (N_a^-) ions. Width $W \approx 0.1$ to $1\ \mu m$.
- **Barrier Potential (V_0):** Built-in potential. Si: $\approx 0.7\ V$, Ge: $\approx 0.3\ V$. Decreases with temperature ($\approx -2\ mV/°C$). ADDED
- **Diffusion vs Drift Current:** Unbiased junction has $I_{diff} = I_{drift}$ $\&implies;$ $I_{net} = 0$.

Bias Mode	Depletion Width	Barrier Height	Resistance
Forward Bias (P → +, N → -)	Decreases	$V_0 - V$	Low ($10 - 100\ \Omega$)
Reverse Bias (P → -, N → +)	Increases	$V_0 + V$	Very High ($10^5 - 10^6\ \Omega$)

Dynamic (AC) Resistance

$$r_d = \frac{\Delta V}{\Delta I} = \frac{1}{\text{Slope of I-V curve}}$$

Diode Equation ADDED

$$I = I_0\left[\exp\left(\frac{e\,V}{\eta\,k_B\,T}\right) - 1\right]$$

I_0 = reverse saturation current, η = ideality factor (1 for Ge, 2 for Si).

5. Special Purpose Optoelectronic Diodes

- **Zener Diode:** Heavily doped P-N junction operated in *reverse breakdown* region. Used as **Voltage Regulator**.
- **Photodiode:** Operated in *reverse bias* under light. Photo-generated carriers increase reverse saturation current ($I \propto \text{Light Intensity}$). Requires photon energy $h\nu \geq E_g$.
- **Light Emitting Diode (LED):** Operated in *forward bias*. Spontaneous recombination emits light. Emitted wavelength $\lambda \approx \frac{hc}{E_g} = \frac{12400}{E_g(\text{eV})}\ \text{Å}$.
- **Solar Cell:** Unbiased P-N junction generating EMF via photovoltaic effect (V_{oc} = Open Circuit Voltage, I_{sc} = Short Circuit Current).

Zener Diode Regulator Equations ADDED

$$I_S = \frac{V_{in} - V_Z}{R_S}, \quad I_L = \frac{V_Z}{R_L}, \quad I_Z = I_S - I_L$$

6. Rectifier Circuits & Parameters

Parameter	Half-Wave Rectifier	Full-Wave Rectifier
Number of Diodes	1	2 (Center-Tap) / 4 (Bridge)
DC / Avg Current (I_{dc})	$\frac{I_0}{\pi} \approx 0.318\ I_0$	$\frac{2\,I_0}{\pi} \approx 0.636\ I_0$
RMS Current (I_{rms})	$\frac{I_0}{2} = 0.5\ I_0$	$\frac{I_0}{\sqrt{2}} \approx 0.707\ I_0$
Max Efficiency (η)	$\frac{0.406\ R_L}{r_f + R_L} \to 40.6\%$	$\frac{0.812\ R_L}{r_f + R_L} \to 81.2\%$
Ripple Factor (γ) ADDED	$\sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = 1.21$	$\sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = 0.48$
Output Frequency (f_{out})	f_{in}	$2\,f_{in}$
Peak Inverse Voltage	V_0	$2V_0$ (Center-Tap), V_0 (Bridge)

7. Junction Transistors (BJT) Fundamentals

- Three Terminals: **Emitter (E)** [Heavily doped, medium size], **Base (B)** [Lightly doped, thin ~ 1 μm], **Collector (C)** [Moderately doped, largest size].
- Operational Rule: Emitter-Base junction is **Forward Biased**, Collector-Base junction is **Reverse Biased** (Active region).

Fundamental Current Equation

$$I_E = I_B + I_C$$

Base current is very small: $I_B \approx (1 \text{ to } 5)\% \text{ of } I_E$, so $I_C \approx I_E$.

8. Transistor Parameters & Gain Relations

Common Base (CB) Configuration

DC Current Gain: $\alpha_{dc} = \frac{I_C}{I_E}$ (α < 1, typically 0.95 to 0.99)

AC Current Gain: $\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}$

Voltage Gain: $A_v = \alpha \times \frac{R_{out}}{R_{in}}$

Power Gain: $A_p = \alpha^2 \times \frac{R_{out}}{R_{in}}$

Common Emitter (CE) Configuration

DC Current Gain: $\beta_{dc} = \frac{I_C}{I_B}$ (β >> 1, typically 20 to 500)

AC Current Gain: $\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}$

Voltage Gain: $A_v = \frac{\Delta V_{CE}}{\Delta V_{BE}} = \beta \times \frac{R_L}{R_{in}}$

Power Gain: $A_p = \beta \times A_v = \beta^2 \times \frac{R_L}{R_{in}}$

Relationship between α and β

$$\beta = \frac{\alpha}{1 - \alpha}, \quad \alpha = \frac{\beta}{1 + \beta}, \quad \frac{1}{\alpha} - \frac{1}{\beta} = 1$$

Transconductance (g_m) ADDED

$$g_m = \frac{\Delta I_C}{\Delta V_{BE}} = \frac{\beta}{R_{in}} \text{ \&implies; } A_v = g_m \cdot R_L$$

Phase Relationship: CE Amplifier introduces a phase shift of **180° (π rad)** between input and output signal. CB Amplifier has **0° phase shift**.

9. Transistor Switch & LC Oscillator

- **As Switch:** Operating in *Cut-off Region* (Switch OFF, $V_{out} \approx V_{CC}$) and *Saturation Region* (Switch ON, $V_{out} \approx 0$).
- **Oscillator:** Converts DC power to AC power without external AC input using *positive feedback*.

Resonance Frequency of LC Oscillator

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Barkhausen Criterion for Oscillation ADDED

$$|A \cdot \beta_{fb}| = 1 \quad \text{\{and\}} \quad \text{Total Phase Shift} = 0 \text{ \{or\}} \quad 2\pi n$$

A = open-loop amplifier gain, β_{fb} = feedback fraction.

10. Logic Gates Summary Table ADDED

Gate	Boolean Eqn	Truth Table Output
OR	$Y = A + B$	0+0=0, 0+1=1, 1+0=1, 1+1=1
AND	$Y = A \cdot B$	0·0=0, 0·1=0, 1·0=0, 1·1=1
NOT	$Y = \bar{A}$	$0 \rightarrow 1, 1 \rightarrow 0$
NAND (Universal)	$Y = \overline{A \cdot B}$	Output 0 ONLY when both inputs are 1.
NOR (Universal)	$Y = \overline{A + B}$	Output 1 ONLY when both inputs are 0.
XOR (Ex-OR)	$Y = A \oplus B = \bar{A}B + A\bar{B}$	Output 1 when inputs are **different**. (0⊕0=0, 0⊕1=1, 1⊕0=1, 1⊕1=0)
XNOR (Ex-NOR)	$Y = \overline{A \oplus B} = AB + \bar{A}\bar{B}$	Output 1 when inputs are **identical**. (0⊙0=1, 0⊙1=0, 1⊙0=0, 1⊙1=1)

11. Boolean Algebra & Identities ADDED

De Morgan's Theorems

$$\overline{A + B} = \bar{A} \cdot \bar{B}$$
$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

Core Boolean Identities

- **OR Laws:** $A + 0 = A, A + 1 = 1, A + A = A, A + \bar{A} = 1$
- **AND Laws:** $A \cdot 0 = 0, A \cdot 1 = A, A \cdot A = A, A \cdot \bar{A} = 0$
- **Double Inversion:** $\overline{\bar{A}} = A$
- **Redundancy / Absorption Laws:**
  - $A + A \cdot B = A$
  - $A \cdot (A + B) = A$
  - $A + \bar{A} \cdot B = A + B$
  - $A \cdot (\bar{A} + B) = A \cdot B$

12. Constants & JEE Speed Revision Tips

- **Elementary Charge:** $e = 1.6 \times 10^{-19}\ C$
- **Boltzmann Constant:** $k_B = 1.38 \times 10^{-23}\ J/K = 8.62 \times 10^{-5}\ eV/K$
- **Thermal Voltage at 300 K:** $V_T = \frac{k_B T}{e} \approx 26\ mV \approx \frac{1}{39}\ V$.
- **Quick Check:** For a CE transistor, if $\beta = 100$, then $\alpha = \frac{100}{101} \approx 0.99$.
- **Logic Gate Shortcut:** Bubbled AND gate is equivalent to a NOR gate ($\bar{A} \cdot \bar{B} = \overline{A + B}$). Bubbled OR gate is equivalent to a NAND gate ($\bar{A} + \bar{B} = \overline{A \cdot B}$).

JEE REVISION: ELECTRIC CHARGES & FIELDS

1. BASIC PROPERTIES OF CHARGE

$q = ne$

→ Quantization of charge (e = 1.6 × 10^{-19} C)

$q = It = \frac{V}{R}t$

→ In terms of current/resistance

$q = W / V = F / E$

→ In terms of potential/field

$a = F / m = qE / m$

→ Acceleration of charged particle

2. COULOMB'S LAW

$F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_1 q_2}{r^2} = \frac{k\, q_1 q_2}{r^2}$

→ Force in vacuum

$\mathbf{F} = \frac{k\, q_1 q_2}{r^2}\,\hat{r}$

→ Vector form

$F_m = \frac{1}{4\pi\varepsilon_0\varepsilon_r} \cdot \frac{q_1 q_2}{r^2} = \frac{F_0}{\varepsilon_r}$ [Added]

→ In medium (ε_r = dielectric constant)

$\mathbf{F_{net}} = \mathbf{F_1} + \mathbf{F_2} + \mathbf{F_3} + \ldots$

→ Principle of Superposition

3. NULL POINT & EQUILIBRIUM

Distance of null point from Q_1 (where $E = 0$):

$x = \frac{r}{\sqrt{(Q_2/Q_1)} \pm 1}$

→ Use (+) for like charges (null point between charges).

→ Use (−) for unlike charges (null point outside, near weaker charge).

Equilibrium of 3 point charges: If Q_1, Q_2 are like charges, for entire system equilibrium, 3rd charge q (unlike) placed at null point must be: $q = -\frac{Q_1 Q_2}{(\sqrt{Q_1} + \sqrt{Q_2})^2}$ [Added]

4. ELECTRIC FIELD (E)

$\mathbf{E} = lim_{q\to 0}\,(\mathbf{F} / q) = \frac{kQ}{r^2}\,\hat{r}$

→ Point charge

$E = -\,dV / dr$

→ Relation with potential

5. ELECTRIC DIPOLE

$\mathbf{p} = q(2\mathbf{l}) = q\mathbf{d}$

→ Dipole moment (Direction: −q to +q)

1. Axial Point (End-on):

$E_{axial} = \frac{1}{4\pi\varepsilon_0}\,\frac{2p}{r^3}$

→ Short dipole ($r \gg a$)

$E_{axial\ exact} = \frac{1}{4\pi\varepsilon_0}\,\frac{2pr}{(r^2-a^2)^2}$ [Added]

2. Equatorial Point (Broad-on):

$E_{eq} = -\,\frac{1}{4\pi\varepsilon_0}\,\frac{p}{r^3}$

→ Short dipole ($r \gg a$)

$E_{eq\ exact} = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{(r^2+a^2)^{3/2}}$ [Added]

3. General Point (r, θ):

$E = \frac{1}{4\pi\varepsilon_0}\,\frac{p}{r^3}\,\sqrt{(1 + 3\cos^2\theta)}$

Direction: $\tan\alpha = ½ \tan\theta$

→ Angle with radial vector

6. DIPOLE IN UNIFORM ELECTRIC FIELD

$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E} = pE\sin\theta$

→ Torque

$W = pE\,(cos\theta_1 - cos\theta_2)$ [Added]

→ Work done in rotating dipole

$U = -\mathbf{p}\cdot\mathbf{E} = -pE\cos\theta$ [Added]

→ Potential energy

$T = 2\pi\sqrt{(I / pE)}$ [Added]

→ Time period of oscillation for small angles (I = moment of inertia)

7. FLUX & GAUSS'S LAW

$\Phi = \int \mathbf{E} \cdot d\mathbf{A} = \mathbf{E} \cdot \mathbf{A} = EA\cos\theta$

→ Electric Flux

$\Phi = \oint \mathbf{E} \cdot d\mathbf{s} = \frac{q_{enclosed}}{\varepsilon_0}$

→ Gauss's Law

→ Flux through closed surface is independent of shape/size.

8. CHARGE DENSITIES

Density Type	Symbol & Formula	Unit
Linear	$\lambda = Q/L = dQ/dl$	C/m
Surface	$\sigma = Q/A = dQ/dA$	C/m^2
Volume	$\rho = Q/V = dQ/dV$	C/m^3

9. FIELD DUE TO LINES & SHEETS

Infinitely Long Line Charge:

$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r}\hat{r}$

Finite Line Charge: [Added]

$E_{\perp} = \frac{k\lambda}{r}(\sin\theta_1 + \sin\theta_2)$

$E_{\parallel} = \frac{k\lambda}{r}(\cos\theta_2 - \cos\theta_1)$

Infinite Non-Conducting Thin Sheet:

$E = \frac{\sigma}{2\varepsilon_0}\hat{n}$

→ Uniform, independent of r

Infinite Conducting Thin Sheet:

$E = \frac{\sigma}{\varepsilon_0}\hat{n}$

Near any charged conductor:

$E = \frac{\sigma}{\varepsilon_0}$

→ Normal to surface

10. FIELD DUE TO RINGS & DISCS

Uniformly Charged Ring (Axis):

$E_{axis} = \frac{kQx}{(R^2 + x^2)^{3/2}}$

→ Max E occurs at $x = R/\sqrt{2}$

$E_{centre} = 0$ (at $x = 0$)

Semi-Circular Ring (Centre): [Added]

$E = \frac{2k\lambda}{R}$

Uniformly Charged Disc (Axis): [Added]

$E = \frac{\sigma}{2\varepsilon_0} \times \left[1 - \frac{x}{\sqrt{(R^2 + x^2)}}\right]$

11. FIELD DUE TO SPHERES & CYLINDERS

Hollow/Solid Conducting Sphere:

For $r \geq R$: $E = \frac{kQ}{r^2}$

→ Acts as point charge

For $r < R$: $E = 0$

→ Inside conductor

Solid Non-Conducting Sphere:

For $r \geq R$: $E = \frac{kQ}{r^2}$

For $r \leq R$: $E = \frac{kQr}{R^3} = \frac{\rho r}{3\varepsilon_0}$

→ Inside sphere, $E \propto r$.

Long Conducting Cylinder:

For $r \geq R$: $E = \frac{\lambda}{2\pi\varepsilon_0 r}$

For $r < R$: $E = 0$

ELECTROMAGNETIC WAVES

JEE Main & Advanced Complete Formula & Revision Sheet

PHYSICS REVISION

1. MAXWELL'S EQUATIONS & DISPLACEMENT CURRENT

Displacement Current (I_D)

Arises due to time-varying electric field / flux:

$$I_D = \varepsilon_0 \cdot (d\Phi_E / dt) = \varepsilon_0 \cdot d/dt \int \mathbf{E} \cdot d\mathbf{A}$$

Parallel Plate Capacitor Charging:

$I_D = \varepsilon_0 \cdot d/dt\ (E \cdot A) = \varepsilon_0 \cdot d/dt\ (qA / \varepsilon_0 A) = dq/dt = I_C$

→ **Inside capacitor gap: Conduction Current $I_C = 0$, Displacement Current $I_D = I$**

→ **Outside capacitor gap: $I_C = I$, $I_D = 0$**

Modified Ampère-Maxwell Law

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_C + I_D) = \mu_0 I_C + \mu_0 \varepsilon_0 (d\Phi_E / dt)$$

Magnetic Field inside Capacitor Gap (Radius R) [Added]:

- At distance $r \le R$: $B(r) = (\mu_0 I_C r) / (2\pi R^2)$
- At distance $r \ge R$: $B(r) = (\mu_0 I_C) / (2\pi r)$

LAW / NAME	INTEGRAL FORM	PHYSICAL SIGNIFICANCE
Gauss's Law (Electrostatics)	$\oint \mathbf{E} \cdot d\mathbf{A} = q_{encl} / \varepsilon_0$	Electric charges act as sources/sinks of **E**.
Gauss's Law (Magnetism)	$\oint \mathbf{B} \cdot d\mathbf{A} = 0$	Magnetic monopoles do not exist; lines form closed loops.
Faraday's Law of Induction	$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B / dt$	Changing magnetic flux induces an electric field.
Ampère-Maxwell Law	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_C + \mu_0 \varepsilon_0 (d\Phi_E / dt)$	Conduction and changing electric flux induce magnetic field.

2. MATHEMATICAL REPRESENTATION OF EM WAVES

Sinusoidal Wave Equations

For wave propagating in **+x direction**:

$$E_y(x,t) = E_0 \sin(kx - \omega t + \varphi)$$
$$B_z(x,t) = B_0 \sin(kx - \omega t + \varphi)$$

E_0, B_0 = Peak amplitudes of Electric and Magnetic fields.

$\omega = 2\pi f = 2\pi/T$ (Angular Frequency)

$k = 2\pi/\lambda = \omega/v$ (Wave Vector / Propagation Constant)

Direction & Vector Relationships

Direction of Propagation: $\hat{n}_{wave} = (\vec{E} \times \vec{B}) / |\vec{E} \times \vec{B}|$

- **Transverse Nature:** $\mathbf{E} \perp \mathbf{B}$, $\mathbf{E} \perp \mathbf{k}$, $\mathbf{B} \perp \mathbf{k}$
- Dot products: $\mathbf{E} \cdot \mathbf{B} = 0$, $\mathbf{E} \cdot \mathbf{k} = 0$, $\mathbf{B} \cdot \mathbf{k} = 0$
- **E** and **B** are in **same phase** (reach maxima/zero simultaneously).
- **Amplitude Relation:** $E_0 / B_0 = E(x,t) / B(x,t) = v$

3. WAVE SPEED & MEDIUM PROPERTIES

Speed in Vacuum & Medium

$$c = 1 / \sqrt{(\mu_0 \varepsilon_0)} \approx 3.00 \times 10^8 \text{ m/s}$$

In any material medium:

$v = 1 / \sqrt{(\mu \varepsilon)} = c / \sqrt{(\mu_r \varepsilon_r)} = c / n$

Where Refractive Index $n = \sqrt{(\mu_r \varepsilon_r)} = \sqrt{(\mu_r K)}$

→ **For non-magnetic media ($\mu_r \approx 1$): $n \approx \sqrt{\varepsilon_r} = \sqrt{K}$**

Wavelength & Frequency in Medium

- Frequency f remains **unchanged** when entering a medium.
- Wavelength changes: $\lambda_{medium} = \lambda_0 / n$
- Wave speed changes: $v = f \cdot \lambda_{medium}$
- $\varepsilon_0 \approx 8.854 \times 10^{-12}$ F/m, $\mu_0 = 4\pi \times 10^{-7}$ T·m/A

4. ENERGY DENSITY OF EM WAVES

Instantaneous Energy Density

- Electric: $u_E = \frac{1}{2} \varepsilon_0 E^2$
- Magnetic: $u_B = B^2 / (2\mu_0)$
- Total Instantaneous: $u = u_E + u_B = \frac{1}{2} \varepsilon_0 E^2 + B^2 / (2\mu_0)$

Average Energy Density ($\langle u \rangle$)

Energy is **equally divided** between **E** and **B** fields:

$$\langle u_E \rangle = \frac{1}{4} \varepsilon_0 E_0^2 = \frac{1}{2} \varepsilon_0 E_{rms}^2$$
$$\langle u_B \rangle = B_0^2 / (4\mu_0) = B_{rms}^2 / (2\mu_0)$$

Total Average Energy Density:

$u_{avg} = \langle u_E \rangle + \langle u_B \rangle = 2\langle u_E \rangle = 2\langle u_B \rangle = \frac{1}{2} \varepsilon_0 E_0^2 = B_0^2 / (2\mu_0) = \varepsilon_0 E_{rms}^2$

5. POYNTING VECTOR & INTENSITY

Poynting Vector (S)

Represents instantaneous energy flux density (Power per unit normal area, unit: W/m^2):

$$\mathbf{S} = (1 / \mu_0) (\&vec;E \times \&vec;B)$$

Direction of $\mathbf{S}$ gives the direction of energy propagation.

Wave Intensity (I)

Average energy crossing unit area per second:

$$I = |\langle \mathbf{S} \rangle| = c \cdot u_{avg} = \tfrac{1}{2} c \varepsilon_0 E_0^2 = (c B_0^2) / (2\mu_0) = c \varepsilon_0 E_{rms}^2$$

Intensity Variation with Distance from Source

Point Source: Isotropic

$I = P / (4\pi r^2) \propto 1/r^2$

$E_0 \propto 1/r$, $B_0 \propto 1/r$

Line Source: Cylindrical

$I = P / (2\pi r L) \propto 1/r$

$E_0 \propto 1/\sqrt{r}$, $B_0 \propto 1/\sqrt{r}$

Plane Wave: Unfocused

$I = \text{Constant}$

$E_0, B_0 = \text{Independent of } r$

6. MOMENTUM, RADIATION PRESSURE & FORCE

Linear Momentum (p)

EM wave delivering total energy U to a surface:

$p = U / c$ **(Perfectly Absorbing)**

$p = 2U / c$ **(Perfectly Reflecting)**

Photon relation: $p = h\nu/c = h / \lambda$

Radiation Pressure (P_{rad}) & Force (F)

For Normal Incidence (Intensity I, Beam area A):

SURFACE TYPE	RADIATION PRESSURE (P_{RAD})	FORCE (F)
100% Absorbing (r=0)	$I / c = u_{avg}$	$IA / c = P_{source} / c$
100% Reflecting (r=1)	$2I / c = 2u_{avg}$	$2IA / c$
Partial Reflection (r)	$(1+r) I / c$	$(1+r) IA / c$

Oblique Incidence (Angle θ with Normal to Surface) [Added]

Perfectly Absorbing:

$P_{rad} = (I / c) \cos^2\theta$ | $F_{normal} = (IA / c) \cos\theta$

Perfectly Reflecting:

$P_{rad} = (2I / c) \cos^2\theta$ | $F_{normal} = (2IA / c) \cos\theta$

7. THE ELECTROMAGNETIC SPECTRUM SUMMARY

TYPE	WAVELENGTH (Λ)	FREQUENCY (F)	PRODUCTION SOURCE	APPLICATIONS & DETECTION
γ-Rays	$< 10^{-11}$ m	$> 3 \times 10^{19}$ Hz	Radioactive nuclear decay, nuclear reactions	Cancer radiotherapy, nuclear structure research
X-Rays	10^{-11} to 3×10^{-8} m	10^{16} to 3×10^{19} Hz	Sudden stopping of fast electrons in metal	Bone radiography, crystal structure, security
Ultraviolet (UV)	3×10^{-8} to 4×10^{-7} m	7.5×10^{14} to 10^{16} Hz	Inner shell atomic transitions, Sun, Hg lamps	Water purification, germicidal lamps, LASIK
Visible Light	4×10^{-7} to 7.8×10^{-7} m	3.8×10^{14} to 7.5×10^{14} Hz	Valence electron transitions in excited atoms	Human vision, photography, optical fibers
Infrared (IR)	7.8×10^{-7} to 10^{-3} m	3×10^{11} to 3.8×10^{14} Hz	Thermal vibrations of molecules & hot bodies	Night vision, TV remotes, physical therapy
Microwaves	10^{-3} to 0.3 m	10^{9} to 3×10^{11} Hz	Klystron / Magnetron tubes, Gunn diodes	RADAR navigation, microwave ovens, cellular
Radio Waves	0.3 to 10^{4} m	500 kHz to 10^{9} Hz	Accelerated charges in LC oscillators/antennas	AM/FM radio, TV broadcast, satellite comm.

8. JEE ADVANCED IMPORTANT SHORTCUTS & KEY CONSTANTS

- **Optical Effect:** Electric vector $\mathbf{E}$ causes optical/retinal effects **Speed Ratio:** $c / v = n = \lambda_{vacuum} / \lambda_{medium}$

JEE MAIN & ADVANCED FORMULA SHEET

Complete Chapter Revision • Physics Desk Reference

PHYSICS

PART I: MOVING CHARGES & MAGNETISM

PRIMARY & EXTENDED JEE RESULTS

1. Lorentz Force & Charged Particle Motion

• **Lorentz Force:** $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

• **Pure Magnetic Force:** $\mathbf{F}_m = q(\mathbf{v} \times \mathbf{B}) = q\,v\,B\,\sin\theta$

→ Work done by $\mathbf{F}_m$ is ALWAYS zero ($\mathbf{F}_m \perp \mathbf{v}$). Speed & Kinetic Energy remain constant.

Trajectory in Uniform B-field ($\mathbf{E} = 0$):

1. **Parallel/Antiparallel ($\theta = 0°, 180°$):** Straight line (Un-deflected).
2. **Perpendicular ($\theta = 90°$):** Circular Motion

Radius: $r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV}}{qB}$

Time Period: $T = \frac{2\pi m}{qB}$; Angular Freq: $\omega = \frac{qB}{m}$; Frequency: $f = \frac{qB}{2\pi m}$

→ T, ω, f are INDEPENDENT of velocity v and radius r.

3. **Oblique Angle ($\theta \neq 0°, 90°, 180°$):** Helical Path

• Parallel component: $v_{\parallel} = v\cos\theta$ (causes linear motion)

• Perpendicular component: $v_{\perp} = v\sin\theta$ (causes circular motion)

• Radius: $r = \frac{m\,v\sin\theta}{qB}$; Pitch: $p = v_{\parallel}T = \frac{2\pi\,m\,v\cos\theta}{qB}$

• **Velocity Selector** ADDED: Crossed fields ($\mathbf{E} \perp \mathbf{B} \perp \mathbf{v}$) → undeflected if $v = \frac{E}{B}$.

2. Cyclotron Dynamics

• **Resonance Condition:** $f_{applied} = f_c = \frac{qB}{2\pi m}$

• **Max Exit Speed:** $v_{max} = \frac{q\,B\,R}{m}$ (where R = radius of Dees)

Max Kinetic Energy: $K_{max} = \frac{1}{2}\,m\,v_{max}^2 = \frac{q^2\,B^2\,R^2}{2m}$

→ Cannot accelerate electrons (relativistic mass gain) or uncharged particles (neutrons).

3. Biot-Savart Law & Magnetic Fields

• **Vector Form:** $d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,(d\mathbf{l} \times \hat{\mathbf{r}})}{r^2} = \frac{\mu_0}{4\pi}\frac{I\,(d\mathbf{l} \times \mathbf{r})}{r^3}$

Where $\frac{\mu_0}{4\pi} = 10^{-7}\ T{\cdot}m/A$; μ_0 = permeability of free space.

• **Straight Current-Carrying Conductor:**

$$B = \frac{\mu_0 I}{4\pi d}(\sin\theta_1 + \sin\theta_2)$$

Where d = perpendicular distance from wire; θ_1, θ_2 = subtended angles.

1. **Infinitely Long Wire ($\theta_1 = \theta_2 = 90°$):** $B = \frac{\mu_0 I}{2\pi d}$
2. **Semi-Infinite Wire ($\theta_1 = 90°$, $\theta_2 = 0°$):** $B = \frac{\mu_0 I}{4\pi d}$
3. **Point on Axial Line of Wire:** $B = 0$

4. Field of Circular Loops, Arcs & Axis

• **Circular Arc of Angle θ at Center:**

$$B_{center} = \frac{\mu_0 I}{4\pi R}\theta \quad (\theta \text{ in radians})$$

→ Full Loop ($\theta = 2\pi$): $B_0 = \frac{\mu_0 I}{2R}$ | Semicircle ($\theta = \pi$): $B = \frac{\mu_0 I}{4R}$

• **Field along Axis of Circular Loop (Radius R, Distance x):**

$$\mathbf{B}(x) = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}\hat{\mathbf{i}}$$

• At center ($x = 0$): $B = \frac{\mu_0 I}{2R}$; For $x \gg R$: $B \approx \frac{\mu_0 I R^2}{2x^3} = \frac{\mu_0 M}{2\pi x^3}$

• **Helmholtz Coils Arrangement** ADDED:

Two coaxial coils of N turns separated by distance R. Point of inflexion at $x = R/2$. Uniform magnetic field:

$$B_{uniform} = \left(\frac{8}{5\sqrt{5}}\right)\frac{\mu_0 N I}{R} \approx 0.716\,\frac{\mu_0 N I}{R}$$

5. Solenoids, Toroids & Regular Polygons

• **Finite Solenoid (Length L, Turns/length $n = N/L$):**

$B = \frac{\mu_0 n I}{2}(\cos\theta_1 - \cos\theta_2)$ where θ_1, θ_2 are end-angles.

• **Long Ideal Solenoid ($L \gg R$):**

Inside Center: $B = \mu_0 n I = \frac{\mu_0 N I}{L}$ | At Ends: $B_{end} = \frac{1}{2}\mu_0 n I$

• **Toroid (Mean Radius R, Total turns N):**

Inside core: $B = \mu_0 n I = \frac{\mu_0 N I}{2\pi R}$; Outside / In open space: $B = 0$

• **Regular n-sided Polygon Center** ADDED:

$$B_{center} = n \times \frac{\mu_0 I}{2\pi r}\tan\left(\frac{\pi}{n}\right) \quad (r = \text{incircle radius})$$

6. Magnetic Force on Current Conductors

• **Straight Wire in Uniform B-field:** $\mathbf{F} = I(\mathbf{L} \times \mathbf{B}) = I\,L\,B\sin\theta$

• **Arbitrary Shaped Wire:** $\mathbf{F} = I\int(d\mathbf{l} \times \mathbf{B}) = I\,(\mathbf{L}_{eff} \times \mathbf{B})$

→ $\mathbf{L}_{eff}$ = vector joining starting point directly to end point.

• **Closed Current Loop in Uniform B-field:** $\mathbf{F}_{net} = 0$.

PART I: MOVING CHARGES & MAGNETISM (CONTD.)

CIRCUITAL LAW, FORCES & INSTRUMENTS

7. Ampere's Circuital Law & Applications

• **Ampere's Law:** $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enclosed}$

• **Solid Long Cylindrical Wire (Radius a, Current I):**

1. Inside ($r < a$): $B = \frac{\mu_0 I r}{2\pi a^2}$ ($B \propto r$)
2. Surface ($r = a$): $B = \frac{\mu_0 I}{2\pi a}$
3. Outside ($r > a$): $B = \frac{\mu_0 I}{2\pi r}$ ($B \propto 1/r$)

• **Thick Hollow Pipe (Inner a, Outer b)** ADDED:

• $r < a$: $B = 0$

• $a \le r \le b$: $B = \frac{\mu_0 I}{2\pi r}\left[\frac{r^2 - a^2}{b^2 - a^2}\right]$

• $r > b$: $B = \frac{\mu_0 I}{2\pi r}$

8. Parallel Conductors & Magnetic Dipole

• **Force Between Parallel Wires (Distance d, Length L):**

$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$ (N/m)

→ Like currents ATTRACT; Opposite currents REPEL.

• **1 Ampere Definition:** Force per unit length = $2 \times 10^{-7}\ N/m$ when $d = 1\ m$.

• **Magnetic Dipole Moment of Current Loop:**

$\mathbf{M} = N I \mathbf{A}$ (Units: $A \cdot m^2$ or J/T)

→ Direction given by Right Hand Thumb Rule.

• **Torque & Potential Energy of Dipole in Uniform B:**

Torque: $\boldsymbol{\tau} = \mathbf{M} \times \mathbf{B} \Rightarrow \tau = M B \sin\theta = N I A B \sin\theta$
Potential Energy: $U = -\mathbf{M} \cdot \mathbf{B} = -M B \cos\theta$
Work Done (Rotation): $W_{ext} = M B (\cos\theta_1 - \cos\theta_2)$

• **Stable Equilibrium:** $\theta = 0^\circ$ ($U = -MB$, $\tau = 0$)
• **Unstable Equilibrium:** $\theta = 180^\circ$ ($U = +MB$, $\tau = 0$)

9. Moving Coil Galvanometer & Conversions

• **Working Principle:** Deflecting Torque = Restoring Torque
$N I A B = k\phi \Rightarrow \phi = \left(\frac{N A B}{k}\right) I$

Where k = torsional constant of spring, ϕ = deflection angle.

• **Sensitivities:**

1. Current Sensitivity: $S_I = \frac{\phi}{I} = \frac{N A B}{k}$
2. Voltage Sensitivity: $S_V = \frac{\phi}{V} = \frac{S_I}{R_g} = \frac{N A B}{k R_g}$

• **Conversion to Ammeter (Range 0 - I):**
Connect low resistance Shunt S in **PARALLEL** with Galvanometer (G).

$S = \frac{I_g G}{I - I_g}$ | $R_A = \frac{G S}{G + S}$ (Ideal Ammeter: $R_A = 0$)

• **Conversion to Voltmeter (Range 0 - V):**
Connect high resistance R in **SERIES** with Galvanometer (G).

$R = \frac{V}{I_g} - G$ | $R_V = G + R$ (Ideal Voltmeter: $R_V = \infty$)

PART II: MAGNETISM & MATTER

10. Bar Magnet & Dipole Fundamentals

• **Magnetic Pole Strength (m):** Unit: $A \cdot m$. Scalar.
• **Magnetic Dipole Moment ($\mathbf{M}$):** Vector from South to North Pole.

$\mathbf{M} = m \cdot (2\mathbf{l})$ (where $2l$ = magnetic length)

Geometric length L_g vs Magnetic length L_m: $L_m \approx 0.84\ L_g$.

• **Cutting / Bending Bar Magnets** ADDED:
1. Cut longitudinally into n strips: $m' = m/n \Rightarrow M' = M/n$.
2. Cut transversely into n pieces: $l' = l/n \Rightarrow M' = M/n$.
3. Bent into semicircular arc: $M' = \frac{2M}{\pi}$.

11. Magnetic Field of Bar Magnet / Dipole

• **1. Axial Point (End-on position, distance r from center):**

Exact: $B_{axial} = \frac{\mu_0}{4\pi}\frac{2 M r}{(r^2 - l^2)^2}$

Short Magnet ($r \gg l$): $\mathbf{B}_{axial} = \frac{\mu_0}{4\pi}\frac{2\mathbf{M}}{r^3}$ (Parallel to **M**)

• **2. Equatorial Point (Broadside-on position):**

Exact: $B_{eq} = \frac{\mu_0}{4\pi}\frac{M}{(r^2 + l^2)^{3/2}}$

Short Magnet ($r \gg l$): $\mathbf{B}_{eq} = -\frac{\mu_0}{4\pi}\frac{\mathbf{M}}{r^3}$ (Antiparallel to **M**)

→ For short dipole: $B_{axial} = 2\ B_{eq}$ at equal distances.

• **3. General Point $P(r, \theta)$ for Short Dipole:**

Magnitude: $B = \frac{\mu_0}{4\pi}\frac{M}{r^3}\sqrt{1 + 3\cos^2\theta}$

Direction Angle ϕ with radius vector: $\tan\phi = \frac{\tan\theta}{2}$

12. Oscillations of Dipole & Instruments

• **Time Period of Oscillations in Uniform Field B:**

$T = 2\pi\sqrt{\left(\frac{I}{M B}\right)} \Rightarrow B = \frac{4\pi^2 I}{M T^2}$

Where I = moment of inertia of magnet about oscillation axis.

• **Vibration Magnetometer (Sum & Diff Method)** ADDED:

• Sum position (same direction): $T_1 = 2\pi\sqrt{\left[\frac{I_1 + I_2}{(M_1 + M_2)B_H}\right]}$

• Diff position (opposite direction): $T_2 = 2\pi\sqrt{\left[\frac{I_1 + I_2}{(M_1 - M_2)B_H}\right]}$

Ratio: $\frac{M_1}{M_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2}$

• **Tangent Law & Tangent Galvanometer** ADDED:

$B = B_H \tan\theta \Rightarrow I = K \tan\theta$ where $K = \frac{2 R B_H}{\mu_0 N}$ (Reduction factor).

PART II: MAGNETISM & MATTER (CONTD.)

ATOMIC MAGNETISM, MATERIALS & GEOMAGNETISM

13. Atomic Magnetism & Bohr Magneton

• **Magnetic Dipole Moment of Revolving Electron:**

$$\mu_l = \frac{e\,v\,r}{2} = \frac{e}{2\,m_e}\,L \quad \text{(in vector form: } \boldsymbol{\mu}_l = -\frac{e}{2\,m_e}\,\mathbf{L}\text{)}$$

Where $L = m_e\,v\,r$ = orbital angular momentum.

• **Gyromagnetic Ratio:** $\frac{\mu_l}{L} = \frac{e}{2\,m_e} \approx 8.8 \times 10^{10}\ C/kg$ (Constant).

• **Bohr Magneton (μ_B):**

Using Bohr's postulate ($L = \frac{n\,h}{2\pi}$):

$$\mu_l = n \cdot \left(\frac{e\,h}{4\pi\,m_e}\right) = n \cdot \mu_B$$

1 Bohr Magneton (μ_B) = $9.27 \times 10^{-24}\ A{\cdot}m^2$ (or J/T)

14. Magnetic Intensity & Material Relations

• **Magnetizing Field / Magnetic Intensity ($\mathbf{H}$):**

$\mathbf{H} = \frac{\mathbf{B}_o}{\mu_o} = n\,I$ (Units: A/m). Independent of medium.

• **Intensity of Magnetization ($\mathbf{I}$ or $\mathbf{M}_z$):**

$\mathbf{I} = \frac{\mathbf{M}}{V} = \frac{m}{A}$ (Units: A/m). Dipole moment per unit volume.

• **Magnetic Susceptibility (χ):** $\chi = \frac{I}{H}$ (Dimensionless).

• **Magnetic Permeability (μ):** $\mu = \frac{B}{H}$ | Relative Permeability: $\mu_r = \frac{\mu}{\mu_o} = \frac{B}{B_o}$

• **Core Fundamental Relation:**

$$\mathbf{B} = \mu_o\,(\mathbf{H} + \mathbf{I}) = \mu_o\,H\,(1 + \chi) = \mu\,\mathbf{H}$$

Relative Permeability Link: $\mu_r = 1 + \chi$

15. Hysteresis Loop & Energy Losses

• **Retentivity / Remanence ($o\ b$):** Value of B (or I) when $H = o$.
• **Coercivity ($o\ c$):** Reverse magnetizing field H required to reduce B to zero.

Hysteresis Energy Loss per unit volume per cycle:
$W = \oint \mathbf{H} \cdot d\mathbf{B}$ = *Area of B-H Hysteresis Loop*

• **Material Selection Guidelines:**

Application	Desirable Properties	Best Material
Electromagnets / Transformer Cores	High μ_r, High Retentivity, Low Coercivity, Small Hysteresis Area (Low loss)	Soft Iron
Permanent Magnets	High Retentivity, High Coercivity, High Permeability	Steel / Alnico

16. Classification of Magnetic Materials

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Behavior in B-field	Feebly repelled from strong to weak field	Feebly attracted from weak to strong field	Strongly attracted into field lines
Susceptibility (χ)	Small & Negative ($-1 \le \chi < 0$)	Small & Positive ($0 < \chi < \varepsilon$)	Very Large Positive ($\chi \gg 10^3$)
Relative Permeability (μ_r)	$0 \le \mu_r < 1$	$\mu_r > 1$ (slightly)	$\mu_r \gg 1000$
Temperature Effect	Independent of Temperature	**Curie's Law:** $\chi \propto \frac{1}{T} \Rightarrow \chi = \frac{C}{T}$	**Curie-Weiss Law ($T > T_c$):** $\chi = \frac{C}{T - T_c}$
Examples	$Bi, Cu, H_2O, NaCl, N_2$	$Al, Na, O_2, CaCl_2$	Fe, Co, Ni, Gd, Fe_3O_4

→ T_c = Curie Temperature (Above T_c, Ferromagnetic becomes Paramagnetic).

17. Earth's Magnetism (Geomagnetism)

• **Elements of Earth's Magnetic Field ($\mathbf{B}_e$):**

1. **Declination (θ):** Angle between Geographic Meridian and Magnetic Meridian.
2. **Angle of Dip / Inclination (δ):** Angle made by total field $\mathbf{B}_e$ with Horizontal.

• **Components Resolution:**

Horizontal Component: $B_H = B_e \cos\delta$
Vertical Component: $B_V = B_e \sin\delta$
Net Earth Field: $B_e = \sqrt{(B_H^2 + B_V^2)}$
Dip Relation: $\tan\delta = \frac{B_V}{B_H}$

• **Special Geographical Locations:**
• **At Magnetic Equator:** $\delta = 0^\circ \Rightarrow B_V = 0,\ B_H = B_e$
• **At Magnetic Poles:** $\delta = 90^\circ \Rightarrow B_H = 0,\ B_V = B_e$

• **Apparent Dip (δ') in Plane at Angle α to Magnetic Meridian** ADDED:

$$\tan\delta' = \frac{\tan\delta}{\cos\alpha}$$

• **Apparent Dips in Two Mutual Perpendicular Planes** ADDED: If δ_1' and δ_2' are apparent dips in two perpendicular planes:

$$\cot^2\delta = \cot^2\delta_1' + \cot^2\delta_2'$$

ALTERNATING CURRENT

JEE Main & Advanced Complete Formula & Concept Revision Sheet

PHYSICS

1. FUNDAMENTAL AC RELATIONS & KEY VALUES

Instantaneous Equations
Current: $I = I_0 \sin(\omega t)$ or $I = I_0 \cos(\omega t)$
Voltage: $E = E_0 \sin(\omega t)$ or $E = E_0 \cos(\omega t)$

Time Period & Frequency
Time Period: $T = 2\pi / \omega$
Frequency: $f = \omega / 2\pi$ (India Mains: $f = 50\ Hz$)

Parameter	Mathematical Formula	Key JEE Notes & Conditions
Mean / Average Value (Half Cycle)	$I_{mean} = I_{avg} = 2I_0 / \pi \approx 0.637\, I_0$	Valid for positive half cycle $(0 \le t \le T/2)$. **Full Cycle Average = 0** (meaningless over complete period).
RMS / Virtual Value (Full Cycle)	$I_{rms} = I_v = I_0 / \sqrt{2} \approx 0.707\, I_0$ $E_{rms} = E_v = E_0 / \sqrt{2} \approx 0.707\, E_0$	DC current producing same heat in same resistor & time. Measured by **Hot-Wire Instruments**. Standard AC meters read RMS.
Form Factor	$\text{Form Factor} = \text{Virtual Value} / \text{Mean Value} = \pi / (2\sqrt{2}) \approx 1.11$	Ratio of RMS value to Mean value (half cycle). Constant for sinusoidal waves.
Meter Response	Galvanometer @ 50 Hz → 0 Galvanometer @ 2 Hz → Oscillates	DC meters read zero average on AC. Hot-wire instruments work on $H \propto I^2_{rms}$ (independent of direction).

2. AC THROUGH PURE COMPONENTS

Circuit Element	Reactance / Opposition	Current & Phase Relationship	Phasor Diagram & Remarks
Pure Resistor (R)	R (Independent of ω)	$I = (E_0/R) \sin(\omega t)$ Phase diff $\varphi = 0$	Voltage and current are **in phase**. Resists current, does not oppose change.
Pure Inductor (L)	Inductive Reactance: $X_L = \omega L = 2\pi f L$	$I = (E_0/X_L) \sin(\omega t - \pi/2)$ Phase diff $\varphi = -\pi/2$	Current **lags voltage by 90°** ($\pi/2$). Blocks high freq AC; passes DC ($X_L = 0$ for DC).
Pure Capacitor (C)	Capacitive Reactance: $X_C = 1 / (\omega C) = 1 / (2\pi f C)$	$I = (E_0/X_C) \sin(\omega t + \pi/2)$ Phase diff $\varphi = +\pi/2$	Current **leads voltage by 90°** ($\pi/2$). Blocks DC ($X_C = \infty$ for DC); passes high freq AC.

3. SERIES COMBINATION AC CIRCUITS

R-L Series Circuit
Impedance: $Z = \sqrt{(R^2 + X_L^2)} = \sqrt{(R^2 + \omega^2 L^2)}$
Phase Angle: $\tan\varphi = X_L / R = \omega L / R$ (Voltage leads)
Total RMS Voltage: $E_v = \sqrt{(V_R^2 + V_L^2)}$

R-C Series Circuit
Impedance: $Z = \sqrt{(R^2 + X_C^2)} = \sqrt{(R^2 + 1/(\omega C)^2)}$
Phase Angle: $\tan\varphi = X_C / R = 1 / (\omega C R)$ (Current leads)
Total RMS Voltage: $E_v = \sqrt{(V_R^2 + V_C^2)}$

Series L-C-R Circuit (General Case)
Net Reactance: $X = X_L - X_C = \omega L - 1/(\omega C)$
Phase Angle: $\tan\varphi = (X_L - X_C) / R = (V_L - V_C) / V_R$
Impedance: $Z = \sqrt{[R^2 + (X_L - X_C)^2]} = \sqrt{[R^2 + (\omega L - 1/\omega C)^2]}$
Vector Voltage: $E_v = \sqrt{[V_R^2 + (V_L - V_C)^2]}$

*Sign Conventions for Phase (φ): **Positive φ** if $X_L > X_C$ (Inductive); **Negative φ** if $X_L < X_C$ (Capacitive); **Zero φ** if $X_L = X_C$ (Resonant).*

4. POWER IN AC CIRCUITS & CHOKE COIL

Power Calculations

- Instantaneous Power: $P(t) = E(t) \cdot I(t)$
- Average Power: $\langle P \rangle = E_{rms} I_{rms} \cos\varphi = (E_0 I_0 / 2) \cos\varphi$
- Power Factor: $\cos\varphi = R / Z = V_R / E_v$
- Apparent Power: $P_{app} = E_{rms} \cdot I_{rms}$ (in VA)
- Real Power Dissipation: $P = I^2_{rms} R$

Power Components & Wattless Current

- Real Power Component: $I_{rms} \cos\varphi$ (In phase with V, consumes power).
- Wattless / Idle Current Component: $I_{rms} \sin\varphi$ (Out of phase by $\pi/2$, zero power loss).
- Virtual Power Component: $\frac{1}{2} E_0 I_0 \cos(2\omega t + \varphi)$ (Frequency 2ω, average over cycle = 0).

Choke Coil (Current Control in AC): High inductance ($L \uparrow$) and negligible resistance ($R \approx 0$). Power factor $\cos\varphi = R/Z \approx 0$. Controls AC current with almost zero energy dissipation ($\langle P \rangle \approx 0$). Preferred over resistance in fluorescent tubes.

5. SERIES RESONANCE, QUALITY FACTOR & BANDWIDTH

Resonance Condition

Occurs when $X_L = X_C \Rightarrow \omega_0 L = 1 / (\omega_0 C)$

Resonant Frequency: $\omega_0 = 1 / \sqrt{(LC)}$ rad/s

Cyclic Frequency: $f_0 = 1 / (2\pi\sqrt{(LC)})$ Hz

State at Resonance

- Impedance: $Z_{min} = R$ (Purely resistive)
- Current: $I_{max} = E_v / R$ (Maximum peak)
- Phase Difference: $\varphi = 0 \Rightarrow \cos\varphi = 1$
- Voltages: $V_L = V_C$, Net $V_{LC} = 0$, $E_v = V_R$

Parameter	Formula	Physical Meaning / JEE Tips
Quality Factor (Q)	$Q = (\omega_0 L) / R = 1 / (\omega_0 CR) = (1/R)\sqrt{(L/C)}$ $Q = V_L / E_v = V_C / E_v$	Magnification factor for voltage across L or C at resonance. High Q ⇒ sharper resonance curve.
Bandwidth (Δω)	$\Delta\omega = \omega_2 - \omega_1 = R / L$	Frequency interval where power drops to half of max value ($P = P_{max}/2$, $I = I_0/\sqrt{2}$).
Sharpness	$Sharpness = \omega_0 / \Delta\omega = \omega_0 L / R = Q$	Selectivity of tuning circuit. Higher Q means narrower bandwidth & higher selectivity.
Half-Power Frequencies Added	$\omega_{1,2} = \omega_0 \pm \Delta\omega/2 = \omega_0 \pm R/(2L)$ $\omega_1 \omega_2 = \omega_0^2$	Frequencies at which circuit power is half of resonant power. ω_0 is geometric mean of ω_1 and ω_2.
Parallel Resonance Added	$f_r = (1/2\pi)\sqrt{[1/(LC) - R^2/L^2]}$ If $R \approx 0 \Rightarrow \omega_0 = 1/\sqrt{(LC)}$	Rejector Circuit. At resonance, impedance is maximum $Z_{max} = L/(CR)$ and current is minimum.

6. TRANSFORMERS, LC OSCILLATIONS & STORED ENERGY

Transformer Relations

Transformation Ratio: $k = N_s / N_p = E_s / E_p$

Efficiency: $\eta = (Output\ Power / Input\ Power) = (E_s I_s) / (E_p I_p)$

Ideal Transformer ($\eta = 100\%$):

$$E_s / E_p = N_s / N_p = I_p / I_s = k$$

- **Step-Up:** $k > 1 \Rightarrow N_s > N_p, E_s > E_p, I_s < I_p$
- **Step-Down:** $k < 1 \Rightarrow N_s < N_p, E_s < E_p, I_s > I_p$

Energy Storage & LC Oscillations

- Energy in Inductor: $U_L = \frac{1}{2} L I^2$ (Max: $\frac{1}{2} L I_0^2$)
- Energy in Capacitor: $U_C = \frac{1}{2} C V^2 = q^2 / (2C)$
- **LC Oscillations:** Added
  Charge: $q(t) = q_0 \cos(\omega t)$
  Current: $I(t) = -dq/dt = I_0 \sin(\omega t)$ where $I_0 = \omega q_0$
  Natural Frequency: $\omega = 1/\sqrt{(LC)}$
  Total Energy Conserved: $U_{total} = q^2/(2C) + \frac{1}{2}LI^2 = q_0^2/(2C)$

Important JEE Advanced Results:

JEE ADVANCED & MAIN PHYSICS SERIES

Mechanical Properties of Solids

Chapter 9 & 10

1. Stress & Strain Fundamentals

Stress (σ): Internal restoring force set up per unit area of cross-section.

$$Stress = \frac{F_{restoring}}{A}$$

Types of Stress: **Longitudinal** (Tensile/Compressive), **Volume** (Hydraulic), and **Tangential** (Shear).

Strain (ε): The ratio of change in dimension to the original dimension.

- **Longitudinal:** $\varepsilon_l = \Delta l / l$
- **Volumetric:** $\varepsilon_v = \Delta V / V$
- **Shear:** $\tan\varphi = x / L \approx \varphi$ (for small angles)

Hooke's Law: Within the elastic limit, stress is directly proportional to strain.

$$Stress = E \times Strain$$

Where E is the Modulus of Elasticity.

2. Thermal Stress

When a rod rigidly fixed at its ends is heated, thermal stress develops.

- **Thermal Strain:** $\alpha\,\Delta\theta$
- **Thermal Stress:** $Y\alpha\,\Delta\theta$
- **Thermal Tension:** $F = YA\,\alpha\,\Delta\theta$

Elastic Moduli & Energy

Young's Modulus (Y)

Ratio of longitudinal stress to longitudinal strain.

$$Y = \frac{F \cdot l}{A \cdot \Delta l} = \frac{MgL}{\pi r^2 \Delta l}$$

Elongation due to own weight:

$$\Delta l = \frac{MgL}{2AY} = \frac{\rho g L^2}{2Y}$$

Bulk (k) & Rigidity (η) Modulus

Bulk Modulus (k): Volume stress / Volume strain.

$$k = -\frac{\Delta P}{\Delta V / V}$$

Compressibility (C): $C = 1 / k$

Modulus of Rigidity (η): Shear stress / Shear strain.

$$\eta = \frac{F_{tangential} / A}{\varphi}$$

Elastic Potential Energy

Work done in stretching a wire is stored as elastic potential energy (U).

$$U = \frac{1}{2} F \Delta l = \frac{1}{2} \times stress \times strain \times V$$

Energy Density (u): Energy stored per unit volume.

$$u = \frac{1}{2} \times stress \times strain = \frac{1}{2} Y$$

Relations Between Moduli

Poisson's Ratio (σ): Lateral strain / Longitudinal strain = $(\Delta D/D) / (\Delta l/l)$

- $Y = 3k(1 - 2\sigma)$
- $Y = 2\eta(1 + \sigma)$
- $\sigma = \frac{3k - 2\eta}{6k + 2\eta}$
- $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{k}$

JEE ADVANCED & MAIN PHYSICS SERIES

Fluids: Hydrostatics & Buoyancy

1. Density & Fluid Pressure

Density (ρ): Mass per unit volume (M/V).

- Variation with Temp: $\rho = \rho_0(1 - \gamma\Delta T)$
- Variation with Pressure: $\rho = \rho_0(1 + \Delta P/B)$
- Relative Density: $\rho_{substance} / \rho_{water\ at\ 4^\circ C}$

Pressure (P): Normal force per unit area. $P = F / A$

1 atm = 1.013×10^5 Pa = 76 cm of Hg = 760 torr.

2. Pressure in a Liquid Column

Pressure at a depth h in a liquid of uniform density:

$$P = P_0 + \rho g h$$

Where P_0 is atmospheric pressure. Gauge pressure is $P_g = P - P_0 = \rho g h$.

Atmospheric Variation: Decreases exponentially with height h: $P = P_0 e^{-ha}$ where $a = \rho_0 g / P_0$.

Accelerating Fluids & Force on Containers

Vertical Acceleration

If a container accelerates upwards with acceleration a_0, the effective gravity increases.

$$P = h\rho(g + a_0)$$

If accelerating downwards: $P = h\rho(g - a_0)$

Horizontal Acceleration

If a container accelerates horizontally with a, the free surface inclines at angle θ:

$$\tan\theta = \frac{a}{g}$$

Force on Vertical Walls: The force exerted by a liquid of height H on a vertical wall of width b due to gauge pressure is: $F = \frac{1}{2}\rho g b H^2$. This total force acts at a depth of $\frac{2H}{3}$ from the free surface.

Buoyancy & Archimedes' Principle

When a body is partially or wholly immersed in a fluid, it experiences an upward buoyant force equal to the weight of the fluid displaced.

$$B = V_{displaced} \times \rho_{liquid} \times g$$

Concept	Formula / Condition
Apparent Weight (W')	$W' = W_{true} - B = V(\rho_{body} - \rho_{liquid})g$ (if fully submerged)
Body Sinks	$\rho_{body} > \rho_{liquid}$ (Weight > Buoyancy)
Body Floats (Fully Submerged)	$\rho_{body} = \rho_{liquid}$ (Weight = Buoyancy anywhere in fluid)
Body Floats (Partially Submerged)	$\rho_{body} < \rho_{liquid}$. Displaced volume $V_{in} = V_{total}(\rho_{body} / \rho_{liquid})$

Fluid Dynamics

Equation of Continuity

Based on the conservation of mass for an incompressible fluid flow:

$$A_1v_1 = A_2v_2 = \text{constant}$$

Bernoulli's Theorem

Conservation of energy per unit volume for streamline flow:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Velocity of Efflux (Torricelli's Law): For a hole at depth h below the free surface: $v = \sqrt{(2gh)}$.
Time to reach ground: $t = \sqrt{[2(H-h)/g]}$.
Horizontal range: $R = 2\sqrt{[h(H-h)]}$. Range is maximum when $h = H/2$.

Viscosity

Newton's Law of Viscosity

Internal viscous drag between fluid layers:

$$F = -\eta A \frac{dv}{dx}$$

Where η is the coefficient of viscosity.

Stokes' Law & Terminal Velocity

Viscous force on a sphere: $F = 6\pi\eta rv$

Constant velocity achieved when falling freely:

$$V_T = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

Poiseuille's Formula: Rate of flow $Q = (dV/dt) = \pi Pr^4 / (8\eta l)$.

Reynolds Number: $R_e = \rho vd / \eta$. (If $R_e < 1000$ flow is laminar; > 2000 flow is turbulent).

Surface Tension

Parameter	Formula	Description
Surface Tension (T)	$T = F / l$	Force per unit length on imaginary line.
Surface Energy (U)	$U = T \times A$	$A = 4\pi R^2$ (drop), $A = 8\pi R^2$ (soap bubble).
Work Done (Splitting)	$W = 4\pi R^2 T (n^{1/3} - 1)$	Splitting one big drop of radius R into n drops.
Excess Pressure (P_{ex})	$P_{ex} = 2T / R$ (Drop) $P_{ex} = 4T / R$ (Bubble)	Pressure difference across curved liquid surface.
Capillary Rise (h)	$h = \frac{2T\cos\theta}{r\rho g}$	Ascent/descent of liquid in a capillary tube.

ELECTROMAGNETIC INDUCTION (EMI)

Comprehensive Revision & Formula Sheet • JEE Main & Advanced

Physics Revision Series
Key Formulas & Concepts

1. Magnetic Flux (Φ_B)

Fundamentals

Uniform Magnetic Field:

$$\Phi = B \cdot A = BA\cos\theta$$

- θ = Angle between B and Area vector A (normal to surface).
- SI Unit: **Weber (Wb)** or $T \cdot m^2$. Dimension: $[M L^2 T^{-2} A^{-1}]$.

Non-Uniform Magnetic Field:

$$\Phi = \int B \cdot dA$$

Sign & Special Values:

- $0^\circ \le \theta < 90^\circ \Rightarrow \Phi > 0$ (Positive flux)
- $\theta = 90^\circ$ (Field parallel to surface plane) $\Rightarrow \Phi = 0$
- $90^\circ < \theta \le 180^\circ \Rightarrow \Phi < 0$ (Negative flux)

Gauss's Law for Magnetism:

$$\Phi_{closed} = \oint B \cdot dA = 0$$

→ Net magnetic flux through any closed surface is **always zero** (Monopoles do not exist).

2. Laws of EMI & Charge Flow

Core Principles

Faraday's & Lenz's Law:

$$e = -N\,(d\Phi / dt)$$

- N = Number of turns in coil.
- Negative sign represents **Lenz's Law** (Induced EMF opposes the change in magnetic flux causing it; based on *Conservation of Energy*).

Induced Current & Power:

$$I = |e| / R = (N/R)\,|d\Phi/dt|$$

- EMF is independent of resistance R, but current depends on R.
- Thermal Power Dissipating: $P = I^2R = e^2/R$.

Induced Charge Flow (Crucial Result):

$$dq = I\,dt = (1/R)\,d\Phi \Rightarrow \Delta q = |\Delta\Phi| / R = |\Phi_1 - \Phi_2| / R$$

→ **Independent of the time interval or rate of flux change!**

3. Motional EMF (Translational & Rotational)

Kinetic Induction

Configuration	Induced EMF Formula	Conditions & Notes
Straight Rod in Motion	$e = (v \times B) \cdot L = BLv\sin\theta$	**v, B, L mutually ⊥** $\Rightarrow e = BLv$ θ = angle between v and B. Higher potential at end of $(v \times B)$.
Arbitrary / Curved Wire	$e = (v \times B) \cdot L_{eff}$	L_{eff} = Vector joining starting point to endpoint of conductor directly.
Rotating Rod about One End	$e = \frac{1}{2} B\omega l^2 = BfA$	**Axis ⊥ B field** ω = angular speed, f = frequency, $A = \pi l^2$ (area swept).
Rotating Disc (Faraday Disc) Added	$e = \frac{1}{2} B\omega R^2$	EMF induced between center axle and outer rim of radius R.
Rotating Coil in B-Field	$e(t) = NAB\omega\sin(\omega t) = e_0\sin(\omega t)$	Peak EMF: $e_0 = NAB\omega$. Principle of AC Generator.

Mechanical Force & Power on Moving Rod:

- Magnetic Force: $F_m = ILB = (B^2L^2v)/R$
- Power required to move rod: $P_{mech} = F_m v = (B^2L^2v^2)/R = P_{thermal}$

Equivalent Battery Representation:

A moving conductor acts as a cell of EMF $e = BLv$ with internal resistance r equal to the rod's resistance. Polarity is given by right-hand rule of $(v \times B)$.

4. Induced Electric Field (Non-Conservative E-Field)

Added for Advanced

Time-Varying B-Field

Integral Form (Maxwell-Faraday Eq):

$$\oint E_{ind} \cdot dr = -d\Phi_B/dt = -A\,(dB/dt)$$

- Induced electric lines of force form **closed continuous loops**.
- Non-conservative nature: Potential cannot be defined ($\oint E \cdot dr \neq 0$).

Cylindrical Region of Changing B-Field (Radius R):

$$\text{Inside } (r \le R): E_{ind} = (r/2)\,(dB/dt)$$

$$\text{Outside } (r \ge R): E_{ind} = (R^2/2r)\,(dB/dt)$$

- Electric force on charge q: $F = qE_{ind}$.

5. Self & Mutual Inductance

Coefficients

A. Self Inductance (L):

$$\Phi_{total} = N\,\Phi = L\,I \Rightarrow e = -\,L\,(dI/dt)$$

- SI Unit: **Henry (H)**. $1\ H = 1\ Wb/A = 1\ V{\cdot}s/A$.
- Depends *only* on geometric shape/size and medium permeability (μ). Independent of current I.

Long Solenoid Self-Inductance:

$$L = \mu_0 \mu_r\, n^2\, A\, l = (\mu_0 \mu_r\, N^2\, A)/l$$

- $n = N/l$ = turns per unit length, A = cross-sectional area.

B. Mutual Inductance (M):

$$\Phi_2 = M\,I_1 \Rightarrow e_2 = -\,M\,(dI_1/dt)$$

- Reciprocity Theorem: $M_{12} = M_{21} = M$.
- Flux Linkage Identity: $I_1\,\Phi_2 = I_2\,\Phi_1$.

Two Coaxial Solenoids:

$$M = \mu_0 \mu_r\, n_1\, n_2\, A\, l = (\mu_0 \mu_r\, N_1\, N_2\, A)/l$$

- A = area of inner coil, l = length of solenoid.

Coupling Coefficient (K):

$$M = K\sqrt{(L_1\,L_2)}\ (0 \le K \le 1)$$

- $K = 1$ (Ideal maximum coupling, standard tight winding).
- $K = 0$ (No flux coupling between coils).

Combinations of Inductors:

- **Series (Far apart, M ≈ 0):** $L_{eq} = L_1 + L_2$
- **Series (With M):** $L_{eq} = L_1 + L_2 \pm 2M$ **(+ same sense, - opp sense)**
- **Parallel (Far apart, M ≈ 0):** $1/L_{eq} = 1/L_1 + 1/L_2 \Rightarrow L_{eq} = (L_1\,L_2)/(L_1 + L_2)$
- **Parallel (With M):** $L_{eq} = (L_1\,L_2 - M^2)/(L_1 + L_2 \mp 2M)$ Added

6. Growth & Decay of Current in L-R Circuits

Transients

A. Growth of Current (Switch Closed):

Circuit Eq: $E - L\,(dI/dt) = I\,R$

$$I(t) = (E/R)\,(1 - e^{-Rt/L}) = I_0\,(1 - e^{-t/\tau})$$

- Time Constant: $\tau = L/R$.
- At $t = 0$: Inductor acts as **Open Circuit** ($I = 0$, $(dI/dt)_{max} = E/L$).
- At $t \to \infty$: Inductor acts as **Short Circuit / Plain Wire** ($I = I_0 = E/R$).
- At $t = \tau$: $I = I_0\,(1 - e^{-1}) \approx 0.632\ I_0$ (63.2% growth).

B. Decay of Current (Switch Opened):

Initial Current = I_0

$$I(t) = I_0\,e^{-Rt/L} = I_0\,e^{-t/\tau}$$

- At $t = \tau$: $I = I_0\,e^{-1} \approx 0.368\ I_0$ (Falls to 36.8%).
- Rate of growth/decay is fast when $\tau = L/R$ is small.
- Voltage across Inductor during decay: $V_L = -\,I_0\,R\,e^{-t/\tau}$.

7. Magnetic Energy & Density

Energy

Energy Stored in Inductor:

$$U = \tfrac{1}{2}\,L\,I^2 = \tfrac{1}{2}\,I\,\Phi = \Phi^2/(2L)$$

Interaction Energy of 2 Coils:

$$U_{int} = M\,I_1\,I_2$$

Magnetic Energy Density (u_B): Added

$$u_B = B^2/(2\,\mu_0)$$ **(Energy per unit volume)**

8. Superconducting Loop & LC Oscillations

Special Cases

Superconducting Loop ($R = 0$):

- EMF $e = 0 \Rightarrow d\Phi_{total}/dt = 0$

$$\Phi_{total} = \Phi_{external} + L\,I = Constant$$

→ **Magnetic flux through a superconducting loop CANNOT change!**

LC Circuit Oscillations: Added

$$f = 1/(2\pi\sqrt{(LC)}),\ \omega_0 = 1/\sqrt{(LC)}$$

- Charge: $q(t) = Q_0\cos(\omega_0 t)$, Current: $i(t) = -I_0\sin(\omega_0 t)$
- Total Energy: $U = q^2/(2C) + \tfrac{1}{2}\,Li^2 = Q_0^2/(2C)$

9. Transformer

AC Devices

Core Transformer Equations:

- Principle: **Mutual Induction** (Works on AC only).

$$V_s/V_p = N_s/N_p$$

- **Ideal Transformer (100% Efficiency):**

$$P_{in} = P_{out} \Rightarrow V_p\,I_p = V_s\,I_s \Rightarrow V_s/V_p = N_s/N_p = I_p/I_s$$

Types & Efficiency:

- **Efficiency (η):** $\eta = (P_{out}/P_{in}) \times 100\% = (V_s\,I_s/V_p\,I_p) \times 100\%$
- **Step-Up Transformer:** $N_s > N_p \Rightarrow V_s > V_p$ and $I_s < I_p$
- **Step-Down Transformer:** $N_s < N_p \Rightarrow V_s < V_p$ and $I_s > I_p$
- **Energy Losses:** Copper loss (I^2R), Eddy current loss (laminated core reduces it), Hysteresis loss, Flux leakage.

PART II

CHEMISTRY

Physical Chemistry | Inorganic Chemistry | Organic Chemistry & Reagents

34 Chapters | Pages 94 to 230

THERMODYNAMICS & CHEMICAL ENERGETICS

JEE Main & Advanced Complete Formula Revision Sheet | Physical Chemistry

1. Basic Terms & Properties

- **System Types:** Open (exchanges mass + energy), Closed (exchanges energy only), Isolated (no exchange of mass/energy).
- **State Functions:** Path independent; depend only on initial/final state: U, H, S, G, P, T, V, n. Note: $\Delta(\text{State Function}) = 0$ for any cyclic process.
- **Path Functions:** Depend on process path: Heat (q), Work (w).
- **Extensive Properties:** Mass-dependent: Mass, Volume (V), U, H, S, G, Heat capacity (C), Area.
- **Intensive Properties:** Mass-independent: T, P, Density (ρ), Refractive index, Viscosity, Surface tension, Molar/Specific properties (V_m, C_m, s), Cell Potential (E_{cell}).

2. First Law of Thermodynamics (FLOT)

Energy of universe is conserved. Mathematical form:

$$\Delta U = q + w \quad | \quad \text{Differential form: } dU = dq + dw$$

IUPAC Sign Conventions (Chemistry):

- Heat added to system: $q > 0$ | Heat released by system: $q < 0$
- Work done *on* system (compression): $w > 0$
- Work done *by* system (expansion): $w < 0$
- Internal Energy increase: $\Delta U > 0$ | Decrease: $\Delta U < 0$

3. Pressure-Volume Work Calculations

General expansion/compression work against external pressure P_{ext}:

$$w = -\int_{V_1}^{V_2} P_{ext}\, dV$$

Irreversible Process (Constant P_{ext})

$$w_{irr} = -P_{ext}(V_2 - V_1) = -P_{ext}\Delta V$$

Free expansion: Against vacuum ($P_{ext} = 0$) $\Rightarrow w = 0$.

Reversible Isothermal Process (Ideal Gas)

$$w_{rev,\,iso} = -nRT\ln(V_2/V_1) = -2.303\, nRT \log_{10}(P_1/P_2)$$

Since $\Delta T = 0 \Rightarrow \Delta U = 0 \Rightarrow q_{rev} = -w_{rev}$

4. Heat Capacities & Molar Heat Capacities

- **Heat Capacity:** $C = dq/dT$ (Extensive) | **Specific Heat:** $s = Q/(m\,\Delta T)$ (Intensive)
- **Molar Heat Capacities:** $C_v = (dq_v/dT)_{molar} = (\partial U/\partial T)_v$; $C_p = (dq_p/dT)_{molar} = (\partial H/\partial T)_p$
- **Mayer's Relation:** $C_p - C_v = R$ | Adiabatic Index: $\gamma = C_p/C_v = 1 + (2/f)$
- **Internal Energy & Enthalpy for Ideal Gas (Any process):** $\Delta U = n C_v \Delta T$ | $\Delta H = n C_p \Delta T$

Gas Atomicity	Translational (f_{tr})	Rotational (f_{rot})	Vibrational (f_{vib})	C_v (Excl. Vib)	C_p (Excl. Vib)	γ (Excl. Vib)	C_v (Incl. Vib)
Monoatomic (He, Ne, Ar)	3	0	0	3/2 R	5/2 R	5/3 ≈ 1.67	3/2 R
Diatomic (H_2, O_2, N_2)	3	2	1	5/2 R	7/2 R	7/5 = 1.40	7/2 R
Linear Polyatomic (CO_2)	3	2	3N − 5	5/2 R	7/2 R	1.40	(5/2 + f_{vib}) R
Non-Linear Polyatomic (H_2O, CH_4)	3	3	3N − 6	3 R	4 R	4/3 ≈ 1.33	(3 + f_{vib}) R

5. Reversible & Irreversible Adiabatic Processes

For any adiabatic process: $q = o \Rightarrow \Delta U = w = n\,C_v\,(T_2 - T_1) = (P_2V_2 - P_1V_1)\,/\,(\gamma - 1)$

Reversible Adiabatic Governing Equations

- $\mathbf{P\,V^{\gamma} = constant}$
- $\mathbf{T\,V^{\gamma - 1} = constant}$
- $\mathbf{T^{\gamma}\,P^{1-\gamma} = constant}$ or $T\,P^{(1-\gamma)/\gamma} = c$

Irreversible Adiabatic Work

Equate $\Delta U = w_{irr}$:

$$\mathbf{n\,C_v\,(T_2 - T_1) = -\,P_{ext}\,(V_2 - V_1) = -\,P_{ext}\,(nRT_2/P_2 - nRT_1/P_1)}$$

6. Polytropic Process (P V n = constant) [Added]

Work done: $\mathbf{w_{poly} = (P_2V_2 - P_1V_1)\,/\,(n-1) = nR(T_2 - T_1)\,/\,(n-1)}$

Molar Heat Capacity: $\mathbf{C_{poly} = C_v + R\,/\,(1-n) = C_m}$

7. Enthalpy (H) & Phase Transformations

Definition: Heat content at constant pressure $\mathbf{H = U + PV \;\Rightarrow\; \Delta H = \Delta U + \Delta(PV)}$

- At constant Pressure ($P_{ext} = P_{sys}$): $\mathbf{\Delta H = q_p}$ | At constant Volume: $\mathbf{\Delta U = q_v}$
- For ideal gas reactions at constant T: $\mathbf{\Delta H = \Delta U + \Delta n_g\,R\,T}$ (where $\Delta n_g = \sum n_{g,\,products} - \sum n_{g,\,reactants}$)
- If gaseous moles change with temperature shift: $\Delta H = \Delta U + n_g\,R\,\Delta T$
- **Phase Enthalpies:** $\Delta_{sub}H^{\circ} = \Delta_{fus}H^{\circ} + \Delta_{vap}H^{\circ}$ (all measured at phase transition temperatures).

8. Thermochemistry & Enthalpy of Reactions

Standard State: Pure substance at 1 bar pressure and specified temperature (usually 298.15 K). Element standard state $\Delta_fH^{\circ} = o$ (e.g., O_2(g), C(graphite), S_8(rhombic), Br_2(l), I_2(s)).

Enthalpy of Reaction (Δ_rH°)

$$\mathbf{\Delta_rH^{\circ} = \sum \Delta_fH^{\circ}(products) - \sum \Delta_fH^{\circ}(reactants)}$$

$$\mathbf{\Delta_rH^{\circ} = \sum \Delta_cH^{\circ}(reactants) - \sum \Delta_cH^{\circ}(products)}$$

Bond Enthalpies (Gaseous Species)

$$\mathbf{\Delta_rH^{\circ} = \sum B.E.(reactants) - \sum B.E.(products)}$$

Resonance Energy = $\Delta_fH^{\circ}(exp) - \Delta_fH^{\circ}(calc) = \Delta_cH^{\circ}(calc) - \Delta_cH^{\circ}(exp)$.

- **Hess's Law of Constant Heat Summation:** Enthalpy change is identical whether a reaction occurs in one step or multiple steps: $\Delta H_{total} = \Delta H_1 + \Delta H_2 + \Delta H_3 + \ldots$
- **Enthalpy of Neutralization ($\Delta_{neu}H^{\circ}$):** Strong Acid + Strong Base $\Rightarrow H^+(aq) + OH^-(aq) \rightarrow H_2O(l); \Delta H^{\circ} = -57.1\ kJ/eq = -13.7\ kcal/eq$. If weak acid/base is involved: $|\Delta_{neu}H^{\circ}| < 57.1\ kJ$ due to ionization energy consumed ($\Delta_{neu}H^{\circ} = -57.1 + \Delta_{ion}H^{\circ}$). *Exception:* HF neutralisation $|\Delta_{neu}H^{\circ}| > 57.1\ kJ$ due to high hydration energy of F^-.

9. Kirchhoff's Law (Temperature Variation of ΔH and ΔU)

$$\mathbf{\Delta H_{T_2} - \Delta H_{T_1} = \int_{T_1}^{T_2} \Delta C_p\,dT = \Delta C_p\,(T_2 - T_1)} \text{ (if } \Delta C_p \text{ is temperature independent)}$$

$$\mathbf{\Delta U_{T_2} - \Delta U_{T_1} = \int_{T_1}^{T_2} \Delta C_v\,dT = \Delta C_v\,(T_2 - T_1)}$$

Where $\Delta C_p = \sum (C_p)_{products} - \sum (C_p)_{reactants}$

10. Second Law of Thermodynamics (SLOT) & Entropy (S)

Entropy (S) measures microscopic randomness/disorder. State function and extensive property.

$$\mathbf{dS = dq_{rev} / T \Rightarrow \Delta S_{sys} = \int_1^2 (dq_{rev} / T)} \quad | \quad \text{Units: } J\,K^{-1}\,mol^{-1}$$

SLOT Criteria for Spontaneity:

- $\mathbf{\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings} > 0} \Rightarrow$ Spontaneous (Irreversible)
- $\mathbf{\Delta S_{total} = 0} \Rightarrow$ Reversible process at equilibrium
- $\mathbf{\Delta S_{total} < 0} \Rightarrow$ Non-spontaneous process

Surroundings Entropy Change: $\Delta S_{surr} = -\,q_{actual} / T_{surr}$ (always treated as reversible heat transfer for surroundings).

11. Entropy Change (ΔS) Formulas for Ideal Gases

General Formula for n moles of ideal gas undergoing state change $(T_1, V_1, P_1) \rightarrow (T_2, V_2, P_2)$:

$$\mathbf{\Delta S_{sys} = n\,C_v \ln(T_2 / T_1) + n\,R \ln(V_2 / V_1)}$$

$$\mathbf{\Delta S_{sys} = n\,C_p \ln(T_2 / T_1) + n\,R \ln(P_1 / P_2)}$$

Specific Process Changes

- **Isothermal:** $\Delta S = n\,R\,ln(V_2 / V_1) = n\,R\,ln(P_1 / P_2)$
- **Isochoric:** $\Delta S = n\,C_v\,ln(T_2 / T_1) = n\,C_v\,ln(P_2 / P_1)$
- **Isobaric:** $\Delta S = n\,C_p\,ln(T_2 / T_1) = n\,C_p\,ln(V_2 / V_1)$

Phase Transitions & Heating

- **Phase Change (Const T, P):** $\mathbf{\Delta S_{trans} = \Delta_{trans}H / T_{trans}}$
- Heating substance $(T_1 \rightarrow T_2)$: $\Delta S = n\,C_m\,ln(T_2 / T_1)$
- **Parametric T Variation:** $\Delta S = \int (n\,C_p / T)\,dT$

12. Gibbs Free Energy (G) & Spontaneity Criteria

Defined at constant T and P to assess spontaneity using system parameters alone: $\mathbf{G = H - TS}$

$$\mathbf{\Delta G = \Delta H - T\,\Delta S} \quad \text{(at constant T, P)}$$

Criteria for Spontaneity (Const T, P)

- $\mathbf{\Delta G < 0} \Rightarrow$ Spontaneous (Irreversible)
- $\mathbf{\Delta G = 0} \Rightarrow$ System at Equilibrium
- $\mathbf{\Delta G > 0} \Rightarrow$ Non-spontaneous (Reverse is spontaneous)
- **Non-PV Work:** $-\Delta G = w_{useful,\,max} = n\,F\,E_{cell}$

Equilibrium Constant & ΔG°

$$\mathbf{\Delta G = \Delta G^\circ + R\,T \ln Q}$$

At equilibrium ($\Delta G = 0$, $Q = K_{eq}$):

$$\mathbf{\Delta G^\circ = -\,R\,T \ln K_{eq} = -\,2.303\,R\,T \log_{10} K_{eq}}$$

13. Temperature Dependence of Spontaneity (ΔG = ΔH − T ΔS)

ΔH	ΔS	$\Delta G = \Delta H - T\,\Delta S$	Spontaneity Condition
− (Exothermic)	**+ (Disorder ↑)**	Always Negative (< 0)	**Spontaneous at all temperatures**
+ (Endothermic)	− (Order ↑)	Always Positive (> 0)	**Non-spontaneous at all temperatures**
− (Exothermic)	− (Order ↑)	Negative at low T, Positive at high T	**Spontaneous at low T** ($T < T_{eq} = \Delta H / \Delta S$)
+ (Endothermic)	+ (Disorder ↑)	Positive at low T, Negative at high T	**Spontaneous at high T** ($T > T_{eq} = \Delta H / \Delta S$)

14. Third Law, Clausius-Clapeyron & Van 't Hoff Equations

- **Third Law of Thermodynamics:** The entropy of a perfectly crystalline substance approaches zero as absolute temperature approaches zero ($S_{0\,K} = 0$). Allows absolute entropy determination: $S_T = \int_0^T (C_p / T)\,dT$.

Clausius-Clapeyron Equation

For liquid-vapor / solid-vapor phase equilibria:

$$\log_{10}(P_2 / P_1) = (\Delta_{vap}H / 2.303\ R)\ [\ (1 / T_1) - (1 / T_2)\ ]$$

Van 't Hoff Equation [Added]

Variation of equilibrium constant with T:

$$\log_{10}(K_2 / K_1) = (\Delta rH^\circ / 2.303\ R)\ [\ (1 / T_1) - (1 / T_2)\ ]$$

15. Master Process Summary Table (Ideal Gas, 1 Mole)

Thermodynamic Process	Condition	Work Done (w)	Heat Exchange (q)	ΔU	ΔH	ΔS_{sys}
Reversible Isothermal	$dT = 0$	$-RT\ln(V_2/V_1)$	$+RT\ln(V_2/V_1)$	0	0	$R\ln(V_2/V_1)$
Irreversible Isothermal	$dT = 0$	$-P_{ext}(V_2 - V_1)$	$+P_{ext}(V_2 - V_1)$	0	0	$R\ln(V_2/V_1)$
Reversible Isochoric	$dV = 0$	0	$C_v(T_2 - T_1)$	$C_v\Delta T$	$C_p\Delta T$	$C_v\ln(T_2/T_1)$
Reversible Isobaric	$dP = 0$	$-P(V_2 - V_1)$	$C_p(T_2 - T_1)$	$C_v\Delta T$	$C_p\Delta T$	$C_p\ln(T_2/T_1)$
Reversible Adiabatic	$dq = 0$	$C_v(T_2 - T_1)$	0	$C_v\Delta T$	$C_p\Delta T$	0
Irreversible Adiabatic	$dq = 0$	$-P_{ext}(V_2 - V_1)$	0	$C_v\Delta T$	$C_p\Delta T$	> 0
Free Expansion	$P_{ext} = 0$	0	0 (if adiabatic)	0	0	$R\ln(V_2/V_1)$

16. JEE Advanced Shortcuts & Key Relations [Added]

Maxwell Relations (Thermodynamic Square)

- $(\partial T / \partial V)_S = -(\partial P / \partial S)_V$
- $(\partial T / \partial P)_S = (\partial V / \partial S)_P$
- $(\partial S / \partial V)_T = (\partial P / \partial T)_V$
- $(\partial S / \partial P)_T = -(\partial V / \partial T)_P$

Joule-Thomson Coefficient & Key Results

- **J-T Coefficient:** $\mu_{JT} = (\partial T / \partial P)_H = -(1 / C_p)\ [\ V - T(\partial V/\partial T)_p\ ]$
- For ideal gas: $\mu_{JT} = 0$ (no cooling/heating on throttling).
- **Inversion Temp:** $T_i = 2a / (R\ b)$ (van der Waals gas).
- Efficiency of Carnot Engine: $\eta = 1 - T_c / T_h = -w / q_h$

SURFACE CHEMISTRY

Comprehensive JEE Main & Advanced Formula & Quick Reference Sheet

CHEMISTRY REVISION TOOL

1. FUNDAMENTALS OF ADSORPTION & THERMODYNAMICS

Adsorption Definitions & Specific Area

- **Adsorption:** Surface phenomenon; higher conc. of adsorbate at surface than bulk.
- **Desorption:** Reverse of adsorption (removal of adsorbate).
- **Sorption:** Simultaneous adsorption + absorption (e.g., dyes in cotton).
- **Specific Surface Area (A_{sp}):** Total surface area per unit mass of adsorbent:

$$\mathbf{A_{sp} = (Total\ Surface\ Area) / m \quad [m^2\ g^{-1}]}$$

- **Specific Surface Area Formula** [Added]:

$$\mathbf{A_{sp} = (N_A \cdot n \cdot a_m) / m}$$

where n = moles adsorbed at monolayer coverage, a_m = cross-sectional area of one adsorbate molecule.

Thermodynamics of Adsorption

$$\mathbf{\Delta G = \Delta H - T\Delta S \quad (\Delta G < 0\ for\ Spontaneous\ Process)}$$

- **ΔS < 0:** Randomness of gas molecules decreases upon adsorption.
- **ΔH < 0:** Process is **ALWAYS Exothermic** (ΔH must be negative so ΔG < 0).
- **Equilibrium Condition:** As adsorption proceeds, ΔH becomes less negative until $|\Delta H| = T|\Delta S|$ (ΔG = 0).
- **Extent of Adsorption (x/m):** x = mass of adsorbate, m = mass of solid adsorbent.

2. COMPARISON: PHYSISORPTION VS. CHEMISORPTION

Property	Physical Adsorption (Physisorption)	Chemical Adsorption (Chemisorption)
Attraction Force	Weak Van der Waals forces	Strong Chemical Bonds (covalent/ionic)
Enthalpy (ΔH)	Low ($20 - 40\ kJ\ mol^{-1}$)	High ($80 - 240\ kJ\ mol^{-1}$)
Reversibility	Reversible ($Gas + Solid \rightleftarrows Gas/Solid + Heat$)	Irreversible
Specificity	Non-specific (occurs on all solid surfaces)	Highly specific (occurs only if bond can form)
Molecular Layers	**Multimolecular** layers at high pressure	**Unimolecular** (monolayer) only
Effect of Temp (T)	Decreases continuously with increase in T	First increases (Activation energy required), then decreases
Ease of Liquefaction	Easily liquefiable gases (high T_c: CO_2, NH_3, HCl) > Permanent gases (H_2, N_2, O_2)	No direct correlation with critical temperature T_c

3. ADSORPTION ISOTHERMS & EQUATIONS

A. Freundlich Adsorption Isotherm (Empirical)

$$\mathbf{x / m = k \cdot P^{1/n} \quad (where\ n > 1,\ at\ constant\ Temp)}$$

$$\mathbf{\log(x/m) = \log(k) + (1/n) \cdot \log(P)}$$

B. Langmuir Adsorption Isotherm (Theoretical)

$$\mathbf{x / m = (a \cdot P) / (1 + b \cdot P)}$$

$$\mathbf{P / (x/m) = (1/a) + (b/a) \cdot P}$$ Linear Form

- **Plot:** $log(x/m)$ vs $log\ P$ gives a straight line.
  - **Slope** = $1/n$ (Range: 0 to 1, typically 0.1 to 0.5)
  - **Y-Intercept** = $log\ k$
- **At Low Pressure:** $1/n = 1 \Rightarrow x/m = k \cdot P$ (Direct dependence).
- **At High Pressure:** $1/n = 0 \Rightarrow x/m = k$ (Independent of P; Saturation level $(x/m)_{max}$).
- **At Intermediate Pressure:** $x/m \propto P^{1/n}$.
- **For Solution Phase:** Replace P with concentration C:

$$\mathbf{x/m = k \cdot C^{1/n} \Rightarrow \log(x/m) = \log k + (1/n) \log C}$$

- **Limitation:** Fails at high pressure (does not account for saturation limits theoretical deviation).

- **Assumptions:** Dynamic equilibrium between adsorption and desorption; uniform surface sites; unimolecular layer; no interaction between adjacent adsorbed molecules.
- **Constants:** a and b are Langmuir parameters ($a = k_a/k_d$).
- **Case I (Very Low Pressure, bP << 1):**

$$\mathbf{x/m = a \cdot P \quad (First\ order\ w.r.t\ P)}$$

- **Case II (Very High Pressure, bP >> 1):**

$$\mathbf{x/m = a / b = Constant \quad (Zero\ order\ w.r.t\ P)}$$

- **Fractional Surface Coverage (θ)** [Added]:

$$\mathbf{\theta = (b \cdot P) / (1 + b \cdot P) \ \Rightarrow\ x/m = (x/m)_{max} \cdot \theta}$$

4. ADSORPTION ISOBARS & ADSORPTION ISOSTERES [ADDED]

Adsorption Isobars (x/m vs T at Constant P)

- **Physisorption Isobar:** Monotonically decreases as T increases (exothermic process equilibrium shifts backward).
- **Chemisorption Isobar:** First increases with T (supplies Activation Energy E_a), reaches a maximum, then decreases.

Adsorption Isosteres (P vs T at Constant x/m)

$$\mathbf{\ln P = -(\Delta H_{ads} / R) \cdot (1 / T) + C \ \Rightarrow\ \partial(\ln P) / \partial(1/T) = -\Delta H_{ads} / R}$$

- Plot of $ln\ P$ vs $1/T$ for a fixed x/m is a straight line with slope = $-\Delta H_{ads} / R$. Used to calculate differential heat of adsorption.

5. CATALYSIS: TYPES, THEORIES & KINETICS

Classification & Terminology of Catalysis

- **Homogeneous Catalysis:** Reactants, products & catalyst are in the *same phase*.
  - $2SO_2(g) + O_2(g) \xrightarrow{[NO(g)]} 2SO_3(g)$ (Lead Chamber)
  - Hydrolysis of ester by aqueous liquid acid $H^+(aq)$.
- **Heterogeneous Catalysis:** Reactants & catalyst in *different phases*.
  - $2SO_2(g) + O_2(g) \xrightarrow{[Pt(s)\ or\ V_2O_5(s)]} 2SO_3(g)$ (Contact Process)
  - $N_2(g) + 3H_2(g) \xrightarrow{[Fe(s) + Mo(promoter)]} 2NH_3(g)$ (Haber's)
  - $4NH_3(g) + 5O_2(g) \xrightarrow{[Pt\text{-}Rh\ gauze]} 4NO(g) + 6H_2O(g)$ (Ostwald's)
- **Promoter:** Enhances catalyst activity (e.g., Mo in Haber's process).
- **Poison (Inhibitor):** Decreases catalyst activity (e.g., CO or As_2O_3 for Pt).
- **Auto-catalysis:** One of the reaction products acts as catalyst.
  - $2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \rightarrow K_2SO_4 + 2MnSO_4 + 10CO_2 + 8H_2O$ (Mn^{2+} acts as autocatalyst).

Enzyme Catalysis & Michaelis-Menten Kinetics

- **Enzymes:** Complex nitrogenous organic compounds (globular proteins), highly specific, optimum pH (5–7), optimum Temp (298–310 K / 37°C).
- **Lock & Key Mechanism Steps:**

$$\mathbf{Step\ 1:\ E + S \rightleftarrows ES^{\ddagger}\ (Fast) \qquad Step\ 2:\ ES^{\ddagger} \rightarrow E + P\ (Slow,\ Rate\ Determining)}$$

- **Michaelis-Menten Rate Equation** JEE Advanced:

$$\mathbf{r_o = (V_{max} \cdot [S]) / (K_m + [S])}$$

where K_m = Michaelis constant (substrate concentration at which $r_o = V_{max}/2$).

- **Shape-Selective Catalysis:** Zeolites (e.g., **ZSM-5** converts alcohols directly into gasoline/hydrocarbons by shape-selective dehydration; honeycomb pore size: 260–740 pm).

6. COLLOIDAL STATE & DETAILED CLASSIFICATION

System Characterization by Particle Size

Property	True Solution	Colloidal Sol	Suspension
Particle Size	< 1 nm (< 10 Å)	**1 nm – 1000 nm** (10 – 10^4 Å)	> 1000 nm (> 10^4 Å)
Nature	Homogeneous	**Heterogeneous** (2 phases)	Heterogeneous
Filterability	Passes SPM & filter paper	Passes filter paper, **retained by SPM**	Retained by ordinary filter paper
Settling	Does not settle	Settles only on **ultracentrifugation**	Settles under gravity

Classification Based on Dispersed Phase Types

- **Multimolecular Colloids:** Aggregates of atoms/small molecules with diameter < 1 nm bound by van der Waals forces. (e.g., Gold sol Au_n, Sulphur sol S_8).
- **Macromolecular Colloids:** Individual molecules of large size in colloidal range. High stability. (e.g., Starch, Cellulose, Proteins, Polymers like Nylon, Polystyrene).
- **Associated Colloids (Micelles):** At low conc. behave as normal strong electrolytes; above **CMC** and **Kraft Temp (T_k)** aggregate to form colloids. (e.g., Soaps $C_{17}H_{35}COONa$, Synthetic detergents).

8 Types of Colloidal Systems (Physical State Combination)

Dispersed Phase	Dispersion Medium	Type of Colloid	Standard JEE Examples
Solid	Solid	Solid Sol	Colored glasses, Gemstones, Alloys
Solid	Liquid	**Sol**	Paints, Cell fluids, Starch sol, Gold sol, Muddy water
Solid	Gas	Aerosol	Smoke, Dust
Liquid	Solid	**Gel**	Cheese, Butter, Jellies, Curd, Boot polish
Liquid	Liquid	**Emulsion**	Milk, Hair cream, Cold cream, Vanishing cream
Liquid	Gas	Liquid Aerosol	Fog, Mist, Cloud, Insecticide sprays
Gas	Solid	Solid Foam	Pumice stone, Foam rubber
Gas	Liquid	Foam	Froth, Whipped cream, Soap lather

***Note: Gas in Gas is ALWAYS a homogeneous true solution, NEVER a colloid.**

7. MICELLE FORMATION & SOAP CLEANSING MECHANISM

Micelle Parameters & Critical Values

- **Kraft Temperature (T_k):** Minimum temperature above which micelle formation takes place.
- **Critical Micelle Concentration (CMC):** Minimum concentration above which micelle formation begins.
  • For Soaps: $CMC = 10^{-4}$ to 10^{-3} $mol\ L^{-1}$.
- **Aggregation Number:** Number of monomeric units in one micelle (typically **100 or more molecules**).
- **Sodium Stearate ($C_{17}H_{35}COONa$):**
  • Hydrocarbon Tail ($C_{17}H_{35}-$): Non-polar, Hydrophobic (Lyophobic).
  • Head Group ($-COO^-Na^+$): Polar, Hydrophilic (Lyophilic).

Lyophilic vs. Lyophobic Sols

Property	Lyophilic (Liquid-loving)	Lyophobic (Liquid-hating)
Preparation	Direct mixing with medium	Special methods required
Reversibility	**Reversible** (resols upon adding medium)	**Irreversible** (coagulated state stays)
Stability	Self-stable (hydration + charge)	Unstable (requires stabilizing agents)
Viscosity	Much higher than medium	Same as dispersion medium

8. PREPARATION & PURIFICATION OF SOLS

Preparation Methods (Lyophobic Sols)

- **A. Bredig's Arc Method (Electrical Dispersion):** Combines dispersion & condensation. Electric arc struck between submerged metal electrodes (Au, Pt, Ag, Cu) in cold bath (ice). Metal vaporizes and instantly condenses into colloidal size. Stabilized by adding trace KOH.
- **B. Peptization:** Process of converting a freshly prepared precipitate into colloidal sol by shaking with medium in presence of a small amount of electrolyte (Peptizing Agent).
  • $Fe(OH)_3\ (ppt) + FeCl_3\ (aq) \rightarrow Fe(OH)_3 \cdot Fe^{3+}$ (Red Positively Charged Sol).
- **C. Chemical Methods:**
  • *Double Decomposition:* $As_2O_3 + 3H_2S \rightarrow As_2S_3\ (yellow\ sol) + 3H_2O$
  • *Hydrolysis:* $FeCl_3 + 3H_2O\ (boiling) \rightarrow Fe(OH)_3\ (red\ sol) + 3HCl$
  • *Oxidation:* $2H_2S + SO_2 \rightarrow 3S\ (white\ sol) + 2H_2O$
  • *Reduction:* $2AuCl_3 + 3SnCl_2 \rightarrow 2Au\ (sol) + 3SnCl_4$ (or with HCHO / hydrazine).

Purification Methods

- **1. Dialysis:** Removal of crystalloids (ions) through SPM (parchment/cellophane). Slow process.
- **2. Electrodialysis:** Electric field applied across SPM to accelerate ionic removal.
- **3. Ultrafiltration:** Filtration using special graded ultrafilter paper impregnated with **Collodion solution** (4% nitrocellulose solution in a 1:1 mixture of alcohol and ether) and hardened with formaldehyde. Pore size reduced to < 1 nm.
- **4. Ultracentrifugation:** High-speed rotation forces colloidal particles to settle down at the bottom of tube; supernatant liquid is replaced with fresh medium.

9. PROPERTIES OF COLLOIDAL SYSTEMS (ELECTROKINETIC & OPTICAL)

Colligative & Optical Properties

- **Colligative Properties:** Colloidal particles are large aggregate clusters ⇒ total number of particles per unit volume is very small ⇒ values of $\pi, \Delta T_b, \Delta T_f, \Delta P$ are extremely small compared to true solutions of same mass concentration.

  > **Osmotic Pressure (π) is the ONLY measurable colligative property for molar mass determination of colloids!**

- **Tyndall Effect:** Scattering of light by colloidal particles causing illumination of light path.
  Conditions:
  1. Particle diameter is not much smaller than wavelength of light ($d \approx \lambda$).
  2. Large difference between Refractive Indices of dispersed phase and dispersion medium ($RI_{DP} \neq RI_{DM}$).
  **Note: Lyophilic sols show weak Tyndall effect due to small RI difference!*
- **Brownian Motion:** Continuous zig-zag motion of colloidal particles caused by unbalanced bombardment of dispersion medium molecules. Independent of particle nature, depends on particle size and viscosity. Provides stability by counteracting gravity.

Electrical Properties & Zeta Potential

- **Preferential Adsorption Theory:** Sol particles adsorb the ion common to their lattice structure:
  • $AgNO_3\ (aq,\ excess) + KI\ (aq) \rightarrow AgI \cdot Ag^+$ (Positive Sol)
  • $KI\ (aq,\ excess) + AgNO_3\ (aq) \rightarrow AgI \cdot I^-$ (Negative Sol)
- **Helmholtz Electrical Double Layer:**
  • *Fixed Layer:* Preferentially adsorbed ions attached to particle.
  • *Diffused Layer:* Mobile counter-ions in the liquid phase.
- **Zeta Potential (ζ) / Electrokinetic Potential:**

  > $\zeta = (4\pi \cdot \eta \cdot u) / \varepsilon$

  Potential difference between fixed layer and diffused layer. High $|\zeta|$ ⇒ Stable sol.
- **Isoelectric Point (pI):** pH at which net charge on colloidal particle is zero (sol is least stable and coagulates easily).
- **Electrokinetic Phenomena Summary:**
  • *Electrophoresis:* Motion of colloidal particles in electric field.
  • *Electro-osmosis:* Motion of dispersion medium when particles are fixed.
  • *Dorn Effect (Sedimentation Potential):* Potential generated when particles settle under gravity/centrifugation (Reverse of Electrophoresis).
  • *Streaming Potential:* Potential generated when liquid is forced through porous plug (Reverse of Electro-osmosis).

10. COAGULATION, HARDY-SCHULZE RULE & PROTECTION

Hardy-Schulze Rule & Coagulation Value

- **Hardy-Schulze Rule:**
  1. Ions carrying charge opposite to sol particles are effective in causing coagulation (Active/Flocculating Ion).
  2. **Coagulating Power $\propto$ (Valency of active ion)^4** (or higher power).
- **For Negatively Charged Sol (e.g., As_2S_3, Au sol, AgI/I^-):**

  > **Coagulating Power: $Sn^{4+} > Al^{3+}$ (or Fe^{3+}) > Ba^{2+} (or Mg^{2+}) > Na^+ (or K^+)**

- **For Positively Charged Sol (e.g., $Fe(OH)_3$, TiO_2, AgI/Ag^+):**

  > **Coagulating Power: $[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^-$**

- **Flocculation / Coagulation Value:** Minimum millimoles of an electrolyte required to cause coagulation of 1 Litre sol in 2 hours.

  > **Coagulation Value = (Millimoles of Electrolyte) / (Volume of Sol in Litres)**

- **Fundamental Inverse Relationship:**

  > **Coagulating Power $\propto$ 1 / (Coagulation Value)**

Protection of Colloids & Gold Number

- **Protection:** Addition of a lyophilic sol to a lyophobic sol prevents its coagulation by forming a protective hydration layer.
- **Gold Number (Zsigmondy):** Minimum mass in **milligrams (mg)** of protective lyophilic colloid required to prevent coagulation of **10 mL standard gold sol** upon addition of **1 mL of 10% NaCl solution**.

  > **Gold Number = [Mass of lyophilic sol (mg) × 10] / [Volume of Gold Sol (mL)]**

- **Protection Rule: Smaller the Gold Number, GREATER is the protective power!**

  > **Protective Power $\propto$ 1 / (Gold Number)**

- **Standard Gold Numbers for Revision:**
  • Gelatin: 0.005 – 0.01 (Highest Protective Power)
  • Egg Albumin: 0.08 – 0.10
  • Gum Arabic: 0.15 – 0.25
  • Potato Starch: 20 – 25 (Lowest Protective Power)
- **Congo Rubin Number** [Added]**:** Mg of protective colloid to prevent color change of 100 mL of 0.01% Congo Rubin dye with 10 mL of 100 mM KCl.

11. EMULSIONS & STANDARD JEE CHARGE TABLE

Emulsion Types & Testing

- **Oil-in-Water (O/W):** Oil dispersed phase, Water medium. (e.g., Milk, Vanishing cream). Conducts electricity, water-soluble dye spreads easily. Emulsifier: Proteins, gums, soaps.
- **Water-in-Oil (W/O):** Water dispersed phase, Oil medium. (e.g., Butter, Cold cream, Cod liver oil). Non-conducting, oil-soluble dye spreads. Emulsifier: Heavy metal salts of fatty acids, long chain alcohols.
- **Demulsification:** Separation into distinct layers by heating, freezing, centrifugation, or chemical treatment.

Standard Charge Table for Sol Identification (JEE Direct Questions)

Positively Charged Sols (+ve)	Negatively Charged Sols (−ve)
Hydrated metallic oxides: $Fe_2O_3 \cdot xH_2O$, $Al_2O_3 \cdot xH_2O$, $CrO_3 \cdot xH_2O$	Metals: *Cu, Ag, Au, Pt sols*
Basic dyes: Methylene blue, Bismarck brown	Metallic sulphides: As_2S_3, Sb_2S_3, CdS
Oxides: TiO_2 *sol*	Acid dyes: Eosin, Congo red
Haemoglobin (blood) in acid medium	Sols of starch, gum, gelatin, clay, charcoal, silicic acid

STRUCTURE OF ATOM

JEE Main & JEE Advanced Complete Formula Revision Sheet

CHEMISTRY / ATOMIC PHYSICS

1. Fundamental Physical Constants & Subatomic Particles

- Planck's const (h): 6.626×10^{-34} J·s
  h = 6.626 × 10^{-27} erg·s
- Speed of light (c): 3×10^{8} m/s
- Rydberg const (R_H): 1.09678×10^{7} m^{-1}
- 1 eV = 1.602×10^{-19} J
- 1 Å = 10^{-10} m = 10^{-8} cm

Particle	Absolute Mass	Relative Mass	Absolute Charge	Charge/Mass (e/m) Ratio
Electron (e^-)	9.109 × 10^{-31} kg (9.11 × 10^{-28} g)	1 / 1837 amu	−1.602 × 10^{-19} C (−4.8 × 10^{-10} esu)	1.7588 × 10^{11} C/kg (Thomson / Millikan)
Proton (p^+)	1.6726 × 10^{-27} kg (1.66 × 10^{-24} g)	1.00727 amu ≈ 1	+1.602 × 10^{-19} C (+4.8 × 10^{-10} esu)	9.58 × 10^{7} C/kg (Depends on gas in tube)
Neutron (n)	1.6749 × 10^{-27} kg	1.00866 amu ≈ 1	0	0 (Chadwick, 1932)

2. Electromagnetic Radiation & Photoelectric Effect

- Wave frequency & wave number: $\nu = c / \lambda$, $\bar{\nu} = 1 / \lambda$
- Energy of Photon: $E = h\nu = hc / \lambda = hc\,\bar{\nu}$
- Quick calculation formula: $E\ (eV) \approx 12400 / \lambda\ (Å)$ *or* *1240 / λ (nm)*
- Einstein's Photoelectric Equation: $h\nu = h\nu_0 + \frac{1}{2} m_e v^2_{max}$
- Work function (W_0 or φ): $W_0 = h\nu_0 = hc / \lambda_0$
- Stopping potential (V_0): $eV_0 = K.E._{max} = \frac{1}{2} m_e v^2_{max} = h(\nu - \nu_0)$

3. Bohr's Model of Hydrogen & Hydrogen-like Species

(Single-electron species: H, He^+, Li^{2+}, Be^{3+})

- Electrostatic force balance: $m_e v^2 / r = K Z e^2 / r^2$ *(K = 1 / 4πε₀)*
- Angular momentum quantization: $L = m_e v r = n h / (2\pi) = n \hbar$ *(n = 1, 2, 3...)*
- Radius of n-th orbit (r_n):

$$r_n = [n^2 h^2 / (4\pi^2 m_e K Z e^2)] = 0.529 \times (n^2 / Z)\ Å$$

- Velocity of electron in n-th orbit (v_n):

$$v_n = [2\pi K Z e^2 / (n h)] = 2.18 \times 10^6 \times (Z / n)\ m/s$$

- Revolution frequency (f) & Time period (T):

$$f = v_n / (2\pi r_n) \propto Z^2 / n^3, \quad T = 1 / f \propto n^3 / Z^2$$

- Kinetic Energy (K.E.): $K.E. = \frac{1}{2} m_e v^2 = K Z e^2 / (2r)$
- Potential Energy (P.E.): $P.E. = -K Z e^2 / r$
- Total Energy (E_n): $E_n = K.E. + P.E. = -K Z e^2 / (2r)$

> **Fundamental Energy Relationship:**
> $E_n = -K.E. = \frac{1}{2} P.E.$ | $P.E. = 2 E_n = -2 K.E.$

- Total Energy Values (E_n):

$$E_n = -[2\pi^2 m_e Z^2 e^4 K^2 / (n^2 h^2)]$$

$$E_n = -13.6 \times (Z^2 / n^2)\ eV/atom$$

$$E_n = -2.18 \times 10^{-18} \times (Z^2 / n^2)\ J/atom$$

$$E_n = -1312 \times (Z^2 / n^2)\ kJ/mol$$

4. Hydrogen Emission Spectrum & Spectral Transitions

- Rydberg Equation for transition from $n_2 \rightarrow n_1$ (where $n_2 > n_1$):

$$\Delta E = E_{n2} - E_{n1} = 2.18 \times 10^{-18} Z^2 (1 / n_1^2 - 1 / n_2^2)\ J$$

$$\bar{\nu} = 1 / \lambda = R_H Z^2 (1 / n_1^2 - 1 / n_2^2) \quad [R_H \approx 1.096778 \times 10^7\ m^{-1},\ 1 / R_H \approx 912\ Å]$$

Spectral Series	Lower Orbit (n_1)	Upper Orbit (n_2)	Spectral Region	Shortest Wavelength (λ_{min}: $\infty \rightarrow n_1$)
Lyman	$n_1 = 1$	$n_2 = 2, 3, 4...$	Ultraviolet (UV)	$\lambda_{min} = 1 / R_H \approx 912$ Å $/ Z^2$
Balmer	$n_1 = 2$	$n_2 = 3, 4, 5...$	Visible (H_α, H_β, H_γ, H_δ)	$\lambda_{min} = 4 / R_H \approx 3646$ Å $/ Z^2$
Paschen	$n_1 = 3$	$n_2 = 4, 5, 6...$	Near Infrared (Infra-Red)	$\lambda_{min} = 9 / R_H \approx 8205$ Å $/ Z^2$
Brackett	$n_1 = 4$	$n_2 = 5, 6, 7...$	Mid Infrared	$\lambda_{min} = 16 / R_H \approx 14588$ Å $/ Z^2$
Pfund	$n_1 = 5$	$n_2 = 6, 7, 8...$	Far Infrared	$\lambda_{min} = 25 / R_H \approx 22793$ Å $/ Z^2$
Humphreys [Added]	$n_1 = 6$	$n_2 = 7, 8, 9...$	Far Infrared	$\lambda_{min} = 36 / R_H$

- **Total Spectral Lines (Sample containing large number of H-atoms):**
  $N = \Delta n(\Delta n + 1) / 2$ where $\Delta n = n_2 - n_1$ (for transition to $n_1 = 1$: $N = n(n - 1) / 2$)
- **Max Lines for a SINGLE Isolated Atom:**
  $N_{single} = n_2 - n_1 = \Delta n$

5. Quantum Mechanics: Matter Waves & Uncertainty Principle

- **de-Broglie Wavelength (λ):**
  $\lambda = h / p = h / (m v)$
- **λ in terms of Kinetic Energy (K.E.):**
  $\lambda = h / \sqrt{(2 m K.E.)}$
- **λ for Charge Particle Accelerated by Potential (V):**
  $\lambda = h / \sqrt{(2 m q V)}$
- **Specific Formula for Electron:**
  $\lambda_e = \sqrt{(150 / V)}$ Å $= 12.27 / \sqrt{V}$ Å *(V in Volts)*
- **Quantization of Orbit Circumference:**
  $2\pi r_n = n \lambda \rightarrow$ *n-th orbit contains n de-Broglie standing waves.*
- **Heisenberg's Uncertainty Principle:**
  $\Delta x \cdot \Delta p \geq h / (4\pi) \rightarrow \Delta x \cdot (m \Delta v) \geq h / (4\pi)$
- **Energy-Time Uncertainty Relation:**
  $\Delta E \cdot \Delta t \geq h / (4\pi)$
- **Special Case (Δx = Δp):**
  $\Delta p = \sqrt{(h / 4\pi)} \rightarrow \Delta v = (1 / 2m) \sqrt{(h / \pi)}$
- **Thermal Gas Particle Wavelength:** [Added]
  $\lambda = h / \sqrt{(3 m k_B T)}$ *(k_B = Boltzmann const)*

6. Quantum Numbers & Rules of Electronic Structure

Quantum Number	Symbol & Allowed Values	Physical Significance / Information
Principal Quantum Number	n = 1, 2, 3, 4... (K, L, M, N...)	Determines shell size, main energy level. Max subshells = n, Max orbitals = n^2, Max e^- = $2n^2$.
Azimuthal / Subsidiary Q.N.	l = 0 to (n − 1) (s=0, p=1, d=2, f=3, g=4)	Determines subshell shape. Max orbitals in subshell = (2l + 1). Max e^- in subshell = 2(2l + 1) = 4l + 2.
Magnetic Quantum Number	m_l = −l to +l through 0 (Total values = 2l + 1)	Determines spatial orientation of orbitals under external magnetic field.
Spin Quantum Number	s = +½ (↑) or −½ (↓)	Determines spin orientation of electron around its own axis.

- **Orbital Angular Momentum ($L_{orbital}$):**
  $L = \sqrt{(l(l + 1))} (h / 2\pi) = \sqrt{(l(l + 1))} \hbar$
  (For s-orbital l=0 → L = 0; p-orbital l=1 → L = √2 ħ; d-orbital l=2 → L = √6 ħ)
- **Spin Angular Momentum (S_{spin}):**
  $S = \sqrt{(s(s + 1))} (h / 2\pi) = \sqrt{(s(s + 1))} \hbar$ *(where s = ½)*
- **Spin Magnetic Moment (μ_s):**
  $\mu_s = \sqrt{(n(n + 2))}$ B.M. *(n = number of unpaired electrons)*

Rules for Filling Orbitals:

- **Aufbau Principle:** Orbitals are filled in increasing order of energy given by the **(n + l) rule**. If two orbitals have same (n + l), lower n has lower energy. (Sequence: 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p...)
- **Pauli's Exclusion Principle:** No two electrons in an atom can have the exact same set of all four quantum numbers (n, l, m_l, m_s). An orbital can hold max 2 electrons with opposite spins.
- **Hund's Rule of Maximum Multiplicity:** Electron pairing in degenerate orbitals (p, d, f) does not take place until each available orbital is singly occupied with parallel spins. Total Max Spin = ½ × (No. of unpaired e^-).

7. Atomic Nodes & Wave Functions (Quantum Mechanical Treatment)

• **Radial Nodes (Spherical Nodes):**

$N_{radial} = n - l - 1$ *(Points where Radial Wavefunction R(r) = 0)*

• **Angular Nodes (Nodal Planes):**

$N_{angular} = l$ *(Planes through nucleus where electron density = 0)*

• **Total Nodes in an Orbital:**

$N_{total} = N_{radial} + N_{angular} = (n - l - 1) + l = n - 1$

• **Schrödinger Wave Equation (1D):**

$d^2\Psi / dx^2 + (8\pi^2 m / h^2)(E - V)\Psi = 0$

• **Probability Density:**

$P = |\Psi|^2 dV$ *(Probability of finding e^- in volume dV)*

• **Radial Probability Distribution Function:**

$4\pi r^2 R^2(r) dr$ *(Max at $r = r_0 = 0.529$ Å for 1s H-atom)*

8. Nuclear Species & Atomic Classifications

Term	Definition & Condition	Examples
Isotopes	Same Atomic Number (Z / Protons), Different Mass Number (A / Neutrons).	^1H, ^2D, ^3T | $^{35}_{17}Cl$, $^{37}_{17}Cl$
Isobars	Same Mass Number (A), Different Atomic Number (Z).	$^{40}_{18}Ar$, $^{40}_{19}K$, $^{40}_{20}Ca$
Isotones	Same number of Neutrons (N = A − Z).	$^{30}_{14}Si$, $^{31}_{15}P$, $^{32}_{16}S$ → N = 16
Isoelectronic	Same total number of electrons.	O^{2-}, F^-, Ne, Na^+, Mg^{2+}, Al^{3+} (10 e^-) NO_3^-, CO_3^{2-} | N_2, CO, CN^-
Isodiaphers	Same Isotopic Number / Neutron Excess value: **(N − Z) = (A − 2Z)**.	$^{238}_{92}U$ and $^{234}_{90}Th$ → (N − Z) = 54
Isosters [Added]	Same number of atoms AND same total number of electrons.	N_2 and CO (2 atoms, 14 e^-) | CO_2 and N_2O (3 atoms, 22 e^-)

• **Average Atomic Mass:** $A_{avg} = (A_1 x_1 + A_2 x_2 + \ldots) / (x_1 + x_2 + \ldots)$ *(where x_i are natural percentage/ratio abundances)*

9. JEE Advanced High-Yield Relations & Shortcuts

• **Equivalent Current of Revolving Electron:** [Added]

$I = e / T = e f = (e v_n) / (2\pi r_n) \propto Z^2 / n^3$

• **Magnetic Field at Nucleus due to Orbiting e^-:** [Added]

$B = (\mu_0 I) / (2 r_n) = (\mu_0 e v_n) / (4\pi r_n^2) \propto Z^3 / n^5$

• **Magnetic Dipole Moment of Orbiting e^-:** [Added]

$M = I \cdot A = I(\pi r_n^2) = (e / 2m_e) L = n \cdot (e\hbar / 2m_e)$

(Bohr Magneton $\mu_B = e\hbar / 2m_e = 9.27 \times 10^{-24}$ A·m^2)

• **Correction for Finite Nuclear Mass (Reduced Mass μ):** [Added]

$\mu = (m_e M_N) / (m_e + M_N)$ *(For Positronium: $\mu = m_e / 2$)*

Replace m_e with μ in Bohr's equations:

$R_{real} = R_\infty / (1 + m_e / M_N)$

• **Nuclear Radius Formula (Rutherford Scattering):** [Added]

$R = R_0 A^{1/3}$ *($R_0 \approx 1.2$ to 1.43×10^{-15} m = 1.2 fm)*

• **Distance of Closest Approach (α-particle scattering):** [Added]

$r_{min} = (4 K Z e^2) / (m_\alpha v^2) = (2 K Z e^2) / (K.E._\alpha)$

10. X-Rays & Moseley's Law

[JEE Advanced Added Section]

- **Moseley's Law for Characteristic X-rays:**

$$\sqrt{\nu} = a\,(Z - b)$$

(a = proportionality const, b = screening/shielding const)

For K_α transition: $b = 1 \rightarrow \nu = R_H\, c\, (3/4)\, (Z - 1)^2$

- **Continuous X-rays & Duane-Hunt Cutoff Wavelength:**

$$\lambda_{min} = hc / (e\,V) = 12400 / V\ \text{Å}$$

(V = Accelerating potential of X-ray tube in Volts)

11. Quick Numerical Shortcuts for JEE Main / Advanced

- **Energy Ratio:** $E_1 : E_2 : E_3 : E_4 = -13.6 : -3.4 : -1.51 : -0.85$ eV (for H-atom).
- **Ionization Energy (I.E.):** I.E. = $E_\infty - E_{ground} = +13.6 \times Z^2$ eV.
- **Separation / Excitation Energy:** Energy required to remove e^- from excited state n to ∞.
- **First Excitation Potential:** Potential to move e^- from n=1 to n=2 (10.2 V for H-atom).
- **Nodes Calculation Quick Check:**
  3p orbital → n=3, l=1 → Radial = 3-1-1 = 1, Angular = 1.
  4d orbital → n=4, l=2 → Radial = 4-2-1 = 1, Angular = 2.
- **Isotopic Mass Defect:** Mass Defect $\Delta m = [Z\, m_p + (A-Z)\, m_n] - M_{actual}$.
- **Binding Energy (B.E.):** B.E. = Δm (in amu) × 931.5 MeV.

STATES OF MATTER: GASEOUS & LIQUID STATE

JEE Main & Advanced Complete Formula & Quick Revision Sheet

CHEMISTRY REVISION

1. FUNDAMENTAL PARAMETERS & UNIT CONVERSIONS

Parameter	Key Units & Standard Conversion Relations
Volume (V)	1 L = 1 dm^3 = 1000 mL = 1000 cm^3 = 10^{-3} m^3 1 m^3 = 10^3 L = 10^6 mL = 10^6 cm^3
Pressure (P)	1 atm = 760 mmHg = 76 cmHg = 760 torr 1 atm = 101,325 N/m^2 (Pa) = 101.325 kPa 1 bar = 10^5 Pa ≈ 0.987 atm | 1 atm = 1.01325 bar
Temp (T)	T(K) = t(°C) + 273.15 | Absolute Zero = 0 K = −273.15 °C
STP / NTP	**Old STP:** 0 °C, 1 atm → V_{molar} = 22.4 L/mol **IUPAC STP:** 0 °C, 1 bar → V_{molar} = 22.7 L/mol

2. UNIVERSAL GAS CONSTANT (R) & BOLTZMANN (K B)

Molar Gas Constant (R):

- $R = 0.0821\ L{\cdot}atm{\cdot}mol^{-1}{\cdot}K^{-1} \approx 1/12\ L{\cdot}atm{\cdot}mol^{-1}{\cdot}K^{-1}$
- $R = 8.314\ J{\cdot}mol^{-1}{\cdot}K^{-1}$ (SI Unit)
- $R = 8.314 \times 10^7\ erg{\cdot}mol^{-1}{\cdot}K^{-1}$ (CGS Unit)
- $R \approx 2.0\ cal{\cdot}mol^{-1}{\cdot}K^{-1}$ (1 cal = 4.184 J)
- $R = 0.083\ bar{\cdot}L{\cdot}mol^{-1}{\cdot}K^{-1}$ (or $bar{\cdot}dm^3{\cdot}K^{-1}{\cdot}mol^{-1}$)

Boltzmann Constant (k_B) [Added] **:**

$$k_B = R / N_A = 1.38 \times 10^{-23}\ J{\cdot}K^{-1}{\cdot}molecule^{-1}$$

3. IDEAL GAS LAWS & STATE EQUATIONS

- **Boyle's Law:** $V \propto 1/P$ (at const T, n) $\Rightarrow P_1V_1 = P_2V_2 = K$
  $\log P + \log V = \text{const} \Rightarrow \log P = -\log V + \text{const}$
- **Charles's Law:** $V \propto T$ (at const P, n) $\Rightarrow V_1/T_1 = V_2/T_2$
  $V_t = V_0\,(1 + t / 273.15)$ (V_0 = vol at 0 °C)
- **Gay-Lussac's Law:** $P \propto T$ (at const V, n) $\Rightarrow P_1/T_1 = P_2/T_2$
  $P_t = P_0\,(1 + t / 273.15)$
- **Avogadro's Law:** $V \propto n$ (at const P, T)

Ideal Gas Equation: $P\,V = n\,R\,T = (w / M)\,R\,T$

Density & Molar Concentration:

$$\text{Density: } d = P\,M / (R\,T) \ \mid\ \text{Molar Conc.: } C = n / V = P / (R\,T)$$

4. DALTON'S LAW OF PARTIAL PRESSURES

For a mixture of non-reacting gases in volume V at temperature T:

$$P_{total} = P_1 + P_2 + P_3 + \ldots = \sum P_i$$

Partial Pressure Relations:

- **Mole Fraction form:** $P_A = X_A \cdot P_{total}$ (where $X_A = n_A / n_{total}$)
- **Volume Fraction form:** $P_A = (V_A / V_{total}) \cdot P_{total}$
- **Moist Gas Correction:**
  $P_{dry\ gas} = P_{moist\ gas} - \text{Aqueous Tension}$
  (Aqueous tension = Vapour pressure of water at given T)
- **Gas Mixing Equation:** When container 1 (P_1, V_1) mixes with container 2 (P_2, V_2) into total volume V_T:
  $P_{final} = (P_1V_1 + P_2V_2) / V_T$

5. GRAHAM'S LAW OF DIFFUSION & EFFUSION

At constant T & P, rate of diffusion/effusion (r) is inversely proportional to square root of gas density (d) or molar mass (M):

$$r \propto 1/\sqrt{d} \propto 1/\sqrt{VD} \propto 1/\sqrt{M}$$

Rate Definitions (r):

$$r = V_{diffused} / t = n_{diffused} / t = d_{distance} / t = (\Delta P) / t$$

Combined Rate Comparison Formula:

$$r_1 / r_2 = (V_1 / t_1) / (V_2 / t_2) = (n_1 / t_1) / (n_2 / t_2) = \sqrt{(d_2 / d_1)} = \sqrt{(M_2 / M_1)}$$

Effusion with Variable Pressure Gradient:

When initial partial pressures P_1 and P_2 are different:

$$r \propto P / \sqrt{M} \ \Rightarrow\ r_1 / r_2 = (P_1 / P_2) \times \sqrt{(M_2 / M_1)}$$

* Note: Vapour Density (VD) = Molar Mass / 2.

6. KINETIC THEORY & KINETIC GAS EQUATION

Kinetic Gas Equation:

$$P\,V = (1/3)\,m\,N\,u_{rms}^{\ 2} = (1/3)\,M_{total}\,u_{rms}^{\ 2}$$

(m = mass of 1 molecule, N = number of molecules, mN = M_{total})

Translational Kinetic Energy:

- **Total K.E. of n moles:** $E_K = (3/2)\,n\,R\,T$
- **Molar K.E. (1 mole or N_A molecules):** $E_{K,\,molar} = (3/2)\,R\,T$
- **Molecular K.E. (1 molecule):** $E_{K,\,mol} = (3/2)\,k_B\,T$

Equipartition of Energy & Degrees of Freedom (f) [Added] **:**

$$E_{total\ per\ mol} = (f/2)\,R\,T \ \mid\ C_v = (f/2)\,R \ \mid\ C_p = (f/2 + 1)\,R$$

Monatomic: f = 3 (3T) | Diatomic (room T): f = 5 (3T + 2R) | Non-linear polyatomic: f = 6 (3T + 3R)

7. MAXWELL-BOLTZMANN SPEED DISTRIBUTION

1. Most Probable Speed (u_{mp} or α): Speed possessed by maximum fraction of molecules.

$$u_{mp} = \sqrt{(2\,RT / M)} = \sqrt{(2\,PV / M)} = \sqrt{(2\,P / d)}$$

2. Average Speed (u_{av} or $\bar{v}$): Arithmetic mean of speeds of all molecules.

$$u_{av} = \sqrt{(8\,RT / (\pi\,M))} = \sqrt{(8\,PV / (\pi\,M))} = \sqrt{(8\,P / (\pi\,d))}$$

3. Root Mean Square Speed (u_{rms} or u): Square root of mean of squared speeds.

$$u_{rms} = \sqrt{(3\,RT / M)} = \sqrt{(3\,PV / M)} = \sqrt{(3\,P / d)}$$

SPEED RATIO COMPARISONS

$u_{mp} : u_{av} : u_{rms} = \sqrt{2} : \sqrt{(8/\pi)} : \sqrt{3}$

$u_{mp} : u_{av} : u_{rms} = 1 : 1.128 : 1.224$

$u_{mp} : u_{av} : u_{rms} = 0.816 : 0.921 : 1.000$

Speed Fraction Differential Formula [Added] **:**

$$dN / N = 4\pi\,(M / (2\pi RT))^{3/2}\,u^2\,\exp(-M\,u^2 / (2RT))\,du$$

Temperature & Mass Dependence Key Rules:

- As T ↑, peak shifts to right (higher speed) and flattens (fraction possessing u_{mp} decreases). Total area under curve stays constant (= 1).
- At same T, lighter gas molecules (smaller M) have broader speed distribution and higher u_{rms}.

8. REAL GASES & VAN DER WAALS EQUATION

Real gases deviate from ideal behavior due to molecular volume and intermolecular attractive forces.

Van der Waals Equation of State:

$$(P + a\,n^2 / V^2)\,(V - n\,b) = n\,R\,T$$

For 1 mole ($V_m = V/n$): $(P + a / V_m^2)\,(V_m - b) = R\,T$

Physical Significance & Units of Constants:

Constant	Physical Meaning	SI & Common Units
'a'	Measures magnitude of intermolecular attraction. Higher 'a' → Easier Liquefaction.	$atm \cdot L^2 \cdot mol^{-2}$ $bar \cdot dm^6 \cdot mol^{-2}$ $N \cdot m^4 \cdot kg^{-2}$ (SI)
'b'	Co-volume / excluded volume per mole. Measures effective size of gas molecules.	$L \cdot mol^{-1}$ $cm^3 \cdot mol^{-1}$ $m^3 \cdot mol^{-1}$ (SI)

Excluded Volume Relation: $b = 4\,N_A\,v_{molecule} = 4\,N_A\,(4/3\,\pi\,r^3)$

9. COMPRESSIBILITY FACTOR (Z) & DEVIATION

$$Z = P\,V_m / (R\,T) = V_{real,\,molar} / V_{ideal,\,molar}$$

- **Ideal Gas:** $Z = 1$ ($P\,V_m = R\,T$) at all P & T.
- **Negative Deviation (Z < 1):** Attractive forces dominate. Gas is more compressible than ideal gas. (Observed at low/moderate P).
- **Positive Deviation (Z > 1):** Repulsive forces / molecular volume dominate. Gas is less compressible than ideal gas. (Observed at high P, and at all P for H_2 & He).

10. VAN DER WAALS EQUATION UNDER LIMITING CONDITIONS

Condition	Simplification / Ignored Term	Resulting Equation & Z Value
Low / Moderate P (Large V_m)	Attractions matter, volume negligible: $b \ll V_m \Rightarrow (V_m - b) \approx V_m$	$(P + a/V_m^2)V_m = RT$ $P\,V_m + a/V_m = RT$ **Z = 1 − a / (V_m R T) < 1**
High Pressure (Small V_m)	Volume non-negligible, attraction negligible: $a/V_m^2 \ll P$	$P\,(V_m - b) = RT$ $P\,V_m - P\,b = RT$ **Z = 1 + P b / (R T) > 1**
Extremely Low P / High T	Both attraction & volume corrections are negligible: $a/V_m^2 \to 0,\ b \to 0$	$P\,V_m = R\,T$ **Z = 1 (Ideal Behavior)**
Exceptional Gases: H_2 & He	Extremely small mass → 'a' is negligible at ordinary T ($a \approx 0$)	$P\,(V_m - b) = RT$ **Z = 1 + P b / (R T) > 1**

11. CRITICAL STATE & CHARACTERISTIC TEMPS [Added]

At the Critical Point, gas transitions to liquid continuously with no phase boundary.

Critical Constants (in terms of a and b):

- **Critical Temp:** $T_c = 8a / (27\,R\,b)$
- **Critical Vol:** $V_c = 3\,b$
- **Critical Press:** $P_c = a / (27\,b^2)$
- **Critical Z:** $Z_c = P_c V_c / (RT_c) = 3/8 = 0.375$

Characteristic Temperatures:

- **Boyle Temperature (T_b):** Temp at which real gas obeys ideal gas law over a wide pressure range.

$$T_b = a / (R\,b) = (27/8)\,T_c = 3.375\,T_c$$

- **Inversion Temperature (T_i):** Temp above which gas shows heating on expansion (Joule-Thomson effect).

$$T_i = 2a / (R\,b) = 2\,T_b = 6.75\,T_c$$

12. COLLISIONAL PARAMETERS & TRANSPORT [JEE Adv]

For gas molecules modeled as hard spheres with collision diameter σ and number density $n^* = N / V$:

1. Mean Free Path (λ): Average distance traveled between two successive collisions.

$$\lambda = 1 / (\sqrt{2}\, \pi\, \sigma^2\, n^*) = k_B\, T / (\sqrt{2}\, \pi\, \sigma^2\, P)$$

• Dependences: $\lambda \propto T / P$ (at const V, λ is independent of T).

2. Collision Frequency (Z_1 & Z_{11}):

- **Collisions made by 1 molecule per sec (Z_1):**

$$Z_1 = \sqrt{2}\, \pi\, \sigma^2\, u_{av}\, n^* = \sqrt{2}\, \pi\, \sigma^2\, u_{av}\, (P / (k_B\, T))$$

- **Total bimolecular collisions per unit vol per sec (Z_{11}):**

$$Z_{11} = (1 / 2)\, \sqrt{2}\, \pi\, \sigma^2\, u_{av}\, (n^*)^2 = (1 / \sqrt{2})\, \pi\, \sigma^2\, u_{av}\, (n^*)^2$$

13. LIQUID STATE — SURFACE TENSION (Γ OR T)

Work done per unit area to expand liquid surface or force per unit length acting perpendicular to line on surface.

$$\gamma = F / l = w / \Delta A \quad [\text{SI Unit: N/m or J/m}^2 \mid \text{CGS: dyne/cm}]$$

Factors Affecting Surface Tension:

- **Temperature:** As T ↑, Intermolecular force ↓ ⇒ Surface Tension ↓.
  $\gamma = \gamma_0\, (1 - T/T_c)^n$ ($\gamma = 0$ at critical temperature T_c).
- **Surfactants / Solutes:** Soluble salts (e.g. NaCl) ↑ γ; Soaps/detergents ↓ γ.

Excess Pressure in Bubbles & Drops [Added] :

- **Liquid Drop (1 surface):** $\Delta P = P_{in} - P_{out} = 2\gamma / R$
- **Soap Bubble (2 surfaces):** $\Delta P = P_{in} - P_{out} = 4\gamma / R$

Capillary Rise Formula (Jurin's Law) [Added] :

$$h = (2\, \gamma \cos \theta) / (r\, \rho\, g)$$

(θ = contact angle, r = capillary radius, ρ = liquid density)

14. LIQUID STATE — VAPOUR PRESSURE

Pressure exerted by vapour in dynamic equilibrium with liquid at a specified temperature.

Key Properties & Factors:

- **Independent of:** Volume of liquid, volume/shape of container, surface area.
- **Depends on:** Nature of liquid (Intermolecular forces ↑ ⇒ V.P. ↓) & Temperature (T ↑ ⇒ V.P. ↑).
- **Boiling Point:** Temp at which Vapour Pressure = External Atmospheric Pressure. (Standard B.P. at 1 bar; Normal B.P. at 1 atm).

Clausius-Clapeyron Equation [Added] :

$$\log_{10}(P_2 / P_1) = (\Delta H_{vap} / (2.303\, R)) \times [\, (T_2 - T_1) / (T_1\, T_2)\,]$$

(P_1, P_2 are vapour pressures at temperatures T_1 and T_2; ΔH_{vap} = Molar enthalpy of vaporization)

15. LIQUID STATE — VISCOSITY (H)

Internal resistance to flow present between adjacent parallel layers of liquid moving at different velocities.

Newton's Law of Viscosity:

$$F = \eta\, A\, (dv / dx)$$

(F = Viscous drag force, A = area, dv/dx = velocity gradient)

Coefficient of Viscosity: $\eta = (F / A) / (dv/dx)$
Units: N·s·m^2 (Pa·s) | CGS Unit: Poise (1 Pa·s = 10 Poise)

Factors Affecting Viscosity:

- **Temperature:** As T ↑ ⇒ Kinetic Energy ↑ ⇒ Viscosity ↓.
  $\eta = A \exp(E_a / RT)$
- **Pressure:** As P ↑ ⇒ Viscosity ↑ (Except for liquid Water where η slightly decreases with P initially).
- **Intermolecular Attraction:** Strong H-bonding/dipole forces ↑ Viscosity (Honey > Water).

Poiseuille's Flow Equation [Added] :

$$V / t = (\pi\, P\, r^4) / (8\, \eta\, l)$$

16. LAST-MINUTE JEE FORMULA SHORTCUTS

- **Ideal Gas density ratio:** $d_1/d_2 = (P_1 M_1 T_2)/(P_2 M_2 T_1)$
- **Payload of Balloon:** Mass of displaced air − Total mass of balloon & gas inside.
- **Barometric Formula [Added] :** $P = P_0 \exp(-M\, g\, z / RT)$
- **Van der Waals liquefaction order:** Matches 'a' values ($a_{NH_3} > a_{CO_2} > a_{O_2} > a_{H_2}$).

SOME BASIC CONCEPTS OF CHEMISTRY

JEE Main & Advanced Revision Sheet

1. FUNDAMENTAL UNITS, CONVERSIONS & DEFINITIONS

Length & Volume Units

1 Å = 10^{-10} m = 10^{-8} cm

1 nm = 10^{-9} m | 1 pm = 10^{-12} m

1 L = 1 dm^3 = 10^{-3} m^3 = 1000 mL

1 mL = 1 cm^3 (cc)

Pressure & Energy Units

1 atm = 760 mmHg = 760 torr

1 atm = 101325 Pa ($N\ m^{-2}$)

1 bar = 10^5 Pa | 1 cal = 4.184 J

1 eV = 1.6022×10^{-19} J

1 J = 10^7 ergs

Atomic Mass Unit (amu / u)

1 amu = $^1/_{12}$ × mass of 1 C-12 atom

1 amu = $^1/_{N_A}$ g = 1.66×10^{-24} g

1 amu = 1.66×10^{-27} kg

Energy equivalency: 1 amu ≈ 931.5 MeV [Added]

Mass Definitions

Atomic/Molecular Mass:	Mass of 1 atom/molecule in amu.	
Actual Mass:	Mass of 1 atom/molecule in grams =	Molar Mass × 1.66×10^{-24} g
Relative Mass:	Unitless ratio w.r.t.	$^1/_{12}$ mass of C-12 atom.
Gram Atomic/Molecular Mass (GAM/GMM):	Mass of 1 mole (N_A entities) in grams.	

Matter & Substance Classification

Matter: Anything occupying space & having mass.

Pure Substance: Fixed composition (Elements: metal, nonmetal, metalloid; Compounds: ionic, covalent).

Mixture: Variable composition (Homogeneous: Solutions; Heterogeneous: Colloids, Suspensions).

2. CORE MOLE CONCEPT & FORMULA INTERCONVERSIONS

Fundamental Mole Relationships

For Elements (Atoms)	For Molecules	For Ionic Compounds
Moles (n) = $^{w\ (g)}/_{GAM}$	Moles (n) = $^{w\ (g)}/_{GMM}$	Moles (n) = $^{w\ (g)}/_{GFM}$
No. of atoms = $n \times N_A$	No. of molecules = $n \times N_A$	No. of formula units = $n \times N_A$
1 g-atom = 1 mole of atoms	1 g-molecule = 1 mole of molecules	1 g-formula unit = 1 mole unit

Gaseous Molar Volume & Constants

Moles of Gas (n) = $^{V\ (L)\ at\ STP}/_{V_m}$

Old STP (1 atm, 273.15 K): $V_m = 22.4\ L\ mol^{-1} = 22400$ mL

IUPAC Standard (1 bar, 273.15 K): $V_m = 22.7\ L\ mol^{-1}$

Avogadro's Number (N_A): $6.022 \times 10^{23}\ mol^{-1}$

Loschmidt Number: 2.687×10^{19} molecules/mL (at STP)

Vapour Density (V.D.) & Gas Density

V.D. = $^{Density\ of\ Gas\ /\ Vapour}/_{Density\ of\ H_2\ gas}$ (at same T & P)

Molar Mass (GMM): $GMM = 2 \times V.D.$

Ideal Gas Equation: $PV = nRT = {}^{w}/_{M}RT$

Gas Density (d): $d = {}^{PM}/_{RT} \Rightarrow M = {}^{dRT}/_{P}$

3. LAWS OF CHEMICAL COMBINATION

Law & Scientist	Core Statement / Mathematical Form	Key Example / Condition
1. Conservation of Mass (Lavoisier, 1789)	Total mass of reactants = Total mass of products. $\sum m_{reactants} = \sum m_{products}$	Holds for chemical changes. **Exception:** Nuclear reactions ($E = \Delta m \cdot c^2$).
2. Definite / Constant Proportions (Proust, 1799)	A pure compound contains the same elements combined in fixed proportion by mass.	CO_2 always has C : O mass ratio = 3 : 8 (12:32). **Exception:** Isotopic variations, non-stoichiometric compounds.
3. Multiple Proportions (Dalton, 1803)	When elements A & B form >1 compound, masses of B reacting with fixed mass of A are in simple whole-number ratio.	CO (12:16) & CO_2 (12:32) ⇒ Ratio of O = 16:32 = 1:2. SO_2 & SO_3 ⇒ Ratio of O = 2:3.
4. Reciprocal Proportions (Richter, 1792)	If A & B react separately with fixed mass of C, the mass ratio of A:B is same or simple multiple of their combining ratio together.	CH_4 (12g C to 4g H) & CO_2 (12g C to 32g O) ⇒ H:O ratio = 4:32 (1:8), matches H_2O (1:8).
5. Gaseous Volumes (Gay-Lussac, 1808)	Gases react/form in simple whole-number volume ratios at constant T & P.	$H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(g)}$ (Volume ratio 1 : 1 : 2).
6. Avogadro's Law (Avogadro, 1811)	Equal volumes of all gases under same T & P contain equal number of molecules. $V \propto n$ (at constant T, P).	Direct foundation for molar volume concept and molecular formula determination.

4. CONCENTRATION TERMS & EXPRESSION FORMULATIONS

Concentration Unit	Mathematical Formula	Definition / Remarks & Temperature Dependency
Mass % (w/w)	$\%\ (w/w) = \text{Mass of solute (g)} / \text{Mass of solution (g)} \times 100$	Grams of solute in 100 g solution. **Temp Independent.**
Volume % (v/v)	$\%\ (v/v) = \text{Volume of solute (mL)} / \text{Volume of solution (mL)} \times 100$	mL of solute in 100 mL solution. **Temp Dependent.**
Mass by Vol % (w/v)	$\%\ (w/v) = \text{Mass of solute (g)} / \text{Volume of solution (mL)} \times 100$	Grams of solute in 100 mL solution. **Temp Dependent.**
Parts Per Million (ppm)	$ppm = \text{Mass of solute} / \text{Mass of solution} \times 10^6$	Used for trace quantities/hardness. **Temp Independent.**
Mole Fraction (χ)	$\chi_A = n_A / (n_A + n_B + \dots) \mid \sum \chi_i = 1$	Ratio of moles of one component to total moles. **Temp Independent.**
Molarity (M)	$M = \text{Moles of solute (n)} / \text{Volume of solution (L)} = w \times 1000 / M_0 \times V(mL)$	Moles of solute in 1 L solution. **Temp Dependent.**
Molality (m)	$m = \text{Moles of solute (n)} / \text{Mass of solvent (kg)} = w \times 1000 / M_0 \times W_{solvent}(g)$	Moles of solute in 1 kg solvent. **Temp Independent.**
Normality (N)	$N = \text{Gram Equivalents of solute} / \text{Volume of solution (L)} = w \times 1000 / E \times V(mL)$	Gram equivalents in 1 L solution. **Temp Dependent.**
Strength (S)	$S\ (g/L) = \text{Mass of solute (g)} / \text{Volume of solution (L)} = M \times M_0 = N \times E$	Grams of solute per Litre of solution. **Temp Dependent.**

5. CRITICAL INTERCONVERSION FORMULAS (JEE SHORTCUTS)

Molarity (M) from Mass % (x) & Density (d)

$$M = {10 \times x \times d} / {M_0}$$

where x = % (w/w), d = density of solution in g/mL, M_0 = molar mass of solute.

Normality (N) from Mass % (x) & Density (d)

$$N = {10 \times x \times d} / {E} = M \times \text{n-factor}$$

where E = equivalent weight of solute ($E = M_0 / \text{n-factor}$).

Molality (m) from Molarity (M) & Density (d)

$$m = {1000 \times M} / {(1000 \times d - M \times M_0)}$$

d in g/mL; M_0 = molar mass of solute in g/mol.

Molality (m) & Mole Fraction (χ_{solute})

$$m = {\chi_{solute} \times 1000} / {(1 - \chi_{solute}) \times M_{solvent}}$$

For aqueous solutions: $m = {1000 \times \chi_{solute}} / {18 \times (1 - \chi_{solute})}$

Molarity of Pure Liquid

$$M = {1000 \times d} / {M_0}$$

For pure water (d = 1 g/mL, M_0 = 18): $M_{H_2O} = 55.55\ M$.

Molarity (M) from Mole Fraction (χ_{solute}) [Added]

$$M = {1000 \times d \times \chi_A} / {[\chi_A M_A + \chi_B M_B]}$$

where A = solute, B = solvent, d = solution density in g/mL.

6. DILUTION & MIXING LAWS

Dilution Law

Moles of solute remain constant before & after dilution:

$$M_1 V_1 = M_2 V_2 \ | \ N_1 V_1 = N_2 V_2$$

Volume of water added: $V_{H_2O} = V_2 - V_1$

Aliquoting property: Taking out a portion of a uniform solution keeps molarity/concentration identical.

Mixing Solutions (Same Solute)

Total Molarity of mixture (M_{mix}):

$$M_{mix} = {(M_1V_1 + M_2V_2 + \ldots)} / {(V_1 + V_2 + \ldots)}$$

Total Normality (N_{mix}): $N_{mix} = {(N_1V_1 + N_2V_2 + \ldots)} / {(V_1 + V_2 + \ldots)}$

7. EQUIVALENT CONCEPT & N-FACTOR CALCULATION

Gram Equivalent & Law of Equivalence

Eq. Weight (E): $E = {\text{Molar Mass } (M_0)} / {\text{n-factor}}$

Gram Equivalents (g.eq): $g.eq = {w\ (g)} / {E} = \text{Moles} \times \text{n-factor} = N \times V(L)$

Milli-equivalents (mEq): $mEq = N \times V\ (mL) = g.eq \times 1000$

Law of Equivalence: In any chemical reaction $aA + bB \rightarrow cC + dD$:
g.eq of A reacted = g.eq of B reacted = g.eq of C formed = g.eq of D formed

Rules for Determining n-Factor (Valency Factor)

Acid: Basicity = Number of replaceable H^+ ions per molecule.
Examples: H_2SO_4 (n = 2); H_3PO_4 (n = 3); H_3PO_3 (n = 2); H_3PO_2 (n = 1); H_3BO_3 (n = 1, monobasic Lewis acid).

Base: Acidity = Number of replaceable OH^- ions or H^+ accepted.
Examples: NaOH (n = 1); $Mg(OH)_2$ (n = 2); NH_3 (n = 1).

Salt: Total magnitude of positive or negative charge per formula unit.
Examples: NaCl (n = 1); Na_2CO_3 (n = 2); $Al_2(SO_4)_3$ (n = 6).

Ion / Radical: Magnitude of charge on the ion.
Examples: SO_4^{2-} (n = 2); PO_4^{3-} (n = 3); Al^{3+} (n = 3).

Oxidising / Reducing Agent: Net change in oxidation state per molecule.
Examples: $KMnO_4$ in acidic (n = 5), neutral (n = 3), basic (n = 1).
$K_2Cr_2O_7 \rightarrow Cr^{3+}$ (n = 6); $C_2O_4^{2-} \rightarrow 2CO_2$ (n = 2).

8. STOICHIOMETRY, LIMITING REAGENT & YIELD

Stoichiometric Relations

For reaction: $\mathbf{a\,A + b\,B \rightarrow c\,C + d\,D}$

$\mathbf{\text{Moles of A reacted}/_{a} = \text{Moles of B reacted}/_{b} = \text{Moles of C formed}/_{c} = \text{Moles of D formed}/_{d}}$

*Coefficients (a, b, c, d) represent mole, volume (gases), and molecule ratios—**NOT mass ratios**.*

Limiting Reagent (L.R.) Identification

Calculate ratio for each reactant: $\mathbf{\text{Ratio} = \text{Available Moles} / \text{Stoichiometric Coefficient}}$

The reactant with the **MINIMUM ratio** is the Limiting Reagent.

All product yields and excess reagent consumptions MUST be calculated using L.R. moles.

Percentage Purity & Yield

% Purity: $\mathbf{\% \text{ Purity} = \text{Actual Mass of Pure Compound} / \text{Total Mass of Impure Sample} \times 100}$

% Yield: $\mathbf{\% \text{ Yield} = \text{Actual Yield (Experimental)} / \text{Theoretical Yield (Calculated)} \times 100}$

Unknown impurities are assumed inert in stoichiometry calculations.

Empirical & Molecular Formula Determination

Empirical Formula (E.F.): Simplest whole number atom ratio.

Molecular Formula (M.F.): Actual number of constituent atoms.

Relation: $\mathbf{\text{Molecular Formula} = (\text{Empirical Formula})_{n}}$

where $\mathbf{n = \text{Molecular Mass} / \text{Empirical Formula Mass} = \text{GMM} / \text{EFM}}$

9. JEE ADVANCED RELATIONS & SPECIALIZED RESULTS

Average Atomic Mass & Average Molar Mass

Average Atomic Mass: $\mathbf{A_{avg} = (A_1x_1 + A_2x_2 + \ldots) / (x_1 + x_2 + \ldots) = \sum (A_i\,\%_i) / 100}$

Average Molar Mass of Gas Mixture: $\mathbf{M_{avg} = \text{Total Mass} / \text{Total Moles} = (M_1n_1 + M_2n_2 + \ldots) / (n_1 + n_2 + \ldots)}$

Vapour Density of Mixture: $\mathbf{V.D._{mix} = M_{avg} / 2}$

Eudiometry (Gas Analysis Laws) [Added]

- **Combustion of Hydrocarbon:** $\mathbf{C_xH_y + (x + {}^{y}/_{4})\,O_2 \rightarrow x\,CO_{2(g)} + {}^{y}/_{2}\,H_2O_{(l)}}$
- Contraction in volume on combustion: $\mathbf{\Delta V = 1 + {}^{y}/_{4} - x}$ (per unit vol hydrocarbon).
- **Absorbers:** KOH absorbs CO_2, SO_2; Turpentine oil absorbs O_3; Pyrogallol absorbs O_2; Anhydrous $CaCl_2$/$CuSO_4$ absorbs H_2O.

Oleum Labeling (% Free SO_3) [Added]

Oleum = $\mathbf{H_2SO_4 + SO_3}$ (Labelled as 100+x %)

% Labeling = $\mathbf{100 + (\text{g of } H_2O \text{ required for 100g oleum})}$

% Free SO_3: $\mathbf{\%\ \text{Free } SO_3 = 80 \times (x) / 18 = 4.44 \times x}$

% H_2SO_4 initial: $\mathbf{100 - \%\ \text{Free } SO_3}$

Volume Strength of H_2O_2 [Added]

Decomposition: $\mathbf{2\,H_2O_2 \rightarrow 2\,H_2O + O_{2(g)}}$

Volume Strength (V): L of O_2 produced at STP by 1 L H_2O_2.

Molarity (M): $\mathbf{M = \text{Volume Strength} / 11.2}$

Normality (N): $\mathbf{N = \text{Volume Strength} / 5.6}$

% Strength (w/v): $\mathbf{\%\ \text{Strength} = 17 \times \text{Volume Strength} / 56}$

Principle of Atom Conservation (POAC) [Added]

In any reaction, total atoms of element K remain conserved.

$$\mathbf{n_K \text{ in Reactants} = n_K \text{ in Products}}$$

Example: For C in $C_xH_y \rightarrow x\,CO_2$: $x \times n_{C_xH_y} = 1 \times n_{CO_2}$.

Extremely helpful when chemical reaction equation is unbalanced or complex.

Percentage Composition Formulas

% of Element in Compound:

$\mathbf{\%\ \text{Element} = (n \times \text{Atomic Mass of Element}) / \text{Molar Mass of Compound} \times 100}$

Minimum Molar Mass Formula:

$\mathbf{\text{Min Molar Mass} = \text{Atomic Mass of Element} \times 100 / \%\ \text{of Element}}$

Assuming compound contains at least 1 atom of that element.

10. SUMMARY OF KEY PHYSICAL CONSTANTS FOR QUICK CALCULATIONS

Constant	Symbol	Standard Value & Units	Calculation Approximation Shortcut
Avogadro's Constant	N_A	$6.02214 \times 10^{23}\ mol^{-1}$	6×10^{23}
Universal Gas Constant	R	$8.314\ J\ K^{-1}\ mol^{-1} = 0.0821\ L\ atm\ K^{-1}\ mol^{-1} = 2\ cal\ K^{-1}\ mol^{-1}$	$0.0821 \approx {}^{1}/_{12}\ L\ atm$ | $8.314 \approx {}^{25}/_{3}\ J$
Standard Molar Volume	V_m	22.414 L (at 1 atm, 273 K) | 22.7 L (at 1 bar, 273 K)	22.4 L
Mass of Electron	m_e	$9.1 \times 10^{-31}\ kg = 9.1 \times 10^{-28}\ g = 0.000548\ amu$	$9.1 \times 10^{-31}\ kg$
Mass of Proton / Neutron	m_p / m_n	$m_p = 1.6726 \times 10^{-24}\ g$ | $m_n = 1.6749 \times 10^{-24}\ g$	$\approx 1\ amu = 1.66 \times 10^{-24}\ g$

S-BLOCK ELEMENTS — COMPLETE JEE FORMULA & REVISION SHEET

Group 1 (Alkali Metals) & Group 2 (Alkaline Earth Metals) | Comprehensive JEE Main & Advanced Reference

JEE ADVANCED EDITION

1. GENERAL PERIODIC TRENDS & FUNDAMENTAL PROPERTIES

PROPERTY	GROUP 1: ALKALI METALS (NS^1) [LI, NA, K, RB, CS, FR]	GROUP 2: ALKALINE EARTH METALS (NS^2) [BE, MG, CA, SR, BA, RA]
Atomic / Ionic Radii	Largest in respective periods. `Li < Na < K < Rb < Cs` (Atomic & M^+)	Smaller than corresponding Group 1 due to higher Zeff. `Be < Mg < Ca < Sr < Ba` (Atomic & M^{2+})
Ionization Energy (IE)	Lowest in respective periods; decreases down group. `Li > Na > K > Rb > Cs`	Higher than Group 1; decreases down group. `Be > Mg > Ca > Sr > Ba` ($IE_2 > IE_1$)
Density (Anomaly Alert!)	Low density (Li, Na, K lighter than H_2O). `Li (0.53) < K (0.86) < Na (0.97) < Rb < Cs` **Note: K is lighter than Na due to vacant 3d orbital expansion.*	Denser than Group 1 due to smaller size. `Ca (1.55) < Mg (1.74) < Be (1.84) < Sr < Ba` **Minimum at Ca due to crystal packing anomaly.*
Melting & Boiling Points	Low values (weak metallic metallic bonding). `Li > Na > K > Rb > Cs` (Regular decrease)	Higher than Group 1 (closer packing + 2 valence e^-). MP: `Be > Ca > Sr > Ba > Mg` (Mg has lowest MP) BP: `Be > Ba > Ca > Sr > Mg`
Standard Electrode Potential (E°) & Reducing Power	Strong reducing agents (gaseous/aq media). E° value: `Li (-3.04V) > Rb (-2.93V) = Cs > K (-2.92V) > Na (-2.71V)` *Li is strongest reducing agent in aq. solution due to high hydration.*	Weaker reducing agents than Group 1. Reducing Power (aq): `Be < Mg < Ca < Sr < Ba` *E° is lowest (most negative) for Ba/Ca.*
Hydration Enthalpy & Hydrated Radii	$\Delta H_hyd \propto (z/r)$. Order: `Li+ > Na+ > K+ > Rb+ > Cs+` Hydrated Size: `Li+(aq) > Na+(aq) > K+(aq) > Rb+(aq) > Cs+(aq)`	Much higher than Group 1 (M^{2+} charge density). Hydration ΔH: `Be2+ > Mg2+ > Ca2+ > Sr2+ > Ba2+` Hydrated Size: `Be2+(aq) > Mg2+(aq) > Ca2+(aq) > Sr2+(aq) > Ba2+(aq)`
Ionic Mobility & Conductance (aq)	Inversely proportional to hydrated radius. `Li+(aq) < Na+(aq) < K+(aq) < Rb+(aq) < Cs+(aq)`	Inversely proportional to hydrated radius. `Be2+(aq) < Mg2+(aq) < Ca2+(aq) > Sr2+(aq) < Ba2+(aq)`
Flame Colouration	Li: Crimson Red | Na: Golden Yellow | K: Pale Violet | Rb: Violet | Cs: Blue	Be, Mg: No colour (High IE, e^- not excited) Ca: Brick Red | Sr: Crimson Red | Ba: Apple Green | Ra: Crimson

2. CHEMICAL REACTIVITY & CORE REACTIONS

A. Reaction with Oxygen / Air

Group 1: Reactivity increases down group. Thermally stable lattice energy controls product:

- **Li:** Forms Monoxide → `4Li + O2 → 2Li2O` (O^{2-} ion)
- **Na:** Forms Peroxide → `2Na + O2 → Na2O2` (O_2^{2-} ion, diamagnetic)
- **K, Rb, Cs:** Form Superoxides → `M + O2 → MO2` (O_2^- ion, **paramagnetic** & coloured)

Group 2:

- **Be, Mg:** Passivated by thin protective oxide layer. Burn to form monoxide: `2M + O2 → 2MO`.
- **Ca, Sr, Ba:** Form monoxide + peroxide (Ba forms `BaO2` readily at 500°C).

B. Solution in Liquid Ammonia (High-Yield JEE Concept)

Dissolve to give deep blue conducting solutions:

`M + (x+y)NH3 → [M(NH3)_x]+ + [e(NH3)_y]-`

- **Dilute Solution (< 3M):** Paramagnetic, intense **Blue colour** due to light absorption by *ammoniated electrons* (absorbs in Red region).
- **Concentrated Solution (> 3M):** Turns **Bronze / Copper coloured**, diamagnetic (electrons pair up to form e^- clusters).
- **Decomposition on standing:** Catalyzed by Fe/impurities:
  `2M + 2NH3(l) → 2MNH2 (Amide) + H2↑`
- **Anomalous Li behavior:** Li forms imide `Li2NH` on heating with NH_3.

C. Reaction with Water

Gp 1: React violently:
`2M + 2H2O → 2MOH + H2↑`

Reactivity: `Li < Na < K < Rb < Cs`

Gp 2: Be does not react. Mg reacts with steam: `Mg + H2O(g) → MgO + H2`. Ca, Sr, Ba react with cold water:
`M + 2H2O → M(OH)2 + H2↑`

D. Reaction with Nitrogen

Only **Li** (Group 1) and **All Group 2 elements** react directly with N_2 to form nitrides:

`6Li + N2 → 2Li3N`

`3M + N2 → M3N2 (Mg3N2, Be3N2...)`

Hydrolysis of Nitrides gives NH_3:

`Li3N + 3H2O → 3LiOH + NH3↑`

`Mg3N2 + 6H2O → 3Mg(OH)2 + 2NH3↑`

E. Reaction with Hydrogen

All form ionic salt-like hydrides `M+H-` or `M2+(H-)2` on heating (except Be).

Stability of Hydrides: `LiH > NaH > KH > RbH > CsH`

Gp 2 Stability: `BeH2 > MgH2 > CaH2 > SrH2 > BaH2`

Preparation of BeH_2:

`2BeCl2 + LiAlH4 → 2BeH2 + LiCl + AlCl3`

3. GENERAL TRENDS IN COMPOUNDS OF S-BLOCK METALS

COMPOUND TYPE	GROUP 1 (ALKALI METALS) TRENDS	GROUP 2 (ALKALINE EARTH METALS) TRENDS
Hydroxides [MOH / $M(OH)_2$]	**Basic Character:** $LiOH < NaOH < KOH < RbOH < CsOH$ **Solubility:** Increases down group. **Thermal Stability:** Highly stable (except LiOH: $2LiOH \xrightarrow{\Delta} Li_2O + H_2O$).	**Basic Character:** $Be(OH)_2$ [Amphoteric] < $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$ **Solubility:** Increases down group (ΔH_hyd remains constant, ΔH_lat decreases).
Carbonates [M_2CO_3 / MCO_3]	**Thermal Stability:** $Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$ *Li_2CO_3 decomposes easily to $Li_2O + CO_2$. Others are stable to redness.*	**Thermal Stability:** $BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$ $MCO_3 \xrightarrow{\Delta} MO + CO_2\uparrow$ ($BeCO_3$ stored in CO_2 atm). **Solubility:** Decreases down group (Be > Mg > Ca > Sr > Ba).
Sulphates [M_2SO_4 / MSO_4]	Soluble in water (except Li_2SO_4 sparingly soluble). Thermally highly stable.	**Solubility:** $BeSO_4 > MgSO_4 >> CaSO_4 > SrSO_4 > BaSO_4$ *$BeSO_4$ & $MgSO_4$ are highly soluble due to high hydration energy of Be^{2+} & Mg^{2+}. $BaSO_4$ is insoluble white ppt.*
Nitrates [MNO_3 / $M(NO_3)_2$]	**Thermal Decomposition:** • $2MNO_3 \xrightarrow{\Delta} 2MNO_2 + O_2\uparrow$ (M = Na, K, Rb, Cs) • $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2\uparrow + O_2\uparrow$ (Anomalous Li)	**Thermal Decomposition:** All decompose to Oxide + NO_2 + O_2: $2M(NO_3)_2 \xrightarrow{\Delta} 2MO + 4NO_2\uparrow + O_2\uparrow$ Nitrates form hydrates (e.g. $Mg(NO_3)_2 \cdot 6H_2O$).
Halides [MX / MX_2]	High MP/BP ionic solids. Covalent character: LiF < LiCl < LiBr < LiI (Fajans' Rule). LiCl is soluble in pyridine, alcohol, and acetone.	BeX_2 & $MgCl_2$ are covalent/partially covalent. **$BeCl_2$ Structure:** • *Solid:* Polymeric chain with 3-centre-2-electron (3c-2e$^-$) bridge bonds. • *Vapour (<1200 K):* Chloro-bridged dimer Be_2Cl_4. • *Vapour (>1200 K):* Linear monomer Cl-Be-Cl (sp hybridization).

Fajans' Rule Application for s-Block: Small cation + Large anion = High Polarization = High Covalent Character.
• Covalent Order: $LiCl > NaCl > KCl > RbCl > CsCl$ | $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$
• Melting Point Order: $NaCl > KCl > RbCl > CsCl > LiCl$ (LiCl is lowest due to high covalent character).

4. ANOMALOUS BEHAVIOURS & DIAGONAL RELATIONSHIPS

Diagonal Relationship: Lithium (Li) ↔ Magnesium (Mg)

Cause: Similar ionic radii (Li^+ = 76pm, Mg^{2+} = 72pm) and charge/radius ratio.

1. **Hardness & MP:** Both Li & Mg are harder and have higher MP/BP than their group members.
2. **Nitride Formation:** Both react directly with $N_2 \rightarrow$ Li_3N & Mg_3N_2.
3. **Decomposition of Carbonates & Nitrates:**
   $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$ | $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$
   $4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$ |
   $2Mg(NO_3)_2 \xrightarrow{\Delta} 2MgO + 4NO_2 + O_2$
4. **Solubility & Covalency:** Chlorides ($LiCl$, $MgCl_2$) are covalent, deliquescent, crystallize from aq. solution as hydrates ($LiCl \cdot 2H_2O$, $MgCl_2 \cdot 6H_2O$), and dissolve in ethanol/ether.
5. **Hydroxides:** $LiOH$ and $Mg(OH)_2$ are weak bases and sparingly soluble.

Diagonal Relationship: Beryllium (Be) ↔ Aluminium (Al)

Cause: Identical charge/radius ratio (Be^{2+} = 0.31, Al^{3+} = 0.36).

1. **Passivation:** Both are rendered passive by conc. HNO_3 due to oxide film.
2. **Amphoteric Character:** Oxides (BeO, Al_2O_3) and Hydroxides [$Be(OH)_2$, $Al(OH)_3$] are amphoteric.
   $Be(OH)_2 + 2NaOH \rightarrow Na_2[Be(OH)_4]$ (Beryllate)
   $Al(OH)_3 + NaOH \rightarrow Na[Al(OH)_4]$ (Aluminate)
3. **Polymeric/Dimeric Chlorides:** Both $BeCl_2$ and $AlCl_3$ have chloro-bridged structure, are Lewis acids, and dissolve in organic solvents.
4. **Carbide Hydrolysis (Methanides):** Both give **Methane gas** on hydrolysis:
   $Be_2C + 4H_2O \rightarrow 2Be(OH)_2 + CH_4\uparrow$
   $Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4\uparrow$
   (Contrast: $CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$ Acetylene)
5. **Complexation:** Max Coordination Number of Be is 4 ($[BeF_4]^{2-}$, sp^3), whereas Al is 6 ($[AlF_6]^{3-}$, sp^3d^2).

5. IMPORTANT S-BLOCK COMPOUNDS (PREPARATION, REACTIONS & PROPERTIES)

1. Sodium Carbonate (Washing Soda) — $Na_2CO_3 \cdot 10H_2O$

Manufacture: Solvay (Ammonia-Soda) Process

$NH_3 + H_2O + CO_2 \rightarrow NH_4HCO_3$

$NaCl + NH_4HCO_3 \rightarrow NaHCO_3\downarrow$ (sparingly soluble) + NH_4Cl

$2NaHCO_3 \xrightarrow{\Delta\ (250°C)} Na_2CO_3 + H_2O + CO_2\uparrow$

Recovery of NH_3:

$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + 2H_2O$

Important JEE Note:* **K_2CO_3 CANNOT be prepared by Solvay process because $KHCO_3$ is highly soluble in water and does not precipitate out.

Properties: Efflorescent. Monohydrate $Na_2CO_3 \cdot H_2O$ on heating above 373K forms anhydrous Na_2CO_3 (Soda Ash, MP = 852°C).

2. Sodium Hydroxide (Caustic Soda) — NaOH

Manufacture: Castner-Kellner Cell (Hg-cathode electrolysis)

- **Cathode (Hg):** $Na^+ + e^- \xrightarrow{(Hg)} Na\text{-amalgam}$
- **Anode (C):** $2Cl^- \rightarrow Cl_2\uparrow + 2e^-$
- **Decomposer:** $2Na\text{-amalgam} + 2H_2O \rightarrow 2NaOH + 2Hg + H_2\uparrow$

Properties: White, deliquescent solid. Absorbs CO_2 from air to form Na_2CO_3.

3. Sodium Thiosulphate (Hypo) — $Na_2S_2O_3 \cdot 5H_2O$

Preparation Methods:

1. $Na_2SO_3 + S$ (flowers of sulphur) $\xrightarrow{Boil} Na_2S_2O_3$
2. Spring's Reaction: $Na_2SO_3 + Na_2S + I_2 \rightarrow Na_2S_2O_3 + 2NaI$

Key JEE Reactions:

- **Iodometric Titration:** $2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6$ (Sodium Tetrathionate) + $2NaI$
  (Starch indicator: Blue to Colourless).
- **Photography (Fixer Action):** Dissolves unreacted AgBr:
  $AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2]$ (Soluble complex) + $NaBr$
- **$AgNO_3$ Test:** Excess $AgNO_3$ → White ppt turning black:
  $Na_2S_2O_3 + 2AgNO_3 \rightarrow Ag_2S_2O_3\downarrow$ (White) $\xrightarrow{H_2O} Ag_2S\downarrow$ (Black) + H_2SO_4

4. Calcium Compounds: Quicklime, Slaked Lime, POP & Gypsum

A. Quicklime (CaO): $CaCO_3 \xrightarrow{1000°C} CaO + CO_2\uparrow$

• Slaking: $CaO + H_2O \rightarrow Ca(OH)_2 + Heat$

• Basic Flux: $CaO + SiO_2 \rightarrow CaSiO_3$ (Slag)

B. Plaster of Paris (POP) & Gypsum:

$CaSO_4 \cdot 2H_2O$ (Gypsum) $\xrightarrow{120°C} CaSO_4 \cdot \frac{1}{2}H_2O$ (POP) + $1\frac{1}{2}H_2O$

$CaSO_4 \cdot \frac{1}{2}H_2O \xrightarrow{200°C} CaSO_4$ (Dead Burnt Plaster) $\xrightarrow{>400°C} CaO + SO_2 + O_2$

Setting of POP: Exothermic two-stage hydration:
POP → Orthorhombic Gypsum (Setting) → Monoclinic Gypsum (Hardening).

C. Mortar: Slaked Lime : Sand (1 : 3) + Water.

6. QUALITATIVE ANALYTICAL DETECTION & SPECIAL RESULTS

A. Mg^{2+} & Ca^{2+} Qualitative Tests

- **Mg^{2+} Phosphate Test:** Salt solution + NH_4Cl + NH_4OH + Na_2HPO_4 → White crystalline ppt of **Magnesium Ammonium Phosphate**:
  $MgSO_4 + Na_2HPO_4 + NH_4OH \rightarrow Mg(NH_4)PO_4\downarrow$ (White) + $Na_2SO_4 + H_2O$
  Ignition of ppt:
  $2Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7$ (Pyrophosphate) + $2NH_3 + H_2O$
- **Mg^{2+} Charcoal Cavity Test:** Heating salt with $Co(NO_3)_2$ gives a **Pink Mass** ($CoO \cdot MgO$ / Thenard's Pink Added).
- **Ca^{2+} Ammonium Oxalate Test:** $Ca^{2+} + (NH_4)_2C_2O_4 \rightarrow CaC_2O_4\downarrow$ (White ppt, insoluble in acetic acid, soluble in dil. HCl).

B. Hardness of Water & Portland Cement Added High-Yield

Portland Cement Composition:

- **CaO:** 50–60% | **SiO_2:** 20–25% | **Al_2O_3:** 5–10% | **MgO:** 2–3% | **Fe_2O_3:** 1–2% | **SO_3:** 1–2%
- **Important Ratios:** $CaO / (SiO_2 + Al_2O_3 + Fe_2O_3) \approx 2.0$ | $SiO_2 / Al_2O_3 = 2.5$ to 4.0
- **Role of Gypsum (2-3%):** Added to slow down initial setting time of cement (retards hydration of tricalcium aluminate).

Crown Ether Complexation:
18-Crown-6 selectively complexates K^+; 15-Crown-5 complexes Na^+.

SOLID STATE — JEE REVISION FORMULA SHEET

COMPREHENSIVE MASTER SHEET

1. 7 CRYSTAL SYSTEMS & BRAVAIS LATTICES (14)

System	Edge Axes	Angles	Bravais Types
Cubic	a = b = c	α = β = γ = 90°	P, I, F (3)
Tetragonal	a = b ≠ c	α = β = γ = 90°	P, I (2)
Orthorhombic	a ≠ b ≠ c	α = β = γ = 90°	P, I, F, C (4)
Monoclinic	a ≠ b ≠ c	α = γ = 90° ≠ β	P, C (2)
Triclinic	a ≠ b ≠ c	α ≠ β ≠ γ ≠ 90°	P (1)
Hexagonal	a = b ≠ c	α = β = 90°, γ = 120°	P (1)
Rhombohedral	a = b = c	α = β = γ ≠ 90°	P (1)

2. UNIT CELL CONTRIBUTIONS & PARAMETERS

- **Corner atom contribution:** 1/8 per unit cell
- **Face-center atom contribution:** 1/2 per unit cell
- **Body-center atom contribution:** 1 per unit cell
- **Edge-center atom contribution:** 1/4 per unit cell

Unit Cell Type	Z (Atoms/Cell)	Edge & Radius Relation (r)
Simple Cubic (SCC)	1	r = a / 2
Body-Centered (BCC)	2	r = (√3 / 4) a
Face-Centered (FCC / CCP)	4	r = (√2 / 4) a = a / (2√2)

3. PACKING FRACTION (PF) & COORDINATION NUMBER (CN)

Type	CN	Packing Fraction (PF)	Empty Space
Simple Cubic	6	π / 6 ≈ **52.4%**	47.6%
BCC	8	(√3 π) / 8 ≈ **68.0%**	32.0%
FCC / CCP / HCP	12	(√2 π) / 6 ≈ **74.0%**	26.0%

Stacking Sequences: *HCP = ABAB... pattern; FCC/CCP = ABCABC... pattern.*

4. DENSITY OF UNIT CELL (D)

$$d = (Z \times M) / (a^3 \times N_A)$$

Where: *Z = effective atoms per unit cell, M = molar mass (g/mol), a = edge length in cm, $N_A = 6.022 \times 10^{23}\ mol^{-1}$. Units of d: g/cm^3.*

5. INTERPLANAR DISTANCE & X-RAY DIFFRACTION

Bragg's Law: Condition for constructive interference of X-rays reflected from crystal planes.

$$2d \sin \theta = n\lambda$$

Where: *d = interplanar distance, θ = glancing/reflection angle, λ = wavelength, n = order of reflection (1, 2, 3...).*

For cubic crystals, interplanar distance for plane (hkl):

$$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$$

SOLID STATE — JEE REVISION FORMULA SHEET

VOIDS, RADIUS RATIO & IONIC LATTICES

6. VOIDS IN CLOSE-PACKED STRUCTURES (FCC/CCP)

If number of close-packed spheres (atoms) = **N**:

- **Number of Tetrahedral Voids (TV)** = 2N (located on body diagonals, 2 per diagonal × 4 diagonals = 8 in FCC).
- **Number of Octahedral Voids (OV)** = N (1 at body center + 12 at edge centers, each shared by 4 unit cells: 1 + 12×(1/4) = 4 in FCC).
- **Distance from Corner:** TV from corner = $(\sqrt{3}/4)a$; OV from corner = a / 2.

7. RADIUS RATIO RULES (R_+ / R_-)

Radius Ratio (r_+/r_-)	CN	Void Geometry	Example
0.155 – 0.225	3	Planar Triangle	BO_3^{3-}, B_2O_3
0.225 – 0.414	4	Tetrahedral	ZnS, CuCl, AgI
0.414 – 0.732	6	Octahedral	NaCl, MgO, NaBr
0.732 – 1.000	8	Body-Centered Cubic	CsCl, CsBr, TlCl

8. IMPORTANT IONIC CRYSTAL STRUCTURES (AB & AB_2)

Z

Structure	Cation / Anion Positions	CN	
NaCl type (AB)	Cl^- in FCC; Na^+ in All Octahedral Voids	6 : 6	4
CsCl type (AB)	Cl^- at Corners; Cs^+ at Body Center	8 : 8	1
Zinc Blende / Sphalerite (ZnS)	S^{2-} in FCC; Zn^{2+} in 50% Tetrahedral Voids	4 : 4	4
Fluorite (CaF_2) [AB_2]	Ca^{2+} in FCC; F^- in All Tetrahedral Voids	8 : 4	4
Antifluorite (Na_2O) [A_2B]	O^{2-} in FCC; Na^+ in All Tetrahedral Voids	4 : 8	4

Distance Relations: *For NaCl: $r_+ + r_- = a/2$. For CsCl: $r_+ + r_- = (\sqrt{3}/2)\,a$.*

SOLID STATE — JEE REVISION FORMULA SHEET

DEFECTS, PROPERTIES & ADVANCED RELATIONS

9. CRYSTAL DEFECTS (IMPERFECTIONS)

Density Effect

Defect Type	Description / Mechanism	
Schottky Defect	Equal number of cations & anions missing from lattice sites. High ionic character, high CN.	Decreases
Frenkel Defect	Cation leaves lattice site and occupies interstitial site. Large size difference between ions (small cation).	Remains Constant

Note: *AgBr exhibits both Schottky and Frenkel defects.*

Non-Stoichiometric & Impurity Defects:

10. ELECTRICAL, MAGNETIC & DOPING PROPERTIES

Doping in Group 14 Elements (Silicon / Germanium):

- **n-Type Semiconductor:** Doping Group 14 with Group 15 (P, As) creates excess free electrons (negative charge carriers).
- **p-Type Semiconductor:** Doping Group 14 with Group 13 (B, Al) creates electron deficiencies / **holes** (positive charge carriers).

11. MAGNETIC PROPERTIES [ADVANCED]

Presence of unpaired electrons (attracted)Domains aligned parallel in same direction (strong)Domains aligned anti-parallel (net moment = 0)Unequal parallel and anti-parallel domain alignment

- **Metal Excess (F-Center):** Anion vacancy occupied by unpaired electron; imparts color & paramagnetism (e.g., NaCl heated in Na vapor turns yellow).
- **Metal Excess (Interstitial):** Extra cation & electron in interstitial site (e.g., ZnO turns yellow on heating).
- **Metal Deficiency:** Cation missing; extra positive charge on nearby metal ion (e.g., $Fe_{0.98}O$, NiO). Shows p-type semiconductor behavior.

Property	Dipole Alignment	Example
Diamagnetic	All electrons paired (repelled by magnetic field)	NaCl, H_2O, C_6H_6
Paramagnetic	O_2, Cu^{2+}, Fe^{3+}	
Ferromagnetic	Fe, Co, Ni, CrO_2	
Antiferromagnetic	MnO, MnO_2	
Ferrimagnetic	Fe_3O_4, Ferrites	

12. IMPORTANT JEE SHORTCUT / FORMULA SUMMARY

- **Formula Units per Unit Cell (Z):** Calculated via stoichiometry from effective coordination and occupancy.
- **Void Fraction in FCC:** Tetrahedral = 32%, Octahedral = 11.5% of total volume.

SOLUTIONS — JEE MASTER FORMULA SHEET

Physical Chemistry | Complete Main & Advanced Revision Guide

JEE MAIN & ADV

1. Concentration Terms & Interconversions

- **Molarity (M):** $M = \frac{n_{solute}}{V_{solution}\text{ (in L)}} = \frac{W_2 \times 1000}{M_2 \times V\text{ (in mL)}}$ *(Temp. Dependent)*
- **Molality (m):** $m = \frac{n_{solute}}{W_{solvent}\text{ (in kg)}} = \frac{W_2 \times 1000}{M_2 \times W_1\text{ (in g)}}$ *(Temp. Independent)*
- **Mole Fraction (X):** $X_A = \frac{n_A}{n_A + n_B}$, $X_A + X_B = 1$
- **Normality (N):** $N = \frac{\text{Gram Equivalents of Solute}}{V_{solution}\text{ (in L)}} = M \times \text{n-factor}$
- **Formality (F):** $F = \frac{n_{\text{formula mass}}}{V_{solution}\text{ (L)}}$ *(For ionic compounds like NaCl)*
- **Mass % (w/w):** $(W_{solute} / W_{solution}) \times 100$
- **Volume % (v/v):** $(V_{solute} / V_{solution}) \times 100$
- **Mass/Volume % (w/v):** $(W_{solute}\text{ (g)} / V_{solution}\text{ (mL)}) \times 100$
- **Parts Per Million (ppm):** $ppm = \frac{\text{Mass of Solute}}{\text{Total Mass of Solution}} \times 10^6$

Key Interconversion Formulas [Added]:

- **Molality from Mole Fraction:** $m = \frac{X_B \times 1000}{X_A \times M_1}$ (M_1 = molar mass of solvent)
- **Molarity from Molality & Density (d in g/mL):** $M = \frac{1000 \times m \times d}{1000 + m \times M_2}$
- **Molarity from % (w/w) & Density (d):** $M = \frac{10 \times \%\text{ (w/w)} \times d}{M_2}$

2. Solubility & Henry's Law

- **Henry's Law Equation:** $p = K_H \cdot X$

where p = partial pressure of gas in vapour phase, X = mole fraction of gas in liquid, K_H = Henry's law constant (depends on nature of gas & temperature).

- **Temperature Effect on K_H:** As Temperature ↑ ⇒ K_H ↑ ⇒ Solubility (X) ↓.
- **Dissolution Thermodynamics:**
  - **Endothermic ($\Delta_{sol}H > 0$):** Temp ↑ ⇒ Solubility ↑
  - **Exothermic ($\Delta_{sol}H < 0$):** Temp ↑ ⇒ Solubility ↓
  - **Pressure Effect:** Significant only for gas-in-liquid; negligible for solids/liquids.

3. Vapour Pressure, Raoult's Law & Liquid Mixtures

- **Clausius-Clapeyron Equation:**

$$\log \frac{P_2}{P_1} = \frac{\Delta H_{vap}}{2.303\ R} \times \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

- **Raoult's Law (Volatile Solute Pair A & B):**

$P_A = P^\circ_A X_A$, $P_B = P^\circ_B X_B$

Total Pressure (P_T): $P_T = P_A + P_B = P^\circ_A X_A + P^\circ_B X_B = P^\circ_B + (P^\circ_A - P^\circ_B) X_A$

- **Composition in Vapour Phase (Dalton's Law):**

$Y_A = \frac{P_A}{P_T} = \frac{P^\circ_A X_A}{P_T}$; $Y_B = \frac{P_B}{P_T} = \frac{P^\circ_B X_B}{P_T}$

JEE Shortcut — P_T from Vapour Mole Fractions (Y_A, Y_B) [Added]:

$$\frac{1}{P_T} = \frac{Y_A}{P^\circ_A} + \frac{Y_B}{P^\circ_B}$$

- **Raoult's Law as Special Case of Henry's Law:**

When solute is volatile gas, $P_i = K_H X_i$. If $K_H = P^\circ_i$, Henry's Law becomes Raoult's Law.

- **Non-Volatile Solute (B) in Volatile Solvent (A):**

$P_{solution} = P_A = P^\circ_A X_A = P^\circ_A (1 - X_B)$

4. Ideal vs Non-Ideal Solutions & Azeotropes

Property	Ideal Solution	Positive Deviation (+ve)	Negative Deviation (-ve)
Intermolecular Forces	A-B = A-A & B-B	A-B < A-A & B-B (Weaker)	A-B > A-A & B-B (Stronger)
Vapour Pressure	$P_T = P^\circ_A X_A + P^\circ_B X_B$	$P_T > P^\circ_A X_A + P^\circ_B X_B$	$P_T < P^\circ_A X_A + P^\circ_B X_B$
Enthalpy of Mixing (ΔH_{mix})	$\Delta H_{mix} = 0$	$\Delta H_{mix} > 0$ (Endothermic)	$\Delta H_{mix} < 0$ (Exothermic)
Volume of Mixing (ΔV_{mix})	$\Delta V_{mix} = 0$	$\Delta V_{mix} > 0$ (Expansion)	$\Delta V_{mix} < 0$ (Contraction)
Entropy & Gibbs Energy	$\Delta S_{mix} > 0$, $\Delta G_{mix} < 0$	$\Delta S_{mix} > 0$, $\Delta G_{mix} < 0$	$\Delta S_{mix} > 0$, $\Delta G_{mix} < 0$
Azeotrope Formed	None	**Minimum Boiling Azeotrope** (Boiling pt < both)	**Maximum Boiling Azeotrope** (Boiling pt > both)
Standard Examples	n-Hexane + n-Heptane, C_6H_6 + Toluene, C_6H_5Cl + C_6H_5Br	Ethanol + Water, Acetone + Ethanol, CCl_4 + $CHCl_3$, Acetone + CS_2	Acetone + Aniline, HNO_3 + Water, HCl + Water, Acetone + $CHCl_3$

5. Colligative Properties (For Non-Volatile & Non-Electrolyte Solutes)

A. Relative Lowering of Vapour Pressure (RLVP):

$$\frac{P^\circ_A - P_S}{P^\circ_A} = X_B = \frac{n_B}{n_A + n_B}$$

For dilute solution ($n_B << n_A$): $\frac{P^\circ_A - P_S}{P^\circ_A} \approx \frac{n_B}{n_A} = \frac{W_B \cdot M_A}{M_B \cdot W_A}$

B. Elevation of Boiling Point (ΔT_b):

$$\Delta T_b = T_b - T^\circ_b = K_b \times m = \frac{K_b \times W_B \times 1000}{M_B \times W_A(g)}$$

Ebullioscopic Constant Formula:

$$K_b = \frac{R \cdot M_A \cdot (T^\circ_b)^2}{1000 \times \Delta H_{vap}} = \frac{R \cdot (T^\circ_b)^2}{1000 \times L_{vap}}$$

(L_{vap} in cal/g or J/g; ΔH_{vap} in cal/mol or J/mol)

C. Depression in Freezing Point (ΔT_f):

$$\Delta T_f = T^\circ_f - T_f = K_f \times m = \frac{K_f \times W_B \times 1000}{M_B \times W_A(g)}$$

Cryoscopic Constant Formula:

$$K_f = \frac{R \cdot M_A \cdot (T^\circ_f)^2}{1000 \times \Delta H_{fusion}} = \frac{R \cdot (T^\circ_f)^2}{1000 \times L_{fusion}}$$

(L_{fusion} in cal/g or J/g; ΔH_{fusion} in cal/mol or J/mol)

D. Osmotic Pressure (π):

$$\pi = C R T = \frac{n_B}{V} R T = \frac{W_B R T}{M_B V}$$

- **Isotonic Solutions:** $\pi_1 = \pi_2 \Rightarrow C_1 = C_2$ (at same T)
- **Hypertonic:** Solution with higher π compared to another.
- **Hypotonic:** Solution with lower π compared to another.

6. Van 't Hoff Factor (i) & Abnormal Molar Mass

• **Definition of i:**

$$i = \frac{\text{Observed Colligative Property}}{\text{Calculated (Normal) Colligative Property}} = \frac{\text{Normal (Theoretical) Molar Mass}}{\text{Observed (Abnormal) Molar Mass}} = \frac{\text{Total moles after dissociation/association}}{\text{Total initial moles before dissociation/association}}$$

• **Dissociation (i > 1):** $A_n \rightarrow n\,A$

Initial: 1 mole | Equilibrium: $(1 - \alpha) + n\alpha$

$$i = 1 + (n - 1)\,\alpha \;\Rightarrow\; \alpha = \frac{i - 1}{n - 1}$$

(α = degree of dissociation, n = no. of ions produced per molecule)

• **Association (i < 1):** $n\,A \rightarrow (A)_n$

Initial: 1 mole | Equilibrium: $(1 - \alpha) + \frac{\alpha}{n}$

$$i = 1 + \left(\frac{1}{n} - 1\right)\alpha \;\Rightarrow\; \alpha = \frac{1 - i}{1 - (1/n)}$$

(α = degree of association, n = number of polymers formed)

Modified Colligative Property Relations (Using i):

• **RLVP (Dilute & Exact Form):** $\frac{P^\circ_A - P_S}{P^\circ_A} = \frac{i\,n_B}{n_A + i\,n_B}$ | $\frac{P^\circ_A - P_S}{P_S} = \frac{i\,n_B}{n_A}$

• **Elevation of BP:** $\Delta T_b = i \cdot K_b \cdot m$

• **Depression of FP:** $\Delta T_f = i \cdot K_f \cdot m$

• **Osmotic Pressure:** $\pi = i \cdot C \cdot R \cdot T$

7. Advanced JEE Main & Advanced Relations [Added Shortcuts & Results]

• Exact RLVP Relation:

$$\frac{P^\circ_A - P_S}{P_S} = \frac{n_B}{n_A} = \frac{W_B \times M_A}{M_B \times W_A}$$

Use this formula directly for concentrated solutions to avoid approximation error!

• Osmotic Pressure of Mixed Solutions:

$$\pi_{net} = \frac{\pi_1 V_1 + \pi_2 V_2 + \ldots}{V_1 + V_2 + \ldots}$$

For mixing non-reacting solutions without volume change.

• Relation Between ΔT_b & ΔT_f:

$$\frac{\Delta T_b}{\Delta T_f} = \frac{K_b}{K_f}$$

For the same molality solution in the same solvent.

• Abnormal Molar Mass Equation:

$$M_{observed} = \frac{M_{calculated}}{i}$$

• Van 't Hoff Factor for Mixed Solutes:

$$i_{net} = \frac{i_1 n_1 + i_2 n_2 + \ldots}{n_1 + n_2 + \ldots}$$

8. Important Physical Constants & Units for Calculations

Gas Constant (R):

- R = 0.0821 L·atm·K^{-1}·mol^{-1}
- R = 8.314 J·K^{-1}·mol^{-1}
- R = 2 cal·K^{-1}·mol^{-1}

Water Constants:

- K_b (H_2O) = 0.52 K·kg·mol^{-1}
- K_f (H_2O) = 1.86 K·kg·mol^{-1}
- Molar Mass (H_2O) = 18 g/mol

Unit Conversions:

- 1 atm = 760 mmHg = 760 Torr
- 1 atm = 1.013 × 10^5 Pa / N·m^{-2}
- 1 L·atm = 101.3 Joules

JEE MAIN & ADVANCED: SALT ANALYSIS

Comprehensive Master Formula & Reaction Revision Sheet

CHEMISTRY

1. SYSTEMATIC CLASSIFICATION OF BASIC RADICALS (CATIONS)

Group	Group Reagent	Cations	Composition & Colour of Ppt	Solubility / Key Condition
Group I	Dilute HCl	Ag^+, Pb^{2+}, Hg_2^{2+}	$AgCl$ (White) $PbCl_2$ (White) Hg_2Cl_2 (White)	$PbCl_2$ soluble in hot water. K_{sp} exceeded by high $[Cl^-]$.
Group II-A (Insoluble in YAS)	H_2S gas in presence of dilute HCl (pH ~ 0.5 – 1)	Hg^{2+}, Pb^{2+}, Bi^{3+}, Cu^{2+}, Cd^{2+}	HgS (Black), PbS (Black) Bi_2S_3 (Black), CuS (Black) CdS (Yellow)	Low $[S^{2-}]$ due to common ion effect of H^+ suppresses precipitation of Group IV cations.
Group II-B (Soluble in YAS)		As^{3+}, As^{5+}, Sb^{3+}, Sb^{5+}, Sn^{2+}, Sn^{4+}	As_2S_3 / As_2S_5 (Yellow) Sb_2S_3 / Sb_2S_5 (Orange) SnS (Brown), SnS_2 (Yellow)	
Group III	NH_4OH in presence of excess NH_4Cl	Fe^{3+}, Cr^{3+}, Al^{3+}	$Fe(OH)_3$ (Red-Brown) $Cr(OH)_3$ (Dirty Green) $Al(OH)_3$ (White Gelatinous)	NH_4Cl suppresses $[OH^-]$ via common ion effect to prevent precipitation of Group IV & Mg^{2+}.
Group IV	H_2S in presence of NH_4OH + NH_4Cl	Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+}	CoS (Black), NiS (Black) ZnS (Bluish White) MnS (Buff / Flesh)	High $[OH^-]$ increases $[S^{2-}]$ sufficient to exceed K_{sp} of Group IV sulfides.
Group V	$(NH_4)_2CO_3$ in presence of NH_4OH + NH_4Cl	Ba^{2+}, Sr^{2+}, Ca^{2+}	$BaCO_3$ (White) $SrCO_3$ (White) $CaCO_3$ (White)	Medium must be alkaline. NH_4Cl prevents precipitation of $MgCO_3$ (soluble).
Group VI	No common group reagent; Na_2HPO_4 + NH_4OH	Mg^{2+} (also Na^+, K^+)	$Mg(NH_4)PO_4$ (White Crystalline)	Tested in filtrate of Group V. Scratching test tube walls accelerates precipitation.
Group Zero	NaOH (Heat)	NH_4^+	NH_3 gas evolved (pungent)	Forms white fumes with HCl; brown ppt with Nessler's Reagent.

2. DRY TESTS & PRELIMINARY IDENTIFICATION

A. Flame Test (Bunsen Non-Luminous Flame)

- **Li:** Crimson Red | **Na:** Golden Yellow
- **K:** Violet / Lilac (Pinkish through cobalt glass)
- **Ca:** Brick Red | **Sr:** Crimson Red
- **Ba:** Apple Green | **Cu:** Greenish Blue / Sky Blue
- **Pb:** Blue-White [Added]

Rule: Avoid flame test if Heavy Metals (Pb, As, Sb, Bi, Sn) are present as they alloy with Pt wire.

B. Dry Heating Test Observations

- **ZnO:** Yellow when hot, White when cold
- **PbO:** Red-Brown when hot, Yellow when cold
- **Bi_2O_3:** Orange-Yellow when hot, Yellow cold
- **Fe_2O_3:** Dark Red / Brown residue

C. Borax Bead Test Colors: $Na_2B_4O_7 \cdot 10H_2O \rightarrow 2NaBO_2 + B_2O_3$

Metal	Oxidising Flame (Hot/Cold)	Reducing Flame (Hot/Cold)
Cu	Green / Blue	Colourless / Brownish Red
Fe	Brown-Yellow / Pale Yellow	Bottle Green (Hot & Cold)
Cr	Green / Green	Green / Green
Co	Deep Blue (Hot & Cold)	Deep Blue (Hot & Cold)
Mn	Violet / Amethyst Red	Colourless / Colourless
Ni	Violet / Red-Brown	Grey (due to metallic Ni)

3. MICROCOSMIC SALT BEAD TEST [$NA(NH_4)HPO_4 \cdot 4H_2O \rightarrow NAPO_3 + NH_3 \uparrow + H_2O$]

Principle: Orthophosphates formed with metallic oxides display characteristic colors.
• **Cu:** Oxidising: Green (hot), Blue (cold) | Reducing: Red (cold) • **Fe:** Oxidising: Yellow/Red-Brown (hot), Yellow (cold)
• **Cr:** Green (hot & cold in both flames) | • **Co:** Blue (all conditions) | • **SiO_2:** Insoluble; forms a characteristic *silica skeleton* in the bead [Added].

4. SYSTEMATIC CONFIRMATORY TESTS & REACTIONS FOR CATIONS

Group Zero: Ammonium Ion (NH_4^+)

1. **NaOH Test:** $NH_4^+ + OH^- \rightarrow NH_3\uparrow + H_2O$ (Pungent fumes; forms white fumes of NH_4Cl with HCl-dipped rod).
2. **Nessler's Reagent Test:** Alkaline $K_2[HgI_4]$ solution.
$2K_2[HgI_4] + NH_3 + 3KOH \rightarrow H_2N{\cdot}HgO{\cdot}HgI\downarrow$ (Brown Ppt) $+ 7KI + 2H_2O$
(Brown ppt is Iodide of Millon's base).

Group I Cations (Ag^+, Pb^{2+}, Hg_2^{2+})

• **Pb^{2+}:** Ppt dissolves in boiling water; on cooling forms yellow crystals (Golden Spangles).
$Pb^{2+} + 2KI \rightarrow PbI_2\downarrow$ (Yellow); Soluble in excess hot KI.
$Pb^{2+} + K_2CrO_4 \rightarrow PbCrO_4\downarrow$ (Yellow); Soluble in NaOH forming $[Pb(OH)_4]^{2-}$.
• **Ag^+:** $AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2]Cl$ (Soluble) $+ 2H_2O$.
Re-precipitates on acidifying: $[Ag(NH_3)_2]Cl + 2HNO_3 \rightarrow AgCl\downarrow$ (White) $+ 2NH_4NO_3$.
• **Hg_2^{2+}:** $Hg_2Cl_2 + 2NH_4OH \rightarrow Hg\downarrow$ (Black) $+ Hg(NH_2)Cl\downarrow$ (White) $+ NH_4Cl + 2H_2O$.

Group II Cations (II-A: Hg^{2+}, Pb^{2+}, Bi^{3+}, Cu^{2+}, Cd^{2+} | II-B: As^{3+}, Sb^{3+}, $Sn^{2+/4+}$)

• **Hg^{2+}:** Aqua regia dissolves HgS → $HgCl_2$.
$2HgCl_2 + SnCl_2 \rightarrow Hg_2Cl_2\downarrow$ (White) $\rightarrow 2Hg\downarrow$ (Grey/Black).
• **Bi^{3+}:** Hydrolysis Test: $BiCl_3 + H_2O \rightarrow BiOCl\downarrow$ (White Turbidity / Bismuthyl Chloride) $+ 2HCl$.
Sodium Stannite Test: $2Bi(OH)_3 + 3Na_2SnO_2 \rightarrow 2Bi\downarrow$ (Black) $+ 3Na_2SnO_3 + 3H_2O$.
• **Cu^{2+}:** Excess $NH_4OH \rightarrow [Cu(NH_3)_4]^{2+}$ (Deep Blue / Azure Blue solution).
Potassium Ferrocyanide Test: $2Cu^{2+} + [Fe(CN)_6]^{4-} \rightarrow Cu_2[Fe(CN)_6]\downarrow$ (Chocolate Brown Ppt).
• **Cd^{2+}:** In presence of KCN (to mask Cu^{2+} as $[Cu(CN)_4]^{3-}$), pass $H_2S \rightarrow CdS\downarrow$ (Bright Yellow Ppt).
• **As^{3+}:** Dissolve As_2S_3 in conc $HNO_3 \rightarrow H_3AsO_4$. Add Ammonium Molybdate + $HNO_3 \rightarrow (NH_4)_3[As(Mo_{12}O_{40})]\downarrow$ (Yellow Ppt).
• **Sb^{3+}:** Dissolve in conc HCl, add water + $H_2S \rightarrow Sb_2S_3\downarrow$ (Orange Ppt).
• **Sn^{2+}:** Reduces $HgCl_2$ to white Hg_2Cl_2 then grey Hg.

Group III Cations (Fe^{3+}, Cr^{3+}, Al^{3+})

Fe^{3+} Tests:
1. $Fe^{3+} + 3SCN^- \rightarrow [Fe(SCN)]^{2+}$ or $Fe(SCN)_3$ (Blood Red Color).
2. $4Fe^{3+} + 3[Fe(CN)_6]^{4-} \rightarrow Fe_4[Fe(CN)_6]_3\downarrow$ (Prussian Blue Ppt).

Al^{3+} Tests:
1. Dissolves in NaOH: $Al(OH)_3 + OH^- \rightarrow [Al(OH)_4]^-$ (Soluble Sodium Aluminate).
2. Boiling with NH_4Cl re-precipitates $Al(OH)_3\downarrow$ (White Gelatinous).
3. **Lake Test:** $Al(OH)_3$ adsorbs Blue Litmus → Floating Blue Lake.

Cr^{3+} Tests:
1. Fuse ppt with Na_2O_2 / NaOH + $H_2O_2 \rightarrow Na_2CrO_4$ (Yellow Solution).
2. Acidify with CH_3COOH + $Pb(CH_3COO)_2 \rightarrow PbCrO_4\downarrow$ (Yellow Ppt).
3. Chrome Peroxide Test: $CrO_4^{2-} + H_2O_2 + H^+$ + Ether → CrO_5 (Deep Blue Ether Layer).

Group IV Cations (Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+})

• **Ni^{2+} (DMG Test):** Add Dimethylglyoxime + $NH_4OH \rightarrow$ $Ni^{2+} + 2DMG \rightarrow Ni(DMG)_2\downarrow$ (Rosy Red / Scarlet Red Ppt).
Structure has H-bonding, square planar, diamagnetic neutral complex.
• **Co^{2+}:** KNO_2 + CH_3COOH test:
$Co^{2+} + 7HNO_2 + 3K^+ \rightarrow K_3[Co(NO_2)_6]\downarrow$ (Fischer's Salt / Yellow Ppt) $+ NO + 3H^+ + H_2O$.
Vogel's Test: Co^{2+} + NH_4SCN in acetone/ether → $(NH_4)_2[Co(SCN)_4]$ (Deep Blue Color in alcohol layer).
• **Zn^{2+}:** Dissolves in excess NaOH:
$Zn(OH)_2 + 2OH^- \rightarrow [Zn(OH)_4]^{2-}$ (Soluble Zincate).
Pass $H_2S \rightarrow ZnS\downarrow$ (White Ppt).
Potassium Ferrocyanide: $2Zn^{2+} + [Fe(CN)_6]^{4-} \rightarrow Zn_2[Fe(CN)_6]\downarrow$ (White/Bluish White Ppt).
• **Mn^{2+} (Sodium Bismuthate Test):**
$2Mn^{2+} + 5NaBiO_3 + 14H^+ \rightarrow 2MnO_4^-$ (Pink / Purple Permanganate) $+ 5Bi^{3+} + 5Na^+ + 7H_2O$.

Group V Cations (Ba^{2+}, Sr^{2+}, Ca^{2+})

Dissolve carbonates in hot CH_3COOH.

1. **Ba^{2+}:** $Ba^{2+} + K_2CrO_4 \rightarrow BaCrO_4\downarrow$ **(Yellow Ppt)**. Insoluble in CH_3COOH.
2. **Sr^{2+}:** (From filtrate of Ba): $Sr^{2+} + (NH_4)_2SO_4 \rightarrow SrSO_4\downarrow$ **(White Ppt)**.
3. **Ca^{2+}:** (From filtrate of Sr): $Ca^{2+} + (NH_4)_2C_2O_4 \rightarrow CaC_2O_4\downarrow$ **(White Ppt)**. Soluble in dil. HCl.

Group VI Cation (Mg^{2+})

1. **Magneson Reagent Test:** $Mg(OH)_2$ adsorbs Magneson dye in alkaline medium → **Blue Precipitate / Lake**.
2. **Phosphate Test:** $Mg^{2+} + NH_4OH + Na_2HPO_4 \rightarrow Mg(NH_4)PO_4\downarrow$ **(White Crystalline Ppt)** $+ 2Na^+ + H_2O$.

5. ANALYSIS OF ACID RADICALS (ANIONS)

A. Dilute H_2SO_4 / Dilute HCl Group (Volatile Products)

Anion	Evolved Gas / Observation	Confirmatory Reaction & Equations
CO_3^{2-} (Carbonate)	$CO_2\uparrow$ (Brisk effervescence, colourless, odourless)	Turns Lime Water milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow$ **(White)** $+ H_2O$. Excess CO_2 turns it clear: $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$ **(Soluble)**.
SO_3^{2-} (Sulphite)	$SO_2\uparrow$ (Suffocating odour of burning sulphur)	Turns acidified $K_2Cr_2O_7$ paper green: $K_2Cr_2O_7 + 3SO_2 + H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3$ **(Green)** $+ H_2O$. $BaCl_2$ gives white ppt of $BaSO_3$ soluble in dil HCl.
S^{2-} (Sulphide)	$H_2S\uparrow$ (Rotten egg smell)	1. Turns Lead Acetate paper black: $Pb(CH_3COO)_2 + H_2S \rightarrow PbS\downarrow$ **(Black)** $+ 2CH_3COOH$. 2. **Sodium Nitroprusside Test:** $S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}$ **(Purple / Violet Color)**.
NO_2^- (Nitrite)	$NO_2\uparrow$ (Reddish-brown pungent gas)	1. Turns starch-iodide paper blue: $2NO_2^- + 2I^- + 4H^+ \rightarrow I_2 + 2NO + 2H_2O$. 2. Acidified $FeSO_4$ gives dark brown/black color instantly without conc. H_2SO_4.
CH_3COO^- (Acetate)	$CH_3COOH\uparrow$ (Vinegar-like smell)	**Neutral $FeCl_3$ Test:** $3CH_3COO^- + Fe^{3+} \rightarrow (CH_3COO)_3Fe$ **(Blood Red Color)**. Boiling gives reddish-brown ppt: $(CH_3COO)_3Fe + 2H_2O \rightarrow Fe(OH)_2(CH_3COO)\downarrow + 2CH_3COOH$.

B. Concentrated H_2SO_4 Group

Anion	Evolved Gas / Observation	Confirmatory Reaction & Key Equations
Cl^- (Chloride)	$HCl\uparrow$ (Pungent fumes, white fumes with NH_4OH)	1. $AgNO_3$ Test: $Ag^+ + Cl^- \rightarrow AgCl\downarrow$ **(White Curdy)**; Soluble in NH_4OH. 2. **Chromyl Chloride Test (Crucial for JEE):** Solid Cl^- + $K_2Cr_2O_7$ + conc. H_2SO_4 (heat) → $CrO_2Cl_2\uparrow$ **(Reddish-Brown Vapours)**. Pass into NaOH → Na_2CrO_4 **(Yellow Sol)**. Add CH_3COOH + $Pb(CH_3COO)_2$ → $PbCrO_4\downarrow$ **(Yellow Ppt)**. *Note: Test negative for Hg_2Cl_2, $HgCl_2$, AgCl, $PbCl_2$, $SnCl_2$ (covalent/insoluble chlorides).*
Br^- (Bromide)	$Br_2\uparrow$ (Reddish-brown fumes, intensifies with MnO_2)	1. $AgNO_3$ Test: $AgBr\downarrow$ **(Pale Yellow)**; Sparingly soluble in NH_4OH. 2. **Layer Test:** Salt sol + Cl_2 water + CS_2 / CCl_4 → Br_2 dissolves giving **Orange-Red Organic Layer**.
I^- (Iodide)	$I_2\uparrow$ (Violet vapours, turns starch paper blue)	1. $AgNO_3$ Test: $AgI\downarrow$ **(Bright Yellow)**; Insoluble in NH_4OH. 2. **Layer Test:** Salt sol + Cl_2 water + CCl_4 → I_2 dissolves giving **Violet / Purple Organic Layer**.

Anion	Evolved Gas / Observation	Confirmatory Reaction & Key Equations
NO_3^- (Nitrate)	$NO_2\uparrow$ (Brown fumes, intensifies on adding Cu turnings)	**Brown Ring Test:** Aqueous salt solution + freshly prepared $FeSO_4$ + conc. H_2SO_4 carefully along test tube wall → **$[Fe(H_2O)_6]^{2+} + NO \rightarrow [Fe(H_2O)_5(NO)]^{2+}$ (Brown Ring at Junction) + H_2O**. *Fe oxidation state in brown ring complex is **+1** ($t_{2g}^5 e_g^2$, paramagnetic, μ = 3.87 BM).*

C. Group 'B' Independent Anions (No gas with Acids)

1. Sulphate (SO_4^{2-}):

• **$BaCl_2$ Test:** **$SO_4^{2-} + BaCl_2 \rightarrow BaSO_4\downarrow$ (White Ppt)**.
Insoluble in conc. HCl and conc. HNO_3.

• **Lead Acetate Test:** **$SO_4^{2-} + Pb(CH_3COO)_2 \rightarrow PbSO_4\downarrow$ (White Ppt)**.
Soluble in ammonium acetate solution: **$PbSO_4 + 2CH_3COONH_4 \rightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$**.

2. Phosphate (PO_4^{3-}):

• **Ammonium Molybdate Test:** Salt solution + conc. HNO_3 + Ammonium Molybdate → Heat → **$PO_4^{3-} + 12(NH_4)_2MoO_4 + 23H^+ \rightarrow (NH_4)_3[P(Mo_{12}O_{40})]\downarrow$ (Canary Yellow Ppt) + $21NH_4^+ + 12H_2O$**.

• **Magnesia Mixture Test:** $MgSO_4 + NH_4Cl + NH_4OH \rightarrow$ **$Mg(NH_4)PO_4\downarrow$ (White Crystalline Ppt)**.

6. CRITICAL SOLUBILITY RULES & JEE ADVANCED MEMORY SHORTCUTS

General Solubility Rules:

1. All Nitrates (NO_3^-), Acetates (CH_3COO^-), and Ammonium (NH_4^+) salts are soluble in water.
2. All Chlorides soluble EXCEPT AgCl, Hg_2Cl_2, CuCl, BiOCl, SbOCl; $PbCl_2$ (soluble in hot water).
3. All Sulphates soluble EXCEPT $BaSO_4$, $PbSO_4$, $SrSO_4$ (Ag_2SO_4, $CaSO_4$ sparingly soluble).
4. All Carbonates, Sulphites, Phosphates insoluble EXCEPT Group 1 metals & NH_4^+.

Key Precipitate Colors Summary:

• **Black sulfides:** HgS, PbS, CuS, Bi_2S_3, NiS, CoS, Ag_2S.
• **Yellow sulfides:** CdS, As_2S_3, SnS_2.
• **Orange sulfide:** Sb_2S_3 | **Brown sulfide:** SnS.
• **White sulfide:** ZnS | **Buff / Flesh sulfide:** MnS.
• **Chromates:** $BaCrO_4$ (Yellow), $PbCrO_4$ (Yellow), Ag_2CrO_4 (Brick Red).

REDOX REACTIONS & STOICHIOMETRY (JEE MAIN / ADVANCED)

QUICK REVISION FORMULA BOOKLET

1. FUNDAMENTAL DEFINITIONS & CORE RELATIONS

Classical vs. Electronic Definitions

- **Oxidation:** Addition of O/electronegative element, removal of H/electropositive element, or *Loss of Electrons* (De-electronation).
- **Reduction:** Addition of H/electropositive element, removal of O/electronegative element, or *Gain of Electrons* (Electronation).
- **Oxidising Agent (OA / Oxidant):** Accepts e^-, undergoes reduction, O.N. decreases.
- **Reducing Agent (RA / Reductant):** Donates e^-, undergoes oxidation, O.N. increases.

Fundamental Concentration Measures

- **Molarity (M):** $M = n / V_{(L)} = (w / M_w) \times (1000 / V_{(mL)})$
- **Normality (N):** $N = Eq / V_{(L)} = (w / E_w) \times (1000 / V_{(mL)})$
- **Relation:** $Normality\ (N) = Molarity\ (M) \times n\text{-}factor$
- **No. of Equivalents (Eq):** $Eq = w / E_w = N \times V_{(L)} = n \times (n\text{-}factor)$
- **No. of Milli-equivalents (mEq):** $mEq = N \times V_{(mL)} = mmoles \times (n\text{-}factor)$

2. RULES FOR ASSIGNING OXIDATION NUMBER (O.N.)

- **Free/Elemental state:** O.N. = 0 (e.g., O_2, S_8, P_4, Fe, O_3).
- **Monoatomic ions:** O.N. = Charge on ion (e.g., Na^+ = +1, Mg^{2+} = +2, Al^{3+} = +3, Cl^- = −1).
- **Fluorine:** Always −1 in all compounds.
- **Hydrogen:** Generally +1 (with non-metals); −1 in binary metal hydrides (e.g., NaH, CaH_2, $LiAlH_4$).
- **Alkali Metals (Grp 1):** Always +1. **Alkaline Earth (Grp 2):** Always +2.
- **Oxygen:**
  - General oxides: −2
  - Peroxides (O_2^{2-}, −O−O− bond): −1 (e.g., H_2O_2, Na_2O_2, BaO_2)
  - Superoxides (O_2^-): −1/2 (e.g., KO_2, RbO_2)
  - Ozonides (O_3^-): −1/3 (e.g., KO_3) [Added]
  - Fluorides: In OF_2 (O = +2); In O_2F_2 (O = +1).
- **Sum of O.N.:** Neutral molecule = 0; Polyatomic ion = net charge.

Special Compounds with Bonding Anomalies & Average vs. Fractional O.N. (JEE Favorite)

Compound Name & Formula	Structural Representation / Bond Info	Individual Atom O.N.	Average O.N. & JEE Note
Caro's Acid H_2SO_5 (Peroxomonosulfuric acid)	H−O−S(=O)$_2$−O−O−H (Contains 1 peroxide bond: −O−O−)	S = +6 3 Oxide O = −2, 2 Peroxide O = −1	**Average O for S = +6** (Not +8, as S max valency = 6). Formula approach gives +8 which is wrong.
Marshall's Acid $H_2S_2O_8$ (Peroxodisulfuric acid)	H−O−S(=O)$_2$−O−O−S(=O)$_2$−O−H (Contains 1 peroxide bridge)	Both S = +6 6 Oxide O = −2, 2 Peroxide O = −1	**Average O.N. of S = +6** (Formula yields +7, invalid).
Chromium Peroxide CrO_5 (Butterfly structure)	Cr(=O)$(O_2)_2$ (Contains 2 peroxide linkages: 4 peroxide O)	Cr = +6 1 Oxide O = −2, 4 Peroxide O = −1	**O.N. of Cr = +6** (Deep blue complex in ether). Formula gives +10 (wrong).
Bleaching Powder $CaOCl_2$ or Ca(OCl)Cl	Ca^{2+} $[Cl]^-$ $[O-Cl]^-$	Direct Chloride Cl = −1 Hypochlorite OCl^- Cl = +1	**Average O.N. of Cl = 0**. Contains chlorine in two different oxidation states (−1 and +1).
Carbon Suboxide C_3O_2	O = C = C = C = O	Terminal C = +2, +2 Central C = 0	**Average O.N. of C = +4/3**. Fractional O.N. arises due to structural non-equivalence.
Tribromooctaoxide Br_3O_8	$O_3Br - Br(=O)_2 - BrO_3$	Terminal Br = +6, +6 Central Br = +4	**Average O.N. of Br = +16/3**. Formula gives +16/3 directly.
Tetrathionate Ion $S_4O_6^{2-}$	$^-O_3S - S - S - SO_3^-$	Terminal S = +5, +5 Central S (S−S bond) = 0, 0	**Average O.N. of S = +2.5** (+5/2). S−S linkages contribute 0 to O.N.

Compound Name & Formula	Structural Representation / Bond Info	Individual Atom O.N.	Average O.N. & JEE Note
Mixed Oxides Fe_3O_4, Pb_3O_4, Mn_3O_4	$Fe_3O_4 = FeO \cdot Fe_2O_3$ $Pb_3O_4 = 2PbO \cdot PbO_2$	Fe: +2 (1 atom), +3 (2 atoms) Pb: +2 (2 atoms), +4 (1 atom)	**Avg Fe = +8/3** **Avg Pb = +8/3**

3. BALANCING REDOX REACTIONS (SYSTEMATIC STEPS)

Ion-Electron (Half-Reaction) Method

1. Split reaction into Oxidation Half and Reduction Half.
2. Balance atoms other than H and O.
3. Balance O atoms by adding **H_2O** molecules.
4. Balance H atoms:
   - **Acidic Medium:** Add **H^+** to deficient side.
   - **Basic Medium:** Add equal **OH^-** to both sides for every H^+ added, forming H_2O.
5. Balance charge by adding electrons (e^-).
6. Multiply half-reactions by integers to equalize e^- and add.

Oxidation Number Change Method

1. Identify atoms undergoing O.N. change.
2. Calculate total increase and decrease in O.N. per formula unit.
3. Multiply species by suitable factors to make net O.N. increase = net O.N. decrease.
4. Balance all other atoms except H and O.
5. Balance charge using **H^+** (acidic) or **OH^-** (basic).
6. Balance O and H using **H_2O**.

4. CALCULATION OF N-FACTOR (VALENCY FACTOR) — COMPREHENSIVE RULES

Equivalents (Eq) = Moles (n) × n-factor | Equivalent Weight (E_w) = Molar Mass (M_w) / n-factor

Category	Formula / Rule	Examples & Key Remarks
Acids	**n-factor = Basicity** (No. of replaceable H^+ ions per acid molecule)	$HCl \rightarrow 1$ | $H_2SO_4 \rightarrow 2$ | $HNO_3 \rightarrow 1$ H_3PO_4 (Orthophosphoric) → 3 H_3PO_3 (Phosphorous) → 2 (1 H linked to P) H_3PO_2 (Phosphinic/Hypophosphorous) → 1 (2 H linked to P) H_3BO_3 (Boric acid) → 1 (Monobasic Lewis acid in H_2O)
Bases	**n-factor = Acidity** (No. of replaceable OH^- ions per base molecule)	$NaOH \rightarrow 1$ | $Ca(OH)_2 \rightarrow 2$ | $Al(OH)_3 \rightarrow 3$ $NH_4OH \rightarrow 1$
Salts (Non-redox)	**n-factor = Total positive or negative charge** per formula unit	$NaCl \rightarrow 1$ (Na^+) $Na_2CO_3 \rightarrow 2$ ($2\times Na^+ = 2$) $Al_2(SO_4)_3 \rightarrow 6$ ($2\times Al^{3+} = 6$ or $3\times SO_4^{2-} = 6$) $KAl(SO_4)_2\cdot 12H_2O$ (Potash Alum) → 8 [Added]
Ions	**n-factor = Magnitude of charge on ion**	$SO_4^{2-} \rightarrow 2$ | $PO_4^{3-} \rightarrow 3$ | $Al^{3+} \rightarrow 3$ | $CO_3^{2-} \rightarrow 2$
Oxidising / Reducing Agents (Single element changes O.N.)	**n-factor = |Change in O.N. per atom| × (No. of such atoms in 1 molecule)**	$KMnO_4$ (Acidic): $Mn^{+7} \rightarrow Mn^{+2} \Rightarrow \|7-2\| \times 1 = \mathbf{5}$ $K_2Cr_2O_7$ (Acidic): $2Cr^{+6} \rightarrow 2Cr^{+3} \Rightarrow \|6-3\| \times 2 = \mathbf{6}$ $FeSO_4$: $Fe^{+2} \rightarrow Fe^{+3} \Rightarrow \|3-2\| \times 1 = \mathbf{1}$ $H_2C_2O_4$: $2C^{+3} \rightarrow 2C^{+4} \Rightarrow \|4-3\| \times 2 = \mathbf{2}$
Salts where multiple elements undergo O.N. change in SAME direction	**n-factor = $\|Change_1\| \times n_1 + \|Change_2\| \times n_2$**	**$FeC_2O_4 \rightarrow Fe^{3+} + 2CO_2$** $Fe^{+2} \rightarrow Fe^{+3}$ ($\Delta = 1 \times 1 = 1$) $2C^{+3} \rightarrow 2C^{+4}$ ($\Delta = 1 \times 2 = 2$) **Total n-factor = 1 + 2 = 3**
Salts where elements change O.N. in OPPOSITE directions	**n-factor = |Net change in electrons lost or gained per mole|**	**$KClO_3 \rightarrow KCl + {}^3/_2O_2$** $Cl^{+5} + 6e^- \rightarrow Cl^{-1}$ (gains $6e^-$) $3O^{-2} \rightarrow 3O^0 + 6e^-$ (loses $6e^-$) **Total n-factor = 6**

Category	Formula / Rule	Examples & Key Remarks
Disproportionation Reactions (Same element oxidised & reduced)	$1/n_{net} = (1/n_1) + (1/n_2)$ $\Rightarrow n_{net} = (n_1 \times n_2)/(n_1 + n_2)$	**$2H_2O_2 \rightarrow 2H_2O + O_2$** Oxidation half: $H_2O_2 \rightarrow O_2$ ($n_1 = 2$) Reduction half: $H_2O_2 \rightarrow H_2O$ ($n_2 = 2$) $n_{net} = (2 \times 2)/(2 + 2) = 1$ **$P_4 \rightarrow PH_3 + H_2PO_2^-$: $n_{net} = 3/2 = 1.5$** [Added]

5. STANDARD MEDIA-DEPENDENT N-FACTORS (MUST REMEMBER FOR JEE)

$KMnO_4$ (Potassium Permanganate)

- **Acidic Medium (pH < 4):**
  $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$
  O.N. change: +7 → +2 | **n-factor = 5** | $E_w = M/5$
- **Neutral / Faintly Alkaline Medium (pH 7–9):**
  $MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2(s) + 4OH^-$
  O.N. change: +7 → +4 | **n-factor = 3** | $E_w = M/3$
- **Strongly Alkaline Medium (pH > 12):**
  $MnO_4^- + e^- \rightarrow MnO_4^{2-}$ *(Manganate)*
  O.N. change: +7 → +6 | **n-factor = 1** | $E_w = M/1$

$K_2Cr_2O_7$ (Potassium Dichromate)

- **Acidic Medium ONLY:**
  $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$
  O.N. change: 2 × (+6 → +3) | **n-factor = 6** | $E_w = M/6$
- **In basic medium:** Dichromate converts to chromate:
  $Cr_2O_7^{2-} + 2OH^- \rightleftharpoons 2CrO_4^{2-} + H_2O$
  (No redox change; O.N. remains +6).

6. LAW OF CHEMICAL EQUIVALENCE & TITRATION FORMULAS

Law of Equivalence: In any chemical reaction, equivalents of all reactants reacted are EQUAL to equivalents of all products formed.
For reaction: $aA + bB \rightarrow cC + dD$ ⇒ Eq of A = Eq of B = Eq of C = Eq of D

1. Acid-Base Titration Neutralization:
Eq of Acid = Eq of Base
$N_1V_1 = N_2V_2 \Rightarrow M_1V_1 \times (\text{n-factor})_1 = M_2V_2 \times (\text{n-factor})_2$

2. Redox Titration Equivalence:
Eq of Oxidising Agent (OA) = Eq of Reducing Agent (RA)
$(N \times V)_{OA} = (N \times V)_{RA} \Rightarrow (n \times \text{n-factor})_{OA} = (n \times \text{n-factor})_{RA}$

3. Resultant Normality of Mixture of non-reacting acids/bases:
$N_{res} = (N_1V_1 + N_2V_2 + N_3V_3 + ...)/(V_1 + V_2 + V_3 + ...)$

7. SUMMARY OF STANDARD REDOX TITRATIONS (EXCLUDING IODOMETRY)

Analyte (RA)	Titrant (OA)	Balanced Half Reactions & Full Stoichiometry	Molar Ratio (RA : OA)	Equivalent Weight
Fe^{2+} (Ferrous)	**MnO_4^-** (Acidic)	$Fe^{2+} \rightarrow Fe^{3+} + e^-$ $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ **Net: $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$**	5 : 1	$E_{Fe^{2+}} = M/1$ $E_{MnO_4^-} = M/5$
Fe^{2+} (Ferrous)	**$Cr_2O_7^{2-}$** (Acidic)	$Fe^{2+} \rightarrow Fe^{3+} + e^-$ $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ **Net: $6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$**	6 : 1	$E_{Fe^{2+}} = M/1$ $E_{Cr_2O_7^{2-}} = M/6$
$C_2O_4^{2-}$ (Oxalate) / $H_2C_2O_4$	**MnO_4^-** (Acidic)	$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$ $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ **Net: $5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$**	5 : 2	$E_{C_2O_4^{2-}} = M/2$ $E_{MnO_4^-} = M/5$
H_2O_2 (Hydrogen Peroxide)	**MnO_4^-** (Acidic)	$H_2O_2 \rightarrow 2H^+ + O_2 + 2e^-$ $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ **Net: $5H_2O_2 + 2MnO_4^- + 6H^+ \rightarrow 5O_2 + 2Mn^{2+} + 8H_2O$**	5 : 2	$E_{H_2O_2} = M/2$ $E_{MnO_4^-} = M/5$
As_2O_3 (Arsenious oxide)	**MnO_4^-** (Acidic)	$As_2O_3 + 5H_2O \rightarrow 2AsO_4^{3-} + 10H^+ + 4e^-$ **Net Equivalence: Eq(As_2O_3) = Eq(MnO_4^-)**	5 : 4	$E_{As_2O_3} = M/4$ $E_{MnO_4^-} = M/5$

Analyte (RA)	Titrant (OA)	Balanced Half Reactions & Full Stoichiometry	Molar Ratio (RA : OA)	Equivalent Weight
AsO_3^{3-} (Arsenite)	**BrO_3^-** (Bromate)	$AsO_3^{3-} + H_2O \rightarrow AsO_4^{3-} + 2H^+ + 2e^-$ (n=2) $BrO_3^- + 6H^+ + 6e^- \rightarrow Br^- + 3H_2O$ (n=6)	3 : 1	$E_{AsO_3^{3-}} = M / 2$ $E_{BrO_3^-} = M / 6$

8. IODIMETRIC & IODOMETRIC TITRATIONS (HIGH-YIELD JEE ADVANCED TOPIC)

Iodimetric Titration (Direct Method)

Direct titration of a reducing agent (RA) with a standard solution of free Iodine (I_2) in neutral/faintly acidic medium.

> **Standard Titrant: I_2 solution ($I_2 + 2e^- \rightarrow 2I^-$, n-factor = 2)**

Common Estimations:

- **H_2S:** $H_2S + I_2 \rightarrow S + 2I^- + 2H^+ \Rightarrow E_{H_2S} = M / 2$
- **SO_3^{2-}:** $SO_3^{2-} + I_2 + H_2O \rightarrow SO_4^{2-} + 2I^- + 2H^+ \Rightarrow E = M / 2$
- **Sn^{2+}:** $Sn^{2+} + I_2 \rightarrow Sn^{4+} + 2I^- \Rightarrow E_{Sn^{2+}} = M / 2$
- **As(III) [pH = 8]:** $H_2AsO_3^- + I_2 + H_2O \rightarrow HAsO_4^{2-} + 2I^- + 3H^+ \Rightarrow E = M / 2$
- **N_2H_4 (Hydrazine):** $N_2H_4 + 2I_2 \rightarrow N_2 + 4H^+ + 4I^- \Rightarrow E = M / 4$

Iodometric Titration (Indirect Method)

An oxidising agent (OA) reacts with excess KI to liberate I_2. Liberated I_2 is then titrated against standard Hypo ($Na_2S_2O_3$) using starch indicator.

> **Hypo Reaction: $I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$**
> **($2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2e^- \Rightarrow$ n-factor of Hypo = 1)**

Fundamental Equivalences:

> **Eq of OA = Eq of Liberated I_2 = Eq of Hypo ($Na_2S_2O_3$)**

Key Estimations:

- **$CuSO_4$ (Cu^{2+}):**
  $2Cu^{2+} + 4I^- \rightarrow Cu_2I_2\downarrow$ (white) + I_2
  $2CuSO_4 \equiv I_2 \equiv 2Na_2S_2O_3 \Rightarrow$ **$E_{CuSO_4} = M / 1$**
- **MnO_2:** $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$
  $Cl_2 + 2I^- \rightarrow I_2 + 2Cl^- \Rightarrow$ **$E_{MnO_2} = M / 2$**
- **$CaOCl_2$ (Bleaching Powder / Available Cl_2):**
  $CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$
  $Cl_2 + 2I^- \rightarrow I_2 + 2Cl^- \Rightarrow$ **$E_{CaOCl_2} = M / 2$**
- **IO_3^- (Iodate):** $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$
  $IO_3^- \equiv 3I_2 \equiv 6Na_2S_2O_3 \Rightarrow$ **$E_{IO_3^-} = M / 6$**

9. VOLUME STRENGTH OF H_2O_2 & OLEUM LABELLING [ADDED JEE STANDARD]

Volume Strength of H_2O_2

Decomposition: $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$ at STP

- **Volume Strength (V):** Volume of O_2 in litres produced at STP from 1 L of H_2O_2 solution.
- **Relation with Molarity (M):** ***Volume Strength = 11.2 × M*** (using 22.4 L/mol) or **11.35 × M** (using 22.7 L bar/mol).
- **Relation with Normality (N):** ***Volume Strength = 5.6 × N***
- **Percentage Strength (% w/v):** *% Strength = (Volume Strength × 68) / 224 =* ***0.3035 × Volume Strength***
- **Strength (g/L):** *Strength (g/L) = N × E_w = N × 17 = M × 34*

Oleum Labelling (% Free SO_3) & Hardness of Water

- **Oleum ($H_2S_2O_7 = H_2SO_4 + SO_3$):**
  Labelled as *(100 + x)%* oleum.
  % Free SO_3 = [(x / 18) × 80] = ***(80 / 18) × (Oleum% − 100)***
  Where x = mass of H_2O required to react completely with free SO_3 in 100g sample.
- **Hardness of Water:** Expressed in ppm of $CaCO_3$ equivalents.
  Hardness (ppm) = (Mass of $CaCO_3$ eq / Total mass of water) × 10^6
  1 mol Ca^{2+} / Mg^{2+} / $Ca(HCO_3)_2$ ≡ 1 mol $CaCO_3$ (100g).

10. LAST-MINUTE JEE SHORTCUTS & COMMON EXAM PITFALLS

1. Equivalent Weight in Acidic vs Basic:
Always check medium! MnO_4^- has n-factor = 5 (acidic), 3 (neutral/faint basic), 1 (strongly basic).

2. Starch Indicator Addition:
In iodometry, add starch near the end point (pale yellow I_2) to prevent starch-iodine complex trapping I_2 permanently.

3. Disproportionation n-factor:
Do not add individual O.N. changes linearly! Use harmonic inverse rule: $1/n_{net} = 1/n_1 + 1/n_2$.

ORGANIC ANALYSIS & PURIFICATION

JEE Main & JEE Advanced Formula Revision Sheet

CHEMISTRY REVISION

1. PURIFICATION METHODS OF ORGANIC COMPOUNDS

Method	Core Principle & Conditions	Standard JEE Examples
Sublimation	Direct transition: Solid ⇌ Vapor without passing through liquid state. Impurities are non-sublimable.	Naphthalene, Camphor, Iodine, Ammonium chloride (NH_4Cl), Benzoic acid, Anthracene.
Crystallisation	Difference in solubility of compound and impurities in a solvent. Compound sparingly soluble at room temp, highly soluble at high temp.	Purification of Sugar, Potash Alum, Benzoic acid from water. *Fractional:* Urea + $CuSO_4$.
Simple Distillation	Large difference in boiling points (**Δb.p. > 40°C or > 30 K**). Liquids must be stable at b.p.	Chloroform (b.p. 334 K) + Aniline (b.p. 458 K); Ether (b.p. 308 K) + Toluene (b.p. 384 K).
Fractional Distillation	Small difference in boiling points (**Δb.p. < 10–15°C / 5–10°C**). Uses fractionating column.	Acetone (329 K) + Methyl Alcohol (338 K); Crude petroleum fractions; Benzene + Toluene.
Vacuum Distillation (Reduced Pressure)	Used for liquids that **decompose at or below their boiling points**. Lowering pressure lowers b.p.	**Glycerol** (boils at 290°C with decomposition; distils undamaged at 180°C under 12 mmHg), Concentrated H_2SO_4.
Steam Distillation	For substances: (1) Insoluble/immiscible in water, (2) Steam volatile, (3) Contain non-volatile impurities.	**Aniline**, *o*-Nitrophenol (steam volatile) vs *p*-Nitrophenol, Nitrobenzene, Essential oils.
Differential Extraction	Based on organic compound being **much more soluble in organic solvent** than in water. Uses separatory funnel.	Extracting Benzoic acid from aqueous solution using Benzene or Ether.

Steam Distillation - Vapour Pressure Relation

$$P = P_w + P_o = P_{atm}$$

Boiling occurs when sum of partial pressures equals atmospheric pressure P (hence mixture boils at T < b.p. of water or organic liquid).

Ratio of Masses in Distillate:

$$w_o / w_w = (P_o \times M_o) / (P_w \times M_w)$$

w = mass, P = vapor pressure at distillation temp, M = molar mass (M_w = 18 g/mol).

Differential Extraction Formula Added

$$W_n = W \times [\,(K \cdot V) / (K \cdot V + V_s)\,]^n$$

W = initial mass of solute in aqueous layer, W_n = mass remaining in water after *n* extractions, V = volume of aqueous layer, V_s = volume of organic solvent per extraction, K = partition/distribution coefficient = $C_{water} / C_{organic}$.

Note: Multiple extractions with smaller solvent volumes are more efficient than single extraction with full volume.

2. CHROMATOGRAPHY PRINCIPLES & FORMULAS

Classification of Chromatographic Techniques:

- **Adsorption Chromatography (Solid Stationary Phase):**
  - *Column Chromatography:* Bulk separation; stationary phase (silica gel/alumina) in glass tube. Highly adsorbed components retained near top.
  - *Thin Layer Chromatography (TLC): Glass plate coated with adsorbent. Used for qualitative purity check & tracking reactions.*
- ***Partition Chromatography (Liquid Stationary Phase):***
  - *Paper Chromatography: Special paper containing trapped water as stationary phase. Separation based on continuous partition between mobile solvent and trapped water.*

Retardation Factor (R_f Value)

$$R_f = \text{(Distance moved by component from baseline)} / \text{(Distance moved by solvent front from baseline)}$$

- R_f value is always **≤ 1.0**.
- Higher R_f → Less strongly adsorbed (moves faster with mobile phase).
- Lower R_f → Strongly adsorbed to stationary phase.

Visualization Techniques: *UV light radiation, Iodine crystals (gives brown spots for unsaturated compounds), Ninhydrin spray (violet/pink spots for amino acids).*

3. QUALITATIVE ANALYSIS (DETECTION OF ELEMENTS)

Element	*Test & Reagents*	*Chemical Reactions*	*Observation / Result*
Carbon (C) & Hydrogen (H)	*Heat organic compound with dry Copper(II) oxide (**CuO**).*	*$C + 2CuO \rightarrow 2Cu + CO_2\uparrow$* *$2H + CuO \rightarrow Cu + H_2O$* *$CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$* *$5H_2O + CuSO_4$ (anhydrous) →*	*CO_2 turns lime water **Milky / Turbid**.* *Water vapor turns anhydrous $CuSO_4$ **White to Blue**.*

3. QUALITATIVE ANALYSIS (LASSAIGNE'S SODIUM FUSION TEST)

Preparation of Sodium Fusion Extract (Lassaigne's Extract): *Fusing organic compound with metallic Sodium converts covalent bonds into ionic sodium salts: Na + C + N → **NaCN**, 2Na + S → **Na_2S**, Na + X → **NaX**.*

Element	Lassaigne Test Reagents	Chemical Equations	Confirmation Color
Nitrogen (N)	*L.E. + freshly prepared **$FeSO_4$** + **NaOH**, boil, cool, acidify with conc. **HCl** + add **$FeCl_3$**.*	*$6CN^- + Fe^{2+} \rightarrow [Fe(CN)_6]^{4-}$* *$3[Fe(CN)_6]^{4-} + 4Fe^{3+} \rightarrow \mathbf{Fe_4[Fe(CN)_6]_3}$ (Ferric Ferrocyanide)*	**Prussian Blue** color / precipitate.
Sulphur (S)	*1. L.E. + acetic acid (CH_3COOH) + **Lead Acetate** [$(CH_3COO)_2Pb$].*	*$S^{2-} + Pb^{2+} \rightarrow \mathbf{PbS}\downarrow$*	**Black Precipitate**
	*2. L.E. + freshly prepared **Sodium Nitroprusside** [$Na_2[Fe(CN)_5NO]$].*	*$S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow \mathbf{[Fe(CN)_5NOS]^{4-}}$ (Sodium thionitroprusside)*	**Violet / Purple** coloration.
Both N & S	*L.E. + **$FeCl_3$** (without excess Na; NaSCN formed instead of NaCN + Na_2S).*	*Na + C + N + S → NaSCN* *$SCN^- + Fe^{3+} \rightarrow \mathbf{[Fe(SCN)]^{2+}}$ (Thiocyanatoiron(III))*	**Blood Red** coloration.
Halogens (X)	*Pre-treatment: Boil L.E. with dil. **HNO_3** first to decompose NaCN and Na_2S as HCN↑ and H_2S↑ (otherwise they interfere by forming AgCN/Ag_2S ppt). Then add **$AgNO_3$**.*		
	*L.E. + dil. HNO_3 + **$AgNO_3$** solution.*	*$Ag^+ + Cl^- \rightarrow \mathbf{AgCl}\downarrow$ (Soluble in aqueous NH_4OH / ammonia)* *$Ag^+ + Br^- \rightarrow \mathbf{AgBr}\downarrow$ (Sparingly soluble in NH_4OH)* *$Ag^+ + I^- \rightarrow \mathbf{AgI}\downarrow$ (Insoluble in NH_4OH)*	**White ppt (Cl)** **Pale Yellow ppt (Br)** **Yellow ppt (I)**
	Layer Test (Br, I): *L.E. + dil. HCl + CS_2 / CCl_4 + Chlorine water.* **Added**	*$2Br^- + Cl_2 \rightarrow Br_2 + 2Cl^-$ (Br_2 dissolves in CCl_4 layer)* *$2I^- + Cl_2 \rightarrow I_2 + 2Cl^-$ (I_2 dissolves in CCl_4 layer)*	Orange/Brown layer (Br) Violet layer (I)
Phosphorus (P)	*Organic compound fused with **Na_2O_2** (Sodium peroxide) → extract with water → boil with conc. **HNO_3** + add **Ammonium Molybdate** [$(NH_4)_2MoO_4$].*	*$Na_3PO_4 + 3HNO_3 \rightarrow H_3PO_4 + 3NaNO_3$* *$H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \rightarrow \mathbf{(NH_4)_3PO_4 \cdot 12MoO_3}\downarrow + 21NH_4NO_3 + 12H_2O$*	**Canary Yellow** precipitate of Ammonium Phosphomolybdate.

4. QUANTITATIVE ANALYSIS (ELEMENTAL ESTIMATION FORMULAS)

1. Estimation of Carbon & Hydrogen (Liebig's Method)

Compound combusted with excess CuO: C → CO_2 (absorbed in KOH tubes), H → H_2O (absorbed in anhydrous $CaCl_2$ tube).

$$\%\ C = (12 / 44) \times (\text{Mass of } CO_2 / \text{Mass of organic compound}) \times 100$$

$$\%\ H = (2 / 18) \times (\text{Mass of } H_2O / \text{Mass of organic compound}) \times 100$$

2. Estimation of Nitrogen — Dumas Method

Organic-N heated with CuO in CO_2 atmosphere → N_2 gas collected over aqueous KOH.

$$P_1 = P_{atm} - \text{Aqueous Tension (at T K)}$$

$$V_{STP} = (P_1 \times V_1 \times 273) / (760 \times T_1)$$

$$\%\ N = (28 / 22400) \times (V_{STP} \text{ in mL} / \text{Mass of org. comp}) \times 100$$

$$\text{Shortcut: } \%\ N = V_{STP}\ (mL) / [8 \times \text{Mass of org. comp (g)}]$$

3. Estimation of Nitrogen — Kjeldahl's Method

Compound digested with conc. H_2SO_4 + $CuSO_4$/K_2SO_4 → $(NH_4)_2SO_4$ → boiled with excess NaOH → NH_3 evolved is absorbed in known excess standard acid (H_2SO_4 or HCl).

$$\%\ N = (1.4 \times N \times V) / \text{Mass of organic compound (g)}$$

- **N** = Normality of neutralized acid solution.
- **V** = Volume of acid neutralized by evolved NH_3 (in mL) = $V_{taken} - V_{back\text{-}titrated}$.

If Molarity M used for H_2SO_4: N = 2M → % N = (2.8 × M × V) / w.

4. Estimation of Halogens (Carius Method)

Heated with fuming HNO_3 + $AgNO_3$ in sealed Carius tube → AgX precipitate.

$$\%\ X = (\text{At. wt of X} / \text{Mol. wt of AgX}) \times (\text{Mass of AgX} / \text{Mass of org. comp}) \times 100$$

- **% Cl:** $(35.5 / 143.5) \times (m_{AgCl} / m_{oc}) \times 100$
- **% Br:** $(80 / 188) \times (m_{AgBr} / m_{oc}) \times 100$
- **% I:** $(127 / 235) \times (m_{AgI} / m_{oc}) \times 100$

5. Estimation of Sulphur (Carius Method)

Heated with fuming HNO_3 (or Na_2O_2) → H_2SO_4 → precipitated as $BaSO_4$ by adding $BaCl_2$.

$$\%\ S = (32 / 233) \times (\text{Mass of } BaSO_4 / \text{Mass of org. comp}) \times 100$$

6. Estimation of Phosphorus

Organic P oxidized to H_3PO_4 with fuming HNO_3. Determined by 2 methods:

$$\text{Method A (as } Mg_2P_2O_7\text{): } \%\ P = (62 / 222) \times (m_{Mg_2P_2O_7} / m_{oc}) \times 100$$

$$\text{Method B (as APM): } \%\ P = (31 / 1877) \times (m_{APM} / m_{oc}) \times 100$$

POLYMERS — JEE FORMULA & REVISION SHEET

Complete Monomer Structures, Polymerization Mechanisms, Reactions & Commercial Data

JEE MAIN + ADVANCED

1. FUNDAMENTALS & CLASSIFICATION OF POLYMERS

Classification by Source & Structure

- **Natural:** Proteins, Nucleic acids, Cellulose, Starch, Natural Rubber.
- **Semi-Synthetic:** Cellulose diacetate (Rayon), Cellulose nitrate, Vulcanized Rubber.
- **Synthetic:** Polythene, PVC, Nylon 6,6, Teflon, Synthetic Rubbers.
- **Linear:** High Density Polythene (HDP), PVC, Nylon, Polyesters (High M.P., High Density).
- **Branched Chain:** Low Density Polythene (LDP), Glycogen, Starch (Lower M.P. & tensile strength).
- **Cross-Linked / Network:** Bakelite, Melamine-formaldehyde, Glyptal (Bi-/Tri-functional monomers).

Classification by Intermolecular Forces

- **Elastomers:** Weakest intermolecular forces (van der Waals). Vulcanized to add cross-links. *e.g., Natural rubber, Buna-S, Buna-N, Neoprene.*
- **Fibres:** Strongest forces (H-bonding / dipole-dipole). High tensile strength & sharp M.P. *e.g., Nylon 6,6, Terylene, Polyacrylonitrile.*
- **Thermoplastics:** Intermediate forces. Linear/branched; soften on heating & harden on cooling. *e.g., Polythene, Polystyrene, PVC.*
- **Thermosetting:** Low M.W. semi-fluids that undergo extensive cross-linking on heating to become infusible hard mass. *e.g., Bakelite, Melamine resin.*

2. ADDITION (CHAIN-GROWTH) POLYMERIZATION MECHANISMS

A. Free Radical Mechanism (Initiator Driven)

Initiators (In•): Benzoyl peroxide [$C_6H_5CO-O-O-COC_6H_5$], *t*-Butyl peroxide [$(CH_3)_3C-O-O-C(CH_3)_3$], Acetyl peroxide.

1. Chain Initiation: In–In → 2 In• | In• + CH_2=CH(G) → In–CH_2–C•H(G)

2. Chain Propagation: In–CH_2–C•H(G) + *n* CH_2=CH(G) → In–[CH_2–CH(G)]$_n$–CH_2–C•H(G)

3. Chain Termination: Coupling of radicals or Disproportionation:
2 R–CH_2–C•H(G) → R–CH_2–CH(G)–CH(G)–CH_2–R (Coupling) OR R–CH=CH(G) + R–CH_2–CH_2(G) (Disproportionation)

Chain Transfer Agents: Regulate M.W. by transferring radical. *e.g., CCl_4, $C_6H_5CH_2SH$.*
Reaction: R• + CCl_4 → R–Cl + •CCl_3 → •CCl_3 + CH_2=CH(G) → $Cl_3C-CH_2-C•H(G)$

Chain Inhibitors: Terminate chain by forming unreactive radicals. *e.g., Benzoquinone, Phenols, Amines, O_2.*

B. Cationic Addition Polymerization

Initiators: Lewis acids + traces of H_2O (H^+, BF_3•H_2O, $AlCl_3$, H_2SO_4).
Favored by: Monomers with Electron-Donating Groups (+I, +M) e.g., Isobutylene [CH_2=$C(CH_3)_2$].

Initiation: $H^+ + CH_2=CR_2 \rightarrow CH_3-C^+R_2$

Propagation: $CH_3-C^+R_2 + CH_2=CR_2 \rightarrow CH_3-CR_2-CH_2-C^+R_2$

Termination: Proton loss or combination with anion (A^-).

C. Anionic Addition Polymerization

Initiators: Strong bases ($NaNH_2$, KNH_2, Organometallics like *n*-BuLi, Grignard reagents).
Favored by: Monomers with Electron-Withdrawing Groups (−I, −M) e.g., Acrylonitrile, Styrene, Methyl methacrylate.

Initiation: $B^- + CH_2=CH(EWG) \rightarrow B-CH_2-C^-H(EWG)$

Propagation: B–CH_2–C^-H(EWG) + Monomer → Carbanion chain.

3. CONJUGATED DIENE POLYMERIZATION (1,2- VS 1,4- ADDITION)

For conjugated diene: $CH_2=C(R)-CH=CH_2$ (where R = H, CH_3, Cl):

- **1,2-Addition Polymerization:** Polymer retains pendant vinyl double bonds: **-[CH_2-C(R)(CH=CH_2)]$_n$-**
- **1,4-Addition Polymerization:** Double bond shifts to 2,3 position in main chain: **-[CH_2-C(R)=CH-CH_2]$_n$-**
  - **Cis-1,4-polyisoprene:** Natural Rubber (coiled structure, elastic).
  - **Trans-1,4-polyisoprene:** Gutta-Percha (zigzag non-elastic structure, crystalline).

4. COMMERCIALLY IMPORTANT ADDITION POLYMERS

Polymer & Type	Monomer(s) & Structure	Repeating Unit	Conditions & Key Uses
LDP (Low Density Polythene)	Ethene $CH_2=CH_2$	$-[CH_2-CH_2]_n-$	1000–2000 atm, 350–570 K, O_2/peroxide. Highly branched, flexible. Insulation, squeeze bottles.
HDP (High Density Polythene)	Ethene $CH_2=CH_2$	$-[CH_2-CH_2]_n-$	6–7 atm, 333–343 K, **Ziegler-Natta Catalyst** [$Et_3Al + TiCl_4$]. Linear, tough. Buckets, pipes.
PVC (Polyvinyl Chloride)	Vinyl Chloride $CH_2=CH-Cl$	$-[CH_2-CH(Cl)]_n-$	Peroxide initiator. Raincoats, vinyl flooring, pipes, leather cloth. Add plasticizer for softness.
Teflon (PTFE)	Tetrafluoroethene $CF_2=CF_2$	$-[CF_2-CF_2]_n-$	High pressure, Persulphate catalyst. Chemically inert, heat resistant. Non-stick cookware, gaskets.
PAN (Orlon / Acrilan)	Acrylonitrile $CH_2=CH-CN$	$-[CH_2-CH(CN)]_n-$	Peroxide catalyst. High resistance to chemicals/light. Commercial substitute for wool (blankets).
Polystyrene	Styrene $CH_2=CH-C_6H_5$	$-[CH_2-CH(C_6H_5)]_n-$	Free radical / benzoyl peroxide. Toys, radio/TV cabinets, insulating material, packaging.
PMMA [Added] (Lucite/Plexiglas)	Methyl Methacrylate $CH_2=C(CH_3)COOCH_3$	$-[CH_2-C(CH_3)(COOCH_3)]_n-$	Unbreakable glass substitute, optical lenses, aircraft windows, light covers.

5. CONDENSATION (STEP-GROWTH) POLYMERS

Polymer & Class	Monomers & Formula	Polymer Structure / Linkage	Key Uses & Notes
Nylon 6,6 (Polyamide)	1. Adipic Acid: $HOOC(CH_2)_4COOH$ 2. Hexamethylenediamine: $H_2N(CH_2)_6NH_2$	$-[NH(CH_2)_6NH-CO(CH_2)_4CO]_n-$ **Linkage:** Amide ($-CONH-$)	553 K, High pressure. High tensile strength fibre. Bristles, sheets, textiles.
Nylon 6 (Polyamide)	**Caprolactam** (7-membered ring with amide linkage)	$-[NH-(CH_2)_5-CO]_n-$ Ring opening polymerization.	Heated with H_2O at 533–543 K. Tyre cords, fabrics, ropes.
Dacron / Terylene (Polyester)	1. Terephthalic acid: $p-HOOC-C_6H_4-COOH$ 2. Ethylene glycol: $HO-CH_2-CH_2-OH$	$-[O-CH_2CH_2-O-CO-C_6H_4-CO]_n-$ **Linkage:** Ester ($-COO-$)	420–460 K, Zinc acetate + Sb_2O_3 catalyst. Crease-resistant, blended with cotton/wool.
Glyptal (Polyester)	1. Phthalic acid: $o-HOOC-C_6H_4-COOH$ 2. Ethylene glycol: $HO-CH_2-CH_2-OH$	$-[O-CH_2CH_2-O-CO-C_6H_4(o)-CO]_n-$ Linear polyester.	Paints, lacquers, building adhesives. Cross-links on heating.
Bakelite (Phenol-Formaldehyde)	Phenol + Formaldehyde ($HCHO$) Ratio depends on reaction condition.	1. **Novolac:** Linear polymer (acid/base cat). 2. **Bakelite:** Novolac + HCHO + Heat → Cross-linked 3D network.	Novolac in paints. Bakelite in electrical switches, handles, combs, phonograph records.
Melamine-Formaldehyde Resin	1. Melamine (2,4,6-triamino-1,3,5-triazine) 2. Formaldehyde ($HCHO$)	Condensation resin intermediate → Cross-linked polymer network on heating.	Unbreakable crockery, dinnerware, laminated sheets.

Polymer & Class	Monomers & Formula	Polymer Structure / Linkage	Key Uses & Notes
Urea-Formaldehyde Resin	1. Urea (NH_2CONH_2) 2. Formaldehyde ($HCHO$)	Cross-linked thermosetting resin.	Unbreakable cups, laminated sheets, adhesives.

6. BAKELITE REACTION STEPS (PHENOL + HCHO)

Step 1: Phenol + HCHO (Acid/Base catalyst) ⟶ *o*- and *p*-Hydroxymethylphenols.
Step 2: Condensation of hydroxymethylphenols with elimination of H_2O yields linear polymer = **Novolac**.
$n\ (o\text{-}HO\text{-}CH_2\text{-}C_6H_4\text{-}OH) \rightarrow -[C_6H_3(OH)\text{-}CH_2]_n-$ (Methylene bridges at *ortho* positions).
Step 3: Novolac heated with excess HCHO undergoes extensive cross-linking at *ortho* & *para* positions to form infusible 3D mass = **Bakelite**.

7. NATURAL & SYNTHETIC RUBBERS

Rubber Type	Monomer(s) & IUPAC Name	Polymer Repeating Unit	Properties & Applications
Natural Rubber	**Isoprene** (2-Methylbuta-1,3-diene) $CH_2=C(CH_3)\text{-}CH=CH_2$	$-[CH_2\text{-}C(CH_3)=CH\text{-}CH_2]_n-$ **cis-1,4-polyisoprene** configuration.	Elastic due to weak van der Waals forces & coiled chains. Soluble in non-polar solvents, high H_2O absorption.
Neoprene (Synthetic)	**Chloroprene** (2-Chlorobuta-1,3-diene) $CH_2=C(Cl)\text{-}CH=CH_2$	$-[CH_2\text{-}C(Cl)=CH\text{-}CH_2]_n-$ trans-1,4-polymer predominates.	Resistant to vegetable/mineral oils, heat, aging. Conveyor belts, gaskets, hoses.
Buna-S (SBR)	1. Buta-1,3-diene (75%) 2. Styrene (25%)	$-[CH_2\text{-}CH=CH\text{-}CH_2\text{-}CH(C_6H_5)\text{-}CH_2]_n-$ Copolymer (Addition).	High abrasion resistance. Auto tyres, floor tiles, footwear, bubble gum base.
Buna-N (NBR)	1. Buta-1,3-diene 2. Acrylonitrile ($CH_2=CH\text{-}CN$)	$-[CH_2\text{-}CH=CH\text{-}CH_2\text{-}CH(CN)\text{-}CH_2]_n-$ Copolymer (Addition).	Resistant to action of petrol, lubricating oil, organic solvents. Oil seals, tank linings.

8. VULCANIZATION OF RUBBER (MECHANISM & PARAMETERS)

Raw rubber has low tensile strength, high water absorption, softens >335 K, becomes brittle <283 K, and is easily oxidized.

- **Process:** Heating raw rubber with **Sulphur (3–5%)** and additives at **373 K to 415 K (110°C)**.
- **Chemistry:** Sulphur forms **-S-S- cross-links** at the reactive allylic positions ($-CH_2\text{-}C(CH_3)=CH\text{-}CH_2-$), tying double bonds of adjacent chains.
- **Effect of % Sulphur:**
  - **1 to 5% Sulphur:** Soft elastic rubber (tyres, rubber bands).
  - **30% Sulphur:** Hard, non-elastic ebonite / vulcanite [Added].

9. BIODEGRADABLE POLYMERS & BIOPOLYMERS

PHBV (Aliphatic Polyester)

Monomers:
1. 3-Hydroxybutanoic acid: $CH_3-CH(OH)-CH_2-COOH$
2. 3-Hydroxypentanoic acid: $CH_3-CH_2-CH(OH)-CH_2-COOH$

Structure:
$-[O-CH(CH_3)-CH_2-CO-O-CH(CH_2CH_3)-CH_2-CO]_n-$

Uses: Speciality packaging, orthopaedic devices, controlled drug release. Degraded by environmental microbes.

Nylon 2-Nylon 6 (Polyamide)

Monomers:
1. Glycine: H_2N-CH_2-COOH (2 carbons)
2. Amino caproic acid: $H_2N-(CH_2)_5-COOH$ (6 carbons)

Structure:
$-[HN-CH_2-CO-NH-(CH_2)_5-CO]_n-$

Type: Alternating polyamide copolymer, fully biodegradable.

Biopolymers vs Biodegradable Synthetic Polymers: Biopolymers (e.g., proteins, nucleic acids, cellulose, starch) are natural polymers that sustain life processes. Synthetic biodegradable polymers (PHBV, Nylon 2-Nylon 6, PGA, PLA) contain functional groups (esters/amides) susceptible to enzymatic bacterial hydrolysis.

10. SPECIALTY COPOLYMERS & COMMERCIAL MATERIALS

Name	Monomeric Composition	Key Properties & Specific JEE Uses
Saran	Vinylidene chloride ($CH_2=CCl_2$) + Vinyl chloride ($CH_2=CHCl$)	High barrier to water, oxygen, & chemicals. Used for wrapping food materials.
ABS Rubber	Acrylonitrile + Buta-1,3-diene + Styrene	High impact strength & toughness. Used in helmets, luggage, auto parts.
Dynel	Vinyl chloride ($CH_2=CHCl$) + Acrylonitrile ($CH_2=CH-CN$)	Flame-resistant synthetic fibre. Used for making human hair wigs, protective clothing.
Thermocol	Foamed plastic obtained by blowing air through molten **Polystyrene** or Polyurethane.	Extremely lightweight, shock-absorbent insulator. Packaging & temporary structure.
Kevlar [Added]	Terephthaloyl chloride + 1,4-phenylenediamine	Aromatic polyamide (Aramid). Extremely high tensile strength-to-weight ratio. Bulletproof vests.
Silk	Natural polyamide fibre obtained from silkworm cocoons.	Thread-like natural protein polymer (fibroin).

11. POLYMER MOLECULAR WEIGHT AVERAGES & FORMULAS [ADDED FOR JEE ADVANCED]

Number-Average Molecular Weight ($\bar{M}_n$)

Determined by **colligative properties** (Osmotic Pressure measurement is best).

$$\bar{M}_n = \sum(N_i M_i) / \sum N_i$$

Where N_i = number of molecules with molecular mass M_i.

Weight-Average Molecular Weight ($\bar{M}_w$)

Determined by **light scattering** or ultracentrifugation.

$$\bar{M}_w = \sum(N_i M_i^2) / \sum(N_i M_i)$$

Where $N_i M_i$ is total mass of species *i*.

Polydispersity Index (PDI):
- Formula: **PDI = $\bar{M}_w$ / $\bar{M}_n$**
- For **Natural Polymers (Monodisperse):** PDI = 1 (since $\bar{M}_w = \bar{M}_n$).
- For **Synthetic Polymers (Polydisperse):** PDI > 1 ($\bar{M}_w$ is always > $\bar{M}_n$).
- **Degree of Polymerization (DP):** DP = $M_{polymer}$ / $M_{monomer}$.

12. JEE QUICK RECALL & KEY DISTINCTIONS

Homopolymers vs Copolymers:
• Homopolymer: Single monomer unit (e.g., Polythene, PVC, Teflon, PAN, Nylon 6).
• Copolymer: 2+ monomer units (e.g., Buna-S, Buna-N, Nylon 6,6, Terylene, PHBV, Saran, Dynel).

Glass Transition Temp (T_g) & Melting Temp (T_m): [Added]
• Below T_g: Polymer is hard, glassy, brittle.
• Above T_g: Polymer becomes rubbery / flexible.
• Elastomers have T_g well *below* room temp.

Order of Tensile Strength / Intermolecular Force:

Fibres > Thermosetting > Thermoplastics > Elastomers

Example: Nylon 6,6 > Bakelite > Polythene > Buna-S.

Ziegler-Natta Catalyst Mechanism:
Triethylaluminium + Titanium tetrachloride: $Al(C_2H_5)_3 + TiCl_4$.
Acts via coordination polymerization leading to linear unbranched HDP chain.

PERIODIC CLASSIFICATION & PERIODICITY

1. GENESIS OF PERIODIC CLASSIFICATION & ELECTRONIC STRUCTURE

HISTORICAL LAWS & FORMULAS

- **Prout's Hypothesis:** $Atomic\ Weight = n \times (Atomic\ weight\ of\ H)$, where $n = 1, 2, 3...$
- **Dobereiner's Triads:** For elements A, B, C with atomic weights W_A, W_B, W_C:

  $$W_B = (W_A + W_C) / 2$$

  Example: Li (7), Na (23), K (39) → $(7 + 39)/2 = 23$.
- **Newland's Octaves:** Properties repeat every 8th element (valid up to Ca).
- **Lothar Meyer's Curve:** Graph of Atomic Volume vs. Atomic Weight:
  - *Peaks:* Alkali metals (Li, Na, K, Rb, Cs).
  - *Descending slope:* Alkaline earth metals (Be, Mg, Ca, Sr, Ba).
  - *Ascending slope:* Halogens & Noble gases (except He).
  - *Troughs (Bottom):* Transition metals (small atomic volumes).
- **Moseley's Law (Modern Basis):**

  $$\sqrt{\nu} = a(Z - b)$$

  ν = X-ray frequency, Z = Atomic Number, a, b = Screening constants.

POSITION DETERMINATION RULES

- **Period Number:** Highest Principal Quantum Number (n) in electronic configuration.
- **Group Number Determination:**
  - *s-block:* $Group = Number\ of\ ns\ electrons$ (Group 1 or 2).
  - *p-block:* $Group = 10 + (ns\ electrons) + (np\ electrons)$ (Groups 13–18).
  - *d-block:* $Group = (ns\ electrons) + (n-1)d\ electrons$ (Groups 3–12).
  - *f-block:* Always **Group 3 (III-B)** (Lanthanoids Z=58-71, Actinoids Z=90-103).
- **Outer Configurations:**
  - *s-block:* ns^{1-2} (n = 1–7)
  - *p-block:* $ns^2\ np^{1-6}$ (n = 2–7)
  - *d-block:* $(n-1)d^{1-10}\ ns^{0-2}$ (n = 4–7)
  - *f-block:* $(n-2)f^{1-14}\ (n-1)d^{0-1}\ ns^2$ (n = 6, 7)
- **Kernel & Shell Terminology:**
  - *Penultimate Shell:* $(n - 1)$ shell.
  - *Pre-penultimate Shell:* $(n - 2)$ shell.
  - *Kernel / Core Charge:* $Z - (Inner\ shell\ electrons)$.

2. EFFECTIVE NUCLEAR CHARGE (Z*) & SCREENING CONSTANT (Σ)

EFFECTIVE NUCLEAR CHARGE FORMULA & SLATER'S RULES [Added]

$$Z^*\ (or\ Z_{eff}) = Z - \sigma$$

Where Z = Actual Atomic Number, σ = Shielding/Screening Constant.

- **Shielding Power Order:** $s > p > d > f$ (for same principal shell n).
- **Period Trend:** Z^* increases left to right across a period (by 0.65 per element for s/p block).
- **Ion Charge Effect:**
  - Cation formation → Z^* increases ($Z_{eff}(A^+) > Z_{eff}(A)$).
  - Anion formation → Z^* decreases ($Z_{eff}(A^-) < Z_{eff}(A)$).

Slater's Rules for Calculating σ:

Electrons Grouping	Same Group (nth)	(n-1)th Shell	(n-2)th & Inner
For ns, np e$^-$	0.35 each (0.30 for 1s)	0.85 each	1.00 each
For nd, nf e$^-$	0.35 each	1.00 each	1.00 each

Note: The electron for which σ is being calculated does not shield itself.

3. ATOMIC & IONIC RADII

TYPES OF RADII & FORMULAS

IONIC RADII & TRENDS

- **Covalent Radius (r_{cov}):** Half internuclear distance in homonuclear covalent molecule:

$$r_A = d_{A-A} / 2$$

- **Heteronuclear Covalent Radius (Schoemaker-Stevenson Equation):** [Added]

$$d_{A-B} = r_A + r_B - 0.09\,|X_A - X_B|$$

Where X_A, X_B are Pauling electronegativities (d in Å).

- **Van der Waals Radius (r_{vdw}):** Half internuclear distance between non-bonded adjacent atoms in solid state.

$$r_{vdw} = d_{non\text{-}bonded} / 2$$

- **Metallic Radius (r_{met}):** Half internuclear distance between two adjacent metal ions in metallic lattice.
- **Comparison Order:**

$$r_{vdw} > r_{met} > r_{cov} \quad (r_{vdw} \approx 1.4 \text{ to } 1.9 \times r_{cov})$$

- **Cationic Radius:** $r_{cation} < r_{parent\ atom}$

Reason: Loss of $e^- \rightarrow Z_{eff}$ increases.

$$\textit{Size of Cation} \propto 1 / \textit{(Positive Charge)}$$

Example: $Fe > Fe^{2+} > Fe^{3+}$

- **Anionic Radius:** $r_{anion} > r_{parent\ atom}$

Reason: Gain of $e^- \rightarrow Z_{eff}$ decreases, inter-electronic repulsion increases.

$$\textit{Size of Anion} \propto \textit{Negative Charge}$$

Example: $O^{2-} > O^- > O$

- **Isoelectronic Series Radius Formula:**

$$\textit{Radius} \propto 1 / Z \quad \textit{(for same number of } e^-\textit{)}$$

Standard Trend: $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

Group 2/15 Isoelectronic: $S^{2-} > Cl^- > K^+ > Ca^{2+}$

4. IONIZATION ENERGY / IONIZATION POTENTIAL (IE / IP)

DEFINITION & FUNDAMENTAL RELATIONS

- **Definition:** Minimum energy required to remove the most loosely bound electron from an isolated gaseous atom in ground state:

$$A(g) + IE_1 \rightarrow A^+(g) + e^-$$

- **Successive Ionization Energies:** Always endothermic ($\Delta H > 0$):

$$IE_1 < IE_2 < IE_3 < IE_4 \ldots$$

Reason: Z_{eff} increases progressively after removal of each electron.

- **Valence Electron Determination from Sudden Jump:** If $IE_{(n+1)} \gg IE_n$, the atom has **n valence electrons** (noble gas core reached).

- **Key Governing Factors:**
  - $IE \propto 1 / (Atomic\ Size)$
  - $IE \propto Z_{eff}$
  - $IE \propto 1 / (Shielding\ Effect)$
  - $IE \propto Penetration\ Power\ (s > p > d > f)$
  - $IE \propto$ *Stability of Half-filled* (p^3, d^5, f^7) *&* *Fully-filled* (s^2, p^6, d^{10}, f^{14}) *subshells.*
- **Penetration Energy Order:** For same quantum shell n:

$$IE(ns) > IE(np) > IE(nd) > IE(nf)$$

CRITICAL IE EXCEPTIONS & JEE ORDER TABLES

Period / Group	Expected Trend	Actual Experimental Order	Governing Cause / Anomaly Explanation
Period 2 (IE_1)	Li < Be < B < C < N < O < F < Ne	**Li < B < Be < C < O < N < F < Ne**	Be ($2s^2$) > B ($2p^1$) due to full 2s stability; N ($2p^3$) > O ($2p^4$) due to half-filled stability.
Period 3 (IE_1)	Na < Mg < Al < Si < P < S < Cl < Ar	**Na < Al < Mg < Si < S < P < Cl < Ar**	Mg ($3s^2$) > Al ($3p^1$); P ($3p^3$) > S ($3p^4$).
Period 2 (IE_2)	Standard left to right	**Be < C < B < N < F < O < Ne < Li**	Li^+ ($1s^2$) has highest IE_2; O^+ ($2p^3$) > F^+ ($2p^4$).
Group 13 (Boron)	B > Al > Ga > In > Tl	**B > Tl > Ga > Al > In**	Ga ≈ Al due to poor d-electron shielding (Transition contraction); Tl > In due to 14f-electron shielding (Lanthanoid contraction).
Group 14 (Carbon)	C > Si > Ge > Sn > Pb	**C > Si > Ge > Pb > Sn**	Pb > Sn due to Lanthanoid contraction in 6th period (Z_{eff} increase).
d-Block Series	3d < 4d < 5d	**3d < 4d << 5d** (Group 4: 3d < 4d < 5d)	5d series elements have very high IE due to Lanthanoid contraction ($4f^{14}$ poor shielding).

5. ELECTRON GAIN ENTHALPY ($\Delta_{EG}H$) & ELECTRON AFFINITY (EA)

DEFINITION & SIGN CONVENTION

- **Definition:** Enthalpy change when a neutral gaseous atom gains an electron:

$$X(g) + e^- \rightarrow X^-(g) \quad \Delta_{eg}H$$

- **Relation with Electron Affinity (EA):**

$$EA = -\Delta_{eg}H \quad (at\ 0\ K)$$

- **Sign Conventions:**
  - *1st Electron Gain Enthalpy ($\Delta_{eg}H_1$):* Usually **exothermic (negative)** for neutral non-metals.
  - *Noble Gases, Be, Mg, N:* Have **positive $\Delta_{eg}H_1$** (endothermic) due to stable configurations.
  - *2nd & Successive $\Delta_{eg}H$:* **ALWAYS Positive (Endothermic)** due to electrostatic repulsion between anion and incoming e^-.

$$O^-(g) + e^- \rightarrow O^{2-}(g);\ \ \Delta_{eg}H_2 = +780\ kJ/mol$$

KEY TRENDS & JEE ANOMALY ORDERS

- **3rd Period vs 2nd Period Anomaly (CRITICAL):**
  3rd period elements (Cl, S, P) have **MORE negative $\Delta_{eg}H$** (higher EA) than 2nd period elements (F, O, N).
  Reason: Small compact size of 2p subshell leads to severe inter-electronic repulsions.

  Halogens EA Order: $Cl > F > Br > I$
  Group 16 EA Order: $S > Se > Te > Po > O$
  Group 15 EA Order: $P > As > Sb > Bi > N$

- **Maximum/Minimum Values in Periodic Table:**
  - *Highest EA / Most negative $\Delta_{eg}H$:* **Chlorine (Cl)**
  - *Highest Positive $\Delta_{eg}H$:* **Helium (He) / Neon (Ne)**

6. ELECTRONEGATIVITY (EN) & QUANTIFICATION SCALES

ELECTRONEGATIVITY SCALES & CORE MATHEMATICAL FORMULAS

1. Pauling Scale

Based on ionic-covalent resonance energy (Δ in eV or kcal/mol):

$$|X_A - X_B| = 0.208\sqrt{(\Delta)} \text{ (in kcal/mol)}$$
$$|X_A - X_B| = 0.102\sqrt{(\Delta)} \text{ (in kJ/mol)}$$
$$\Delta = E_{A-B} - \sqrt{(E_{A-A} \times E_{B-B})}$$

Reference: $X_F = 4.0$ (highest in PT).

2. Mulliken Scale

Average of Ionization Potential and Electron Affinity (in eV):

$$X_{Mulliken} = (IE + EA)/2$$

Inter-conversion with Pauling:

$$X_{Pauling} = X_{Mulliken}/2.8 \text{ (in eV)}$$
$$X_{Pauling} = (IE + EA)/540 \text{ (in kJ/mol)}$$

3. Allred-Rochow Scale [Added]

Electrostatic force exerted by nucleus on valence electrons at radius r (in Å):

$$X_{AR} = 0.359\,(Z^*/r^2) + 0.744$$

Z^* calculated using Slater's rules.

FACTORS AFFECTING ELECTRONEGATIVITY & KEY APPLICATION TRENDS

- **Oxidation State:** Higher oxidation state → higher EN.
  Example: $EN\ of\ Fe^{3+} > Fe^{2+}$; $EN\ of\ O^{-} > O^{2-}$
- **Hybridization State:** Higher % s-character → higher EN (s-orbitals closer to nucleus).

  $$sp\ (50\%\ s) > sp^2\ (33.3\%\ s) > sp^3\ (25\%\ s)$$

  EN Values: $C(sp) \approx 3.25$, $C(sp^2) \approx 2.75$, $C(sp^3) \approx 2.50$.
- **Group Exceptions in EN:**
  • *Group 12 (Zn, Cd, Hg):* EN increases down the group.
  • *Group 13 (Al to Tl):* EN increases from Al to Tl (B=2.0, Al=1.5, Ga=1.6, In=1.7, Tl=1.8).
  • *Group 14 (Si to Pb):* EN remains nearly constant (Si=1.8, Ge=1.8, Sn=1.8, Pb=1.9).
- **Inert Gases:** EN is zero (do not form standard covalent bonds).

7. CHEMICAL PERIODICITY, OXIDE CHARACTER & BOND PROPERTIES

NATURE OF OXIDES & GALLIS RULE

- **Electronegativity Difference Criterion ($X_O - X_A$):**
  • If $X_O - X_A \geq 2.3$ → Oxide is **Strongly Basic**.
  • If $X_O - X_A < 2.3$ → Oxide is **Amphoteric / Acidic**.
- **Period 3 Oxide Trend:**

Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
ΔX	2.6	2.3	2.1	1.8	1.5	1.1	0.5
Nature	Str. Basic	Basic	Amphoteric	Wk. Acidic	Acidic	Str. Acidic	Str. Acidic

- **Gallis Rule for Hydroxides (AOH):**
  • If $X_A < 1.7$ → Hydroxide is **Basic** (e.g., NaOH, X_{Na} = 0.9).
  • If $X_A > 1.7$ → Hydroxide is **Acidic** (e.g., ClOH, X_{Cl} = 3.0).
- **Oxidation State Effect on Oxide Acidity:**

  $$Acidic\ Strength \propto Oxidation\ Number\ of\ Central\ Atom$$

  Example: $N_2O < N_2O_3 < N_2O_4 < N_2O_5$
- **Size Effect on Oxide Acidity:**

  $$Acidic\ Strength \propto 1\ /\ (Size\ of\ Central\ Atom)$$

  Example: $N_2O_5 > P_2O_5 > As_2O_5$

PERCENTAGE IONIC CHARACTER & BOND FORMULAS [Added]

- **Hannay-Smith Equation:** Predicts % ionic character in covalent bond A–B:

  $$\%\ Ionic\ Character = 16|X_A - X_B| + 3.5|X_A - X_B|^2$$

  Rule: If $|X_A - X_B| = 1.9$, bond is 50% ionic, 50% covalent.
- **Pauling Equation for % Ionic Character:**

  $$\%\ Ionic\ Character = (1 - e^{-0.25(X_A - X_B)^2}) \times 100$$

- **Acid-Base Nature of Elements across Periods:**
  • *Metallic Character:* Increases down group, decreases across period.
  • *Non-Metallic Character:* Decreases down group, increases across period.
  • *Reducing Power:* Increases down the group for metals (Alkali/Alkaline earth).

8. HIGH-YIELD FACTS, CONSTANTS & EXCEPTIONS SUMMARY

JEE DIRECT FACTS & RAPID ONE-LINER MEMORY MATRIX

Most Electronegative Element	Fluorine (F, EN = 4.0)
Least Electronegative / Most Electropositive	Cesium (Cs) [Stable], Francium (Fr) [Overall]
Highest Electron Affinity (Negative $\Delta_{eg}H$)	Chlorine (Cl)
Highest Ionization Energy	Helium (He)
Lowest Ionization Energy	Cesium (Cs)
Largest Atomic Radius	Cesium (Cs)
Heaviest & Densest Solid Metal	Osmium (Os, ρ = 22.6 g/cm^3) / Iridium (Ir)
Lightest Solid Metal	Lithium (Li)
Liquid Elements (RT ~ 25°C - 30°C)	Hg, Br_2, Ga, Cs, Fr (5 elements)
Gaseous Elements in Periodic Table	11 (H_2, N_2, O_2, F_2, Cl_2 + 6 Inert Gases)
Submetals / Metalloids	B, Si, Ge, As, Sb, Te, At

Volatile d-block metals	Zn, Cd, Hg (closed shell d^{10} configuration)
Noble Metals	Ru, Rh, Pd, Ag, Re, Os, Ir, Pt, Au
Most Abundant Element on Earth Crust	Oxygen (O) followed by Silicon (Si)
First Artificial Element	Technetium (Tc, Z = 43)
Transuranic Elements	Elements with Z > 92 (after Uranium)
Strongest Base / Alkali	CsOH / Cs_2CO_3 (most stable carbonate)
Elements Kept in Kerosene	Na, K, Rb, Cs
Element Kept under Water	White Phosphorus (P_4)
Natural Explosive	NCl_3
Poorest Electrical Conductors	Lead (Pb - Metal), Sulfur (S - Non-metal)
Bridge Elements	2nd Period elements showing diagonal relationship with 3rd Period (Li-Mg, Be-Al, B-Si).

METALLURGY & EXTRACTION OF ELEMENTS

Comprehensive JEE Main & Advanced Formula & Quick Revision Sheet

JEE Main & Advanced

Primary Source: Class Notes & Mind Maps

1. IMPORTANT ORES & CHEMICAL COMPOSITIONS

Ore Category	Ore Name	Chemical Formula	Key Metals Extracted
Native State	Gold, Platinum, Silver	Au, Pt, Ag, Cu	Au, Pt, Ag
Oxides	Bauxite / Diaspore	$Al_2O_3 \cdot 2H_2O$ / $Al_2O_3 \cdot H_2O$	Aluminium (Al)
	Haematite / Magnetite	Fe_2O_3 / Fe_3O_4	Iron (Fe)
	Pyrolusite / Cuprite	MnO_2 / Cu_2O (Ruby Copper)	Mn, Cu
Sulphides	Copper Pyrites (Chalcopyrite)	$CuFeS_2$	Copper (Cu)
	Copper Glance (Chalcocite)	Cu_2S	Copper (Cu)
	Galena / Zinc Blende	PbS / ZnS	Pb, Zn
Carbonates	Malachite / Azurite [Added]	$CuCO_3 \cdot Cu(OH)_2$ / $2CuCO_3 \cdot Cu(OH)_2$	Copper (Cu)
	Calamine / Siderite / Dolomite	$ZnCO_3$ / $FeCO_3$ / $CaCO_3 \cdot MgCO_3$	Zn, Fe, Mg/Ca
Halides & Sulphates	Horn Silver / Cryolite / Barite	AgCl / Na_3AlF_6 / $BaSO_4$	Ag, Al, Ba

2. TERMINOLOGY, FLUX & SLAG CHEMISTRY

Slag Formation Equations

Flux + Infusible Gangue → Fusible Slag

• **Acidic Flux** (removes Basic Gangue like FeO, CaO):
SiO_2 (Acidic) + FeO (Basic) → **$FeSiO_3$** (Ferrous silicate slag)
SiO_2 (Acidic) + CaO (Basic) → **$CaSiO_3$** (Calcium silicate slag)

• **Basic Flux** (removes Acidic Gangue like SiO_2):
CaO / MgO / $MgCO_3$ + SiO_2 → **$CaSiO_3$ / $MgSiO_3$**

• **Properties of Slag:** Immiscible with molten metal, lower melting point, lower density (floats on top, prevents re-oxidation).

Refractory Materials [JEE Key Concept]

Substances withstanding high temperatures without melting/decomposing:

• **Acidic Refractory:** Silica (SiO_2), Quartz, Sandstone.

• **Basic Refractory:** Lime (CaO), Magnesite ($MgCO_3$/MgO), Dolomite.

• **Neutral Refractory:** Graphite, Chromite ($FeCr_2O_4$), Bone ash ($Ca_3(PO_4)_2$).

3. ORE CONCENTRATION (BENEFICIATION) METHODS

Method	Operating Principle & Reagents	Applicability & Examples
Gravity Separation (Levigation)	Difference in specific gravities of heavy ore and lighter gangue particles using a running stream of water.	Oxides & Carbonates (e.g., Cassiterite SnO_2, Haematite Fe_2O_3, Native Au).
Froth Floatation	Preferential wetting: Sulphide ore wetted by oil, gangue wetted by water. • **Frothers:** Pine oil, Eucalyptus oil, Camphor oil. • **Collectors:** Sodium/Potassium Ethyl Xanthate. • **Froth Stabilizers:** Cresol, Aniline. • **Depressants:** NaCN (Depresses ZnS in Galena mixture: NaCN + ZnS → $[Zn(CN)_4]^{2-}$ soluble). • **Activator:** $CuSO_4$.	Mainly Sulphide Ores (ZnS, PbS, $CuFeS_2$, Cu_2S).

Method	Operating Principle & Reagents	Applicability & Examples
Magnetic Separation	Difference in magnetic properties of ore and impurities using magnetic roller belt.	• Wolframite ($FeWO_4$, magnetic) from Cassiterite (SnO_2, non-magnetic). • Chromite ($FeO \cdot Cr_2O_3$), Magnetite (Fe_3O_4), Pyrolusite (MnO_2).
Leaching (Chemical)	Ore dissolves in a chemical reagent while impurities remain insoluble.	• **Bayer's Process (Al_2O_3):** NaOH leaching. • **MacArthur-Forrest Cyanide Process (Au, Ag):** NaCN solution leaching.

4. CONVERSION OF CONCENTRATED ORE TO OXIDE

Calcination (Absence / Limited Air)

• **Condition:** Heated below m.p. in limited supply of air.

• **Purpose:** Used for Hydroxides, Carbonates, Hydrated oxides.

• **Reactions:**

$ZnCO_3$ (Calamine) $\rightarrow ZnO + CO_2\uparrow$

$CaCO_3 \cdot MgCO_3 \rightarrow CaO + MgO + 2CO_2\uparrow$

$CuCO_3 \cdot Cu(OH)_2 \rightarrow 2CuO + CO_2\uparrow + H_2O\uparrow$

$Al_2O_3 \cdot 2H_2O \rightarrow Al_2O_3 + 2H_2O\uparrow$

• **Result:** Removes moisture & organic impurities; makes ore porous.

Roasting (Excess Air)

• **Condition:** Heated below m.p. in excess supply of air.

• **Purpose:** Used primarily for Sulphide ores.

• **Reactions:**

$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2\uparrow$

$2PbS + 3O_2 \rightarrow 2PbO + 2SO_2\uparrow$

$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2\uparrow$

$S + O_2 \rightarrow SO_2\uparrow$ | $4As + 3O_2 \rightarrow 2As_2O_3\uparrow$

• **Result:** Converts sulphides to oxides; removes volatile S, As, Sb oxides.

5. THERMODYNAMICS OF REDUCTION & ELLINGHAM DIAGRAM

Thermodynamic Principles & Governing Formulas

• **Gibbs Free Energy Equation:** $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

• **Feasibility Condition:** Reduction is spontaneous only when overall $\Delta G^\circ < 0$ (Negative).

• **Equilibrium Constant Relation:** $\Delta G^\circ = -R\,T\,ln\,K_p = -2.303\,R\,T\,log\,K_p$ **[Added]**

• **Coupled Reactions:** If a reduction reaction (positive ΔG°) is coupled with a oxidation reaction (large negative ΔG°), overall reaction proceeds if $\Delta G_1^\circ + \Delta G_2^\circ < 0$.

Key Features of Ellingham Diagram (ΔG° vs T)

1. Plots ΔG° per mole of O_2 consumed against Temperature T.

2. Most metal oxidation lines slope **upward** ($\Delta S^\circ < 0$ because gas O_2 is consumed).

3. **Sudden change in slope:** Indicates phase change (melting or boiling of metal/oxide).

4. **Carbon lines:**

• **$2C + O_2 \rightarrow 2CO$**: Slopes **downward** ($\Delta S^\circ > 0$, CO gas produced). Active at high T.

• **$C + O_2 \rightarrow CO_2$**: Nearly **horizontal** ($\Delta S^\circ \approx 0$).

• **$2CO + O_2 \rightarrow 2CO_2$**: Slopes **upward** ($\Delta S^\circ < 0$).

Golden Rules for JEE Problems

• **Reducing Ability Rule:** A metal whose line lies **LOWER** on the Ellingham diagram can reduce the oxide of any metal whose line lies **ABOVE** it at that specific temperature.

• **Example:** Mg line is below Al_2O_3 line below 1350°C → **Mg can reduce Al_2O_3** below 1350°C.

• Above 1350°C, Al line goes below Mg line → **Al can reduce MgO** above 1350°C.

• **Blast Furnace (Iron):** Below 1073 K, **CO** is a better reducing agent than C (CO line is lower). Above 1073 K, **C** is a better reducing agent (C → CO line is lower).

6. METHODS OF REDUCTION TO FREE METAL

Reduction Method	Chemical Mechanism & Key Reactions	Target Metals
1. Smelting (Carbon / CO Reduction)	Heating oxide ore with coke (C) or CO in presence of flux in a furnace. • $MO + C \rightarrow M + CO\uparrow$ • $MO + CO \rightarrow M + CO_2\uparrow$ • For Sn: $SnO_2 + 2C + CaO \rightarrow Sn + 2CO\uparrow + CaSiO_3$ (Slag)	Fe, Pb, Zn, Sn, Cu
2. Gold-Schmidt Thermite Process (Aluminothermic)	Highly exothermic reduction using metallic Al powder for oxides with high melting points. • $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr + Heat$ • $3MnO_2 + 4Al \rightarrow 2Al_2O_3 + 3Mn + Heat$ • $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + Heat$ (Thermite welding)	Cr, Mn, Fe
3. Kroll & IMI Process (Active Metal Reduction)	Reduction of halide with active metals (Na, Mg) in inert Ar atmosphere. • Titanium: $TiO_2 + 2C + 2Cl_2 \rightarrow TiCl_4 + 2CO\uparrow$ • $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$ (Kroll Process) • $TiCl_4 + 4Na \rightarrow Ti + 4NaCl$ (Hunter Process)	Ti, Zr, Hf, Ta
4. Auto-Reduction (Self-Reduction)	Sulphide ore is partially roasted to oxide/sulphate, which then reacts with remaining unreacted sulphide without external agent. • Copper: $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2\uparrow$ then $\mathbf{Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2\uparrow}$ • Lead: $2PbS + 3O_2 \rightarrow 2PbO + 2SO_2\uparrow$ then $\mathbf{PbS + 2PbO \rightarrow 3Pb + SO_2\uparrow}$ • Mercury: $HgS + O_2 \rightarrow Hg + SO_2\uparrow$	Cu, Pb, Hg (Less electropositive metals)
5. Electrolytic Reduction (Electrometallurgy)	Electrolysis of fused/molten salts (chlorides/oxides) for highly electropositive metals. Water cannot be used as solvent due to higher discharge potential of H^+.	Alkali metals (Na, K), Alkaline earths (Mg, Ca), Al
6. Hydrometallurgy (Displacement)	Dissolving metal into soluble complex followed by displacement with more electropositive metal. • $4Au + 8NaCN + O_2 + 2H_2O \rightarrow 4Na[Au(CN)_2] + 4NaOH$ • $2Na[Au(CN)_2] + Zn \rightarrow \mathbf{Na_2[Zn(CN)_4] + 2Au}\downarrow$	Au, Ag

7. EXTRACTION OF IRON (BLAST FURNACE DETAILS)

Blast Furnace Temperature Zones

1. **Combustion Zone (1900 K – 2200 K):**
$C + O_2 \rightarrow CO_2$ ($\Delta H = -393.5$ kJ)

2. **Fusion Zone (1400 K – 1600 K):**
$CO_2 + C \rightarrow 2CO$ ($\Delta H = +172$ kJ, Endothermic)

3. **Slag Formation Zone (900 K – 1200 K):**
$CaCO_3 \rightarrow CaO + CO_2\uparrow$
$CaO + SiO_2 \rightarrow CaSiO_3$ (Slag)

4. **Reduction Zone (500 K – 900 K):**
$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$
$Fe_3O_4 + CO \rightarrow 3FeO + CO_2$
$FeO + CO \rightarrow Fe + CO_2$ (Spongy Iron)

Iron Types & Carbon % Comparison

Type of Iron	Carbon %	Key Features
Pig Iron	~ 4%	Direct from blast furnace, extremely brittle.
Cast Iron	~ 3%	Made by melting pig iron with scrap iron. Extremely hard & brittle.
Wrought Iron	< 0.2%	**Purest commercial form of iron.** Made in Reverberatory furnace lined with Fe_2O_3 (Hematite).
Steel	0.2 – 1.5%	Intermediate purity, highly ductile & malleable.

8. REFINING & PURIFICATION METHODS

Refining Process	Mechanism & Reactions	Metals Refined
Liquation	Low melting metal flows down a sloping hearth leaving behind high melting impurities (dross).	**Sn, Pb, Bi, Hg** (Metals with low m.p.)
Distillation	Volatile metals are evaporated and condensed, leaving behind non-volatile impurities.	**Zn, Cd, Hg** (Metals with low b.p.)
Zone Refining (Fractional Crystallisation)	Based on principle that impurities are **more soluble in molten state (melt)** than in solid metal. Circular mobile heater moves slowly.	Semiconductors & Ultra-pure metals: **Si, Ge, Ga, B, In**
Vapour Phase Refining	Metal converted to volatile compound at lower T, which then decomposes at higher T to give pure metal. • **Mond's Process (for Ni):** $Ni + 4CO \xrightarrow{50^oC} Ni(CO)_4$ (volatile) $Ni(CO)_4 \xrightarrow{200^oC}$ **Ni (Pure) + 4CO↑** • **Van-Arkel Process (for Zr, Ti):** $Zr + 2I_2 \xrightarrow{523\ K} ZrI_4$ (volatile) $ZrI_4 \xrightarrow{1800\ K,\ W\text{-}Filament}$ **Zr (Pure) + $2I_2$**	Ni (Mond) Zr, Ti, Hf, Th (Van-Arkel)
Electrolytic Refining	• **Anode:** Impure crude metal. • **Cathode:** Thin strip of pure metal. • **Electrolyte:** Aqueous metal salt + acid. • **Anode Mud:** Insoluble noble impurities (Au, Ag, Pt, Sb, Se, Te) settle below anode.	Cu, Zn, Ag, Au, Al, Pb, Ni
Chromatography	Based on differential adsorption of components on stationary phase (e.g., Al_2O_3 column).	Trace elements & lanthanides.
Cupellation	Lead impurity in Ag is oxidized to volatile Litharge (PbO) by hot air blast on bone ash cupel.	Ag containing Pb impurity.

9. DETAILED EXTRACTION FLOWCHARTS: AL & CU

Hall-Héroult & Baeyer's Process (Aluminium)

1. **Leaching (Baeyer's Process):**

$Al_2O_3 \cdot 2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$

$2NaAlO_2 + CO_2 + 3H_2O \rightarrow 2Al(OH)_3\downarrow + Na_2CO_3$

$2Al(OH)_3 \xrightarrow{1470\ K} Al_2O_3 + 3H_2O$

2. **Hall-Héroult Electrolysis:**

• **Electrolyte:** Purified Al_2O_3 + **Na_3AlF_6 (Cryolite) + CaF_2 (Fluorspar)**.

Role of Cryolite/CaF_2: Lowers m.p. of Al_2O_3 (from 2323K to ~1140K) and increases electrical conductivity.

• **Cathode (Carbon lining):** $Al^{3+} + 3e^- \rightarrow Al(l)$

• **Anode (Graphite rods):** $C + O^{2-} \rightarrow CO + 2e^-$ | $C + 2O^{2-} \rightarrow CO_2 + 4e^-$

Note: Graphite anodes burn away continuously and must be replaced.

Copper Metallurgy (Copper Pyrites)

1. **Roasting:**

$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2\uparrow$

2. **Smelting in Blast Furnace:**

FeS oxidized to FeO: $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2\uparrow$

FeO removed by Silica flux: **$FeO + SiO_2 \rightarrow FeSiO_3$ (Slag)**.

Molten mass collected is **Copper Matte (Cu_2S + FeS)**.

3. **Bessemerisation (Bessemer Converter):**

Self-reduction produces **Blister Copper** (98% pure, blistered appearance due to escaping SO_2 bubbles):

$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2\uparrow$

$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2\uparrow$

10. IMPORTANT ALLOYS & QUICK REFERENCE TABLE

Alloy Name	Composition	Key JEE Applications & Characteristics
Brass	Cu (60-80%) + Zn (20-40%)	Utensils, condenser tubes, musical instruments.
Bronze / Bell Metal	Cu (75-90%) + Sn (10-25%)	Statues, coins, bells (harder than pure Cu).
German Silver	Cu (25-50%) + Zn (25-35%) + Ni (10-35%)	**Contains NO Silver (0% Ag)!** Used for decorative items.
Duralumin	Al (95%) + Cu (4%) + Mg (0.5%) + Mn (0.5%)	Aircraft bodies, light structural parts.
Alnico	Al + Ni + Co + Fe	Making powerful permanent magnets.
Stainless Steel	Fe + Cr (12-20%) + Ni (8-10%) + C	Resists corrosion due to oxide layer formation.
Monel Metal / Constantan	Ni + Cu (+ Fe) / Cu (60%) + Ni (40%)	Corrosion-resistant alloy / Electrical measuring devices.

JEE Advanced Super-Quick Revision Checklist

- **Cyanide Process Leaching (Au/Ag):** Requires oxygen (O_2) as an oxidizing agent. Reaction is aerobic!
- **Froth Floatation Collector:** Xanthates increase water repellency (hydrophobicity) of sulphide particles.
- **Serpeck's Process:** Used for Bauxite containing **Silica (SiO_2)** impurity using Nitrogen gas (**AlN** intermediate).
- **Hoope's Process:** Electrolytic refining of Aluminium using three molten layers of different densities.

P-BLOCK ELEMENTS — JEE REVISION SHEET

Complete Formulae, Structural Parameters, Reactions & Trends | Groups 13 to 18

JEE Main & Advanced
High-Density Quick Ref

General p-Block Characteristics: Outer Config: $ns^2\ np^{1-6}$ (He: $1s^2$). **Inert Pair Effect:** Reluctance of ns^2 electron pair to participate in bonding down the group due to poor shielding of *d/f* orbitals. Higher O.S. stability decreases; Lower O.S. stability increases down the group.

1. GROUP 13: BORON FAMILY ($NS^2\ NP^1$)

Periodic Trends & Physical Properties

- **Atomic Radius:** **B < Ga < Al < In < Tl** *(Anomalous Ga < Al due to poor screening of 3d-e⁻; σ_{3d} poor)*.
- **Ionization Enthalpy (IE_1):** **B > Tl > Ga > Al > In**.
- **Electronegativity:** **B > Tl > In > Ga > Al** (B=2.0, Al=1.5).
- **Density:** Increases down the group (**B < Al < Ga < In < Tl**).
- **Melting Point:** **B > Al > Tl > In > Ga** (Ga M.P. = 29.67°C / 303 K; liquid up to 2000°C).
- **Oxidation State Stability:**
  +3 O.S.: **$B^{+3} > Al^{+3} > Ga^{+3} > In^{+3} > Tl^{+3}$** ($Tl^{+3}$ is strong oxidant).
  +1 O.S.: **$Tl^{+1} > In^{+1} > Ga^{+1} > Al^{+1} > B^{+1}$** ($Tl^{+1}$ most stable).
- **Reducing Character:** **Al > Ga > In > Tl**.
- **Lewis Acid Strength of Trihalides:** **$BF_3 < BCl_3 < BBr_3 < BI_3$** *(Back-bonding: 2pπ–2pπ in BF_3 strongest → reduces Lewis acidity)*.

Important Boron Compounds & Reactions

1. Diborane (B_2H_6):
• Prep: $4BF_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3LiF + 3AlF_3$
• Prep: $2NaBH_4 + I_2 \rightarrow B_2H_6 + 2NaI + H_2\uparrow$
• Prep (Ind): $2BF_3 + 6NaH \xrightarrow{\textbf{450 K}} B_2H_6 + 6NaF$
• Combustion: $B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$ ($\Delta H = -1976$ kJ/mol)
• Hydrolysis: $B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 6H_2\uparrow$
• **Structure:** 4 terminal B–H bonds (2c–2e⁻, regular). 2 bridge B–H–B bonds (3c–2e⁻ banana bonds). B is sp^3 hybridized. Terminal H–B–H angle = 120°, Bridge H–B–H = 97°.
• **Ammonia Rxn:**
$B_2H_6 + 2NH_3$ (excess, low T) $\rightarrow [BH_2(NH_3)_2]^+[BH_4]^-$
$[BH_2(NH_3)_2]^+[BH_4]^- \xrightarrow{\textbf{Heat (450 K)}} 2B_3N_3H_6$ (Borazine) $+ 12H_2$

2. Borazine ($B_3N_3H_6$) — "Inorganic Benzene":
• Isoelectronic & isostructural with benzene (C_6H_6).
• Planar 6-membered ring, alternating B and N. Both B and N are sp^2.
• More reactive than benzene toward addition reactions due to polar $B^{\delta+}$–$N^{\delta-}$ bonds.

3. Borax ($Na_2B_4O_7{\cdot}10H_2O$):
• Real structural formula: **$Na_2[B_4O_5(OH)_4]{\cdot}8H_2O$**
• Contains 5 B–O–B bridges, two sp^2 B and two sp^3 B atoms.
• Aqueous solution is alkaline: $Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + 4H_3BO_3$.
• **Borax Bead Test:** $Na_2B_4O_7{\cdot}10H_2O \xrightarrow{\textbf{Δ (−10H₂O)}} Na_2B_4O_7 \xrightarrow{\textbf{Δ (740°C)}} 2NaBO_2 + B_2O_3$ (glassy bead). Metaborates show distinct colors: **$CoO + B_2O_3 \rightarrow Co(BO_2)_2$ (Blue bead)**.

4. Orthoboric Acid (H_3BO_3 / $B(OH)_3$):
• Layered structure linked by hydrogen bonds.
• Weak **monobasic Lewis acid** (not protonic): $B(OH)_3 + 2H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$ ($pK_a = 9.25$).
• **Polyhydroxy Titration:** Cannot be titrated directly with NaOH (end-point indefinite). Titrated in presence of *cis-1,2-diols* (glycerol, mannitol, catechol) which form stable chelate complexes with $[B(OH)_4]^-$, increasing acidity.
• **Action of Heat:** $H_3BO_3 \xrightarrow{\textbf{100°C (−H₂O)}} HBO_2$ (Metaboric) $\xrightarrow{\textbf{160°C}} H_2B_4O_7$ (Tetraboric) $\xrightarrow{\textbf{Red Hot}} B_2O_3$.
• **Ethyl Borate Test:** $H_3BO_3 + 3C_2H_5OH \xrightarrow{\textbf{conc. H₂SO₄}} B(OC_2H_5)_3$ (Triethyl borate, burns with green-edged flame) $+ 3H_2O$.

5. Aluminium Compounds & Alums:
• **Anhydrous $AlCl_3$:** Covalent dimer Al_2Cl_6 in vapor/non-polar solvent (4 terminal Cl, 2 bridge 3c–4e⁻ Cl–Al bonds, sp^3 Al). In H_2O: dissociates into $[Al(H_2O)_6]^{3+} + 3Cl^-$ (hydration energy exceeds lattice energy). Fumes in moist air due to HCl gas generation.
• **Alums:** General formula $M_2SO_4{\cdot}M'_2(SO_4)_3{\cdot}24H_2O$ [M = Na^+, K^+, NH_4^+; M' = Al^{3+}, Fe^{3+}, Cr^{3+}]. Potash Alum: $K_2SO_4{\cdot}Al_2(SO_4)_3{\cdot}24H_2O$. Aqueous solution is acidic due to hydrolysis of Al^{3+}.

2. GROUP 14: CARBON FAMILY ($NS^2\ NP^2$)

Periodic Trends & General Properties

Allotropes of Carbon

- **Catenation Tendency:** **C >> Si > Ge ≈ Sn >> Pb** (Directly proportional to M–M bond energy: C–C = 348 kJ/mol, Si–Si = 222, Ge–Ge = 167, Sn–Sn = 155).
- **Oxidation State Stability:**
  +4 O.S.: **$C^{+4} > Si^{+4} > Ge^{+4} > Sn^{+4} > Pb^{+4}$** ($Pb^{+4}$ compounds are strong oxidants: PbO_2, $PbCl_4$).
  +2 O.S.: **$Pb^{+2} > Sn^{+2} > Ge^{+2} > Si^{+2} > C^{+2}$** ($Sn^{+2}$ is a reducing agent).
- **Oxides Nature:**
  Dioxides: CO_2, SiO_2, GeO_2 (Acidic); SnO_2, PbO_2 (Amphoteric).
  Monoxides: CO (Neutral); GeO (Acidic); SnO, PbO (Amphoteric).
- **Halides:** MX_4 mostly covalent (Exceptions: SnF_4, PbF_4 ionic). Stability of MX_2: **$GeX_2 < SnX_2 < PbX_2$**. CCl_4 cannot be hydrolysed (no d-orbitals on C); $SiCl_4$ hydrolyses rapidly: $SiCl_4 + 4H_2O \rightarrow Si(OH)_4 + 4HCl$. PbI_4 does not exist Added (Pb^{4+} oxidizes I^- to I_2).

• **Diamond:** 3D rigid framework, sp^3 C (C–C = 154 pm), bond angle 109.5°. Electrical insulator, high thermal conductivity.
• **Graphite:** Layered hexagonal sheet, sp^2 C (C–C = 141.5 pm), inter-layer distance = 340 pm (held by weak van der Waals forces). Good electrical conductor (delocalized π-electrons). Thermodynamically most stable allotrope ($\Delta_f H° = 0$).
• **Fullerenes (C_{60} - Buckminsterfullerene):** Soccer ball shape, sp^2 C. Contains **20 six-membered rings and 12 five-membered rings**. Six-membered ring fused to 5 or 6-membered rings; 5-membered ring fused *only to 6-membered rings*. C–C single = 143.5 pm, C=C double = 138.3 pm.

Silicones & Silicates:
• **Silicones:** Organosilicon polymers with repeating unit **$(R_2SiO)_n$** with Si–O–Si linkages.
- Chain Propagating: $R_2SiCl_2 + 2H_2O \rightarrow R_2Si(OH)_2 \rightarrow$ Linear Silicone.
- Cross-linking: $RSiCl_3 \rightarrow$ Cross-linked Silicone.
- Chain Terminating/Stopping Agent: R_3SiCl (monochlorosilane).
• **Silicates:** Basic structural unit = **$[SiO_4]^{4-}$** (Tetrahedral, sp^3 Si).
- Orthosilicates: Discrete $[SiO_4]^{4-}$ (0 shared corners, e.g., Mg_2SiO_4).
- Pyrosilicates: $[Si_2O_7]^{6-}$ (1 shared corner, e.g., Thortveitite).
- Cyclic/Chain Silicates: 2 shared corners per tetrahedron ($(SiO_3)_n^{2n-}$).
- 3D Silicates: All 4 oxygen corners shared (e.g., Quartz, Feldspar).
• **Zeolites:** Aluminosilicates (3D network where some Si^{4+} replaced by Al^{3+}). ZSM-5 converts alcohol directly into gasoline.

Compounds of Lead, Tin & Carbides:
• **Red Lead (Pb_3O_4):** Mixed oxide ($2PbO·PbO_2$). Rxn with HNO_3: $Pb_3O_4 + 4HNO_3 \rightarrow 2Pb(NO_3)_2 + PbO_2\downarrow$ (brown) + $2H_2O$.
• **Tin Reactions with HNO_3:**
- Dilute HNO_3: $4Sn + 10HNO_3 \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$
- Conc. HNO_3: $Sn + 4HNO_3 \rightarrow H_2SnO_3$ (Metastannic acid)$\downarrow + 4NO_2 + H_2O$
• **Stannous Chloride ($SnCl_2$):** Reducing agent. Reduces $HgCl_2$ (white ppt Hg_2Cl_2) → Hg (black ppt): $2HgCl_2 + SnCl_2 \rightarrow Hg_2Cl_2\downarrow + SnCl_4$; $Hg_2Cl_2 + SnCl_2 \rightarrow 2Hg\downarrow + SnCl_4$.
• **Carbides Classification:**
- Methanides (yield CH_4 on H_2O): Be_2C, Al_4C_3 (contain C^{4-}).
- Acetylides (yield C_2H_2 on H_2O): CaC_2, MgC_2 (contain C_2^{2-}).
- Allylides (yield propyne C_3H_4): Mg_2C_3 (contains C_3^{4-}).

3. GROUP 15: NITROGEN FAMILY / PNICTOGENS ($NS^2\ NP^3$)

Periodic Trends & Hydrides

- **Covalency Limit:** N max covalency = 4 (no vacant 2d-orbitals). Others (P, As, Sb, Bi) extend to 5 or 6 (e.g., PCl_5, $[PF_6]^-$).
- **Hydrides (MH_3):** NH_3, PH_3, AsH_3, SbH_3, BiH_3.
  • **Thermal Stability / Bond Dissociation Energy:** **$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$**.
  • **Reducing Character:** **$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$** ($BiH_3$ strongest RA).
  • **Basic Character:** **$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$** (Lone pair concentration).
  • **Bond Angle:** **NH_3 (107.8°) > PH_3 (93.5°) > AsH_3 (91.8°) > SbH_3 (91.3°)** (Drago's Rule: PH_3 to BiH_3 involve almost pure p-orbitals).
  • **Boiling Point:** **$PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$** ($NH_3$ high due to H-bonding).
  • **Melting Point:** **$PH_3 < AsH_3 < SbH_3 < NH_3$**.

Oxides & Halides of Group 15

• **Oxides of N:** N_2O (+1, Neutral, Linear), NO (+2, Neutral, Paramagnetic gas), N_2O_3 (+3, Acidic, Blue solid, N–N bond), NO_2 (+4, Acidic, Brown gas, Paramagnetic dimerizes to N_2O_4), N_2O_5 (+5, Acidic, Solid exists as $[NO_2]^+[NO_3]^-$).
• **Acidic Character of Oxides:** Increases with Higher O.S. (**$N_2O_3 < N_2O_5$**). Down group: **N_2O_3, P_2O_3 (Acidic) > As_2O_3, Sb_2O_3 (Amphoteric) > Bi_2O_3 (Basic)**.
• **Halides:** Trihalides (MX_3, sp^3, Pyramidal). Pentahalides (MX_5, sp^3d, Trigonal bipyramidal). N cannot form pentahalides.
• **PCl_5 Behavior:** Axial bonds longer & weaker than equatorial bonds. Hydrolysis: $PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$; $POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$. **Solid PCl_5 is ionic: $[PCl_4]^+[PCl_6]^-$** (Tetrahedral + Octahedral). PBr_5(s) exists as $[PBr_4]^+Br^-$.

Nitric Acid (HNO_3) & Disproportionation:

• **Ostwald's Process:**

1. $4NH_3 + 5O_2 \xrightarrow{\text{Pt/Rh, 500 K, 9 bar}} 4NO + 6H_2O$
2. $2NO + O_2 \rightleftharpoons 2NO_2$
3. $3NO_2 + H_2O \rightarrow 2HNO_3 + NO$

• **Disproportionation in Acid:** $3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$ (+3 to +5 and +2).

• **Nitrate Brown Ring Test:** $NO_3^- + 3Fe^{2+} + 4H^+ \rightarrow NO + 3Fe^{3+} + 2H_2O$; $[Fe(H_2O)_6]^{2+} + NO \rightarrow$ **$[Fe(H_2O)_5(NO)]^{2+}$ (Brown ring, O.S. of Fe = +1, NO = +1)** + H_2O.

• **Reactions with Metals:**

- **Cu + Dil. HNO_3 (30%):** $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO\uparrow + 4H_2O$
- **Cu + Conc. HNO_3:** $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2\uparrow + 2H_2O$
- **Zn + Dil. HNO_3:** $4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O\uparrow + 5H_2O$
- **Zn + Conc. HNO_3:** $Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2NO_2\uparrow + 2H_2O$
- Passivity: Fe, Cr, Al, Ni rendered passive by conc. HNO_3.

Phosphorus Allotropes & Compounds:

• **White P (P_4):** Discrete tetrahedral P_4 ($\angle P-P-P = 60°$, high angle strain). Translucent waxy solid, poisonous, soluble in CS_2, chemiluminescent (glows in dark). Burns in air: $P_4 + 5O_2 \rightarrow P_4O_{10}$. Reacts with alkali: $P_4 + 3NaOH + 3H_2O \rightarrow$ **$PH_3 + 3NaH_2PO_2$** (Disproportionation).

• **Red P:** Polymeric chain of P_4 tetrahedra linked via covalent bonds. Non-poisonous, insoluble in CS_2, does not glow in dark. Obtained by heating White P at 573 K in inert atmosphere.

• **Black P:** Thermodynamically most stable. α-black (from Red P at 803 K), β-black (from White P at 473 K under high P). Layered structure.

• **Phosphine (PH_3):** Prep: $Ca_3P_2 + 6H_2O \rightarrow 2PH_3 + 3Ca(OH)_2$. Rotten fish smell. Holme's Signals & Smoke Screens use $Ca_3P_2 + CaC_2$.

Oxoacids of Phosphorus (Key Structural Parameters)

Name	Formula	Oxid. State	No. of P–OH (Basicity)	No. of P–H bonds	No. of P=O bonds	Reducing / Characteristic Features
Hypophosphorous acid	H_3PO_2	+1	1 (Monobasic)	2	1	Strong RA (2 P–H bonds). $AgNO_3 \rightarrow Ag\downarrow$
Orthophosphorous acid	H_3PO_3	+3	2 (Dibasic)	1	1	Good RA. Disproportionates on Δ: $4H_3PO_3 \rightarrow 3H_3PO_4 + PH_3$
Pyrophosphorous acid	$H_4P_2O_5$	+3	2 (Dibasic)	2	2	Contains 1 P–O–P bridge
Hypophosphoric acid	$H_4P_2O_6$	+4	4 (Tetrabasic)	0	2	Contains direct **P–P bond**
Orthophosphoric acid	H_3PO_4	+5	3 (Tribasic)	0	1	Non-reducing, stable acid
Pyrophosphoric acid	$H_4P_2O_7$	+5	4 (Tetrabasic)	0	2	Contains 1 P–O–P bridge ($2\,H_3PO_4 - H_2O$)
Metaphosphoric acid	$(HPO_3)_n$	+5	n	0	n	Trimer $(HPO_3)_3$ cyclic has 3 P–O–P bridges, 3 P=O, 3 P–OH

4. GROUP 16: OXYGEN FAMILY / CHALCOGENS ($NS^2\ NP^4$)

Periodic Trends & Hydrides

- **Electron Gain Enthalpy ($\Delta_{eg}H$): S > Se > Te > Po > O** (O is least negative due to small size & inter-electronic repulsion in 2p).
- **Electronegativity: O (3.5) > S > Se > Te > Po.**
- **Hydrides (H_2E):** H_2O, H_2S, H_2Se, H_2Te, H_2Po.
  - **Acidic Character: $H_2O < H_2S < H_2Se < H_2Te$** (H–E bond energy decreases down group).
  - **Thermal Stability: $H_2O > H_2S > H_2Se > H_2Te$.**
  - **Reducing Power: $H_2O < H_2S < H_2Se < H_2Te$.**
  - **Boiling Point: $H_2S < H_2Se < H_2Te < H_2O$** ($H_2O$ abnormally high due to H-bonding).
  - **Bond Angle: H_2O (104.5°) > H_2S (92.1°) > H_2Se (91°) > H_2Te (90°).**

Ozone (O_3) & Halides of Sulphur

• **Ozone Prep:** $3O_2 \rightleftharpoons 2O_3$ ($\Delta H = +142$ kJ/mol, endothermic). Silent electric discharge used to prevent decomposition.

• **O_3 Properties:** Angular structure (resonance hybrid of 2 canonical forms, O–O = 128 pm, 117°). Powerful oxidant: PbS (black) + $4O_3 \rightarrow PbSO_4$ (white) + $4O_2$. $2I^- + H_2O + O_3 \rightarrow I_2 + 2OH^- + O_2$ (Quantitative estimation of O_3 using standard $Na_2S_2O_3$).

• **Tailing of Mercury:** $2Hg + O_3 \rightarrow Hg_2O + O_2$ (Hg loses meniscus, sticks to glass; restored by H_2O). Absorbed by turpentine oil.

• **SF_4 vs SF_6:** SF_4 (sp^3d, See-saw geometry, 1 lone pair, reactive, hydrolyses easily). SF_6 (sp^3d^2, Octahedral, sterically protected S atom, thermodynamically & kinetically inert, used as gaseous insulator).

Allotropes of Sulphur:
• **Rhombic Sulphur (α-S):** Yellow, M.P. 385.8 K, insoluble in H_2O, soluble in CS_2. Stable below 369 K. Contains crown-shaped S_8 rings.
• **Monoclinic Sulphur (β-S):** M.P. 393 K, soluble in CS_2. Stable above 369 K.
• **Transition Temperature = 369 K (96°C):** Both α-S and β-S coexist stably.
• **Cyclo-S_6:** Adopts chair conformation.
• **S_2 Molecule:** Dominant species at high temperature (~1000 K), **paramagnetic** (2 unpaired electrons in π* antibonding orbitals, like O_2).

Sulphuric Acid (H_2SO_4) — Contact Process:
• 1. $S + O_2 \rightarrow SO_2$
• 2. $2SO_2 + O_2 \xrightleftharpoons{V_2O_5,\ 720\ K,\ 2\ bar} 2SO_3$ ($\Delta H = -196.6$ kJ/mol, exothermic).
• 3. $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$ (Oleum).
• 4. $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$ (96–98%).
• **Properties:** Dehydrating agent: $C_{12}H_{22}O_{11} \xrightarrow{\text{conc. } H_2SO_4} 12C + 11H_2O$ (Charring). $HCOOH \rightarrow CO + H_2O$; $H_2C_2O_4 \rightarrow CO + CO_2 + H_2O$. Moderate Oxidant: $Cu + 2H_2SO_4(conc) \rightarrow CuSO_4 + SO_2 + 2H_2O$.

Oxoacids of Sulphur (Structures & Oxidation States)

Acid Name	Formula	Oxid. State of S	Key Structural Features
Sulphurous acid	H_2SO_3	+4	1 S=O, 2 S–OH, 1 lone pair on S
Sulphuric acid	H_2SO_4	+6	2 S=O, 2 S–OH (Dibasic)
Thiosulphuric acid	$H_2S_2O_3$	+6 (central), −2 (terminal)	1 S=S bond, 1 S=O, 2 S–OH (Average O.S. = +2)
Pyrosulphuric acid (Oleum)	$H_2S_2O_7$	+6	Contains **1 S–O–S bridge**, 4 S=O, 2 S–OH
Peroxomonosulphuric acid (Caro's acid)	H_2SO_5	+6 (not +8!)	Contains **1 peroxo (–O–O–) linkage**: HO–S(=O)$_2$–O–O–H
Peroxodisulphuric acid (Marshall's acid)	$H_2S_2O_8$	+6	Contains **1 peroxo (–O–O–) bridge**: HO–S(=O)$_2$–O–O–S(=O)$_2$–OH
Dithionic acid	$H_2S_2O_6$	+5	Contains direct **S–S bond**, 4 S=O, 2 S–OH

5. GROUP 17: HALOGENS ($NS^2\ NP^5$)

Periodic & Physical Trends

- **Bond Dissociation Enthalpy (X_2):** **$Cl_2 > Br_2 > F_2 > I_2$**
  (F_2 bond is abnormally weaker than Cl_2 & Br_2 due to large lone pair-lone pair electronic repulsion in small F_2 molecule).
  Cl_2 (242.6 kJ) > Br_2 (192.8) > F_2 (158.8) > I_2 (151.1).
- **Electron Gain Enthalpy ($\Delta_{eg}H$):** **Cl > F > Br > I** (Cl = −349 kJ/mol, F = −333 kJ/mol; F lower than Cl due to small 2p orbital space).
- **Electronegativity:** **F (4.0) > Cl (3.0) > Br (2.8) > I (2.5)**.
- **Oxidizing Power:** **$F_2 > Cl_2 > Br_2 > I_2$** (F_2 oxidizes all halide ions: $F_2 + 2X^- \rightarrow 2F^- + X_2$).
- **Hydrogen Halides (HX):**
  • **Acidic Strength:** **HF < HCl < HBr < HI**.
  • **Thermal Stability:** **HF > HCl > HBr > HI**.
  • **Reducing Power:** **HF < HCl < HBr < HI**.
  • **Boiling Point:** **HCl < HBr < HI < HF** (HF highest due to strong intermolecular H-bonding).

Reactions & Bleaching Powder

• **Rxn of Cl_2 with Alkalies:**
  - Cold & Dilute NaOH: $2NaOH + Cl_2 \rightarrow$ **NaCl + NaOCl** (Sodium hypochlorite, +1) + H_2O.
  - Hot & Concentrated NaOH: $6NaOH + 3Cl_2 \rightarrow$ **5NaCl + $NaClO_3$** (Sodium chlorate, +5) + $3H_2O$.
• **Bleaching Powder:** $Ca(OH)_2 + Cl_2 \rightarrow$ **$Ca(OCl)_2 \cdot CaCl_2 \cdot Ca(OH)_2 \cdot 2H_2O$** (Composition of dry bleaching powder). Active ingredient = $Ca(OCl)_2$. Bleaching action is permanent due to oxidation: $Cl_2 + H_2O \rightarrow HOCl + HCl$; $HOCl \rightarrow HCl + [O]$.
• **Aqua Regia:** 3 parts conc. HCl + 1 part conc. HNO_3. Dissolves Au & Pt:
$Au + 4H^+ + NO_3^- + 4Cl^- \rightarrow [AuCl_4]^- + NO + 2H_2O$
$Pt + 16H^+ + 4NO_3^- + 18Cl^- \rightarrow 3[PtCl_6]^{2-} + 4NO + 8H_2O$.

Interhalogen Compounds (AX_n):
• Formed between two different halogens (A is less electronegative, larger size). n = 1, 3, 5, 7.
• Interhalogens are **more reactive than parent halogens** (except F_2) because A–X bond is polar and weaker than X–X bond.
• Types & Geometries:
  - **AX (ClF, BrF, ICl):** Linear.
  - **AX_3 (ClF_3, BrF_3, IF_3):** sp^3d, **Bent T-shape** (2 lone pairs on central atom). ClF_3 used for enrichment of U: $U + 3ClF_3 \rightarrow UF_6(g) + 3ClF$.
  - **AX_5 (ClF_5, BrF_5, IF_5):** sp^3d^2, **Square Pyramidal** (1 lone pair).
  - **AX_7 (IF_7):** sp^3d^3, **Pentagonal Bipyramidal** (0 lone pairs).

Oxoacids of Halogens & Polyhalides:
• Fluorine forms only ONE oxoacid: **HOF** (Hypofluorous acid, O.S. of F = −1).
• **Acidic Strength of Chlorine Oxoacids:**
$HOCl < HClO_2 < HClO_3 < HClO_4$
(Reason: Conjugate base ClO_4^- is stabilized by resonance over 4 oxygen atoms).
• Acidic strength for same O.S.: **HOCl > HOBr > HOI**.
• **Polyhalides:** I_2 is sparingly soluble in H_2O, but highly soluble in KI due to formation of **I_3^-** ion: $KI + I_2 \rightarrow K^+[I_3]^-$. Geometry of I_3^-: sp^3d, **Linear** (3 lone pairs at equatorial positions).

6. GROUP 18: NOBLE GASES ($NS^2\ NP^6$)

Physical Trends & General Properties

- **Atmospheric Abundance:** Ar > Ne > Kr > He > Xe (Rn does not occur naturally, decay product of Radium).
- **Boiling Point & Melting Point:** He < Ne < Ar < Kr < Xe < Rn (He has lowest B.P. of any known substance = 4.2 K). Held only by weak dispersion forces.
- **Ionization Enthalpy:** He > Ne > Ar > Kr > Xe > Rn.
- **First Noble Gas Compound:** Synthesized by Neil Bartlett (1962): $O_2^+[PtF_6]^-$ led to $Xe^+[PtF_6]^-$ (red solid) because IE of Xe (1170 kJ/mol) ≈ IE of O_2 molecule (1175 kJ/mol).
- **Dewar's Charcoal Method:** Separation of noble gases using coconut charcoal based on differential adsorption at different temperatures (He, Ne not adsorbed at 173 K; Ar, Kr, Xe adsorbed).

Preparation of Xenon Fluorides

• **XeF_2 (Linear):** $Xe + F_2$ (1 : 1 ratio) **673 K, 1 bar**→ $XeF_2(s)$
• **XeF_4 (Square Planar):** $Xe + F_2$ (1 : 5 ratio) **873 K, 7 bar**→ $XeF_4(s)$
• **XeF_6 (Distorted Octahedral):** $Xe + F_2$ (1 : 20 ratio) **573 K, 60–70 bar**→ $XeF_6(s)$
• $XeF_4 + O_2F_2$ **143 K**→ $XeF_6 + O_2$
• **Fluoride Ion Donor:**
$XeF_2 + PF_5 \rightarrow [XeF]^+[PF_6]^-$
$XeF_4 + SbF_5 \rightarrow [XeF_3]^+[SbF_6]^-$
• **Fluoride Ion Acceptor:**
$XeF_6 + MF$ (M = Na, K, Rb, Cs) → $M^+[XeF_7]^-$

Xenon Compounds Geometry & Hydrolysis Summary

Compound	Hybridization	Lone Pairs	Bond Pairs	Molecular Geometry	Hydrolysis Reactions
XeF_2	sp^3d	3	2	Linear	$2XeF_2 + 2H_2O \rightarrow 2Xe + 4HF + O_2$ (Complete)
XeF_4	sp^3d^2	2	4	Square Planar	$6XeF_4 + 12H_2O \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$ (Disproportionation)
XeF_6	sp^3d^3	1	6	Distorted Octahedral	• Complete: $XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$ • Partial (1 eq H_2O): $XeF_6 + H_2O \rightarrow$ **$XeOF_4$** + 2HF • Partial (2 eq H_2O): $XeF_6 + 2H_2O \rightarrow$ **XeO_2F_2** + 4HF
XeO_3	sp^3	1	3 (double)	Trigonal Pyramidal	Colourless explosive solid. Rxn with base: $XeO_3 + OH^- \rightarrow HXeO_4^-$ (Xenate)
$XeOF_4$	sp^3d^2	1	5 (1 O + 4 F)	Square Pyramidal	Colourless volatile liquid. Hydrolyses to XeO_2F_2 and XeO_3
XeO_2F_2 Added	sp^3d	1	4 (2 O + 2 F)	See-saw	Intermediate in partial hydrolysis of XeF_6

Key JEE Advanced Exam Traps & Special Highlights:
1. **Paramagnetic Species in p-Block:** NO (gas), NO_2 (gas), S_2 (vapors at ~1000 K), O_2 gas.
2. **Solid State Structures:** $PCl_5(s) = [PCl_4]^+[PCl_6]^-$; $PBr_5(s) = [PBr_4]^+Br^-$; $N_2O_5(s) = [NO_2]^+[NO_3]^-$; $XeF_6(s) = [XeF_5]^+F^-$ Added.
3. **Back Bonding Effect:** Order of Lewis acidity of $BF_3 < BCl_3 < BBr_3 < BI_3$ ($2p\pi-2p\pi$ overlap in BF_3 is maximum).
4. **Inertness of SF_6:** Thermodynamically unstable toward hydrolysis but kinetically extremely inert due to steric crowding around sulphur.

JEE FORMULA & QUICK REVISION SHEET

Organic Chemistry: Some Basic Principles & Techniques (IUPAC Naming)

JEE MAIN + ADVANCED

1. Fundamental Concepts & Classification

Vital Force Theory & Syntheses

- **Vital Force Theory:** Berzelius proposed organic compounds only originate from living sources.
- **Wöhler Synthesis (1828):** Synthesized first organic compound (Urea) from inorganic precursor:
  $NH_4Cl + KCNO \rightarrow KCl + NH_4CNO$
  $NH_4CNO \xrightarrow{\Delta \text{ (rearrange)}} H_2N{-}CO{-}NH_2$ (Urea)
- **Kolbe's Synthesis:** Acetate → Methane/Acetic Acid [Added]
- **Berthelot Synthesis:** Synthesis of Methane ($C + 2H_2 \rightarrow CH_4$) [Added]

Structural Classification of Organic Compounds

- **Acyclic / Aliphatic:** Open-chain compounds.
  - **Saturated:** Only σ-bonds (Alkanes, C_nH_{2n+2}).
  - **Unsaturated:** At least one π-bond (Alkenes C_nH_{2n}, Alkynes C_nH_{2n-2}).
- **Cyclic / Ring Chain:** Contains at least one ring.
  - **Homocyclic / Carbocyclic:** Ring composed of *only Carbon*.
  - **Heterocyclic:** Ring contains heteroatoms (O, S, N).

2. AROMATICITY & HÜCKEL'S RULE FORMULAE

Hückel's Rule Formula for Aromaticity: Total π-electrons = (4n + 2) π-e⁻ (where n = 0, 1, 2, 3...)

Aromatic Criteria:

- Cyclic & Completely Planar (sp^2/sp hybridized)
- Complete cyclic delocalization of π-electrons
- Follows `(4n + 2) π-e⁻` rule
- Exhibits high resonance energy & resists addition reactions

Antiaromatic Criteria [Added] :

- Cyclic, Planar, Conjugated ring system
- Follows `(4n) π-e⁻` rule (n = 1, 2, 3...)
- Highly unstable at room temperature
- *e.g.*, Cyclobutadiene (4πe⁻), Cyclopropenyl anion (4πe⁻)

Standard Aromatic Examples:

- **n=0 (2π e⁻):** Cyclopropenyl cation
- **n=1 (6π e⁻):** Benzene, Pyrrole, Furan, Thiophene, Pyridine, Tropylium cation
- **n=2 (10π e⁻):** Naphthalene, Azulene
- **n=3 (14π e⁻):** Anthracene, Phenanthrene, Pyrene

3. IUPAC NAME ARCHITECTURE & PRIMARY RULES

IUPAC NAME = Secondary Prefix + Primary Prefix + Word Root + Primary Suffix + Secondary Suffix

Component	Function / Meaning	Examples
Secondary Prefix	Indicates side-chains / Substituents / non-principal functional groups	-alkyl (-CH_3: methyl), -halo (-Cl: chloro), -nitro (-NO_2), -alkoxy (-OCH_3)
Primary Prefix	Indicates cyclic nature of parent chain	**cyclo** (for monocyclic), **bicyclo** [Added], **spiro** [Added]
Word Root	Number of carbons in Principal Carbon Chain (PCC)	1: Meth, 2: Eth, 3: Prop, 4: But, 5: Pent, 6: Hex, 7: Hept, 8: Oct, 9: Non, 10: Dec
Primary Suffix	Indicates saturation/unsaturation level of PCC	**-ane** (C-C single), **-ene** (C=C double), **-yne** (C≡C triple), **-adiene** (2 C=C)
Secondary Suffix	Indicates Principal Functional Group (PFG)	**-oic acid** (-COOH), **-al** (-CHO), **-one** (>C=O), **-ol** (-OH), **-amine** (-NH_2)

Rule 1: Longest Chain Rule (PCC Selection)

Rule 2: Numbering Rules & Locants

- Select chain with maximum number of Principal Functional Groups.
- If PFG absent, select chain with max number of multiple bonds (C=C or C≡C).
- If multiple bonds equal, select chain with maximum length (max C-atoms).
- If lengths equal, select chain with **maximum number of substituents**.
- If substituents equal, select chain giving **lowest locant set to substituents**.

- **Lowest Locant Rule:** Number from end giving lowest number to first substituent at first point of difference.
- **Lowest Sum Rule:** Used historically; superseded by First Point of Difference.
- **First Point of Difference Rule:** Compare locant sets position-by-position; select set with lower number at first distinction (*e.g.*, locant set 2,3,6 preferred over 2,4,5).
- **Alphabetical Priority:** If identical locants from both ends, lower locant goes to substituent earlier in alphabet (*e.g.*, 3-ethyl-4-methylhexane).

4. MASTER SENIORITY PRIORITY TABLE FOR POLYFUNCTIONAL COMPOUNDS

Priority: -COOH > -SO_3H > Anhydride > Ester > Acid Halide > Amide > Nitrile > Aldehyde > Ketone > Alcohol > Thiol > Amine > C=C > C≡C > Substituents

Priority	Functional Group	Structure	Prefix Name (Substituent)	Suffix Name (Principal Group)	Special Special Suffix (Carbon not in PCC)
1	Carboxylic Acid	−COOH	Carboxy−	−oic acid	−carboxylic acid
2	Sulphonic Acid	−SO_3H	Sulpho−	−sulphonic acid	−sulphonic acid
3	Acid Anhydride	−CO−O−CO−	—	−oic anhydride	−carboxylic anhydride
4	Ester	−COOR	Alkoxycarbonyl− / Carbalkoxy−	alkyl ... −oate	alkyl ... −carboxylate
5	Acid Chloride	−COX	Haloformyl− / Halocarbonyl−	−oyl halide	−carbonyl halide
6	Amide	−$CONH_2$	Carbamoyl−	−amide	−carboxamide
7	Nitrile (Cyanide)	−C≡N	Cyano−	−nitrile	−carbonitrile
8	Isonitrile [Added]	−N≡C	Isocyano−	−carbylamine / −isonitrile	−carbonitrile
9	Aldehyde	−CHO	Formyl− / Oxo− / Aldo−	−al	−carbaldehyde
10	Ketone	>C=O	Keto− / Oxo−	−one	—
11	Alcohol	−OH	Hydroxy−	−ol	—
12	Thiol / Thioalcohol	−SH	Mercapto−	−thiol	—
13	Amine	−NH_2	Amino−	−amine	—
14	Alkene	>C=C<	—	−ene	—
15	Alkyne	−C≡C−	—	−yne	—
16	Ethers / Halides / Nitro	−OR / −X / −NO_2	Alkoxy− / Halo− / Nitro−	*(Always Substituents)*	—

5. SPECIALIZED IUPAC RULES & ADVANCED CASES

Rule for Special Suffix (3+ Identical Terminal Groups)

When an unbranched carbon chain is directly attached to **3 or more identical carbon-containing terminal functional groups** (−COOH, −CHO, −CN, −$CONH_2$, −COX):

Unsaturated Hydrocarbons: Double vs Triple Bonds

- Parent chain must include maximum multiple bonds, even if shorter than saturated chain.
- **Numbering priority:** Double Bond > Triple Bond (when at symmetrical locants).
- **Symmetrical case:** CH≡C−CH_2−CH=CH_2 → Pent-1-en-4-yne (Double bond gets lower number 1).
- **Unsymmetrical case:** CH≡C−CH=CH−CH_3 → Pent-3-en-1-yne (Terminal triple bond gets lower locant 1).
- **Name Order:** Always write "-ene" before "-yne". Drop 'e' from 'ene' if followed by vowel (*e.g.*, -en-yne).

- None of these carbon atoms are included in the parent carbon chain.
- Use special suffixes: *-tricarboxylic acid, -tricarbaldehyde, -tricarbonitrile*.
- **Example:** $NC-CH_2-CH(CN)-CH_2-CN$ → **Propane-1,2,3-tricarbonitrile**.
- **Example:** $HOOC-CH_2-CH(COOH)-CH_2-COOH$ → **Propane-1,2,3-tricarboxylic acid**.
- *Condition:* If groups are NOT all directly attached to main chain, include 2 in parent chain and treat the 3rd as carboxyalkyl prefix (*e.g.*, 3-(carboxymethyl)hexane-1,2,5-tricarboxylic acid).

Complex Substituents & Multiplying Prefixes

- Carbon attached directly to main chain is always assigned **C-1'**.
- Enclose complex substituent name in brackets: *e.g.*, 3-(2'-methylpropyl)nonane.
- **Multiplying Prefixes:** Use **bis-** (2), **tris-** (3), **tetrakis-** (4) for multiple identical complex substituents.
- *Example:* $C(CH_2Cl)_4$ → 3,3-Bis(chloromethyl)-1,5-dichloropentane.
- Alphabetical ordering: Prefixes **iso, neo, cyclo** are considered in alphabetization; **sec, tert, di, tri** are IGNORED.

Replacement Nomenclature (Oxa / Thia / Aza)

- Heteroatoms (O, S, N) replacing C-atoms in a chain are named as **Oxa** (O), **Thia** (S), **Aza** (N).
- Total count includes carbon atoms + heteroatoms to form root name.
- *Example 1:* $CH_3-CH_2-NH-CH_2-CH(C_2H_5)-CH_2-CH_3$ → **5-Ethyl-3-azaheptane**.
- *Example 2:* $CH_3-O-CH_2-CH_2-O-CH_2-CH_2-O-CH_3$ → **2,5,8-Trioxanonane**.
- *Example 3:* Heterocyclic ring with S atom → **Thiacyclohexane**.

6. ALICYCLIC, AROMATIC & POLYCYCLIC SYSTEMS

Alicyclic (Cyclic) Naming Rules

- **Ring vs Chain Priority:**
  - If Ring Carbons ≥ Chain Carbons → Named as **Cycloalkane** (chain is substituent).
  - If Chain Carbons > Ring Carbons → Named as **Alkane** (ring is substituent).
  - **EXCEPTION:** If chain contains a **multiple bond or Functional Group**, chain becomes PARENT regardless of ring size.
- **Substituted Cycloalkenes:** Double bond gets position 1,2. Number towards substituent to give lowest locant (*e.g.*, 3-Ethylcyclopentene).
- **Cyclic Ketones/Alcohols:** Carbon with -OH or =O gets locant 1 (*e.g.*, 5,6-dimethylcyclohex-2-en-1-one).

Bicyclo & Spiro Compounds Rules [Added]

- **Bicyclo Compounds:** Two fused rings sharing 2 bridgehead carbons.
  Formula: `Bicyclo[a.b.c]alkane` (where a ≥ b ≥ c are carbons in bridges).
  Numbering: Start at bridgehead → longest path → second longest path → shortest bridge.
- **Spiro Compounds:** Two rings sharing exactly 1 quaternary carbon.
  Formula: `Spiro[a.b]alkane` (where a ≤ b are carbons in rings).
  Numbering: Start in smaller ring adjacent to spiro atom → around smaller ring → through spiro carbon → larger ring.

7. ESSENTIAL COMMON / TRIVIAL NAMES FOR JEE QUESTIONS

Common Name	IUPAC Name	Chemical Structure / Formula	JEE Relevance / Note
Toluene	Methylbenzene	$C_6H_5-CH_3$	Standard aromatic base
An interior / Ortho-Xylene	1,2-Dimethylbenzene	$C_6H_4(CH_3)_2$	o-, m-, p- isomers
Phenol	Benzenol	C_6H_5OH	Aromatic alcohol
Aniline	Benzenamine	$C_6H_5NH_2$	Aromatic amine
Anisole	Methoxybenzene	$C_6H_5OCH_3$	Aromatic ether
Acetophenone	1-Phenylethan-1-one	$C_6H_5COCH_3$	Aromatic ketone
Benzaldehyde	Benzenecarbaldehyde	C_6H_5CHO	Aromatic aldehyde
Benzoic Acid	Benzenecarboxylic acid	C_6H_5COOH	Aromatic acid

Common Name	IUPAC Name	Chemical Structure / Formula	JEE Relevance / Note
Picric Acid [Added]	2,4,6-Trinitrophenol	$C_6H_2(NO_2)_3OH$	Highly acidic without -COOH group
Neopentane [Added]	2,2-Dimethylpropane	$C(CH_3)_4$	Quaternary carbon test
Isoprene [Added]	2-Methylbuta-1,3-diene	$CH_2=C(CH_3)-CH=CH_2$	Monomer of Natural Rubber
Glycerol / Glycerin	Propane-1,2,3-triol	$CH_2OH-CHOH-CH_2OH$	Trihydric alcohol
Acrolein [Added]	Prop-2-enal	$CH_2=CH-CHO$	Unsaturated aldehyde

8. JEE Problem Solving Strategy & Common Traps

Critical Examination Pitfalls & Rapid Verification Steps

- **Trap 1 (Locant set comparison):** Do NOT add locants when applying First Point of Difference. Compare sets element by element (*e.g.*, (2, 2, 6, 6) vs (2, 3, 5, 6) → 2,2... wins at second locant despite higher sum).
- **Trap 2 (Double vs Triple bond priority):** Numbering priority gives preference to double bond OVER triple bond *only when they are at identical positions from opposite ends*. Otherwise, lower locant rule applies strictly.
- **Trap 3 (Carboxylic carbon inclusion):** If -COOH carbon is counted in parent chain, suffix is *-oic acid*. If carbon is NOT counted (or directly on ring), suffix MUST be *-carboxylic acid*.
- **Trap 4 (Alphabetical prefixes):** Ignore *di-, tri-, tetra-, sec-, tert-* when ordering alphabetically. DO inclusion of *iso-, neo-, cyclo-*.
- **Trap 5 (Bond-Line Notation):** Always count free ends and bends as C-atoms with filled valencies by implicit Hydrogens. Ensure correct degree (1°, 2°, 3°, 4°) assignment.

ISOMERISM — MASTER JEE FORMULA & REVISION SHEET

Comprehensive compilation covering Structural Isomerism, Stereoisomerism, Conformational Analysis, and Counting Formulas [JEE Main & Advanced]

1. FUNDAMENTAL DEFINITIONS & DEGREE OF UNSATURATION (DBE)

Isomers: Compounds having the same molecular formula but different physical, chemical, or biological properties.

Degree of Unsaturation (DBE / Index of Hydrogen Deficiency):

$$DBE = \sum n(v - 2) / 2 + 1$$

= (No. of π bonds) + (No. of rings) in the molecule.

Where n = number of atoms of a specific element, v = valency. (H & Halogens count as valency 1, O & S count as valency 2, N & P count as valency 3).

2. STRUCTURAL (CONSTITUTIONAL) ISOMERISM

Arises due to differences in connectivity of atoms. Main types:

- **Chain Isomerism:** Difference in length of parent carbon chain or side-chain skeleton. (e.g., n-Pentane, isopentane, neopentane).
- **Positional Isomerism:** Same carbon skeleton, but different positions of functional groups, multiple bonds, or substituents (e.g., 1-propanol vs. 2-propanol; ortho, meta, para isomers).
- **Functional Isomerism:** Same molecular formula, different functional groups.

Molecular Formula	Homologous Series Pair
$C_nH_{2n+2}O$	Alcohol & Ether
$C_nH_{2n}O$	Aldehyde & Ketone
$C_nH_{2n}O_2$	Carboxylic Acid & Ester
$C_nH_{2n+3}N$	1°, 2°, 3° Amines
$C_nH_{2n+1}N$	Cyanide & Isocyanide

3. METAMERISM & RING-CHAIN ISOMERISM

Metamerism: Arises due to difference in alkyl groups attached to the same polyvalent heteroatom ($-O-$, $-S-$, $-NH-$, $-C(=O)-$).

Priority order: Metamerism takes precedence over chain/positional isomerism when alkyl groups vary.

Ring-Chain Isomerism: Open-chain and cyclic structures sharing the same molecular formula (e.g., Propene C_3H_6 and Cyclopropane).

4. TAUTOMERISM (DESMOTROPISM)

Dynamic equilibrium between two functional isomers due to migration of an acidic H-atom (prototropy). Requires α-H attached to saturated sp^3 carbon adjacent to $>C=O$, $-NO_2$, $-N=O$, or $-CN$.

Important Tautomeric Systems:

- **Keto-Enol:** $-C(=O)-CH_2- \rightleftharpoons -C(OH)=CH-$
- **Nitro-Aci:** $R_2CH-NO_2 \rightleftharpoons R_2C=N(OH)O$
- **Nitroso-Oximino:** $R_2CH-N=O \rightleftharpoons R_2C=N-OH$

Enol Content Enhancement Factors:

1. Intramolecular Hydrogen Bonding (e.g., β-diketones like acetylacetone have very high enol content).
2. Extended Conjugation & Aromatization in enol form (e.g., Phenol exists almost 100% in enol form).
3. High acidity of α-H .

Enol Content Order:
Acetylacetone > Ethyl acetoacetate > Acetone > Acetaldehyde

5. STEREOISOMERISM & GEOMETRICAL ISOMERISM (GI)

Stereoisomers have the same constitution but different 3D spatial arrangements.

Geometrical Isomerism (CIS / TRANS or E / Z): Arises due to restricted rotation about carbon-carbon double bonds (`>C=C<`), cyclic rings, or `>C=N–` (oximes) and `–N=N–` (azo) systems.

Physical Properties Comparison (Cis vs. Trans of But-2-ene):

Property	Cis-Isomer	Trans-Isomer
Stability	Lower (Steric repulsion)	**Higher**
Dipole Moment (μ)	**Higher ($\mu > 0$)**	Lower ($\mu \approx 0$ if symmetric)
Boiling Point	**Higher** (due to dipole forces)	Lower
Melting Point	Lower (poor crystal packing)	**Higher** (symmetrical)

Counting Geometrical Isomers (n = number of GI sites):

- **Unsymmetrical System:** 2^n
- **Symmetrical System (n is even):** $2^{n-1} + 2^{(n/2)-1}$
- **Symmetrical System (n is odd):** $2^{n-1} + 2^{(n+1)/2} - 1$

6. OPTICAL ISOMERISM & CHIRALITY

Molecules rotating Plane Polarized Light (PPL) are optically active. Requires absence of elements of symmetry (POS, COS).

- **Chiral Carbon (Asymmetric C):** Carbon bonded to 4 different groups. *Chirality is necessary/sufficient for point chirality, but optical activity depends on net molecular dissymmetry (meso compounds have chiral centers but are optically inactive due to internal compensation).*
- **Enantiomers:** Non-superimposable mirror images. Identical physical properties except rotation of PPL.
- **Diastereomers:** Stereoisomers not mirror images. Different physical/chemical properties.

7. ABSOLUTE (R/S) & RELATIVE (D/L, E/Z) CONFIGURATIONS

C.I.P. Sequence Rules for Priority:

1. Higher atomic number gets higher priority (`1 > 2 > 3 > 4`).
2. If first atom ties, check subsequent atoms along the path.
3. Isotopes: higher atomic mass has priority (`T > D > H`).
4. Multiple bonds: treated as bonded to equivalent duplicate/triplicate single-bonded atoms.

Fischer Projection R/S Determination:

• If lowest priority group `(4)` is on **Vertical line**: Clockwise = `R` , Anticlockwise = `S` .
• If group `(4)` is on **Horizontal line**: Clockwise = `S` , Anticlockwise = `R` .

Counting Optical Isomers (n = number of chiral carbons):

System Type	Active Forms (a)	Meso Forms (m)	Total Stereoisomers
Unsymmetrical	2^n	0	2^n
Symmetrical (n = even)	2^{n-1}	$2^{(n/2)-1}$	$2^{n-1} + 2^{(n/2)-1}$
Symmetrical (n = odd)	$2^{n-1} - 2^{(n-1)/2}$	$2^{(n-1)/2}$	2^{n-1}

8. CONFORMATIONAL ISOMERISM (NEWMAN PROJECTIONS)

Rotation about σ C–C single bonds. [Added: bond lengths and bond angles remain unchanged across conformations; only dihedral angle changes].

n-Butane Stability Order:
`Anti-Staggered (Most Stable, 180°) > Gauche (60°) > Partially Eclipsed (120°) > Fully Eclipsed (0°, Least Stable)`

Cyclohexane Conformation Stability:
`Chair > Twist-Boat > Boat > Half-Chair (Least Stable)`

HYDROGEN & ITS COMPOUNDS

Comprehensive JEE Main & Advanced Formula Revision Sheet

JEE CHEMSHEET

1. POSITION & FUNDAMENTALS

Electronic Config: **$1s^1$** (Dual behavior: Alkali metals & Halogens).

- **Alkali Metal Analogy:** Forms H^+, forms oxides (H_2O), sulfides (H_2S), and acts as reducing agent. *(I.E. higher than alkali metals)*.
- **Halogen Analogy:** High I.E., forms hydride anion H^-, diatomic (H_2), forms covalent compounds with non-metals.
- **Abundance:** 70% of universe mass; 9th element in earth's crust (0.9% by mass).

2. ISOTOPES OF HYDROGEN

Property	Protium (^1H)	Deuterium (^2H / D)	Tritium (^3H / T)
Composition	1p, 0n, $1e^-$	1p, 1n, $1e^-$	1p, 2n, $1e^-$
Abundance	99.985%	0.0156%	~10^{-15}% (Trace)
Stability	Stable	Stable	Radioactive (β^-)
Half-life ($t_{1/2}$)	–	–	**12.33 Years**
B.D.E. (kJ/mol)	435.88	443.35	446.9 [Added]

***Isotope Effect:** Reaction rate of H > D > T due to higher C–H bond dissociation enthalpy vs C–D/C–T.

3. PREPARATION OF DIHYDROGEN (H_2)

Laboratory Methods:

- $Zn + 2H^+ \rightarrow Zn^{2+} + H_2\uparrow$ (Granulated Zn + dil. H_2SO_4)
- $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$ (Sodium Zincate)
- $2Al + 2KOH + 2H_2O \rightarrow 2KAlO_2 + 3H_2\uparrow$ (Uyeno's)
- $CaH_2 + 2H_2O \rightarrow Ca(OH)_2 + 2H_2\uparrow$ (Hydrolith process)

Commercial / Industrial Methods:

- **Water Electrolysis:** $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$ (Traces of acid/base). High purity (>99.95%) H_2 obtained using warm aq. $Ba(OH)_2$ with Ni electrodes.
- **Lane's Process:** $3Fe + 4H_2O(steam) \rightarrow Fe_3O_4 + 4H_2\uparrow$ (1025–1075 K).
- **Coal Gasification:** $C(s) + H_2O(g) \rightarrow CO(g) + H_2(g)$ (1270 K).
  Syngas / Water Gas = CO + H_2 (1:1 ratio).
- **Water-Gas Shift Reaction (Bosch Process):** $CO + H_2O(g) \rightarrow CO_2 + H_2$ (673 K, Fe_2O_3 / Cr_2O_3 catalyst). CO_2 is scrubbed using aq. Na_2CO_3 or ethanolamine solution.
- **Hydrocarbon Cracking:** $CH_4 \rightarrow C + 2H_2$ (1270 K, Ni catalyst).

4. FORMS & ALLOTROPES OF HYDROGEN

(A) Ortho and Para Hydrogen:

- **Ortho-H_2:** Nuclear spins in *SAME* direction (↑↑).
- **Para-H_2:** Nuclear spins in *OPPOSITE* direction (↑↓).
- **Equilibrium:** At 0 K → 100% Para (more stable low T). At room temp → **75% Ortho + 25% Para (3:1 ratio)**. Cannot get >75% Ortho.
- Chemical properties are identical; thermal conductivity and heat capacity differ.

(B) Reactive Forms of Hydrogen:

- **Nascent H:** Freshly generated in-situ (atomic state). Reactive: Decolorizes acidic $KMnO_4$, $K_2Cr_2O_7$ (orange → green), and $FeCl_3$ (yellow → colorless).
- **Atomic H:** Generated via electric arc ($H_2 \rightarrow 2H$; $\Delta H = +435.9\ kJ/mol$). Half-life = 0.3 s. Used in atomic hydrogen torch (3000 K). Recombination heat yields extremely high temperature.
- **Occluded H:** Hydrogen gas adsorbed in metal cavities (Pd, Pt, Ni). Pd absorbs up to 900 times its volume of H_2. Occlusion decreases with temperature rise.

5. CLASSIFICATION OF HYDRIDES

Type	Formed by	Key Properties & Formulas
Ionic / Saline	s-Block metals (except Be, Mg)	Stoichiometric, crystalline, non-volatile, ionic (M^+H^-). High m.p. Conduct electricity in molten state releasing H_2 at anode: $2H^- \rightarrow H_2(g) + 2e^-$. *Reactivity with H_2O:* $NaH + H_2O \rightarrow NaOH + H_2\uparrow$
Covalent / Molecular	p-Block non-metals (Group 13 to 17)	Volatile, low m.p./b.p. Classified by electron count: • *Electron Deficient:* Group 13 (e.g., B_2H_6, AlH_3). Act as Lewis acids. • *Electron Precise:* Group 14 (e.g., CH_4, SiH_4). Exact octet. • *Electron Rich:* Group 15–17 (e.g., NH_3, H_2O, HF). Have lone pairs (Lewis bases).
Metallic / Interstitial	d-Block (Grp 3, 4, 5, 10, 11, 12) & f-Block metals	Non-stoichiometric (e.g., $TiH_{1.5\text{-}1.8}$, $ZrH_{1.3\text{-}1.75}$, $PdH_{0.6\text{-}0.8}$, $VH_{0.56}$). Metallic luster, electrical conductors. **Hydride Gap:** Groups 7, 8, and 9 metals do NOT form hydrides. In Grp 6, only Chromium forms CrH.

6. WATER (H_2O) & HARDNESS CALCULATIONS

Physical & Structural Properties:

- Bent shape, sp^3 hybridized, Bond Angle = 104.5°, O–H length = 95.7 pm.
- Maximum density of water at **4 °C (398 K) = 1.000 g/cm^3**. Ice has open cage-like structure due to H-bonding (density of ice < density of water).
- High dielectric constant ($\varepsilon_r = 78.39$), strong ionizing solvent.

Hardness Types & Cause:

- Hardness due to dissolved soluble salts of Ca^{2+} and Mg^{2+}.
- **Temporary Hardness:** Presence of Bicarbonates: $Ca(HCO_3)_2$, $Mg(HCO_3)_2$.
- **Permanent Hardness:** Presence of Chlorides & Sulfates: $CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$.

Degree of Hardness Formula (in ppm of $CaCO_3$ equivalent):

$$\textbf{Hardness (ppm)} = \frac{\text{Mass of Hardness Causing Salt (g)}}{\text{Molar Mass of Salt (g/mol)}} \times 100\ \text{g/mol (Molar Mass of } CaCO_3) \times \frac{10^6}{\text{Mass of Water (g)}}$$

Molar Mass Equivalents for Quantitative JEE Problems:

Salt Molecule	Molar Mass (g/mol)	$CaCO_3$ Equivalent Ratio
$CaCO_3$	100.0	1 mole Salt ≡ 1 mole $CaCO_3$ (100 g)
$Mg(HCO_3)_2$	146.0	146 g $Mg(HCO_3)_2$ ≡ 100 g $CaCO_3$
$Ca(HCO_3)_2$	162.0	162 g $Ca(HCO_3)_2$ ≡ 100 g $CaCO_3$
$MgCl_2$	95.0	95 g $MgCl_2$ ≡ 100 g $CaCO_3$
$MgSO_4$	120.0	120 g $MgSO_4$ ≡ 100 g $CaCO_3$
$CaCl_2$	111.0	111 g $CaCl_2$ ≡ 100 g $CaCO_3$
$CaSO_4$	136.0	136 g $CaSO_4$ ≡ 100 g $CaCO_3$

Softening & Water Treatment Methods:

(A) Temporary Hardness Removal:

- **Boiling:**

  $Ca(HCO_3)_2 \rightarrow CaCO_3\downarrow + H_2O + CO_2\uparrow$

  $Mg(HCO_3)_2 \rightarrow Mg(OH)_2\downarrow + 2CO_2\uparrow$

  (Note: $Mg(OH)_2$ precipitates instead of $MgCO_3$ due to lower K_{sp}).
- **Clark's Process (Calculated Lime):**

  $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$

  $Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + Mg(OH)_2\downarrow + 2H_2O$

(B) Permanent Hardness Removal:

- **Washing Soda:** $MCl_2 + Na_2CO_3 \rightarrow MCO_3\downarrow + 2NaCl$ (M = Ca, Mg)
- **Calgon Process:** Sodium hexametaphosphate $Na_6P_6O_{18}$ or $Na_2[Na_4(PO_3)_6]$.

  $2M^{2+} + Na_2[Na_4(PO_3)_6] \rightarrow Na_2[M_2(PO_3)_6] + 4Na^+$
- **Permutit / Zeolite Method:** Hydrated Na-Al-silicate ($Na_2Al_2Si_2O_8 \cdot xH_2O$ or Na_2Z).

  $Na_2Z + M^{2+} \rightarrow MZ + 2Na^+$ (Regenerated using brine aq. NaCl).
- **Synthetic Resin (De-mineralization):**

  • Cation exchanger: $2R\text{-}SO_3H + M^{2+} \rightarrow (RSO_3)_2M + 2H^+$

  • Anion exchanger: $R\text{-}NH_3^+OH^- + X^- \rightarrow R\text{-}NH_3^+X^- + OH^-$

  • Combined: $H^+ + OH^- \rightarrow H_2O$ (Gives pure de-ionized water).

7. HEAVY WATER (D_2O)

- Discovered by Urey (1932). Prepared by **exhaustive electrolysis of water** (containing NaOH) using Ni electrodes (Urey and Washburn method).
- Used as a **moderator** and coolant in nuclear reactors to slow down fast neutrons.
- Used as a tracer compound for reaction mechanisms.

Comparison of H_2O and D_2O Properties:

Property	H_2O	D_2O
Molecular Mass (g/mol)	18.015	**20.028**
Melting Point (K)	273.15	**276.83**
Boiling Point (K)	373.15	**374.42**
Density at 298K (g/cm^3)	0.997	**1.106**
Dielectric Constant (ε)	78.39	**78.06**
Viscosity (cP)	0.890	**1.107**

8. HYDROGEN PEROXIDE (H_2O_2) — PREPARATION & STRUCTURE

Methods of Preparation:

- **Merck's Method (Barium Peroxide):** $BaO_2 \cdot 8H_2O + H_2SO_4(conc,\ cold) \rightarrow BaSO_4\downarrow + H_2O_2 + 8H_2O$.

  Note: Hydrated BaO_2 must be used; anhydrous BaO_2 forms a protective coating of $BaSO_4$ preventing further reaction. Excess H_2SO_4 is removed using $BaCO_3$. H_3PO_4 is preferred over H_2SO_4: $3BaO_2 + 2H_3PO_4 \rightarrow Ba_3(PO_4)_2\downarrow + 3H_2O_2$.

- **Electrolytic Oxidation (Peroxydisulfuric Acid):** Electrolysis of 50% H_2SO_4 at high current density yields peroxydisulfuric acid ($H_2S_2O_8$) at anode:

  $2HSO_4^- \rightarrow H_2S_2O_8 + 2e^-$; followed by hydrolysis: $H_2S_2O_8 + 2H_2O \rightarrow 2H_2SO_4 + H_2O_2$.

  [Added] *D_2O_2 preparation:* $K_2S_2O_8(s) + 2D_2O(l) \rightarrow 2KDSO_4(aq) + D_2O_2(l)$.

- **Auto-oxidation of 2-Ethylanthraquinol (Industrial Method):** Air oxidized to 2-ethylanthraquinone + H_2O_2. Reduced back using H_2/Pd for continuous cycle.

Non-Planar Open-Book Structure:

- **Gas Phase:** O–O length = 147.5 pm, O–H length = 95.0 pm. Dihedral Angle (θ) = **111.5°**, H–O–O Angle = 94.8°.
- **Solid/Crystalline Phase (110 K):** O–O length = 145.8 pm, O–H length = 98.8 pm. Dihedral Angle (θ) = **90.2°**, H–O–O Angle = 101.9°.

Physical & Storage Notes:

- Syrupy pale blue liquid. High boiling point (423.2 K) due to hydrogen bonding.
- **Decomposition:** $2H_2O_2 \rightarrow 2H_2O + O_2$ ($\Delta H = -196$ kJ).

  Catalyzed by Pt, Ag, MnO_2, light, or glass alkali.
- **Stabilizers:** Glycerol, acetanilide, or H_3PO_4 added to retard decomposition. Stored in wax-lined plastic bottles in dark.

9. H_2O_2 CONCENTRATION & VOLUME STRENGTH FORMULAS

Definition: *"x Volume H_2O_2"* means 1 mL of H_2O_2 solution yields **x mL of O_2 gas at STP (NTP)**.

1. Volume Strength vs Molarity (M):

$$\text{Volume Strength} = 11.2 \times \text{Molarity (M)}$$

(Assuming 1 mol gas = 22.4 L at STP)

$$\text{Volume Strength} = 11.35 \times \text{Molarity (M)}$$ [Added]

(STP at 1 bar pressure: 1 mol = 22.7 L)

2. Volume Strength vs Normality (N):

$$\text{Volume Strength} = 5.6 \times \text{Normality (N)}$$

$$\text{Strength (g/L)} = \text{Normality} \times 17$$

$$\text{Strength (g/L)} = \text{Molarity} \times 34$$

3. Percentage Strength (% w/v):

$$\%\ (w/v)\ \text{Strength} = \frac{\text{g of } H_2O_2}{100\ \text{mL solution}} = \frac{17 \times \text{Vol. Strength}}{56} = 0.3035 \times \text{Volume Strength}$$

Standard JEE Ready Reference Values:

Volume Strength	Molarity (M)	Normality (N)	Strength (g/L)	% (w/v) Strength
10 V H_2O_2	0.89 M	1.78 N	30.35 g/L	3.035 %
20 V H_2O_2	1.78 M	3.57 N	60.70 g/L	6.070 %
30 V ("100 Volume / Perhydrol")	2.68 M (or 8.9 M for 100V)	5.36 N	91.05 g/L (303.5 g/L)	30.0 % (w/v)

10. CHEMICAL REACTIONS & DUAL REDOX NATURE OF H_2O_2

Medium	Oxidizing Action of H_2O_2 (H_2O_2 reduced to H_2O or OH^-)	Reducing Action of H_2O_2 (H_2O_2 oxidized to O_2)
Acidic Medium	• $2Fe^{2+} + 2H^+ + H_2O_2 \rightarrow 2Fe^{3+} + 2H_2O$ • $PbS(black) + 4H_2O_2 \rightarrow PbSO_4(white) + 4H_2O$ • $2KI + H_2SO_4 + H_2O_2 \rightarrow I_2 + K_2SO_4 + 2H_2O$ • $H_2S + H_2O_2 \rightarrow S\downarrow + 2H_2O$	• $2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2\uparrow$ • $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2\uparrow$ • $O_3 + H_2O_2 \rightarrow H_2O + 2O_2\uparrow$

Medium	Oxidizing Action of H_2O_2 (H_2O_2 reduced to H_2O or OH^-)	Reducing Action of H_2O_2 (H_2O_2 oxidized to O_2)
Basic Medium	• $2Fe^{2+} + H_2O_2 \rightarrow 2Fe^{3+} + 2OH^-$ • $Mn^{2+} + H_2O_2 \rightarrow Mn^{4+}\ (MnO_2) + 2OH^-$	• $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2\uparrow$ • $2MnO_4^- + 3H_2O_2 \rightarrow 2MnO_2 + 3O_2\uparrow + 2H_2O + 2OH^-$

Key Analytical Tests for H_2O_2 (JEE Advanced Special):

- **Chromic Acid Test (Ether Layer Test):** Acidified $K_2Cr_2O_7$ + H_2O_2 in presence of ether forms **Chromium Peroxide (CrO_5)**, giving a *deep blue color in ether layer*.
  $Cr_2O_7^{2-} + 2H^+ + 4H_2O_2 \rightarrow 2CrO_5\ (\text{butterfly structure}) + 5H_2O$.
  (Cr in CrO_5 has oxidation state +6 with two peroxide linkage: $CrO(O_2)_2$).
- **Titanium Sulfate Test:** $Ti^{4+} + H_2O_2 + 2H_2O \rightarrow H_2TiO_4\ (\text{Pertitanic acid, orange-yellow}) + 4H^+$.

HYDROCARBONS — JEE MAIN & ADVANCED REVISION SHEET

Comprehensive Reactions, Mechanisms, Stereochemistry & Direct Result Shortcuts

JEE REVISION SHEET

1. FUNDAMENTALS, HYBRIDIZATION & BOND PARAMETERS

Alkanes: C_nH_{2n+2} (Paraffins)

- C-atom: sp^3 hybridized (109.5° / 109°28′)
- C–C distance: 1.54 Å; C–H: 1.09 Å
- Non-polar, inert to electrophiles/nucleophiles.

Alkenes: C_nH_{2n} (Olefins)

- C-atom: sp^2 hybridized (120° bond angle)
- C=C distance: 1.34 Å; C–H: 1.08 Å
- Electrophilic addition reactive due to π-cloud.

Alkynes: C_nH_{2n-2} (Acetylenes)

- C-atom: sp hybridized (180° linear geometry)
- C≡C distance: 1.20 Å; C–H: 1.06 Å
- Terminal sp-H is acidic ($pK_a \approx 25$).

2. ALKANES: METHODS OF PREPARATION & KEY REACTIONS

Reaction Name / Type	Reaction Equation / Reagents	JEE Key Notes & Limitations
Decarboxylation (Soda-Lime)	$R\text{-}COONa + NaOH \xrightarrow{(CaO,\ \Delta)} R\text{-}H + Na_2CO_3$	Product has **1 less Carbon**. CaO keeps mixture dry & catalyst. Rate: 3° > 2° > 1° acid salt.
Sabatier-Senderens	$R\text{-}CH{=}CH_2 + H_2 \xrightarrow{(Ni\ /\ 200\text{-}300^\circ C)} R\text{-}CH_2\text{-}CH_3$	Ni, Pt, Pd, or Raney-Ni can be used. *CH_4 cannot be prepared.*
Wurtz Reaction	$2\ R\text{-}X + 2\ Na \xrightarrow{(dry\ ether,\ \Delta)} R\text{-}R + 2\ NaX$	Prepares **symmetrical alkanes** only. Mixture formed for un-symmetrical X. Fails with 3° alkyl halides (gives alkene via E2). CH_4 cannot be made.
Frankland Method	$2\ R\text{-}X + Zn \xrightarrow{(Ether,\ \Delta)} R\text{-}R + ZnX_2$	Similar to Wurtz; dialkyl zinc intermediate (R_2Zn).
Corey-House Synthesis	$R\text{-}X + 2Li \rightarrow R\text{-}Li \xrightarrow{(CuI)} LiR_2Cu \xrightarrow{(R'\text{-}X)} R\text{-}R'$	Best for **unsymmetrical alkanes**. R'-X must be 1° or methyl for good yield; 2°/3° gives alkenes.
Clemmensen Reduction	$>C{=}O \xrightarrow{(Zn\text{-}Hg\ +\ conc.\ HCl)} >CH_2 + H_2O$	Reduces Aldehydes/Ketones to Alkanes. *Unsuitable for acid-sensitive groups* (OH, 3° alcohol, double bonds).
Wolff-Kishner Reduction	$>C{=}O \xrightarrow{(NH_2NH_2,\ KOH\ /\ Ethylene\ Glycol,\ \Delta)} >CH_2$	Base-stable reduction. *Unsuitable for base-sensitive groups* (esters, haloalkanes).
Red P + HI Reduction	$R\text{-}OH$ / $R\text{-}CHO$ / $R\text{-}COOH \xrightarrow{(Red\ P\ +\ HI,\ 150\text{-}250^\circ C)} R\text{-}CH_3$	Universal powerful reducing system converting oxygenated organic compounds to alkanes.
Kolbe's Electrolytic Method	$2\ R\text{-}COOK + 2\ H_2O \xrightarrow{(Electrolysis)} R\text{-}R + 2\ CO_2$ (Anode) + H_2 + 2 KOH (Cathode)	Anode: R-R + CO_2 (Free radical mechanism). Solution becomes **basic** (pH increases). CH_4 cannot be prepared.
Grignard Hydrolysis	$R\text{-}MgX + H\text{-}OH$ (or ROH, RNH_2) $\longrightarrow R\text{-}H + Mg(OH)X$	Zerewitinoff reaction: Active hydrogen quantitates alkane release.
Hydrolysis of Carbides	$Al_4C_3 + 12\ H_2O \rightarrow 3\ CH_4 + 4\ Al(OH)_3$ $Be_2C + 4\ H_2O \rightarrow CH_4 + 2\ Be(OH)_2$	Used specifically for **Methane** synthesis. (CaC_2 yields C_2H_2; Mg_2C_3 yields C_3H_4 [Added]).

Chemical Properties & Reactions of Alkanes

1. Free-Radical Halogenation: $R\text{-}H + X_2 \xrightarrow{(h\nu\ /\ \Delta)} R\text{-}X + HX$

- Reactivity order: F_2 (explosive) > Cl_2 > Br_2 > I_2 (reversible, needs HIO_3/HNO_3).
- Hydrogen abstraction rate: 3° H (5.0) > 2° H (3.8) > 1° H (1.0) for Chlorination at 298 K.
- Bromination is highly selective: 3° (1600) > 2° (82) > 1° (1).

2. Nitration (Vapour Phase): $R\text{-}H + HNO_{3(g)} \xrightarrow{(400\text{-}500^\circ C)} R\text{-}NO_2 + H_2O$

- Involves C-H and C-C cleavage; yields a mixture of nitroalkanes (e.g., Ethane → Nitroethane + Nitromethane).

3. Reed Reaction: $R\text{-}H + SO_2 + Cl_2 \xrightarrow{(h\nu)} R\text{-}SO_2Cl + HCl$

4. Controlled Catalytic Oxidation:

- $CH_4 + O_2 \xrightarrow{(Cu\ tube\ /\ 200^\circ C\ /\ 100\ atm)} 2\ CH_3OH$
- $CH_4 + O_2 \xrightarrow{(Mo_2O_3,\ \Delta)} HCHO + H_2O$
- $2\ R\text{-}CH_3 + 3\ O_2 \xrightarrow{((CH_3COO)_2Mn)} 2\ R\text{-}COOH + 2\ H_2O$
- Alkane with 3° H (e.g., Isobutane) + $KMnO_4$ → 3° Alcohol (tert-butanol).

5. Isomerization: n-Alkane $\xrightarrow{(Anhy.\ AlCl_3\ +\ HCl,\ 25^\circ C)}$ Branched Alkane

6. Aromatization: n-Hexane $\xrightarrow{(Cr_2O_3\text{-}Al_2O_3\ /\ 500^\circ C,\ 10\text{-}20\ atm)}$ Benzene + 4 H_2

3. ALKENES: PREPARATION, ADDITIONS & STEREOCHEMISTRY

Methods of Preparation of Alkenes

Method	Reagents & Reaction	Regio/Stereo Rules & Remarks
Dehydration of Alcohols	$R\text{-}CH_2\text{-}CH_2\text{-}OH \xrightarrow{(Conc.\ H_2SO_4\ /\ 160\text{-}180^\circ C)} Alkene + H_2O$	E1 mechanism (via Carbocation). Ease: 3° > 2° > 1°. **Saytzeff Rule:** More substituted alkene is major. Rearrangement possible!
Dehydrohalogenation	$R\text{-}CH_2\text{-}CH_2\text{-}X$ + Alc. KOH $\xrightarrow{(\Delta)} R\text{-}CH{=}CH_2$	E2 mechanism. Anti-periplanar elimination. Bulkier base (e.g., t-BuOK) gives **Hofmann alkene** (less substituted) as major.
Dehalogenation of Vic-Dihalides	$R\text{-}CH(Br)\text{-}CH(Br)\text{-}R' + Zn \xrightarrow{(Alcohol,\ \Delta)} R\text{-}CH{=}CH\text{-}R' + ZnBr_2$	Also accomplished using **NaI / Acetone** (Anti-elimination). meso-dihalide gives trans-alkene.
Partial Reduction of Alkynes	1. $R\text{-}C{\equiv}C\text{-}R' + H_2 \xrightarrow{(Pd/CaCO_3\ +\ Quinoline\ /\ Lindlar)}$ **cis-alkene** 2. $R\text{-}C{\equiv}C\text{-}R'$ + Na / Liq. $NH_3 \xrightarrow{(Birch\ Reduction)}$ **trans-alkene**	Lindlar catalyst or Ni_2B (P-2) gives **cis-alkene (syn)**. Birch reduction gives **trans-alkene (anti)** via radical-anion.
Pyrolysis of Amine Oxides / Esters	1. Cope Elimination: Amine oxide $\xrightarrow{(\Delta)}$ Alkene (Syn) 2. Ester Pyrolysis: Acetate ester $\xrightarrow{(\Delta)}$ Alkene (Syn)	Ei mechanism (Cyclic transition state). Always yields **Hofmann product** with zero rearrangement.

Electrophilic & Free-Radical Addition Reactions

Markovnikov's Rule: In electrophilic addition of H-X to an unsymmetrical alkene, the positive part (H^+) attaches to the double-bonded carbon possessing the greater number of hydrogen atoms (forming the more stable carbocation intermediate).

Kharasch Peroxide Effect (Anti-Markovnikov): HBr in the presence of peroxides (R-O-O-R) adds via **free radical mechanism** to yield Anti-Markovnikov product. *Valid ONLY for HBr* (not for HCl or HI due to endothermic propagation steps).

Reaction	Reagent & Intermediate	Stereochemistry & Product Result
Halogen Addition	X_2 (Br_2 or Cl_2) in CCl_4 *Intermediate:* Cyclic Halonium ion	**Anti addition**. No carbocation rearrangement. Test for unsaturation (decolorizes brown Br_2).
Halohydrin Formation	$X_2 + H_2O$ (HOX)	**Anti addition**. OH attaches to more substituted carbon (Markovnikov orientation via bromonium ion).
Oxymercuration-Demercuration (OMDM)	1. $(CH_3COO)_2Hg$ / H_2O, THF 2. $NaBH_4$, NaOH	**Markovnikov addition** of H_2O. **No Carbocation Rearrangement!** Overall Anti-stereochemistry.
Hydroboration-Oxidation (HBO)	1. B_2H_6 / THF 2. H_2O_2 / OH^-	**Anti-Markovnikov addition** of H_2O. **Syn addition** of H and OH. No rearrangement.
Hydroboration-Protonolysis	1. B_2H_6 / THF 2. CH_3COOH (or CH_3COOD)	Replaces boron with H (or D). Yields alkane (or deuterated alkane) with **Syn stereochemistry**.

Oxidation, Hydroxylation & Cleavage of Alkenes

1. Syn-Hydroxylation (Vicinal cis-Diol):

• **Baeyer's Reagent:** Cold dil. 1% alkaline $KMnO_4$ (Pink color discharged → Test for unsaturation).

• **OsO_4 / H_2O_2 (or $NaHSO_3$):** Yields cis-1,2-diol via cyclic osmate ester (Syn addition).

2. Anti-Hydroxylation (Vicinal trans-Diol):

• Peracid (mCPBA / RCO_3H) followed by H_3O^+ cleavage (Prileschajew reaction) → Epoxide → **trans-1,2-diol**.

3. Reductive Ozonolysis (O_3 / Zn + H_2O or Me_2S):

• Cleaves C=C double bond into carbonyls (>C=O).

• Aldehydes remain intact (Zn prevents oxidation by H_2O_2).

4. Oxidative Ozonolysis (O_3 / H_2O_2 or $KMnO_4$ / H^+):

• $=CH_2 \rightarrow CO_2 + H_2O$ | $=CHR \rightarrow R\text{-}COOH$ | $=CR_2 \rightarrow$ Ketone

5. Wacker Process: $CH_2{=}CH_2 + O_2 \xrightarrow{(PdCl_2,\ CuCl_2\ /\ H_2O)} CH_3CHO$

Allylic Oxidation & Halogenation Shortcuts

• **Allylic Halogenation:** NBS (N-Bromosuccinimide) / CCl_4, hv OR Cl_2 / 500°C → Allylic bromide/chloride (Free radical).

• **Allylic Oxidation:** SeO_2 (Selenium Dioxide) oxidizes allylic methyl/methylene to allylic alcohol ($-CH{=}CH\text{-}CH_2OH$).

4. ALKYNES: CHEMICAL REACTIVITY, REDUCTIONS & NUCLEOPHILIC ADDITIONS

Acidic Nature of Terminal Alkynes

Terminal hydrogen attached to sp-hybridized C (50% s-character, high EN) is acidic:

1. **$NaNH_2$ in liq. NH_3:** R-C≡C-H + $NaNH_2$ → R-C≡C$^-Na^+$ + NH_3↑
2. **Tollens' Test:** + $[Ag(NH_3)_2]OH$ → R-C≡C-Ag↓ (White ppt)
3. **Ammoniacal Cu_2Cl_2:** + $[Cu(NH_3)_2]Cl$ → R-C≡C-Cu↓ (Red ppt)
4. **Grignard Reaction:** R-C≡C-H + R'-MgX → R-C≡C-MgX + R'-H↑

Electrophilic & Nucleophilic Additions

Alkynes are *less reactive than alkenes towards Electrophilic Addition* because vinyl cation intermediate is less stable than alkyl cation.

• **Hydration (Kucherov Reaction):**

R-C≡C-H + H_2O —(1% $HgSO_4$ / 20% H_2SO_4, 60°C)→ [R-C(OH)=CH_2] —(Tautomerism)→ R-CO-CH_3 (Ketone).

Exception: Acetylene (HC≡CH) yields **Acetaldehyde** (CH_3CHO).

• **Nucleophilic Addition of HCN:** CH≡CH + HCN —($Ba(CN)_2$)→ CH_2=CH-CN (Acrylonitrile).

Reduction & Cleavage Matrix for Alkynes

Reaction / Reagent	Substrate & Products	Result & Stereochemistry
Complete Hydrogenation	R-C≡C-R' + 2 H_2 —(Ni / Pt / Pd, Δ)→ R-CH_2-CH_2-R'	Alkane formed.
Lindlar Catalyst / Ni_2B	R-C≡C-R' + H_2 —(Pd/$BaSO_4$/Quinoline)→ cis-Alkene	**cis-Alkene** (Syn addition).
Birch Reduction	R-C≡C-R' + Na / liq. NH_3 → trans-Alkene	**trans-Alkene** (Anti addition via radical anion).
Hydroboration-Reduction	1. R_2BH / THF 2. CH_3COOH → cis-Alkene	**cis-Alkene**.
Ozonolysis (Reductive)	R-C≡C-R' —(1. O_3 2. Zn / H_2O)→ R-CO-CO-R'	1,2-Diketone (α-Diketone).
Ozonolysis (Oxidative)	R-C≡C-R' —(1. O_3 2. H_2O or H_2O_2)→ R-COOH + R'-COOH	Carboxylic acid cleavage. Terminal alkynes yield CO_2 + H_2O.
Cyclic Polymerization	3 HC≡CH —(Red hot Fe / Cu tube, 500°C)→ Benzene (C_6H_6) 3 CH_3-C≡CH —(Red hot Cu tube)→ Mesitylene (1,3,5-Trimethylbenzene)	Trimerization to aromatic rings. High yield shortcut in conversions.

5. AROMATIC HYDROCARBONS: BENZENE & TOLUENE REACTIONS

Electrophilic Aromatic Substitution (EAS) Mechanism

Overall: Ar-H + E^+ → [Arenium Ion / σ-complex (Wheland intermediate)] → Ar-E + H^+. *Step 1 (formation of σ-complex) is Rate Determining (RDS).*

EAS Reaction	Reagent System	Active Electrophile	Product with Benzene
Nitration	Conc. HNO_3 + Conc. H_2SO_4 (1:1)	**NO_2^+** (Nitronium ion)	Nitrobenzene
Sulphonation	Fuming H_2SO_4 ($H_2S_2O_7$ / Oleum)	**SO_3** (Neutral electrophile)	Benzene sulphonic acid
Halogenation	X_2 + Anhy. FeX_3 / AlX_3 (or Fe)	**X^+** (Halonium ion)	Halobenzene (Ph-X)
Friedel-Crafts Alkylation	R-X + Anhy. $AlCl_3$ (or ROH/H^+, Alkene/H^+)	**R^+** (Carbocation)	Alkylbenzene (Rearrangements!)
Friedel-Crafts Acylation	RCOCl or $(RCO)_2O$ + Anhy. $AlCl_3$	**R-C≡O$^+$** (Acylium ion)	Acylbenzene (No rearrangement)
Gattermann-Koch Synthesis	CO + HCl + Anhy. $AlCl_3$ / CuCl	**HC≡O$^+$** (Formyl cation)	Benzaldehyde (Ph-CHO)
Gattermann Aldehyde	HCN + HCl + Anhy. $ZnCl_2$ followed by H_2O/Δ	**HN=CH$^+$**	Benzaldehyde (Ph-CHO)
Chloromethylation (Blanc)	HCHO + HCl + Anhy. $ZnCl_2$	**$^+CH_2OH$**	Benzyl chloride (Ph-CH_2Cl)

Friedel-Crafts Limitations (JEE Advanced): 1. Polyalkylation occurs because alkyl groups activate the ring.
2. Rearrangement of carbocations occurs (e.g., n-propyl chloride + $AlCl_3$ + Benzene gives **Cumene / Isopropylbenzene** as major).
3. Heavily deactivating groups (-NO_2, -SO_3H, -COOH, -COR, -CN) or aniline (-NH_2 which forms complex with $AlCl_3$) **do not undergo Friedel-Crafts reaction**.

Directing Effects & Ring Activation in Substituent Benzene

Ortho/Para Directors (Activating):

- Strong Activators (+M/+I): $-O^-$, $-NH_2$, $-OH$, $-OR$
- Moderate Activators: $-NHCOR$, $-OCOR$
- Weak Activators (+I/Hyperconjugation): $-CH_3$, $-C_2H_5$, $-R$, $-Ph$
- **Halogens (-F, -Cl, -Br, -I):** Deactivating (-I > +M) BUT **Ortho/Para directing**!

Meta Directors (Deactivating):

- Strong Deactivators (-M/-I): $-NO_2$, $-NR_3^+$, $-CF_3$, $-CCl_3$
- Moderate Deactivators (-M/-I): $-CN$, $-SO_3H$, $-CHO$, $-COR$, $-COOH$, $-COOR$
- Direct incoming electrophile to Meta position due to less electron deficiency at meta compared to o/p.

Toluene (Methylbenzene): Reactions & Side-Chain vs Ring Reactions

Reaction Type	Reagents & Conditions	Products Formed
Free-Radical Side-Chain Chlorination	Cl_2 / hv (or Δ) • 1 eq Cl_2 → Benzyl Chloride • 2 eq Cl_2 → Benzal Chloride • 3 eq Cl_2 → Benzo Trichloride	Ph-CH_2Cl $\xrightarrow{(aq.\ NaOH)}$ Benzyl Alcohol Ph-$CHCl_2$ $\xrightarrow{(aq.\ NaOH)}$ Benzaldehyde Ph-CCl_3 $\xrightarrow{(aq.\ NaOH)}$ Benzoic Acid
Side-Chain Oxidation ($KMnO_4$)	$KMnO_4$ / OH^-, Δ (or $K_2Cr_2O_7$ / H^+)	**Benzoic Acid (Ph-COOH)**. *Requires at least one benzylic Hydrogen!* (tert-Butylbenzene fails to oxidize).
Etard Reaction	1. CrO_2Cl_2 / CS_2 (or CCl_4) 2. H_3O^+	Chromium complex [Ph-CH($OCrOHCl_2$)$_2$] hydrolyzed to **Benzaldehyde (Ph-CHO)**.
Side-Chain Oxidation (CrO_3)	CrO_3 in $(CH_3CO)_2O$, then H_3O^+	Benzylidene diacetate intermediate hydrolyzed to **Benzaldehyde**.
Nuclear Ring Substitution	Cl_2 / Fe (Dark, Low Temp)	**o-Chlorotoluene + p-Chlorotoluene** (p-isomer is major).
Nitration of Toluene	Conc. HNO_3 + Conc. H_2SO_4, Δ	o- and p-Nitrotoluene. Under drastic conditions yields **TNT (2,4,6-Trinitrotoluene)**.
Birch Reduction of Ring	• Benzene + Na / liq. NH_3 + EtOH → 1,4-Cyclohexadiene • Toluene + Na / liq. NH_3 + EtOH → 1-Methyl-1,4-cyclohexadiene	Electron donating group ($-CH_3$) preserves double bond at attached carbon (sp^3 at non-substituted C). [Added]

6. JEE ADVANCED REACTIONS, STEREOSPECIFICITY RULES & SHORTCUTS

CAR & TAM Stereospecific Addition Rules:

- **C**is-alkene + **A**nti addition → **R**acemic mixture (±)
- **T**rans-alkene + **A**nti addition → **M**eso compound
- **C**is-alkene + **S**yn addition → **M**eso compound (CSM)
- **T**rans-alkene + **S**yn addition → **R**acemic mixture (TSR)

Pyrolysis & Elimination Stereospecificity:

- **meso**-Dihalide + Anti-elimination (Zn/NaI) → **trans-alkene**
- **racemic**-Dihalide + Anti-elimination → **cis-alkene**
- Cope Elimination & Ester Pyrolysis: **100% Syn-elimination** via 5/6-membered cyclic transition state.

Master Reagents & Key Conversions Summary

Reagent / Catalyst	Functional Transformation	Target Product & Notes
Lindlar ($Pd/CaCO_3$ + Quinoline)	Alkynes → cis-Alkenes	Stereospecific Syn-hydrogenation.
Na / Liquid NH_3 (Birch)	Alkynes → trans-Alkenes / Benzene → 1,4-diene	Anti-reduction via radical anion.
$(CH_3COO)_2Hg$ / H_2O, $NaBH_4$	Alkene → Alcohol (Markovnikov)	OMDM: Zero carbocation rearrangement.
B_2H_6 / THF + H_2O_2 / OH^-	Alkene → Alcohol (Anti-Markovnikov)	HBO: Syn-addition, zero rearrangement.
mCPBA followed by H_3O^+	Alkene → trans-1,2-Diol	Anti-hydroxylation via epoxide ring opening.
Cold Dilute Alkaline $KMnO_4$	Alkene → cis-1,2-Diol	Baeyer's Test: Pink color disappears.
O_3 followed by Zn / H_2O	Alkene → Aldehydes & Ketones	Reductive Ozonolysis (C=C cleaves cleanly).
CrO_2Cl_2 / CS_2 (Etard Reagent)	Toluene → Benzaldehyde	Selective side-chain oxidation.
Red Hot Fe/Cu tube (500°C)	Ethyne → Benzene | Propyne → Mesitylene	Cyclic trimerization of alkynes.
V_2O_5 / 450°C + O_2	Benzene → Maleic Anhydride + CO_2 + H_2O	Vapour phase ring oxidation.

HALOALKANES & HALOARENES

JEE Main & Advanced Complete Formula & Quick Revision Sheet

JEE REVISION SHEET

1. PHYSICAL PROPERTIES & PHYSICAL CONSTANTS TRENDS

Dipole Moment (μ) Order

- **Alkyl Halides:** CH_3Cl (1.86 D) > CH_3F (1.85 D) > CH_3Br (1.82 D) > CH_3I (1.62 D)
  (Note: C-Cl bond length compensates for lower electronegativity vs F)
- **Haloarenes:** Chlorobenzene (1.69 D) > Bromobenzene (1.61 D) > Iodobenzene
- **Dihalobenzenes:** *o*-dichlorobenzene (2.54 D) > *m*-dichlorobenzene (1.48 D) > *p*-dichlorobenzene (0 D, symmetrical cancellation)

Boiling Point & Density Trends

- **B.P. vs Halogen:** R–I > R–Br > R–Cl > R–F (Van der Waals forces increase with molar mass & size)
- **B.P. vs Isomers:** Primary > Secondary > Tertiary (Branching reduces surface area & B.P.)
- **B.P. Dihalobenzenes:** *o*-isomer > *p*-isomer > *m*-isomer
- **M.P. Dihalobenzenes:** *p*-isomer >> *o*-isomer > *m*-isomer *(P-isomer fits symmetrically into crystal lattice)*
- **Density:** n-C_3H_7I > n-C_3H_7Br > n-C_3H_7Cl

Bond Enthalpy & C–X Bond Length

Bond Length: C–I > C–Br > C–Cl > C–F

Bond Strength: C–F > C–Cl > C–Br > C–I

Solubility: Slightly soluble in H_2O; readily soluble in organic solvents.

2. MECHANISM COMPARISON: S_N1 VS S_N2

Feature	S_N1 (Unimolecular)	S_N2 (Bimolecular)
Kinetics & Rate Law	1st Order: Rate = k[RX]	2nd Order: Rate = k[RX][Nu^-]
Steps & Intermediate	2 Steps; Form planar Carbocation (sp^2) intermediate (Rearrangement possible!)	1 Step; Pentacoordinate Transition State (sp^3d hybrid), NO intermediate
Reactivity Order	3° > 2° > 1° > CH_3X (governed by Carbocation stability)	CH_3X > 1° > 2° > 3° (governed by Steric Hindrance)
Stereochemistry	Racemisation (Inversion + Retention) via planar carbocation	100% Inversion of Configuration (Walden Inversion)
Solvent & Nucleophile	Favoured by **Polar Protic Solvents** (H_2O, EtOH) & Weak Neutral Nu	Favoured by **Polar Aprotic Solvents** (DMSO, DMF, Acetone) & Strong Nu^-
Leaving Group Rate	R–I > R–Br > R–Cl > R–F (Weaker conjugate base = better leaving group)	

3. MECHANISM COMPARISON: E1 VS E2 VS E1CB

E1 Mechanism

- 2 Steps; Rate = k[RX]
- Carbocation intermediate forms (rearrangements occur).
- Reactivity: 3° > 2° > 1°
- Weak base, heat, polar protic solvent.

E2 Mechanism

- 1 Step concerted; Rate = k[RX][Base]
- Anti-periplanar transition state (stereospecific).
- Reactivity: 3° > 2° > 1°
- Strong bulky/unbulky base + heat.

E1cB Mechanism

- 2 Steps via Carbanion intermediate; Rate = k[RX][Base]
- Requires **poor leaving group** (–F, –OH) & highly acidic β–H.
- Hoffmann product dominates.

4. REGIOSELECTIVITY & NUCLEOPHILICITY RULES

Saytzeff (Zaitsev) vs Hoffmann Elimination

- **Saytzeff Rule:** Highly substituted, thermodynamically stable alkene is major product. (Standard unhindered bases like alc. KOH, EtO^-).
- **Hoffmann Rule:** Less substituted, kinetically controlled alkene is major product. Occurs when: (i) Base is bulky (t-BuO^-K^+), (ii) Leaving group is poor (–F, $-NR_3^+$), (iii) Steric crowding at β-carbon.

Nucleophilicity Trends & Solvent Effects

- **In Polar Protic Solvents:** I^- > Br^- > Cl^- > F^- (Solvation cage shields smaller ions; I^- is large & polarizable).
- **In Polar Aprotic Solvents:** F^- > Cl^- > Br^- > I^- (Anions bare; basicity determines nucleophilicity).
- **Steric Effect:** EtO^- > t-BuO^- (Nucleophilicity); t-BuO^- > EtO^- (Basicity).
- **Negatively charged > Neutral:** OH^- > H_2O, NH_2^- > NH_3.

5. METHODS OF PREPARATION OF ALKYL HALIDES (RX)

A. From Alcohols (R–OH)

1. Lucas Reagent: $R–OH + HCl$ (conc.) $\xrightarrow{[Anhyd.\ ZnCl_2]}$ $R–Cl + H_2O$ Turbidity time: 3° (Instantaneous), 2° (5 mins), 1° (No turbidity at room temp, requires heat).

2. PCl_5: $R–OH + PCl_5 \rightarrow R–Cl + POCl_3 + HCl$ (Inversion / S_N2)

3. PCl_3: $3\ R–OH + PCl_3 \rightarrow 3\ R–Cl + H_3PO_3$ (Inversion / S_N2)

4. Thionyl Chloride (Darzen's Process - Best method as SO_2 & HCl are gases): a. Pure $SOCl_2$: $R–OH + SOCl_2 \rightarrow R–Cl + SO_2\uparrow + HCl\uparrow$ (Retention of config via S_Ni mechanism) b. $SOCl_2$ + Pyridine: $R–OH + SOCl_2 + C_5H_5N \rightarrow R–Cl + SO_2\uparrow + HCl$ (Inversion of config via S_N2)

5. Red P + X_2 ($X_2 = Br_2, I_2$): $3\ R–OH + P + X_2 \rightarrow 3\ R–X + H_3PO_3$

B. Halide Exchange Reactions & Named Preparations

1. Finkelstein Reaction (Iodide Synthesis): $R–Cl$ / $R–Br + NaI \xrightarrow{[Dry\ Acetone]} R–I + NaCl\downarrow$ / $NaBr\downarrow$ (Driven by precipitation of NaCl/NaBr in acetone via Le Chatelier's Principle).

2. Swarts Reaction (Fluoride Synthesis): $R–Cl$ / $R–Br + AgF$ (or Hg_2F_2, CoF_2, SbF_3) $\rightarrow R–F + AgCl\downarrow$ / $AgBr\downarrow$

3. Hunsdiecker Reaction (Decarboxylation Halogenation): $R–COOAg + Br_2 \xrightarrow{[CCl_4,\ \Delta]} R–Br + CO_2\uparrow + AgBr\downarrow$ (Free radical mechanism; Yield: 1° > 2° > 3°)

C. From Hydrocarbons (Alkanes, Alkenes, Alkynes)

1. Free Radical Halogenation of Alkanes: $R–H + Cl_2 \xrightarrow{[h\nu\ or\ \Delta]} R–Cl + HCl$ (Reactivity of H: 3° > 2° > 1°; Selectivity: Br_2 is highly selective [160:82:1], Cl_2 is unselective [5:3.8:1])

2. Addition of HX to Alkenes: $R–CH=CH_2 + HX \rightarrow R–CH(X)–CH_3$ (Markovnikov addition via carbocation intermediate; rearrangement possible) $R–CH=CH_2 + HBr \xrightarrow{[Peroxide\ /\ Kharasch\ Effect]} R–CH_2–CH_2–Br$ (Anti-Markovnikov via free radicals; ONLY valid for HBr!)

3. Allylic & Benzylic Halogenation: $R–CH_2–CH=CH_2$ + NBS (N-Bromosuccinimide) $\xrightarrow{[h\nu\ /\ CCl_4]} R–CH(Br)–CH=CH_2$ Alternatively: Cl_2 / Br_2 at 500-600°C (800 K) gives allylic substitution.

4. Addition of X_2 to Alkenes & Alkynes: $CH_2=CH_2 + Br_2/CCl_4 \rightarrow BrCH_2-CH_2Br$ (Anti-addition via cyclic bromonium ion; decolorizes reddish-brown Br_2 water) $R-C\equiv C-H + 2\ HX \rightarrow R-CX_2-CH_3$ (Geminal dihalide)

6. COMPREHENSIVE CHEMICAL REACTIONS OF ALKYL HALIDES (R–X)

Reagent / Nucleophile	Product Formed	Reaction Type & JEE Key Remarks
aq. KOH / Moist Ag_2O	Alcohol (R–OH)	Nucleophilic Substitution (S_N2 for 1°, S_N1 for 3°)
alc. KOH / Δ	Alkene	β-Elimination (E2); Saytzeff product major
KCN (Ionic)	Alkyl Cyanide / Nitrile (R–C≡N)	Cyanide attack via C-atom (Ambident nucleophile)
AgCN (Covalent)	Alkyl Isocyanide / Isonitrile (R–N≡C)	Attack via lone pair on N-atom
KNO_2 / $NaNO_2$	Alkyl Nitrite (R–O–N=O)	Attack via Oxygen atom (Ambident nucleophile)
$AgNO_2$	Nitroalkane (R–NO_2)	Attack via Nitrogen lone pair
NaOR' (Sodium Alkoxide)	Ether (R–O–R')	**Williamson Ether Synthesis:** Best with 1° RX; 3° RX gives alkene (E2)
NH_3 (Excess)	1° Amine (R–NH_2)	**Hofmann Ammonolysis;** Excess RX gives 2°, 3° & Quaternary salt $R_4N^+X^-$
R'–C≡CNa	Higher Alkyne (R–C≡C–R')	Alkylation of terminal alkynes (S_N2)
$LiAlH_4$ / Et_2O	Alkane (R–H)	Reduction (S_N2 hydride transfer; 1° & 2° RX reduced, 3° RX gives alkene) [Added]
$NaBH_4$ / EtOH	Alkane (R–H)	Reduces 2° & 3° RX only; 1° RX does not react [Added]
Na / Dry Ether	Symmetrical Alkane (R–R)	**Wurtz Reaction:** Free radical mechanism; fails for methane synthesis
R'$_2$CuLi (Gilman Reagent)	Unsymmetrical Alkane (R–R')	**Corey-House Synthesis:** Works cleanly on 1°, 2°, alkenyl & aryl halides!
Mg / Dry Ether	Grignard Reagent (R–MgX)	Organometallic formation; must be strictly anhydrous!

7. HALOARENES (ARYL HALIDES) — PREPARATION & REACTIONS

Methods of Preparation of Ar–X

1. Direct Electrophilic Halogenation: $Ar-H + X_2 \xrightarrow{[Fe\ or\ FeX_3\ /\ Dark]} Ar-X + HX$ (X = Cl, Br)

2. Sandmeyer Reaction (Best for Ar–Cl, Ar–Br): $Ar-NH_2 \xrightarrow{[NaNO_2 + HCl,\ 0\text{-}5°C]} Ar-N_2^+Cl^-$ (Diazotization) $Ar-N_2^+Cl^- \xrightarrow{[Cu_2Cl_2/HCl]} Ar-Cl + N_2\uparrow$ $Ar-N_2^+Cl^- \xrightarrow{[Cu_2Br_2/HBr]} Ar-Br + N_2\uparrow$

3. Gattermann Reaction: $Ar-N_2^+Cl^- \xrightarrow{[Cu\ powder\ /\ HCl]} Ar-Cl + N_2\uparrow + CuCl$

Nucleophilic Aromatic Substitution (S_NAr vs Benzyne)

Aryl halides are *extremely unreactive* towards ordinary S_N1/S_N2 due to: (i) Partial double bond character from resonance, (ii) sp^2 carbon electronegativity, (iii) Instability of phenyl cation.

1. Dow's Process (Drastic Conditions): $C_6H_5Cl + 2\ NaOH \xrightarrow{[623K,\ 300\ atm]} C_6H_5ONa \xrightarrow{[H^+]} C_6H_5OH$

2. Activating Effect of $-NO_2$ Groups (S_NAr via Meisenheimer Complex): $-NO_2$ at *ortho/para* positions drastically increases S_NAr rate! • 1 $-NO_2$ (*p*): 15% NaOH, 443 K • 2 $-NO_2$ (*o,p*): aq. Na_2CO_3, 368 K • 3 $-NO_2$ (2,4,6-trinitrochlorobenzene): Warm H_2O → Picric Acid

4. Schiemann Reaction (Fluoroarene Synthesis): $Ar-NH_2$ --[$NaNO_2$ + HBF_4]--> $Ar-N_2^+ BF_4^-$ --[Δ]--> $Ar-F + N_2\uparrow + BF_3$

5. Iodobenzene Synthesis: $Ar-N_2^+Cl^-$ + KI (aq) --[Warm]--> $Ar-I + KCl + N_2\uparrow$

3. Elimination-Addition (Benzyne Mechanism): Chlorobenzene + KNH_2 / Liq. NH_3 --> Aniline via Benzyne intermediate (Triply bonded reactive species; gives cine substitution).

Electrophilic Aromatic Substitution (EAS) in Haloarenes

Halogens are **deactivating** due to strong −I effect, but **ortho/para-directing** due to +M (+R) resonance stabilization of intermediate. *Para product is generally major due to lower steric hindrance.*

• Chlorination: Cl_2 / $FeCl_3$ --> 1,4-Dichlorobenzene (Major) + 1,2-isomer • Nitration: Conc. HNO_3 + Conc. H_2SO_4 --> 1-Chloro-4-nitrobenzene (Major)

• Sulfonation: Conc. H_2SO_4, Δ --> 4-Chlorobenzenesulfonic acid (Major) • Friedel-Crafts Alkylation: CH_3Cl / Anhyd. $AlCl_3$ --> 1-Chloro-4-methylbenzene • Friedel-Crafts Acylation: CH_3COCl / Anhyd. $AlCl_3$ --> 4-Chloroacetophenone

Coupling Reactions of Aryl Halides

1. Fitting Reaction: 2 Ar−X + 2 Na --[Dry Ether]--> Ar−Ar (Biphenyl) + 2 NaX

2. Wurtz-Fitting Reaction: Ar−X + R−X + 2 Na --[Dry Ether]--> Ar−R (Alkylbenzene) + 2 NaX

3. Ullmann Reaction: 2 Ar−I + Cu powder --[Δ in sealed tube]--> Ar−Ar + Cu_2I_2 [Added]

8. SPECIAL HALIDES & POLYHALOGEN COMPOUNDS

Geminal vs Vicinal Dihalides

- **Geminal (Alkyline):** R−CHX_2 (e.g., CH_3CHCl_2)
  • Hydrolysis: CH_3CHX_2 + aq. KOH --> CH_3CHO (Aldehyde)
  • $CH_3CX_2CH_3$ + aq. KOH --> CH_3COCH_3 (Ketone)
- **Vicinal (Alkylene):** RCHX−CH_2X (e.g., $ClCH_2-CH_2Cl$)
  • Hydrolysis: $ClCH_2CH_2Cl$ + aq. KOH --> $HOCH_2CH_2OH$ (Glycol)
- **Dehalogenation:** Both Gem & Vic dihalides + Zn dust / MeOH --> Alkene
- **Dehydrohalogenation:** Both + alc. KOH / $NaNH_2$ --> Alkyne

Reactions of Chloroform ($CHCl_3$)

- **Oxidation (Phosgene formation):** 2 $CHCl_3$ + O_2 --[Light/Air]--> 2 $COCl_2$ (Poisonous Phosgene) + 2 HCl.
  Prevention: Stored in dark amber bottles; 1% EtOH added to convert phosgene to harmless diethyl carbonate.
- **Carbylamine Reaction (Isocyanide Test):** R−NH_2 / $C_6H_5NH_2$ + $CHCl_3$ + 3 KOH (alc.) --> R−NC (Offensive smell) + 3 KCl + 3 H_2O. *Test for 1° amines only!*
- **Reimer-Tiemann Reaction:** Phenol + $CHCl_3$ + 3 aq. NaOH --> Salicylaldehyde + 3 NaCl. *Electrophile: Dichlorocarbene ($:CCl_2$)*
- **Action of Ag powder:** 2 $CHCl_3$ + 6 Ag --[Δ]--> HC≡CH (Acetylene) + 6 AgCl↓
- **Nitration:** $CHCl_3$ + HNO_3 --> CCl_3NO_2 (Chloropicrin / Tear gas) + H_2O

9. IODOFORM TEST, CCL_4 & POLYHALOGEN COMPOUNDS

Iodoform Test (Haloform Reaction)

Reagents: I_2 + NaOH (or NaOI / I_2 + Na_2CO_3)

Positive Test Condition: Compounds containing either **CH_3–C=O** group or **CH_3–CH(OH)–** group.

General Reaction: $CH_3COR + 3\ I_2 + 4\ NaOH \rightarrow CHI_3\downarrow$ (Yellow Precipitate, MP 119°C) + RCONa + 3 NaI + 3 H_2O

- **Positive Test Giving Compounds:** Ethanol (CH_3CH_2OH), Acetaldehyde (CH_3CHO), All Methyl Ketones (CH_3COR), Secondary methyl alcohols ($CH_3CH(OH)R$).
- **Negative Test Compounds:** Methanol, Propan-1-ol, Benzaldehyde, Pentan-3-one, Acetic acid/Esters (carbonyl oxygen involved in resonance).

Carbon Tetrachloride (CCl_4) & Pyrene

- **Preparation:** $CH_4 + 4\ Cl_2 \xrightarrow{hv} CCl_4 + 4\ HCl$
  $CS_2 + 3\ Cl_2 \xrightarrow{[AlCl_3]} CCl_4 + S_2Cl_2$
- **Hydrolysis:** $CCl_4 + 4\ KOH \rightarrow C(OH)_4 \xrightarrow{[-2\ H_2O]} CO_2 \xrightarrow{[2\ KOH]} K_2CO_3 + H_2O$
- **Etched Action with HF:** $CCl_4 + 2\ HF \xrightarrow{[SbF_5]} CCl_2F_2$ (Freon-12) + 2 HCl
- **Use:** Commercial fire extinguisher under trade name **Pyrene**.

Vinyl, Allyl & Benzyl Halides Comparison

- **Vinyl Chloride (CH_2=CHCl):** Extremely unreactive in S_N due to C–Cl double bond character from resonance (CH_2=CH–Cl ↔ $^-CH_2$–CH=Cl^+).
- **Allyl Chloride (CH_2=CH–CH_2Cl):** Highly reactive in S_N1 (Allyl carbocation stabilized by resonance) and S_N2 (unhindered sp^3 carbon).
- **Benzyl Chloride (Ph–CH_2Cl):** Highly reactive in S_N1 (Benzyl carbocation resonance stabilized) and S_N2.

10. COMPLETE GRIGNARD REAGENT (R–MGX) REACTION SYNTHESIS CHART

Grignard Reagent Synthetic Applications

Grignard reagents behave as strong **Bases** (react with active H) and strong **Nucleophiles** (attack polar carbonyls/epoxides).

1. Acid-Base Reactions (Active H containing compounds give Alkanes R–H): R–MgX + H_2O / ROH / RNH_2 / RCOOH / R–C≡C–H → R–H + Mg(OH/OR/NHR)X

2. Nucleophilic Addition to Carbonyls (followed by H_3O^+ hydrolysis): • Formaldehyde (HCHO) + R–MgX → 1° Alcohol (R–CH_2OH) • Any other Aldehyde (R'CHO) + R–MgX → 2° Alcohol (R'–CH(OH)–R) • Ketone (R'COR'') + R–MgX → 3° Alcohol (R'R''C(OH)R) • Acid Chloride (R'COCl) + 1 eq R–MgX → Ketone (R'COR); Excess R–MgX → 3° Alcohol • Esters (R'COOR'') + 1 eq R–MgX → Ketone; Excess R–MgX → 3° Alcohol • Carbon Dioxide (O=C=O) + R–MgX → Carboxylic Acid (R–COOH) • Cyanides (R'C≡N) + R–MgX $\xrightarrow{[H_3O^+]}$ Ketone (R'COR) + NH_3 • Oxiranes / Epoxides + R–MgX → Primary Alcohol with chain elongated by 2 carbons (R–CH_2CH_2OH)

11. JEE ADVANCED KEY SHORTCUTS & STEREOCHEMICAL RULES

Stereospecific Addition / Elimination Shortcut

- **CAR Rule: C**is alkene + **A**nti addition → **R**acemic / Threo mixture
- **TAM Rule: T**rans alkene + **A**nti addition → **M**eso / Erythro compound
- **Cis + Syn Addition:** Meso compound
- **Trans + Syn Addition:** Racemic mixture
- *E2 elimination is strict Anti-Elimination!*

Common JEE Conceptual Traps

- **Bredt's Rule:** Double bond or carbocation CANNOT exist at a bridgehead position in small bicyclic systems.
- **Neopentyl Halide (Me_3C–CH_2X):** Inert to S_N2 due to extreme steric hindrance; reacts via S_N1 with methyl shift rearrangement.
- **DDT (p,p'-Dichlorodiphenyltrichloroethane):** Chloral + 2 Chlorobenzene $\xrightarrow{[Conc.\ H_2SO_4]}$ DDT + H_2O. Non-biodegradable insecticide.

GENERAL ORGANIC CHEMISTRY (GOC)

JEE MAIN & ADVANCED COMPLETE REVISION & FORMULA SHEET

Target: JEE Main / Advanced
Topic: Core Principles & Effects

1. REACTION MECHANISM FUNDAMENTALS & CLEAVAGE

Reaction Framework

Substrate (under attack) + **Reagent** (attacking species) → [Transition State] → **Intermediate** → **Product**

Homolytic Bond Fission: Shared e^- pair distributed equally forming free radicals.

$A:B \rightarrow A^{\bullet} + B^{\bullet}$

Conditions: High Temp (Δ), Light ($h\nu$), Peroxides (ROOR), Non-polar solvents/phase.

Heterolytic Bond Fission

Shared e^- pair taken by one atom forming Cation & Anion.

$A:B \rightarrow A^+ + :B^-$ (where EN of B > A)

Conditions: Polar protic solvents, presence of acids/bases, lower temperatures.

Bond Energy Relation: Bond Cleavage Energy $\propto$ 1 / (Stability of formed intermediate).

2. CLASSIFICATION OF REAGENTS

Reagent Type	Positively Charged / Electron Deficient	Neutral / Electron Rich
Electrophiles (E^+) e^- pair seekers (Lewis Acids)	H^+, H_3O^+, Cl^+, Br^+, NO_2^+, NO^+, SO_3H^+, R^+, R_3C^+, CH_3CO^+	SO_3, BF_3, $AlCl_3$, $ZnCl_2$, $FeCl_3$, $SnCl_4$, PCl_5, CO_2, CX_4, $:CCl_2$ (carbene), $:NH_2$ (nitrene) *Note:* NH_4^+ and H_3O^+ are NOT electrophiles [Added].
Nucleophiles (Nu^-) e^- pair donors (Lewis Bases)	Cl^-, Br^-, I^-, OH^-, OR^-, CN^-, HS^-, $RCOO^-$, NH_2^-, R_3C^-, $[CH(COOC_2H_5)_2]^-$	$H_2O:$, $:NH_3$, RNH_2, ROH, R-O-R, R-S-H, RMgX, $LiAlH_4$, $R_3P:$, PH_3, $CH_2=CH_2$, $HC\equiv CH$
Free Radicals	Species with at least one unpaired e^-. E.g., $H^{\bullet}$, $^{\bullet}CH_3$, $Cl^{\bullet}$, $C_6H_5^{\bullet}$. Complete octet incomplete (7 valence e^-).	

3. ELECTRONIC DISPLACEMENT EFFECTS

A. Inductive Effect (±I) — Permanent, Distance Dependent ($\propto 1/d^2$), Operates via σ-bonds

Polarization of a σ-bond induces polarization in neighboring bonds. Negligible after 3rd C-atom.

−I Order (Electron Withdrawing):

$-NF_3^+ > -NR_3^+ > -NH_3^+ > -NO_2 > -SO_3H > -CN > -COOH > -F > -Cl > -Br > -I > -OAr > -OR > -OH > -C\equiv CH > -NH_2 > -C_6H_5 > -CH=CH_2 > -H$

+I Order (Electron Releasing):

$-CH_2^- > -NH^- > -O^- > -COO^- > -C(CH_3)_3$ (3°) $> -CH(CH_3)_2$ (2°) $> -CH_2CH_3$ (1°) $> -CH_3 > -D > -H$

B. Mesomeric Effect (±M / ±R) — Permanent, Distance Independent, Operates via π-bonds

Delocalization of π-electrons in a conjugated system leading to permanent polarity.

+M / +R Effect (Electron Donating to Conjugated System):

Groups with lone pair or negative charge. Increases e^- density at *ortho* and *para* positions (o/p directors, ring activators).

$-O^- > -NH_2 > -NHR > -NR_2 > -OH > -OR > -NHCOCH_3 > -OCOCH_3 > -Ph > -F > -Cl > -Br > -I$

−M / −R Effect (Electron Withdrawing from Conjugated System):

Groups containing multiple bonds with electronegative atoms. Decreases e^- density at *o/p* positions (*meta* directors, ring deactivators).

$-NO_2 > -CN > -SO_3H > -CHO > -COR > -COOH > -COOR > -CONH_2$

Stability Rules for Resonating Structures (Canonical Forms)

1. **Neutral Structures:** Neutral species are more stable than charged/dipolar species.
2. **Complete Octet:** Structures with complete octets for all atoms are significantly more stable than incomplete octet structures.
3. **Covalent Bond Count:** Structure with maximum number of covalent bonds is more stable.
4. **Electronegativity vs Charge:** Negative charge on more EN atom (O, N, F) and positive charge on more electropositive atom (C) is more stable.
5. **Charge Separation:** Like charges close together → unstable; Opposite charges farther apart → less stable than closer opposite charges.

Resonance Key Quantitative Formulae

Bond Order = (Total No. of Bonds between 2 atoms across all RS) / (Total No. of RS)

Example: Benzene C-C Bond Order = (2 + 1) / 2 = 1.5

Resonance Energy = E_{exp} (Real Hybrid) − E_{calc} (Most stable RS)

Resonance Effects on Properties

- Bond Length ∝ 1 / (Number of Resonating Structures)
- Bond Strength ∝ Number of Resonating Structures
- Reactivity (Bond Cleavage) ∝ 1 / (Number of RS)
- Stability ∝ Resonance Energy ∝ Number of RS

C. Hyperconjugation (σ-π Conjugation / No-Bond Resonance / Baker-Nathan Effect)

Delocalization of $\sigma_{C\text{-}H}$ or $\sigma_{C\text{-}D}$ electrons of an α-carbon (sp^3) into an adjacent vacant p-orbital or π*-orbital of an sp^2 C-atom.

Conditions Required:

- Must have at least one sp^2-hybridized carbon attached to an sp^3 α-carbon.
- The α-carbon must carry at least one α-Hydrogen (α-H).
- No. of Hyperconjugative Structures = **(No. of α-H) + 1** (including original structure).

Application Formulae:

Alkene Stability ∝ No. of α-H ∝ 1 / Heat of Hydrogenation (ΔH_{hyd})

Heat of Combustion (ΔH_{comb}) ∝ 1 / Stability (for isomers)

Isotopic Effect: Hyperconjugation order: **$-CH_3 > -CD_3 > -CT_3$** (C-H bond weaker than C-D/C-T).

D. Electromeric Effect (±E) — Temporary Effect

Complete transfer of a shared π-electron pair to one of the bonded atoms in the presence of an attacking reagent. Reversed upon reagent removal.

- **+E Effect:** π-e^- transferred to the atom to which reagent gets attached. (e.g., Electrophilic addition of H^+ to alkenes).
- **−E Effect:** π-e^- transferred away from the atom to which reagent gets attached. (e.g., Nucleophilic addition of CN^- to carbonyls).

Priority Order of Electronic Effects for Stability/Reactivity Assessment:

`Aromaticity > Resonance/Mesomeric (M) > Hyperconjugation (H) > Inductive Effect (I)`

4. REACTIVE INTERMEDIATES — DETAILED COMPARISON

Property	Carbocation (C^+)	Carbon Free Radical ($C^\bullet$)	Carbanion (C^-)
Formation	Heterolytic fission (C → X)	Homolytic fission (C • X)	Heterolytic fission (C ← Z)
Valence e^- count	6 e^- (Incomplete octet)	7 e^- (Incomplete octet)	8 e^- (Complete octet)
Nature & Charge	Electrophilic, Positively charged	Electrophilic, Neutral (1 unpaired e^-)	Nucleophilic, Negatively charged
Hybridization & Shape	sp^2, Trigonal Planar (120°)	sp^2, Trigonal Planar (120°)	sp^3, Pyramidal (lone pair present)
Stabilized by	+M, +H, +I groups (e^- donors)	+M, +H, +I groups (e^- donors)	−M, −I groups (e^- acceptors)
Destabilized by	−M, −I groups	−M, −I groups	+M, +I groups

Stability Sequences of Intermediates

Carbocation Stability Order:

$(Ph)_3C^+ > (Ph)_2CH^+ > PhCH_2^+ > CH_2{=}CH\text{-}CH_2^+$ (Allyl) $> (CH_3)_3C^+$ (3°) $> (CH_3)_2CH^+$ (2°) $> CH_3CH_2^+$ (1°) $> CH_3^+ > CH_2{=}CH^+$ (Vinyl) $> HC{\equiv}C^+$

Note on Carbocation Rearrangement: 1° or 2° carbocations spontaneously rearrange via **1,2-hydride shift**, **1,2-methyl shift**, or **1,2-phenyl shift** to yield a more stable carbocation [Added] .

Free Radical Stability Order:

$(Ph)_3C^\bullet > (Ph)_2CH^\bullet > PhCH_2^\bullet > CH_2{=}CH\text{-}CH_2^\bullet > (CH_3)_3C^\bullet$ (3°) $> (CH_3)_2CH^\bullet$ (2°) $> CH_3CH_2^\bullet$ (1°) $> CH_3^\bullet > CH_2{=}CH^\bullet$

Carbanion Stability Order:

$(Ph)_3C^- > (Ph)_2CH^- > PhCH_2^- > CH_2{=}CH\text{-}CH_2^- > HC{\equiv}C^- > CH_3^- > CH_3CH_2^-$ (1°) $> (CH_3)_2CH^-$ (2°) $> (CH_3)_3C^-$ (3°)

Carbenes (:CH_2) — Neutral 6 e^- Intermediate

Generated by photolysis/thermolysis of Diazomethane (CH_2N_2) or Ketene ($CH_2{=}C{=}O$) or via α-elimination of $CHCl_3$ with base.

Feature	Singlet Carbene (:CH_2)	Triplet Carbene (•CH_2•)
Unshared e^- State	Paired in one sp^2 orbital (spin = 0)	Unpaired in two different p-orbitals (spin = 1)
Hybridization & Geometry	sp^2 hybridised, Bent shape (∠H-C-H ≈ 103°)	sp hybridised, Linear (∠H-C-H = 180°)
C-H Bond Length	1.12 Å	1.03 Å
Relative Stability	Less stable	More stable (Hund's Rule favored)
Chemical Behavior	Acts as an Electrophile	Acts as a Diradical

5. AROMATICITY & HUCKEL'S RULE [ADDED]

Aromatic Species

- Cyclic, Planar (sp^2/sp C).
- Complete π-conjugation.
- **(4n + 2) π-electrons** (n = 0, 1, 2...).
- Exceptionally stable.
- *E.g.:* Benzene (6π), Cyclopropenyl cation (2π), Cyclopentadienyl anion (6π), Tropylium cation (6π), Pyrrole, Furan, Pyridine.

Anti-Aromatic Species

- Cyclic, Planar (sp^2/sp C).
- Complete π-conjugation.
- **(4n) π-electrons** (n = 1, 2...).
- Highly unstable.
- *E.g.:* Cyclobutadiene (4π), Cyclopropenyl anion (4π), Cyclopentadienyl cation (4π).

Non-Aromatic Species

- Non-cyclic OR non-planar OR lacks complete π-conjugation (contains sp^3 C).
- Stability: Aromatic > Non-Aromatic > Anti-Aromatic.
- *E.g.:* Cyclooctatetraene (COT is tub-shaped non-planar → Non-aromatic).

6. ACIDIC & BASIC STRENGTH PRINCIPLES [ADDED]

Acidic Strength ($HA \rightleftharpoons H^+ + A^-$)

Acidic Strength ∝ K_a ∝ 1 / pK_a ∝ Stability of Conjugate Base (A^-)

• **Direct Factors:** Stability of A^- ∝ −M ∝ −I ∝ Electronegativity ∝ % s-character (sp > sp^2 > sp^3).

• **Standard Organic Acid Strength Order:**

RSO_3H > CF_3COOH > $PhCOOH$ > $HCOOH$ > CH_3COOH > H_2CO_3 > $PhOH$ > H_2O > ROH > $HC{\equiv}CH$ > NH_3 > $CH_2{=}CH_2$ > $CH_3{-}CH_3$

• **Ortho Effect (Benzylic/Benzoic Acids):** Nearly all *ortho*-substituted benzoic acids are stronger acids than benzoic acid regardless of +I/−I nature due to steric inhibition of resonance (SIR).

Basic Strength ($B: + H^+ \rightleftharpoons BH^+$)

Basic Strength ∝ K_b ∝ 1 / pK_b ∝ Availability of lone pair ∝ Stability of BH^+

• **Direct Factors:** Basic Strength ∝ +M ∝ +H ∝ +I ∝ 1 / % s-character.

• **Amine Basicity in Aqueous Medium:**

  • For **$-CH_3$** group: **2° > 1° > 3° > NH_3** (213 order; $(CH_3)_2NH$ > CH_3NH_2 > $(CH_3)_3N$ > NH_3)

  • For **$-C_2H_5$** group: **2° > 3° > 1° > NH_3** (231 order)

  • In Gas Phase: **3° > 2° > 1° > NH_3** (purely +I driven).

7. STEREOCHEMISTRY SUMMARY: ENANTIOMERS VS DIASTEREOMERS

Property	Enantiomers	Diastereomers
Molecular & Structural Formula	Same	Same
Stereochemical Formula	Non-superimposable mirror images	Not mirror images of each other
Dipole Moment & Physical Constants	Identical (m.p., b.p., density, refractive index)	Different (m.p., b.p., solubility, dipole moment)
Optical Rotation	Equal in magnitude, opposite in sign (±α)	Different magnitude and/or sign
Reactivity with Inactive Reagents	Identical	Different
Reactivity with Active Reagents	Different	Different

8. CLASSIFICATION OF ORGANIC REACTIONS

1. Addition Reactions (π-bond → 2 σ-bonds)

• **Electrophilic Addition (EAR):** E^+ attacks in rate-determining slow step. Common in alkenes/alkynes.

$$CH_3\text{-}CH{=}CH_2 + H^+ \rightarrow [CH_3\text{-}CH^+\text{-}CH_3] \xrightarrow{Cl^-} CH_3\text{-}CHCl\text{-}CH_3$$

• **Nucleophilic Addition (NAR):** Nu^- attacks in slow step. Common in aldehydes/ketones.

NAR Reactivity Order: **$HCHO > CH_3CHO > (CH_3)_2CO$**

Halogenated Aldehyde Reactivity: **$CCl_3CHO > CHCl_2CHO > CH_2ClCHO$**

• **Free Radical Addition:** Anti-Markovnikov addition of HBr in presence of peroxides (Kharasch Effect).

2. Elimination Reactions (2 σ-bonds → π-bond)

• **α-Elimination:** Both groups lost from the same atom → forms Carbene.

$$CHCl_3 + OH^- \rightarrow :CCl_2 + H_2O + Cl^-$$

• **β-Elimination:** Groups lost from adjacent atoms → forms Alkene. (E1, E2, E1cB mechanisms).

$$CH_3\text{-}CH_2\text{-}Cl + NaNH_2 \rightarrow CH_2{=}CH_2 + NaCl + NH_3$$

• **γ-Elimination:** Groups lost from 1,3-positions → forms 3-membered ring (Cyclopropane).

$$Cl\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}Cl + Zn\ (dust) \rightarrow Cyclopropane + ZnCl_2$$

3. Substitution Reactions

• **Nucleophilic Substitution (S_N):** S_N1 (via Carbocation intermediate) & S_N2 (via concerted Transition State).

$$CH_3\text{-}Cl + OH^-\ (aq) \rightarrow CH_3\text{-}OH + Cl^-$$

Acyl Nucleophilic Substitution Reactivity Order:

`Acid Chloride > Acid Anhydride > Ester > Amide`

• **Electrophilic Substitution (S_E/S_EAr):** Characteristic of aromatic rings.

$$C_6H_6 + Cl^+\ (Cl_2/AlCl_3) \rightarrow C_6H_5Cl + H^+$$

• **Free Radical Substitution (S_R):** Free radical halogenation of alkanes.

4. Condensation, Rearrangement & Polymerization

• **Condensation:** Two molecules join with loss of simple molecules (H_2O, HCl, NH_3, ROH).

$$CH_3CHO + CH_3COCH_3 \xrightarrow{dil.\ NaOH} CH_3\text{-}CH{=}CH\text{-}CO\text{-}CH_3 + H_2O$$

• **Rearrangement:** Structural skeleton reorganizes.

$$CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_3 \xrightarrow{Anhyd.\ AlCl_3/HCl} CH_3\text{-}CH(CH_3)\text{-}CH_3$$

• **Polymerization:** Small monomers link into macromolecules.

$$n\ CH_2{=}CH_2 \rightarrow (-CH_2\text{-}CH_2-)_n$$

EQUILIBRIUM FORMULA & REVISION SHEET

Chemical & Ionic Equilibrium • Ultimate JEE Main & Advanced Notes

JEE MAIN & ADVANCED

1. FUNDAMENTAL RELATIONS & EQUILIBRIUM CONSTANTS

General Reversible Reaction:

$aA + bB \rightleftharpoons cC + dD$

$K_c = [C]^c[D]^d / ([A]^a[B]^b)$ | $K_p = (P_C)^c(P_D)^d / ((P_A)^a(P_B)^b)$

$K_x = (X_C)^c(X_D)^d / ((X_A)^a(X_B)^b)$

Active Mass: Molar concentration $[A] = n / V$. For pure solids & liquids, active mass = 1 (intensive property).

Units: $\text{Unit}(K_c) = (\text{mol L}^{-1})^{\Delta n_g}$, $\text{Unit}(K_p) = (\text{atm or bar})^{\Delta n_g}$. (Dimensionless if standard state = 1 bar / 1 M).

Core Master Relation:

$$K_p = K_c(RT)^{\Delta n_g} = K_x P^{\Delta n_g}$$

Δn_g = (moles of gaseous products) − (moles of gaseous reactants)

- **If $\Delta n_g = 0$:** $K_p = K_c = K_x$ (Independent of pressure/volume)
- **If $\Delta n_g > 0$:** $K_p > K_c$ (at room temp where RT > 1)
- **If $\Delta n_g < 0$:** $K_p < K_c$

2. PROPERTIES & TRANSFORMATIONS OF K

Reaction Operation	New Equilibrium Constant
Reverse reaction ($cC+dD \rightleftharpoons aA+bB$)	$K' = 1 / K$
Multiply by factor n	$K' = (K)^n$
Divide by factor n	$K' = K^{1/n} = \sqrt[n]{K}$
Add two reactions (Rxn 1 + Rxn 2)	$K_{net} = K_1 \times K_2$
Subtract reactions (Rxn 1 − Rxn 2)	$K_{net} = K_1 / K_2$

3. REACTION QUOTIENT (Q) & THERMODYNAMICS

Reaction Quotient (Q): Same expression as K at any arbitrary non-equilibrium state.

- **Q < K:** Proceeds in **Forward** direction (to form more products).
- **Q = K:** System is in **Dynamic Equilibrium**.
- **Q > K:** Proceeds in **Backward** direction (to form reactants).

$\Delta G = \Delta G^\circ + RT \ln Q$ | At Equilibrium ($\Delta G = 0$, $Q = K$):

$$\Delta G^\circ = -RT \ln K_c = -2.303\ RT \log K_c \Rightarrow K_c = e^{-\Delta G^\circ / RT}$$

4. TEMPERATURE DEPENDENCE & DEGREE OF DISSOCIATION

[Added JEE Advanced Details]

Van't Hoff Isochore Equation:

$$d(\ln K) / dT = \Delta H^\circ / (RT^2) \Rightarrow \log(K_2 / K_1) = (\Delta H^\circ / 2.303R) \times [(T_2 - T_1) / (T_1T_2)]$$

- **Endothermic ($\Delta H > 0$):** $T \uparrow \Rightarrow K \uparrow$ (Forward shift)
- **Exothermic ($\Delta H < 0$):** $T \uparrow \Rightarrow K \downarrow$ (Backward shift)

Vapour Density & Degree of Dissociation (α):

Consider $A_n(g) \rightleftharpoons n\,A(g)$: Initial moles = 1, at eq = $1 - \alpha + n\alpha$.

$$\alpha = (D - d) / ((n - 1)\,d) = (M_{obs}^{calc} - M_{obs}) / ((n - 1)\,M_{obs})$$

D = Initial/Theoretical Vapour Density, d = Observed Vapour Density at eq.

5. LE CHATELIER'S PRINCIPLE SUMMARY

Disturbance / Constraint	Shift Direction ($\Delta n_g > 0$)	Shift Direction ($\Delta n_g < 0$)	Shift Direction ($\Delta n_g = 0$)
Increase Concentration of Reactant	Forward	Forward	Forward
Increase Pressure / Decrease Volume	Backward (fewer gas moles)	Forward (fewer gas moles)	No Effect
Decrease Pressure / Increase Volume	Forward (more gas moles)	Backward (more gas moles)	No Effect
Inert Gas Addition at Constant Volume	No Shift	No Shift	No Shift
Inert Gas Addition at Constant Pressure	Forward (shifts to gas side)	Backward (shifts to gas side)	No Shift

Disturbance / Constraint	Shift Direction ($\Delta n_g > 0$)	Shift Direction ($\Delta n_g < 0$)	Shift Direction ($\Delta n_g = 0$)
Add Catalyst	No Shift (Only speeds up rate)	No Shift	No Shift

6. ACID-BASE THEORIES & CONJUGATE PAIRS

Arrhenius Theory:

Acid: H^+ donor in water.

Base: OH^- donor in water.

Limitation: Only aqueous solutions.

Brønsted-Lowry Theory:

Acid: H^+ donor.

Base: H^+ acceptor.

Conjugate Base = Acid − H^+

Conjugate Acid = Base + H^+

Lewis Theory:

Acid: Lone pair acceptor (vacant orbitals: BF_3, $AlCl_3$, cations).

Base: Lone pair donor (NH_3, H_2O, anions).

7. OSTWALD DILUTION LAW & WATER AUTO-IONIZATION

Ostwald Dilution Law (Weak Electrolytes):

For weak acid $HA \rightleftharpoons H^+ + A^-$ with concentration C and degree of dissociation α:

$K_a = C\alpha^2 / (1 - \alpha) \Rightarrow$ If $\alpha \ll 1$ ($\alpha < 0.05$): $\boldsymbol{\alpha = \sqrt{(K_a / C)} = \sqrt{(K_a V)}}$

$\boldsymbol{[H^+] = C\alpha = \sqrt{(K_a C)}}$ | For Weak Base: $\boldsymbol{[OH^-] = C\alpha = \sqrt{(K_b C)}}$

Note: As dilution $V \to \infty$, $\alpha \to 1$ (100% ionization).

Ionic Product of Water (K_W):

$2\,H_2O \rightleftharpoons H_3O^+ + OH^- \Rightarrow K_w = [H^+][OH^-]$

At 25°C: $K_w = 1.0 \times 10^{-14} \Rightarrow pK_w = pH + pOH = 14$

For Pure Neutral Water: $[H^+] = [OH^-] = \sqrt{(K_w)} \Rightarrow pH = pK_w / 2$

$T \uparrow \Rightarrow K_w \uparrow$ (e.g., at 60°C, $K_w \approx 10^{-13}$, pH of neutral water = 6.5).

8. PH CALCULATION RULES & STANDARD CASES

Definitions: $pH = -\log_{10}[H^+]$, $pOH = -\log_{10}[OH^-]$, $pK_a = -\log_{10}K_a$

- **Strong Acid (SA):** $[H^+] = N = M \times$ n-factor. If $[H^+] < 10^{-6}$ M, include water ionization: $[H^+]_{total} = [H^+]_{acid} + 10^{-7}$.
- **Mixture of Two Strong Acids:** $[H^+]_{mix} = (N_1V_1 + N_2V_2) / (V_1 + V_2)$
- **Mixture of Strong Acid & Strong Base:** If $N_aV_a > N_bV_b$: $[H^+] = (N_aV_a - N_bV_b) / (V_a + V_b)$

Polyprotic Weak Acid (e.g., H_2S, H_3PO_4):

Stepwise constants: $K_{a1} \gg K_{a2} \gg K_{a3}$

• Main hydrogen ion conc: $\boldsymbol{[H^+] \approx \sqrt{(K_{a1} C)}}$

• Conc of divalent anion $[A^{2-}] \approx K_{a2}$ (Independent of C!)

• Conc of trivalent anion $[A^{3-}] \approx (K_{a2} K_{a3}) / [H^+]$

9. SALT HYDROLYSIS MASTER TABLE

Salt Type & Examples	Hydrolysis Type	Hydrolysis Constant (K_h)	Degree of Hydrolysis (h)	pH Formula
Strong Acid + Strong Base (NaCl, KNO_3, Na_2SO_4)	No Hydrolysis (Neutral solution)	—	$h = 0$	**pH = 7.0** *(at 25°C)*
Weak Acid + Strong Base (CH_3COONa, KCN, Na_2CO_3)	Anionic Hydrolysis (Basic solution, pH > 7)	$K_h = K_w / K_a$	$h = \sqrt{(K_w / (K_a C))}$	$\boldsymbol{pH = ½ [pK_w + pK_a + \log C]}$
Strong Acid + Weak Base (NH_4Cl, $FeCl_3$, $CuSO_4$)	Cationic Hydrolysis (Acidic solution, pH < 7)	$K_h = K_w / K_b$	$h = \sqrt{(K_w / (K_b C))}$	$\boldsymbol{pH = ½ [pK_w - pK_b - \log C]}$

Salt Type & Examples	Hydrolysis Type	Hydrolysis Constant (K_h)	Degree of Hydrolysis (h)	pH Formula
Weak Acid + Weak Base (CH_3COONH_4, $(NH_4)_2CO_3$)	Both Cationic & Anionic (Independent of C)	$K_h = K_w / (K_a K_b)$	$h = \sqrt{(K_w / (K_a K_b))}$	$\mathbf{pH = ½ [pK_w + pK_a - pK_b]}$

Amphiprotic Salt Species (e.g., $NaHCO_3$, $NaHSO_3$, $NaHPO_4^-$): $\mathbf{pH = ½ [pK_{a1} + pK_{a2}]}$ *(Independent of Concentration C)*

10. BUFFER SOLUTIONS & HENDERSON-HASSELBALCH EQUATIONS

Acidic Buffer (Weak Acid + Salt with SB):
e.g., $CH_3COOH + CH_3COONa$
$\mathbf{pH = pK_a + \log_{10}([Salt] / [Acid]) = pK_a + \log_{10}([Conjugate Base] / [Acid])}$

Basic Buffer (Weak Base + Salt with SA):
e.g., $NH_4OH + NH_4Cl$
$\mathbf{pOH = pK_b + \log_{10}([Salt] / [Base]) = pK_b + \log_{10}([Conjugate Acid] / [Base])}$
$pH = pK_w - pOH$

Buffer Range & Maximum Capacity:
- Effective Acidic Buffer Range: $pH = pK_a \pm 1$
- Effective Basic Buffer Range: $pOH = pK_b \pm 1$
- **Maximum Buffer Capacity Condition:** $[Salt] = [Acid] \Rightarrow pH = pK_a$

Buffer Capacity (β / Φ):
$\mathbf{\beta = (Moles\ of\ Acid/Base\ added\ per\ Litre) / (Change\ in\ pH) = d\ B / d(pH)}$
$\beta = 2.303 \times [(C_{acid} \times C_{salt}) / (C_{acid} + C_{salt})]$

11. SOLUBILITY PRODUCT (K_{SP}) & PRECIPITATION EQUILIBRIUM

General Electrolyte Dissociation:

$$A_xB_y(s) \rightleftharpoons x\ A^{y+}(aq) + y\ B^{x-}(aq)$$

Solubility Product: $\mathbf{K_{sp} = [A^{y+}]^x\ [B^{x-}]^y = (x\ S)^x\ (y\ S)^y = x^x \cdot y^y \cdot S^{(x+y)}}$
$\mathbf{S = [\ K_{sp} / (x^x \cdot y^y)\ ]^{1/(x+y)}}$

Ionic Product (Q_{sp}) Criteria:

- $\mathbf{Q_{sp} < K_{sp}}$: Unsaturated solution (No precipitate, more salt can dissolve).
- $\mathbf{Q_{sp} = K_{sp}}$: Saturated solution (Dynamic equilibrium).
- $\mathbf{Q_{sp} > K_{sp}}$: Supersaturated solution (**Precipitation occurs**).

Salt Type	Example Salts	Dissociation Ions	K_{sp} in terms of Molar Solubility (S)	Solubility Formula (S)
AB (1:1)	AgCl, $BaSO_4$, $CaCO_3$	$A^+ + B^-$	$K_{sp} = S^2$	$S = \sqrt{(K_{sp})}$
AB_2 or A_2B (1:2 / 2:1)	PbI_2, Ag_2CrO_4, $CaCl_2$	$A^{2+} + 2\ B^-$	$K_{sp} = 4\ S^3$	$S = (K_{sp} / 4)^{1/3}$
AB_3 or A_3B (1:3)	$Al(OH)_3$, $FeCl_3$	$A^{3+} + 3\ B^-$	$K_{sp} = 27\ S^4$	$S = (K_{sp} / 27)^{1/4}$
A_3B_2 (3:2)	$Ca_3(PO_4)_2$, $Ba_3(PO_4)_2$	$3\ A^{2+} + 2\ B^{3-}$	$K_{sp} = 108\ S^5$	$S = (K_{sp} / 108)^{1/5}$

12. COMMON ION & SIMULTANEOUS EQUILIBRIUM

Common Ion Effect: Adding an ion common to a weak electrolyte/sparingly soluble salt **suppresses dissociation / decreases solubility S'**.

AgCl in presence of C M NaCl:
$[Cl^-] \approx C \Rightarrow \mathbf{S' = K_{sp} / C}$ *($S' \ll S$)*

Simultaneous Solubility [JEE Advanced]**:**
For AgCl (K_{sp1}) and AgBr (K_{sp2}) in same solution:
$\mathbf{[Ag^+] = \sqrt{(K_{sp1} + K_{sp2})}}$

13. TITRATION CURVES & INDICATOR SELECTION

Titration Type	pH Range at Eq. Point	Suitable Indicator
Strong Acid vs Strong Base	3.5 – 10.5	Phenolphthalein, Methyl Orange
Weak Acid vs Strong Base	7.5 – 10.5	Phenolphthalein (HPh)
Strong Acid vs Weak Base	3.0 – 6.5	Methyl Orange (MeOH), Methyl Red
Weak Acid vs Weak Base	6.5 – 7.5	Phenol Red (No sharp endpoint)

Indicator Working Equation: $HIn \rightleftharpoons H^+ + In^-$

$\mathbf{pH = pK_{In} + \log_{10}([In^-]/[HIn])}$ | Useful range: $\mathbf{pK_{In} \pm 1}$

14. JEE ADVANCED CORNER CASES & EXAM SHORTCUTS [Added]

1. Complex Formation & Solubility:

$AgCl(s) + 2\ NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Cl^-$

$\mathbf{K_{net} = K_{sp} \times K_f}$

K_f = Stability constant of complex.

2. Hydrolysis of Polyvalent Cations:

For Fe^{3+}, Al^{3+} salts:

$M^{3+} + H_2O \rightleftharpoons M(OH)^{2+} + H^+$

$\mathbf{[H^+] = \sqrt{(K_h\ C)}}$

3. Mixture of Weak Acids:

For HA_1 (K_{a1}, C_1) and HA_2 (K_{a2}, C_2):

$\mathbf{[H^+] = \sqrt{(K_{a1}\ C_1 + K_{a2}\ C_2)}}$

1. ATMOSPHERIC STRUCTURE & REGIONS BASICS

Region	Altitude (km)	Temp. Range	Major Species / Features
Troposphere	0 – 10 km	15°C to −56°C	N_2, O_2, H_2O vapour, CO_2. Turbulence, clouds.
Stratosphere	10 – 50 km	−56°C to −2°C	O_3, N_2, O_2. Contains protective Ozone layer.
Mesosphere [Added]	50 – 85 km	−2°C to −92°C	N_2^+, O_2^+, NO^+. Temp decreases with altitude.
Thermosphere [Added]	85 – 500 km	−92°C to 1200°C	O_2^+, O^+, NO^+, e^- (Ionosphere). Gas ionization.

2. GASEOUS AIR POLLUTANTS & REACTIONS

- **Oxides of Sulphur (SO_x):** Major source: Combustion of S-containing fossil fuels.

$$2SO_2\ (g) + O_2\ (g) \rightarrow 2SO_3\ (g)\ \text{(Catalysed by particulate matter)}$$

$$SO_2\ (g) + O_3\ (g) \rightarrow SO_3\ (g) + O_2\ (g)$$

$$SO_2\ (g) + H_2O_2\ (l) \rightarrow H_2SO_4\ (aq)$$

  Effects: Severe respiratory irritation (asthma, bronchitis, emphysema), eye irritation, acid rain.
- **Oxides of Nitrogen (NO_x):** At high temp (>1210°C / 1483 K or lightning):

$$N_2\ (g) + O_2\ (g) \triangleq 2NO\ (g)$$

$$2NO\ (g) + O_2\ (g) \rightarrow 2NO_2\ (g)\ |\ NO\ (g) + O_3\ (g) \rightarrow NO_2\ (g) + O_2\ (g)$$

  Effects: Damage to lung tissue, retards photosynthesis in plants, causes leaf damage.
- **Carbon Monoxide (CO):** Incomplete combustion of carbon fuels.
  • Binds to hemoglobin to form **Carboxyhemoglobin**.
  • Stability ratio: **Carboxyhemoglobin is ~300 times more stable than Oxyhemoglobin**.
  • At **3–4%** carboxyhemoglobin, oxygen-carrying capacity of blood is severely reduced → hypoxia, dizziness, death.
- **Carbon Dioxide (CO_2):** Respiration, fossil fuels, limestone decomposition ($\mathbf{CaCO_3 \rightarrow CaO + CO_2}$). Mild narcotic; causes asphyxiation at high levels.
- **Hydrocarbons:** Incomplete combustion in automobiles. Polynuclear aromatic hydrocarbons (PAH) are **carcinogenic**.

5. SMOG: CLASSICAL VS PHOTOCHEMICAL

Property	Classical / London Smog	Photochemical / LA Smog
Climate	Cool, humid climate	Warm, dry, sunny climate
Components	Smoke + Fog + SO_2	O_3, NO_x, Acrolein, Formaldehyde, PAN
Nature	**Reducing Smog**	**Oxidising Smog**
Primary Source	Fossil fuel burning (coal, S-containing fuel)	Automobile exhaust (hydrocarbons + NO)

Photochemical Smog Mechanism:

$$NO_2\ (g) \overset{h\nu}{\rightsquigarrow} NO\ (g) + O\ (g)$$

$$O\ (g) + O_2\ (g) \triangleq O_3\ (g)$$

$$NO\ (g) + O_3\ (g) \rightarrow NO_2\ (g) + O_2\ (g)$$

Unburnt hydrocarbons react with O_3 and NO_2 to produce toxic compounds:

- **Formaldehyde:** $\mathbf{HCHO}$
- **Acrolein:** $\mathbf{CH_2{=}CHCHO}$
- **Peroxyacetyl Nitrate (PAN):** $\mathbf{CH_3CO\text{-}O\text{-}O\text{-}NO_2}$

Toxicity Note: **PAN** has the highest toxicity to plants, causing **'bronzing'** and **'glazing'** of younger leaves.

6. OZONE DEPLETION MECHANISM

Dynamic Equilibrium in Stratosphere:

$$O_2\ (g) \overset{UV}{\rightsquigarrow} O\ (g) + O\ (g)$$

$$O_2\ (g) + O\ (g) \triangleq O_3\ (g)\ \text{[Thermodynamically unstable]}$$

Depletion by Chlorofluorocarbons (CFCs / Freons):
• Photodissociation of Freon-12 ($\mathbf{CF_2Cl_2}$):

$$CF_2Cl_2\ (g) \overset{UV}{\rightsquigarrow} \bullet Cl\ (g) + \bullet CF_2Cl\ (g)$$

$$\bullet Cl\ (g) + O_3\ (g) \rightarrow ClO\bullet\ (g) + O_2\ (g)$$

$$ClO\bullet\ (g) + O\ (g) \rightarrow \bullet Cl\ (g) + O_2\ (g)$$

Chain propagation: Single •Cl radical can destroy **~100,000 O_3 molecules**.

Antarctic Ozone Hole Mechanics:
• **Summer (Chlorine Sinks):**

$$ClO\bullet\ (g) + NO_2\ (g) \rightarrow ClONO_2\ (g)$$

$$\bullet Cl\ (g) + CH_4\ (g) \rightarrow \bullet CH_3\ (g) + HCl\ (g)$$

• **Winter (Polar Stratospheric Clouds - PSCs):** PSCs contain $\mathbf{HNO_3{\bullet}3H_2O}$ (type I) or ice (type II).

$$ClONO_2\ (g) + H_2O\ (g) \rightarrow HOCl\ (g) + HNO_3\ (g)$$

$$ClONO_2\ (g) + HCl\ (g) \rightarrow Cl_2\ (g) + HNO_3\ (g)$$

• **Spring (Sunlight returns - rapid ozone depletion):**

$$HOCl\ (g) \overset{h\nu}{\rightsquigarrow} \bullet OH\ (g) + \bullet Cl\ (g)$$

$$Cl_2\ (g) \overset{h\nu}{\rightsquigarrow} 2\bullet Cl\ (g)$$

3. GREENHOUSE EFFECT & ACID RAIN

Greenhouse Gases (GHGs): **CO_2 (50-60%), CH_4 (20%), CFCs (14%), O_3, N_2O (6%), H_2O vapour**.

• Traps outgoing infrared radiation (heat) radiated by the Earth's surface.

• *Consequences:* Polar ice caps melt, global temperature rise, spread of infectious diseases (malaria, dengue, yellow fever).

Acid Rain Chemistry:

• Normal rain water pH = **5.6** (due to dissolved atmospheric CO_2):

$$H_2O\ (l) + CO_2\ (g) \triangleq H_2CO_3\ (aq) \triangleq H^+\ (aq) + HCO_3^-\ (aq)$$

• **Acid Rain** definition: **pH < 5.6** (due to H_2SO_4 and HNO_3).

$$2SO_2\ (g) + O_2\ (g) + 2H_2O\ (l) \rightarrow 2H_2SO_4\ (aq)$$

$$4NO_2\ (g) + O_2\ (g) + 2H_2O\ (l) \rightarrow 4HNO_3\ (aq)$$

• **Taj Mahal Degradation (Marble Cancer):**

$$CaCO_3\ (s) + H_2SO_4\ (aq) \rightarrow CaSO_4\ (aq) + H_2O\ (l) + CO_2\ (g)\uparrow$$

Result: Marble loses lustre, turns yellow and pitted.

4. PARTICULATE POLLUTANTS

Type	Description / Examples
Viable Particulates	Minute living organisms dispersed in air: Bacteria, fungi, moulds, algae. Fungi cause plant diseases.
Non-Viable Particulates	Formed by disintegration of solid matter or vapor condensation:
• *Smoke*	Solid/liquid mixture from combustion of organic matter (e.g. oil smoke, cigarette smoke, combustion).
• *Dust*	Fine solid particles **> 1 µm** diameter from crushing/grinding (e.g. cement, fly ash, sand dust).
• *Mists*	Liquid spray droplets & vapor condensation (e.g. H_2SO_4 mist, herbicide/insecticide spray).
• *Fumes*	Condensation of vapors during sublimation, distillation, boiling, metallurgical processes (metals/oxides).

• **Lead (Pb) Particulates:** Major air pollutant released by leaded petrol exhausts. Impairs RBC development, damages brain/nerves.

• **Pneumoconiosis:** Lung tissue scarring/fibrosis caused by prolonged inhalation of small mineral dust particles (silica, asbestos, coal).

7. WATER QUALITY, BOD & COD FORMULAS

Dissolved Oxygen (DO): Cold water DO limit = **10 ppm**. If DO < **6 ppm**, fish growth is inhibited.

Biochemical Oxygen Demand (BOD): Amount of oxygen required by bacteria to decompose organic matter present in a given volume of water sample over 5 days at 20°C.

$$BOD\ (in\ ppm) = (DO_{initial} - DO_{final})\ in\ mg/L$$

Water Sample Type	BOD Value (ppm)
Clean Water	**< 5 ppm**
Highly Polluted Water	**≥ 17 ppm**

Chemical Oxygen Demand (COD): Measure of total oxidisable matter (both biodegradable and non-biodegradable) using acidic $K_2Cr_2O_7$.

$$COD > BOD\ (Always\ for\ a\ given\ polluted\ water\ sample)$$ [Added]

Eutrophication: Nutrient enrichment (phosphates, nitrates) leading to dense algal bloom, oxygen depletion, and loss of aquatic biodiversity.

8. DRINKING WATER QUALITY STANDARDS (MUST MEMORISE)

Permissible Limits & Health Effects:

Species	Max. Permissible Limit	Harmful Effects / Symptoms of Excess
Fluoride (F^-)	**1.0 ppm** (1 mg/dm^3)	• **> 2 ppm:** Brown mottling of teeth. • **> 10 ppm:** Severe damage to bones & teeth.
Lead (Pb)	**50 ppb** (0.05 ppm)	Damage to kidney, liver, brain, reproductive system.
Nitrate (NO_3^-)	**50 ppm**	**Methemoglobinemia** ('Blue Baby' Syndrome).
Sulphate (SO_4^{2-})	**500 ppm**	Pulls water into intestines, causing severe **laxative effect**.

Maximum Concentration of Metals in Drinking Water:

Metal Symbol	Max. Limit (ppm or mg/dm^3)	Metal Symbol	Max. Limit (ppm or mg/dm^3)
Fe (Iron)	0.2 ppm	**Cu** (Copper)	3.0 ppm
Mn (Manganese)	0.2 ppm	**Zn** (Zinc)	5.0 ppm
Al (Aluminium)	0.2 ppm	**Cd** (Cadmium)	0.005 ppm

9. PESTICIDES, HERBICIDES & FUNGICIDES

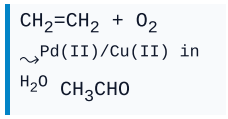
- **Pesticides:** Synthetic chemicals used to control or destroy unwanted organisms.
- **Insecticides:**
  • *Organochlorines:* **DDT** (Dichlorodiphenyltrichloroethane), **BHC** (Benzene Hexachloride / Gammaxane / 666), **Aldrin**, **Dieldrin**. Persistent, lipophilic, non-biodegradable; accumulate in food chain and root crops (potatoes, carrots).
  • *Organophosphates / Carbamates:* Malathion, Parathion (more toxic to nerve channels, but less persistent) [Added].
- **Herbicides:** Chemical weedkillers.
  • *Inorganic:* $NaClO_3$ (Sodium Chlorate), Na_3AsO_3 (Sodium Arsenite).
  • *Organic:* Triazines, 2,4-D (2,4-Dichlorophenoxyacetic acid) [Added].
- **Fungicides:** Organomercury compounds. Leave toxic mercury residues in soil.

10. WASTE MANAGEMENT & RECYCLING [ADDED]

• **Biodegradable Waste:** Plant origin, domestic sewage, food waste → converted to compost / biogas.
• **Non-biodegradable Waste:** Plastics, metals, glass, toxic chemicals → incineration or recycling.

11. GREEN CHEMISTRY PRINCIPLES & REACTIONS

Green Chemistry focuses on reducing/eliminating hazardous substances and waste in chemical processes.

Process	Conventional (Hazardous) Reagent	Green / Modern Eco-friendly Alternative
Dry Cleaning of Clothes	Tetrachloroethene ($Cl_2C{=}CCl_2$) - Suspected Carcinogen & groundwater pollutant.	Liquefied CO_2 with suitable detergent OR H_2O_2 for bleaching action.
Bleaching of Paper	Chlorine gas (Cl_2) - Forms toxic chlorinated pollutants.	Hydrogen Peroxide (H_2O_2) in presence of suitable catalyst.
Synthesis of Ethanal (Acetaldehyde)	Two-step oxidation involving toxic heavy metal salts.	One-step catalytic oxidation of ethene using $Pd(II)$ / $Cu(II)$ in water (Wacker process yield ~90%): 

Atom Economy Formula: [Added]

```
% Atom Economy = [(Mol. Mass of Desired Product) / (Total Mol. Mass of All Reactants)] × 100
```

12. HIGH-YIELD JEE NUMERICAL VALUES & SHORTCUTS

Quantity / Parameter	JEE Target Threshold Value
Rain Water Acidic pH Cutoff	pH < 5.6
Clean Water BOD Limit	< 5 ppm
Polluted Water BOD Limit	≥ 17 ppm
F^- Mottling Teeth Threshold	> 2.0 ppm
F^- Bone Damage Threshold	> 10.0 ppm
Lead (Pb) Max Limit in Water	50 ppb (0.05 ppm)
Nitrate Max Limit in Water	50 ppm
Sulphate Max Laxative Limit	500 ppm
Cadmium (Cd) Max Limit	0.005 ppm
Carboxyhemoglobin Stability	300× vs Oxyhemoglobin

13. COMMON JEE CONCEPT TRAPS [ADDED]

- **Trap 1:** *Is CO_2 a primary air pollutant?* No, CO_2 is a natural component of atmosphere; excess CO_2 causes greenhouse effect, not direct atmospheric toxicity like CO or SO_2.
- **Trap 2:** *London Smog vs LA Smog nature:* London = **Reducing** (due to SO_2); LA = **Oxidising** (due to O_3, NO_2).
- **Trap 3:** *Chlorine Sinks in Stratosphere:* $ClONO_2$ and HCl do NOT deplete ozone directly; they are chlorine sinks that release active •Cl in spring.

ELECTROCHEMISTRY

JEE Main & Advanced Complete Formula & Revision Sheet

Target: JEE 2025/2026
Physics & Chemistry Core

1. Conductance & Resistance Parameters in Electrolytes

Resistance & Resistivity:

$$R = \rho (l / A) = (1 / \kappa) \cdot (l / A)$$

R in Ω, ρ in Ω·cm (c.g.s) or Ω·m (SI). Depend on temperature & electrolyte nature.

Conductance & Conductivity (Specific Conductance):

$$G = 1 / R, \quad \kappa = 1 / \rho = G \cdot (l / A) = G \cdot G^*$$

G in Siemens (S = Ω^{-1} = mho). κ in S cm^{-1} or S m^{-1}. (1 S cm^{-1} = 100 S m^{-1})

Cell Constant (G*):

$$G^* = l / A = \kappa \cdot R = \kappa / G$$

Units: cm^{-1} or m^{-1}. Determined using standard KCl solution.

Molar Conductivity (Λ_m):

$$\Lambda_m = (1000 \times \kappa) / M$$

κ in S cm^{-1}, M = Molarity (mol L^{-1}). Units: S cm^2 mol^{-1}. [SI: $\Lambda_m = \kappa / (1000 \times M)$ in S m^2 mol^{-1}]

Equivalent Conductivity (Λ_{eq}):

$$\Lambda_{eq} = (1000 \times \kappa) / N \quad | \quad \Lambda_{eq} = \Lambda_m / \text{n-factor}$$

N = Normality (eq L^{-1}). Units: S cm^2 eq^{-1}. n-factor = total cation/anion charge.

Concentration Variation (Debye-Hückel-Onsager):

$$\Lambda_m = \Lambda_m^0 - B \sqrt{C} \quad \textit{(Strong Electrolyte)}$$

Λ_m^0 = Limiting Molar Conductivity at infinite dilution (C → 0). B = Constant (depends on solvent & temp).

2. Kohlrausch's Law of Independent Migration & Ionic Mobility

Kohlrausch's Law Statement: At infinite dilution, total conductivity of an electrolyte is the sum of individual contributions of its constituent ions.

$$\text{For } A_xB_y: \Lambda_m^0 = x \lambda_{m, A^{y+}}^0 + y \lambda_{m, B^{x-}}^0$$

$$\text{For } A_xB_y: \Lambda_{eq}^0 = \lambda_{eq, A^{y+}}^0 + \lambda_{eq, B^{x-}}^0$$

1. Degree of Dissociation (α):

$\alpha = \Lambda_m^C / \Lambda_m^0 = \Lambda_{eq}^C / \Lambda_{eq}^0$

(For weak electrolytes at conc. C)

2. Dissociation Constant (K_a / K_b):

$K = (C \alpha^2) / (1 - \alpha) = [C (\Lambda_m / \Lambda_m^0)^2] / [1 - (\Lambda_m / \Lambda_m^0)]$

If $\alpha << 0.1 \Rightarrow K \approx C \alpha^2$

3. Solubility of Sparingly Soluble Salt:

$S = (1000 \times \kappa_{salt}) / \Lambda_m^0$ (mol L^{-1})

$K_{sp} = x^x y^y S^{(x+y)}$ *($\kappa_{salt} = \kappa_{solution} - \kappa_{water}$)*

Ionic Mobility (μ): Velocity of ion under unit potential gradient.

$$\mu = \text{Ionic Velocity / Potential Gradient} = v / (E / l) \quad (cm^2 V^{-1} s^{-1})$$

$$\lambda_m^0 = z \cdot F \cdot \mu \quad | \quad \lambda_{eq}^0 = F \cdot \mu$$

Transport / Transference Number (t):

$$t_c = l_c / l_{total} = \mu_c / (\mu_c + \mu_a) = \lambda_c^0 / \Lambda_m^0 \quad | \quad t_c + t_a = 1$$

Current fraction carried by specific ion. Conc. ↑ ⇒ t ↓.

3. Galvanic / Voltaic Cells & Nernst Equation

Cell Notation & Standard EMF:

Convention: **Anode (Oxidation) || Cathode (Reduction)**
[LOAN: Left Oxidation Anode Negative]

$$E^0_{cell} = E^0_{cathode, RP} - E^0_{anode, RP} = E^0_{OP, anode} + E^0_{RP, cathode}$$

Nernst Equation (Single Electrode Potential):

For $M^{n+} + n\ e^- \rightarrow M(s)$:

$$E_{RP} = E^0_{RP} - (RT / nF) \ln [1 / [M^{n+}]] = E^0_{RP} + (0.0591 / n) \log_{10} [M^{n+}] \quad \textit{(at 298 K)}$$

Nernst Equation (Full Cell Reaction):

For $a\ A + b\ B \rightleftharpoons c\ C + d\ D$ (Reaction Quotient $Q = [C]^c [D]^d / ([A]^a [B]^b)$):

$$E_{cell} = E^0_{cell} - (2.303 RT / nF) \log_{10} Q = E^0_{cell} - (0.0591 / n) \log_{10} Q \quad \textit{(at 298 K)}$$

Standard Hydrogen Electrode (SHE) & pH Relation:

Half Cell: $2\ H^+(aq) + 2\ e^- \rightleftharpoons H_2(g, P\ bar)$ *($E^0 = 0.00$ V)*

$$E_{H^+/H_2} = -0.0591\ pH - (0.0591 / 2) \log_{10} P_{H_2} \quad \Rightarrow \quad \text{If } P_{H_2} = 1 \text{ atm}, E_{RP} = -0.0591\ pH$$

4. Thermodynamics of Galvanic Cells

Gibbs Free Energy Change:

$\Delta G = -n F E_{cell}$

$\Delta G^0 = -n F E^0_{cell}$

Spontaneous: $E_{cell} > 0$, $\Delta G < 0$

Non-spontaneous: $E_{cell} < 0$, $\Delta G > 0$

Equilibrium Constant (K_{eq}):

At eq, $E_{cell} = 0 \Rightarrow \Delta G = 0$

$\log_{10} K_{eq} = (n E^0_{cell}) / 0.0591$

$\Delta G^0 = -2.303 RT \log_{10} K_{eq}$

Entropy & Enthalpy of Cell:

$\Delta S = n F (\partial E / \partial T)_P$

$\Delta H = -n F E + n F T (\partial E / \partial T)_P$

$(\partial E / \partial T)_P$ = Temp. Coefficient ($V K^{-1}$)

If $(\partial E / \partial T)_P > 0$, rxn is endothermic.

5. Concentration Cells ($E^0_{cell} = 0$)

1. Electrolyte Concentration Cells:

Electrodes identical, immersed in solutions of different concentrations.

$M | M^{n+}(C_1) || M^{n+}(C_2) | M$

$E_{cell} = (0.0591 / n) \log_{10} (C_2 / C_1)$ *(at 298 K)*

Spontaneous if $C_2 > C_1$ (C_2 = cathode conc., C_1 = anode conc.).

2. Electrode Concentration Cells:

Electrodes at different gas pressures/amalgam concentrations in same electrolyte.

$Pt, H_2(P_1) | H^+(C) || H^+(C) | H_2(P_2), Pt$

$E_{cell} = (0.0591 / 2) \log_{10} (P_1 / P_2)$ *(at 298 K)*

Spontaneous if $P_1 > P_2$.

6. Electrolysis & Quantitative Laws (Faraday's Laws)

Faraday's 1st Law of Electrolysis:

Mass of substance deposited/liberated (w) $\propto$ Charge passed (Q).

$w = Z \cdot Q = Z \cdot I \cdot t = (E / 96500) \cdot I \cdot t$

Z = Electrochemical Equivalent = E / F = M / (n · F).

1 Faraday (F) = 96500 C mol^{-1} (charge of 1 mole of e^-).

Faraday's 2nd Law of Electrolysis:

For same charge passed through different electrolytes in series:

$w_1 / w_2 = E_1 / E_2 \Rightarrow w_1 / Z_1 = w_2 / Z_2$

Number of equivalents deposited of all species is equal.

Equivalents & Volume Shortcuts for Gases (at STP):

Product / Ion	n-factor	Mass per Faraday (1 F)	Gas Vol. at STP per 1 F
$Na^+ \rightarrow Na$	1	23 g (1 mole)	–
$Mg^{2+} \rightarrow Mg$	2	12 g (0.5 mole)	–
$Al^{3+} \rightarrow Al$	3	9 g (1/3 mole)	–
$2 H^+ + 2e^- \rightarrow H_2(g)$	2	1 g (0.5 mole)	11.2 L
$2 OH^- \rightarrow H_2O + ½ O_2 + 2e^-$	4	8 g (0.25 mole)	5.6 L
$2 Cl^- \rightarrow Cl_2 + 2e^-$	2	35.5 g (0.5 mole)	11.2 L

Products of Electrolysis (Preferential Discharge Theory):

Cathodic Discharge Order (Decreasing Discharge Potential / Increasing SRP):

$Li^+ < K^+ < Ca^{2+} < Na^+ < Mg^{2+} < Al^{3+} < Zn^{2+} < Fe^{2+} < H^+ < Cu^{2+} < Hg^{2+} < Ag^{2+} < Au^{3+}$

Anodic Discharge Order (Decreasing Discharge Potential / Increasing SOP):

$SO_4^{2-} < NO_3^- < OH^- < Cl^- < Br^- < I^-$

7. Commercial Cells & Batteries

Cell Type	Anode (–)	Cathode (+)	Electrolyte	Anodic & Cathodic Reactions / Remarks
Dry Cell (Leclanché) *Primary (1.5 V)*	Zinc container	Graphite rod	Moist paste of NH_4Cl + $ZnCl_2$	Anode: $Zn(s) \rightarrow Zn^{2+} + 2e^-$ Cathode: $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$
Mercury Cell *Primary (1.35 V)*	Zn-Hg amalgam	Paste of HgO and carbon	Moist KOH + ZnO	Anode: $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$ Cathode: $HgO(s) + H_2O + 2e^- \rightarrow Hg(l) + 2OH^-$ (Overall potential constant)

Cell Type	Anode (−)	Cathode (+)	Electrolyte	Anodic & Cathodic Reactions / Remarks
Lead Storage Battery *Secondary (Rechargeable)*	Spongy Lead (Pb)	Grid of Pb packed with PbO_2	38% H_2SO_4 soln (d = 1.28 g cm^{-3})	**Discharging:** Anode: $Pb + SO_4^{2-} \rightarrow PbSO_4(s) + 2e^-$ Cathode: $PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \rightarrow PbSO_4(s) + 2H_2O$ Overall: $Pb + PbO_2 + 2H_2SO_4 \rightleftharpoons 2PbSO_4 + 2H_2O$
Ni-Cd Battery *Secondary (1.4 V)*	Cadmium (Cd)	$NiO(OH)$ / NiO_2	Moist KOH	$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O$
H_2-O_2 Fuel Cell *Continuous Supply*	Porous Carbon with Pt/Pd	Porous Carbon with Pt/Pd	Aqueous KOH / NaOH	Anode: $2H_2(g) + 4OH^- \rightarrow 4H_2O + 4e^-$ Cathode: $O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-$ (Efficiency ≈ 70%)

8. Corrosion (Electrochemical Theory of Rusting)

Mechanism of Rusting:

- Anodic Spot: $2\,Fe(s) \rightarrow 2\,Fe^{2+} + 4e^-$ ($E^0_{Fe^{2+}/Fe} = -0.44$ V)
- Cathodic Spot: $O_2(g) + 4H^+ + 4e^- \rightarrow 2\,H_2O$ ($E^0 = +1.23$ V)
- Overall Rxn: $2Fe + O_2 + 4H^+ \rightarrow 2Fe^{2+} + 2H_2O$ ($E^0_{cell} = 1.67$ V)
- Atmospheric Oxidation: $4Fe^{2+} + O_2 + 4H_2O \rightarrow 2Fe_2O_3 + 8H^+$
- Rust Composition: $Fe_2O_3 \cdot x\,H_2O$ (Hydrated Ferric Oxide)

Prevention Methods:

1. **Barrier Protection:** Paints, grease, electroplating (Ni, Cr).
2. **Sacrificial Protection:** Galvanization (coating Fe with more electropositive Zn).
3. **Cathodic/Electrical Protection:** Connecting underground Fe pipes to Mg or Zn blocks via wire.
4. **Anti-rust solutions:** Alkaline phosphate/chromate solutions.

9. JEE Advanced Level Useful Relations & Multi-Concept Formulas [Added]

1. Combining Half-Cell Potentials (Non-Additive Potentials):

Potentials (E^0) are intensive and non-additive; Free energies (ΔG^0) are extensive and additive.

$$\mathbf{n_3\,E^0_3 = n_1\,E^0_1 + n_2\,E^0_2 \quad \Rightarrow \quad E^0_3 = (n_1\,E^0_1 + n_2\,E^0_2)\,/\,n_3}$$

Example: $Fe^{3+} + 3e^- \rightarrow Fe$ (n=3). If $Fe^{3+} + e^- \rightarrow Fe^{2+}$ (E^0_1) and $Fe^{2+} + 2e^- \rightarrow Fe$ (E^0_2), then $E^0_3 = (E^0_1 + 2E^0_2) / 3$.

2. Relation Between K_{sp} and EMF of Concentration Cell:

For cell: Ag | Ag^+ (satd. AgX solution) || Ag^+ (C M) | Ag

$$\mathbf{E_{cell} = 0.0591\,\log_{10}\,[\,C\,/\,[Ag^+]_{satd}\,] \quad \text{where } [Ag^+]_{satd} = K_{sp}\,/\,[X^-]}$$

Allows direct determination of K_{sp} from measured E_{cell}.

4. Conductometric Titration Curves Summary:

Titration Pair	Conductance Trend before & after Eq. Point
Strong Acid vs Strong Base (e.g. HCl vs NaOH)	Sharp decrease (H^+ replaced by Na^+), then sharp increase after endpoint (excess OH^-). V-shape.
Weak Acid vs Strong Base (e.g. CH_3COOH vs NaOH)	Initial slight drop, then slow rise (formation of CH_3COONa salt), sharp rise after endpoint (excess OH^-).
Strong Acid vs Weak Base (e.g. HCl vs NH_4OH)	Sharp decrease, then almost flat/horizontal after endpoint (un-ionized NH_4OH).
Weak Acid vs Weak Base (e.g. CH_3COOH vs NH_4OH)	Slow rise during neutralization, flat/constant after endpoint.

3. Potentiometric pH Determination Formulas:

Cell Setup	Formula for pH at 298 K
Pt, H_2(1 atm) | H^+(unknown) || SHE	pH = E_{cell} / 0.0591
Pt, H_2(1 atm) | H^+(unknown) || Calomel ($E_{calomel}$ = 0.2415V)	pH = (E_{cell} − 0.2415) / 0.0591
Quinhydrone Electrode || SHE	pH = (E^0_{Q/H_2Q} − E_{cell}) / 0.0591 *(E^0 = 0.6996 V)*

5. Liquid Junction Potential (LJP) [Added]:

Potential generated across interface of two electrolyte solutions due to difference in ionic speeds.

$$E_{LJP} = (t_a - t_c) \cdot (RT / F) \ln (a_2 / a_1) = (2\,t_a - 1) \cdot (0.0591 / n) \log_{10} (C_2 / C_1)$$

Salt bridge (KCl in Agar-Agar) eliminates LJP because $t_{K^+} \approx t_{Cl^-}$.

6. Simultaneous Electrolysis & Current Efficiency:

$$\% \text{ Current Efficiency} = (\text{Actual Mass} / \text{Theoretical Mass}) \times 100$$

$$Q_{total} = I \times t = Q_1 + Q_2 = (w_1 / Z_1) + (w_2 / Z_2)$$

Used when multiple metals deposit simultaneously or side reactions (e.g., H_2 evolution) occur.

10. Standard Electrochemical Series (SRP Values at 298 K) & Fundamental Constants

Reduction Half-Reaction	E^0 (Volts)	Property Trend
$Li^+ + e^- \rightleftharpoons Li(s)$	−3.05 V	**Strongest Reducing Agent**
$K^+ + e^- \rightleftharpoons K(s)$	−2.93 V	High Reactivity / Low SRP
$Ca^{2+} + 2e^- \rightleftharpoons Ca(s)$	−2.87 V	Displaces H_2 from acids
$Na^+ + e^- \rightleftharpoons Na(s)$	−2.71 V	Strong Reducer
$Mg^{2+} + 2e^- \rightleftharpoons Mg(s)$	−2.37 V	Sacrificial Anode
$Al^{3+} + 3e^- \rightleftharpoons Al(s)$	−1.66 V	Easily Oxidized
$Zn^{2+} + 2e^- \rightleftharpoons Zn(s)$	−0.76 V	Anode in Galvanization
$Fe^{2+} + 2e^- \rightleftharpoons Fe(s)$	−0.44 V	Rusting Anode
$2H^+ + 2e^- \rightleftharpoons H_2(g)$	**0.00 V**	**Reference Standard (SHE)**
$Cu^{2+} + 2e^- \rightleftharpoons Cu(s)$	+0.34 V	Cathode in Daniell Cell
$O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-$	+0.40 V	Alkaline O_2 Reduction
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.54 V	Mild Oxidizing Agent
$Ag^+ + e^- \rightleftharpoons Ag(s)$	+0.80 V	Noble Metal
$Br_2 + 2e^- \rightleftharpoons 2Br^-$	+1.09 V	Strong Oxidizer
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33 V	Acidic Dichromate
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36 V	Strong Oxidizer
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	+1.51 V	Acidic Permanganate
$F_2(g) + 2e^- \rightleftharpoons 2F^-$	+2.87 V	**Strongest Oxidizing Agent**

Fundamental Constants for Calculation:

- Faraday Constant (F) = $N_A \cdot e \approx$ **96485 C mol^{-1} (Use 96500 C)**
- Gas Constant (R) = 8.314 J K^{-1} mol^{-1}
- 2.303 RT / F at 298 K = **0.0591 V** (0.06 V for quick estimations)
- Charge of 1 e^- = 1.602×10^{-19} C
- 1 A = 1 C s^{-1} | 1 V = 1 J C^{-1} | 1 W = 1 J s^{-1} = 1 V · A

Quick Revision Checklist & Common Pitfalls:

1. ***Sign Convention:*** *E^0_{cell} is always positive for spontaneous galvanic cells. If $E^0_{cell} < 0$, cell works in reverse or as an electrolytic cell.*
2. ***Stoichiometric Coefficients:*** *Multiplying a half-reaction equation by n does not change its E^0 value (intensive property), but does change ΔG^0 by a factor of n.*
3. ***Dilution Effect:*** *On dilution, Conductivity (κ) decreases because number of ions per unit volume decreases. However, Molar Conductivity (Λ_m) and Equivalent Conductivity (Λ_{eq}) increase due to increase in total volume and degree of dissociation (α).*
4. ***Anode/Cathode Polarity:***
   - *Galvanic Cell: Anode = Negative (−), Cathode = Positive (+)*
   - *Electrolytic Cell: Anode = Positive (+), Cathode = Negative (−)*
   - *Oxidation ALWAYS occurs at Anode; Reduction ALWAYS at Cathode!*

d & f-BLOCK ELEMENTS

JEE MAIN & ADVANCED COMPLETE FORMULA & REACTION REVISION SHEET

Target: JEE 2025/2026
Focus: Core Formulas & Inorganic Reactions

1. FUNDAMENTAL DEFINITIONS & ELECTRONIC CONFIGURATIONS

General Outer Configurations

- **d-Block Elements:** $(n-1)d^{1-10}\ ns^{0-2}$
- **f-Block Elements:** $(n-2)f^{1-14}\ (n-1)d^{0-1}\ ns^{2}$
- **Transition Elements:** Partially filled (n-1)d orbitals in ground state or any common oxidation state.

Key Anomalous Configurations

- **3d Series:** Cr = $3d^5\ 4s^1$, Cu = $3d^{10}\ 4s^1$
- **4d Series:** Pd = $4d^{10}\ 5s^0$ (Zero s-electrons)
- **5d / f-Block:** Gd (Z=64) = [Xe] $4f^7\ 5d^1\ 6s^2$, Lu (Z=71) = [Xe] $4f^{14}\ 5d^1\ 6s^2$

Non-Typical Transition Elements **Zn, Cd, Hg:** Fully filled d-subshell $(n-1)d^{10}\ ns^2$ in elemental and ionic forms (d-electrons do not participate in bonding). Soft metals with low MP.

Sc, Y, La: Form only tripositive ions (d^0, colourless, diamagnetic).

2. PERIODIC TRENDS & PHYSICAL FORMULAS

Atomic & Ionic Radii Trends

- **3d Series Radius Order:**
  Sc > Ti > V > Cr > Mn ≥ Fe ≈ Co ≈ Ni ≤ Cu < Zn
  Initial decrease due to Z_{eff}; constancy in middle due to Z_{eff} ≈ Screening; slight rise at end due to e^--e^- repulsion.
- **Group Trend (Lanthanoid Contraction):**
  Radius: 3d < 4d ≈ 5d (e.g., Ti < Zr ≈ Hf; Nb ≈ Ta)
  Due to poor shielding of 4f electrons (Z_{eff} increases significantly).
- **Extreme Radii:** Smallest in 3d = **Ni**, Largest = **La**.

Density, MP & Ionization Enthalpy

- **Density Group Trend:** 3d < 4d << 5d (due to mass increase & contraction).
  Highest density = **Os** (22.59 g/cm³) & **Ir** (22.56 g/cm³). Lowest in 3d = **Sc** (3.1 g/cm³).
- **Melting Point:** Peaks at Cr, Mo, W (max unpaired d-e^-).
  Exceptions: Mn & Tc have abnormally low MP (stable half-filled d^5 localizes electrons).
  Group 12 (Zn, Cd, Hg): Very low MP (Hg is liquid).
- **IE Exception:** IE_3 order: 5d > 3d > 4d (5d higher due to Lanthanoid contraction).

3. MAGNETIC MOMENT, COLOUR & CATALYTIC PROPERTIES

Magnetic Moment Formula

$$\mu_{eff} = \sqrt{n(n+2)}\ \text{B.M.}$$

Where **n** = number of unpaired electrons. (1 B.M. = eh / 4πmc)

Unpaired e^- (n)	1	2	3	4	5
$\mu_{calculated}$ (BM)	1.73	2.84	3.87	4.90	5.92

*d^0 and d^{10} configurations are diamagnetic (μ = 0).

Origin of Colour

- **d-d Transitions:** Presence of unpaired d-electrons in t_{2g} / e_g sets.
  Colourless: $Sc^{3+}(d^0)$, $Ti^{4+}(d^0)$, $V^{5+}(d^0)$, $Zn^{2+}(d^{10})$, $Cu^+(d^{10})$.
- **Charge Transfer Spectra:** Intense colour despite d^0:
  • CrO_4^{2-} (Yellow), $Cr_2O_7^{2-}$ (Orange)
  • MnO_4^- (Purple/Dark Pink)

Key Industrial Catalysts

- **V_2O_5:** Contact Process ($SO_2 \rightarrow SO_3$)
- **$Fe_2O_3 + Al_2O_3 + K_2O$:** Haber's Process
- **$TiCl_4 + Al(C_2H_5)_3$:** Ziegler-Natta Catalyst
- **$RhCl(PPh_3)_3$:** Wilkinson's Catalyst (Alkene Hydrogenation)
- **$FeSO_4 + H_2O_2$:** Fenton's Reagent

4. OXIDATION STATES & ELECTRODE POTENTIALS

- **Highest OS in 3d Series: +7** (Mn in MnO_4^-, Mn_2O_7).
- **Highest OS in d-Block: +8** (Os in OsO_4, Ru in RuO_4).
- **Zero OS:** Metal carbonyls e.g., $Ni(CO)_4$, $Fe(CO)_5$ (due to synergic bonding).
- **Oxide vs Fluoride Stability:** Higher OS stabilized more by Oxygen than Fluorine due to oxygen's ability to form multiple (π) bonds. (e.g., Mn_2O_7 exists, but highest fluoride is MnF_4).

Standard Reduction Potential E° Trends [Added]

- **$E°(M^{2+}/M)$ for 3d:** Generally negative, but **$E°(Cu^{2+}/Cu) = +0.34$ V** (Positive due to high enthalpy of atomization & ionization not balanced by hydration energy).
- **$E°(M^{3+}/M^{2+})$ Exceptions:**
  - **Cr^{3+}/Cr^{2+} (-0.41 V):** Cr^{2+} is a strong reducing agent ($d^4 \rightarrow d^3\ t_{2g}^3$ extra stability).
  - **Mn^{3+}/Mn^{2+} (+1.57 V):** Mn^{3+} is a strong oxidizing agent ($d^4 \rightarrow d^5$ half-filled stability).

5. IMPORTANT COMPOUNDS OF D-BLOCK ELEMENTS

A. POTASSIUM DICHROMATE ($K_2Cr_2O_7$)

Orange Crystals

Preparation from Chromite Ore ($FeOCr_2O_3$):

1. **$4\ FeCr_2O_4 + 8\ Na_2CO_3 + 7\ O_2 \rightarrow 8\ Na_2CrO_4$ (Yellow) $+ 2\ Fe_2O_3 + 8\ CO_2$**
2. **$2\ Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7$ (Orange) $+ Na_2SO_4 + H_2O$**
3. **$Na_2Cr_2O_7 + 2\ KCl \rightarrow K_2Cr_2O_7$ (Less soluble, crystallizes out) $+ 2\ NaCl$**

pH-Dependent Interconversion Equilibrium:

$$2\ CrO_4^{2-}\ (\text{Yellow}) + 2\ H^+ \rightleftharpoons Cr_2O_7^{2-}\ (\text{Orange}) + H_2O \quad [\text{Acidic pH: } Cr_2O_7^{2-} \mid \text{Basic pH: } CrO_4^{2-}]$$

Oxidizing Reactions in Acidic Medium (n-factor = 6):

Half Reaction: **$Cr_2O_7^{2-} + 14\ H^+ + 6\ e^- \rightarrow 2\ Cr^{3+}$ (Green) $+ 7\ H_2O$** ($E° = +1.33$ V)

- $Fe^{2+} \rightarrow Fe^{3+}$
- $I^- \rightarrow I_2$
- $H_2S \rightarrow S$ (Yellow ppt)
- SO_3^{2-} / $SO_2 \rightarrow SO_4^{2-}$
- $Sn^{2+} \rightarrow Sn^{4+}$
- $C_2H_5OH \rightarrow CH_3COOH$

Special Test Reactions:

- **Chromyl Chloride Test (for Cl^-):**
  $K_2Cr_2O_7 + 4\ NaCl + 6\ H_2SO_4$(conc) $\rightarrow 2\ CrO_2Cl_2\uparrow$ (Red-orange vapor) $+ 2\ KHSO_4 + 4\ NaHSO_4 + 3\ H_2O$
  Vapors + NaOH → Yellow Na_2CrO_4 solution + $Pb(CH_3COO)_2$ → Yellow $PbCrO_4$ ppt.
- **Reaction with H_2O_2 (Ether medium):**
  $Cr_2O_7^{2-} + 2\ H^+ + 4\ H_2O_2 \rightarrow 2\ CrO_5$ (Deep Blue, Butterfly Structure) $+ 5\ H_2O$

B. POTASSIUM PERMANGANATE ($KMnO_4$)

Dark Purple Crystals

Preparation from Pyrolusite Ore (MnO_2):

1. **$2\ MnO_2 + 4\ KOH + O_2 \rightarrow 2\ K_2MnO_4$ (Dark Green) + $2\ H_2O$**
2. Electrolytic oxidation or Disproportionation by $CO_2/Cl_2/O_3$:
   $3\ MnO_4^{2-} + 4\ H^+ \rightarrow 2\ MnO_4^-$ (Purple) + MnO_2 (Brown ppt) + $2\ H_2O$

Oxidizing Behavior across Different Media:

Medium	Half-Reaction Equation	Equivalent Weight / n-factor
Acidic (pH < 7)	**$MnO_4^- + 8\ H^+ + 5\ e^- \rightarrow Mn^{2+}$ (Colorless) + $4\ H_2O$**	**n = 5** (E = M/5)
Neutral / Faintly Alkaline	**$MnO_4^- + 2\ H_2O + 3\ e^- \rightarrow MnO_2$ (Brown ppt) + $4\ OH^-$**	**n = 3** (E = M/3)
Strongly Alkaline (pH > 13)	**$MnO_4^- + e^- \rightarrow MnO_4^{2-}$ (Green)**	**n = 1** (E = M/1)

Key Oxidation Products of Specific Substrates:

- **In Acidic Medium:**
  $Fe^{2+} \rightarrow Fe^{3+}$ | $C_2O_4^{2-} \rightarrow CO_2$ | $S^{2-} \rightarrow S$
  $SO_3^{2-} \rightarrow SO_4^{2-}$ | $NO_2^- \rightarrow NO_3^-$ | $I^- \rightarrow I_2$
- **In Neutral/Faintly Alkaline Medium:**
  $I^- \rightarrow IO_3^-$ (KIO_3 formed; critical distinction from acidic!)
  $S_2O_3^{2-} \rightarrow SO_4^{2-}$ | $Mn^{2+} \rightarrow MnO_2$

6. KEY TRANSITION METAL SALTS & REACTIONS

Copper Compounds

- **Copper Sulphate (Blue Vitriol - $CuSO_4 \cdot 5H_2O$):**
  Structure: $[Cu(H_2O)_4]SO_4 \cdot H_2O$ (4 H_2O coordinated, 1 hydrogen bonded).
  Thermal Decomposition:
  $CuSO_4 \cdot 5H_2O \rightarrow^{100°C} CuSO_4 \cdot H_2O \rightarrow^{230°C} CuSO_4$ (White) $\rightarrow^{720°C} CuO + SO_2 + ½\ O_2$
- **Reactions of $CuSO_4$:**
  • With KI: **$2\ CuSO_4 + 4\ KI \rightarrow Cu_2I_2\downarrow$ (White ppt) + I_2 + $2\ K_2SO_4$**
  • With KCN: **$2\ CuSO_4 + 10\ KCN \rightarrow 2\ K_3[Cu(CN)_4] + (CN)_2\uparrow$ (Cyanogen) + $2\ K_2SO_4$**
  • Schwizer's Reagent: **$[Cu(NH_3)_4]SO_4$** (Deep blue solution dissolves cellulose).

Mercury & Iron Compounds

- **Calomel (Hg_2Cl_2):** Diamagnetic containing Hg_2^{2+} (Hg-Hg bond).
  Reaction with NH_4OH: **$Hg_2Cl_2 + 2\ NH_4OH \rightarrow Hg\downarrow$ (Black) + $Hg(NH_2)Cl\downarrow$ (White) + NH_4Cl + $2\ H_2O$** (Turns black).
- **Corrosive Sublimate ($HgCl_2$):** Soluble, highly poisonous.
  Reaction with KI: **$HgCl_2 + 2\ KI \rightarrow HgI_2\downarrow$ (Scarlet Red) $\rightarrow^{excess\ KI} K_2[HgI_4]$** (Nessler's Reagent).
- **Iron Salts:**
  • Mohr's Salt: $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ (Primary standard).
  • Brown Ring Complex: **$[Fe(H_2O)_5(NO)]SO_4$** (Fe in +1 OS, NO in +1 OS, μ = 3.87 BM).

7. EXTRACTION SUMMARY OF IMPORTANT D-BLOCK METALS

Metal	Chief Ores	Key Extraction Steps & Principles	Important Chemical Equations
Iron (Fe)	Haematite (Fe_2O_3) Magnetite (Fe_3O_4)	Smelting in Blast Furnace. Charge = Ore + Coke + Limestone (8:4:1). Flux: $CaCO_3 \rightarrow CaO + SiO_2 \rightarrow CaSiO_3$ (Slag).	$\mathbf{Fe_2O_3 + 3\ CO \xrightarrow{500\text{-}800\ K} 2\ Fe + 3\ CO_2}$ Cast Iron: 2.5-5% C | Wrought Iron: Purest (0.1-0.25% C)
Copper (Cu)	Copper Pyrites ($CuFeS_2$) Copper Glance (Cu_2S)	Froth Floatation → Roasting → Smelting → Bessemerisation → Self-Reduction. Product: Blister Copper (98% pure, SO_2 bubbles).	$\mathbf{2\ FeS + 3\ O_2 \rightarrow 2\ FeO + 2\ SO_2}$ $\mathbf{FeO + SiO_2 \rightarrow FeSiO_3}$ **(Slag)** $\mathbf{2\ Cu_2O + Cu_2S \rightarrow 6\ Cu + SO_2\uparrow}$ (Self Reduction)
Silver (Ag)	Argentite (Ag_2S) Horn Silver (AgCl)	MacArthur-Forrest Cyanide Process. Leaching with NaCN in presence of air (O_2 acts as oxidant).	$\mathbf{4\ Ag_2S + 8\ NaCN + 2\ H_2O + O_2 \rightleftharpoons 4\ Na[Ag(CN)_2] + 4\ NaOH + 2\ S}$ $\mathbf{2\ Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2\ Ag\downarrow}$

8. F-BLOCK ELEMENTS: LANTHANOIDS VS ACTINOIDS

Lanthanoids (4f Series: Ce ^{58} to Lu ^{71})

- **Electronic Configuration:** [Xe] $4f^{1\text{-}14}\ 5d^{0\text{-}1}\ 6s^2$
- **Common Oxidation State: +3** (Most stable).
  - **Ce^{4+}:** Noble gas conf. ($4f^0$), strong oxidant in aq solution.
  - **Eu^{2+}:** Half-filled ($4f^7$), strong reducing agent.
  - **Yb^{2+}:** Fully filled ($4f^{14}$), reducing agent.
- **Lanthanoid Contraction:** Regular decrease in atomic & ionic radii (Ln^{3+}) from La to Lu.
  Cause: Imperfect shielding of 4f electrons.
  Consequences:
  1. Radii of 4d & 5d elements match (Zr ≈ Hf).
  2. Separation of Lanthanoids becomes difficult.
  3. Basic character of hydroxides decreases: $La(OH)_3$ (Most basic) > $Lu(OH)_3$ (Least basic).
- **Misch Metal:** Alloy of Ln (~95%), Fe (~5%), traces of S, C, Ca, Al. Used in bullet shells/flints.

Actinoids (5f Series: Th ^{90} to Lr ^{103})

- **Electronic Configuration:** [Rn] $5f^{1\text{-}14}\ 6d^{0\text{-}1}\ 7s^2$
- **Oxidation States:** Show a much wider range of OS (+3, +4, +5, +6, +7) because **5f, 6d, and 7s levels have comparable energies**.
  - Maximum OS increases up to Np & Pu (+7), then decreases.
- **Actinoid Contraction:** Greater contraction from element to element than Lanthanoid contraction due to **poorer shielding by 5f electrons** compared to 4f.
- **Reactivity & Nature:** All are radioactive. Early actinoids are far more reactive than lanthanoids. Lower IE than early lanthanoids.

9. JEE ADVANCED HIGH-YIELD RESULTS & SHORTCUTS

Alloys & Composition

- **Brass:** Cu (60-80%) + Zn (20-40%)
- **Bronze:** Cu (75-90%) + Sn (10-25%)
- **German Silver:** Cu (25-50%) + Zn (25-35%) + Ni (10-35%) [No Ag!]
- **Gun Metal:** Cu + Sn + Zn

Interstitial Compounds

- Non-stoichiometric (e.g., $TiH_{1.7}$, $VH_{0.56}$, Fe_3H).
- Retain metallic conductivity.
- Higher MP than pure parent metal.
- Extremely hard (TiC approaches diamond).
- Chemically inert.

Photographic Chemistry

- **Exposure:** AgBr + hv → Ag + Br
- **Fixing Step:** Removal of unreacted AgBr using Hypo:
  $\mathbf{AgBr + 2\ Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2]}$ **(Soluble) + NaBr**

Quick Revision Checklist **Key Colors for Qualitative Analysis:** Cr^{3+} (Green), CrO_4^{2-} (Yellow), $Cr_2O_7^{2-}$ (Orange), Mn^{2+} (Light Pink/Colorless), MnO_4^- (Purple), MnO_4^{2-} (Dark Green), Fe^{2+} (Pale Green), Fe^{3+} (Yellowish Brown), Cu^{2+} (Blue), Co^{2+} (Pink), Ni^{2+} (Green), CrO_5 (Deep Blue), $Cu_2[Fe(CN)_6]$ (Chocolate Brown ppt), $Fe_4[Fe(CN)_6]_3$ (Prussian Blue).

COORDINATION COMPOUNDS FORMULA SHEET

JEE Main & JEE Advanced Rapid Revision & Reference Guide

CHEMISTRY

1. ADDITION COMPOUNDS & WERNER'S COORDINATION THEORY

Double Salts vs Coordination Compounds:

- **Double Salts:** Exist only in solid state; completely dissociate into simple constituent ions in aqueous solution (e.g., Carnallite: $KCl{\cdot}MgCl_2{\cdot}6H_2O$, Mohr's Salt: $FeSO_4{\cdot}(NH_4)_2SO_4{\cdot}6H_2O$, Potash Alum: $K_2SO_4{\cdot}Al_2(SO_4)_3{\cdot}24H_2O$). Retain original ionic identities.
- **Coordination Compounds:** Retain identity in solid and aqueous solution; do not dissociate fully into constituent simple ions (e.g., $K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$). Properties differ from individual constituents.

Werner's Postulates & Valencies:

- **Primary Valency (1°):** Ionizable, non-directional; corresponds to **Oxidation State (O.S.)** of central metal; satisfied by negative anions.
- **Secondary Valency (2°):** Non-ionizable, directional; corresponds to **Coordination Number (C.N.)**; satisfied by neutral molecules or anions; determines 3D geometry (polyhedra).
- *$AgNO_3$ Precipitation Test Formula:* Moles of AgCl ppt per mole of complex = (Number of ionizable Cl^- ions outside coordination sphere).

2. EAN RULE, LIGANDS & DENTICITY

Effective Atomic Number (EAN) = Z − X + 2Y | *Sidgwick EAN Rule:* Stable if EAN = Atomic number of next noble gas (36, 54, 86).

Where: **Z** = Atomic number of central metal, **X** = Oxidation state of metal, **Y** = Coordination Number (number of donor electron pairs).

Type	Denticity	Ligands, Formulas & Names
Monodentate	1	Anionic: F^- (fluorido), Cl^- (chlorido), Br^- (bromido), CN^- (cyanido), OH^- (hydroxido), NO_2^- (nitrito-N), ONO^- (nitrito-O), O^{2-} (oxido), S^{2-} (sulphido). Neutral: H_2O (aqua), NH_3 (ammine), CO (carbonyl), NO (nitrosyl), py (pyridine). Cationic: NO^+ (nitrosonium), NO_2^+ (nitronium), $NH_2NH_3^+$ (hydrazinium).
Bidentate	2	en (ethane-1,2-diamine: $H_2NCH_2CH_2NH_2$), ox^{2-} (oxalato: $C_2O_4^{2-}$), gly^- (glycinato: $H_2NCH_2COO^-$), dipy / bpy (2,2'-bipyridine), dmg^- (dimethylglyoximato), $acac^-$ (acetylacetonato).
Polydentate	3, 4, 5, 6	dien (diethylenetriamine - 3), terpy (terpyridine - 3), trien (triethylenetetramine - 4), $EDTA^{3-}$ (ethylenediaminetriacetate - 5), **$EDTA^{4-}$** (ethylenediaminetetraacetate - 6: 2 N + 4 O donors, 40 total atoms).
Ambidentate	1 (Flexi)	Ligands with ≥2 donor atoms but attach via only ONE atom at a time: NO_2^- (M–NO_2 vs M–ONO), SCN^- (thiocyanato-S vs isothiocyanato-N: M–NCS), CN^- (M–CN vs M–NC).

Flexidentate Ligands: Polydentate ligands capable of showing variable denticity depending on complex environment (e.g., $EDTA^{4-}$ can act as 4-, 5-, or 6-dentate; SO_4^{2-} can act as monodentate or bidentate).

3. STRUCTURAL & STEREOISOMERISM

A. Structural Isomerism:

- **Ionisation:** Exchange of ligands between sphere & ionization zone (e.g., $[Co(NH_3)_5SO_4]Br$ vs $[Co(NH_3)_5Br]SO_4$).
- **Hydrate/Solvate:** Exchange of water/solvent inside/outside sphere (e.g., $[Cr(H_2O)_6]Cl_3$ (violet) vs $[Cr(H_2O)_5Cl]Cl_2{\cdot}H_2O$ (green)).
- **Linkage:** Ambidentate ligand donor atom variation (e.g., $[Co(NH_3)_5(NO_2)]Cl_2$ vs $[Co(NH_3)_5(ONO)]Cl_2$).
- **Coordination:** Exchange of ligands between complex cation & anion (e.g., $[Co(NH_3)_6][Cr(CN)_6]$ vs $[Cr(NH_3)_6][Co(CN)_6]$).
- **Coordination Position:** Non-bridging ligand distribution across metal centers in bridged complexes.

B. Geometrical Isomerism (GI):

- **C.N. = 4 (Tetrahedral):** NO GI due to all positions being adjacent.
- **C.N. = 4 (Square Planar):**
  - $[Ma_2b_2] \rightarrow 2$ (cis + trans) | $[Ma_2bc] \rightarrow 2$ (cis + trans)
  - $[Mabcd] \rightarrow 3$ GI (Hold 'a' fixed, arrange b,c,d opposite)
  - $[M(AB)_2]$ (unsym. bidentate) $\rightarrow 2$ (cis + trans)
- **C.N. = 6 (Octahedral):**
  - $[Ma_4b_2]$ or $[Ma_4bc] \rightarrow 2$ (cis + trans)
  - $[Ma_3b_3] \rightarrow 2$ (**fac** [facial] & **mer** [meridional])
  - $[Ma_2b_2c_2] \rightarrow 5$ GI | [Mabcdef] $\rightarrow$ 15 GI
  - $[M(AA)_2b_2] \rightarrow 2$ GI (cis + trans)

C. Optical Isomerism (OI): Requires molecule to lack Plane of Symmetry (POS) / Center of Symmetry (COS).
- **Square Planar:** Generally optically inactive due to molecular plane POS (except chiral ligand attached).
- **Tetrahedral:** [Mabcd] and $[M(AB)_2]$ show OI (chiral center).
- **Octahedral:** $[M(AA)_3] \rightarrow 2$ Optical Isomers (d-cis + l-cis); $[M(AA)_2b_2] \rightarrow 3$ Stereoisomers (trans is meso/inactive due to POS; cis is chiral giving d + l); $[M(AA)b_2c_2] \rightarrow$ cis-b-cis-c is active.

4. VALENCE BOND THEORY (VBT) & MAGNETIC MOMENTS

Spin-Only Magnetic Moment: $\mu = \sqrt{[n(n+2)]}$ BM (Bohr Magneton)

Where **n** = number of unpaired d-electrons on metal atom/ion.

n (Unpaired e^-)	1	2	3	4	5
μ (BM) = √[n(n+2)]	√3 = 1.73 BM	√8 = 2.84 BM	√15 = 3.87 BM	√24 = 4.90 BM	√35 = 5.92 BM

VBT Hybridization & Geometry Summary:

C.N.	Hybridization	Geometry	d-orbital involved	Examples
4	sp^3	Tetrahedral	-	$[NiCl_4]^{2-}$ (n=2), $[Ni(CO)_4]$ (n=0), $[Zn(NH_3)_4]^{2+}$ (n=0)
4	dsp^2	Square Planar	$d_{x^2-y^2}$	$[Ni(CN)_4]^{2-}$ (n=0), $[Cu(NH_3)_4]^{2+}$ (n=1), $[PtCl_4]^{2-}$ (n=0)
4	d^3s [Added]	Tetrahedral	d_{xy}, d_{yz}, d_{zx}	MnO_4^-, CrO_4^{2-}, $Cr_2O_7^{2-}$, CrO_2Cl_2 (d^0, diamagnetic)
5	dsp^3 / sp^3d	TBP / Square Pyramidal	d_{z^2} (TBP) / $d_{x^2-y^2}$	$Fe(CO)_5$ (dsp^3, TBP, n=0), $[Cu(NH_3)_5]^{2+}$
6	d^2sp^3	Octahedral (Inner Orbital / Low Spin)	d_{z^2}, $d_{x^2-y^2}$	$[Fe(CN)_6]^{4-}$ (n=0), $[Co(NH_3)_6]^{3+}$ (n=0), $[Cr(NH_3)_6]^{3+}$ (n=3)
6	sp^3d^2	Octahedral (Outer Orbital / High Spin)	d_{z^2}, $d_{x^2-y^2}$	$[Fe(H_2O)_6]^{2+}$ (n=4), $[CoF_6]^{3-}$ (n=4), $[Ni(NH_3)_6]^{2+}$ (n=2)

Key VBT Exceptions & Special Behaviors:

- **Transference of Electron:** In $[Cu(NH_3)_4]^{2+}$ (Cu^{2+}, $3d^9$), one 3d electron is promoted to 4p, undergoing *dsp²* hybridization (Square Planar, paramagnetic, n=1).
- **4d & 5d Transition Series Metals:** ALL ligands (even weak field like Cl^-, Br^-, I^-) act as **Strong Field Ligands (SFL)** and cause pairing. E.g., $[PtCl_4]^{2-}$, $[PdCl_4]^{2-}$ are square planar (dsp^2, diamagnetic).
- **Co^{3+} ($3d^6$) Special SFL Rules:** H_2O and ox^{2-} act as SFL with Co^{3+} → forms low spin inner orbital d^2sp^3 diamagnetic complexes (e.g., $[Co(H_2O)_6]^{3+}$, $[Co(ox)_3]^{3-}$).
- **Ni^{2+} with F^-:** $[NiF_6]^{2-}$ (Ni^{4+}, $3d^6$) forms d^2sp^3 diamagnetic complex [Added].
- **$[Cr(NH_3)_6]^{2+}$, $[Mn(NH_3)_6]^{2+}$, $[Fe(NH_3)_6]^{2+}$:** NH_3 acts as WFL with Cr^{2+}, Mn^{2+}, Fe^{2+} → outer orbital sp^3d^2 paramagnetic complexes.

5. CRYSTAL FIELD THEORY (CFT) & CFSE FORMULAS

Spectrochemical Series (Increasing Δ / Field Strength):
$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

A. Octahedral Field (Δ_o):

Ligands approach along the axes (x, y, z). 5 d-orbitals split into higher energy e_g ($d_{x^2-y^2}$, d_{z^2}) and lower energy t_{2g} (d_{xy}, d_{yz}, d_{zx}).

$CFSE_{oct} = [\ -0.4\ n(t_{2g}) + 0.6\ n(e_g)\ ]\ \Delta_o + m\ P$

where $n(t_{2g})$ & $n(e_g)$ are number of e^- in t_{2g} & e_g; **m** = number of *extra* pairs formed compared to free ion, **P** = pairing energy.

High Spin vs Low Spin Conditions:

- **Low Spin (SFL):** $\Delta_o > P$ → Pairing occurs prior to filling e_g level.
- **High Spin (WFL):** $\Delta_o < P$ → Electrons enter e_g before pairing in t_{2g}.

B. Tetrahedral Field (Δ_t):

Ligands approach between the axes. Splitting inverted: higher energy t_2 (d_{xy}, d_{yz}, d_{zx}) and lower energy e ($d_{x^2-y^2}$, d_{z^2}). No 'g' subscript.

$CFSE_{tet} = [\ -0.6\ n(e) + 0.4\ n(t_2)\ ]\ \Delta_t + m\ P$

$\Delta_t = (4/9)\ \Delta_o$ &implies; Always High Spin ($\Delta_t < P$)

Factors Affecting Splitting Energy (Δ_o):

- **O.S. of Metal:** Higher oxidation state → Larger Δ_o ($Pt^{4+} > Pt^{2+}$).
- **Transition Series:** 5d series > 4d series > 3d series (5d has ~50% greater Δ_o than 3d).
- **Number of Ligands:** $\Delta_o > \Delta_t$ ($\Delta_t = 4/9\ \Delta_o$).

Color & d-d Transitions: Complementary color absorbed/emitted. Energy absorbed: $E = h\nu = hc / \lambda$ &implies; $\Delta_o \propto 1 / \lambda_{absorbed}$. Greater ligand field strength → larger Δ_o → shorter absorbed wavelength (λ).

6. ORGANOMETALLIC COMPOUNDS & SYNERGIC BONDING

Compounds containing at least one direct **Metal–Carbon (M–C) bond** (or metalloid/non-metal like B, Si, P less electronegative than C).

Class	Description & Bonding	Examples & Formulas
σ-bonded OMC	M–C single σ-bond; formed by main group / non-transition metals.	Grignard Reagent (R–Mg–X), Dialkyl zinc R_2Zn ($(C_2H_5)_2Zn$ - Frankland reagent), TEL ($Pb(CH_3)_4$), $Al_2(CH_3)_6$.
π-bonded OMC	Bonding involves π-electrons of unsaturated organic ligands. Hapticity (η^n) = no. of C atoms bound to metal.	Zeise's Salt: $K[PtCl_3(\eta^2\text{-}C_2H_4)]$ (Pt in +2 state) Ferrocene: $Fe(\eta^5\text{-}C_5H_5)_2$ (Fe in +2 state) Dibenzene Chromium: $Cr(\eta^6\text{-}C_6H_6)_2$ (Cr in 0 state).
σ- & π-bonded OMC	Metal Carbonyls. Contain both M–C σ donor and M → C π back-donation. O.S. of metal = 0.	$Ni(CO)_4$ (Tetrahedral), $Fe(CO)_5$ (TBP), $Cr(CO)_6$ (Octahedral), $Mn_2(CO)_{10}$, $Co_2(CO)_8$.

Synergic Bonding Mechanism in Carbonyls:

- **σ-bond:** Lone pair on carbonyl carbon → empty d-orbital of metal (L → M).
- **π-back bond:** Filled metal d-orbital → vacant anti-bonding π* orbital of CO (M → L).
- **Synergic Effect:** σ-donation increases electron density on metal → strengthens π-back bonding → strengthens M–C bond and weakens C–O bond.

JEE Advanced Relationships (Bond Order & Length):

> **Back Bonding ↑ &implies; M–C Bond Order ↑ &implies; M–C Length ↓**
> **Back Bonding ↑ &implies; C–O Bond Order ↓ &implies; C–O Length ↑**

Charge Effect on Carbonyl Complex: Higher negative charge on complex ion → Metal is more electron rich → Greater back-bonding → Stronger M–C bond → Longest/weakest C–O bond.

C–O Bond Length Order: $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^- < [Ti(CO)_6]^{2-}$

7. STABILITY CONSTANTS, APPLICATIONS & CATALYSTS

Stability Constants Formulas:

Stepwise formation: $M + L \rightleftharpoons ML$ (K_1); $ML + L \rightleftharpoons ML_2$ (K_2) ... $ML_{n-1} + L \rightleftharpoons ML_n$ (K_n).

> **Overall Stability Constant (β_n) = $K_1 \times K_2 \times ... \times K_n$**

> **Instability / Dissociation Constant (K_d) = $1 / \beta_n$**

Chelate Effect: Complexes with chelating (ring-forming) ligands have higher thermodynamic stability due to positive entropy change ($\Delta S^\circ > 0$). 6-membered > 5-membered ring stability (if resonance present).

Important Catalysts & Medical Applications:

Wilkinson's Catalyst	$[(C_6H_5)_3P]_3RhCl$ I(Rh^I, Sq. Planar) → Alkene Hydrogenation
Ziegler-Natta Catalyst	$TiCl_4 + Al(C_2H_5)_3$ → Polymerization of Alkenes (HDPE)
Cisplatin	cis-$[Pt(NH_3)_2Cl_2]$ → Anti-cancer chemotherapy drug
$EDTA^{4-}$ Uses	Treatment of Lead (Pb) poisoning, estimation of Ca^{2+}/Mg^{2+} hardness.
Biological Complexes	Chlorophyll (Mg), Hemoglobin (Fe^{2+}), Vitamin B-12 (Co), Carbonic anhydrase / Carboxypeptidase (Zn).

8. JEE ADVANCED CONCEPTS, DISTORTION & SPECIAL RESULTS

A. Jahn-Teller Distortion (Geometric Distortion):

Non-linear complex in an orbitally degenerate electronic state undergoes distortion to lower symmetry and remove degeneracy.

- **Strong Distortion:** Asymmetrically filled e_g orbitals: d^4 (high spin: $t_{2g}^3 e_g^1$), d^7 (low spin: $t_{2g}^6 e_g^1$), **d^9** ($t_{2g}^6 e_g^3$ - e.g., Cu^{2+} octahedral complexes).
- **Weak / No Distortion:** Asymmetrically filled t_{2g} gives weak distortion. Symmetrically filled orbitals (d^3, d^5-HS, d^6-LS, d^8, d^{10}) show **NO distortion**.
- **Tetragonal Elongation (z-out):** M–L bonds along z-axis become longer than xy plane bonds.

B. Special Color & Charge Transfer Spectra [Added]:

Intense colors in d^0 complexes are NOT due to d-d transition, but **Ligand to Metal Charge Transfer (LMCT)**:

- **MnO_4^- (Permanganate):** Intense Purple (d^0 Mn^{VII}, LMCT).
- **$Cr_2O_7^{2-}$ / CrO_4^{2-}:** Orange / Yellow (d^0 Cr^{VI}, LMCT).
- **Fe^{4+} / Prussian Blue:** $Fe_4[Fe(CN)_6]_3$ → Intervalence Charge Transfer (IVCT) between Fe^{II} and Fe^{III}.
- **$Fe^{3+} + SCN^-$:** $[Fe(H_2O)_5(SCN)]^{2+}$ → Blood red color.

CHEMISTRY IN EVERYDAY LIFE

Complete JEE Main & Advanced Quick Formula & Reference Sheet

JEE REVISION SHEET

1. DRUG-TARGET INTERACTIONS & ACTION MECHANISMS

- **Drug Definition:** Low molecular mass chemicals (~100–500 u) producing biological response.
- **Enzyme Inhibitors:** Block binding site of enzyme to prevent natural substrate binding.
- **Competitive Inhibitors:** Compete directly with natural substrate for active site binding.
- **Allosteric Inhibitors:** Bind to a site *other than active site* (allosteric site), changing enzyme active site shape.
- **Receptors:** Proteins crucial for body's communication system.
- **Antagonists:** Drugs that bind to receptor site and inhibit its natural signal function.
- **Agonists:** Drugs that mimic natural chemical messenger by switching on the receptor.
- **Chemical Messengers:** Chemicals transmitting message between neurons or neuron to muscle.

2. THERAPEUTIC CLASSES & KEY DRUGS FOR JEE

Drug Category	Key Function / Mode of Action	Examples & Structure Details
Antipyretics	Brings down body temperature during high fever.	Paracetamol, Aspirin, Analgin, Phenacetin.
Analgesics (Non-narcotic)	Pain relievers (Non-addictive), anti-inflammatory, anti-blood clotting (prevents heart attacks).	**Aspirin** (2-Acetoxybenzoic acid: $C_9H_8O_4$) **Paracetamol** (4-Acetamidophenol), Ibuprofen, Diclofenac sodium, Novalgin.
Analgesics (Narcotic / Opioid)	Relieve severe pain (post-operative/cancer), addictive, sleep-inducing. Action on CNS.	Morphine, Heroin, Codeine.
Tranquilizers (NAD)	Treatment of stress, mild/severe mental diseases. Neurologically active drugs.	*Barbiturates (Hypnotics):* Veronal, Amytal, Nembutal, Luminal, Seconal. *Non-hypnotic:* Chlordiazepoxide, Meprobamate, Equanil, Valium, Serotonin.
Antacid	Neutralize excess HCl secretion in stomach to raise pH to appropriate level.	*Inorganic:* $NaHCO_3$, $Al(OH)_3$, $Mg(OH)_2$, $MgCO_3$, Mg-trisilicate. *Histamine receptor blockers:* Cimetidine (Tagamet), Ranitidine (Zantac).
Antihistamines	Block action of histamine causing allergies (skin rashes, conjunctivitis, rhinitis, sneezing).	Diphenylhydramine, Chlorpheniramine, Brompheniramine (Dimetapp), Terfenadine (Seldane), Promethazine.
Antifertility Drugs	Synthetic progesterone & estrogen derivatives; suppress ovulation.	Norethindrone (synthetic progesterone), Ethynylestradiol / Novestrol / Mestranol (estrogen derivative), Mifepristone (morning after pill).

3. ANTIMICROBIALS: ANTISEPTICS & DISINFECTANTS

Type	Application & Characteristics	Key Examples
Antiseptics	Applied to *living tissues* (wounds, cuts, skin). Kills/prevents microbial growth without damaging tissue.	• **Dettol:** Mixture of Chloroxylenol + Terpineol. • **Bithionol:** Added to soaps to reduce odor. • **Tincture of Iodine:** 2-3% Iodine in alcohol-water. • Iodoform (CHI_3), Furacine, Soframicine. • Boric acid (weak antiseptic for eyes).
Disinfectants	Applied to *inanimate objects* (floors, drainage, instruments). Toxic to living tissue.	• **0.2–0.4 ppm Chlorine** in aqueous solution. • Low concentration of SO_2. • DDT.
Dual Concentration	Concentration determines antiseptic vs disinfectant behavior.	• **0.2% Phenol** → Antiseptic. • **1.0% Phenol** → Disinfectant.

4. ANTIBIOTICS CLASSIFICATION & SPECTRUM

Mode of Action:
- **Bactericidal (Kills bacteria):** Penicillin, Aminoglycosides, Ofloxacin.
- **Bacteriostatic (Inhibits growth):** Erythromycin, Tetracycline, Chloramphenicol.

Spectrum of Action:
- **Broad Spectrum:** Acts on Gram-positive AND Gram-negative bacteria.
  Examples: Chloramphenicol (Chloromycetin - treats typhoid), Ampicillin, Amoxycillin, Vancomycin, Ofloxacin, Tetracycline.
- **Narrow Spectrum:** Penicillin G, Clindamycin, Metronidazole.
- **Limited Spectrum:** Dysidazirine (effective against single organism/disease).

Special JEE Notes: **Salvarsan** (Arsphenamine, first effective treatment for syphilis, contains As=As link) & **Prontosil** (azo dye converting to sulphanilamide in body). Ampicillin & Amoxycillin are synthetic derivatives of Penicillin (contains β-lactam ring). Streptomycin is a broad spectrum antibiotic.

5. CHEMICALS IN FOOD & ARTIFICIAL SWEETENERS

Artificial Sweetener	Sweetness vs Sucrose	Key Conditions & Features
Aspartame	100 ×	Methyl ester of dipeptide (Aspartic acid + Phenylalanine). *Unstable at cooking T* (used in cold food/soft drinks).
Saccharin	550 ×	o-Sulphobenzimide. Excreted unchanged in urine; zero calories. First popular sweetener.
Sucralose	600 ×	Trichloro derivative of sucrose. *Stable at cooking T*, appearance & taste like sugar, provides no calories.
Alitame	2000 ×	High potency sweetener. *Stable at cooking T*, control of sweetness is difficult.

Food Preservatives: Prevent spoilage due to microbial growth.
- Table salt, sugar, vegetable oils.
- Sodium benzoate (C_6H_5COONa) **[Metabolized in body]**.
- Salts of Sorbic acid & Propanoic acid.

Antioxidants in Food: Retard action of oxygen on food.
- **BHT** (Butylated Hydroxytoluene) & **BHA** (Butylated Hydroxyanisole) → Increases shelf life of butter from months to years.
- **SO_2 & Sulphites** → Used in wine, beer, & sugar syrups.

6. SOAPS, DETERGENTS & SAPONIFICATION FORMULAS

Saponification Reaction:
Glyceryl ester of Stearic Acid (Fat) + 3 NaOH → Glycerol + 3 $C_{17}H_{35}COONa$ (Sodium Stearate - Soap)

Cleansing Agent	Chemical Composition & Reaction / Structure	JEE Characteristics & Uses
Soaps	Na or K salts of long-chain fatty acids (C_{12}–C_{18}) like Stearic, Oleic, Palmitic acids. $C_{17}H_{35}COOH$ (Stearic), $C_{15}H_{31}COOH$ (Palmitic).	Insoluble in hard water due to formation of insoluble Ca^{2+} and Mg^{2+} soaps (scum): $2\ RCOONa + CaCl_2 \rightarrow (RCOO)_2Ca \downarrow + 2\ NaCl$
Anionic Detergents	Anionic part of molecule involved in cleansing action. • Sodium lauryl sulphate: $CH_3(CH_2)_{10}CH_2OSO_3^-\ Na^+$ • Sodium dodecylbenzenesulphonate: $CH_3(CH_2)_{11}-C_6H_4-SO_3^-\ Na^+$	Used in household work & **toothpastes**. Hydrocarbon chain can be branched or unbranched.
Cationic Detergents	Quaternary ammonium salts of amines with acetates, chlorides, or bromides as anions. • **Cetyltrimethylammonium bromide:** $[CH_3(CH_2)_{15}N(CH_3)_3]^+\ Br^-$	Cationic part possesses long hydrocarbon chain. Possess **germicidal properties**; used in hair conditioners. Expensive.

Cleansing Agent	Chemical Composition & Reaction / Structure	JEE Characteristics & Uses
Non-ionic Detergents	Esters of high molecular mass formed by reaction of Stearic acid with Polyethylene glycol: $CH_3(CH_2)_{16}COOH + HO(CH_2CH_2O)_nCH_2CH_2OH \rightarrow CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH + H_2O$	Do not contain any ions. Used in **liquid dishwashing detergents**. Mechanism of cleansing is same as soaps.

• **Biodegradability:** Detergents with *straight (unbranched) hydrocarbon chains* are biodegradable. Highly branched chains lead to environmental pollution as microbes cannot degrade them.

7. DYES: CLASSIFICATION BY CONSTITUTION & FUNCTIONAL GROUPS

Class of Dye	Characteristic Chromophore / Group	Examples & JEE Key Facts
Azo Dyes	$-N = N-$ (Azo linkage)	Aniline yellow, Orange-I, Methyl orange, Congo red.
Nitro Dyes	$-NO_2$ group	Martius yellow.
Phthalein Dyes	Phthalein core structure	Phenolphthalein (Acid-base indicator, colorless in acid, pink in alkali).
Triphenylmethane Dyes	$(C_6H_5)_3C$ core	Magenta, Malachite green.
Indigoid Dyes	Carbonyl-alkene conjugated core	Indigo (Vat dye; earlist known natural blue dye).
Anthraquinoid Dyes	Anthraquinone skeleton	Alizarin (Red dye; mordant dye).

8. JEE HIGH-YIELD ONE-LINERS & MISCELLANEOUS FORMULAS

- **Diuretic:** Medicine promoting urine secretion to remove excess Na^+ and water.
- **Adrenaline:** Hormone stimulating glycogenolysis (primarily in liver).
- **Cocaine:** Historically used as an antidepressant in the USA.
- **Chloroquine:** Synthetic drug used specifically for malaria treatment.
- **Aspirin Synthesis:** Salicylic Acid + Acetic Anhydride (H^+ catalyst) → Aspirin + Acetic Acid.
- **Bithionol Function:** Prevents bacterial decomposition of sweat on skin.
- **Artificial Sweetener Potency Order:** Alitame (2000) > Sucralose (600) > Saccharin (550) > Aspartame (100).

CHEMICAL KINETICS & NUCLEAR CHEMISTRY

JEE MAIN & ADVANCED COMPLETE FORMULA SHEET

JEE MAIN | JEE ADVANCED

Comprehensive Revision Sheet

1. Reaction Rate & Stoichiometric Relations

Average Rate (r_{avg}): $r_{avg} = -\Delta[A]/\Delta t = +\Delta[P]/\Delta t$

Instantaneous Rate (r_{inst}): $r_{inst} = Lim_{\Delta t \to 0}\,(\Delta x/\Delta t) = dx/dt = -d[A]/dt$

Units of Rate: mol L^{-1} s^{-1} or atm s^{-1} (gaseous system)

Stoichiometric Rate Relation: For $aA + bB \rightarrow cC + dD$:

$$r = -{}^{1}/_{a} \cdot d[A]/dt = -{}^{1}/_{b} \cdot d[B]/dt = +{}^{1}/_{c} \cdot d[C]/dt = +{}^{1}/_{d} \cdot d[D]/dt$$

• Rate of disappearance of A = $-d[A]/dt$ | Rate of appearance of C = $+d[C]/dt$

2. Rate Law, Order & Molecularity

Differential Rate Law: $r = k\,[A]^{x}\,[B]^{y}$

• **Overall Order (n):** $n = x + y$ (Experimental; can be 0, +ve, −ve, fraction).

• **Molecularity:** Total colliding species in elementary step (Theoretical, integer 1, 2, 3 only; never 0 or fractional).

Rate Constant (k) Units:

$$Unit\ of\ k = (mol\ L^{-1})^{1-n}\ time^{-1} = (conc)^{1-n}\ time^{-1}$$

• Gaseous reaction unit: $(atm)^{1-n}\ time^{-1}$

3. Master Table of Integrated Rate Laws & Kinetics Parameters

Order (n)	Differential Rate Law	Integrated Rate Equation ($[A]_t$)	Half-Life ($t_{1/2}$)	Completion Time ($t_{100\%}$)	Unit of k
0	$-d[A]/dt = k_0$	$[A]_t = [A]_0 - k_0 t \Rightarrow x = k_0 t$	$t_{1/2} = [A]_0 / (2k_0)$	$t_{100\%} = [A]_0 / k_0$	mol L^{-1} s^{-1}
1	$-d[A]/dt = k_1[A]$	$k_1 = {}^{2.303}/_{t} \log_{10}({}^{[A]_0}/_{[A]_t})$ $[A]_t = [A]_0\, e^{-k_1 t}$	$t_{1/2} = {}^{\ln 2}/_{k_1} = {}^{0.693}/_{k_1}$ (Independent of $[A]_0$)	$t_{100\%} = \infty$	s^{-1}
2 (1 reactant)	$-d[A]/dt = k_2[A]^2$	${}^{1}/_{[A]_t} - {}^{1}/_{[A]_0} = k_2 t \Rightarrow k_2 = {}^{x}/_{t \cdot a(a-x)}$	$t_{1/2} = {}^{1}/_{k_2[A]_0}$	$t_{100\%} = \infty$	L mol^{-1} s^{-1}
2 (2 reactants)	$r = k_2[A][B]$	$k_2 = {}^{2.303}/_{t(a-b)} \log_{10}[{}^{b(a-x)}/_{a(b-x)}]$ (a ≠ b)	–	–	L mol^{-1} s^{-1}
3	$-d[A]/dt = k_3[A]^3$	$k_3 = {}^{1}/_{2t}[{}^{1}/_{[A]_t^2} - {}^{1}/_{[A]_0^2}]$	$t_{1/2} = {}^{3}/_{2\,k_3[A]_0^2}$	$t_{100\%} = \infty$	L^2 mol^{-2} s^{-1}
n (n ≥ 2)	$-d[A]/dt = k_n[A]^n$	$k_n = {}^{1}/_{(n-1)t}[{}^{1}/_{[A]_t^{n-1}} - {}^{1}/_{[A]_0^{n-1}}]$	$t_{1/2} = {}^{2^{n-1}-1}/_{(n-1)\,k_n [A]_0^{n-1}}$	$t_{100\%} = \infty$	L^{n-1} mol^{1-n} s^{-1}

Stoichiometric Factor Rule: For $nA \rightarrow Products$, replace rate constant in formulas with $n{\cdot}k$ (e.g., Zero order: $k_0 = x/(nt)$, $t_{1/2} = a/(2nk_0)$; First order: $t_{1/2} = 0.693/(nk_1)$).

4. First-Order Kinetics: Shortcuts & Special Relations

Fractional Completion & Half-Lives:

• Amount remaining after n half-lives:

$$[A]_t = [A]_0 \cdot (1/2)^n \quad \text{where } n = t / t_{1/2}$$

• **Standard Time Ratio Rules:**

- $t_{75\%} = 2 \cdot t_{50\%} = 2 \cdot t_{1/2}$
- $t_{87.5\%} = 3 \cdot t_{1/2}$
- $t_{90\%} = {}^{2.303}/_{k} = {}^{t_{1/2}}/_{\log_{10}2} \approx 3.32 \cdot t_{1/2}$
- $t_{99\%} = 2 \cdot t_{90\%} \approx 6.64 \cdot t_{1/2}$
- $t_{99.9\%} = 10 \cdot t_{1/2} \approx 3.32 \cdot t_{99\%}$

General Half-Life Dependency Method (Determination of Order n):

$$t_{1/2} \propto {}^{1}/_{[A]_0^{n-1}} \Rightarrow {}^{(t_{1/2})_1}/_{(t_{1/2})_2} = ({}^{[A]_{0,2}}/_{[A]_{0,1}})^{n-1}$$

$$n = 1 + {}^{\log_{10}[(t_{1/2})_1 / (t_{1/2})_2]}/_{\log_{10}([A]_{0,2} / [A]_{0,1})}$$

Wilhelmy's Constant & Concentration at time t:

$[A]_t = [A]_0\, e^{-k\,t}$ | Amount reacted: $x = [A]_0\,(1 - e^{-k\,t})$

5. Experimental Methods for First-Order Reactions

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Experimental Method	Measured Variable	Integrated Rate Expression (k)
Gaseous Phase Reaction $A(g) \rightarrow B(g) + C(g)$	Total Pressure P_t at time t, Initial Pressure P_o	$k = {}^{2.303}/_{t} \log_{10}[{}^{P_o}/_{2P_o - P_t}]$ (where $P_t = P_0 + x$)
Gaseous Reaction with Inert Gas [Added] $A(g) \rightarrow bB(g) + cC(g)$	Total Pressure P_t, Infinite Pressure P_∞	$k = {}^{2.303}/_{t} \log_{10}[{}^{P_\infty - P_o}/_{P_\infty - P_t}]$
Titration Method (Reagent V) $A \rightarrow B + C$	Volume of reagent V_o, V_t, V_∞ reacting with A or Pdt	• Reactant Titration: $k = {}^{2.303}/_{t} \log_{10}({}^{V_o}/_{V_t})$ • Product Titration: $k = {}^{2.303}/_{t} \log_{10}[{}^{V_\infty}/_{V_\infty - V_t}]$ • Reactant & Product both react: $k = {}^{2.303}/_{t} \log_{10}[{}^{V_\infty - V_o}/_{V_\infty - V_t}]$
Optical Rotation (Polarimeter) Inversion of Cane Sugar	Optical Rotation Angle r_o, r_t, r_∞	$k = {}^{2.303}/_{t} \log_{10}[{}^{r_o - r_\infty}/_{r_t - r_\infty}]$
Decomposition of H_2O_2	Volume of $KMnO_4$ standard solution V_t	$k = {}^{2.303}/_{t} \log_{10}({}^{V_o}/_{V_t})$

6. Complex Reactions & Parallel / Consecutive Kinetics

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A. Parallel / Side Reactions

$$A \xrightarrow{k_1} B \quad \& \quad A \xrightarrow{k_2} C$$

- **Overall Rate Constant:** $\boldsymbol{k_{overall} = k_1 + k_2}$
- **Overall Half-Life:** $t_{1/2(overall)} = {}^{(t_{1/2})_1 \cdot (t_{1/2})_2}/_{(t_{1/2})_1 + (t_{1/2})_2}$
- **Yield / Mole Fractions of Products:**

$$\%\ Yield\ of\ B = {}^{k_1}/_{k_1 + k_2} \times 100 \quad | \quad \%\ Yield\ of\ C = {}^{k_2}/_{k_1 + k_2} \times 100$$

- **Ratio of Product Concentrations at any time t:** ${}^{[B]_t}/_{[C]_t} = {}^{k_1}/_{k_2}$
- **Overall Activation Energy (E_a):** $E_{a,\ overall} = {}^{k_1 E_{a1} + k_2 E_{a2}}/_{k_1 + k_2}$

B. Consecutive / Sequential Reactions

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

- **Concentration Profiles:**

$$[A]_t = [A]_o\, e^{-k_1 t}$$

$$[B]_t = {}^{k_1 [A]_o}/_{k_2 - k_1} \left(e^{-k_1 t} - e^{-k_2 t} \right)$$

$$[C]_t = [A]_o - [A]_t - [B]_t$$

- **Time for Maximum Concentration of Intermediate B (t_{max}):**

$$\boldsymbol{t_{max} = {}^{\ln(k_2/k_1)}/_{k_2 - k_1} = {}^{2.303 \log_{10}(k_2/k_1)}/_{k_2 - k_1}}$$

- **Maximum Concentration of B ($[B]_{max}$):**

$$[B]_{max} = [A]_o \left({}^{k_2}/_{k_1} \right)^{k_2/(k_1 - k_2)}$$

C. Reversible First-Order Reactions [Added]

$$A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B$$

- **Equilibrium Constant:** $K_{eq} = k_f / k_b = [B]_{eq} / [A]_{eq}$
- **Integrated Rate Law:** $(k_f + k_b)\, t = \ln [{}^{x_{eq}}/_{x_{eq} - x}]$ where x is conc. of B formed.

D. Steady-State Approximation (SSA) [Added]

- For a reactive intermediate I: $d[I]/dt \approx 0$
- **Pre-equilibrium Approximation:** Fast reversible step followed by slow rate-determining step (RDS). Overall rate is governed by RDS.

7. Temperature Dependence & Arrhenius Equation

Temperature Coefficient (μ): Ratio of rate constants at temperatures differing by 10 °C.

$$\mu = {}^{k_{T+10}}/_{k_T} \approx 2 \text{ to } 3 \quad | \quad {}^{k_{T2}}/_{k_{T1}} = \mu^{\Delta T / 10}$$

Arrhenius Equation: $\boldsymbol{k = A\, e^{-E_a/(RT)}}$

• A = Arrhenius Frequency / Pre-exponential Factor (same units as k) • = Activation Energy (or) • = Absolute temperature in Kelvin (), = 8.314 J K mol

Logarithmic Forms of Arrhenius Equation:

- Natural Log: $\ln k = \ln A - {}^{E_a}/_{RT}$
- Log base 10: $\log_{10} k = \log_{10} A - {}^{E_a}/_{2.303\, R\, T}$
- **Two-Temperature Form (Comparison):**

E_a J mol^{-1} kJ mol^{-1}
TKR^{-1-1}

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303\,R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1\,T_2}\right]$$

• Valid when $T_2 > T_1$. If E_a is higher, rate constant is more sensitive to T.

Graphical Interpretation of Arrhenius Plot:

• Plot of $\log_{10} k$ vs $1/T$ yields a straight line with:

- **Slope:** $m = -\frac{E_a}{2.303\,R} \Rightarrow E_a = -2.303\,R \times Slope$
- **Y-Intercept:** $c = \log_{10} A$

Thermodynamic Activation Energy Relations:

• $E_a = E_{threshold} - E_{reactants,\,avg}$

• Enthalpy of reaction: $\Delta H = E_{a,\,forward} - E_{a,\,backward}$

• For Endothermic ($\Delta H > 0$): $E_{a,f} > E_{a,b}$ | Exothermic ($\Delta H < 0$): $E_{a,f} < E_{a,b}$

8. Collision Theory & Catalysis

Collision Theory Formula:

$$k = P \cdot Z_{AB} \cdot e^{-E_a/(RT)}$$

• Z_{AB} = Collision frequency of reactants A & B.

• P = Steric / Orientation factor (probability of proper orientation).

• $e^{-E_a/(RT)}$ = Fraction of molecules with energy $E \geq E_a$.

Effect of Catalyst:

• Provides an alternate reaction path with lower activation energy ($E_a' < E_a$).

• $k_{catalyzed} / k_{uncatalyzed} = e^{(E_a - E_a')/(RT)}$

• **Catalyst does NOT alter:** ΔH, ΔG, K_{eq}, or equilibrium composition. Increases forward and backward rates equally.

9. Photochemistry Formulas

Energy of Absorbed Photon:

$$E = h\nu = \frac{hc}{\lambda}$$

• h = 6.626 × 10^{-34} J s, c = 3 × 10^8 m/s, λ = wavelength in m.

• Energy per mole of photons (1 Einstein): $E = N_A\,h\nu = N_A\,\frac{hc}{\lambda}$

Quantum Yield / Quantum Efficiency (ϕ):

$$\phi = \frac{\text{No. of molecules reacting}}{\text{No. of photons (quanta) absorbed}} = \frac{\text{Moles reacted}}{\text{Einsteins absorbed}}$$

• High quantum yield ($\phi >> 1$): Chain reactions (e.g., $H_2 + Cl_2 \rightarrow 2HCl$).

• Low quantum yield ($\phi < 1$): Deactivation / Luminescence / Secondary recomb.

10. Nuclear Chemistry & Radioactive Decay Kinetics

Radioactive decay strictly follows **First-Order Kinetics** (unaffected by T, P, concentration, or catalysts).

Decay Law & Amount Remaining:

• Decay Rate (Activity A): $A = -dN/dt = \lambda N$

• Radioactive Decay Law: $N_t = N_0\,e^{-\lambda t}$

• In terms of mass: $m_t = m_0\,e^{-\lambda t}$ | Activity: $A_t = A_0\,e^{-\lambda t}$

• Decay Constant (λ): $\lambda = \frac{2.303}{t}\log_{10}\left(\frac{N_0}{N_t}\right)$

• Amount remaining after n half-lives: $N = N_0\,(1/2)^n$ where $n = t / t_{1/2}$

Half-Life ($t_{1/2}$) & Average Life (τ):

• **Half-Life:** $t_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$

• **Average Life / Mean Life (τ):** Sum of lives of all nuclei / total nuclei.

$$\tau = \frac{1}{\lambda} = \frac{t_{1/2}}{0.693} \approx 1.44 \cdot t_{1/2}$$

• **Units of Activity:**

- 1 Becquerel (Bq) = 1 dps (disintegration per sec)
- 1 Curie (Ci) = 3.7 × 10^{10} dps
- 1 Rutherford (Rd) = 10^6 dps

Mass Defect (Δm) & Binding Energy (B.E.):

• Mass defect: $\Delta m = [\, Z\, m_p + (A - Z)\, m_n\,] - M_{observed}$

• Einstein's Energy Relation: $E = \Delta m \cdot c^2$

• Binding Energy:

$$\boldsymbol{B.E.\ (MeV) = \Delta m\ (in\ amu) \times 931.5\ MeV}$$

• **Binding Energy per Nucleon:** $B.E. / A = Total\ B.E. / Mass\ Number$

• Higher B.E. per nucleon implies greater nuclear stability (Peak at $^{56}Fe \approx 8.8$ MeV).

Packing Fraction (P.F.) & Parallel Decay:

• **Packing Fraction:**

$$P.F. = \frac{(Isotopic\ Atomic\ Mass - Mass\ Number\ A)}{Mass\ Number\ A} \times 10^4$$

• **Parallel Decay of Radionuclide $A \xrightarrow{\lambda_1} B$ & $A \xrightarrow{\lambda_2} C$:**

- Total decay constant: $\lambda_{total} = \lambda_1 + \lambda_2$
- Overall Half-Life: $t_{1/2(eff)} = \frac{t_1 \cdot t_2}{t_1 + t_2}$
- Fractional yield of B: $Y_B = \lambda_1 / (\lambda_1 + \lambda_2)$

11. Key Graphical Relationships Summary for JEE

Graph Type	Zero Order	First Order	Second Order
Concentration vs Time	$[A]_t$ vs t: Linear, Slope = $-k$	$ln[A]_t$ vs t: Linear, Slope = $-k$	$1/[A]_t$ vs t: Linear, Slope = $+k$
Rate vs Concentration	Rate vs $[A]$: Horizontal Line (Slope = 0)	Rate vs $[A]$: Linear through origin, Slope = k	Rate vs $[A]^2$: Linear through origin, Slope = k
Half-life vs Initial Conc.	$t_{1/2}$ vs $[A]_0$: Linear, Slope = $1/(2k)$	$t_{1/2}$ vs $[A]_0$: Horizontal Line (Slope = 0)	$t_{1/2}$ vs $1/[A]_0$: Linear, Slope = $1/k$

12. Crucial JEE Advanced Results & Special Conditions

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• **Pseudo-First-Order Reaction:** Reaction of higher order behaving as 1st order when one reactant is in large excess.

E.g., Ester hydrolysis: $CH_3COOCH_3 + H_2O \xrightarrow{H^+} CH_3COOH + CH_3OH$

$Rate = k'\,[Ester][H_2O] = k\,[Ester]$ where $k = k'\,[H_2O]$.

• **Ostwald's Isolation Method:** To find order w.r.t each reactant, take all other reactants in large excess so their concentrations remain constant.

• **Temperature Dependence of Activation Energy [Added]:**

From Collision Theory: $A \propto \sqrt{T} \Rightarrow k = C\, T^{1/2}\, e^{-E_a/(RT)}$

Differentiating: $d(\ln k)/dT = (E_a + \frac{1}{2} R\, T)/(R\, T^2)$

• **Radioactive Equilibrium (Secular Equilibrium) [Added]:**

For decay chain $A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C$ when parent A is very long-lived ($\lambda_A << \lambda_B$):

$$N_A\, \lambda_A = N_B\, \lambda_B \Rightarrow \frac{N_A}{t_{1/2,\,A}} = \frac{N_B}{t_{1/2,\,B}}$$

CHEMICAL BONDING & MOLECULAR STRUCTURE

JEE Main & Advanced Comprehensive Formula & Revision Sheet

CHEMISTRY

1. CHEMICAL BOND & ENERGY DYNAMICS

Basic Definitions & Energetics

- **Chemical Bond:** Force of attraction binding atoms/ions together. Always an **exothermic process** ($\Delta H < 0$).
- **Ionic Bond:** Electrostatic attraction formed by transfer of e^- (Electropositive + Electronegative).
- **Covalent Bond:** Equal sharing of e^- between electronegative atoms.
- **Co-ordinate (Dative) Bond:** One-sided sharing of a lone pair (Donor → Acceptor). E.g., $NH_3 \rightarrow BF_3$.
- **Interaction Strength:** Chemical Bond (~42 kJ/mol) > H-Bond (~10-40 kJ/mol) > Van der Waals (~8 kJ/mol).

Factors Favouring Ionic Bond & Properties

- **Low Ionization Energy (IE):** Metal atom $M_{(g)} \rightarrow M^+_{(g)} + e^-$ (Endothermic).
- **High Electron Affinity (EA):** Non-metal $X_{(g)} + e^- \rightarrow X^-_{(g)}$ (Exothermic).
- **High Lattice Energy (U):** $M^+_{(g)} + X^-_{(g)} \rightarrow MX_{(s)} + U$ (Exothermic).
- **Born-Haber Cycle [Added]:** $\Delta H_f = \Delta H_{sub} + \frac{1}{2}\Delta H_{diss} + IE + EA + U$.
- **Properties:** Hard, brittle, non-directional, high MP/BP, soluble in polar solvents, conducts in molten/aq. state.

2. FORMAL CHARGE, BOND PARAMETERS & DIPOLE MOMENT

Formal Charge (F.C.) Calculation

F.C. = V − L − ½(B)
V = valence e^- in free atom, *L* = non-bonding (lone pair) e^-, *B* = total bonding (shared) e^-.

Example: Ozone (O_3) [Structures: O_1=Double bond end, O_2=Central, O_3=Single bond end]:

- **O_1 (Terminal =):** 6 − 4 − ½(4) = **0**
- **O_2 (Central):** 6 − 2 − ½(6) = **+1**
- **O_3 (Terminal −):** 6 − 6 − ½(2) = **−1**

Dipole Moment (μ) & % Ionic Character

μ = q × d (Vector quantity; Directed from + to − / less EN to more EN atom)
Units: 1 Debye (D) = 10^{-18} e.s.u cm = **3.33×10^{-30} C·m**

- **Percentage Ionic Character:** $\% \, Ionic = (\mu_{observed} / \mu_{ionic}) \times 100$
- Where $\mu_{ionic} = q_e \times d$ = (4.8 × 10^{-10} e.s.u) × d (Å).
- **Net Dipole Moment (Vector Addition):** $\mu_{net} = \sqrt{(\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 cos\ \theta)}$ [Added]
- **Zero Dipole (μ = 0):** Symmetrical molecules (BeF_2, CO_2, BF_3, CH_4, CCl_4, SF_6, XeF_4, p-dichlorobenzene).
- **μ ≠ 0:** Unsymmetrical/Angular (H_2O, NH_3, NF_3, SO_2, o-dichlorobenzene).
- **Special Comparison: μ(NH_3) > μ(NF_3)** (In NH_3, orbital dipole of LP reinforces N-H bond dipoles; in NF_3, LP opposes N-F dipoles).

3. FAJAN'S RULE (COVALENT CHARACTER IN IONIC BONDS)

Polarisation ∝ Covalent Character ∝ 1 / Ionic Character
Cation distorts the electron cloud of Anion. Higher polarisation → Higher covalent character → Lower MP/BP & solubility.

1. Size of Cation:
Smaller Cation → Higher Polarising Power.
Order: LiCl > NaCl > KCl > RbCl > CsCl

2. Size of Anion:
Larger Anion → Higher Polarisability.
Order: AgI > AgBr > AgCl > AgF

3. Charge on Ions:
Higher Charge (+ or −) → Greater Polarisation.
Order: CCl_4 > BCl_3 > $BeCl_2$ > LiCl

4. Pseudo Inert Gas Configuration: Cations with 18 e^- in outer shell ($ns^2np^6nd^{10}$, e.g., Cu^+, Ag^+, Zn^{2+}) have greater polarising power than cations with 8 e^- (inert gas ns^2np^6, e.g., Na^+, K^+, Ca^{2+}) of similar size. *Example:* CuCl > NaCl (Covalent character).

4. VSEPR THEORY & STERIC NUMBER FORMULA

Steric Number (SN) / Electron Pairs (EP) = ½ [V + M − C + A]
V = Valence e^- of central atom, *M* = Monovalent surrounding atoms (H, F, Cl, Br, I), *C* = Positive charge, *A* = Negative charge.
Repulsion Order: Lone Pair - Lone Pair (LP-LP) > Lone Pair - Bond Pair (LP-BP) > Bond Pair - Bond Pair (BP-BP).

SN	Hybrid.	BP	LP	Expected Geometry	Actual Shape / Geometry	Bond Angle	Examples
2	sp	2	0	Linear	Linear	180°	BeF_2, $BeCl_2$, CO_2, HCN, C_2H_2, $[Ag(NH_3)_2]^+$
3	sp^2	3	0	Trigonal Planar	Trigonal Planar	120°	BF_3, BCl_3, $AlCl_3$, SO_3, C_2H_4, NO_3^-, CO_3^{2-}
3	sp^2	2	1	Trigonal Planar	V-shaped / Bent / Angular	<120° (~119°)	SO_2, $SnCl_2$, NO_2^-, O_3
4	sp^3	4	0	Tetrahedral	Tetrahedral	109°28'	CH_4, CCl_4, $SiCl_4$, NH_4^+, SO_4^{2-}, ClO_4^-, PO_4^{3-}
4	sp^3	3	1	Tetrahedral	Trigonal Pyramidal	~107°	NH_3, NF_3, PCl_3, PH_3, H_3O^+, XeO_3
4	sp^3	2	2	Tetrahedral	V-shaped / Bent / Angular	~104.5°	H_2O, H_2S, OF_2, SCl_2, NH_2^-, Cl_2O
4	dsp^2	4	0	Square Planar	Square Planar	90°	$[Ni(CN)_4]^{2-}$, $[PtCl_4]^{2-}$, $[Cu(NH_3)_4]^{2+}$
5	sp^3d	5	0	Trigonal Bipyramidal	Trigonal Bipyramidal (TBP)	90°, 120°	PCl_5, PBr_5, AsF_5, $SbCl_5$
5	sp^3d	4	1	Trigonal Bipyramidal	See-Saw / Distorted Tetrahedral	<90°, <120°	SF_4, SeF_4, IF_4^+
5	sp^3d	3	2	Trigonal Bipyramidal	T-Shape	~90° (~87.5°)	ClF_3, BrF_3, ICl_3
5	sp^3d	2	3	Trigonal Bipyramidal	Linear	180°	XeF_2, I_3^-, Br_3^-, ICl_2^-
6	sp^3d^2	6	0	Octahedral	Octahedral / Square Bipyramidal	90°	SF_6, SeF_6, PF_6^-, $[SnCl_6]^{2-}$
6	sp^3d^2	5	1	Octahedral	Square Pyramidal	<90°	IF_5, BrF_5, $XeOF_4$
6	sp^3d^2	4	2	Octahedral	Square Planar	90°	XeF_4, ICl_4^-
7	sp^3d^3	7	0	Pentagonal Bipyramidal	Pentagonal Bipyramidal (PBP)	72°, 90°	IF_7
7	sp^3d^3	6	1	Pentagonal Bipyramidal	Distorted Octahedral / Capped Oct.	Varies	XeF_6 (gas)
7	sp^3d^3	5	2	Pentagonal Bipyramidal	Pentagonal Planar	72°	XeF_5^-, IF_5^{2-}

5. VALENCE BOND THEORY (VBT) & ORBITAL OVERLAP

Sigma (σ) vs. Pi (π) Bonds

Property	Sigma (σ) Bond	Pi (π) Bond
Overlap	Head-on / Axial overlap	Sideways / Lateral overlap
Orbitals	s-s, s-p_z, p_z-p_z, hybrid	p_x-p_x, p_y-p_y, d-orbitals
Strength	Stronger (Extent of overlap high)	Weaker (Extent of overlap low)
Rotation	Free rotation permitted	Restricted rotation
Existence	Can exist independently	Exists only along with σ-bond
Shape	Determines molecular shape	Does not affect shape

Bent's Rule & Hybridization Rules [Added]

- **Bent's Rule:** More electronegative substituents prefer hybrid orbitals with *less s-character* (more p-character, axial in sp^3d). Lone pairs and multiple bonds prefer orbitals with *more s-character* (equatorial in sp^3d).
- **d-orbitals involved in Hybridization:**
  - sp^3d → **d_{z^2}** (Axial bonds longer than equatorial in PCl_5).
  - sp^3d^2 → **$d_{x^2-y^2}$ & d_{z^2}**
  - dsp^2 → **$d_{x^2-y^2}$**
  - sp^3d^3 → **d_{z^2}, $d_{x^2-y^2}$ & d_{xy}**
- **Hybridization in Hydrocarbons:** CH_3–CH_3 (sp^3), CH_2=CH_2 (sp^2), CH≡CH (sp), =C= (sp).

6. MOLECULAR ORBITAL THEORY (MOT)

LCAO & Energy Level Orders

Bond Order (B.O.) = ½ [N_b – N_a]

N_b = bonding electrons, N_a = anti-bonding electrons.

Properties: B.O. ∝ Bond Energy (B.E.) ∝ Stability ∝ 1 / Bond Length (B.L.)

- If B.O. > 0 → Species exists (Stable). If B.O. ≤ 0 → Species does NOT exist.
- Unpaired e^- → **Paramagnetic** | All paired e^- → **Diamagnetic**

Case 1: Total e^- ≤ 14 (H_2, Li_2, B_2, C_2, N_2)

$\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^*2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^*2p_x = \pi^*2p_y) < \sigma^*2p_z$

(s-p mixing occurs; $\sigma 2p_z$ is pushed higher than $\pi 2p_{x,y}$)

Case 2: Total e^- > 14 (O_2, F_2, Ne_2)

$\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^*2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^*2p_x = \pi^*2p_y) < \sigma^*2p_z$

(No s-p mixing; normal energy ordering)

MOT Quick Reference Table (Total e^- = 8 to 20)

Total e^-	8	9	10	11	12	13	14	15	16	17	18	19	20
Bond Order	0	0.5	1.0	1.5	2.0	2.5	**3.0**	2.5	2.0	1.5	1.0	0.5	0
Mag. Prop.	Dia	Para	**Para**	Para	Dia	Para	Dia	Para	**Para**	Para	Dia	Para	Dia

CRITICAL JEE EXCEPTIONS & SHORTCUTS:

- **B_2 ($10e^-$):** B.O. = 1, Paramagnetic (2 unpaired e^- in $\pi 2p_x$ and $\pi 2p_y$). Bonds are purely **π-bonds**.
- **C_2 ($12e^-$):** B.O. = 2, Diamagnetic. Both bonds in C_2 are **pure π-bonds** (4 e^- in $\pi 2p_{x,y}$).
- **O_2 ($16e^-$):** B.O. = 2, Paramagnetic (2 unpaired e^- in π^*2p_x and π^*2p_y).
- **CO ($14e^-$) & CO^+ ($13e^-$) [Added Exception]:** For CO, B.O. = 3. For CO^+, e^- is removed from non-bonding/weakly antibonding orbital, so **B.O. of CO^+ = 3.5** (B.L. of CO^+ < B.L. of CO).
- **Oxygen Species Stability Order:** O_2^+ (B.O.=2.5) > O_2 (B.O.=2.0) > O_2^- (B.O.=1.5) > O_2^{2-} (B.O.=1.0).
- **Bond Length Order:** O_2^{2-} > O_2^- > O_2 > O_2^+.

7. INTERMOLECULAR FORCES & HYDROGEN BONDING

Types of Intermolecular Forces & Energy Dependence

Hydrogen Bonding Details & Trends

Force Type	Relative Strength	Energy (E) vs Distance (r)
Ion-Ion	Very Strong	$E \propto 1/r$
Ion-Dipole	Moderate	$E \propto 1/r^2$
Dipole-Dipole (Keesom)	Weak (5-25 kJ/mol)	$E \propto 1/r^3$ (stationary) | $E \propto 1/r^6$ (rotating)
Dipole-Induced Dipole (Debye)	Weak	$E \propto 1/r^6$
London / Dispersion Force	Very Weak (0.05-20 kJ)	$E \propto 1/r^6$ (instantaneous dipole-induced)
Hydrogen Bonding	Moderate (10-100 kJ)	$E \propto 1/r^1$ to $1/r^2$

- **Condition:** H attached to highly EN atoms (F, O, N). Strength: F–H···F > O–H···O > N–H···N.
- **Intermolecular H-Bonding:** Between different molecules. Increases BP, MP, viscosity, surface tension, water solubility.
  Examples: H_2O, HF, NH_3, p-nitrophenol.
  • BP order: **H_2O > HF > NH_3** (H_2O forms 4 H-bonds/molecule, HF forms 2).
  • Hydrides BP order: $H_2O > H_2Te > H_2Se > H_2S$; HF > HI > HBr > HCl.
- **Intramolecular H-Bonding:** Within the same molecule (chelation / ring formation). Lowers BP/MP, increases volatility, decreases water solubility.
  Examples: o-nitrophenol, salicylaldehyde, o-hydroxybenzoic acid.
  • Volatility: o-nitrophenol (steam volatile) > p-nitrophenol.

8. SOLID-STATE AUTO-IONIZATION EXCEPTIONS & ODD-ELECTRON SPECIES

Solid-State Hybridization Table

Species	Cationic Part (Hybrid.)	Anionic Part (Hybrid.)
PCl_5 (s)	PCl_4^+ (sp^3 - Tetrahedral)	PCl_6^- (sp^3d^2 - Octahedral)
PBr_5 (s)	PBr_4^+ (sp^3 - Tetrahedral)	Br^-
PI_5 (s) [Added]	PI_4^+ (sp^3 - Tetrahedral)	I^-
XeF_6 (s)	XeF_5^+ (sp^3d^2 - Square Pyr.)	F^-
N_2O_5 (s)	NO_2^+ (sp - Linear)	NO_3^- (sp^2 - Trigonal Planar)
I_2Cl_6 (l)	ICl_2^+ (sp^3 - Bent)	ICl_4^- (sp^3d^2 - Square Planar)
Cl_2O_6 (s)	ClO_2^+ (sp^2 - Bent)	ClO_4^- (sp^3 - Tetrahedral)
I_2 (molten)	I_3^+ (sp^3 - Bent)	I_3^- (sp^3d - Linear)
N_2O_4 (s) [Added]	NO^+ (sp)	NO_3^- (sp^2)

Odd-Electron Species (Paramagnetic)

Species	Hybridization	Shape	Mag. Behaviour
NO_2	sp^2	V-shaped / Bent	Paramagnetic
ClO_2	sp^2	V-shaped / Bent	Paramagnetic
ClO_3	sp^3	Trigonal Pyramidal	Paramagnetic
$\cdot CH_3$	sp^2	Trigonal Planar	Paramagnetic
$\cdot CF_3$ / $\cdot CHF_2$	sp^3	Trigonal Pyramidal	Paramagnetic

**Note:* On dimerization, odd-e⁻ species usually become diamagnetic (e.g., $2NO_2 \rightarrow N_2O_4$).

9. RESONANCE & FRACTIONAL BOND ORDER FORMULAS

Resonance Fractional Bond Order Formula

B.O. = 1 + (No. of π-bonds / No. of σ-bonds)
Or: B.O. = Total no. of bonds between two atoms in all resonating structures / Total resonating structures.

Standard Examples:

Bond Length & Bond Angle Trends

- **Carbonate (CO_3^{2-}):** B.O. = 1 + 1/3 = **1.33**
- **Nitrate (NO_3^-):** B.O. = 1 + 1/3 = **1.33**
- **Sulphate (SO_4^{2-}):** B.O. = 1 + 2/4 = **1.50**
- **Perchlorate (ClO_4^-):** B.O. = 1 + 3/4 = **1.75**
- **Benzene (C_6H_6):** B.O. = 1 + 3/6 = **1.50**
- **Ozone (O_3):** B.O. = (2 + 1)/2 = **1.50**

- **Bond Length (B.L.):** B.L. $\propto$ Atomic Size | B.L. $\propto$ 1 / Bond Order.
  • Single Bond > Double Bond > Triple Bond (C–C: 1.54Å > C=C: 1.34Å > C≡C: 1.20Å).
  • C–H < C=C < C–O < C–C.
  • O–O B.L. order: H_2O_2 (1.48Å) > O_3 (1.28Å) > O_2 (1.21Å).
- **Bond Angle Trends:**
  1. *Hybridization:* sp (180°) > sp^2 (120°) > sp^3 (109.5°).
  2. *Lone Pairs (same hybrid.):* CH_4 (0 LP, 109.5°) > NH_3 (1 LP, 107°) > H_2O (2 LP, 104.5°).
  3. *Electronegativity of Central Atom ↑ → Bond Angle ↑:* $NH_3 > PH_3 > AsH_3 > SbH_3$; $H_2O > H_2S > H_2Se > H_2Te$.
  4. *Electronegativity of Surrounding Atom ↑ → Bond Angle ↓:* $NF_3 < NCl_3 < NBr_3 < NI_3$; $OF_2 < Cl_2O < Br_2O$.

10. JEE ADVANCED HIGH-YIELD RULES & EXCEPTIONS

Drago's Rule [Added]

If central atom is in **Period 3 or below** (P, S, As, Sb, etc.) bonded to less EN elements (like H), hybridisation does *not* take place. Bonding occurs via pure p-orbitals.
• Bond angle ≈ **90°** (PH_3 ~93.5°, H_2S ~92°).
• PH_3, AsH_3, H_2S, H_2Se do NOT undergo normal hybridization.

Back Bonding [Added]

Occurs when one atom has a lone pair (2p) and adjacent atom has an empty orbital (2p or 3d).
• **BF_3:** 2pπ-2pπ back bonding reduces Lewis acid strength: **$BF_3 < BCl_3 < BBr_3 < BI_3$**.
• **$N(SiH_3)_3$:** Planar (sp^2) due to 2pπ-3dπ back bonding; $N(CH_3)_3$ is Pyramidal (sp^3).

Silicates & Polymeric Species [Added]

Basic Unit: SiO_4^{4-} (Tetrahedral, sp^3).
• *Orthosilicates:* 0 shared O (SiO_4^{4-})
• *Pyrosilicates:* 1 shared O ($Si_2O_7^{6-}$)
• *Cyclic/Chain:* 2 shared O ($(SiO_3^{2-})_n$)
• *Sheet:* 3 shared O ($Si_2O_5^{2-})_n$
• *3D-Network:* 4 shared O (SiO_2)

Quick Formula Master-Checklist for JEE Numerical Problems

1. **Steric Number:** SN = ½[V + M − C + A]
2. **Formal Charge:** F.C. = V − L − ½(B)
3. **Dipole Moment:** μ = q × d ; % Ionic = (μ_{obs} / μ_{ionic}) × 100
4. **Bond Order (MOT):** B.O. = ½(N_b − N_a)
5. **Resonance B.O.:** B.O. = 1 + (No. of π-bonds / No. of σ-bonds)
6. **Fajan's Rule:** Polarisation $\propto$ Charge / Size(Cation) $\propto$ Size(Anion)
7. **Born-Haber Cycle:** $\Delta H_f = \Delta H_{sub} + ½\Delta H_{diss} + IE + EA + U$
8. **Bond Energy Relation:** B.E. $\propto$ B.O. $\propto$ 1 / B.L. $\propto$ Δ(EN)

BIOMOLECULES — JEE FORMULA & REVISION SHEET

Comprehensive Reference Sheet for JEE Main & Advanced | Chemistry

JEE MAIN / ADVANCED

1. CARBOHYDRATES: DEFINITION & CLASSIFICATION

Modern Definition: Optically active polyhydroxy aldehydes or polyhydroxy ketones, or substances which yield these on hydrolysis. Must possess at least one chiral carbon atom.
Old Definition & Limitations: Hydrates of carbon $C_n(H_2O)_y$. *Failures:* Formaldehyde (CH_2O) & Acetic Acid ($C_2H_4O_2$) fit formula but are NOT carbohydrates. Rhamnose ($C_6H_{12}O_5$) & Deoxyribose ($C_5H_{10}O_4$) are carbohydrates but do NOT fit.

Monosaccharides
Simplest sugars, cannot be hydrolysed further. General formula: $(CH_2O)_n$ where n = 3 to 7. All are **reducing sugars**.

Oligosaccharides
Yield 2 to 10 monosaccharide units on hydrolysis. Joined by *Glycosidic linkage* (C-O-C). E.g., Sucrose, Maltose, Lactose.

Polysaccharides
Non-sugars, amorphous, tasteless, insoluble in water. Yield >10 (up to 3000) units. E.g., Starch, Cellulose, Glycogen.

2. GLUCOSE (ALDOHEXOSE / DEXTROSE, $C_6H_{12}O_6$)

Preparation Methods

- **Laboratory (from Sucrose):**
  $C_{12}H_{22}O_{11} + H_2O \xrightarrow{(H^+/Alcohol)} C_6H_{12}O_6\ (Glucose) + C_6H_{12}O_6\ (Fructose)$
- **Commercial (from Starch):**
  $(C_6H_{10}O_5)_n + n\ H_2O \xrightarrow{(H^+,\ 393\ K,\ 2\text{-}3\ atm)} n\ C_6H_{12}O_6$

Structural Proofs (Open Chain)

- Straight chain of 6 carbons → HI / Δ gives **n-hexane**.
- Presence of >C=O group → Reacts with **NH_2OH** (oxime) & **HCN** (cyanohydrin).
- Carbonyl is Aldehyde (−CHO) → Mild oxidation with **Br_2 water** gives **Gluconic acid**.
- Five −OH groups → Acetylation with **Ac_2O** gives **Glucose Pentaacetate**.
- Presence of 1° Alcohol ($-CH_2OH$) → Oxidation with **conc. HNO_3** yields **Saccharic acid**.

Summary of Chemical Reactions of D-Glucose

Reagent / Conditions	Reaction Product	Structural Inference / Significance
HI / Red P, Δ	$CH_3-(CH_2)_4-CH_3$ (n-Hexane)	All 6 Carbon atoms are linked in a straight unbranched chain.
NH_2OH (Hydroxylamine)	Glucose Oxime: $CH{=}NOH-(CHOH)_4-CH_2OH$	Confirms presence of a Carbonyl (>C=O) group.
HCN (Hydrogen Cyanide)	Glucose Cyanohydrin: $CH(OH)CN-(CHOH)_4-CH_2OH$	Confirms Carbonyl group addition reaction.
Br_2 water (Mild Oxidising)	Gluconic Acid: $COOH-(CHOH)_4-CH_2OH$	Confirms Carbonyl group is an **Aldehyde** (−CHO).
Conc. HNO_3 (Strong Oxidising)	Saccharic Acid (Glucaric Acid): $COOH-(CHOH)_4-COOH$	Confirms presence of a primary alcohol ($-CH_2OH$) at terminal C-6.
Acetic Anhydride (Ac_2O) / Pyridine	Glucose Pentaacetate	Confirms 5 −OH groups attached to 5 different carbon atoms.
Tollen's Reagent / Fehling's Sol.	Gluconic Acid + Ag mirror / Red Cu_2O ppt	Proves Glucose is a **Reducing Sugar**.
3 $PhNHNH_2$ (Phenylhydrazine)	**Glucosazone** + $PhNH_2$ + NH_3 + 2 H_2O	Consumes C-1 and C-2. D-Glucose & D-Fructose give **SAME** Osazone!
Concentrated NaOH (Alkali)	Lobry de Bruyn-van Ekenstein Transformation	Reversible isomerization: D-Glucose • D-Fructose • D-Mannose via enediol.

3. CYCLIC STRUCTURES, ANOMERS & MUTAROTATION

Limitations of Open-Chain Structure

- Does NOT form 2,4-DNP adduct, Schiff's base, or $NaHSO_3$ adduct.
- Pentaacetate of glucose does NOT react with NH_2OH (indicates absence of free −CHO).
- Glucose exists in two crystalline forms: α (m.p. 419 K) and β (m.p. 423 K).

Anomers & Glycosidic Carbon

Anomers: Diastereomers differing in configuration ONLY at C-1 (anomeric/hemiacetal carbon).
• **α-D-Glucose:** −OH at C-1 is on the RIGHT (Fischer) / DOWN (Haworth).
• **β-D-Glucose:** −OH at C-1 is on the LEFT (Fischer) / UP (Haworth).

Mutarotation

Spontaneous change in specific rotation of an optically active compound in solution until equilibrium is reached.
• α-D-Glucose: initial $[a]_D$ = **+111.5°** (Δ 112°)
• β-D-Glucose: initial $[a]_D$ = **+19.5°** (Δ 19°)
• Equilibrium Mixture: **+52.7°** (≈64% β, 36% α, <0.5% open-chain).

Equilibrium Ratio Calculation:
% α = [(52.7 - 19.5) / (111.5 - 19.5)] × 100 ≈ 36%

4. EPIMERS & OSAZONE FORMATION [JEE ADVANCED]

Epimers: Monosaccharides differing in stereochemistry at a single chiral center *other than C-1*.
- **C-2 Epimers:** D-Glucose and D-Mannose.
- **C-4 Epimers:** D-Glucose and D-Galactose.

Note: All epimers are diastereomers, but not all diastereomers are epimers.

Osazone Formation Ratio:
1 mole Aldose/Ketose + **3 moles** $PhNHNH_2$ → **1 mole** Osazone + **1 mole** $PhNH_2$ + **1 mole** NH_3 + 2 H_2O.
C-1 and C-2 stereocenters are destroyed → D-Glucose, D-Fructose, and D-Mannose yield identical osazone crystals.

5. DISACCHARIDES & POLYSACCHARIDES

Carbohydrate	Monomeric Units	Glycosidic Linkage Type	Reducing / Non-Reducing	Key Features & Specific Rotation
Sucrose (Cane / Table Sugar)	α-D-Glucose + β-D-Fructose	**C_1 (α) – O – C_2 (β)** (Head-to-head)	**Non-Reducing Sugar** (Free hemiacetal absent)	Shows **Inversion of Sugar**. Hydrolysis changes rotation from Dextro (+66.5°) to Laevo (-39.9° net mixture).
Maltose (Malt Sugar)	α-D-Glucose + α-D-Glucose	**C_1 (α) – O – C_4**	**Reducing Sugar** (Free hemiacetal at C-1)	Produced by hydrolysis of starch by *Diastase* enzyme. Shows mutarotation.
Lactose (Milk Sugar)	β-D-Galactose + β-D-Glucose	**C_1 (β) – O – C_4**	**Reducing Sugar** (Free C-1 on glucose)	Galactose is C-4 epimer of glucose. Hydrolysed by *Lactase*.
Amylose (Starch fraction 15-20%)	α-D-Glucose linear chain	**C_1 (α) – O – C_4**	**Non-Reducing**	Water soluble. Gives intense **deep blue color** with Iodine (helical structure traps I_3^-).
Amylopectin (Starch fraction 80-85%)	α-D-Glucose branched polymer	Main chain: **C_1–C_4** Branching: **C_1–C_6** (every 20-25 units)	**Non-Reducing**	Water insoluble. Highly branched network.
Cellulose (Plant Cell Wall)	β-D-Glucose linear polymer	**C_1 (β) – O – C_4**	**Non-Reducing**	Human enzymes cannot hydrolyse β-linkages. Digested by ruminants via cellulase. High tensile H-bonding.
Glycogen (Animal Starch)	α-D-Glucose branched polymer	Main chain: **C_1–C_4** Branching: **C_1–C_6** (every 8-12 units)	**Non-Reducing**	Stored in liver and muscles. More highly branched than amylopectin.

Inversion of Cane Sugar (Quantitative Aspect)

Optical Rotation Data:
- Sucrose: $[a]_D$ = **+66.5°**
- D-Glucose product: $[a]_D$ = **+52.7°**
- D-Fructose product: $[a]_D$ = **-92.4°**

Net Mixture Rotation: (52.7 - 92.4)/2 = **-19.85°** (Laevorotatory). Because sign changes from (+) to (-), the hydrolyzed mixture is called **Invert Sugar**.

Kinetics of Inversion (Kinetics Connection):
Pseudo-first order reaction: $k = (2.303 / t) \log [(r_0 - r_\infty) / (r_t - r_\infty)]$
Where: r_0 = initial rotation, r_t = rotation at time t, r_∞ = final rotation at completion.

6. COMPREHENSIVE QUALITATIVE TESTS FOR CARBOHYDRATES

Test Name	Reagents & Mechanism	Positive Observation	Specificity / Target Carbohydrates
Molisch's Test	α-naphthol in ethanol + conc. H_2SO_4 (slowly added along sides)	**Purple / Violet ring** at junction of two liquid layers	**General Test for ALL Carbohydrates** (Dehydration to furfural derivatives).
Fehling's Test	Fehling A (aq. $CuSO_4$) + Fehling B (Rochelle salt + NaOH), heat	**Red precipitate of Cu_2O**	**All Reducing Sugars** (Monosaccharides & reducing disaccharides). Failed by Sucrose/Starch.
Tollen's Test	Ammoniacal Silver Nitrate $[Ag(NH_3)_2]^+$, warm	**Shining Silver Mirror (Ag)**	**All Reducing Sugars** (Aldoses & Ketoses via enediol rearrangement in alkali).
Benedict's Test	$CuSO_4$ + Sodium Citrate + Na_2CO_3 solution	**Red / Orange / Green ppt** of Cu_2O	**Reducing Sugars** (More stable reagent than Fehling's, used in urine glucose testing).
Barfoed's Test	Copper acetate in dilute acetic acid (Mild acidic medium)	**Red ppt of Cu_2O within 2-3 mins**	Differentiates **Monosaccharides** (Fast +ve) from **Reducing Disaccharides** (Slow/Negative).
Seliwanoff's Test	Resorcinol + conc. HCl, heat	**Cherry-red color** rapidly	Differentiates **Ketohexoses (Fructose)** from Aldohexoses (Glucose).
Iodine Test	Iodine solution in KI (I_3^- ions)	**Intense Deep Blue / Purple color**	Specific for **Starch (Amylose)**. Disappears on heating, reappears on cooling.

Test Name	Reagents & Mechanism	Positive Observation	Specificity / Target Carbohydrates
Bial's Test [JEE Adv]	Orcinol + $FeCl_3$ + conc. HCl	**Blue-green color**	Specific for **Pentoses** (RNA ribose sugar vs Hexoses).

7. CARBOHYDRATE CHAIN ALTERATION REACTIONS [JEE ADVANCED]

Kiliani-Fischer Synthesis (Chain Lengthening):
Aldose (n carbons) $-1.$ HCN$\rightarrow$ Cyanohydrins $-2.$ $Ba(OH)_2$, $H_2O\rightarrow$ Glyconic Acids $-3.$ Δ ($-H_2O$)$\rightarrow$ γ-Lactones $-4.$ Na-Hg / $H^+\rightarrow$ Next Higher Aldose (n+1 carbons).
E.g., D-Erythrose → D-Ribose + D-Arabinose.

Ruff Degradation (Chain Shortening):
Aldose (n carbons) $-1.$ $Br_2/H_2O\rightarrow$ Aldonic Acid $-2.$ $Ca(OH)_2\rightarrow$ Calcium salt $-3.$ $H_2O_2 + Fe^{3+}$ (Fenton's Reagent)$\rightarrow$ Next Lower Aldose (n-1 carbons) + CO_2.
E.g., D-Glucose → D-Arabinose.

8. AMINO ACIDS & ZWITTERION CHEMISTRY

General Formula of α-Amino Acids: $H_2N-CH(R)-COOH$. All natural α-amino acids are **L-configuration** at α-carbon (except **Glycine** which is achiral/optically inactive).
Zwitterion Equilibrium: In aqueous solution, carboxyl group loses H^+ and amino group accepts H^+.

$$H_3N^+-CH(R)-COOH \text{ (Cation in Acidic pH)} \bullet H_3N^+-CH(R)-COO^- \text{ (Zwitterion at pI)} \bullet H_2N-CH(R)-COO^- \text{ (Anion in Basic pH)}$$

Isoelectric Point (pI) Definition & Formulas:
The pH at which net electric charge of the amino acid is zero and it does not migrate under electric field.
- **Neutral AA (1 –COOH, 1 $-NH_2$):** $pI = (pK_{a1} + pK_{a2}) / 2$
- **Acidic AA (extra –COOH in R group):** $pI = (pK_{a1} + pK_{aR}) / 2$ (averages two lowest pK_a's)
- **Basic AA (extra $-NH_2$ in R group):** $pI = (pK_{a2} + pK_{aR}) / 2$ (averages two highest pK_a's)

Properties at pI: Minimum solubility, maximum precipitation, zero electrophoretic mobility.

Electrophoretic Migration:
- **pH < pI:** Amino acid exists predominantly as **Cation** → Migrates toward **Cathode (–)**.
- **pH > pI:** Amino acid exists predominantly as **Anion** → Migrates toward **Anode (+)**.
- **pH = pI:** No net migration towards any electrode.

9. COMPLETE TABLE OF 20 A-AMINO ACIDS (JEE STANDARD SET)

Name	3-Letter	1-Letter	Side Chain (–R) Group	Nature / Type	pI (approx)
Glycine	Gly	G	–H (Achiral, non-optically active)	Neutral / Non-essential	6.0
Alanine	Ala	A	$-CH_3$	Neutral / Non-essential	6.0
Valine*	Val	V	$-CH(CH_3)_2$	Neutral / **Essential**	6.0
Leucine*	Leu	L	$-CH_2-CH(CH_3)_2$	Neutral / **Essential**	6.0
Isoleucine*	Ile	I	$-CH(CH_3)-CH_2CH_3$ (2 chiral centers)	Neutral / **Essential**	6.0
Phenylalanine*	Phe	F	$-CH_2-C_6H_5$	Aromatic / **Essential**	5.5
Tryptophan*	Trp	W	$-CH_2-$(Indole ring)	Aromatic / **Essential**	5.9
Proline	Pro	P	Secondary amine cyclic ring (Pyrrolidine)	Imino acid / Non-essential	6.3
Methionine*	Met	M	$-CH_2-CH_2-S-CH_3$ (Thioether)	Sulfur-cont. / **Essential**	5.7
Cysteine	Cys	C	$-CH_2-SH$ (Forms Disulfide -S-S- bonds)	Sulfur-cont. / Non-essential	5.1
Serine	Ser	S	$-CH_2-OH$	Alcoholic / Non-essential	5.7
Threonine*	Thr	T	$-CH(OH)-CH_3$ (2 chiral centers)	Alcoholic / **Essential**	5.6
Tyrosine	Tyr	Y	$-CH_2-p-C_6H_4-OH$ (Phenolic)	Aromatic / Non-essential	5.7
Asparagine	Asn	N	$-CH_2-CONH_2$	Amide / Non-essential	5.4
Glutamine	Gln	Q	$-CH_2-CH_2-CONH_2$	Amide / Non-essential	5.7
Aspartic Acid	Asp	D	$-CH_2-COOH$	**Acidic** / Non-essential	**3.0**
Glutamic Acid	Glu	E	$-CH_2-CH_2-COOH$	**Acidic** / Non-essential	**3.2**
Lysine*	Lys	K	$-(CH_2)_4-NH_2$	**Basic** / **Essential**	**9.7**
Arginine*	Arg	R	$-(CH_2)_3-NH-C(=NH)NH_2$ (Guanidino)	**Basic** / **Essential**	**10.8**
Histidine*	His	H	$-CH_2-$(Imidazole ring)	**Basic** / **Essential**	**7.6**

* **10 Essential Amino Acids Mnemonic:** *"TV TILL AM PM"* (Trp, Val, Thr, Ile, Leu, Lys, Arg, Met, Phe, His). Must be supplied in diet.

10. PEPTIDES & PROTEINS ARCHITECTURE

Peptide Bond & Polypeptides

Peptide Linkage: Amide linkage **–CO–NH–** formed between –COOH of one AA and $-NH_2$ of another with loss of H_2O.

• Partial double bond character (40%) due to resonance → Planar rigid geometry.

• **N-Terminal:** Free $-NH_3^+$ group (written on LEFT).

• **C-Terminal:** Free $-COO^-$ group (written on RIGHT).

• A peptide with **n** amino acid residues contains **(n - 1)** peptide bonds.

Four Levels of Protein Structure

- **Primary Structure:** Specific sequence of amino acids linked by peptide covalent bonds. Defines identity.
- **Secondary Structure:** Spatial folding of polypeptide chain. Held by **Intramolecular H-bonds** (α-helix, right-handed) or **Intermolecular H-bonds** (β-pleated sheet).
- **Tertiary Structure:** Overall 3D folding of secondary structures. Stabilization: H-bonds, Disulfide (-S-S-) links, Van der Waals, Electrostatic forces.
- **Quaternary Structure:** Spatial arrangement of two or more polypeptide subunits (e.g., Hemoglobin).

Fibrous vs Globular Proteins:

• **Fibrous:** Polypeptide chains parallel, insoluble in water, held by strong H and disulfide bonds. E.g., *Keratin* (hair, wool), *Myosin* (muscles), *Collagen*.

• **Globular:** Chains coil into spherical shapes, soluble in water. E.g., *Insulin*, *Albumin*, *Hemoglobin*.

Denaturation of Proteins:

Disruption of 2° and 3° native structures by heat, pH change, or heavy metal salts. **Primary structure remains INTACT!**

Examples: Coagulation of egg white on boiling; curdling of milk (lactobacillus converting lactose to lactic acid).

11. COMPREHENSIVE QUALITATIVE TESTS FOR PROTEINS

Test Name	Reagents Used	Positive Observation	Structural Requirement / Specificity
Biuret Test	Dilute $CuSO_4$ + Alkaline NaOH	**Violet / Purple color**	Requires **at least 2 peptide bonds** (–CO–NH–). Free AA (except Thr) give negative test.
Xanthoproteic Test	Concentrated HNO_3, warm → add excess NaOH	**Yellow color** → turns **Orange** with base	Specific for **Aromatic amino acids** (Tyr, Phe, Trp) via nitration of benzene ring.
Ninhydrin Test	Ninhydrin reagent (triketohydrindene hydrate), heat	**Intense Blue / Purple color** (Ruhemann's Purple)	General test for **ALL α-amino acids**. *Proline gives Yellow color.*
Millon's Test	Mercuric nitrate + Mercurous nitrate in HNO_3	White ppt turning **Brick-Red** on heating	Specific for **Phenolic –OH group** (Tyrosine residue).
Lead Sulfide Test	NaOH + Lead Acetate $(Pb(OAc)_2)$, boil	**Black precipitate of PbS**	Specific for **Sulfur-containing amino acids** (Cysteine, Cystine). Methionine gives weak/negative.

12. NUCLEIC ACIDS (DNA & RNA)

Chemical Components

- **Pentose Sugar:**
  • **RNA:** β-D-Ribose.
  • **DNA:** β-D-2-Deoxyribose (lacks –OH at C-2').
- **Nitrogenous Bases:**
  • **Purines (Double ring):** Adenine (A), Guanine (G) [Both DNA & RNA].
  • **Pyrimidines (Single ring):** Cytosine (C), Thymine (T - *DNA only*), Uracil (U - *RNA only*).
- **Phosphoric Acid:** H_3PO_4 (forms phosphodiester bonds).

Linkages & Chargaff's Rules

Nucleoside vs Nucleotide:

• **Nucleoside** = Sugar + Nitrogenous Base (linked at **C-1'** via *N-Glycosidic bond*).

• **Nucleotide** = Nucleoside + Phosphate group (esterified at **C-5'** of sugar).

• **Polymer Backbone: 3',5'-Phosphodiester linkages** link nucleotides.

Chargaff's Rule (Molar Equivalences in Double-Stranded DNA):

[A] = [T] & [G] = [C] ⇒ **[Purines] = [Pyrimidines]** i.e., [A + G] = [T + C]

• A • T held by **2 Hydrogen Bonds**.

• G • C held by **3 Hydrogen Bonds** (higher stability & melting temp T_m).

13. COMPLETE VITAMINS SUMMARY TABLE

Vitamin	Chemical Name	Solubility Type	Deficiency Disease / Symptoms
Vitamin A	Retinol	Fat Soluble	Night Blindness (Nyctalopia), Xerophthalmia (hardening of cornea).
Vitamin B_1	Thiamine	Water Soluble	**Beri-Beri** (loss of appetite, nerve damage).
Vitamin B_2	Riboflavin	Water Soluble	Cheilosis (fissuring at corners of mouth), digestive disorders.
Vitamin B_3	Niacin / Nicotinic Acid	Water Soluble	**Pellagra** (dermatitis, diarrhea, dementia).
Vitamin B_6	Pyridoxine	Water Soluble	Convulsions, anemia.
Vitamin B_{12}	Cyanocobalamin	Water Soluble	**Pernicious Anemia** (RBC hemoglobin deficiency). Contains Cobalt metal.
Vitamin C	Ascorbic Acid	Water Soluble	**Scurvy** (bleeding gums), joint pain. Destroyed by heating.
Vitamin D	Ergocalciferol / Cholecalciferol	Fat Soluble	**Rickets** (in children, bone deformities), Osteomalacia (in adults).

Vitamin	Chemical Name	Solubility Type	Deficiency Disease / Symptoms
Vitamin E	Tocopherol	Fat Soluble	Increased fragility of RBCs, muscular weakness, loss of fertility.
Vitamin K	Phylloquinone / Menaquinone	Fat Soluble	Increased blood clotting time (hemorrhage).
Vitamin H	Biotin	**Neither** Fat nor Water soluble	Dermatitis, loss of hair.

14. HORMONES & JEE ADVANCED QUICK EXAM TIPS

Hormones Classification:
• **Steroid Hormones:** Estrogen, Progesterone, Testosterone, Cortisol, Aldosterone.
• **Peptide / Protein Hormones:** Insulin, Glucagon, Oxytocin, Vasopressin.
• **Amino Acid Derivatives:** Epinephrine (Adrenaline), Norepinephrine, Thyroxine (contains Iodine).

Must-Remember JEE Tricks:
1. *Reducing Sugars:* All Monosaccharides (Aldose + Ketose) + Maltose + Lactose. **Sucrose & Polysaccharides are Non-reducing!**
2. *Mutarotation:* Shown by compounds with a **free hemiacetal –OH** group.
3. *Isoelectric Point:* pI < 7 for Asp/Glu; pI > 7 for Lys/Arg/His.
4. *RNA Base Difference:* Uracil replaces Thymine. Sugar has –OH at C-2'.

AMINES & NITROGEN COMPOUNDS

JEE Main & Advanced High-Density Formula & Reaction Sheet

JEE REVISION SHEET

1. NITRO COMPOUNDS (R-NO_2 & AR-NO_2)

EAS & REDUCTIONS

A. Preparations of Nitro Compounds

Direct Nitration of Benzene:
$C_6H_6 + HNO_3$ **(conc. H_2SO_4)** $\rightarrow C_6H_5NO_2 + H_2O$
• *Electrophile:* ***NO_2^+*** *(Nitronium Ion)*
• *HNO_3 acts as a* ***base****, H_2SO_4 as an* ***acid****:*
$HNO_3 + 2H_2SO_4 \rightleftharpoons NO_2^+ + 2HSO_4^- + H_3O^+$
• *Solute particles increase (i > 1)* $\Rightarrow$ *Freezing Point decreases.*

Vapour Phase Nitration (Alkanes):
R-H + HNO_3 **(475 K – 675 K)** $\rightarrow$ R-NO_2 (Free radical, C-C cleavage occurs, gives mixture of mono-nitroalkanes).

From Alkyl Halides (Ambident Nucleophile):
R-X + $AgNO_2$ (alc.) $\rightarrow$ **R-NO_2** (Major) + R-O-N=O (Minor) + AgX
R-X + KNO_2 (aq/alc) $\rightarrow$ **R-O-N=O** (Major, Alkyl nitrite) + R-NO_2

From 3° Amines Oxidation:
$(CH_3)_3C\text{-}NH_2$ **$KMnO_4$** $\rightarrow (CH_3)_3C\text{-}NO_2 + H_2O$

B. Selective & Medium-Dependent Reductions of Nitrobenzene

Medium / Reagent	Product Obtained	Electrons / [H]
Acidic: Sn/HCl, Fe/HCl, Zn/HCl	**Aniline ($C_6H_5NH_2$)**	6 [H]
Neutral: Zn + NH_4Cl (aq.)	**Phenylhydroxylamine (C_6H_5NHOH)**	4 [H] (Baker-Mulliken Test)
Alkaline: Na_3AsO_3 + NaOH	**Azoxybenzene** (C_6H_5-N=N$^+$(O^-)-C_6H_5)	Coupled
Alkaline: Zn + KOH (alc.)	**Hydrazobenzene** (C_6H_5-NH-NH-C_6H_5)	Coupled
Alkaline: Zn + NaOH	**Azobenzene** (C_6H_5-N=N-C_6H_5)	Coupled
Selective Reduction: Na_2S / $(NH_4)_2S$	*m*-Dinitrobenzene $\rightarrow$ ***m*-Nitroaniline**	Zinin Reduction

Acidic α-H Reactions & Henry Condensation:
R-CH_2-NO_2 has acidic α-H due to strong -I/-M effect of -NO_2.
$CH_3NO_2 + CH_3CHO$ **dil. Base** $\rightarrow CH_3\text{-}CH(OH)\text{-}CH_2NO_2$ **$-H_2O$** $\rightarrow CH_3\text{-}CH{=}CH\text{-}NO_2$
$CH_3NO_2 + 3Cl_2 + NaOH \rightarrow$ **CCl_3NO_2** (Chloropicrin / Tear Gas)

2. CYANIDES (NITRILES) & ISOCYANIDES (ISONITRILES)

FUNCTIONAL ISOMERS

A. Preparations

From Alkyl Halides (Ambident Nucleophile):
R-X + KCN (alc.) $\rightarrow$ **R-C≡N** (Major) + R-N≡C (Minor)
R-X + AgCN (alc.) $\rightarrow$ **R-N≡C** (Major) + R-C≡N (Minor)

Dehydration of Amides & Oximes:
R-$CONH_2$ **P_4O_{10} / $SOCl_2$ / PCl_5, Δ** $\rightarrow$ R-C≡N + H_2O
R-CH=N-OH **P_4O_{10}, Δ** $\rightarrow$ R-C≡N + H_2O

From Grignard Reagent:
R-MgX + Cl-C≡N (Cyanogen Chloride) $\rightarrow$ **R-C≡N** + MgXCl

Carbylamine Reaction (R-NC only):
R-NH_2 + $CHCl_3$ + 3KOH (alc.) **Δ** $\rightarrow$ **R-NC** + 3KCl + $3H_2O$

B. Chemical Transformations & Reductions

Hydrolysis:
• R-CN **Partial Hydrolysis (dil. H_2SO_4/H_2O_2)** $\rightarrow$ **R-$CONH_2$** (Amide)
• R-CN **Complete Hydrolysis (H_3O^+ or OH^-, Δ)** $\rightarrow$ **R-COOH + NH_3**
• R-NC **Acidic Hydrolysis (H_3O^+)** $\rightarrow$ **R-NH_2 + HCOOH**
Note: R-NC is inert to alkali hydrolysis! HCOOH gives Silver Mirror test.

Reductions:
• **Mendius Reaction:** R-C≡N **Na / C_2H_5OH** $\rightarrow$ R-CH_2-NH_2 (1° amine)
• **Catalytic / $LiAlH_4$:** R-CN **$LiAlH_4$ or H_2/Ni** $\rightarrow$ R-CH_2-NH_2
• **Stephen Reduction:** R-CN **1. $SnCl_2$/HCl 2. H_2O, Δ** $\rightarrow$ R-CHO (Aldehyde)
• **Isocyanide Reduction:** R-N≡C **H_2/Pt or $LiAlH_4$** $\rightarrow$ **R-NH-CH_3** (2° amine)

Thermal Rearrangement & Grignard:
• R-N≡C **250 °C** $\rightarrow$ R-C≡N
• R'-C≡N + R-MgX $\rightarrow$ [R'-C(R)=NMgX] **H_3O^+** $\rightarrow$ **R'-CO-R** (Ketone) + NH_3

3. AMINES: CLASSIFICATION, STRUCTURE & BASICITY RULES

CORE CONCEPTS & TRENDS

A. Classification, Structure & Inversion

- **Structure:** Nitrogen is sp^3 hybridized, Pyramidal shape (bond angle ~108° for $(CH_3)_3N$).
- **Amine Inversion (Walden/Umbrella Inversion):** Amines with 3 different groups are chiral in theory, but optically inactive/unresolvable due to rapid pyramidal inversion (energy barrier ~25 kJ/mol).
- **Quaternary Ammonium Salts:** $[NR_1R_2R_3R_4]^+X^-$ have no lone pair to invert; thus optically active enantiomers **can** be resolved.

Basicity Constant (K_b & pK_b):
$R\text{-}NH_2 + H_2O \rightleftharpoons R\text{-}NH_3^+ + OH^- \Rightarrow K_b = [RNH_3^+][OH^-] / [RNH_2]$
Basic Strength ∝ K_b ∝ 1 / pK_b ∝ pK_a(conjugate acid)

B. Gas Phase vs. Aqueous Basicity Orders

1. Gas Phase / Non-polar Solvents (Inductive effect only):
3° Amine > 2° Amine > 1° Amine > NH_3

2. Aqueous Phase (Combined: Inductive + Solvation + Steric):
- **Methyl series (R = $-CH_3$):** 2° > 1° > 3° > NH_3
  ($Me_2NH > MeNH_2 > Me_3N > NH_3$) [pK_b: 3.27, 3.38, 4.19, 4.75]
- **Ethyl series (R = $-C_2H_5$):** 2° > 3° > 1° > NH_3
  ($Et_2NH > Et_3N > EtNH_2 > NH_3$) [pK_b: 3.00, 3.25, 3.29, 4.75]
- **Isopropyl series (R = $-CH(CH_3)_2$):** 2° > 1° > NH_3 > 3° [Added]
- ***tert*-Butyl series (R = $-C(CH_3)_3$):** NH_3 > 1° > 2° > 3°

Key Structural Basicity Rules for JEE:
1. **Delocalization:** Amides ($R\text{-}CONH_2$) & Aniline are weaker bases than aliphatic amines because N lone pair is in resonance.
2. **% s-character:** $R\text{-}CH_2\text{-}NH_2$ (sp^3, 25% s) > $R\text{-}CH{=}NH$ (sp^2, 33% s) > $R\text{-}C{\equiv}N$ (sp, 50% s). Higher % s-character ⇒ higher EN ⇒ weaker base.
3. **Guanidine Exception:** Guanidine $HN{=}C(NH_2)_2$ is extremely strong base (pK_a ~13.6) due to symmetric resonance stabilization of conjugate acid.
4. **Substituent effect on Aniline:** EDG ($-NH_2$, -OH, $-OCH_3$, $-CH_3$) increases basicity; EWG ($-NO_2$, -CN, -CHO, -X) decreases basicity.

4. PREPARATIONS OF AMINES & NAME REARRANGEMENTS

COMPREHENSIVE METHODS

A. Standard Preparations

1. Hofmann Ammonolysis of Alkyl Halides:
$R\text{-}X + NH_3$ (excess) **Sealed tube, Δ** → $R\text{-}NH_2$ (1° amine major)
If R-X is excess: $R\text{-}NH_2 \rightarrow R_2NH \rightarrow R_3N \rightarrow R_4N^+X^-$ (Quaternary salt).

2. Reduction Methods:
- Nitro: $R\text{-}NO_2$ **Fe/HCl or H_2/Pd or Sn/HCl** → $R\text{-}NH_2$
- Amides: $R\text{-}CONH_2$ **1. $LiAlH_4$ / Ether 2. H_2O** → **$R\text{-}CH_2\text{-}NH_2$** (Same C count)
- Nitriles: R-CN **$LiAlH_4$ or H_2/Ni** → $R\text{-}CH_2\text{-}NH_2$
- Oximes: $R\text{-}CH{=}NOH$ **$LiAlH_4$ or Na/EtOH** → $R\text{-}CH_2\text{-}NH_2$

3. Gabriel Phthalimide Synthesis (Pure 1° Aliphatic Amines):
Phthalimide **1. KOH (aq) 2. R-X 3. NaOH (aq), Δ or N_2H_4** → **R-NH_2** + Phthalated Salt.
Limitation: Ar-X cannot be used (aromatic halides fail SN2 reaction).

4. Reductive Amination:
$>C{=}O + NH_3$ **[$-H_2O$]** → $>C{=}NH$ (Imine) **H_2/Ni or $NaBH_3CN$** → $>CH\text{-}NH_2$

B. Carbon-Loss Rearrangements (1° Amines)

1. Hofmann Bromamide Degradation:
$R\text{-}CONH_2 + Br_2 + 4KOH$ **Δ** → **R-NH_2** + $K_2CO_3 + 2KBr + 2H_2O$
- *Intermediate: Isocyanate (**R-N=C=O**)*
- *Migrating R group retains stereochemical configuration.*
- *Rate determining step: Intramolecular rearrangement of nitrene/bromamide.*

2. Curtius Rearrangement:
$R\text{-}COCl + NaN_3$ **Δ, H_2O** → $[R\text{-}CO\text{-}N_3]$ **$-N_2$** → $R\text{-}N{=}C{=}O$ **H_2O** → **R-NH_2** + CO_2

3. Schmidt Reaction:
$R\text{-}COOH + HN_3$ (Hydrazoic acid) **conc. H_2SO_4, Δ** → **R-NH_2** + $CO_2 + N_2$

4. Lossen Rearrangement [Added]:
$R\text{-}CONH\text{-}OH$ (Hydroxamic acid) **Base/Heat** → $R\text{-}N{=}C{=}O$ → **R-NH_2**

5. CHEMICAL REACTIONS & DISTINCTION TESTS OF AMINES

REACTIONS & QUAL ANALYSIS

A. Distinguishing Tests (1°, 2°, 3° Amines)

1. Hinsberg's Test (Benzene Sulphonyl Chloride - $PhSO_2Cl$):
- **1° Amine:** $PhSO_2Cl + R\text{-}NH_2 \rightarrow PhSO_2NHR$ (Acidic H present) ⇒ **Soluble in KOH**.
- **2° Amine:** $PhSO_2Cl + R_2NH \rightarrow PhSO_2NR_2$ (No acidic H) ⇒ **Insoluble in KOH**.
- **3° Amine:** No reaction with $PhSO_2Cl$.

Note: p-toluenesulphonyl chloride can also be used.

2. Carbylamine Test (Isocyanide Test):
$R\text{-}NH_2$ / $Ar\text{-}NH_2 + CHCl_3 + 3KOH$ **Δ** → **R-NC / Ar-NC** (Foul smelling) + $3KCl + 3H_2O$
- *Intermediates: Dichlorocarbene (**$:CCl_2$**). Test given ONLY by 1° amines!*

B. Reactions with Nitrous Acid (HNO_2 / $NaNO_2$ + HCl)

- **1° Aliphatic Amine:** $R\text{-}NH_2$ **HNO_2 (0-5°C)** → $[R\text{-}N_2^+Cl^-]$ **H_2O** → **R-OH + $N_2\uparrow$** (quantitative evolution) + Alkyl halides + Alkenes.
- **1° Aromatic Amine:** $Ar\text{-}NH_2$ **$NaNO_2$ + HCl (0-5°C)** → **Ar-$N_2^+Cl^-$** (Stable Diazonium Salt).
- **2° Amine (Aliph/Arom):** $R_2NH + HONO \rightarrow$ **$R_2N\text{-}N{=}O$** (N-Nitrosamine, Yellow oily liquid, Liebermann's Nitroso test).
- **3° Aliphatic Amine:** $R_3N + HNO_2 \rightarrow [R_3NH]^+ONO^-$ (Clear soluble salt).
- **3° Aromatic Amine:** $C_6H_5N(CH_3)_2$ **HNO_2** → ***p*-Nitroso-N,N-dimethylaniline** (EAS product).

3. Hofmann Mustard Oil Test:
$R\text{-}NH_2 + CS_2$ **$HgCl_2$, Δ** → **R-N=C=S** (Alkyl isothiocyanate - mustard oil smell) + HgS (Black ppt) + 2HCl.

Hofmann Exhaustive Methylation & Elimination (E2):
$R\text{-}CH_2\text{-}CH_2\text{-}NH_2$ **1. CH_3I (excess) 2. Ag_2O/H_2O (Moist Ag_2O) 3. Δ** → **Hofmann Alkene** (Least substituted alkene major).

6. ANILINE ($AR\text{-}NH_2$): ELECTROPHILIC AROMATIC SUBSTITUTIONS

RING ACTIVATION & PROTECTION

A. Direct Electrophilic Reactions

1. Bromination (Unprotected):
Aniline + $3Br_2$ **(aq. Bromine water)** → **2,4,6-Tribromoaniline** (White ppt)
Note: $-NH_2$ is strongly activating (+M effect).

2. Nitration (Direct - Anomaly):
Aniline **conc. HNO_3 + conc. H_2SO_4 (288 K)** →
• ***p*-nitroaniline: 51%**
• ***m*-nitroaniline: 47%** (Due to formation of meta-directing Anilinium ion $C_6H_5NH_3^+$ in strongly acidic medium)
• ***o*-nitroaniline: 2%**

3. Sulphonation & Zwitter Ion:
Aniline + conc. H_2SO_4 → Anilinium hydrogen sulphate **453–473 K** → **Sulfanilic acid** ⇌ **Zwitterion** ($^+H_3N\text{-}C_6H_4\text{-}SO_3^-$).

B. Protection of $-NH_2$ Group (Monosubstitution)

Acetylation (Protecting Strategy):
$C_6H_5NH_2 + (CH_3CO)_2O$ **Pyridine** → **Acetanilide** ($C_6H_5NHCOCH_3$)
Lone pair of N is delocalized into carbonyl group; reduces activating power.

Mono-Bromination / Mono-Nitration Path:
1. Aniline **Ac_2O / Pyridine** → Acetanilide
2. Acetanilide **Br_2 / CH_3COOH or HNO_3/H_2SO_4 (288 K)** → *p*-Bromoacetanilide / *p*-Nitroacetanilide (Major)
3. Hydrolysis: **H^+ or OH^- / H_2O** → ***p*-Bromoaniline** or ***p*-Nitroaniline**.

Friedel-Crafts Reaction Failure:
Aniline does **NOT** undergo Friedel-Crafts alkylation or acylation because $AlCl_3$ (Lewis acid catalyst) forms a complex with basic - NH_2 group ($C_6H_5NH_2^+\text{-}AlCl_3^-$), deactivating the benzene ring.

7. ARENE DIAZONIUM SALTS ($AR\text{-}N_2^+CL^-$) & SYNTHETIC UTILITY

JEE MASTER FLOWCHART

Preparation (Diazotization): $C_6H_5NH_2 + NaNO_2 + 2HCl$ **273 – 278 K (0 – 5 °C)** → **$C_6H_5N_2^+Cl^-$** + NaCl + $2H_2O$

Target Product	Reagents & Reaction Name	Key Conditions / Notes
Chlorobenzene (C_6H_5Cl)	**Cu_2Cl_2 / HCl** (Sandmeyer) OR **Cu / HCl** (Gattermann)	Sandmeyer gives better yields than Gattermann.
Bromobenzene (C_6H_5Br)	**Cu_2Br_2 / HBr** (Sandmeyer) OR **Cu / HBr** (Gattermann)	Cu(I) salts act as radical catalysts.
Iodobenzene (C_6H_5I)	**KI, warm**	No catalyst needed; I^- is strong nucleophile.
Fluorobenzene (C_6H_5F)	**1. HBF_4 2. Δ** (Balz-Schiemann Reaction)	Isolates $ArN_2^+BF_4^-$ salt first, then thermolyzed.
Benzonitrile (C_6H_5CN)	**CuCN / KCN** (Sandmeyer Cyanation)	Useful for increasing C chain / acid precursor.
Nitrobenzene ($C_6H_5NO_2$)	**1. HBF_4 2. $NaNO_2$ / Cu, Δ**	Direct conversion of diazonium to nitro compound.
Benzene (C_6H_5H)	**$H_3PO_2 + H_2O$** OR **CH_3CH_2OH, Δ**	Reductive deamination (gives CH_3CHO / H_3PO_3 side product).
Phenol (C_6H_5OH)	**H_2O, Δ (warm to 283K - 333K)**	Evolves N_2 gas cleanly.

A. Azo Coupling Reactions (EAS mechanism)

1. Phenol Coupling (Orange Dye):
$Ar\text{-}N_2^+Cl^- + C_6H_5OH$ **pH = 9 – 10 (mildly basic)** → ***p*-hydroxyazobenzene** (Orange dye)
• *Coupling occurs at para-position relative to -OH.*

2. Aniline Coupling (Yellow Dye):
$Ar\text{-}N_2^+Cl^- + C_6H_5NH_2$ **pH = 4 – 5 (mildly acidic)** → ***p*-aminoazobenzene** (Yellow dye)

3. β-Naphthol Coupling Test:
$Ar\text{-}N_2^+Cl^-$ + 2-Naphthol **NaOH (aq)** → **Scarlet Red Dye**.

B. Basicity Order Summary [JEE Master List]

Decreasing Basicity Order (Aromatic & Heterocyclic):
1. **Guanidine** > Aliphatic 2°/1°/3° Amines > **NH_3**
2. **Imidazoline / Amidine** > **Pyridine** (sp^2, lone pair free) > **Aniline** (sp^3, delocalized)
3. **Aniline** > **Pyrrole** (lone pair part of 6π aromatic ring system)
4. **Substituted Anilines:**
p-anisidine > *p*-toluidine > Aniline > *p*-chloroaniline > *p*-nitroaniline
5. ***ortho*-effect in Anilines:** Almost all *ortho*-substituted anilines are **weaker bases** than aniline regardless of +I/+M effect of substituent due to steric strain and protonation inhibition!

Aldehydes, Ketones & Carboxylic Acids

JEE MAIN & ADVANCED REVISION FORMULA & REACTION SHEET

JEE REVISION SHEET

1. STRUCTURE & HYBRIDIZATION OF CARBONYL GROUP

Carbonyl C: sp^2 hybridized, trigonal planar (~120° bond angles).
Carbonyl O: sp^2 hybridized (contains 2 lone pairs in sp^2 orbitals).
Bonding: 1 σ-bond (C_{sp^2} – O_p) + 1 π-bond (C_p – O_p sidewise overlap).
Polarity: Dipolar structure $>C^{\delta+}=O^{\delta-} \leftrightarrow >C^{+}-O^{-}$ (μ = 2.3 – 2.8 D). High electrophilicity at C.

Boiling Point Trend:
Carboxylic Acids > Alcohols > Ketones > Aldehydes > Ethers > Alkanes
Reasoning: Acids form stable H-bonded dimers; Alcohols form H-bonds; Ketones/Aldehydes have strong dipole-dipole interactions.
Isomers: Among isomeric carbonyls, **Ketones have slightly higher B.P. than Aldehydes** due to higher dipole moment and polarizability.

2. GENERAL PREPARATIONS (ALDEHYDES & KETONES)

Reaction Name / Starting Material	Reagents & Conditions	Product & Important JEE Notes
Dry Distillation of Ca Salts	(i) $(HCOO)_2Ca \rightarrow \Delta$ (ii) $(HCOO)_2Ca + (RCOO)_2Ca \rightarrow \Delta$ (iii) $(RCOO)_2Ca \rightarrow \Delta$	(i) Formaldehyde (**HCHO**) + $CaCO_3$ (ii) Aldehyde (**RCHO**) + 2 $CaCO_3$ (iii) Symmetric Ketone (**R-CO-R**) + $CaCO_3$
Rosenmund Reduction	**R-CO-Cl** + H_2 over **Pd-BaSO$_4$**, S / Quinoline in boiling Xylene	**R-CHO** + HCl. Note: HCHO & Ketones CANNOT be prepared. $BaSO_4$ acts as catalyst poison.
From Nitriles & Esters	(i) Grignard: **R'-CN** + **RMgX** → (i) Ether (ii) H_3O^+ (ii) Stephen's: **R-CN** + $SnCl_2/HCl$ → H_3O^+ (iii) DIBAL-H: **R-CN** or **R-COOR'** → -78°C, H_3O^+	(i) Ketone (**R-CO-R'**) + NH_3 (ii) Aldehyde (**R-CHO**) via aldimine salt intermediate (iii) Aldehyde (**R-CHO**). DIBAL-H reduces esters/nitriles without reducing alkene bonds [Added]
Etard Reaction	Toluene + 2 CrO_2Cl_2 in CS_2 or CCl_4, then H_3O^+	Benzaldehyde (C_6H_5CHO). Brown chromium complex intermediate: $C_6H_5CH[OCrCl_2OH]_2$.
Oxidation of Methylbenzene	**Ar-CH$_3$** + CrO_3 in $(CH_3CO)_2O$ (273–278 K), then H_3O^+ / Δ	Benzaldehyde via Benzylidene diacetate intermediate: **Ar-CH(OCOCH$_3$)$_2$**. Prevents further oxidation to acid.
Wacker's Process	$CH_2=CH_2 + H_2O + PdCl_2 \rightarrow CuCl_2$ / Air	Acetaldehyde (CH_3CHO) + Pd + 2 HCl. (Higher alkenes give Ketones).
Hydroboration-Oxidation of Alkynes	(i) Terminal Alkyne: **R-C≡CH** → (i) $BH_3 \cdot THF$ (ii) H_2O_2/OH^- (ii) Internal Alkyne: **R-C≡C-R'** → Hydroboration	(i) Aldehyde (**R-CH$_2$CHO**) via Anti-Markovnikov enol tautomerization. (ii) Ketone (**R-CO-CH$_2$R'**).
Gattermann-Koch Reaction [Added]	C_6H_6 + CO + HCl in presence of anhydrous $AlCl_3$ / CuCl	Benzaldehyde (C_6H_5CHO). Electrophile is formyl cation HCO^+.

3. NUCLEOPHILIC ADDITION REACTIONS & AMMONIA DERIVATIVES

Reactivity Order (Nucleophilic Addition)

Order: HCHO > RCHO > ArCHO > R-CO-R > Ar-CO-R > Ar-CO-Ar
Key Factors:

- **Inductive (+I):** Alkyl groups release electrons, reducing δ+ on carbonyl carbon.
- **Steric Hindrance:** Bulkier groups block approach of nucleophile to tetrahedral intermediate.
- **Acid Catalysis:** Protonation of carbonyl O (**>C=O$^+$-H**) dramatically increases carbon's electrophilicity.

Addition of Alcohols, Diols & $NaHSO_3$

• **Alcohol Addition:** **RCHO** + **R'OH** ⇌ $HCl(g)$ Hemiacetal ⇌ $R'OH/H^+$ Acetal (**R-CH(OR')$_2$**). Used as carbonyl protecting group!
• **Ethylene Glycol:** Forms cyclic ketals/acetals (stable to base, hydrolyzed by dil. H^+).
• **$NaHSO_3$ Reaction:** Forms crystalline bisulfite adduct **>C(OH)(SO$_3$Na)**. Equilibrium lies right for aldehydes, left for bulky ketones (Purification test).

Nucleophilic Addition-Eliminated Derivatives: $>C=O + H_2N-Z \rightarrow >C=N-Z + H_2O$ (pH = 3.5 – 4.5)

Reagent Name	Reagent Structure (H_2N-Z)	Product Structure (>C=N-Z)	Product Name & JEE Test Significance
Hydroxylamine	NH_2OH	$>C=N-OH$	**Oxime** (Shows Geometrical Isomerism if unsymmetrical)
Hydrazine	NH_2-NH_2	$>C=N-NH_2$	**Hydrazone**
Phenylhydrazine	$C_6H_5-NH-NH_2$	$>C=N-NH-C_6H_5$	**Phenylhydrazone**
2,4-DNP (Brady's Reagent)	$2,4-(NO_2)_2C_6H_3-NH-NH_2$	$>C=N-NH-C_6H_3(NO_2)_2$	**2,4-DNP Hydrazone** (Yellow/Orange/Red ppt - Confirms >C=O group)
Semicarbazide	$NH_2-NH-CO-NH_2$	$>C=N-NH-CO-NH_2$	**Semicarbazone** (Only 1 NH_2 is nucleophilic; non-conjugated one!)

4. NAME REACTIONS & ALPHA-HYDROGEN CONDENSATIONS

Aldol Condensation

Requirement: Carbonyl compound with at least 1 α-H + Dilute Base ($dil.\ NaOH,\ Ba(OH)_2$).

Product: β-hydroxy aldehyde/ketone (Aldol) $\rightarrow \Delta\ (-H_2O)$ α,β-unsaturated carbonyl.

Mechanism Steps:

1. Base abstracts α-H → Resonance stabilized Enolate ion (Nucleophile).
2. Enolate attacks 2nd carbonyl molecule → Tetrahedral intermediate.
3. Dehydration on heating via E1cB mechanism (driven by conjugation).

Crossed Aldol: 2 different carbonyls give up to 4 products. If one component has NO α-H (e.g., $HCHO$, $PhCHO$), it acts exclusively as electrophile!

Cannizzaro Reaction

Requirement: Carbonyl compound with NO α-H + Concentrated Base ($50\%\ NaOH$ / KOH).

Reaction: Disproportionation (Self Redox).

$2\ ArCHO + 50\%\ NaOH \rightarrow ArCOONa\ (Oxidized) + ArCH_2OH\ (Reduced)$

Rate-Determining Step: Hydride ion (H^-) transfer from dianion intermediate to 2nd carbonyl carbon.

Crossed Cannizzaro: $PhCHO + HCHO + Conc.\ NaOH \rightarrow HCOONa + PhCH_2OH$ ($HCHO$ is always oxidized due to higher nucleophilic attack rate!).

5. OXIDATION REACTIONS & QUALITATIVE IDENTIFICATION TESTS

Test / Reagent Name	Reagent Composition & Reaction Formula	Observations & Specificity
Tollen's Test (Silver Mirror)	Ammoniacal Silver Nitrate: $[Ag(NH_3)_2]^+ + OH^-$ $RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$	**Bright Silver Mirror / Black ppt.** Positive for ALL Aldehydes (Aliphatic + Aromatic) & α-hydroxy ketones & Formic Acid. Ketones do NOT react.
Fehling's Test	Fehling A ($aq.\ CuSO_4$) + Fehling B ($Alk.\ Na\text{-}K\ Tartrate\ /\ Rochelle\ salt$) $RCHO + 2Cu^{2+} + 5OH^- \rightarrow RCOO^- + Cu_2O\downarrow + 3H_2O$	**Red-Brown precipitate (Cu_2O).** Positive for **Aliphatic Aldehydes ONLY**. Aromatic aldehydes ($PhCHO$) DO NOT reduce Fehling's!
Benedict's Test [Added]	$CuSO_4 + Sodium\ Citrate + Na_2CO_3$ $RCHO + 2Cu^{2+}\ (citrate\ complex) + 5OH^- \rightarrow RCOO^- + Cu_2O\downarrow$	**Red-Brown ppt**. Same behavior as Fehling's (Aliphatic aldehydes positive, aromatic negative).
Haloform Reaction (Iodoform Test)	$X_2 + NaOH$ (or $NaOX$) where $X = Cl, Br, I$ $R\text{-}CO\text{-}CH_3 + 3\ NaOX \rightarrow RCOONa + CHX_3\downarrow + 2\ NaOH$	**Yellow crystalline ppt (CHI_3, m.p. 119°C).** Specific for $CH_3\text{-}C=O$ group or $CH_3\text{-}CH(OH)\text{-}$ group (Ethanol, Acetaldehyde, Methyl Ketones, Isopropanol).
Popoff's Rule (Ketone Oxidation)	Strong Oxidizers: $KMnO_4/H^+$ or $K_2Cr_2O_7/H^+$ or conc. HNO_3 at high Δ	C-C bond cleavage in unsymmetrical ketones: **Carbonyl group remains with the smaller alkyl group** during oxidation cleavage.

6. COMPREHENSIVE REDUCTION REAGENTS MATRIX

Functional Group	LAH ($LiAlH_4$ in Ether)	$NaBH_4$ in H_2O/EtOH	B_2H_6 in THF	H_2 / Catalyst (Ni, Pd, Pt)	Clemmensen / Wolff-Kishner
Aldehyde (-CHO)	1° Alcohol (RCH_2OH)	1° Alcohol (RCH_2OH)	1° Alcohol (RCH_2OH)	1° Alcohol (RCH_2OH)	Alkane (RCH_3)
Ketone (>C=O)	2° Alcohol (R_2CHOH)	2° Alcohol (R_2CHOH)	2° Alcohol (R_2CHOH)	2° Alcohol (R_2CHOH)	Alkane (R_2CH_2)
Carboxylic Acid (-COOH)	1° Alcohol (RCH_2OH)	NO REACTION	1° Alcohol (Fast!)	1° Alcohol (Severe Δ)	Alkane (Red P + HI ONLY)
Ester (-COOR')	$RCH_2OH + R'OH$	NO REACTION	$RCH_2OH + R'OH$	$RCH_2OH + R'OH$	No direct reaction
Acid Chloride (-COCl)	1° Alcohol (RCH_2OH)	1° Alcohol (RCH_2OH)	1° Alcohol	1° Alcohol (or RCHO via Pd-$BaSO_4$)	Alkane
Alkene / Alkyne (C=C / C≡C)	NO REACTION (except Ph-CH=CH-CHO)	NO REACTION	Reduces to Alkane/Alcohol	Alkane ($-CH_2-CH_2-$)	Unaffected by Clemmensen

Clemmensen Reduction

$>C=O + Zn\text{-}Hg / conc.\ HCl \rightarrow >CH_2 + H_2O$

Condition: Acid-stable substrates.

Fails for acid-sensitive groups like -OH, -O-R.

Wolff-Kishner Reduction

$>C=O + NH_2NH_2 \rightarrow KOH / Ethylene\ Glycol,\ \Delta\ >CH_2 + N_2$

Condition: Base-stable substrates.

Fails for haloalkanes, esters, alkyl halides.

Mozingo / Red P + HI

- **Mozingo:** $>C=O + Dithiol \rightarrow Cyclic\ Dithioacetal \rightarrow Raney\ Ni / H_2\ >CH_2$ (Neutral condition!).
- **Red P + HI:** Converts EVERYTHING (-OH, -CHO, -COOH, >C=O) to Alkane!

7. CARBOXYLIC ACIDS: STRUCTURE, PREPARATION & PHYSICAL PROPERTIES

General Formula: Saturated Monocarboxylic Acids: $C_nH_{2n}O_2$.

Dimerization: Form stable 8-membered cyclic hydrogen-bonded dimers in gas phase & aprotic solvents (benzene). Increases M.W. determination by factor of 2!

Boiling Points: Exceptionally high due to extensive 2 H-bonds per dimer unit. Higher than corresponding alcohols!

Key Methods of Preparation

1. **Oxidation:** 1° Alcohol / Aldehyde $\rightarrow$ $Alk.\ KMnO_4$ or $K_2Cr_2O_7/H^+$ $RCOOH$.
2. **Grignard Carbonation:** $R\text{-}MgX + O=C=O \rightarrow (i)\ Dry\ Ether\ (ii)\ H_3O^+$ $R\text{-}COOH$ (+1 Carbon expansion!).
3. **Nitrile Hydrolysis:** $R\text{-}X \rightarrow alc.\ KCN$ $R\text{-}CN \rightarrow H_3O^+ / \Delta$ $R\text{-}COOH + NH_4^+$.
4. **Alkylbenzene Oxidation:** $Ar\text{-}R \rightarrow KMnO_4 / KOH, \Delta$ $Ar\text{-}COOK \rightarrow H^+$ $Ar\text{-}COOH$ (Requires at least 1 α-H on side chain!).

8. QUANTITATIVE ACIDIC STRENGTH & SUBSTITUENT EFFECTS

Acid Dissociation Equilibrium: $R\text{-}COOH + H_2O \rightleftharpoons R\text{-}COO^- + H_3O^+$ | $K_a = [RCOO^-][H_3O^+] / [RCOOH]$ | $pK_a = -\log_{10} K_a$

Fundamental Rule: Acidic strength $\propto$ Stability of Carboxylate Anion ($R\text{-}COO^-$).

- **Electron Withdrawing Groups (EWG: −I, −M):** Increase acidity by dispersing negative charge.
- **Electron Donating Groups (EDG: +I, +M):** Decrease acidity by intensifying negative charge.

Standard JEE Acidity Sequences (Must-Memorize for JEE Main & Advanced)

- **Aliphatic Acids:** $HCOOH > CH_3COOH > C_2H_5COOH > (CH_3)_2CHCOOH > (CH_3)_3CCOOH$ (Due to +I of alkyls).
- **Halogen-Substituted Acids:** $F\text{-}CH_2COOH > Cl\text{-}CH_2COOH > Br\text{-}CH_2COOH > I\text{-}CH_2COOH$ (−I power effect).
- **Number of Halogens:** $Cl_3C\text{-}COOH$ (pKa 0.65) $> Cl_2CH\text{-}COOH$ (pKa 1.29) $> ClCH_2\text{-}COOH$ (pKa 2.86) $> CH_3COOH$ (pKa 4.76).
- **Distance of EWG:** $CH_3\text{-}CH_2\text{-}CH(Cl)\text{-}COOH > CH_3\text{-}CH(Cl)\text{-}CH_2\text{-}COOH > CH_2(Cl)\text{-}CH_2\text{-}CH_2\text{-}COOH$ (−I decreases with distance).
- **Comparison Across Classes:** $R\text{-}COOH > H_2CO_3 > Ph\text{-}OH > H_2O > R\text{-}OH > HC{\equiv}CH > NH_3$.

The Ortho Effect in Benzoic Acids

Definition: Nearly ALL **ortho-substituted benzoic acids are STRONGER acids than benzoic acid**, regardless of whether the ortho substituent is electron-withdrawing or electron-donating!
Cause: Steric Inhibition of Resonance (SIR). The bulky ortho group forces the **-COOH** group out of the benzene ring plane, preventing cross-conjugation of benzene with carbonyl, thereby maximizing carboxylate anion resonance stability.
Specific Benzoic Acid Acidity Orders:
1. **Nitrobenzoic Acids:** ortho > meta > para > Benzoic acid
2. **Hydroxybenzoic Acids:** ortho (Salicylic acid, intramolecular H-bond in anion) > meta > para > Benzoic acid
3. **Chlorobenzoic Acids:** ortho > meta > para > Benzoic acid
4. **Methylbenzoic Acids (Toluic Acids):** ortho > Benzoic acid > meta > para

9. IMPORTANT CHEMICAL REACTIONS OF CARBOXYLIC ACIDS

Hell-Volhard-Zelinsky (HVZ) Reaction

$R\text{-}CH_2\text{-}COOH + X_2 \xrightarrow{\text{Red P (cat.)}} R\text{-}CH(X)\text{-}COOH + HX$ (X = Cl, Br)

- Substitutes α-hydrogen with halogen.
- Requires presence of α-H (Formic & Benzoic acid do NOT give HVZ!).
- Intermediate is acyl halide / enol form.

Schmidt Reaction & Decarboxylation

- **Schmidt Reaction:** $R\text{-}COOH + HN_3$ (Hydrazoic acid) $\xrightarrow{\text{conc. } H_2SO_4} R\text{-}NH_2 + N_2\uparrow + CO_2\uparrow$ (1° amine with 1 carbon less via Nitrene intermediate).
- **Soda Lime Decarboxylation:** $R\text{-}COONa + NaOH + CaO \xrightarrow{\Delta} R\text{-}H + Na_2CO_3$ (CaO keeps NaOH dry & raises reaction temp).
- **Gem-dicarboxylic acid heating:** $R\text{-}CH(COOH)_2 \xrightarrow{\Delta} R\text{-}CH_2COOH + CO_2\uparrow$.

10. CARBOXYLIC ACID DERIVATIVES: STRUCTURE & NUCLEOPHILIC ACYL SUBSTITUTION

Nucleophilic Acyl Substitution (S_N2th / Addition-Elimination) Mechanism

$R\text{-}CO\text{-}L + Nu^- \rightleftarrows [R\text{-}C(O^-)(Nu)(L)]$ (Tetrahedral Intermediate) $\rightarrow R\text{-}CO\text{-}Nu + L^-$
Reactivity Order Towards Nucleophilic Sub: Acyl Chlorides > Acid Anhydrides > Esters > Amides
Reasoning: Dependent on leaving group ability (Cl^- > $RCOO^-$ > $R'O^-$ > NH_2^-), basicity of leaving group, and resonance stabilization of carbonyl carbon by substituent.

11. REACTIONS & INTERCONVERSIONS OF ACID DERIVATIVES

Derivative Class	Preparation Method	Hydrolysis (H_2O)	Alcoholysis (R'OH)	Ammonolysis (NH_3)	Special Name Reactions / JEE Notes
Acyl Halides (RCOCl)	$RCOOH + SOCl_2$ or PCl_5 or PCl_3	$RCOOH + HCl$ (White fumes)	Ester (RCOOR') + HCl	Amide ($RCONH_2$) + HCl	**Rosenmund:** $RCOCl \rightarrow RCHO$ **Gilman:** $RCOCl + R'_2CuLi \rightarrow R\text{-}CO\text{-}R'$ (Ketone).
Acid Anhydrides ($(RCO)_2O$)	$2\ RCOOH \xrightarrow{P_2O_5/\Delta} (RCO)_2O + H_2O$	$2\ RCOOH$	Ester + RCOOH	Amide + RCOOH	**Friedel-Crafts Acylation:** $Ph\text{-}H + (RCO)_2O \xrightarrow{AlCl_3} Ph\text{-}CO\text{-}R$. Higher B.P. than acid chlorides.
Esters (RCOOR')	$RCOOH + R'OH \rightleftarrows[\text{conc. } H_2SO_4] RCOOR'$	$RCOOH + R'OH$ (Acid/Base saponification)	Trans-esterification (RCOOR")	Amide ($RCONH_2$)	**Claisen Condensation:** 2 Ethyl acetate $\xrightarrow{EtONa}$ Ethyl acetoacetate. **Bouvault-Blanc:** Esters $\xrightarrow{Na/EtOH}$ 1° Alcohols.
Amides ($RCONH_2$)	$RCOOH + NH_3 \xrightarrow{\Delta} RCONH_2$	$RCOOH + NH_4^+$ (Slow, needs Δ)	Slow esterification	Self exchange	**Hofmann Bromamide:** $RCONH_2 + Br_2 + 4\ NaOH \rightarrow R\text{-}NH_2 + Na_2CO_3$ via Isocyanate. **Dehydration:** $RCONH_2 \xrightarrow{P_4O_{10}/\Delta} R\text{-}CN$.

Boiling Point Order of Acid Derivatives

Sequence: $Amides\ (RCONH_2) > Acid\ Anhydrides > Carboxylic\ Acids > Esters > Acid\ Chlorides$

• **Amides:** Highest B.P. due to strong interstitial, 3D intermolecular H-bonding networks.
• **Acyl Chlorides:** Lowest B.P. due to lack of H-bonding & weak C-Cl dipole attraction.

Amphoteric Nature of Amides

• **As Base:** Protonation occurs on **OXYGEN** (not Nitrogen!) to yield resonance stabilized cation: $[R\text{-}C(OH)=NH_2^+]$.
• **As Acid:** Reacts with strong bases: $RCONH_2 + NaOH \rightarrow RCONNa + H_2O$.

12. JEE ADVANCED SPECIAL REACTIONS & STRUCTURAL SHORTCUTS

Wittig Reaction [Added]

$>C=O + Ph_3P=CH\text{-}R \rightarrow >C=CH\text{-}R + Ph_3P=O$

Converts carbonyl directly into alkene with complete control over double bond position without rearrangement!

Baeyer-Villiger Oxidation [Added]

$Ketone + mCPBA\ (or\ RCO_3H) \rightarrow Ester$

Migratory Aptitude: 3° alkyl > 2° alkyl ~ Cyclohexyl > Benzyl > Phenyl > 1° alkyl > Methyl.

Formic Acid Special Behavior

HCOOH has BOTH aldehyde (**-CHO**) and acid (**-COOH**) groups!
Gives positive **Tollen's, Fehling's, Benedict's, & $HgCl_2$ tests** (reduces $HgCl_2$ to white Hg_2Cl_2 ppt then gray **Hg** ppt).

JEE Revision Checklist & Formula Traps:
1. **Aldol vs Cannizzaro:** Check for α-H! If α-H present → Aldol; No α-H → Cannizzaro. (Exception: $(CH_3)_3C\text{-}CHO$ gives Cannizzaro despite bulky alkyls).
2. **Semicarbazide Trap:** $H_2N^{(a)}\text{-}NH\text{-}CO\text{-}NH_2^{(b)}$. Only $NH_2^{(a)}$ acts as nucleophile because $NH_2^{(b)}$ is in resonance with C=O!
3. **Formic Acid vs Sulphuric Acid:** $HCOOH \xrightarrow{conc.\ H_2SO_4,\ \Delta} CO\uparrow + H_2O$ | $(COOH)_2 \xrightarrow{conc.\ H_2SO_4,\ \Delta} CO\uparrow + CO_2\uparrow + H_2O$.

ALCOHOLS, PHENOLS & ETHERS

Complete JEE Main & Advanced Formula, Reaction & Concept Revision Sheet

JEE ULTIMATE REVISION

1. STRUCTURE, PHYSICAL PROPERTIES & ACIDITY OF ALCOHOLS

CORE CONCEPT

Physical Properties & Trends

- **Boiling Point:** Alcohols > Haloalkanes (due to intermolecular H-bonding).
  - Homologous series: B.P. ↑ as C-atoms ↑
  - Isomeric alcohols: B.P. ↓ as Branching ↑ (surface area ↓, van der Waals forces ↓).
- **Solubility:** Lower alcohols are completely miscible with H_2O due to H-bonding.
  - Homologous series: Solubility ↓ as alkyl chain length ↑
  - Isomeric alcohols: Branching ↑ ⇒ Solubility ↑ (t-butanol > n-butanol).

Acidic Strength & Bond Cleavage

- **Acidity:** Alcohols are weaker acids than H_2O (O–H bond less polar).
  - Order of Acidity: **1° (CH_3OH) > 1° (RCH_2OH) > 2° (R_2CHOH) > 3° (R_3COH)**
  - Reason: +I effect of alkyl groups increases e^- density on Oxygen, weakening H^+ release.
- **O–H Bond Strength Order:** 1° < 2° < 3°
- **C–O Bond Strength Order:** 3° < 2° < 1°
- **Reactivity towards C–O cleavage:** 3° > 2° > 1° (requires Lewis acid as OH^- is a poor leaving group).

2. METHODS OF PREPARATION OF ALCOHOLS

REACTIONS & MECHANISMS

Method / Reaction	Reagents & Conditions	Regioselectivity / Stereochemistry	Mechanism & Key JEE Notes
Acid Catalyzed Hydration	dil. H_2SO_4 / H^+, H_2O	Markovnikov's Addition	Proceeds via **Carbocation Intermediate**. Rearrangement (1,2-H^- or 1,2-alkyl shift) takes place if more stable carbocation is possible.
Oxymercuration-Demercuration (OMDM)	(i) $Hg(OOCCH_3)_2$, H_2O / THF (ii) $NaBH_4$, OH^-	Markovnikov's Addition (Overall)	**No Carbocation Rearrangement!** Cyclic mercurinium ion intermediate. OH comes from H_2O, H comes from $NaBH_4$.
Hydroboration-Oxidation (HBO)	(i) B_2H_6 / THF or BH_3·THF (ii) H_2O_2, OH^-	Anti-Markovnikov's Addition	**Syn-addition** of H and OH across alkene. Trialkylborane intermediate. No rearrangement. OH comes from H_2O_2.
Reduction of Carbonyls & Acids	$LiAlH_4$ / H_3O^+ OR $NaBH_4$ OR H_2/Ni, Pd, Pt	Product: 1° or 2° Alcohols	• Aldehydes ⇒ 1° Alcohol | Ketones ⇒ 2° Alcohol • $NaBH_4$ reduces ONLY aldehydes, ketones, and acid chlorides (NOT acids/esters). • $LiAlH_4$ reduces aldehydes, ketones, acids, esters, and acid chlorides.
Bouveault-Blanc Reduction	Na / C_2H_5OH	R–COOR' ⇒ RCH_2OH + R'OH	Reduction of esters, aldehydes, or ketones using Na in alcohol.
Grignard Addition	(i) R"MgX / Dry Ether (ii) H_3O^+	Nu-addition to C=O	• Formaldehyde (HCHO) + RMgX ⇒ **1° Alcohol** • Other Aldehydes (R'CHO) + RMgX ⇒ **2° Alcohol** • Ketones (R'COR") + RMgX ⇒ **3° Alcohol** • Esters (RCOOR') + 2 eq. RMgX ⇒ **3° Alcohol** (or 2° if Formate ester).
Industrial Preparation	• $CO + 2H_2 \rightarrow CH_3OH$ (Cu-ZnO-Cr_2O_3 catalyst, high T, P) • Fermentation: $C_{12}H_{22}O_{11} + H_2O$ &xrightarrow{Invertase} Glucose + Fructose &xrightarrow{Zymase} 2 C_2H_5OH + 2 CO_2	Commercial Methanol & Ethanol	• CH_3OH = Wood Spirit (highly poisonous). • Denatured Alcohol: Ethanol mixed with $CuSO_4$ (color) & Pyridine (foul smell).

3. CHEMICAL REACTIONS & TESTS OF ALCOHOLS

REAGENTS & DISTINCTIONS

C–OH Cleavage & Substitution Reactions

- **HX Reaction:**
  $R-OH + HX \xrightarrow{ZnCl_2} R-X + H_2O$
  Reactivity of HX: HI > HBr > HCl. Reactivity of ROH: 3° > 2° > 1°.
- **PCl_5 / PCl_3 / PBr_3:**
  - $ROH + PCl_5 \rightarrow RCl + POCl_3 + HCl$
  - $3ROH + PX_3 \rightarrow 3RX + H_3PO_3$ (X = Cl, Br; P+I_2 for RI).
- **Darzen's Process (Best Method):**
  - $ROH + SOCl_2 \xrightarrow{Pyridine} RCl + SO_2\uparrow + HCl\uparrow$
    (S_N2 mechanism, inversion of configuration).
  - In Ether/Dioxane (No Pyridine): $ROH + SOCl_2 \rightarrow RCl$ (S_Ni mechanism, retention of configuration).
- **Tosylation:** $ROH + TsCl / Pyridine \rightarrow R-OTs$ (Retains C–O stereochemistry; converts OH into excellent leaving group).

Lucas Test & Dehydration

- **Lucas Reagent:** Conc. HCl + Anhydrous $ZnCl_2$ (Distinguishes 1°, 2°, 3° ROH via turbidity of RCl).
  - **3° Alcohol:** Immediate turbidity at room temperature.
  - **2° Alcohol:** Turbidity within 5–10 minutes.
  - **1° Alcohol:** No turbidity at room temperature (appears only on heating).
- **Acidic Dehydration:** Conc. H_2SO_4 or H_3PO_4 / Δ
  - Order of ease: **3° > 2° > 1°** (via Carbocation intermediate ⇒ Saytzeff Alkene as major product).
  - Temperature Dependence (Ethanol):
    25°C ⇒ $CH_3CH_2OSO_3H$ (Ethyl hydrogen sulphate)
    140°C (413 K) ⇒ $CH_3CH_2OCH_2CH_3$ (Diethyl ether, S_N2)
    170°C (443 K) ⇒ $CH_2=CH_2$ (Ethene, E1 elimination)

Oxidation & Dehydrogenation Summary Table

Reagent / Conditions	Primary (1°) Alcohol	Secondary (2°) Alcohol	Tertiary (3°) Alcohol
PCC / PDC / Swern Oxidation [Anhydrous]	Aldehyde (RCHO) — *Stops at aldehyde stage*	Ketone (RCOR')	No Reaction
$K_2Cr_2O_7$ / H^+ OR $KMnO_4$ / H^+ OR Jones Reagent (CrO_3/aq H_2SO_4)	Carboxylic Acid (RCOOH) — *Same # of C atoms*	Ketone (RCOR') &xrightarrow{Vigorous [O]} Acids with *fewer C atoms* (Popoff's Rule)	No oxidation in neutral/alkaline. Vigorous acidic oxidation gives ketone/acids with *fewer C atoms*.
Cu / 573 K (300°C) (Vapour Phase Dehydrogenation)	Aldehyde (RCHO) + $H_2\uparrow$	Ketone (RCOR') + $H_2\uparrow$	**Dehydration occurs:** Alkene + H_2O (e.g., t-butanol ⇒ 2-Methylpropene)

4. POLYHYDRIC ALCOHOLS: ETHYLENE GLYCOL & GLYCEROL

SPECIAL REACTIONS

Ethylene Glycol (Ethane-1,2-diol)

Glycerol (Propane-1,2,3-triol)

- **Preparation:** Saponification of Triglycerides (Fats/Oils) with NaOH ⇒ Glycerol + 3 RCOONa (Soap).
- **Key Reactions:**
  - $Glycerol + Conc.\ H_2SO_4\ /\ \Delta \rightarrow Acrolein\ (CH_2=CH-CHO)$
    (Pungent smell test for glycerol).
  - $Glycerol + 3\ HNO_3\ /\ Conc.\ H_2SO_4 \rightarrow Glyceryl\ Trinitrate\ (Nitroglycerine)$
  - **With HI:**
    - *Limited HI:* Allyl Iodide ($CH_2=CH-CH_2I$).
    - *Excess HI:* Isopropyl Iodide ($CH_3-CHI-CH_3$).
  - $Glycerol + Fenton's\ Reagent\ (FeSO_4 + H_2O_2) \rightarrow Glycerose$
    (Dihydroxyacetone + Glyceraldehyde).
  - $Glycerol + HIO_4 \rightarrow 2\ HCHO + HCOOH$ (Cleaves 2 C–C bonds).

- **Preparation:**
  • $CH_2=CH_2$ + Cold dil. $KMnO_4$ (Bayer's Reagent) ⇒ Glycol (Syn-hydroxylation).
  • $CH_2=CH_2$ + HOCl ⇒ Cl–CH_2–CH_2–OH &xrightarrow{aq. NaHCO_3} Glycol.
  • $CH_2=CH_2$ + O_2/Ag (400 K) ⇒ Ethylene Oxide &xrightarrow{H_3O^+} Glycol (Anti-hydroxylation).
- **Key Reactions:**
  • `Glycol + 2 HI → CH2I-CH2I [Unstable] → CH2=CH2 + I2`
  • `Glycol + HIO4 or Pb(OAc)4 → 2 HCHO` (Periodic oxidation / C–C cleavage of 1,2-diols).
  • `Glycol + Conc. HNO3/H2SO4 → Glycol Dinitrate` (Explosive).
  • `Glycol &xrightarrow{Cu / 573 K} Glyoxal (CHO-CHO)`
  • `Glycol &xrightarrow{Conc. K_2Cr_2O_7 / H^+} Oxalic Acid (COOH-COOH)`
- **Pinacol-Pinacolone Rearrangement** [JEE Adv] :
  Vicinal diol (Pinacol) + H^+ ⇒ Protonation → Loss of H_2O → Carbocation Formation → 1,2-Shift (Me/Ph shift) → Protonated Ketone → Pinacolone.
  Migratory Aptitude: Ar > H > Alkyl (p-anisyl > phenyl > p-nitrophenyl; 3° > 2° > 1° alkyl).

5. PHENOLS: PREPARATION & ACIDIC NATURE

HIGH PRIORITY JEE TOPIC

Preparation of Phenol (C_6H_5OH)

- **1. From Cumene (Cumene Hydroperoxide Process - Industrial):**
  `Benzene + CH3CH=CH2 &xrightarrow{H_3PO_4} Cumene &xrightarrow{O_2 / h\nu} Cumene Hydroperoxide &xrightarrow{H_3O^+} Phenol + Acetone (CH3COCH3)`
  Note: Acetone is an important byproduct!
- **2. Dow's Process (From Chlorobenzene):**
  `C6H5Cl + NaOH &xrightarrow{623 K, 300 atm} C6H5ONa &xrightarrow{dil. HCl} C6H5OH`
  (Benzyne intermediate mechanism).
- **3. From Benzene Sulphonic Acid:**
  `C6H6 &xrightarrow{Oleum} C6H5SO3H &xrightarrow{NaOH (fuse), 573 K} C6H5ONa &xrightarrow{H^+} Phenol`
  .
- **4. From Diazo Salts:**
  `C6H5NH2 &xrightarrow{NaNO_2 + HCl, 0-5°C} C6H5N2+Cl- &xrightarrow{H_2O / \Delta} Phenol + N2↑ + HCl`
  .
- **5. From Grignard Reagent:**
  `PhMgBr + 1/2 O2 → PhOMgBr &xrightarrow{H_3O^+} Phenol`
  .
- **6. From Decarboxylation of Salicylic Acid:**
  `Salicylic acid + NaOH + CaO / Δ → Phenol` .

Acidity of Phenols & Substituent Effects

- Phenol is more acidic than Alcohols & Water due to resonance stabilization of the **Phenoxide Ion ($C_6H_5O^-$)**.
  pKa values: Alcohols (~16-18) > Water (15.7) > Phenol (10.0) > Carbonic Acid (6.4) > Benzoic Acid (4.2).
  Phenol reacts with Na and aq. NaOH, but does **NOT** react with $NaHCO_3$ (no CO_2 effervescence).
- **Substituent Effects on Phenol Acidity:**
  • **Electron Withdrawing Groups (EWG: $-NO_2$, $-CN$, $-X$, $-CHO$):** Increase acidity by stabilizing phenoxide ion (−I, −M effect).
  • **Electron Donating Groups (EDG: $-R$, $-OCH_3$, $-NH_2$):** Decrease acidity (+I, +M effect).
- **Acidity Trends (Must Remember for JEE):**
  • **Nitrophenols > Phenol > Cresols (Methylphenols)**
  • Nitrophenol Order: **p-Nitrophenol > o-Nitrophenol > m-Nitrophenol > Phenol**
  Reason: p-Nitrophenol has strong −M & −I. o-Nitrophenol has lower acidity than para due to **Intramolecular H-bonding** (makes H^+ removal slightly harder & increases volatility).
  • **2,4,6-Trinitrophenol (Picric acid, pKa = 0.38) > 2,4-Dinitrophenol > p-Nitrophenol**. Picric acid DOES give effervescence with $NaHCO_3$!

6. ELECTROPHILIC AROMATIC SUBSTITUTION (EAS) OF PHENOL

DIRECTING & REACTIVITY

–OH group is strongly Activating and Ortho/Para-directing

Reaction	Reagents & Conditions	Products & Key Features
Nitration (Dilute)	dil. HNO_3 at 298 K (25°C)	**o-Nitrophenol + p-Nitrophenol**. Separated by steam distillation (o-nitrophenol is steam volatile due to *intramolecular H-bonding*; p-nitrophenol has *intermolecular H-bonding*, high B.P.).
Nitration (Concentrated)	Conc. HNO_3 + Conc. H_2SO_4	**2,4,6-Trinitrophenol (Picric Acid)** [Yellow solid]. Yield is poor due to oxidation of ring. (Modern method: Phenol &xrightarrow{conc. H_2SO_4} Phenol-2,4-disulphonic acid &xrightarrow{conc. HNO_3} Picric acid).
Bromination (Non-polar solvent)	Br_2 in CS_2 or $CHCl_3$ or CCl_4 at 273 K	**p-Bromophenol (Major)** + o-Bromophenol (Minor). Monobromination occurs because non-polar solvent prevents ionization of phenol.
Bromination (Aqueous)	Br_2 water (Br_2 / H_2O)	**2,4,6-Tribromophenol (White Precipitate)**. In H_2O, phenol ionizes into highly reactive phenoxide ion, activating all o-, p-positions.
Sulphonation	Conc. H_2SO_4	• At 25°C (298 K) ⇒ **o-Phenolsulphonic acid** (Kinetically Controlled Product). • At 100°C (373 K) ⇒ **p-Phenolsulphonic acid** (Thermodynamically Controlled Product).

7. NAMED REACTIONS & SPECIAL TESTS OF PHENOL

HIGH FREQUENCY JEE QUESTIONS

Kolbe's & Reimer-Tiemann Reactions

- **Kolbe's Reaction (Kolbe-Schmitt):**
  `Phenol &xrightarrow{(i) NaOH} Phenoxide &xrightarrow{(ii) CO_2, 400 K, 4-7 atm} &xrightarrow{(iii) H^+} Salicylic Acid (2-hydroxybenzoic acid) [Major]`
  • Electrophile: Neutral **CO_2** (weak electrophile, requires strongly activated phenoxide).
  • Acetylation of Salicylic acid with Ac_2O/H^+ gives **Aspirin** (Acetylsalicylic acid).
  • Reaction with $MeOH/H^+$ gives **Oil of Wintergreen** (Methyl salicylate).
  • Reaction with $PhOH/POCl_3$ gives **Salol** (Phenyl salicylate - Antiseptic).
- **Reimer-Tiemann Reaction (RTR):**
  `Phenol + CHCl_3 + 3 aq. NaOH &xrightarrow{343 K} &xrightarrow{H^+} Salicylaldehyde (2-hydroxybenzaldehyde) [Major]`
  • Electrophile: Dichlorocarbene **$:CCl_2$** (generated by α-elimination of $CHCl_3$ by base).
  • *Abnormal Riemer-Tiemann Reaction* *[JEE Adv]*: Pyrrole + $CHCl_3$/KOH ⇒ 3-Chloropyridine (Ring expansion).
  • If **CCl_4 + NaOH** is used instead of $CHCl_3$ ⇒ Product is **Salicylic Acid**.

Fries Rearrangement, Coupling & Other Tests

- **Fries Rearrangement:**
  `Phenyl Acetate (C_6H_5OCOCH_3) &xrightarrow{Anhyd. AlCl_3} o- & p-Hydroxyacetophenone`
  • Low T (60°C) favors **p-isomer** | High T (160°C) favors **o-isomer**.
- **Diazo Coupling Reaction:**
  `C_6H_5N_2^+Cl^- + C_6H_5OH &xrightarrow{pH 9-10 (Mildly basic)} p-Hydroxyazobenzene`
  (Orange/Red Dye).
- **Schotten-Baumann Reaction:**
  `Phenol + Benzoyl Chloride (PhCOCl) &xrightarrow{aq. NaOH} Phenyl Benzoate (PhOCOPh)`
  .
- **Neutral $FeCl_3$ Test for Phenols:**
  Phenol + Neutral $FeCl_3$ ⇒ **Violet/Purple Complex** `[Fe(OC_6H_5)_6]^{3-} + 3 HCl` (Distinguishes Phenol from Alcohols).
- **Libermann's Nitroso Test:**
  Phenol + $NaNO_2$ + conc. H_2SO_4 ⇒ Deep Blue/Green color &xrightarrow{H_2O} Red &xrightarrow{aq. NaOH} Blue color.
- **Phthalein Reaction:**
  Phenol (2 moles) + Phthalic Anhydride &xrightarrow{conc. H_2SO_4 / \Delta} **Phenolphthalein** (Pink in basic solution).
- **Reduction with Zn dust:**
  `C_6H_5OH + Zn dust &xrightarrow{\Delta} Benzene + ZnO`
  .
- **Oxidation of Phenol:**
  `Phenol &xrightarrow{Na_2Cr_2O_7 / H_2SO_4 or Air} benzoquinone`
  (conjugated diketone, dark colored).

8. ETHERS: PREPARATION, PROPERTIES & CLEAVAGE

CRITICAL REACTIONS

Structure & Synthesis of Ethers

- **Structure:** C–O–C bond angle is ~111.7° (greater than tetrahedral 109.5° due to steric repulsion between bulky alkyl groups). Bent geometry ⇒ Dipole moment μ > 0 (polar).
- **1. Williamson Ether Synthesis:**
  $R-X + R'-ONa \rightarrow R-O-R' + NaX$ (**S_N2 mechanism**)
  • R–X MUST be **1° alkyl halide** (or CH_3X).
  • If R–X is **3° alkyl halide**, **Elimination (E2)** dominates ⇒ **Alkene is formed as major product!** (e.g., t-butyl bromide + NaOMe ⇒ Isobutylene).
  • Aryl halides (Ar–X) and Vinylic halides do NOT react due to double bond character.
- **2. Dehydration of Alcohols:**
  $2\ R-OH \xrightarrow{H_2SO_4,\ 413\ K\ (140°C)} R-O-R + H_2O$
  (Best for simple symmetrical 1° ethers; proceeds via S_N2).
- **3. Dry Ag_2O method:**
  $2\ R-X + Dry\ Ag_2O \xrightarrow{\Delta} R-O-R + 2\ AgX$
  .
- **4. Addition to Alkenes:**
  $R-CH=CH_2 + ROH \xrightarrow{H^+} R-CH(OR)-CH_3$.

Acidic Cleavage of Ethers with HX (HI/HBr)

- **General Reaction:** $R-O-R' + HX \rightarrow R-X + R'-OH$
  (Reactivity order of HX: HI > HBr > HCl).
- **Cleavage Rules for Mixed Ethers (R–O–R'):**
  • Case 1: Both R & R' are 1° or 2° alkyl groups (S_N2 Mechanism)
  Halide ion (I^-) attacks the **smaller / less hindered alkyl group**.
  Example: $CH_3-O-CH_2CH_3 + HI \rightarrow CH_3I + CH_3CH_2OH$.
  • Case 2: One alkyl group is 3°, Benzyl, or Allyl (S_N1 Mechanism)
  Halide ion (I^-) attaches to the **group forming the more stable carbocation**.
  Example: $(CH_3)_3C-O-CH_3 + HI \rightarrow (CH_3)_3C-I + CH_3OH$.
  • Case 3: Alkyl Aryl Ethers (e.g., Anisole Ph–O–CH_3)
  Cleavage ALWAYS gives **Phenol + Alkyl Iodide** because Ar–O bond has partial double bond character and cannot undergo S_N substitution.
  $Ph-O-CH_3 + HI \rightarrow Phenol\ (PhOH) + CH_3I$.
  • With Excess HI & Heat: The alcohol formed reacts further ⇒
  $R-O-R' + 2\ HI \xrightarrow{\Delta} R-I + R'-I + H_2O$
  (Phenol does NOT react further with HI).

9. ELECTROPHILIC SUBSTITUTION OF ANISOLE & EPOXIDES

JEE ADVANCED SPECIAL

EAS Reactions of Anisole (Methoxybenzene)

- $-OCH_3$ group is activating and o-, p-directing (+M > −I effect).
- **Halogenation:**
  $Anisole + Br_2\ in\ CH_3COOH \rightarrow p\text{-}Bromoanisole\ (90\%\ Major) + o\text{-}Bromoanisole$
  (No $FeBr_3$ catalyst needed).
- **Nitration:**
  $Anisole + conc.\ HNO_3\ /\ conc.\ H_2SO_4 \rightarrow p\text{-}Nitroanisole\ (Major) + o\text{-}Nitroanisole$
  .
- **Friedel-Crafts Alkylation:**
  $Anisole + CH_3Cl\ /\ Anhyd.\ AlCl_3 \rightarrow 4\text{-}Methoxytoluene\ (Major) + 2\text{-}Methoxytoluene$
  .
- **Friedel-Crafts Acylation:**
  $Anisole + CH_3COCl\ /\ Anhyd.\ AlCl_3 \rightarrow 4\text{-}Methoxyacetophenone\ (Major)$
  .

Epoxides / Oxiranes Ring Opening

[JEE Adv]

- **Acid-Catalyzed Ring Opening (H^+ / NuH):**
  Protonation of oxygen occurs first. Nucleophile attacks the **more substituted C-atom** (which holds more partial positive charge / carbocation character). Regioselectivity = S_N1-like; Stereochemistry = Anti.
- **Base-Catalyzed Ring Opening (Nu^- / e.g., RMgX, NaOMe, $LiAlH_4$):**
  Nucleophile attacks the **less hindered C-atom** via pure S_N2 mechanism. Stereochemistry = Anti-addition.
- **Autoxidation of Ethers:** Ethers on standing in presence of air/light form explosive **Peroxides** ($CH_3-CH(OOH)-O-CH_2CH_3$). Tested with $FeSO_4$ + KCNS (red color). Removed by washing with $FeSO_4$ solution.

JEE Last-Minute Quick Test & Distinctions Summary

1° vs 2° vs 3° Alcohols:
• Lucas Test: Instant (3°), 5 min (2°), Heating (1°).
• Cu / 573 K: Aldehyde (1°), Ketone (2°), Alkene (3°).
• Victor Meyer Test: Red color (1°), Blue color (2°), Colorless (3°).

Alcohol vs Phenol:
• Neutral $FeCl_3$: Violet color (Phenol only).
• Br_2 Water: White ppt (Phenol only).
• Litmus paper: Red (Phenol only; Alcohols neutral).
• $NaHCO_3$: Neither reacts (except Picric acid).

Ether Key Checks:
• HI cleavage gives smallest I (S_N2) UNLESS 3°/Benzyl cation is formed (S_N1).
• Ar–O–R + HI ALWAYS yields Phenol + R–I.
• Williamson failed with 3° R–X (gives alkene).

JEE ORGANIC CHEMISTRY — THE REAGENT ATLAS

Subtitle: Reagents • Conversions • Named Reactions • Conditions • Identification Tests | Target: JEE Main & Advanced Master Revision

1. REDUCING AGENTS

Reagent	Transformation & Selectivity	Conditions & JEE Notes
H_2 / Ni, Pt or Pd	Alkenes, alkynes → Alkanes; reducible groups	Catalytic hydrogenation; syn addition; heterogeneous catalysis.
$NaBH_4$	Aldehyde → 1° Alcohol; Ketone → 2° Alcohol	Mild reducing agent; does *not* reduce esters, acids, or amides under standard conditions.
$LiAlH_4$ (LAH)	Acids, esters, acyl chlorides, amides, nitriles, carbonyls	Powerful reducing agent. Acid/ester/acyl chloride/aldehyde/ketone → alcohols; Amide/nitrile → amines. Requires anhydrous ether work-up.
DIBAL-H	Ester / Nitrile → Aldehyde	Controlled low temp (**−78°C**) followed by hydrolysis. Stoichiometry and temp determine selectivity.
Zn(Hg) / conc. HCl	Aldehyde / Ketone C=O → $-CH_2-$	**Clemmensen Reduction**. Strongly acidic conditions; unsuitable for acid-sensitive substrates.
NH_2NH_2 / KOH / Heat	Aldehyde / Ketone C=O → $-CH_2-$	**Wolff–Kishner Reduction**. Strongly basic conditions (anhydrous high-boiling solvent like ethylene glycol).
Sn/HCl or Fe/HCl	Nitrobenzene → Aniline ($Ar-NO_2 \rightarrow Ar-NH_2$)	Standard aromatic nitro reduction. Fe/HCl preferred industrially (catalytic HCl formed).
Na / liquid NH_3	Alkyne → *trans*-Alkene	Dissolving metal reduction (Birch reduction variant for alkynes). Anti-addition.
H_2 / Lindlar Catalyst	Alkyne → *cis*-Alkene	Poisoned palladium ($Pd/BaSO_4$ or quinoline). Syn addition yielding *cis* alkenes.
BH_3·THF ; H_2O_2/OH^-	Alkene → Anti-Markovnikov Alcohol; Terminal Alkyne → Aldehyde	Hydroboration–oxidation. Syn addition of water across double bond without carbocation rearrangement.

2. OXIDISING AGENTS

Reagent	Transformation & Selectivity	Conditions & JEE Notes
PCC (Pyridinium Chlorochromate)	1° Alcohol → Aldehyde; 2° Alcohol → Ketone	Mild anhydrous oxidant in CH_2Cl_2. Stops at aldehyde stage (does not over-oxidize to acid).
$K_2Cr_2O_7$ / H_2SO_4 or Jones (CrO_3/H_2SO_4)	1° Alcohol → Carboxylic Acid; 2° Alcohol → Ketone	Strong acidic aqueous oxidation. Primary alcohols over-oxidized directly to carboxylic acids.
$KMnO_4$ (hot / acidic)	Strong oxidation; Alkene oxidative cleavage	Cleaves alkenes to acids/ketones depending on substitution. Benzoic acid from alkylbenzenes with α-H.
$KMnO_4$ (cold / dilute / alkaline)	Alkene → Vicinal Diol (syn)	**Baeyer's Test** for unsaturation (purple $KMnO_4$ decolorizes, brown MnO_2 precipitate).
O_3 / Zn, H_2O (Reductive Ozonolysis)	Alkene → Carbonyl fragments (Aldehydes/Ketones)	Cleaves C=C completely. Zn destroys H_2O_2 byproduct preventing further oxidation.
O_3 / H_2O_2 (Oxidative Ozonolysis)	Alkene cleavage; aldehydes oxidized to acids	Stronger work-up than reductive ozonolysis.
Tollens' / Fehling's / Benedict's	Aldehyde → Carboxylate; $Cu^{2+} \rightarrow Cu_2O$ (red ppt)	Diagnostic tests for aliphatic/aromatic aldehydes (Tollens gives silver mirror).

3. ALKENES AND ALKYNES

A. Alkene Additions & Reactions

- **H_2 / Ni, Pt, Pd:** Alkene → Alkane (syn addition).
- **Br_2 / CCl_4:** Vicinal dibromide (anti addition, red color discharge).
- **HBr:** Markovnikov addition (carbocation intermediate, rearrangement possible).
- **HBr / ROOR:** Anti-Markovnikov addition (peroxide effect, radical mechanism applicable *only* to HBr).
- **HCl / HI:** Markovnikov addition.
- **H_2O / H^+:** Acid-catalyzed hydration (Markovnikov alcohol, rearrangement).
- **$Hg(OAc)_2$ / H_2O ; $NaBH_4$:** Oxymercuration-demercuration (Markovnikov alcohol, *no* rearrangement).
- **mCPBA (Peracid):** Alkene → Epoxide (concerted syn addition).

B. Alkyne Reactions & Derivatives

- **$HgSO_4$ / H_2SO_4 / H_2O:** Terminal alkyne (except ethyne) → Methyl ketone (Markovnikov hydration via enol). Ethyne → Acetaldehyde.
- **$NaNH_2$ / liq. NH_3:** Terminal alkyne deprotonation → Strong nucleophilic acetylide anion ($R-C\equiv C^-\ Na^+$).
- **Ammoniacal $AgNO_3$ / Cu_2Cl_2:** Tests for terminal alkynes. Form white (Ag) or red/yellow (Cu) acetylide precipitates.
- **O_3 or hot $KMnO_4$:** Oxidative cleavage of alkynes to carboxylic acids ($R-COOH + CO_2$ if terminal).

4. HALOALKANES AND HALOARENES

Reagent / Condition	Reaction / Product	Key Mechanism & Distinction
Aqueous KOH + heat	Haloalkane → Alcohol	Nucleophilic Substitution (S_N1 or S_N2 depending on substrate).
Alcoholic KOH + heat	Haloalkane → Alkene	β-Elimination (E2 mechanism favored; Zaitsev rule applies).
NaI / Dry Acetone	R–Cl / R–Br → R–I	**Finkelstein Reaction**. Precipitation of NaCl/NaBr in acetone drives equilibrium.
AgF / Hg_2F_2 / CoF_2	R–Cl / R–Br → R–F	**Swarts Reaction**. Halogen exchange for alkyl fluorides.
KCN vs AgCN	KCN → Alkyl Nitrile (R–CN) AgCN → Alkyl Isocyanide (R–NC)	KCN is ionic (ambident CN^- attacks via C); AgCN is covalent (attacks via lone pair on N).
Mg / Dry Ether	Haloalkane → Grignard Reagent (RMgX)	Organometallic formation; extremely sensitive to moisture/protic sources.
2 Na / Dry Ether	R–X + 2Na + X–R → R–R	**Wurtz Reaction**. Alkyl-alkyl coupling (best for symmetrical alkanes with odd/even carbons). Fittig = Ar–Ar; Wurtz-Fittig = Ar–R.
NaOH (623K, 300 atm) then H^+	Chlorobenzene → Phenol	**Dow Process** via benzyne intermediate under drastic conditions.

5. ALCOHOLS, PHENOLS AND ETHERS

- **Na Metal:** Alcohol/Phenol → Alkoxide/Phenoxide + H_2 (test for active hydrogen).
- **HX + $ZnCl_2$ (Lucas Reagent):** Differentiates 1°, 2°, 3° alcohols. 3° reacts instantly (turbidity); 2° in 5 mins; 1° no reaction at room temp.
- **$SOCl_2$ / Pyridine:** Alcohol → Alkyl chloride (cleanest method; gaseous byproducts SO_2 and HCl escape).
- **Conc. H_2SO_4 (413 K vs 443 K):** At 413 K (EtOH) → Diethyl ether (intermolecular). At 443 K → Ethene (intramolecular dehydration).
- **$CHCl_3$ / NaOH (aq) then H^+: Reimer–Tiemann Reaction**. Phenol → Salicylaldehyde (ortho-formylation via dichlorocarbene $:CCl_2$).
- **CO_2 / NaOH then H^+ (398K, 4-7 atm): Kolbe–Schmitt Reaction**. Phenol → Salicylic acid (2-hydroxybenzoic acid).
- **Br_2 Water:** Phenol → 2,4,6-Tribromophenol (white precipitate instantly formed without catalyst).
- **Williamson Ether Synthesis:** $R-O^-\ Na^+ + R'-X \rightarrow R-O-R' + NaX$ (S_N2 mechanism; best with 1° alkyl halide).

6. ALDEHYDES AND KETONES

Reagent / Reaction	Product / Result	Key Notes & Mechanism
HCN / KCN	Carbonyl → Cyanohydrin	Nucleophilic addition of cyanide ion to C=O.
NH_2OH / NH_2NH_2 / Semicarbazide	Oxime / Hydrazone / Semicarbazone	Nucleophilic addition followed by elimination (condensation/derivative formation).
2,4-DNP Reagent	Yellow/Orange/Red Crystalline Precipitate	Diagnostic test for carbonyl group (Aldehydes & Ketones).
RMgX then H_3O^+	HCHO → 1° Alcohol; RCHO → 2° Alcohol; R_2CO → 3° Alcohol	Grignard nucleophilic addition to carbonyl carbon followed by acid work-up.
I_2 / NaOH (aq)	Yellow precipitate of Iodoform (CHI_3)	**Iodoform Test**. Positive for methyl ketones (CH_3–C=O) and 2° alcohols with CH_3–CH(OH)– structure (or ethanol).
Dilute Base (NaOH / $Ba(OH)_2$)	β-Hydroxy aldehyde/ketone → α,β-unsaturated carbonyl	**Aldol Condensation**. Requires at least one acidic α-hydrogen.
Conc. NaOH / KOH (no α-H)	Self-oxidation & reduction (Disproportionation)	**Cannizzaro Reaction**. Aldehydes lacking α-H (e.g., HCHO, PhCHO) yield alcohol + carboxylate salt.
Peracid (e.g., mCPBA)	Ketone → Ester (Migration aptitude: 3° > 2° > 1° > Me)	**Baeyer–Villiger Oxidation**.

7. CARBOXYLIC ACIDS AND DERIVATIVES

- **$SOCl_2$ / PCl_5 / PCl_3:** R–COOH → Acyl chloride (R–COCl).
- **ROH / Conc. H_2SO_4:** Fischer Esterification (reversible reaction to ester + H_2O).
- **Soda Lime (NaOH / CaO, Heat):** Decarboxylation of sodium carboxylate → Alkane with **one fewer** carbon atom.
- **Br_2 / Red P then H_2O: Hell–Volhard–Zelinsky (HVZ) Reaction**. α-Halogenation of carboxylic acids having α-hydrogens.

Functional Group Identification Box

- **Carboxylic Acid:** R–C(=O)–OH (contains –COOH).
- **Ester:** R–C(=O)–OR' (contains –COOR').
- **Aldehyde:** R–CHO | **Ketone:** R–CO–R'
- **Amide:** R–$CONH_2$ | **Nitrile:** R–C≡N
- Caution C=O alone does *not* identify an acid or ester; check adjacent heteroatoms.

8. AMINES AND DIAZONIUM SALTS

Reagent / Condition	Transformation	Key Details & Named Reactions
Br_2 / NaOH (aq)	Amide → Primary Amine (R–NH_2)	**Hofmann Bromamide Degradation**. Chain shortening by **one carbon atom** via isocyanate intermediate.
$CHCl_3$ / Alc. KOH (Heat)	1° Amine → Isocyanide (Carbylamine)	**Carbylamine Test**. Foul-smelling foul gas; diagnostic test for 1° aliphatic/aromatic amines.
Hinsberg Reagent ($C_6H_5SO_2Cl$)	Distinguishes 1°, 2°, 3° Amines	1° amine sulfonamide is soluble in alkali; 2° is insoluble; 3° does not react.
$NaNO_2$ + HCl (0–5°C)	Primary Aromatic Amine → Diazonium Salt	**Diazotisation**. Must be maintained strictly at 0–5°C to prevent decomposition to phenol.
CuCl/HCl, CuBr/HBr, CuCN	Ar–$N_2^+Cl^-$ → Aryl Halide / Nitrile (Ar–Cl/Br/CN)	**Sandmeyer Reaction**. Catalytic copper(I) halide insertion. (Gattermann uses Cu powder).
HBF_4 then heat	Ar–N_2^+ → Aryl Fluoride (Ar–F)	**Balz–Schiemann Reaction**. Thermal decomposition of diazonium fluoroborate.
H_3PO_2 or EtOH	Ar–N_2^+ → Benzene ring (Ar–H)	Deamination / replacement of diazonium group by hydrogen.

9. AROMATIC COMPOUNDS & ELECTROPHILIC SUBSTITUTION

A. Nitration & Halogenation

- **Conc. HNO_3 / H_2SO_4:** Benzene → Nitrobenzene (Electrophile: NO_2^+).
- **Br_2 / $FeBr_3$ (or Cl_2 / $FeCl_3$):** Ring halogenation (Electrophile: Br^+ or Cl^+).
- **SO_3 / Conc. H_2SO_4:** Sulphonation (reversible electrophilic substitution).

B. Friedel–Crafts Reactions

- **R–Cl / Anhydrous $AlCl_3$:** Alkylation (forms alkylbenzene). *Watch for carbocation rearrangement!*
- **RCOCl / Anhydrous $AlCl_3$:** Acylation (forms aryl ketone). No rearrangement.

C. Side-Chain Chemistry

- **$KMnO_4$ / Heat, then H^+:** Alkylbenzene side chain with α-H → Benzoic acid (regardless of chain length).
- **Cl_2 / hv or NBS / hv:** Benzylic / allylic radical halogenation.

10. COMPREHENSIVE NAMED REACTIONS QUICK-REFERENCE

Rosenmund: H_2, $Pd/BaSO_4$ (Acyl Cl → Aldehyde)	**Stephen:** $SnCl_2/HCl$, H_2O (Nitrile → Aldehyde)	**Etard:** CrO_2Cl_2, H_2O (Toluene → Benzaldehyde)
Gattermann-Koch: CO, HCl, $AlCl_3$ (Benzene → Benzaldehyde)	**Gabriel Phthalimide:** Potassium phthalimide → Pure 1° Aliphatic Amine	**Schotten-Baumann:** Benzoylation of amines/phenols with benzoyl chloride + NaOH

11. QUALITATIVE TESTS & MISCELLANEOUS REAGENTS

- **Lucas Reagent:** conc. HCl + anhydrous $ZnCl_2$ (distinguishes 1°, 2°, 3° alcohols).
- **Neutral $FeCl_3$:** Violet/colored complex with phenols.
- **Baeyer's Test:** Cold dilute alkaline $KMnO_4$ (purple → colorless, unsaturation test).
- **Tollens' Test:** Ammoniacal $AgNO_3$ (silver mirror for aldehydes/reducing sugars).
- **Fehling's Test:** Alkaline Cu^{2+} + Rochelle salt (brick-red Cu_2O precipitate for aliphatic aldehydes).
- **LDA:** Strong, bulky, non-nucleophilic base (kinetic enolate formation).

12. TOP 10 HIGH-FREQUENCY CONFUSIONS

Comparison Pair	Key Distinction for JEE
$NaBH_4$ vs $LiAlH_4$ vs DIBAL-H	$NaBH_4$ is mild (aldehydes/ketones). $LiAlH_4$ is powerful (reduces acids, esters, amides, nitriles). DIBAL-H at −78°C selectively stops esters/nitriles at aldehydes.
Clemmensen vs Wolff–Kishner	Both convert C=O → $-CH_2-$. Clemmensen uses Zn(Hg)/HCl (**acidic**); Wolff–Kishner uses NH_2NH_2/KOH/heat (**basic**).
Lindlar vs Na / liq. NH_3	Lindlar catalyst gives *cis*-alkene (syn addition); Sodium in liquid ammonia gives *trans*-alkene (anti addition).
Aqueous KOH vs Alcoholic KOH	Aq. KOH promotes **substitution** (R–OH); Alc. KOH + heat promotes **elimination** (Alkene via E2).
KCN vs AgCN	KCN is ionic → C-attack → Nitrile (R–CN). AgCN is covalent → N-attack → Isocyanide (R–NC).
HBr vs HBr / Peroxide	HBr alone gives Markovnikov addition. HBr + Peroxide gives Anti-Markovnikov addition (radical mechanism; **does not apply to HCl or HI**).

PART III

MATHEMATICS

Calculus | Algebra | Coordinate Geometry | Trigonometry | Vectors & 3D

19 Chapters | Pages 232 to 289

JEE REVISION FORMULA SHEET

Basic Mathematics, Logarithms & Linear Inequalities

JEE MAIN & ADVANCED

1. Standard Algebraic Identities & Factorization

- $(a+b)^2 = a^2 + 2ab + b^2 = (a-b)^2 + 4ab$
- $(a-b)^2 = a^2 - 2ab + b^2 = (a+b)^2 - 4ab$
- $a^2 - b^2 = (a+b)(a-b)$
- $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$
- $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
- $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2) = (a+b)^3 - 3ab(a+b)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2) = (a-b)^3 + 3ab(a-b)$
- $a^4 - b^4 = (a-b)(a+b)(a^2+b^2)$
- $a^4 + a^2 + 1 = (a^2 + a + 1)(a^2 - a + 1)$

Identities & Cyclic Sums (JEE Special)

- $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$
- $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$
- *Special Case:* If $a+b+c = 0$ or $a = b = c$, then: $a^3 + b^3 + c^3 = 3abc$
- $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$ [Added]
- $a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + \dots + b^{n-1})$ *(for odd n)* [Added]

2. LAWS OF INDICES, SURDS & RATIO-PROPORTION

Laws of Indices

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = (a^n)^m = a^{mn}$
- $(ab)^n = a^n b^n$; $(a/b)^n = a^n / b^n$
- $(a/b)^{-n} = (b/a)^n$
- $a^m \div b^{-n} = a^m \times b^n$
- $a^0 = 1$ *(a ≠ 0)*

Surds & Radicals

- $(\sqrt[n]{a})^n = a$ *(n ∈ ℕ, n ≥ 2, a > 0)*
- $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{(ab)}$
- $\sqrt[n]{a} / \sqrt[n]{b} = \sqrt[n]{(a/b)}$
- $\sqrt{(a)} \times \sqrt{(b)} = \sqrt{(ab)}$ *(at least one a, b ≥ 0)*
- $(\sqrt{a} \pm \sqrt{b})^2 = a + b \pm 2\sqrt{(ab)}$
- $\sqrt{(a \pm 2\sqrt{b})} = \sqrt{x} \pm \sqrt{y}$ *where x+y=a, xy=b* [Added]

Ratio & Proportion

If a/b = c/d, then:

- $a/c = b/d$ *(Alternendo)*
- $b/a = d/c$ *(Invertendo)*
- $(a+b)/b = (c+d)/d$ *(Componendo)*
- $(a-b)/b = (c-d)/d$ *(Dividendo)*
- $(a+b)/(a-b) = (c+d)/(c-d)$ *(Comp. & Div.)*
- If $a/b = c/d = e/f = \dots = k$, then: $k = (a+c+e+\dots)/(b+d+f+\dots)$

3. MODULUS FUNCTION (ABSOLUTE VALUE) & PROPERTIES

Definition & Elementary Inequalities

$|x| = x$ *(if x ≥ 0)* | $|x| = -x$ *(if x < 0)*

- $|x| = a\ (a > 0) \Rightarrow x = \pm a$
- $|x| \le a \Rightarrow -a \le x \le a \Rightarrow x \in [-a, a]$
- $|x| < a \Rightarrow -a < x < a \Rightarrow x \in (-a, a)$
- $|x| \ge a \Rightarrow x \le -a$ or $x \ge a \Rightarrow x \in (-\infty, -a] \cup [a, \infty)$
- $|x| > a \Rightarrow x < -a$ or $x > a \Rightarrow x \in (-\infty, -a) \cup (a, \infty)$
- $a \le |x| \le b \Rightarrow x \in [-b, -a] \cup [a, b]$
- $a < |x| < b \Rightarrow x \in (-b, -a) \cup (a, b)$
- $a \le |x - c| \le b \Rightarrow x \in [-b+c, -a+c] \cup [a+c, b+c]$

Triangle Inequalities & Exact Equality Conditions

- $|x+y| \le |x| + |y| \Rightarrow$ Equality holds when $xy \ge 0$
- $|x-y| \le |x| + |y| \Rightarrow$ Equality holds when $xy \le 0$
- $|x-y| \ge ||x| - |y|| \Rightarrow$ Equality holds when $xy \ge 0$
- $|x+y| \ge ||x| - |y|| \Rightarrow$ Equality holds when $xy \le 0$
- $|x| = \sqrt{(x^2)}$ *(Always non-negative)* [Added]
- $|f(x)| + |g(x)| = |f(x) + g(x)| \Rightarrow f(x) \cdot g(x) \ge 0$ [Added]
- $|f(x)| - |g(x)| = |f(x) - g(x)| \Rightarrow f(x) \cdot g(x) \ge 0$ [Added]

4. QUADRATIC EQUATIONS & ROOTS QUICK REFERENCE

General Form: $ax^2 + bx + c = 0$ *(a ≠ 0)*

Roots Formula: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a)$

Discriminant: $D = b^2 - 4ac$

Sum of Roots (α + β): $-b / a$

Product of Roots (αβ): c / a

Difference of Roots (|α − β|): $\sqrt{D} / |a|$ [Added]

Nature of Roots:

- $D > 0 \Rightarrow$ Real and Distinct
- $D = 0 \Rightarrow$ Real and Equal ($x = -b/2a$)
- $D < 0 \Rightarrow$ Complex Conjugate pairs

5. LOGARITHMS (FUNDAMENTAL PROPERTIES & CHANGE OF BASE)

Definition: $\log_a N = b \Leftrightarrow a^b = N$ | ***Domain Conditions:*** $N > 0, a > 0, a \neq 1$

- $\log_a 1 = 0$; $\log_a a = 1$
- $\log_a(mn) = \log_a m + \log_a n$ *(m, n > 0)*
- $\log_a(m/n) = \log_a m - \log_a n$ *(m, n > 0)*
- $\log_a(m^k) = k \cdot \log_a m$
- $\log_{(a^k)} m = (1/k) \cdot \log_a m$ *(k ≠ 0)*
- $\log_{(a^k)} (m^p) = (p/k) \cdot \log_a m$
- $\log_b a = 1 / \log_a b$ *(a, b > 0; a, b ≠ 1)*
- $\log_b a = (\log_m a) / (\log_m b)$ *(Base Change Theorem)*
- $a^{\log_a m} = m$ *(Fundamental Logarithmic Identity)*
- $a^{\log_c b} = b^{\log_c a}$ *(Power Base Swap)*
- $\log_a b \cdot \log_b c \cdot \log_c d = \log_a d$ **[Added]**

Logarithmic Equations & Inequalities (Crucial JEE Rules)

Type	Base Condition: Base m > 1	Base Condition: 0 < Base m < 1
$\log_m x > \log_m y$	$x > y > 0$ (Sign remains same)	$0 < x < y$ (Sign REVERSES)
$\log_m a > b$	$a > m^b$ (and $a > 0$)	$0 < a < m^b$
$\log_m a < b$	$0 < a < m^b$	$a > m^b$ (and $a > 0$)
$\log_a f(x) > \log_a g(x)$	$f(x) > g(x) > 0$	$0 < f(x) < g(x)$

** Note: Always take the intersection of solution sets with domain restrictions: Argument > 0 and Base > 0, Base ≠ 1.*

6. WAVY CURVE METHOD (METHOD OF INTERVALS)

Algorithm for Solving Rational Inequalities: P(x) / Q(x) > 0 or < 0

1. Factorize numerator and denominator into linear factors: $f(x) = [(x-a_1)^{k_1}(x-a_2)^{k_2}...] / [(x-b_1)^{p_1}...]$. Make coefficient of x positive in all factors.
2. Find **Critical Points** by equating each factor to zero. Plot all critical points on a real number line in strictly increasing order.
3. Assign a **Plus (+)** sign to the rightmost interval $(\max_root, \infty)$.
4. Move right-to-left across critical points:
   - If factor exponent is **ODD**: Sign **CHANGES** when crossing that root.
   - If factor exponent is **EVEN**: Sign **REMAINS SAME** when crossing that root.
5. Select intervals with **(+)** sign for > 0 or **(−)** sign for < 0. Exclude roots of denominator $Q(x) = 0$.

7. LINEAR INEQUALITIES & LINEAR PROGRAMMING PROBLEMS (LPP)

Types of Inequalities & Operations

- **Strict:** $x < a, x > a$ | **Slack:** $x \leq a, x \geq a$
- **Double Inequality:** $a < x < b \Rightarrow x \in (a, b)$
- **Reversal Rule:** Multiplying/dividing both sides by a *negative number* reverses inequality ($a > b \Rightarrow -a < -b$).
- **Number Line Representation:** Open circle for strict (<, >), Dark filled circle for slack (≤, ≥).

Mathematical Form of LPP

- **Objective Function:** $Z = ax + by$ (to maximize/minimize)
- **Decision Variables:** x, y
- **Constraints:** Linear inequalities e.g., $a_1 x + b_1 y \leq d_1$
- **Non-negative Restrictions:** $x \geq 0, y \geq 0$ (1st Quadrant)

Corner Point Method & Optimality Theorem

Step 1: Graph all linear constraints & find the **Feasible Region** (intersection area).
Step 2: Find coordinates of all **Corner Points (Vertices)** of the feasible region.
Step 3: Evaluate $Z = ax + by$ at each corner point. Let maximum = M and minimum = m.

Bounded Region: M and m are guaranteed to be the maximum and minimum values.
Unbounded Region:
- M is Max if open half-plane $ax + by > M$ has *no common points* with feasible region.
- m is Min if open half-plane $ax + by < m$ has *no common points* with feasible region.

8. SETS, CARDINALITY & DE MORGAN'S LAWS

Set Inclusion & Cardinal Formulae

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$ [Added]
- If A and B are disjoint ($A \cap B = \varnothing$): $n(A \cup B) = n(A) + n(B)$
- Complement: $A' = U - A$

De Morgan's Laws & Set Identities

- $(A \cup B)' = A' \cap B'$
- $(A \cap B)' = A' \cup B'$
- $A - B = A \cap B'$ [Added]
- $n(A - B) = n(A) - n(A \cap B)$ [Added]
- Number of subsets of a set with n elements = 2^n [Added]

MATHEMATICAL REASONING & PMI

JEE Main & Advanced High-Yield Formula Reference Sheet

JEE REVISION

1. MATHEMATICAL STATEMENTS & LOGICAL OPERATIONS

Core Definitions

- **Statement:** A sentence that is either strictly True (T) or False (F), but not both.
- **Simple Statement:** Cannot be broken down further.
- **Compound Statement:** Formed by combining two or more simple statements using logical connectives.
- **Quantifiers:**
  • Universal: *"For every" / "For all"* (∀)
  • Existential: *"There exists"* (∃)

Basic Logical Connectives

- **Negation (NOT):** $\sim p$ or $\neg p$ (Denial of statement p).
- **Conjunction (AND):** $p \wedge q$ (True ONLY when both p and q are T).
- **Disjunction (OR):** $p \vee q$ (False ONLY when both p and q are F).
- **Implication (IF...THEN):** $p \Rightarrow q$ (False ONLY when p is T and q is F).
- **Biconditional (IF AND ONLY IF):** $p \Leftrightarrow q$ (True when both have identical truth values).

2. MASTER TRUTH TABLE FOR CONNECTIVES

p	q	Negation $\sim p$	Conjunction $p \wedge q$	Disjunction $p \vee q$	Conditional $p \Rightarrow q$	Biconditional $p \Leftrightarrow q$
T	**T**	F	**T**	**T**	**T**	**T**
T	**F**	F	F	**T**	**F**	F
F	**T**	T	F	**T**	**T**	F
F	**F**	T	F	F	**T**	**T**

3. CONDITIONAL STATEMENT TRANSFORMATIONS

Implication Equivalent Formulations ($p \Rightarrow q$)

- p implies q
- p is a sufficient condition for q
- q is a necessary condition for p
- p only if q
- $\sim q \Rightarrow \sim p$

Converse, Inverse & Contrapositive

- **Original Statement:** $p \Rightarrow q$
- **Converse:** $q \Rightarrow p$
- **Inverse:** $\sim p \Rightarrow \sim q$
- **Contrapositive:** $\sim q \Rightarrow \sim p$
- *Note:* Statement ≡ Contrapositive; Converse ≡ Inverse.

4. ALGEBRA OF STATEMENTS & IMPORTANT EQUIVALENCES

Standard Equivalence Laws

- **Idempotent:** $p \vee p \equiv p$, $p \wedge p \equiv p$
- **Commutative:** $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$
- **Associative:** $(p \vee q) \vee r \equiv p \vee (q \vee r)$
  $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- **Distributive:** $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
  $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
- **De Morgan's Laws:** $\sim(p \wedge q) \equiv \sim p \vee \sim q$
  $\sim(p \vee q) \equiv \sim p \wedge \sim q$
- **Involution (Double Negation):** $\sim(\sim p) \equiv p$

Identity, Complement & Critical Identities

- **Identity Laws:** $p \wedge T \equiv p$, $p \vee F \equiv p$
  $p \wedge F \equiv F$, $p \vee T \equiv T$
- **Complement Laws:** $p \vee \sim p \equiv T$, $p \wedge \sim p \equiv F$
  $\sim T \equiv F$, $\sim F \equiv T$
- **Absorption Laws:** $p \vee (p \wedge q) \equiv p$, $p \wedge (p \vee q) \equiv p$
- **Conditional Identity:** $p \Rightarrow q \equiv \sim p \vee q$
- **Biconditional Identity:** $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$
  $p \Leftrightarrow q \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$

5. NEGATION OF COMPOUND STATEMENTS & DUALITY

Negation Rules

- **Negation of Conjunction:** $\sim(p \wedge q) \equiv \sim p \vee \sim q$
- **Negation of Disjunction:** $\sim(p \vee q) \equiv \sim p \wedge \sim q$
- **Negation of Implication:** $\sim(p \Rightarrow q) \equiv p \wedge \sim q$
- **Negation of Biconditional:** $\sim(p \Leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

Principle of Duality & Evaluation

- **Dual Statement:** Obtained by replacing $\vee$ with $\wedge$, $\wedge$ with $\vee$, T with F, and F with T.
- **Tautology (t):** Compound statement always True for all truth values.
- **Fallacy / Contradiction (c):** Compound statement always False for all truth values.
- **Equivalence Trick:** $(p \Rightarrow q) \vee (q \Rightarrow p) \equiv T$ (Tautology).

6. METHODS OF PROOF & SET THEORY ANALOGY [ADDED SHORTCUTS]

Validation Methods

- **Direct Method:** Assume p is true, show q is true.
- **Contrapositive Method:** Assume $\sim q$ is true, show $\sim p$ is true.
- **Method of Contradiction:** Assume $\sim p$ is true, derive a logical absurdity.
- **Counter Example:** Find a single case where statement fails.

Set Operations Mapping (JEE Shortcut)

Logic	Set Equivalent	Venn Mapping
$p \wedge q$	$A \cap B$	Intersection
$p \vee q$	$A \cup B$	Union
$\sim p$	A' / A^c	Complement
$p \Rightarrow q$	$A \subseteq B$	Subset ($A' \cup B$)

7. PRINCIPLE OF MATHEMATICAL INDUCTION (PMI)

Core Steps for Proving $P(n)$ for all $n \in \mathbb{N}$

- **Base Case (Step 1):** Verify that the statement $P(n)$ is true for the first natural number, i.e., check $P(1)$ is true.
- **Inductive Step (Step 2):** Assume $P(k)$ is true for some natural number $n = k$. Using this inductive hypothesis, prove that $P(k+1)$ is also true.
- **Conclusion:** If both steps hold, $P(n)$ is true for all natural numbers $n \in \mathbb{N}$.

8. STANDARD SUMMATION IDENTITIES & SERIES FORMULAS

Powers of Natural Numbers

- $\sum n = 1 + 2 + 3 + \ldots + n = {}^{n(n+1)}/_{2}$
- $\sum n^2 = 1^2 + 2^2 + \ldots + n^2 = {}^{n(n+1)(2n+1)}/_{6}$
- $\sum n^3 = 1^3 + 2^3 + \ldots + n^3 = [\, {}^{n(n+1)}/_{2} \,]^2 = (\sum n)^2$
- $\sum n^4 = {}^{n(n+1)(2n+1)(3n^2 + 3n - 1)}/_{30}$

Arithmetic, Geometric & Even/Odd Series

- **Even Sum:** $2 + 4 + 6 + \ldots + 2n = \sum 2n = n(n + 1)$
- **Odd Sum:** $1 + 3 + 5 + \ldots + (2n - 1) = \sum (2n - 1) = n^2$
- **AP Sum:** $S_n = {}^{n}/_{2} \,[\, 2a + (n - 1)d \,]$
- **GP Sum:** $S_n = {}^{a(r^n - 1)}/_{(r - 1)} \quad (r \neq 1)$

9. HIGH-YIELD TRIGONOMETRIC & FACTORIZATION IDENTITIES

Algebraic Factorization Formulas

- $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \ldots + y^{n-1})$ *(Valid for all $n \in \mathbb{N}$)*
- $x^n + y^n = (x + y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 - \ldots + y^{n-1})$ *(Valid ONLY when n is an odd positive integer)*
- **Binomial Theorem:** $(a + b)^n = \sum_{r=0}^{n} {}^{n}C_r \, a^{n-r} \, b^r$

Trigonometric Sums & Products

- $\cos(a) \cdot \cos(2a) \cdot \cos(4a) \cdot \ldots \cdot \cos(2^{n-1}a) = {}^{\sin(2^n a)}/_{(2^n \sin a)}$
- $\sin(a) + \sin(a+b) + \ldots + \sin(a+(n-1)b) = {}^{\sin(nb/2)}/_{\sin(b/2)} \sin(a + {}^{(n-1)b}/_{2})$
- $\cos(a) + \cos(a+b) + \ldots + \cos(a+(n-1)b) = {}^{\sin(nb/2)}/_{\sin(b/2)} \cos(a + {}^{(n-1)b}/_{2})$

SOLUTION OF TRIANGLES

Comprehensive JEE Main & Advanced Formula Revision Sheet

Standard Notation: ΔABC, sides a, b, c
Semi-perimeter: s = (a+b+c)/2

1. FUNDAMENTAL TRIANGLE RELATIONS

- **Triangle Inequality:** $a + b > c,\quad b + c > a,\quad c + a > b$
- **Side-Angle Relation:** $|b - c| < a < b + c$
- **Angle Sum:** $A + B + C = \pi\ (180^\circ)$
- **Perimeter & Semi-perimeter:** $P = a + b + c,\quad s = \frac{1}{2}(a + b + c)$

2. SINE RULE & COSINE RULE

- **Sine Rule:** $a / \sin A = b / \sin B = c / \sin C = 2R$

→ Express form: $a = 2R \sin A,\quad b = 2R \sin B,\quad c = 2R \sin C$

- **Cosine Rule:**

$$\cos A = (b^2 + c^2 - a^2) / (2bc)$$
$$\cos B = (c^2 + a^2 - b^2) / (2ca)$$
$$\cos C = (a^2 + b^2 - c^2) / (2ab)$$

3. PROJECTION FORMULAE

$$a = b \cos C + c \cos B$$
$$b = c \cos A + a \cos C$$
$$c = a \cos B + b \cos A$$

4. NAPIER'S ANALOGY (TANGENT RULE)

$$\tan[(B - C) / 2] = [(b - c) / (b + c)] \cot(A / 2)$$
$$\tan[(C - A) / 2] = [(c - a) / (c + a)] \cot(B / 2)$$
$$\tan[(A - B) / 2] = [(a - b) / (a + b)] \cot(C / 2)$$

5. AREA OF TRIANGLE (Δ)

1. **Basic / Trigonometric Form:**

$$\Delta = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B = \frac{1}{2} ab \sin C$$

2. **Heron's Formula:**

$$\Delta = \sqrt{[s(s - a)(s - b)(s - c)]}$$

3. **Circumradius & Inradius Form:**

$$\Delta = abc / (4R) = r \cdot s$$

4. **Side-Angle Form:**

$$\Delta = \frac{1}{2} [a^2 \sin B \sin C / \sin(B + C)] = \frac{1}{2} [b^2 \sin C \sin A / \sin(C + A)]$$

5. **Sine Relations from Area:**

$$\sin A = 2\Delta / (bc),\quad \sin B = 2\Delta / (ca),\quad \sin C = 2\Delta / (ab)$$

6. TRIGONOMETRIC RATIOS OF HALF ANGLES

Angle	sin(θ / 2)	cos(θ / 2)	tan(θ / 2)
A / 2	$\sqrt{[((s - b)(s - c)) / (bc)]}$	$\sqrt{[(s(s - a)) / (bc)]}$	$\sqrt{[((s - b)(s - c)) / (s(s - a))]} = r / (s - a)$
B / 2	$\sqrt{[((s - c)(s - a)) / (ca)]}$	$\sqrt{[(s(s - b)) / (ca)]}$	$\sqrt{[((s - c)(s - a)) / (s(s - b))]} = r / (s - b)$
C / 2	$\sqrt{[((s - a)(s - b)) / (ab)]}$	$\sqrt{[(s(s - c)) / (ab)]}$	$\sqrt{[((s - a)(s - b)) / (s(s - c))]} = r / (s - c)$

7. CIRCUMCIRCLE (RADIUS R)

- $R = a / (2 \sin A) = b / (2 \sin B) = c / (2 \sin C)$
- $R = abc / (4\Delta)$
- **Side expansion:** $R = \sqrt{[(a+b+c)(-a+b+c)(a-b+c)(a+b-c)]} / (4\Delta)$

8. INCIRCLE (RADIUS R)

1. $r = \Delta / s$
2. $r = (s - a) \tan(A/2) = (s - b) \tan(B/2) = (s - c) \tan(C/2)$
3. $r = 4R \sin(A/2) \sin(B/2) \sin(C/2)$
4. $r = [a \sin(B/2) \sin(C/2)] / \cos(A/2)$ (cyclic)

9. EX-CIRCLES / ESCAPED CIRCLES (RADII R_1, R_2, R_3)

1. Area Form:

$r_1 = \Delta / (s - a)$

$r_2 = \Delta / (s - b)$

$r_3 = \Delta / (s - c)$

2. Semi-perimeter & Tan:

$r_1 = s \tan(A/2)$

$r_2 = s \tan(B/2)$

$r_3 = s \tan(C/2)$

3. Circumradius Form:

$r_1 = 4R \sin(A/2) \cos(B/2) \cos(C/2)$

$r_2 = 4R \cos(A/2) \sin(B/2) \cos(C/2)$

$r_3 = 4R \cos(A/2) \cos(B/2) \sin(C/2)$

4. Side-Cosine Form: $r_1 = [a \cos(B/2) \cos(C/2)] / \cos(A/2)$, $r_2 = [b \cos(C/2) \cos(A/2)] / \cos(B/2)$, $r_3 = [c \cos(A/2) \cos(B/2)] / \cos(C/2)$

10. IMPORTANT LINE SEGMENTS (CEVIANS, MEDIANS, BISECTORS, ALTITUDES)

- **Median Length (m_a):**
  $m_a = \frac{1}{2} \sqrt{(2b^2 + 2c^2 - a^2)}$
- **Apollonius' Theorem:**
  $b^2 + c^2 = 2(m_a^2 + (a/2)^2)$ *(Median AD on BC)*
- **Sides in terms of Medians:**
  $a = \frac{2}{3} \sqrt{(2m_b^2 + 2m_c^2 - m_a^2)}$
- **Altitude Length (h_a):**
  $h_a = 2\Delta / a = b \sin C = c \sin B$
- **Internal Angle Bisector Ratio & Length (l_a):**
  Ratio: $BD / DC = c / b$
  Length: $l_a = [2bc \cos(A/2)] / (b + c) = \sqrt{[bc (1 - a^2 / (b + c)^2)]}$
- **External Angle Bisector Ratio & Length:**
  Ratio: $BD / DC' = -c / b$
  Length: $l_a^{(ext)} = [2bc \sin(A/2)] / |b - c|$
- **Stewart's Theorem (General Cevian d):**
  $b^2m + c^2n = a(d^2 + mn)$ *(BD = m, DC = n, m+n = a)*

11. M-N THEOREM

If point D on BC divides BC in ratio $BD:DC = m:n$ and $\angle ADC = \theta$, $\angle BAD = \alpha$, $\angle CAD = \beta$:

$$(m + n) \cot \theta = m \cot \alpha - n \cot \beta$$
$$(m + n) \cot \theta = n \cot B - m \cot C$$

12. CEVA'S & MENELAUS' THEOREMS

- **Ceva's Theorem (Concurrence):**
  If AD, BE, CF concur at a point: $(BD/DC) \cdot (CE/EA) \cdot (AF/FB) = 1$
- **Menelaus' Theorem (Collinearity):**
  If line cuts sides BC, CA, AB at P, Q, R: $(BP/PC) \cdot (CQ/QA) \cdot (AR/RB) = -1$

13. DISTANCES OF SPECIAL POINTS FROM VERTICES AND SIDES

Point	Distance to Vertex A	Distance to Side BC (a)
Circumcenter (O)	$OA = R$	$O_a = R \cos A$
Incenter (I)	$IA = r \operatorname{cosec}(A / 2)$	$I_a = r$
Excenter (I_1)	$I_1A = r_1 \operatorname{cosec}(A / 2)$	$I_{1a} = r_1$
Orthocenter (H)	$HA = 2R \cos A$	$H_a = 2R \cos B \cos C$
Centroid (G)	$GA = \frac{1}{3} \sqrt{(2b^2 + 2c^2 - a^2)}$	$G_a = 2\Delta / (3a)$

14. PEDAL TRIANGLE (DEF)

Formed by joining feet of altitudes D, E, F.

- **Angles:** $\pi - 2A$, $\pi - 2B$, $\pi - 2C$
- **Sides:** $a \cos A = R \sin 2A$ (cyclic)
- **Area:** $\Delta_p = 2\Delta \cos A \cos B \cos C$
- **Circumradius:** $R_p = R / 2$

15. EXCENTRAL TRIANGLE ($I_1I_2I_3$)

Formed by joining 3 excenters I_1, I_2, I_3.

- **Angles:** $\frac{1}{2}(\pi - A)$, $\frac{1}{2}(\pi - B)$, $\frac{1}{2}(\pi - C)$
- **Sides:** $4R \cos(A/2)$, $4R \cos(B/2)$, $4R \cos(C/2)$
- **Segment Lengths:** $I I_1 = 4R \sin(A/2)$ (cyclic)

• **Inradius:** $r_p = 2R \cos A \cos B \cos C$ [Added]

• Orthocenter H of ΔABC is *Incenter* of Pedal ΔDEF.

• **Area:** $\Delta_e = 8R^2 \cos(A/2) \cos(B/2) \cos(C/2)$ [Added]

• Incenter I of ΔABC is *Orthocenter* of Excentral $\Delta I_1 I_2 I_3$.

16. DISTANCES BETWEEN SPECIAL POINTS

• **Circumcenter - Incenter (OI):**
$OI^2 = R^2 - 2Rr = R^2(1 - 8 \sin(A/2) \sin(B/2) \sin(C/2))$ *(Euler's Relation)*

• **Circumcenter - Orthocenter (OH):**
$OH^2 = R^2(1 - 8 \cos A \cos B \cos C)$

• **Circumcenter - Centroid (OG):**
$OG^2 = R^2 - \text{Ninth}(a^2 + b^2 + c^2) = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$

• **Incenter - Excenter ($I I_1$):**
$I I_1 = 4R \sin(A/2)$

• **Euler Line Ratio:**
O, G, H are collinear with ratio $OG : GH = 1 : 2$

• **Nine-Point Circle Radius:**
$R_9 = R / 2$

17. FREQUENTLY USED EXPRESS IDENTITIES & SPECIAL CASES

1. $\tan(A/2) \tan(B/2) = (s - c) / s$
2. $\cot(A/2) \cot(B/2) = s / (s - c)$
3. $\tan(A/2) + \tan(B/2) = (c / s) \cot(C/2) = (c / \Delta)(s - c)$
4. $\cot(A/2) + \cot(B/2) = [c / (s - c)] \cot(C/2)$
5. $\tan(A/2) - \tan(B/2) = [(a - b) / \Delta](s - c)$
6. $r_1 + r_2 + r_3 - r = 4R$ [Added]
7. $r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2$ [Added]
8. $1/r_1 + 1/r_2 + 1/r_3 = 1/r$ [Added]
9. **Right Triangle** ($\angle C = 90°$): $R = c / 2$, $r = (a + b - c) / 2 = s - c$
10. **Equilateral Triangle** (side a): $\Delta = (\sqrt{3}/4)a^2$, $R = a/\sqrt{3}$, $r = a/(2\sqrt{3})$, $r : R : r_1 = 1 : 2 : 3$

BINOMIAL THEOREM

JEE Main & Advanced Revision Sheet • Comprehensive Formula Bank

6. STANDARD BINOMIAL COEFFICIENT SUMS

Notation: $C_r \equiv {}^nC_r$

$$C_0 + C_1 + C_2 + \ldots + C_n = 2^n$$

$$C_0 - C_1 + C_2 - C_3 + \ldots + (-1)^n C_n = 0$$

$$C_0 + C_2 + C_4 + \ldots = C_1 + C_3 + C_5 + \ldots = 2^{n-1}$$

Odd Index Total Sum: ${}^{2n+1}C_0 + {}^{2n+1}C_1 + \ldots + {}^{2n+1}C_n = 2^{2n}$

7. NUMERICALLY GREATEST TERM (NGT)

To find the numerically greatest term in expansion of $(1 + x)^n$:

Calculate: $m = \dfrac{|x|\,(n+1)}{|x| + 1}$

- **If m is an INTEGER:** Two equal greatest terms exist: T_m and T_{m+1} ($|T_m| = |T_{m+1}|$).
- **If m is NOT an integer:** Only one greatest term exists: $T_{[m]+1}$, where $[\bullet]$ is GIF.

For $(A + B)^n$: Rewrite as $A^n(1 + B/A)^n$, then apply NGT with $x = B/A$.

8. TERM INDEPENDENT OF X & RADICAL TERMS

Term Independent of x: In $(a\,x^{\alpha} \pm b/x^{\beta})^n$:

General exponent of x: $n\alpha - r(\alpha + \beta) = 0 \Rightarrow r = \dfrac{n\alpha}{\alpha + \beta}$

If r is non-integer, no independent term exists.

Rational Terms in $(a^{1/p} + b^{1/q})^N$ (a, b are primes):

General term: $T_{r+1} = {}^NC_r\, a^{(N-r)/p}\, b^{r/q}$

Term is rational iff both $\dfrac{N-r}{p}$ and $\dfrac{r}{q}$ are integers $(0 \le r \le N)$.

No. of Irrational Terms = (Total terms) - (No. of rational terms).

BINOMIAL THEOREM

Advanced Series, Summation Identities & Multinomial Theorem

9. SERIES WITH LINEAR FACTORS (R • $^{N}C_{R}$)

Method: Differentiation of $(1+x)^n$ or using $r \cdot {}^nC_r = n \cdot {}^{n-1}C_{r-1}$

$$\sum_{r=1}^{n} r \cdot {}^nC_r = 1 \cdot C_1 + 2 \cdot C_2 + \ldots + n \cdot C_n = n \cdot 2^{n-1}$$

$$\sum_{r=1}^{n} (-1)^{r-1} r \cdot {}^nC_r = 1 \cdot C_1 - 2 \cdot C_2 + 3 \cdot C_3 - \ldots = 0$$

$$\sum_{r=1}^{n} r^2 \cdot {}^nC_r = 1^2 \cdot C_1 + 2^2 \cdot C_2 + \ldots + n^2 \cdot C_n = n(n+1) \cdot 2^{n-2}$$

$$\sum_{r=1}^{n} r^3 \cdot {}^nC_r = n^2(n+3) \cdot 2^{n-3}$$ [Added]

Arithmetic-Binomial Series:

$$\sum_{r=0}^{n} (a + r\,d) \, {}^nC_r = (2a + n\,d) \cdot 2^{n-1}$$

10. SERIES WITH RECIPROCAL FACTORS ($^{N}C_{R}$/ (R+1))

Method: Integration of $(1+x)^n$ or using $\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

$$C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + \ldots + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$$

$$\sum_{r=0}^{n} \frac{{}^nC_r}{(r+1)(r+2)} = \frac{2^{n+2} - n - 3}{(n+1)(n+2)}$$ [Added]

11. SUM OF SQUARES & PRODUCT COEFFICIENTS

Obtained by equating coefficients in $(1+x)^n (1+x)^n = (1+x)^{2n}$

$$C_0^2 + C_1^2 + C_2^2 + \ldots + C_n^2 = {}^{2n}C_n = \frac{(2n)!}{(n!)^2}$$

$$C_0C_r + C_1C_{r+1} + C_2C_{r+2} + \ldots + C_{n-r}C_n = {}^{2n}C_{n-r} = \frac{(2n)!}{(n-r)!(n+r)!}$$

$$C_0^2 - C_1^2 + C_2^2 - C_3^2 + \ldots =$$

- 0 *(if n is ODD)*
- $(-1)^{n/2} \cdot {}^nC_{n/2}$ *(if n is EVEN)*

12. DOUBLE SUMMATION ($\sum\sum C_i C_j$) [JEE Adv]

For independent variables i and j running from 0 to n:

$$\sum_{0 \le i < j \le n} \sum (C_i + C_j) = n \cdot 2^n$$

$$\sum_{0 \le i < j \le n} \sum (C_i \cdot C_j) = \frac{1}{2}\left[\left(\sum_{i=0}^{n} C_i\right)^2 - \sum_{i=0}^{n} C_i^2\right] = \frac{1}{2}\left[2^{2n} - {}^{2n}C_n\right]$$

$$\sum_{0 \le i < j \le n} \sum (i + j)\, C_i\, C_j = n \cdot 2^{2n-1}$$

13. MULTINOMIAL THEOREM (POSITIVE INTEGRAL INDEX)

$$(a_1 + a_2 + \ldots + a_m)^n = \sum \frac{n!}{n_1!\, n_2! \ldots n_m!} a_1^{n_1} a_2^{n_2} \ldots a_m^{n_m}$$

Conditions: $n_1, n_2, \ldots, n_m \in W$ and $n_1 + n_2 + \ldots + n_m = n$.

- **General Term:** $\frac{n!}{n_1!\, n_2! \ldots n_m!} a_1^{n_1} a_2^{n_2} \ldots a_m^{n_m}$
- **Total Number of Distinct Terms:** $^{n+m-1}C_{m-1}$
- **Trinomial Expansion $(x+y+z)^n$:** Number of terms = $^{n+2}C_2 = \frac{(n+1)(n+2)}{2}$
- **Greatest Coefficient in Multinomial:**
  Value = $\frac{n!}{(q!)^{m-r}((q+1)!)^r}$, where q = quotient & r = remainder when n is divided by m ($n = mq + r$).
- **Coefficient of x^r in $(a_0 + a_1x + a_2x^2 + \ldots + a_kx^k)^n$:**
  $\sum \frac{n!}{n_0!\, n_1! \ldots n_k!} a_0^{n_0} a_1^{n_1} \ldots a_k^{n_k}$
  Subject to: $\sum n_i = n$ and $n_1 + 2n_2 + 3n_3 + \ldots + k\,n_k = r$.

14. POLYNOMIAL EXPANSION PRODUCTS

In expansion of $(x-1)(x-2)(x-3)\ldots(x-n)$:

- Coefficient of $x^{n-1} = -\sum i = -\frac{n(n+1)}{2}$
- Coefficient of $x^{n-2} = \sum_{i<j} i\,j = \frac{(n-1)n(n+1)(3n+2)}{24}$ [Added]

In expansion of $(x+1)(x+2)(x+3)\ldots(x+n)$:

- Coefficient of $x^{n-1} = +\frac{n(n+1)}{2}$

15. BINOMIAL THEOREM FOR ANY INDEX (N ∈ Q)

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

STRICT CONDITION: Valid ONLY if **|x| < 1**.

- Number of terms is **INFINITE**.
- Symbol ${}^nC_r$ cannot be used directly; use full fraction form.
- **General Term:** $T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r$
- If expression is $(x+y)^n$, pull out larger term: $x^n(1+y/x)^n$ if $|y/x| < 1$.

Standard Infinite Expansions (|x| < 1):

1. $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r + \dots$
2. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r + \dots$
3. $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots + (r+1)x^r + \dots$
4. $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots + (-1)^r(r+1)x^r + \dots$
5. $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots + \frac{(r+1)(r+2)}{2}x^r + \dots$

[Added]

16. APPROXIMATION FORMULAS (|X| ≪ 1)

If $|x| \ll 1$, then $(1+x)^n \approx 1 + nx$

Higher powers (x^2, x^3, ...) are neglected.

Examples: $(1-x)^n \approx 1 - nx$ | $(1+x)^{-n} \approx 1 - nx$ | $(1-x)^{-n} \approx 1 + nx$

17. DIVISIBILITY & REMAINDER PROBLEMS

To prove expression E is divisible by k or find remainder when divided by k:

Express base as $(1+a)^n$ or $(k \pm 1)^n$ and expand.

$(1+a)^n = 1 + {}^nC_1a + {}^nC_2a^2 + \dots + {}^nC_na^n$

$= 1 + na + a^2[{}^nC_2 + {}^nC_3a + \dots] = 1 + na + k \cdot \lambda$

Key JEE Result: $(1+a)^n - 1 - na$ is always divisible by a^2.

18. FRACTIONAL PART & INTEGER METHOD (I + F)

For expressions of the type $(A+\sqrt{B})^n = I + f$, where $I \in Z^+$, $0 \le f < 1$: Define complementary fraction $f' = (A-\sqrt{B})^n$ or $(\sqrt{B}-A)^n$ such that $0 < f' < 1$.

- If $I + f + f'$ becomes an EVEN integer $2k$:
  Then $f + f' = 1 \Rightarrow f' = 1 - f$
  Product Identity: $(I+f)(1-f) = (A^2 - B)^n$

- If $I + f - f'$ becomes an ODD integer:
  Then $f = f' \Rightarrow (I+f) \cdot f = (A^2 - B)^n$

Property: Integer part I is odd/even based on cancellation of irrational terms.

19. EXPONENTIAL SERIES

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty \quad (\forall x \in R \text{ or } C)$$

$$a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 (\ln a)^2}{2!} + \dots \infty \quad (a > 0)$$

- **Euler Constant e:** $e = \lim_{n\to\infty}(1+1/n)^n = 2.71828\dots$
- $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \infty$
- $e^{-1} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots \infty$
- $e + e^{-1} = 2\left[1 + \frac{1}{2!} + \frac{1}{4!} + \dots \infty\right]$
- $e - e^{-1} = 2\left[\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots \infty\right]$

20. LOGARITHMIC SERIES

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty \quad (-1 < x \le 1)$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots \infty \quad (-1 \le x < 1)$$

$$\ln\frac{1+x}{1-x} = 2\left[x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \infty\right] \quad (|x| < 1)$$

Special Values: $\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

21. COMPLEX NUMBERS & BINOMIAL [JEE Adv]

$$C_0 + C_4 + C_8 + \dots = 2^{n-2} + 2^{(n-2)/2}\cos\frac{n\pi}{4}$$

$$C_1 + C_5 + C_9 + \dots = 2^{n-2} + 2^{(n-2)/2}\sin\frac{n\pi}{4}$$

Derived by expanding $(1+i)^n$ using De Moivre's Theorem.

CALCULUS & INTEGRALS — JEE REVISION SHEET

Differentiation • Indefinite Integration • Definite Integration • Area Under Curves

JEE MAIN & ADVANCED

1. DIFFERENTIATION & CALCULUS FOUNDATIONS

CORE DIFFERENTIATION RULES

Rule	Formula
Linearity	$d/dx\,[c\cdot f(x) \pm g(x)] = c\cdot f'(x) \pm g'(x)$
Product Rule	$d/dx\,[u\cdot v] = u\cdot v' + v\cdot u'$
Triple Product	$d/dx\,[u\cdot v\cdot w] = u'vw + uv'w + uvw'$
Quotient Rule	$d/dx\,[u/v] = (v\cdot u' - u\cdot v') / v^2$
Chain Rule	$dy/dx = (dy/du)\cdot(du/dx)$
Parametric	$dy/dx = (dy/dt)/(dx/dt)$; $d^2y/dx^2 = [d/dt(dy/dx)]/(dx/dt)$
Logarithmic	$y = [f(x)]^{g(x)} \Rightarrow dy/dx = y\cdot[g'(x)\ln f(x) + g(x)f'(x)/f(x)]$

STANDARD DERIVATIVES REFERENCE

Function f(x)	f'(x)	Conditions
x^n, a^x, e^x	$n\,x^{n-1}$, $a^x \ln a$, e^x	$a > 0, a \neq 1$
$\ln\|x\|$, $\log_a\|x\|$	$1/x$, $1/(x \ln a)$	$x \neq 0$
$\sin x$, $\cos x$	$\cos x$, $-\sin x$	$x \in \mathbb{R}$
$\tan x$, $\cot x$	$\sec^2 x$, $-\text{cosec}^2 x$	Domain limits
$\sec x$, $\text{cosec}\ x$	$\sec x \tan x$, $-\text{cosec}\ x \cot x$	Domain limits
$\sin^{-1}x$, $\cos^{-1}x$	$1/\sqrt{(1-x^2)}$, $-1/\sqrt{(1-x^2)}$	$\|x\| < 1$
$\tan^{-1}x$, $\cot^{-1}x$	$1/(1+x^2)$, $-1/(1+x^2)$	$x \in \mathbb{R}$
$\sec^{-1}x$, $\text{cosec}^{-1}x$	$1/(\|x\|\sqrt{(x^2-1)})$, $-1/(\|x\|\sqrt{(x^2-1)})$	$\|x\| > 1$
$\|x\|$	$\|x\|/x = \text{sgn}(x)$	$x \neq 0$

Determinant Differentiation:
If $F(x) = |f_i(x)\ g_i(x)\ h_i(x)|$ (for 3×3 matrix), then:
$F'(x) = |R_1'\ R_2\ R_3| + |R_1\ R_2'\ R_3| + |R_1\ R_2\ R_3'|$ (Row-wise or Column-wise differentiation).

Standard Trigonometric Substitutions:

- $\sqrt{(a^2-x^2)} \Rightarrow x = a\sin\theta$ or $a\cos\theta$
- $\sqrt{(a^2+x^2)} \Rightarrow x = a\tan\theta$ or $a\sinh\theta$
- $\sqrt{(x^2-a^2)} \Rightarrow x = a\sec\theta$ or $a\cosh\theta$
- $\sqrt{((a-x)/(a+x))}$ or $\sqrt{((a^2-x^2)/(a^2+x^2))} \Rightarrow x = a\cos 2\theta$ or $x^2 = a^2\cos 2\theta$

2. INDEFINITE INTEGRATION — STANDARD INTEGRALS & IDENTITIES

ALGEBRAIC FORMS

- $\int x^n\,dx = (x^{n+1}/(n+1)) + C$ *(n ≠ -1)*
- $\int (1/x)\,dx = \ln|x| + C$
- $\int e^{ax+b}\,dx = (1/a)e^{ax+b} + C$
- $\int a^{px+q}\,dx = (1/p)(a^{px+q}/\ln a) + C$

TRIGONOMETRIC FORMS

- $\int \sin x\,dx = -\cos x + C$
- $\int \cos x\,dx = \sin x + C$
- $\int \sec^2 x\,dx = \tan x + C$
- $\int \text{cosec}^2 x\,dx = -\cot x + C$
- $\int \sec x \tan x\,dx = \sec x + C$
- $\int \text{cosec}\ x \cot x\,dx = -\text{cosec}\ x + C$

LOG-TRIG FORMS

- $\int \tan x\,dx = \ln|\sec x| = -\ln|\cos x|$
- $\int \cot x\,dx = \ln|\sin x| + C$
- $\int \sec x\,dx = \ln|\sec x + \tan x| = \ln|\tan(\pi/4 + x/2)|$
- $\int \text{cosec}\ x\,dx = \ln|\text{cosec}\ x - \cot x| = \ln|\tan(x/2)|$

9 CORE STANDARD INTEGRALS (MUST-KNOW FOR JEE MAIN/ADVANCED)

No.	Integral Form	Evaluated Result	No.	Integral Form	Evaluated Result
1	$\int dx/(x^2+a^2)$	$(1/a)\tan^{-1}(x/a) + C$	4	$\int dx/\sqrt{(a^2-x^2)}$	$\sin^{-1}(x/a) + C$
2	$\int dx/(x^2-a^2)$	$(1/(2a))\ln\|(x-a)/(x+a)\| + C$	5	$\int dx/\sqrt{(x^2+a^2)}$	$\ln\|x + \sqrt{(x^2+a^2)}\| + C$
3	$\int dx/(a^2-x^2)$	$(1/(2a))\ln\|(a+x)/(a-x)\| + C$	6	$\int dx/\sqrt{(x^2-a^2)}$	$\ln\|x + \sqrt{(x^2-a^2)}\| + C$
7	$\int \sqrt{(a^2-x^2)}\,dx$	$(x/2)\sqrt{(a^2-x^2)} + (a^2/2)\sin^{-1}(x/a) + C$			
8	$\int \sqrt{(x^2+a^2)}\,dx$	$(x/2)\sqrt{(x^2+a^2)} + (a^2/2)\ln\|x + \sqrt{(x^2+a^2)}\| + C$			
9	$\int \sqrt{(x^2-a^2)}\,dx$	$(x/2)\sqrt{(x^2-a^2)} - (a^2/2)\ln\|x + \sqrt{(x^2-a^2)}\| + C$			

INTEGRATION BY PARTS (ILATE RULE)

$\int u\cdot v\,dx = u\int v\,dx - \int[(du/dx)\cdot\int v\,dx]\,dx$

Priority Order (ILATE): Inverse Trig > Logarithmic > Algebraic > Trigonometric > Exponential.

SPECIAL EXPONENTIAL FORMS

- $\int e^x[f(x) + f'(x)]\,dx = e^x f(x) + C$
- $\int [f(x) + x f'(x)]\,dx = x f(x) + C$

3. INTEGRATION TECHNIQUES & SPECIAL STANDARD FORMS

QUADRATIC FORMS: $AX^2 + BX + C$

- **Form 1:** $\int dx / (ax^2 + bx + c)$ or $\int dx / \sqrt{(ax^2 + bx + c)}$
  Method: Express $ax^2 + bx + c = a [(x + b/(2a))^2 + (4ac - b^2)/(4a^2)]$, then substitute $x + b/(2a) = t$.
- **Form 2:** $\int (px + q)\, dx / (ax^2 + bx + c)$ or $\int (px + q)\, dx / \sqrt{(ax^2 + bx + c)}$
  Method: Put $px + q = A \cdot d/dx\,(ax^2 + bx + c) + B = A(2ax + b) + B$. Compare coefficients for A & B.

TRIGONOMETRIC RATIONAL FORMS

- **Form 1:** $\int dx / (a + b \sin^2 x + c \cos^2 x)$
  Method: Divide numerator and denominator by $\cos^2 x$, substitute $\tan x = t$, $\sec^2 x\, dx = dt$.
- **Form 2:** $\int dx / (a + b \sin x + c \cos x)$
  Method: Substitute half-angle identities:
  $\sin x = (2 \tan(x/2)) / (1 + \tan^2(x/2))$, $\cos x = (1 - \tan^2(x/2)) / (1 + \tan^2(x/2))$, then put $\tan(x/2) = t$.
- **Form 3:** $\int (p \cos x + q \sin x + r) / (a \cos x + b \sin x + c)\, dx$
  Method: Put $\text{Num} = A(\text{Denom}) + B \cdot d/dx(\text{Denom}) + C$.

IRRATIONAL RADICAL INTEGRATION FORMS (JEE ADVANCED SPECIAL)

Integral Form	Substitution / Method
$\int dx / ((ax + b)\sqrt{(px + q)})$	Put $px + q = t^2$
$\int dx / ((ax^2 + bx + c)\sqrt{(px + q)})$	Put $px + q = t^2$
$\int dx / ((ax + b)\sqrt{(px^2 + qx + r)})$	Put $ax + b = 1/t$
$\int dx / ((ax^2 + b)\sqrt{(px^2 + q)})$	Put $x = 1/t$, then substitute $c + dt^2 = u^2$
$\int \sqrt{((x - \alpha) / (\beta - x))}\, dx$ or $\int \sqrt{((x - \alpha)(\beta - x))}\, dx$	Put $x = \alpha \cos^2\theta + \beta \sin^2\theta$
$\int \sqrt{((x - \alpha) / (x - \beta))}\, dx$ or $\int \sqrt{((x - \alpha)(x - \beta))}\, dx$	Put $x = \alpha \sec^2\theta - \beta \tan^2\theta$

ALGEBRAIC TWINS & EULER FORMS

- **Algebraic Twins:**

$\int (x^2 \pm 1) / (x^4 + Kx^2 + 1)\, dx$

Divide Ndr & Ddr by x^2; write Ddr as $(x \mp 1/x)^2 + C$, put $x \mp 1/x = t$.

- **Key Manipulations:**

$2 / (x^4 + 1) = (x^2 + 1)/(x^4 + 1) - (x^2 - 1)/(x^4 + 1)$

$2x^2 / (x^4 + 1) = (x^2 + 1)/(x^4 + 1) + (x^2 - 1)/(x^4 + 1)$

TRIGONOMETRIC TWINS & PRODUCTS

- **Trig Twins:**

$\int \sqrt{(\tan x)}\, dx$ or $\int \sqrt{(\cot x)}\, dx$

Put $\tan x = t^2 \Rightarrow dx = 2t\, dt / (1 + t^4)$. Converts into algebraic twin form!

- **Product Forms e^{ax} sin/cos bx:**

$\int e^{ax} \sin bx\, dx = (e^{ax} / (a^2+b^2)) [a \sin bx - b \cos bx] + C$

$\int e^{ax} \cos bx\, dx = (e^{ax} / (a^2+b^2)) [a \cos bx + b \sin bx] + C$

4. DEFINITE INTEGRATION & PROPERTIES

Fundamental Theorems of Calculus:

FTC-1: If $g(x) = \int_a^x f(t)\, dt$, then $g'(x) = f(x)$.

FTC-2: $\int_a^b f(x)\, dx = F(b) - F(a)$, where $F'(x) = f(x)$.

PROPERTIES P0 TO P8 (JEE FUNDAMENTAL PROPERTIES)

Prop	Mathematical Statement	Key Application / Condition
P0	$\int_a^b f(x)\, dx = \int_a^b f(t)\, dt$	Change of dummy variable
P1	$\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx$; $\int_a^a f(x)\, dx = 0$	Reversal of integration limits
P2	$\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$	Splitting intervals (Essential for $\lvert f(x)\rvert$, $[x]$, $\{x\}$)
P3 (King)	$\int_a^b f(x)\, dx = \int_a^b f(a + b - x)\, dx$	**Most Used Property!** Eliminates x, simplifies trig numerators
P4	$\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$	King's Property for lower limit = 0
P5 (Even/Odd)	$\int_{-a}^a f(x)\, dx = 2 \int_0^a f(x)\, dx$ if $f(-x)=f(x)$ (Even); $= 0$ if $f(-x)=-f(x)$ (Odd)	Check symmetry of integrand first
P6 (Queen)	$\int_0^{2a} f(x)\, dx = \int_0^a [f(x) + f(2a - x)]\, dx$ $= 2 \int_0^a f(x)\, dx$ if $f(2a-x) = f(x)$; $= 0$ if $f(2a-x) = -f(x)$	Halving upper limit for periodic or symmetric functions
P7 (Periodic)	If $f(x+T) = f(x)$: (i) $\int_0^{nT} f(x)\, dx = n \int_0^T f(x)\, dx$ (ii) $\int_a^{a+nT} f(x)\, dx = n \int_0^T f(x)\, dx$; (iii) $\int_{mT}^{nT} f(x)\, dx = (n - m) \int_0^T f(x)\, dx$	$n, m \in \mathbb{Z}$, T = fundamental period

LEIBNITZ RULE (DIFFERENTIATION UNDER INTEGRAL SIGN)

If $F(x) = \int_{g(x)}^{h(x)} f(t)\,dt$, then:
$dF/dx = h'(x)\cdot f(h(x)) - g'(x)\cdot f(g(x))$
$d/dx \int_{g(x)}^{h(x)} f(x,t)\,dt = \int_{g(x)}^{h(x)} \partial f/\partial x\,dt + h'f(x,h) - g'f(x,g)$
[Added]

DEFINITE INTEGRAL AS LIMIT OF A SUM

$\lim_{n\to\infty} (1/n) \sum_{r=1}^{pn} f(r/n) = \int_0^p f(x)\,dx$
Conversion Steps: Replace $r/n \to x$, $1/n \to dx$, $\lim \sum \to \int$. Lower limit = $\lim_{n\to\infty}$ (lower r / n); Upper limit = $\lim_{n\to\infty}$ (upper r / n).

5. DEFINITE INTEGRAL RESULTS, INEQUALITIES & REDUCTION FORMULAS

WALLI'S FORMULA

$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = [\,(n-1)(n-3)\ldots(1 \text{ or } 2)\,] / [\,n(n-2)\ldots(1 \text{ or } 2)\,]\cdot K$
where K = π/2 if n is even, K = 1 if n is odd.
$\int_0^{\pi/2} \sin^m x \cos^n x\,dx = [\,(m-1)(m-3)\ldots\,][\,(n-1)(n-3)\ldots\,] / [\,(m+n)(m+n-2)\ldots\,]\cdot K$
where K = π/2 if BOTH m and n are even, else K = 1.

GAMMA & BETA FUNCTIONS [ADDED]

$\Gamma(n) = \int_0^\infty e^{-t} t^{n-1}\,dt$; $\Gamma(n+1) = n\,\Gamma(n) = n!$ *(n ∈ ℕ)*
$\Gamma(1/2) = \sqrt{\pi}$; $\Gamma(n)\Gamma(1-n) = \pi / \sin(n\pi)$
$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx = (\Gamma(m)\Gamma(n)) / \Gamma(m+n)$
$\int_0^{\pi/2} \sin^p x \cos^q x\,dx = 1/2\, B((p+1)/2, (q+1)/2)$

STANDARD DEFINITE VALUES (DIRECT JEE SHORTCUTS)

- $\int_0^{\pi/2} \ln(\sin x)\,dx = \int_0^{\pi/2} \ln(\cos x)\,dx = -(\pi/2)\ln 2$
- $\int_0^{\pi/2} \ln(\tan x)\,dx = 0$
- $\int_0^{\pi} x \sin x / (1+\cos^2 x)\,dx = \pi^2/4$
- $\int_a^b (|x|/x)\,dx = |b| - |a|$

INEQUALITIES & ESTIMATION OF INTEGRALS

- If $m \le f(x) \le M$ for $x \in [a,b] \Rightarrow m(b-a) \le \int_a^b f(x)\,dx \le M(b-a)$
- If $f(x) \le g(x) \Rightarrow \int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$
- $|\int_a^b f(x)\,dx| \le \int_a^b |f(x)|\,dx$
- **Cauchy-Schwarz:** $(\int_a^b f(x)g(x)\,dx)^2 \le (\int_a^b f^2(x)\,dx)(\int_a^b g^2(x)\,dx)$

KEY REDUCTION FORMULAS FOR INDEFINITE / DEFINITE INTEGRALS

Function	Reduction Formula
$I_n = \int \sin^n x\,dx$	$I_n = -(\sin^{n-1}x \cos x)/n + ((n-1)/n)\, I_{n-2}$
$I_n = \int \cos^n x\,dx$	$I_n = (\cos^{n-1}x \sin x)/n + ((n-1)/n)\, I_{n-2}$
$I_n = \int \tan^n x\,dx$	$I_n = (\tan^{n-1}x)/(n-1) - I_{n-2}$
$I_n = \int x^n e^{mx}\,dx$	$I_n = (1/m)\, x^n e^{mx} - (n/m)\, I_{n-1}$
$I_n = \int (\ln x)^n\,dx$	$I_n = x(\ln x)^n - n\, I_{n-1}$
$I_n = \int dx / (x^2+a^2)^n$	$I_n = x / (2(n-1)a^2(x^2+a^2)^{n-1}) + ((2n-3)/(2(n-1)a^2))\, I_{n-1}$

6. APPLICATION OF INTEGRALS (AOI) & AREA UNDER CURVES

AREA FORMULATIONS (CARTESIAN & PARAMETRIC)

Region Bounded By	Integral Formula
y = f(x), x-axis, x = a to x = b	Area = $\int_a^b \lvert y\rvert\,dx = \int_a^b \lvert f(x)\rvert\,dx$
x = g(y), y-axis, y = c to y = d	Area = $\int_c^d \lvert x\rvert\,dy = \int_c^d \lvert g(y)\rvert\,dy$
Between y = f(x) and y = g(x)	Area = $\int_a^b \lvert f(x) - g(x)\rvert\,dx$
Parametric x = f(t), y = g(t)	Area = $\int_{t_1}^{t_2} g(t)\,f'(t)\,dt$
Average Value of f(x) on [a, b]	$f_{avg} = (1/(b-a)) \int_a^b f(x)\,dx$

CURVE TRACING RULES FOR QUICK AREA SKETCHING

- **Symmetry:**
  - Even powers of y ⇒ Symmetric about x-axis.
  - Even powers of x ⇒ Symmetric about y-axis.
  - Interchanging x and y unchanged ⇒ Symmetric about line y = x.
  - (x, y) → (−x, −y) unchanged ⇒ Symmetric in opposite quadrants.
- **Tangents & Stationary Points:** Solve $dy/dx = 0$ for horizontal tangents.
- **Asymptotes & Intercepts:** Find points where curve cuts axes (x=0 or y=0). Check y as $x \to \pm\infty$.

STANDARD AREA SHORTCUTS & FORMULAS (HIGH-YIELD FOR JEE SPEED)

Curves Involved	Enclosed Area Formula	Conditions / Notes
Ellipse $x^2/a^2 + y^2/b^2 = 1$	$\text{Area} = \pi\, a\, b$	Total enclosed region
Parabolas $y^2 = 4ax$ and $x^2 = 4by$	$\text{Area} = (16 / 3)\, a\, b$	Standard intersection region
Parabola $y^2 = 4ax$ and line $y = mx$	$\text{Area} = 8\, a^2 / (3\, m^3)$	$m \neq 0$
Parabola $y^2 = 4ax$ and double ordinate $x = x_1$	$\text{Area} = (4/3)\, x_1\, \sqrt{(4a\, x_1)} = (2/3) \cdot (\text{Base} \times \text{Height})$	Arch of parabola
Parabola $x^2 = 4ay$ and line $y = mx + c$ [Added]	$\text{Area} = (a / 6)\, (m^2 + c/a)^{3/2}$	Enclosed segment
Circle $x^2 + y^2 = r^2$ and Parabola $y^2 = 4ax$ [Added]	Solve intersection $x = x_0$; sum circular sector + parabolic segment	Break at intersection x_0

JEE Revision Warning: Always find point(s) of intersection of bounding curves first. If curves cross each other at $x = c \in (a, b)$, split the integral: $\text{Area} = \int_a^c (f(x)-g(x))\, dx + \int_c^b (g(x)-f(x))\, dx$. Area is NEVER negative!

DIFFERENTIAL CALCULUS

Methods of Differentiation (MOD) & Applications of Derivatives (AOD)

JEE MAIN & ADVANCED REVISION SHEET

1. FUNDAMENTAL DERIVATIVES & DIFFERENTIATION RULES

Standard Functions

Constant:	$d(k)/dx = 0$
Linear:	$d(kx)/dx = k$
Power Rule:	$d(x^n)/dx = n\,x^{n-1}$
Exponential:	$d(e^x)/dx = e^x$
General Exp:	$d(a^x)/dx = a^x \ln a$
Natural Log:	$d(\ln x)/dx = 1/x$
General Log:	$d(\log_a x)/dx = 1/(x \ln a)$
Variable Power:	$d(x^x)/dx = x^x(1 + \ln x)$

Trigonometric Functions

d(sin x)/dx	$= \cos x$
d(cos x)/dx	$= -\sin x$
d(tan x)/dx	$= \sec^2 x$
d(cot x)/dx	$= -\operatorname{cosec}^2 x$
d(sec x)/dx	$= \sec x \tan x$
d(cosec x)/dx	$= -\operatorname{cosec} x \cot x$

Inverse Trigonometric Functions

$d(\sin^{-1}x)/dx$	$= 1/\sqrt{(1-x^2)}$
$d(\cos^{-1}x)/dx$	$= -1/\sqrt{(1-x^2)}$
$d(\tan^{-1}x)/dx$	$= 1/(1+x^2)$
$d(\cot^{-1}x)/dx$	$= -1/(1+x^2)$
$d(\sec^{-1}x)/dx$	$= 1/(\lvert x\rvert\sqrt{(x^2-1)})$
$d(\operatorname{cosec}^{-1}x)/dx$	$= -1/(\lvert x\rvert\sqrt{(x^2-1)})$

Core Differentiation Rules & First Principle

- **First Principle:** $f'(x) = \lim_{h\to 0} [f(x+h) - f(x)]/h$
- **Product Rule:** $d/dx\,[f(x)\cdot g(x)] = f'(x)g(x) + f(x)g'(x)$
- **Quotient Rule:** $d/dx\,[f(x)/g(x)] = [f'(x)g(x) - f(x)g'(x)]/[g(x)]^2$
- **Chain Rule:** $d/dx\,[f(g(x))] = f'(g(x))\cdot g'(x)$
- **Derivative of Inverse Function** Added: $g'(x) = 1/f'(g(x))$ where $g(x) = f^{-1}(x)$

Advanced Differentiation Methods Added

- **Logarithmic Differentiation:** For $y = [f(x)]^{g(x)} \to$ Take ln on both sides: $\ln y = g(x)\ln f(x) \to dy/dx = y[g'(x)\ln f(x) + g(x)f'(x)/f(x)]$
- **Parametric Differentiation:** If $x = f(t)$, $y = g(t) \to dy/dx = (dy/dt)/(dx/dt) = g'(t)/f'(t)$
- **Second Parametric Derivative:** $d^2y/dx^2 = [d/dt\,(dy/dx)]/(dx/dt) = [g''(t)f'(t) - g'(t)f''(t)]/[f'(t)]^3$
- **Implicit Differentiation:** For $F(x, y) = 0 \to dy/dx = -(\partial F/\partial x)/(\partial F/\partial y)$

2. RATE OF CHANGE, ERRORS & APPROXIMATIONS

Rate Measurer & Instantaneous Rate

- **Average Rate of Change:** $\Delta y/\Delta t = [f(t+\Delta t) - f(t)]/\Delta t$
- **Instantaneous Rate of Change:** $dy/dt = \lim_{\Delta t\to 0}(\Delta y/\Delta t) = f'(t)$
- **Rate at a specific point $x = x_0$:** $dy/dx|_{x=x_0} = f'(x_0)$

Errors & Approximations

- **Approximation Formula:** $f(a+\Delta x) \approx f(a) + f'(a)\cdot\Delta x$ or $\Delta y \approx (dy/dx)\cdot\Delta x$
- **Absolute Error in x:** Δx
- **Relative Error in x:** $\Delta x/x$
- **Percentage Error in x:** $(\Delta x/x)\times 100\%$

3. TANGENTS, NORMALS & GEOMETRICAL LENGTHS

Equations of Tangent & Normal

- **Slope of Tangent:** $m = (dy/dx)|_{(x_1, y_1)} = f'(x_1)$
- **Tangent Equation at (x_1, y_1):** $y - y_1 = m(x - x_1)$
- **Normal Equation at (x_1, y_1):** $y - y_1 = -(1/m)(x - x_1)$ (if $m \neq 0$)
- **Parametric Forms ($x = g(t)$, $y = \varphi(t)$):**
  - Tangent: $y - \varphi(t) = [\varphi'(t)/g'(t)]\cdot[x - g(t)]$
  - Normal: $y - \varphi(t) = [-g'(t)/\varphi'(t)]\cdot[x - g(t)]$
- **Tangent from External Point P(a, b):** Solve for contact point Q(h, k): $f'(h) = [f(h) - b]/(h - a)$

Lengths of Tangent, Normal, Subtangent & Subnormal

For point P(h, k) with slope $m = (dy/dx)|_P$:

Quantity	Formula	Projection / Meaning
Length of Tangent (PT)	$\lvert k\rvert \sqrt{(1 + 1/m^2)} = \lvert y_1\rvert \sqrt{(1 + (dx/dy)^2)}$	Segment of tangent between point & x-axis
Length of Normal (PN)	$\lvert k\rvert \sqrt{(1 + m^2)} = \lvert y_1\rvert \sqrt{(1 + (dy/dx)^2)}$	Segment of normal between point & x-axis
Length of Subtangent (TM)	$\lvert k / m\rvert = \lvert y_1 (dx/dy)\rvert$	Projection of tangent segment on x-axis
Length of Subnormal (MN)	$\lvert k \cdot m\rvert = \lvert y_1 (dy/dx)\rvert$	Projection of normal segment on x-axis

Angle of Intersection & Orthogonality

- **Angle of Intersection (θ):** $\tan\theta = \lvert(m_1 - m_2) / (1 + m_1m_2)\rvert$ where m_1, m_2 are slopes at intersection.
- **Orthogonal Curves (θ = 90°):** $m_1 \cdot m_2 = -1 \Rightarrow f'(x_0) \cdot g'(x_0) = -1$
- **Condition for Touch (Parallel Tangents):** $m_1 = m_2$
- **Shortest Distance between non-intersecting curves:** Lies along their *common normal*.

Special Orientation Conditions & Parametric Equations

- **Tangent || x-axis (θ = 0°):** $dy/dx = 0 \Rightarrow$ Equation: $y = y_0$
- **Tangent ⊥ x-axis / || y-axis (θ = 90°):** $dy/dx \to \infty$ or $dx/dy = 0 \Rightarrow$ Equation: $x = x_0$
- **Standard Parametric Substitutions for Tangents:**
  1. $x^{2/3} + y^{2/3} = a^{2/3} \Rightarrow x = a\cos^3\theta, y = a\sin^3\theta$
  2. $\sqrt{x} + \sqrt{y} = \sqrt{a} \Rightarrow x = a\cos^4\theta, y = a\sin^4\theta$
  3. $x^n/a^n + y^n/b^n = 1 \Rightarrow x = a(\cos\theta)^{2/n}, y = b(\sin\theta)^{2/n}$
  4. $c^2(x^2 + y^2) = x^2y^2 \Rightarrow x = c\sec\theta, y = c\,\mathrm{cosec}\,\theta$

4. MEAN VALUE THEOREMS & SPECIAL DIFFERENTIATION RULES

Rolle's Theorem

Hypothesis: Function $f(x)$ defined on $[a, b]$ such that:

- 1. Continuous on closed interval $[a, b]$
- 2. Differentiable on open interval (a, b)
- 3. $f(a) = f(b)$

Conclusion: ∃ at least one $c \in (a, b)$ such that $\mathbf{f'(c) = 0}$.

Geometrical: Tangent is parallel to x-axis at x = c.
Algebraic: Between two roots of f(x)=0, there lies at least one root of f'(x)=0.

Lagrange's Mean Value Theorem (LMVT)

Hypothesis: Function $f(x)$ defined on $[a, b]$ such that:

- 1. Continuous on closed interval $[a, b]$
- 2. Differentiable on open interval (a, b)

Conclusion: ∃ at least one $c \in (a, b)$ such that $\mathbf{f'(c) = [f(b) - f(a)] / (b - a)}$

Geometrical: Tangent at x = c is parallel to the chord joining endpoints A(a, f(a)) and B(b, f(b)).

Leibnitz Rule of Differentiation Under Integral Sign

$$d/dx\left[\int_{\varphi(x)}^{\psi(x)} f(t)\,dt\right] = f(\psi(x)) \cdot \psi'(x) - f(\varphi(x)) \cdot \varphi'(x)$$

Special Case (Constant lower limit a):

$$d/dx\left[\int_a^{\psi(x)} f(t)\,dt\right] = f(\psi(x)) \cdot \psi'(x)$$

Cauchy's Mean Value Theorem Added

If $f(x)$ and $g(x)$ are continuous on $[a, b]$ and differentiable on (a, b), and $g'(x) \neq 0$ on (a, b), then ∃ $c \in (a, b)$:

$$[f(b) - f(a)] / [g(b) - g(a)] = f'(c) / g'(c)$$

5. MONOTONICITY (INCREASING & DECREASING FUNCTIONS)

Definitions & Interval Tests

- **Increasing on (a, b):** $x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)\ \forall\ x_1, x_2 \in (a, b)$. Test: $f'(x) \ge 0$
- **Strictly Increasing:** $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$. Test: $f'(x) > 0$ (points where $f'(x)=0$ must not form an interval).
- **Decreasing on (a, b):** $x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2)$. Test: $f'(x) \le 0$
- **Strictly Decreasing:** $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$. Test: $f'(x) < 0$
- **Constant Function:** $f'(x) = 0\ \forall\ x \in (a, b)$

Properties of Monotonic Functions

- **Invertibility:** If $f(x)$ is continuous and strictly monotonic on $[a, b]$, then f^{-1} exists and maintains the same monotonicity.
- **Reciprocal:** If $f(x) > 0$ is strictly increasing, then $1/f(x)$ is strictly decreasing.
- **Composition Rules:**

f(x)	g(x)	fog(x) / gof(x)
Increasing (+)	Increasing (+)	**Increasing (+)**
Decreasing (−)	Decreasing (−)	**Increasing (+)**
Increasing (+)	Decreasing (−)	**Decreasing (−)**

6. MAXIMA AND MINIMA (EXTREMA)

Definitions & Classification of Critical Points

- **Critical Points:** Points in domain where $f'(x) = 0$ OR $f'(x)$ does not exist.
- **Stationary Points:** Domain points where $f'(x) = 0$. (Stationary $\subset$ Critical).
- **Local Maxima at x = c:** $\exists\ h > 0$ such that $f(c) \ge f(x)\ \forall\ x \in (c - h, c + h)$.
- **Local Minima at x = c:** $\exists\ h > 0$ such that $f(c) \le f(x)\ \forall\ x \in (c - h, c + h)$.
- **Extrema of Discontinuous Functions:**
  - Local Min at $x = a$: $f(a) \le f(a + h)$ AND $f(a) \le f(a - h)$
  - Local Max at $x = a$: $f(a) \ge f(a + h)$ AND $f(a) \ge f(a - h)$

Tests for Local Extrema

- **First Derivative Test:** Let $f'(c) = 0$.
  - *Local Maxima:* $f'(x)$ changes sign from **+ to −** as x increases through c.
  - *Local Minima:* $f'(x)$ changes sign from **− to +** as x increases through c.
  - *Point of Inflection:* $f'(x)$ does not change sign across c.
- **Second Derivative Test:** If $f'(c) = 0$:
  - $f''(c) < 0 \Rightarrow x = c$ is a **Local Maxima**.
  - $f''(c) > 0 \Rightarrow x = c$ is a **Local Minima**.
  - $f''(c) = 0 \Rightarrow$ Test fails; use 1st derivative or n-th derivative test.

Higher Order Derivative Test (N-th Derivative Test)

If $f'(c) = f''(c) = f'''(c) = \ldots = f^{n-1}(c) = 0$ and $f^n(c) \ne 0$:

- **If n is EVEN:**
  - $f^n(c) < 0 \Rightarrow x = c$ is a **Local Maxima**
  - $f^n(c) > 0 \Rightarrow x = c$ is a **Local Minima**
- **If n is ODD:** $x = c$ is *neither maxima nor minima* (Point of Inflection).

Concavity, Convexity & Point of Inflection

- **Concave Upward (Convex):** $f''(x) > 0\ \forall\ x \in (a, b)$.
- **Concave Downward (Concave):** $f''(x) < 0\ \forall\ x \in (a, b)$.
- **Point of Inflection:** Point where graph changes concavity ($f''(x)$ changes sign). Necessary condition: $f''(c) = 0$ or does not exist.

7. GLOBAL EXTREMA & JEE ADVANCED SHORTCUTS

Absolute (Global) Maxima and Minima

For a continuous function $f(x)$ on closed interval $[a, b]$:

- **Step 1:** Find all critical points $c_1, c_2, ..., c_n \in (a, b)$ where $f'(x) = 0$ or $\nexists$.
- **Step 2:** Evaluate functional values at all critical points and end points: $\{f(a), f(c_1), f(c_2), ..., f(c_n), f(b)\}$.
- **Absolute Maxima:** $\text{Max} = \max \{ f(a), f(c_1), ..., f(c_n), f(b) \}$
- **Absolute Minima:** $\text{Min} = \min \{ f(a), f(c_1), ..., f(c_n), f(b) \}$

JEE Advanced Special Results & Inequalities Added

- **Standard Inequalities via Monotonicity:**
  - For $x > 0$: $x - x^3/6 < \sin x < x$
  - For $x > 0$: $1 - x^2/2 < \cos x < 1$
  - For $x > 0$: $x < \tan x$
  - For $x > 0$: $x / (1 + x) < \ln(1 + x) < x$
- **Tangent & Secant Inequality:** Concavity determines $f(x)$ relation to tangent $T(x)$ and secant $S(x)$:
  - If $f''(x) > 0$: Graph lies *above tangent* and *below chord*.

8. ESSENTIAL MENSURATION FORMULAS FOR AOD OPTIMIZATION

Useful Mensuration Formulas for Maxima & Minima Problems

Shape / Solid	Volume (V)	Surface Area (Curved / Total)	Key Variables / Relations
Cuboid	$V = l \cdot b \cdot h$	$S = 2(lb + bh + hl)$	l, b, h = dimensions
Cube	$V = a^3$	$S = 6a^2$	a = side length
Right Circular Cone	$V = (1/3)\,\pi\, r^2\, h$	$CSA = \pi\, r\, l$ $TSA = \pi\, r\, l + \pi\, r^2$	Slant height $l = \sqrt{(r^2 + h^2)}$
Right Circular Cylinder	$V = \pi\, r^2\, h$	$CSA = 2\pi\, r\, h$ $TSA = 2\pi\, r\, h + 2\pi\, r^2$	r = radius, h = height
Sphere	$V = (4/3)\,\pi\, r^3$	$S = 4\pi\, r^2$	r = radius
Circular Sector	$\text{Area} = (1/2)\, r^2\, \theta$	Arc Length $s = r\, \theta$	θ in radians
Right Prism	$V = (\text{Area of Base}) \times h$	$LSA = (\text{Perimeter of Base}) \times h$ $TSA = LSA + 2(\text{Area of Base})$	Lateral faces are rectangles
Right Pyramid	$V = (1/3)\,(\text{Area of Base}) \times h$	$LSA = (1/2)\,(\text{Perimeter of Base}) \times l$	l = slant height; faces are triangles

9. JEE REVISION CHECKLIST & IMPORTANT SHORTCUTS

Common Optimization Maxima/Minima Results Added

- **Cylinder in Sphere:** Max volume of cylinder inscribed in a sphere of radius R occurs when height $h = 2R / \sqrt{3}$.
- **Cone in Sphere:** Max volume of cone inscribed in sphere of radius R occurs when height $h = 4R / 3$ (Volume = 8/27 of sphere volume).
- **Cylinder in Cone:** Max volume of cylinder inscribed in a cone occurs when height of cylinder $h_{cyl} = (1/3)\, h_{cone}$.
- **Rectangle in Circle:** Max area rectangle inscribed in a circle is a **square** of side $a = R\sqrt{2}$.

Key Pitfalls & JEE Tricks

- **Non-Differentiable Points:** Extrema can occur at points where derivative does not exist (e.g., $f(x) = |x|$ at $x = 0$ has local min even though $f'(0)$ DNE). Always test critical points where f' DNE!
- **LMVT & Rolle's Continuity:** Check endpoint continuity and interval differentiability before applying LMVT/Rolle's.
- **Orthogonality Trap:** Don't just set $m_1 m_2 = -1$; first verify the curves actually *intersect* at that point!

SEQUENCES AND SERIES

JEE Main & Advanced Complete Revision Formula Sheet

MATHEMATICS

1. FUNDAMENTAL CONCEPTS & SEQUENCE TYPES

Sequence & Series Definitions

Sequence: Function f: $\mathbb{N} \to \mathbb{R}$, terms denoted as $f_n = a_n$.

Series: Algebraic sum of sequence terms: $S_n = \sum_{k=1}^{n} a_k$.

General Relation: $a_n = S_n - S_{n-1}$ *(for $n \geq 2$), $a_1 = S_1$*

Arithmetic Progression (A.P.)

Condition: $a_{n+1} - a_n = d$ *(constant common difference)*

General Term: $T_n = a + (n-1)d = \ell$ *(from start)*

T_n from end: $T_n' = \ell - (n-1)d$

Sum of n terms: $S_n = (n/2)[2a + (n-1)d] = (n/2)[a + \ell]$

2. A.P. PROPERTIES & STANDARD SELECTIONS

Standard Selection of Terms

- **3 terms in A.P.:** $a - d, a, a + d$ *(common diff = d)*
- **4 terms in A.P.:** $a - 3d, a - d, a + d, a + 3d$ *(diff = 2d)*
- **5 terms in A.P.:** $a - 2d, a - d, a, a + d, a + 2d$ *(diff = d)*
- **6 terms in A.P.:** $a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d$ *(diff = 2d)*

Core A.P. Properties

- If a, b, c are in A.P. $\Rightarrow$ **$2b = a + c$** or **$a + c = b + d$** *(for a,b,c,d in AP)*
- Sum of terms equidistant from start & end is constant: $T_k + T_{n-k+1} = a + \ell$.
- If $\{a_n\}$ is A.P. $\Rightarrow \{a_n \pm k\}$ is A.P. (same d), $\{k \cdot a_n\}$ is A.P. (diff = kd).
- If $\{a_n\}$, $\{b_n\}$ are A.P.s $\Rightarrow \{a_n \pm b_n\}$ is also an A.P.
- S_n of A.P. is a quadratic in n with zero constant term: $S_n = An^2 + Bn$ *($d = 2A$)*.

3. GEOMETRIC PROGRESSION (G.P.)

Concept / Formula	Mathematical Expression	Conditions / Remarks
Condition & General Term	$a_{k+1} / a_k = r$ (constant) $T_n = a \cdot r^{n-1}$	$a \neq 0$, $r \neq 0$. T_n from end: $T_n' = \ell / r^{n-1}$
Sum of First n Terms	$S_n = a(1 - r^n) / (1 - r)$ *for $r < 1$* $S_n = a(r^n - 1) / (r - 1)$ *for $r > 1$* $S_n = n \cdot a$ *for $r = 1$*	Derived directly from algebraic identity $1 - r^n$.
Sum of Infinite G.P.	$S_\infty = a / (1 - r)$	Valid **only if $\lvert r \rvert < 1$** ($-1 < r < 1$). If $\lvert r \rvert \geq 1$, S_∞ diverges.
Selection of Terms	3 terms: $a/r, a, ar$ | 4 terms: $a/r^3, a/r, ar, ar^3$ 5 terms: $a/r^2, a/r, a, ar, ar^2$	Use symmetric terms when product of terms is given in question.
G.P. Properties	If a, b, c in G.P. $\Rightarrow \mathbf{b^2 = a \cdot c}$ $T_k \cdot T_{n-k+1} = a \cdot \ell =$ constant	If $\{a_n\}$ in G.P. $\Rightarrow \{1/a_n\}, \{a_n^k\}, \{c \cdot a_n\}$ are in G.P. If $\{a_n\}, \{b_n\}$ in G.P. $\Rightarrow \{a_n b_n\}$ and $\{a_n / b_n\}$ are in G.P.

4. HARMONIC PROGRESSION (H.P.)

Definition & General Term

A sequence $a_1, a_2, a_3, \ldots$ is in H.P. if $1/a_1, 1/a_2, 1/a_3, \ldots$ is in A.P.

General Term: $T_n = 1 / [(1/a_1) + (n - 1)d] = 1 / [a + (n - 1)d]$

Three terms a, b, c in H.P. $\Rightarrow$ **b = 2ac / (a + c)**

Alternatively: a/c = (a − b) / (b − c)

Key Notes for H.P.

- There is **no general formula** for the sum of n terms of an H.P.
- No term of an H.P. can be zero.
- Questions on H.P. should generally be solved by converting terms into corresponding A.P.

5. MEANS (A.M., G.M., H.M.) & INSERTION OF MEANS

Single Means Between Two Numbers (a and b)

Arithmetic Mean: $A = (a + b) / 2$

Geometric Mean: $G = \sqrt{(a \cdot b)}$ *(a, b > 0)*

Harmonic Mean: $H = 2ab / (a + b)$

Mean Type	Insertion of n Means Between a and b	Important Properties / Formulas
n A.M.s $(A_1, A_2, \ldots, A_n)$	Common diff: $d = (b - a) / (n + 1)$ $A_k = a + k \cdot (b - a)/(n + 1)$	Sum of n A.M.s = n × (Single A.M.) $\sum_{k=1}^{n} A_k = n \cdot [(a + b) / 2]$
n G.M.s $(G_1, G_2, \ldots, G_n)$	Common ratio: $r = (b / a)^{1/(n+1)}$ $G_k = a \cdot (b / a)^{k/(n+1)}$	Product of n G.M.s = (Single G.M.)^n $\prod_{k=1}^{n} G_k = (a \cdot b)^{n/2} = G^n$
n H.M.s $(H_1, H_2, \ldots, H_n)$	$1/H_k = (1/a) + k \cdot d'$ where $d' = (a - b) / [a \cdot b \cdot (n + 1)]$	$H_k = [a \cdot b \cdot (n + 1)] / [b(n + 1) + k(a - b)]$

Important Relations Between Means (for positive real numbers):

- **A ≥ G ≥ H** (Equality holds if and only if $a_1 = a_2 = \ldots = a_n$)
- For two positive numbers a and b: $\mathbf{G^2 = A \cdot H}$ (A, G, H form a G.P.)
- Quadratic equation having a, b as roots: $\mathbf{x^2 - 2Ax + G^2 = 0} \Rightarrow$ roots a, b = $A \pm \sqrt{(A^2 - G^2)}$
- For any n positive numbers: A.M. = $(\sum a_i)/n$, G.M. = $(\prod a_i)^{1/n}$, H.M. = $n / (\sum 1/a_i)$

6. ARITHMETICO-GEOMETRIC PROGRESSION (A.G.P.)

General Form: a, (a + d)r, (a + 2d)r^2, (a + 3d)r^3, ...

n^{th} Term: $T_n = [a + (n - 1)d] \cdot r^{n-1}$

Sum of First n Terms (S_n):

$$S_n = [a / (1 - r)] + [d \cdot r(1 - r^{n-1}) / (1 - r)^2] - [\{a + (n - 1)d\} \cdot r^n / (1 - r)] \quad (r \neq 1)$$

Sum to Infinity (S_∞):

$$S_\infty = [a / (1 - r)] + [d \cdot r / (1 - r)^2] \quad \textit{(Valid only when } |r| < 1\textit{)}$$

Standard Method for Solving A.G.P.: *Multiply S_n by common ratio 'r', shift terms by one position to the right, and subtract ($S_n - rS_n$).*

7. SIGMA NOTATION & SUM OF FIRST N NATURAL NUMBERS

Properties of Summation Operator (Σ)

- $\sum_{r=1}^{n} (a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r$
- $\sum_{r=1}^{n} k \cdot a_r = k \cdot \sum_{r=1}^{n} a_r$ *(k is constant)*
- $\sum_{r=1}^{n} k = k \cdot n$
- $\sum_{r=1}^{n} a_r \cdot b_r \neq (\sum a_r) \cdot (\sum b_r)$

Standard Finite Sums

$\sum r = 1 + 2 + \ldots + n = \mathbf{n(n + 1) / 2}$

$\sum r^2 = 1^2 + 2^2 + \ldots + n^2 = \mathbf{n(n + 1)(2n + 1) / 6}$

$\sum r^3 = 1^3 + 2^3 + \ldots + n^3 = \mathbf{[n(n + 1) / 2]^2 = (\sum r)^2}$

$\sum r^4 = \mathbf{n(n + 1)(2n + 1)(3n^2 + 3n - 1) / 30}$ [Added]

8. METHOD OF DIFFERENCE & TELESCOPING SERIES

1. General Telescoping Principle

If general term T_k can be written as $\mathbf{T_k = V_k - V_{k-1}}$ or $\mathbf{T_k = V_{k+1} - V_k}$:

$S_n = \sum_{k=1}^{n} T_k = (V_1 - V_0) + (V_2 - V_1) + \ldots + (V_n - V_{n-1}) = \mathbf{V_n - V_0}$

2. Factors in A.P. in Numerator

$T_n = [a + (n-1)d][a + n{\cdot}d] \ldots [a + (n + r - 2)d]$

Express T_n using next & previous terms in A.P.:

$T_n = [1 / \{(r+1)d\}] \cdot [\, V_n - V_{n-1} \,]$

where $V_n = [a + (n-1)d] \ldots [a + (n + r - 1)d]$

$\mathbf{S_n = [1 / \{(r+1)d\}] \cdot (V_n - V_0)}$

3. Factors in A.P. in Denominator

$T_n = 1 / [\, \{a + (n-1)d\}\{a + n{\cdot}d\} \ldots \{a + (n + r - 2)d\} \,]$

Split by taking diff of extreme factors:

$T_n = [1 / \{(r-1)d\}] \cdot [\, V_{n-1} - V_n \,]$

where $V_n = 1 / [\, \{a + n{\cdot}d\} \ldots \{a + (n + r - 2)d\} \,]$

$\mathbf{S_n = [1 / \{(r-1)d\}] \cdot (V_0 - V_n)}$

9. JEE ADVANCED HIGH-YIELD RESULTS & SHORTCUTS

Double Summation Identities

$\sum_{1 \le i < j \le n} a_i a_j = \mathbf{(1/2)\,[\,(\sum_{i=1}^{n} a_i)^2 - \sum_{i=1}^{n} a_i^2\,]}$

$\sum_{1 \le i < j \le n} 1 = \mathbf{n(n-1)/2}$

$\sum_{1 \le i < j \le n} (a_i + a_j) = \mathbf{(n-1)\sum_{i=1}^{n} a_i}$ [Added]

Special Trigonometric & Inverse Series

$\mathbf{\sum_{k=1}^{n} \tan^{-1}[\,1/(1 + k + k^2)\,]}$

$= \sum_{k=1}^{n} [\, \tan^{-1}(k+1) - \tan^{-1}(k) \,] = \mathbf{\tan^{-1}(n+1) - \pi/4}$

$S_\infty = \mathbf{\pi/4}$ *(as $n \to \infty$)*

Standard High-Frequency Series Results

[Added JEE Advanced]

Odd Natural Numbers:
$1 + 3 + 5 + \ldots + (2n-1) = \mathbf{n^2}$

Even Natural Numbers:
$2 + 4 + 6 + \ldots + 2n = \mathbf{n(n+1)}$

Squares of Odd Numbers:
$1^2 + 3^2 + \ldots + (2n-1)^2 = \mathbf{n(4n^2 - 1)/3}$

10. IMPORTANT INEQUALITIES FOR JEE ADVANCED

- **Weighted A.M. – G.M. Inequality:** If $w_1, w_2, \ldots, w_n > 0$, then:
  $(w_1a_1 + w_2a_2 + \ldots + w_na_n) / (w_1 + w_2 + \ldots + w_n) \ge (a_1^{w_1} \cdot a_2^{w_2} \ldots a_n^{w_n})^{1/(\sum w_i)}$
- **Cauchy-Schwarz Inequality:** For any real numbers $(a_1,\ldots,a_n)$ and $(b_1,\ldots,b_n)$:
  $(a_1^2 + a_2^2 + \ldots + a_n^2)(b_1^2 + b_2^2 + \ldots + b_n^2) \ge (a_1b_1 + a_2b_2 + \ldots + a_nb_n)^2$ *(Equality holds if $a_1/b_1 = a_2/b_2 = \ldots = a_n/b_n$)* [Added]
- **m^{th} Power Theorem:** If $a_1, a_2, \ldots, a_n > 0$ and a_i are not all equal, then:
  $[\,(a_1^m + a_2^m + \ldots + a_n^m)/n\,] > [\,(a_1 + a_2 + \ldots + a_n)/n\,]^m$ if $m < 0$ or $m > 1$.
  $[\,(a_1^m + a_2^m + \ldots + a_n^m)/n\,] < [\,(a_1 + a_2 + \ldots + a_n)/n\,]^m$ if $0 < m < 1$. [Added]

DIFFERENTIAL EQUATIONS

JEE Main & Advanced Complete Revision & Formula Sheet

MATHEMATICS

1. FUNDAMENTALS & CORE DEFINITIONS

Concept	Definition / Core Formula	JEE Note / Condition
Differential Equation (DE)	An equation involving independent variable (x), dependent variable (y), and differential coefficients (dy/dx, d^2y/dx^2, ...).	Relation between variables and derivatives.
Order	The order of the *highest order derivative* appearing in the differential equation.	Always a positive integer. Defined for all DEs.
Degree	The power/degree of the *highest order derivative* after clearing all radicals and fractional powers of derivatives.	Must be a polynomial in derivatives. If not reducible to polynomial form, degree is **Not Defined**.
General Solution	Solution containing number of independent arbitrary constants equal to the **order** of the DE.	Order $n \Rightarrow n$ arbitrary constants.
Particular Solution	Solution obtained by giving specific values to the arbitrary constants in the general solution.	Contains **zero** arbitrary constants.

2. FORMATION OF DIFFERENTIAL EQUATIONS

Procedure for a Family of Curves with n Parameters:

1. Identify the number of arbitrary constants (n) in the family equation $f(x, y, c_1, c_2, ..., c_n) = 0$.
2. Differentiate the equation successively n times w.r.t. x to obtain n additional equations.
3. Eliminate all n arbitrary constants ($c_1, c_2, ..., c_n$) using the original equation and the derived equations.
4. The resulting relation of order n is the required differential equation.

3. STANDARD METHODS FOR FIRST ORDER, FIRST DEGREE DES

3.1 Variables Separable Form

Standard Form: $f(x)\,dx = g(y)\,dy \;\Rightarrow$ General Solution: $\int f(x)\,dx = \int g(y)\,dy + C$

3.2 Equations Reducible to Variable Separable Form

Form: $\mathbf{dy/dx = f(ax + by + c)}$
Substitution: Put $\mathbf{t = ax + by + c} \;\Rightarrow\; \mathbf{dt/dx = a + b(dy/dx)}$
Transformed DE: $\mathbf{dt / (a + b\,f(t)) = dx} \;\Rightarrow$ Integrate both sides w.r.t variables t and x.

3.3 Homogeneous Differential Equations

A function $F(x, y)$ is homogeneous of degree n if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$. A DE $dy/dx = F(x, y)$ is homogeneous if $F(x, y)$ is of degree 0.

Type 1: $dy/dx = g(y/x)$
Put $y = vx \Rightarrow dy/dx = v + x\,(dv/dx)$
Reduces to: $dv / (g(v) - v) = dx / x$

Type 2: $dx/dy = h(x/y)$
Put $x = vy \Rightarrow dx/dy = v + y\,(dv/dy)$
Reduces to: $dv / (h(v) - v) = dy / y$

3.4 Equations Reducible to Homogeneous Form

Form: $dy/dx = (a_1x + b_1y + c_1) / (a_2x + b_2y + c_2)$

Condition	Substitution / Method	Transformed Equation
Case 1: $a_1/a_2 \neq b_1/b_2$	Put $x = X + h$, $y = Y + k$. Choose h, k such that: $a_1h + b_1k + c_1 = 0$, $a_2h + b_2k + c_2 = 0$	$dY/dX = (a_1X + b_1Y) / (a_2X + b_2Y)$ (Homogeneous in X and Y; solve via $Y = vX$).
Case 2: $a_1/a_2 = b_1/b_2 = m$	Let $a_1x + b_1y = t \Rightarrow a_2x + b_2y = t/m$	Reduces directly to **Variable Separable** form.
Case 3: $b_1 + a_2 = 0$ [Added]	Cross multiply and regroup terms directly as exact differentials.	Direct integration by inspection.

4. FIRST ORDER LINEAR DIFFERENTIAL EQUATIONS (LDE)

Standard Form 1 (in y):

$$dy/dx + P(x)\,y = Q(x)$$

- Integrating Factor: $\text{I.F.} = e^{\int P(x)\,dx}$
- General Solution:

$$y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.})\,dx + C$$

Standard Form 2 (in x):

$$dx/dy + P_1(y)\,x = Q_1(y)$$

- Integrating Factor: $\text{I.F.} = e^{\int P_1(y)\,dy}$
- General Solution:

$$x \cdot (\text{I.F.}) = \int Q_1(y) \cdot (\text{I.F.})\,dy + C$$

5. EXTENDED LINEAR FORMS & BERNOULLI'S EQUATION

5.1 Bernoulli's Equation

Form: $dy/dx + P(x)\,y = Q(x)\,y^n$ (where $n \neq 0, 1$)

1. Divide entire equation by y^n: $\Rightarrow y^{-n}\,(dy/dx) + P(x)\,y^{1-n} = Q(x)$
2. Substitute $v = y^{1-n} \Rightarrow dv/dx = (1 - n)\,y^{-n}\,(dy/dx)$
3. Transformed LDE in v: $\Rightarrow dv/dx + (1 - n)\,P(x)\,v = (1 - n)\,Q(x)$
4. Solution: $v \cdot e^{\int (1-n)P\,dx} = \int [\,(1-n)\,Q \cdot e^{\int (1-n)P\,dx}\,]\,dx + C$

5.2 Generalized Function Form [Added]

Form: $f'(y)\,(dy/dx) + P(x)\,f(y) = Q(x) \Rightarrow$ Substitute $v = f(y) \Rightarrow$ Transformed LDE: $dv/dx + P(x)\,v = Q(x)$.

6. EXACT DIFFERENTIAL EQUATIONS & INTEGRATING FACTORS

A differential equation of form $M(x, y)\,dx + N(x, y)\,dy = 0$ is exact if LHS is the total differential du of some function $u(x, y)$.

Condition for Exactness:

$$\partial M / \partial y = \partial N / \partial x$$

General Solution Formula:

$$\int_{(y \text{ const})} M\,dx + \int_{(\text{terms of N free from x})} N\,dy = C$$

Essential Exact Differential Relations (Must-Remember Shortcuts for Inspection)

Exact Differential Expression	Equivalent Differential $d(u)$
$x\,dy + y\,dx$	$d(x\,y)$
$(x\,dy - y\,dx) / x^2$	$d(y / x)$
$(y\,dx - x\,dy) / y^2$	$d(x / y)$
$(x\,dy + y\,dx) / (x\,y)$	$d(\ln(xy))$
$(x\,dy - y\,dx) / (x\,y)$	$d(\ln(y / x))$
$(y\,dx - x\,dy) / (x\,y)$	$d(\ln(x / y))$
$(dx + dy) / (x + y)$	$d(\ln(x + y))$
$(x\,dy - y\,dx) / (x^2 + y^2)$	$d(\tan^{-1}(y / x))$
$(y\,dx - x\,dy) / (x^2 + y^2)$	$d(\tan^{-1}(x / y))$
$(x\,dx + y\,dy) / (x^2 + y^2)$	$d(\frac{1}{2}\ln(x^2 + y^2)) = d(\ln\sqrt{(x^2 + y^2)})$
$(x\,dy + y\,dx) / (x^2\,y^2)$	$d(-1 / (xy))$
$(y\,e^x\,dx - e^x\,dy) / y^2$	$d(e^x / y)$
$(x\,e^y\,dy - e^y\,dx) / x^2$	$d(e^y / x)$
$(2x\,y\,dx - x^2\,dy) / y^2$	$d(x^2 / y)$
$(2x\,y\,dy - y^2\,dx) / x^2$	$d(y^2 / x)$
$f'(x, y) / [f(x, y)]^n$	$d[(f(x, y))^{1-n} / (1 - n)] \quad (n \neq 1)$

7. GEOMETRIC & CARTESIAN APPLICATIONS OF FIRST ORDER DES

Let $P(x, y)$ be a point on a curve $y = f(x)$. Let $y' = dy/dx$ denote the slope of the tangent at P.

Geometric Quantity	Cartesian Formula	JEE Shortcut / Note
Slope of Tangent	$m_T = dy/dx$	Angle with x-axis: $\tan\theta = dy/dx$
Slope of Normal	$m_N = -dx/dy$	Perpendicular to tangent.
Equation of Tangent	$Y - y = (dy/dx)(X - x)$	(X, Y) are current coordinates.
Equation of Normal	$Y - y = -(dx/dy)(X - x)$	Normal at point $P(x, y)$.
Length of Tangent (PT)	$\lvert y\sqrt{(1 + (dx/dy)^2)}\rvert = \lvert (y\sqrt{(1 + (y')^2)}) / y'\rvert$	Distance from P to x-intercept of tangent.
Length of Normal (PN)	$\lvert y\sqrt{(1 + (dy/dx)^2)}\rvert = \lvert y\sqrt{(1 + (y')^2)}\rvert$	Distance from P to x-intercept of normal.
Length of Subtangent (ST)	$\lvert y \cdot (dx/dy)\rvert = \lvert y / y'\rvert$	Projection of tangent on x-axis.
Length of Subnormal (SN)	$\lvert y \cdot (dy/dx)\rvert = \lvert y \cdot y'\rvert$	Projection of normal on x-axis.

Geometric Quantity	Cartesian Formula	JEE Shortcut / Note
X-intercept of Tangent	$x - y\,(dx/dy) = x - y / y'$	Set $Y = 0$ in tangent equation.
Y-intercept of Tangent	$y - x\,(dy/dx) = y - x \cdot y'$	Set $X = 0$ in tangent equation.

8. ORTHOGONAL TRAJECTORIES (OT)

Definition: A curve which cuts every member of a given family of curves at right angles ($90°$).

Procedure for Cartesian Curves $f(x, y, c) = 0$:

1. Differentiate $f(x, y, c) = 0$ w.r.t. x and eliminate parameter c to obtain the DE of the given family: $F(x, y, dy/dx) = 0$.
2. Replace **dy/dx with $-dx/dy$** in the differential equation.
3. Solve the resulting DE: **$F(x, y, -dx/dy) = 0$** to get the family of orthogonal trajectories.

Example: Straight lines $y = mx$ through origin are orthogonal trajectories to circles $x^2 + y^2 = a^2$.

9. ADVANCED DIFFERENTIAL EQUATION FORMS (JEE ADVANCED)

9.1 Clairaut's Equation

Standard Form: $\mathbf{y = p\,x + f(p)}$ where $\mathbf{p = dy/dx}$

Differentiating w.r.t x gives: $\mathbf{dp/dx\,[\,x + f'(p)\,] = 0}$

- **General Solution:** Set $dp/dx = 0 \Rightarrow p = c$ (constant). Replace p with c:

  $$\mathbf{y = c\,x + f(c)} \quad \text{(Family of straight lines)}$$

- **Singular Solution:** Eliminate p between $\mathbf{y = p\,x + f(p)}$ and $\mathbf{x + f'(p) = 0}$.

  Represents the envelope of the family of straight lines (contains no arbitrary constant).

9.2 Second Order DE Reducible to First Order

Form: $\mathbf{d^2y/dx^2 = f(x)}$

1. Integrate once w.r.t x: $dy/dx = \int f(x)\,dx + C_1$
2. Integrate again: $y = \int [\,\int f(x)\,dx\,]\,dx + C_1x + C_2$

Form: $\mathbf{d^2y/dx^2 = f(y)}$ [Added]

1. Multiply by $2\,(dy/dx)$: $2\,(dy/dx)(d^2y/dx^2) = 2\,f(y)\,(dy/dx)$
2. Integrate: $(dy/dx)^2 = 2\int f(y)\,dy + C_1$

10. STANDARD PHYSICAL & GROWTH APPLICATIONS

Physical Law / Phenomenon	Differential Equation	Integrated Solution Form
Radioactive Decay / Population Growth	$dN/dt = \pm k\,N$	$N(t) = N_0\,e^{\pm k\,t}$
Newton's Law of Cooling	$dT/dt = -k\,(T - T_0)$	$T(t) = T_0 + (T_{initial} - T_0)\,e^{-k\,t}$
Orthogonal Trajectories in Polar [Added]	$f(r, \theta, dr/d\theta) = 0$	Replace $dr/d\theta$ with $\mathbf{-r^2\,(d\theta/dr)}$

STATISTICS

JEE Main & JEE Advanced Master Formula & Revision Sheet

1. Measures of Central Tendency

1.1 Arithmetic Mean ($\bar{x}$ or A)

- **Ungrouped / Raw Data:** For n observations $x_1, x_2, \ldots, x_n$:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

- **Discrete Frequency Distribution:** For values x_i with frequencies f_i:

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N}, \quad \text{where } N = \sum_{i=1}^{n} f_i$$

- **Continuous Grouped Data:** Same as discrete formula, where x_i represents the **class mark** (mid-point) of the i-th class interval:

$$x_i = \frac{\text{Lower limit} + \text{Upper limit}}{2}$$

- **Combined Mean:** If groups of sizes $n_1, n_2, \ldots, n_r$ have arithmetic means $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_r$ respectively:

$$\bar{x}_{\text{combined}} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \cdots + n_r\bar{x}_r}{n_1 + n_2 + \cdots + n_r} = \frac{\sum_{i=1}^{r} n_i\bar{x}_i}{\sum_{i=1}^{r} n_i}$$

- **Weighted Mean [Added]:** For weights w_i assigned to x_i:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

1.2 Median (M_d or M)

Data must be arranged in **ascending or descending order**.

- **Ungrouped Data (n observations):**
  - If n is **odd**: $M_d = \left(\frac{n+1}{2}\right)$ th observation.
  - If n is **even**: M_d = Mean of $\left(\frac{n}{2}\right)$ th and $\left(\frac{n}{2}+1\right)$ th observations.
- **Continuous Grouped Frequency Distribution:**

$$M_d = l + \left(\frac{\frac{N}{2} - C}{f}\right) \times h$$

where:
- l = lower limit of median class (class where cumulative frequency reaches $\geq N/2$).
- $N = \sum f_i$ = total frequency.
- C = cumulative frequency of class preceding median class.
- f = frequency of median class.
- h = class width (length of median class interval).

1.3 Mode (M_o)

- **Ungrouped / Discrete Data:** The observation with the maximum frequency.
- **Continuous Grouped Frequency Distribution:**

$$M_o = l + \left(\frac{f_0 - f_1}{2f_0 - f_1 - f_2}\right) \times h \quad \text{or} \quad l + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) \times h$$

where:
- l = lower limit of modal class (class with maximum frequency).
- f_0 (or f_m) = frequency of modal class.
- f_1 = frequency of pre-modal class (class preceding modal class).
- f_2 = frequency of post-modal class (class succeeding modal class).
- h = width of modal class interval.

1.4 Empirical Relationship

For moderately skewed distributions:

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$

$$\text{Mean} - \text{Mode} = 3 \times (\text{Mean} - \text{Median})$$

2. Measures of Dispersion

Degree of spread of data around a central value. Key measures: Range, Mean Deviation, Variance, Standard Deviation.

2.1 Range

$$\text{Range} = x_{\max} - x_{\min}$$

2.2 Mean Deviation (M.D.)

Average of absolute deviations from a central measure (Mean, Median, or Mode).

- **Ungrouped Data (n terms):**
  - About Mean ($\bar{x}$): $\text{M.D.}(\bar{x}) = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$
  - About Median (M): $\text{M.D.}(M) = \frac{\sum_{i=1}^{n} |x_i - M|}{n}$
- **Grouped Frequency Distribution (Discrete / Continuous):**
  - About Mean ($\bar{x}$): $\text{M.D.}(\bar{x}) = \frac{\sum_{i=1}^{n} f_i|x_i - \bar{x}|}{N}$
  - About Median (M): $\text{M.D.}(M) = \frac{\sum_{i=1}^{n} f_i|x_i - M|}{N}$

where $N = \sum f_i$, and x_i is class mark for continuous distributions.

2.3 Variance (σ^2 or $\text{Var}(X)$) & Standard Deviation (σ)

- **Ungrouped Data:**

$$\text{Var}(\sigma^2) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$= \frac{\sum x_i^2}{n} - (\bar{x})^2 = \frac{1}{n}\sum x_i^2 - \left(\frac{\sum x_i}{n}\right)^2$$

$$= \frac{1}{n^2}\left[n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2\right]$$

$$\text{S.D.}(\sigma) = \sqrt{\text{Var}(\sigma^2)} = \sqrt{\frac{1}{n}\sum(x_i - \bar{x})^2}$$

- **Discrete & Continuous Frequency Distributions:**

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{n} f_i(x_i - \bar{x})^2$$
$$= \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2$$
$$= \frac{1}{N^2}\left[N\sum_{i=1}^{n} f_i x_i^2 - \left(\sum_{i=1}^{n} f_i x_i\right)^2\right]$$

$$\text{S.D.}(\sigma) = \frac{1}{N}\sqrt{N\sum f_i x_i^2 - \left(\sum f_i x_i\right)^2}$$

3. Shortcut & Step-Deviation Methods

Useful for large numerical values or continuous uniform class intervals.

Let A = Assumed Mean, h = Class width / common factor. Define step-deviation:

$$y_i = d_i = \frac{x_i - A}{h}$$

- **Mean via Step-Deviation:**

$$\bar{x} = A + \left(\frac{\sum f_i y_i}{N}\right) \times h$$

- **Variance via Step-Deviation:**

$$\sigma_x^2 = h^2 \cdot \sigma_y^2 = \frac{h^2}{N^2}\left[N\sum f_i y_i^2 - \left(\sum f_i y_i\right)^2\right]$$

- **Standard Deviation via Step-Deviation:**

$$\sigma_x = h \cdot \sigma_y = \frac{h}{N}\sqrt{N\sum f_i y_i^2 - \left(\sum f_i y_i\right)^2}$$

4. Coefficients & Relative Measures

- **Coefficient of Standard Deviation:**

$$\text{Coeff. of S.D.} = \frac{\sigma}{\bar{x}}$$

- **Coefficient of Variation (C.V.):**

$$\text{C.V.} = \left(\frac{\sigma}{\bar{x}}\right) \times 100 \quad (\bar{x} \neq 0)$$

- **Consistency Rule (JEE Main Favorite):**
  - Series with **smaller C.V.** is **more consistent**, stable, or uniform.
  - Series with **larger C.V.** has greater variability / dispersion.
  - If two distributions have equal means, the one with smaller σ is more consistent.

5. Combined Statistics

For two sub-groups of sizes n_1, n_2 with means $\bar{x}_1, \bar{x}_2$ and variances σ_1^2, σ_2^2:

1. **Combined Mean:**

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

2. **Deviations of sub-group means from combined mean:**

$$d_1 = \bar{x}_1 - \bar{x}, \quad d_2 = \bar{x}_2 - \bar{x}$$

3. **Combined Variance (σ^2):**

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

4. **Combined Standard Deviation (σ):**

$$\sigma = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

5. **Generalization for k groups [Added]:**

$$\sigma^2 = \frac{\sum_{i=1}^{k} n_i(\sigma_i^2 + d_i^2)}{\sum_{i=1}^{k} n_i}, \quad d_i = \bar{x}_i - \bar{x}$$

6. Linear Transformation Properties

Let new variable be $y_i = ax_i + b$, where $a, b \in \mathbb{R}$.

Statistic	Transformation Formula	Dependence
Mean	$\bar{y} = a\bar{x} + b$	Origin & Scale
Median	$M_y = aM_x + b$	Origin & Scale
Mode	$(M_o)_y = a(M_o)_x + b$	Origin & Scale
Range	$\text{Range}_y = \lvert a\rvert \cdot \text{Range}_x$	Scale ONLY
M.D.	$\text{M.D.}(y) = \lvert a\rvert \cdot \text{M.D.}(x)$	Scale ONLY
S.D. (σ)	$\sigma_y = \lvert a\rvert \cdot \sigma_x$	Scale ONLY
Variance (σ^2)	$\sigma_y^2 = a^2 \cdot \sigma_x^2$	Scale ONLY

Key Concept:

- S.D. and Variance are **INDEPENDENT of change of origin** (b), but **DEPENDENT on change of scale** (a).
- Adding/subtracting a constant b to all observations leaves σ and σ^2 unchanged.

7. Standard Results for First n Natural Numbers

For the dataset $\{1, 2, 3, \ldots, n\}$:

1. **Mean ($\bar{x}$):**

$$\bar{x} = \frac{n+1}{2}$$

2. **Sum of Squares ($\sum x_i^2$):**

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

3. **Variance (σ^2):**

$$\sigma^2 = \frac{n^2 - 1}{12}$$

4. **Standard Deviation (σ):**

$$\sigma = \sqrt{\frac{n^2 - 1}{12}}$$

5. **First n Even Natural Numbers $\{2, 4, 6, \ldots, 2n\}$ [Added]:**

$$\bar{x} = n + 1, \quad \text{Var} = \frac{n^2 - 1}{3}$$

6. **First n Odd Natural Numbers $\{1, 3, 5, \ldots, 2n - 1\}$ [Added]:**

$$\bar{x} = n, \quad \text{Var} = \frac{n^2 - 1}{3}$$

8. Mathematical Properties & JEE Tricks

- **Minimum Sum of Absolute Deviations:**

$$\sum_{i=1}^{n} |x_i - a| \quad \text{is MINIMUM when } a = \text{Median } (M)$$

- **Minimum Sum of Squared Deviations:**

$$\sum_{i=1}^{n} (x_i - a)^2 \quad \text{is MINIMUM when } a = \text{Mean } (\bar{x})$$

- **Algebraic Sum of Deviations from Mean:**

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

- **Correcting Incorrect Observations (Frequent JEE Problem):**

$$\sum x_{\text{correct}} = \sum x_{\text{incorrect}} - (\text{Wrong items}) + (\text{Correct items})$$

$$\bar{x}_{\text{correct}} = \frac{\sum x_{\text{correct}}}{n}$$

$$\sum x^2_{\text{correct}} = \sum x^2_{\text{incorrect}} - (\text{Wrong items})^2 + (\text{Correct items})^2$$

$$\sigma^2_{\text{correct}} = \frac{\sum x^2_{\text{correct}}}{n} - (\bar{x}_{\text{correct}})^2$$

- **S.D. Bounds Relation [Added]:**

$$\text{M.D. about Mean} \leq \text{Standard Deviation } (\sigma)$$

- **Variance of AP Series [Added]:** For an A.P. with common difference d having n terms:

$$\text{Var} = d^2 \left(\frac{n^2 - 1}{12} \right)$$

LIMITS, CONTINUITY & DIFFERENTIABILITY

JEE MAIN + ADVANCED

Comprehensive JEE Main & Advanced Formula Revision Sheet

1. LIMITS OF FUNCTIONS & EXISTENCE

Existence of Limit & Indeterminate Forms

Limit $\lim_{x \to a} f(x) = L$ exists finitely iff:

$$\mathbf{LHL = RHL = L}$$

LHL: $\lim_{h \to 0^+} f(a - h)$, RHL: $\lim_{h \to 0^+} f(a + h)$

7 Standard Indeterminate Forms:

$[\ 0/0\ ]$, $[\ \infty/\infty\ ]$, $[\ 0 \times \infty\ ]$, $[\ \infty - \infty\ ]$, $[\ 1^{\infty}\ ]$, $[\ 0^0\ ]$, $[\ \infty^0\ ]$

Algebra & Fundamental Theorems of Limits

If $\lim_{x \to a} f(x) = l$ and $\lim_{x \to a} g(x) = m$ (both finite):

- Sum / Diff: $\lim_{x \to a} [f(x) \pm g(x)] = l \pm m$
- Product: $\lim_{x \to a} [f(x) \cdot g(x)] = l \cdot m$
- Quotient: $\lim_{x \to a} [f(x) / g(x)] = l / m$ *($m \neq 0$)*
- Power: $\lim_{x \to a} [f(x)]^{m/n} = l^{m/n}$
- Composite: $\lim_{x \to a} f(g(x)) = f(\lim_{x \to a} g(x)) = f(m)$ *(if f continuous at m)*

2. STANDARD LIMIT FORMULAS

Algebraic & Exponential

1. $\lim_{x \to a} (x^n - a^n) / (x - a) = n\,a^{n-1}$
2. $\lim_{x \to 0} (e^x - 1) / x = 1$
3. $\lim_{x \to 0} (a^x - 1) / x = \ln a$ *($a > 0$)*
4. $\lim_{x \to 0} (a^x - b^x) / x = \ln(a/b)$ *($a, b > 0$)*
5. $\lim_{x \to 0} ((1+x)^n - 1) / x = n$

Trigonometric & Inverse

1. $\lim_{x \to 0} (\sin x) / x = 1$
2. $\lim_{x \to 0} (\tan x) / x = 1$
3. $\lim_{x \to 0} (\sin^{-1} x) / x = 1$
4. $\lim_{x \to 0} (\tan^{-1} x) / x = 1$
5. $\lim_{x \to 0} (1 - \cos x) / x^2 = 1/2$ Added
6. $\lim_{x \to 0} (\sin x^\circ) / x = \pi / 180$

Logarithmic & Base-e

1. $\lim_{x \to 0} \ln(1 + x) / x = 1$
2. $\lim_{x \to 0} \log_a(1 + x) / x = \log_a e$
3. $\lim_{x \to \infty} (1 + 1/x)^x = e$
4. $\lim_{x \to 0} (1 + x)^{1/x} = e$
5. $\lim_{h \to 0} (1 + \alpha\, h)^{1/h} = e^{\alpha}$
6. $\lim_{x \to \infty} (\ln x) / x^m = 0$ *($m > 0$)*

3. ADVANCED LIMIT EVALUATION TECHNIQUES

Evaluating $[\ 1^{\infty}\ ]$ Form

If $\lim_{x \to a} f(x) = 1$ and $\lim_{x \to a} g(x) = \infty$, then:

$$\mathbf{\lim_{x \to a} [f(x)]^{g(x)} = e^{\lim_{x \to a} g(x) \cdot [f(x) - 1]}}$$

Special form: $\lim_{x \to 0} [1 + f(x)]^{1/g(x)} = e^{\lim_{x \to 0} f(x)/g(x)}$

L'Hôpital's Rule & Sandwich Theorem

L'Hôpital's Rule: For $[\ 0/0\ ]$ or $[\ \infty/\infty\ ]$ forms:

$$\lim_{x \to a} f(x) / g(x) = \lim_{x \to a} f'(x) / g'(x)$$

Repeat if form persists and derivatives exist.

Sandwich (Squeeze) Theorem: If $f(x) \le g(x) \le h(x)$ and $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$, then $\mathbf{\lim_{x \to a} g(x) = L}$.

Polynomial / Rational Function at Infinity

$\lim_{x \to \infty} (a_0 x^p + a_1 x^{p-1} + \ldots) / (b_0 x^q + b_1 x^{q-1} + \ldots) = \mathbf{(a_0 / b_0)}$ if $p = q$; $\mathbf{0}$ if $p < q$; $\pm\infty$ if $p > q$ *(sign depends on a_0/b_0)*.

4. STANDARD SERIES EXPANSIONS (ESSENTIAL FOR JEE LIMITS)

- $e^x = 1 + x + x^2/2! + x^3/3! + \ldots$
- $a^x = 1 + x \ln a + (x \ln a)^2/2! + \ldots$
- $\ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + \ldots$ *($-1 < x \le 1$)*
- $\sin x = x - x^3/3! + x^5/5! - x^7/7! + \ldots$
- $\cos x = 1 - x^2/2! + x^4/4! - x^6/6! + \ldots$
- $\tan x = x + x^3/3 + 2x^5/15 + \ldots$
- $\sin^{-1} x = x + 1^2 \cdot x^3/3! + 1^2 \cdot 3^2 \cdot x^5/5! + \ldots$
- $\tan^{-1} x = x - x^3/3 + x^5/5 - x^7/7 + \ldots$
- $(1+x)^n = 1 + nx + n(n-1)x^2/2! + \ldots$ *($|x| < 1$)*

- $(1+x)^{1/x} = e\,[\,1 - x/2 + 11x^2/24 - 21x^3/48 + \ldots\,]$ | • Growth Hierarchy: $\ln(x) << x^{\alpha} << a^x << x! << x^x$ *(as $x \to \infty$, $a > 1$, $\alpha > 0$)*

5. CONTINUITY OF FUNCTIONS

Continuity at a Point & in an Interval

A function $f(x)$ is continuous at $x = a$ if:

$$\lim_{x \to a} f(x) = f(a) \Rightarrow \text{LHL} = \text{RHL} = f(a)$$

- **Open Interval (a, b):** Continuous at every $x \in (a, b)$.
- **Closed Interval [a, b]:** Continuous in (a, b) AND $\lim_{x \to a^+} f(x) = f(a)$ AND $\lim_{x \to b^-} f(x) = f(b)$.

Algebra & Composition of Continuous Functions

If f and g are continuous at $x = c$:

- $f \pm g$, $f \cdot g$, and $k \cdot f$ are continuous at $x = c$.
- f / g is continuous at $x = c$ provided $g(c) \neq 0$.
- **Composite Theorem:** If g is continuous at c and f is continuous at $g(c)$, then $(f \circ g)(x) = f(g(x))$ is continuous at $x = c$.

6. CLASSIFICATION OF DISCONTINUITIES

Type of Discontinuity	Mathematical Condition	JEE Example / Description
1. Removable: Missing Point	$\lim_{x \to a} f(x)$ exists finitely, but $f(a)$ is undefined.	$f(x) = (x^2 - 1)/(x - 1)$ at $x = 1$
2. Removable: Isolated Point	$\lim_{x \to a} f(x)$ exists finitely and $f(a)$ defined, but $L \neq f(a)$.	Piecewise function with redefining point
3. Non-Removable: Finite (Jump)	LHL and RHL exist finitely, but LHL ≠ RHL. Jump Size = $\lvert RHL - LHL \rvert$	$f(x) = [x]$ (GIF) or $\text{sgn}(x)$ at $x = 0$
4. Non-Removable: Infinite	At least one of LHL or RHL tends to $+\infty$ or $-\infty$.	$f(x) = 1/(x - a)$ or $\tan x$ at $x = \pi/2$
5. Non-Removable: Oscillatory	Limit oscillates infinitely between two finite limits.	$f(x) = \sin(1/x)$ as $x \to 0$

7. CORE THEOREMS ON CONTINUOUS FUNCTIONS

Boundedness & Intermediate Value Theorem (IVT)

If $f(x)$ is continuous in $[a, b]$:

1. **Boundedness:** $f(x)$ is bounded and attains its Min (m) & Max (M) on $[a, b]$.
2. **IVT:** Takes every value between $f(a)$ and $f(b)$ at least once.
3. **Root Existence:** If $f(a) \cdot f(b) < 0$, then $f(c) = 0$ for at least one $c \in (a, b)$.

Differentiability vs Continuity Hierarchy

Differentiable ⇒ Continuous

(Converse is NOT true! e.g., $f(x) = |x|$ at $x = 0$)

- Not Continuous ⇒ Not Differentiable.
- Continuous ⇒ May or may not be Differentiable.

8. DIFFERENTIABILITY OF FUNCTIONS

First Principle of Derivative (LHD & RHL)

Function $f(x)$ is differentiable at $x = a$ iff LHD = RHD = \text{finite}:

$$\textbf{LHD: } f'(a^-) = \lim_{h \to 0^+} [f(a - h) - f(a)] / (-h)$$

$$\textbf{RHD: } f'(a^+) = \lim_{h \to 0^+} [f(a + h) - f(a)] / h$$

Derivative at x: $f'(x) = \lim_{h \to 0} [f(x+h) - f(x)] / h$

Product & Sum Combination Rules for Non-Differentiable Functions

Let $f(x)$ be diff. and $g(x)$ non-diff. at $x = a$:

- Product $F(x) = f(x) \cdot g(x)$ CAN be differentiable at $x = a$ if $f(a) = 0$.
- If both f, g non-diff at $x = a$, $f + g$ or $f \cdot g$ MAY still be diff at $x = a$.
- If $f(x)$ is diff at $x = a$, $f'(x)$ is not necessarily continuous at $x = a$.

9. ROLLE'S THEOREM & LAGRANGE'S MEAN VALUE THEOREM (LMVT)

Rolle's Theorem

If $f: [a, b] \to \mathbb{R}$ is:

1. Continuous on $[a, b]$
2. Differentiable on (a, b)
3. $f(a) = f(b)$

Then $\exists\ c \in (a, b)$ such that $\mathbf{f'(c) = 0}$.

(Geometrically: Tangent at c is parallel to X-axis)

Lagrange's Mean Value Theorem (LMVT)

If $f: [a, b] \to \mathbb{R}$ is:

1. Continuous on $[a, b]$
2. Differentiable on (a, b)

Then $\exists\ c \in (a, b)$ such that:

$$\mathbf{f'(c) = [\, f(b) - f(a) \,] / (b - a)}$$

(Geometrically: Tangent at c is parallel to chord AB)

10. STANDARD DERIVATIVES & FUNDAMENTAL RULES

Algebra of Derivatives

- **Sum/Diff:** $(u \pm v)' = u' \pm v'$
- **Product (Leibnitz):** $(u \cdot v)' = u'v + uv'$
- **Quotient Rule:** $(u / v)' = (u'v - uv') / v^2$ *($v \neq 0$)*
- **Chain Rule:** If $y = f(u)$, $u = g(v)$, $v = h(x)$:
  $$\mathbf{dy/dx = (dy/du) \cdot (du/dv) \cdot (dv/dx)}$$

Standard Elementary Derivatives

- $d(x^n)/dx = n\,x^{n-1}$ | $d(e^x)/dx = e^x$
- $d(a^x)/dx = a^x \ln a$ | $d(\ln|x|)/dx = 1/x$
- $d(\log_a x)/dx = 1 / (x \ln a)$
- $d(\sin x)/dx = \cos x$ | $d(\cos x)/dx = -\sin x$
- $d(\tan x)/dx = \sec^2 x$ | $d(\cot x)/dx = -\operatorname{cosec}^2 x$
- $d(\sec x)/dx = \sec x \tan x$ | $d(\operatorname{cosec} x)/dx = -\operatorname{cosec} x \cot x$

11. DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS

Function f(x)	Derivative f'(x)	Valid Domain of x / Restrictions
$\sin^{-1}x$	$1 / \sqrt{(1 - x^2)}$	$x \in (-1, 1)$
$\cos^{-1}x$	$-1 / \sqrt{(1 - x^2)}$	$x \in (-1, 1)$
$\tan^{-1}x$	$1 / (1 + x^2)$	$x \in \mathbb{R}$
$\cot^{-1}x$	$-1 / (1 + x^2)$	$x \in \mathbb{R}$
$\sec^{-1}x$	$1 / (\|x\| \sqrt{(x^2 - 1)})$	$\|x\| > 1 \Rightarrow x \in (-\infty, -1) \cup (1, \infty)$
$\operatorname{cosec}^{-1}x$	$-1 / (\|x\| \sqrt{(x^2 - 1)})$	$\|x\| > 1 \Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

12. ADVANCED DIFFERENTIATION METHODS (JEE ADVANCED)

1. Trigonometric Substitution

- $\sqrt{(a^2 - x^2)} \Rightarrow x = a \sin\theta$ or $a\cos\theta$
- $\sqrt{(a^2 + x^2)} \Rightarrow x = a \tan\theta$ or $a\cot\theta$
- $\sqrt{(x^2 - a^2)} \Rightarrow x = a \sec\theta$ or $a\operatorname{cosec}\theta$
- $\sqrt{((a+x)/(a-x))} \Rightarrow x = a\cos 2\theta$

2. Implicit Differentiation

For $f(x, y) = 0$:
Differentiate termwise w.r.t x, applying chain rule to y terms ($d/dx[g(y)] = g'(y)\,dy/dx$).
Shortcut **$dy/dx = -(\partial f/\partial x) / (\partial f/\partial y)$**

3. Logarithmic Differentiation

For $y = [u(x)]^{v(x)}$:
Take $\ln y = v(x) \ln[u(x)]$.
$$\mathbf{dy/dx = y \cdot [\, v'/u + v \cdot u'/u \,]}$$

4. Parametric Differentiation & 2nd Order

If $x = f(t)$ and $y = g(t)$:
- **First Derivative:** $dy/dx = (dy/dt) / (dx/dt) = g'(t) / f'(t)$
- **Second Derivative:** $d^2y/dx^2 = d/dx(dy/dx) = [\, d/dt(dy/dx) \,] / (dx/dt)$

5. Derivative of Inverse Function & Determinants

- **Inverse Function Derivative:** If $g(x) = f^{-1}(x)$, then:
  $$\mathbf{g'(x) = 1 / f'(g(x))} \;|\; \mathbf{d^2g/dx^2 = -f''(g(x)) / [f'(g(x))]^3}$$
  Added
- **Determinant Differentiation:** Differentiate one row (or column) at a time, keeping others constant, and sum the results.

13. ESSENTIAL LOGARITHM PROPERTIES & FUNCTIONAL EQUATIONS

Logarithmic Identities

1. $\log_b(pq) = \log_b p + \log_b q$
2. $\log_b(p/q) = \log_b p - \log_b q$
3. $\log_b(p^k) = k \log_b p$ | 4. $\log_a p = (\log_b p) / (\log_b a)$ *(Base change)*

Standard Functional Equations JEE Adv

- $f(x+y) = f(x)\,f(y) \Rightarrow f(x) = a^x$
- $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$
- $f(x+y) = f(x) + f(y) \Rightarrow f(x) = kx$
- $f(x)\,f(1/x) = f(x) + f(1/x) \Rightarrow f(x) = 1 \pm x^n$

PERMUTATIONS & COMBINATIONS

Comprehensive JEE Main & Advanced Revision Formula Booklet

1. Fundamental Counting Principles

- **Multiplication Principle:** If operation 1 can be done in m ways and operation 2 in n ways in succession, total ways = $m \times n$.
- **Addition Principle:** If either of two independent operations can be done in m or n ways respectively, total ways = $m + n$.

2. Factorial Notation

- **Definition:** $n! = n(n-1)(n-2)\ldots 3 \cdot 2 \cdot 1 = n \cdot (n-1)!$
- **Base values:** $0! = 1! = 1$ (Factorials of negative integers are not defined).
- **Even Factorial Identity:** $(2n)! = 2^n \cdot n!\,[1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)]$

3. Permutations (Arrangements)

- **Basic Formula:** ${}^nP_r = \dfrac{n!}{(n-r)!} = n(n-1)\ldots(n-r+1)$ where $0 \le r \le n$.
- **Special Values:** ${}^nP_n = n!$, ${}^nP_0 = 1$, ${}^nP_1 = n$.
- **Non-distinct Objects:** Permutations of n things where p_1 are alike of 1st kind, p_2 of 2nd kind, …: $\dfrac{n!}{p_1!\,p_2!\ldots p_k!}$
- **Repetition Allowed:** Number of arrangements of n things taken r at a time = n^r.
- **Specific Inclusion:** Always include 1 particular object: $r \cdot {}^{n-1}P_{r-1}$.
- **Together/Separated:** m specified things together = $m! \times (n-m+1)!$. Separated = $n! - m! \times (n-m+1)!$.

4. Circular Permutations

- **Table / Seating (Clockwise & Anticlockwise distinct):** $(n-1)!$
- **Necklace / Garland / Beads (Clockwise & Anticlockwise same):** $\dfrac{(n-1)!}{2}$
- **Taking r at a time:** $\dfrac{{}^nP_r}{r}$ (distinct orders) or $\dfrac{{}^nP_r}{2r}$ (indistinct orders).

5. Combinations (Selections)

- **Formula:** ${}^nC_r = \binom{n}{r} = \dfrac{n!}{r!(n-r)!} = \dfrac{{}^nP_r}{r!}$
- **Complementary Property:** ${}^nC_r = {}^nC_{n-r}$
- **Equality Identity:** ${}^nC_a = {}^nC_b \Rightarrow a = b$ or $a + b = n$.
- **Pascal's Rule:** ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$
- **Ratio Identity:** $\dfrac{{}^nC_r}{{}^nC_{r-1}} = \dfrac{n-r+1}{r}$
- **Absorption Identity:** $n \cdot {}^{n-1}C_{r-1} = (n-r+1) \cdot {}^nC_{r-1}$
- **At least one selection:** ${}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n - 1$.
- **Restricted Selection:** p included = ${}^{n-p}C_{r-p}$; p excluded = ${}^{n-p}C_r$.

6. Divisors of a Natural Number

Let $N = p_1^{a_1}\, p_2^{a_2} \ldots p_k^{a_k}$ (prime factorization):

- **Total Divisors:** $(a_1+1)(a_2+1)\ldots(a_k+1)$
- **Sum of Divisors:** $\dfrac{p_1^{a_1+1}-1}{p_1-1} \times \dfrac{p_2^{a_2+1}-1}{p_2-1} \ldots$
- **Two Factor Products:** $\dfrac{1}{2}\prod(a_i+1)$ [if not square]; $\dfrac{1}{2}[\prod(a_i+1)+1]$ [if square].
- **Coprime Factors:** Number of ways = 2^{k-1} (where k is distinct prime factors).

7. Division & Distribution of Objects

- **Unequal Groups (m, n, p items):** $\dfrac{(m+n+p)!}{m!\,n!\,p!}$
- **Equal Groups (l groups of n items):** $\dfrac{(ln)!}{(n!)^l \cdot l!}$
- **Distinct Balls in Distinct Boxes:** Empty allowed = r^n; Empty not allowed = $r^n - {}^rC_1(r-1)^n + {}^rC_2(r-2)^n - \ldots$
- **Identical Balls in Distinct Boxes:** Empty allowed = ${}^{n+r-1}C_{r-1}$; Empty not allowed = ${}^{n-1}C_{r-1}$.

8. Derangements D(n)

Number of ways n items in correct envelopes are placed such that none goes to its original place:

$$\mathbf{D(n) = n!\,\sum_{r=0}^{n}(-1)^r}$$

$$r! = n!\,\left[1 - \frac{1}{1!} + \frac{1}{2!} - \ldots + (-1)^n\frac{1}{n!}\right]$$

Standard values: D(1)=0, D(2)=1, D(3)=2, D(4)=9, D(5)=44.

9. Multinomial Theorem & Equations

- **Non-negative Integer Solutions** ($x_1 + x_2 + \ldots + x_m = n$): ${}^{n+m-1}C_{m-1}$
- **Positive Integer Solutions** ($x_i \geq 1$): ${}^{n-1}C_{m-1}$
- **General Coefficient:** Coefficient of $x_1^{p_1}x_2^{p_2}\ldots x_k^{p_k}$ in $(a_1x_1 + \ldots + a_kx_k)^n$ is $\frac{n!}{p_1!p_2!\ldots p_k!}a_1^{p_1}\ldots a_k^{p_k}$.

10. Geometrical Counting Results

- **Rectangles / Squares in n × n Square:** Rectangles = $\frac{n(n+1)}{2}^2$; Squares = $\frac{n(n+1)(2n+1)}{6}$
- **Points & Lines:** Given n points (m collinear), Straight lines = ${}^nC_2 - {}^mC_2 + 1$; Triangles = ${}^nC_3 - {}^mC_3$.
- **Intersections:** Max intersection points of n circles = ${}^nP_2$; n lines = ${}^nC_2$.

PROBABILITY — JEE REVISION FORMULA SHEET

Comprehensive Formula & Reference Sheet for JEE Main & JEE Advanced

JEE MAIN & ADVANCED

1. Classical Definition & Basic Axioms

Classical Probability: $P(A) = \frac{m}{n} = \frac{n(A)}{n(S)} = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$

Axiomatic Bounds: $0 \le P(A) \le 1$; $P(\emptyset) = 0$ (Impossible); $P(S) = 1$ (Sure event)

Complementary Event: $P(A') = P(\bar{A}) = P(A^c) = 1 - P(A)$

Odds in Favour & Against:

• Odds in favour of A = $\frac{m}{n} = \frac{P(A)}{P(\bar{A})}$ | • Odds against A = $\frac{n}{m} = \frac{P(\bar{A})}{P(A)}$

(where m = favourable ways, n = unfavourable ways)

2. Addition Theorems & Set Identities

Two Events Union:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Three Events Union:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

De Morgan's Laws:

• $P(\bar{A} \cap \bar{B}) = P((A \cup B)') = 1 - P(A \cup B)$

• $P(\bar{A} \cup \bar{B}) = P((A \cap B)') = 1 - P(A \cap B)$

Difference of Events:

• $P(A - B) = P(A \cap \bar{B}) = P(A) - P(A \cap B)$

• $P(A) = P(A \cap B) + P(A \cap \bar{B})$

3. Event Classifications & Key Properties

Event Type	Mathematical Condition / Formula	Key JEE Notes
Mutually Exclusive (Disjoint)	$A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$ $P(A \cup B) = P(A) + P(B)$	Events cannot happen together. Occurrence of one precludes others.
Exhaustive System	$E_1 \cup E_2 \cup \dots \cup E_n = S$ $P(E_1 \cup E_2 \cup \dots \cup E_n) = 1$	Together, events cover the entire sample space.
Mutually Exclusive & Exhaustive (Partition)	(i) $E_i \cap E_j = \emptyset$ $(\forall\, i \ne j)$ (ii) $\bigcup_{i=1}^{n} E_i = S$ (iii) $P(E_i) > 0$	$\sum_{i=1}^{n} P(E_i) = 1$. Essential foundation for Total Probability & Baye's Rule.
Independent Events	$P(A \cap B) = P(A) \cdot P(B)$ $P(A / B) = P(A)$, $P(B / A) = P(B)$	If A, B are independent $\Rightarrow$ (A, $\bar{B}$), ($\bar{A}$, B), and ($\bar{A}$, $\bar{B}$) are also pairwise independent.
Mutually Independent (n Events)	Simultaneous multiplication holds for ALL subsets of events (e.g., triplets $P(A \cap B \cap C) = P(A)P(B)P(C)$).	Pairwise independence does *NOT* guarantee mutual independence.

4. Conditional Probability

Definition: Probability of E given F has already occurred:

$$P(E / F) = \frac{P(E \cap F)}{P(F)} = \frac{n(E \cap F)}{n(F)} \quad (\text{provided } P(F) \ne 0)$$

Properties of Conditional Probability:

1. $P(S / F) = P(F / F) = 1$
2. $P((A \cup B) / F) = P(A / F) + P(B / F) - P((A \cap B) / F)$
3. If A and B are disjoint: $P((A \cup B) / F) = P(A / F) + P(B / F)$
4. $P(E' / F) = 1 - P(E / F)$

5. Multiplication Theorem & At Least One

Multiplication Rule:

$$P(E \cap F) = P(E) \cdot P(F / E) = P(F) \cdot P(E / F)$$

Extension to 3 Events:

$P(A \cap B \cap C) = P(A) \cdot P(B / A) \cdot P(C / (A \cap B))$

Probability of "At Least One" Independent Event:

If $p_1, p_2, \dots, p_n$ are individual success probabilities:

• P(None happens) = $(1 - p_1)(1 - p_2)\dots(1 - p_n)$

$$P(\text{At least one}) = 1 - (1 - p_1)(1 - p_2)\dots(1 - p_n)$$

6. Law of Total Probability & Baye's Theorem

Law of Total Probability:
Let $\{E_1, E_2, ..., E_n\}$ be a partition of sample space S. For any event A:

$$P(A) = \sum_{j=1}^{n} P(E_j) \cdot P(A / E_j)$$

Use when an event A can occur through multiple mutually exclusive paths E_j.

Baye's Theorem (Posteriori Probability):
If event A has occurred, the probability it arose from path E_i:

$$P(E_i / A) = \frac{P(E_i) \cdot P(A / E_i)}{\sum_{j=1}^{n} P(E_j) \cdot P(A / E_j)}$$

Terminologies: $P(E_i)$ = Priori Probabilities; $P(E_i / A)$ = Posteriori Probabilities.

7. Random Variable, Mean (Expectation) & Variance

Probability Distribution: A discrete random variable X takes values $x_1, x_2, ..., x_n$ with probabilities $p_1, p_2, ..., p_n$ such that $p_i \geq 0$ and $\sum p_i = 1$.

Mean / Expected Value E(X):

$$\mu = E(X) = \sum_{i=1}^{n} x_i p_i$$

Variance Var(X) or σ^2:

$$Var(X) = E(X^2) - [E(X)]^2$$

Expectation of a Function g(X):
$E(g(X)) = \sum_{i=1}^{n} g(x_i) p_i$

$Var(X) = \sum x_i^2 p_i - (\sum x_i p_i)^2 = \sum (x_i - \mu)^2 p_i$
Standard Deviation: $\sigma_X = \sqrt{Var(X)}$

Properties of Expectation & Variance: [Added]
• $E(aX + b) = a E(X) + b$ | • $Var(aX + b) = a^2 Var(X)$ | • $Var(b) = 0$
• If X and Y are independent: $E(XY) = E(X) E(Y)$ and $Var(X \pm Y) = Var(X) + Var(Y)$

8. Bernoulli Trials & Binomial Distribution B(n, p)

Bernoulli Trial Conditions: (i) Finite fixed n trials, (ii) Independent trials, (iii) Exactly 2 outcomes (Success/Failure), (iv) Constant P(Success) = p, P(Failure) = $q = 1 - p$.

Probability Mass Function (PMF):

$$P(X = r) = {}^{n}C_r \, q^{n-r} \, p^r \quad (r = 0, 1, 2, ..., n)$$

Key Parameters of B(n, p):
• **Mean (μ):** $E(X) = np$
• **Variance (σ^2):** $Var(X) = npq$
• **Standard Deviation (σ):** $\sqrt{npq}$

Binomial Distribution Properties & Recurrence:
• Mean > Variance for Binomial Distribution ($np > npq$ since $0 < q < 1$).
• **Recurrence Relation: [Added]** $\frac{P(X = r)}{P(X = r-1)} = \frac{n - r + 1}{r} \cdot \frac{p}{q}$
• **Most Likely Outcome (Mode of B(n,p)): [Added]**
  – If $(n + 1)p$ is an integer $k \Rightarrow$ Dual modes at $r = k$ and $r = k - 1$.
  – If $(n + 1)p$ is not an integer $\Rightarrow$ Unique mode at $r = [(n + 1)p]$ (integral part).

9. Standard Inequalities & Bounds

Boole's Inequality:

$$P(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} P(A_i)$$

Bonferroni's Inequality:

$$P(\bigcap_{i=1}^{n} A_i) \geq \sum_{i=1}^{n} P(A_i) - (n - 1)$$

10. JEE Advanced Shortcuts & High-Yield Results [Added]

1. Derangement Formula (D_n): Number of ways to rearrange n objects such that none occupies its original position:

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + (-1)^n \frac{1}{n!}\right] \Rightarrow P(\text{Derangement}) = \frac{D_n}{n!} \approx \frac{1}{e} \quad \textit{(for large } n\textit{)}$$

2. Geometric Distribution (First Success):
Probability that 1st success occurs at trial r:

- $P(X = r) = q^{r-1} p$ *(r = 1, 2, 3, ...)*
- **Mean E(X):** $\frac{1}{p}$ | **Variance Var(X):** $\frac{q}{p^2}$

3. Multinomial Probability:
Probability of obtaining $r_1, r_2, \ldots, r_k$ outcomes of type 1, 2, ..., k in n trials ($\sum r_i = n$):

$$P = \frac{n!}{r_1!\, r_2! \ldots r_k!} \cdot p_1^{r_1} p_2^{r_2} \ldots p_k^{r_k}$$

4. Law of Total Variance (EVVE Formula):

$$\text{Var}(X) = E[\text{Var}(X / Y)] + \text{Var}[E(X / Y)]$$

5. Geometrical Probability:

$$P(\text{Event}) = \frac{\text{Favourable Length / Area / Volume}}{\text{Total Length / Area / Volume}}$$

COMPLEX NUMBERS — JEE MAIN & ADVANCED

Complete Revision Sheet (Parts I & II) • Combined Extracted & Core JEE Formulas

JEE SHORTCUT SHEET

1. Fundamentals & Basic Algebra

Definition: $z = a + ib = r(\cos\theta + i\sin\theta) = re^{i\theta}$

a = Re(z), b = Im(z), i = √−1 (where $i^2 = -1$).

Powers of iota (i):

$i^{4n} = 1,\ i^{4n+1} = i,\ i^{4n+2} = -1,\ i^{4n+3} = -i$

Sum of 4 consecutive powers: $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$. Also $1/i = -i$.

Algebraic Operations:

- Add/Sub: $(a+ib) \pm (c+id) = (a\pm c) + i(b\pm d)$
- Multiplication: $(a+ib)(c+id) = (ac-bd) + i(ad+bc)$
- Division: $(a+ib)/(c+id) = [(ac+bd) + i(bc-ad)] / (c^2+d^2)$

Equality: $z_1 = z_2 \Leftrightarrow \mathrm{Re}(z_1) = \mathrm{Re}(z_2)$ & $\mathrm{Im}(z_1) = \mathrm{Im}(z_2)$

Types of z:

- Purely Real: $b = 0 \Leftrightarrow z = \bar{z}$
- Purely Imaginary: $a = 0 \Leftrightarrow z = -\bar{z}$ (provided z ≠ 0)

2. Conjugate of a Complex Number

If $z = a + ib$, then conjugate $\bar{z} = a - ib$.

Basic Relations:

$z + \bar{z} = 2\,\mathrm{Re}(z)$ | $z - \bar{z} = 2i\,\mathrm{Im}(z)$ | $z\,\bar{z} = a^2 + b^2 = |z|^2$

Algebraic Properties of Conjugate:

1. $\overline{(\bar{z})} = z$
2. $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$
3. $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2 \Rightarrow \overline{z^n} = (\bar{z})^n$
4. $\overline{(z_1/z_2)} = \bar{z}_1 / \bar{z}_2$ (for $z_2 \neq 0$)
5. $\mathrm{Re}(z) = (z + \bar{z})/2,\ \mathrm{Im}(z) = (z - \bar{z})/(2i)$
6. $z_1\bar{z}_2 + \bar{z}_1 z_2 = 2\,\mathrm{Re}(\bar{z}_1 z_2) = 2\,\mathrm{Re}(z_1\bar{z}_2) = 2|z_1||z_2|\cos(\theta_1 - \theta_2)$
7. If $z = f(z_1)$, then $\bar{z} = f(\bar{z}_1)$

3. Modulus & Inequalities

Definition: $|z| = \sqrt{a^2 + b^2} \geq 0$. Note: $|z| = 0 \Leftrightarrow z = 0$.

Core Properties:

- $|z| = |\bar{z}| = |-z| = |-\bar{z}|$
- $-|z| \leq \mathrm{Re}(z) \leq |z|$ and $-|z| \leq \mathrm{Im}(z) \leq |z|$
- $|z_1 z_2| = |z_1|\,|z_2| \Rightarrow |z^n| = |z|^n$
- $|z_1 / z_2| = |z_1| / |z_2|$ ($z_2 \neq 0$)

Triangle Inequalities:

$$||z_1| - |z_2|| \leq |z_1 \pm z_2| \leq |z_1| + |z_2|$$

- Max of $|z_1 + z_2|$ is $|z_1| + |z_2|$ (when $\arg(z_1) = \arg(z_2)$)
- Min of $|z_1 + z_2|$ is $||z_1| - |z_2||$ (when $\arg(z_1) - \arg(z_2) = \pi$)

Modulus Identities:

1. $|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm (z_1\bar{z}_2 + \bar{z}_1 z_2)$
2. $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$ *(Parallelogram Law)*
3. Shortcut: If $|z + 1/z| = a$ (a > 0), then:
   Max $|z| = (a + \sqrt{a^2+4})/2$, Min $|z| = (\sqrt{a^2+4} - a)/2$

4. Argument (Amplitude)

Definition: $\arg(z) = \theta = \tan^{-1}(b/a)$.
Principal Argument $\mathrm{Arg}(z) = \theta \in (-\pi, \pi]$. Note: $\mathrm{Arg}(0)$ is undefined.

Quadrant	Condition	Principal Arg θ
1st Quad	a > 0, b > 0	$\alpha = \tan^{-1}\lvert b/a\rvert$
2nd Quad	a < 0, b > 0	$\pi - \alpha$
3rd Quad	a < 0, b < 0	$-\pi + \alpha$
4th Quad	a > 0, b < 0	$-\alpha$

Argument Properties: *($k \in \mathbb{Z}$ for principal branch)*

- $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) + 2k\pi$
- $\arg(z_1 / z_2) = \arg(z_1) - \arg(z_2) + 2k\pi$
- $\arg(z^n) = n\arg(z) + 2k\pi$
- $\arg(\bar{z}) = -\arg(z)$
- $\arg(z_2/z_1) = \theta \Leftrightarrow \arg(z_1/z_2) = 2k\pi - \theta$
- If $\arg(z) = \pm\pi/2 \Rightarrow$ z is purely imaginary.
- If $\arg(z) = 0$ or $\pi \Rightarrow$ z is purely real.

5. Euler's Form & Square Root

Euler's Formula: $e^{i\theta} = \cos\theta + i\sin\theta$

- $e^{-i\theta} = \cos\theta - i\sin\theta$, $|e^{i\theta}| = 1$, $\arg(e^{i\theta}) = \theta$
- $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$, $\sin\theta = (e^{i\theta} - e^{-i\theta})/(2i)$

Important Euler Identities:

- $e^{i\theta} + e^{i\phi} = 2\cos((\theta-\phi)/2) \cdot e^{i(\theta+\phi)/2}$
- $e^{i\theta} - e^{i\phi} = 2i\sin((\theta-\phi)/2) \cdot e^{i(\theta+\phi)/2}$
- $|e^{i\theta} - e^{i\phi}| = 2\,|\sin((\theta-\phi)/2)|$
- $\arg(e^{i\theta} - e^{i\phi}) = (\theta + \phi)/2 + \pi/2$

Square Root of z = a + ib:

$$\sqrt{(a+ib)} = \pm\,[\,\sqrt{((|z|+a)/2)} + i\cdot \text{sign}(b)\cdot\sqrt{((|z|-a)/2)}\,]$$

where $\text{sign}(b) = 1$ if b > 0 and -1 if b < 0.

Standard Square Roots:

- $\sqrt{i} = \pm\,(1+i)/\sqrt{2}$ | $\sqrt{(-i)} = \pm\,(1-i)/\sqrt{2}$
- $\sqrt{\omega} = \pm\,\omega^2$ | $\sqrt{(\omega^2)} = \pm\,\omega$

6. De Moivre's Theorem & Logarithm

De Moivre's Theorem (DMT): 1. If $n \in \mathbb{Z}$: 2. If $n = p/q \in \mathbb{Q}$: One value of is . General roots: , .

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

$$(\cos\theta + i\sin\theta)^{p/q} \cos(p\theta/q) + i\sin(p\theta/q)$$

$$\cos((2k\pi + p\theta)/q) + i\sin((2k\pi + p\theta)/q) \quad k = 0, 1, \ldots, q-1$$

Product Expansion:

$$\prod_{k=1}^{n}(\cos\theta_k + i\sin\theta_k) = \cos(\Sigma\theta_k) + i\sin(\Sigma\theta_k)$$

Logarithm of Complex Number:

$$\log(z) = \log(re^{i\theta}) = \log|z| + i\arg(z)$$

General value: $\log(z) = \log|z| + i(\theta + 2n\pi),\ n \in \mathbb{Z}$.

7. Cube Roots of Unity

Roots of $z^3 = 1$ are $1, \omega, \omega^2$ where:

$$\omega = (-1+i\sqrt{3})/2 = e^{i2\pi/3},\quad \omega^2 = (-1-i\sqrt{3})/2 = e^{i4\pi/3}$$

Core Properties:

1. $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$
2. In general: $1 + \omega^r + \omega^{2r} = 3$ (if r is multiple of 3), else 0.
3. $\omega^{3n} = 1,\ \omega^{3n+1} = \omega,\ \omega^{3n+2} = \omega^2$
4. $\omega^n + \omega^{n+1} + \omega^{n+2} = 0$
5. Conjugates: $\bar{\omega} = \omega^2$ and $(\bar{\omega})^2 = \omega$
6. Modulus: $|\omega| = |\omega^2| = 1$
7. Cube roots of -1 are $-1, -\omega, -\omega^2$.

Algebraic Factorization Identities:

- $a^2 + ab + b^2 = (a - b\omega)(a - b\omega^2)$
- $a^2 - ab + b^2 = (a + b\omega)(a + b\omega^2)$
- $a^3 + b^3 = (a+b)(a+b\omega)(a+b\omega^2)$
- $a^3 - b^3 = (a-b)(a-b\omega)(a-b\omega^2)$
- $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$
- $a^2 + b^2 + c^2 - ab - bc - ca = (a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$
- $x^2 + x + 1 = (x-\omega)(x-\omega^2)$

Special JEE Tricks:

- If $a,b,c \in \mathbb{R}$ & $a + b\omega + c\omega^2 = 0 \Rightarrow a = b = c$.
- If $a+b+c=0=a^2+b^2+c^2 \Rightarrow a:b:c = 1:\omega:\omega^2$.
- Ratios $1:\sqrt{3}$ or $\sqrt{3}:1$ expressible via ω:

  $1 + i\sqrt{3} = -2\omega^2,\quad \sqrt{3} + i = 2\omega/i = -2i\omega$.

8. n-th Roots of Unity

Roots of $z^n = 1$ are $z = e^{i(2k\pi/n)}$ for $k = 0, 1, 2, \ldots, n-1$.

Let $\alpha = e^{i(2\pi/n)} = \cos(2\pi/n) + i\sin(2\pi/n)$.

The n roots are $1, \alpha, \alpha^2, \alpha^3, \ldots, \alpha^{n-1}$ (in G.P. with common ratio α).

Properties:

1. **Sum of roots:** $1 + \alpha + \alpha^2 + \ldots + \alpha^{n-1} = 0$
2. **Product of roots:** $1\cdot\alpha\cdot\alpha^2\ldots\alpha^{n-1} = (-1)^{n-1}$
3. **Sum of p-th powers:**

   $1^p + \alpha^p + \alpha^{2p} + \ldots + \alpha^{(n-1)p} = n$ (if p is div by n), else 0.
4. **Polynomial Factorization:**

   $z^n - 1 = (z-1)(z-\alpha)(z-\alpha^2)\ldots(z-\alpha^{n-1})$

   $1 + z + z^2 + \ldots + z^{n-1} = \prod_{k=1}^{n-1}(z-\alpha^k)$

Geometry of Roots: The n roots represent vertices of a regular polygon of n sides inscribed in a unit circle $|z| = 1$ with one vertex at $(1, 0)$.

9. Rotation Theorem (Conformal Mapping)

If vector $AB\ (z_1 \to z_2)$ is rotated about $A(z_0)$ by angle θ anticlockwise to give vector $AC\ (z_0 \to z_3)$:

$$(z_3 - z_0)\,/\,(z_1 - z_0) = |(z_3 - z_0)/(z_1 - z_0)|\cdot e^{i\theta}$$

- $\arg((z_3 - z_0)/(z_1 - z_0)) = \theta$ (taken in anticlockwise direction).

Collinearity & Angle between Lines:

Angle θ between line segment z_1z_2 and z_3z_4:

$\arg((z_1 - z_2) / (z_3 - z_4)) = \theta$

• **Parallel/Collinear:** $(z_1 - z_2)/(z_3 - z_4)$ is purely real ⇒ $\theta = 0$ or π.

• **Perpendicular:** $(z_1 - z_2)/(z_3 - z_4)$ is purely imaginary ⇒ $\theta = \pm\pi/2$.

10. Geometry of Triangles in Argand Plane

Vertices $A(z_1), B(z_2), C(z_3)$ with side lengths a, b, c:

• **Centroid (G):** • • • Or • $z = (z_1 + z_2 + z_3) / 3$

Incentre (I): $z = (az_1 + bz_2 + cz_3) / (a + b + c)$

Excentre (I_A): $z = (-az_1 + bz_2 + cz_3) / (-a + b + c)$ [Added]

Orthocentre (H):

$z = (z_1 \tan A + z_2 \tan B + z_3 \tan C) / (\tan A + \tan B + \tan C)$

$z = (a \sec A\, z_1 + b \sec B\, z_2 + c \sec C\, z_3)/(a \sec A + b \sec B + c \sec C)$

Circumcentre (O):

$z = (z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C) / (\sin 2A + \sin 2B + \sin 2C)$

Area of Triangle ΔABC:

$\text{Area} = (1/4) \left| \begin{matrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{matrix} \right|$ (in determinant form)

Condition for Equilateral Triangle:

> $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$
> Or $1/(z_1 - z_2) + 1/(z_2 - z_3) + 1/(z_3 - z_1) = 0$

• If origin O is circumcentre: $z_1^2 + z_2^2 + z_3^2 = 0$ [Added]

11. Straight Lines in Complex Plane

Line through two points z_1, z_2:

• Parametric: $z = z_1 + t(z_1 - z_2), \quad t \in \mathbb{R}$

• Non-parametric: $(z - z_1)/(\bar{z} - \bar{z}_1) = (z_1 - z_2)/(\bar{z}_1 - \bar{z}_2)$

• Determinant Form: $\left| \begin{matrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{matrix} \right| = 0$

General Equation of Straight Line:

> $\bar{\alpha} z + \alpha \bar{z} + r = 0$ (where $r \in \mathbb{R}$, $\alpha \in \mathbb{C}$, $\alpha \neq 0$)

Complex Slope (ω):

• Line $\bar{\alpha} z + \alpha \bar{z} + r = 0$: $\omega = -\alpha / \bar{\alpha}$

• Line joining z_1, z_2: $\omega = (z_1 - z_2) / (\bar{z}_1 - \bar{z}_2)$

• Line making inclination θ with real axis: $\omega = e^{i2\theta}$

Parallel & Perpendicular Lines:

• Parallel: $\omega_1 = \omega_2$

• Perpendicular: $\omega_1 + \omega_2 = 0$

Perpendicular Distance: Distance from z_0 to line $\bar{\alpha} z + \alpha \bar{z} + r = 0$:

$p = |\bar{\alpha} z_0 + \alpha \bar{z}_0 + r| / (2|\alpha|)$

12. Circles in Complex Plane

Center & Radius Form: $|z - z_0| = r$

General Equation of Circle:

$$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + k = 0 \quad (\text{where } k \in \mathbb{R})$$

• **Center:** $-\alpha$ | **Radius:** $r = \sqrt{(|\alpha|^2 - k)}$

• Condition for real circle: $|\alpha|^2 - k \geq 0$.

Diameter Form: Circle with diameter endpoint z_1, z_2:

• $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$

• Or $\arg((z - z_2)/(z - z_1)) = \pm\pi/2$

Circle through 3 Non-collinear Points:
Cross ratio $[(z - z_2)(z_3 - z_1)] / [(z - z_1)(z_3 - z_2)]$ is purely real.

Concyclic Points Condition: 4 points z_1, z_2, z_3, z_4 are concyclic if:
$[(z_3 - z_1)(z_4 - z_2)] / [(z_3 - z_2)(z_4 - z_1)] \in \mathbb{R}$

Appolonius Circle:
• $|(z - z_1)/(z - z_2)| = k$ represents a circle if $k \neq 1, 0$; and a perpendicular bisector line if $k = 1$.

Equation: $|z - z_1|^2 + |z - z_2|^2 = k$
Represents circle if $k \geq (1/2)|z_1 - z_2|^2$.

13. Vector Products, Reflection & Inverse Points

Dot Product (Scalar Product):
$z_1 \cdot z_2 = |z_1||z_2| \cos\theta = x_1x_2 + y_1y_2 = Re(\bar{z}_1z_2) = (1/2)(\bar{z}_1z_2 + z_1\bar{z}_2)$
• If $z_1 \perp z_2 \Rightarrow z_1 \cdot z_2 = 0 \Rightarrow z_1/\bar{z}_1 + z_2/\bar{z}_2 = 0$ (Sum of slopes = 0).

Cross Product (Vector Magnitude):
$z_1 \times z_2 = |z_1||z_2| \sin\theta = x_1y_2 - y_1x_2 = Im(\bar{z}_1z_2) = (1/2i)(\bar{z}_1z_2 - z_1\bar{z}_2)$
• If $z_1 \parallel z_2 \Rightarrow z_1 \times z_2 = 0 \Rightarrow z_1/\bar{z}_1 = z_2/\bar{z}_2$ (Slopes are equal).

Reflection Point w.r.t. Straight Line:
Points z_1, z_2 are reflection points w.r.t. line $\bar{\alpha}z + \alpha\bar{z} + r = 0$ iff:

$$\bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0 \quad (r \in \mathbb{R})$$

Inverse Points w.r.t. Circle:
Two points z_1, z_2 are inverse w.r.t. circle with center O and radius ρ if O, z_1, z_2 are collinear & $OP \cdot OQ = \rho^2$.
Condition for circle $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$:

$$z_1\bar{z}_2 + \bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$$

14. Standard Loci & Conics in Complex Plane

Equation	Condition	Locus / Shape
$\|z - z_1\| + \|z - z_2\| = 2k$	$2k > \|z_1 - z_2\|$	Ellipse (foci z_1, z_2)
	$2k = \|z_1 - z_2\|$	Line segment joining z_1, z_2
	$2k < \|z_1 - z_2\|$	Empty Set (No Solution)
$\|\|z - z_1\| - \|z - z_2\\| = 2k$	$2k < \|z_1 - z_2\|$	Hyperbola (foci z_1, z_2)
	$2k = \|z_1 - z_2\|$	Union of two rays
	$2k > \|z_1 - z_2\|$	Empty Set (No Solution)
$\|z - z_0\| = \|\bar{\alpha}z + \alpha\bar{z} + r\| / (2\|\alpha\|)$	$\bar{\alpha}z_0 + \alpha\bar{z}_0 + r \neq 0$	Parabola (focus z_0, directrix line)

Circular Arcs & Rays:
$\arg((z - z_1)/(z - z_2)) = \theta$:
• If $\theta = 0 \Rightarrow$ Pair of rays outside segment z_1z_2.
• If $\theta = \pi \Rightarrow$ Line segment joining z_1 and z_2.
• If $0 < \theta < \pi$ $(\theta \neq \pi/2) \Rightarrow$ Segment of a circle passing through z_1, z_2.
• If $\theta = \pi/2 \Rightarrow$ Major/minor semicircle on z_1z_2 as diameter.

15. JEE Advanced Useful Results & Shortcuts

1. **Distance from Origin to Line:** Minimum distance of $|z - z_1| + |z - z_2|$ is $|z_1 - z_2|$.

2. **Maximum & Minimum Value Loci:**
If $|z - z_0| = r$, then:
• Max $|z| = |z_0| + r$ | Min $|z| = ||z_0| - r|$
• Max $|z - z_1| = |z_0 - z_1| + r$ | Min $|z - z_1| = ||z_0 - z_1| - r|$

3. **Equilateral Triangle Center Identity:** [Added]
For equilateral Δ with centroid G(z_0): $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$.

4. **Pisot / Homographic Transformation:** [Added]
Transformation $w = (az + b)/(cz + d)$ maps circles/lines to circles/lines in $\mathbb{C}$.

5. **Lagrange's Identity for Complex Numbers:** [Added]
$|1 - \bar{z}_1z_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$
• If $|z_1| < 1$ and $|z_2| < 1 \Rightarrow |(z_1 - z_2)/(1 - \bar{z}_1z_2)| < 1$.

QUADRATIC EQUATIONS & POLYNOMIALS

Complete JEE Main & Advanced Formula Revision Sheet

Mathematics Sheet
High Density Edition

1. STANDARD FORM & NATURE OF ROOTS

Main & Advanced

Standard Form & Roots Formula

Quadratic Equation: $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$ and $a \neq 0$.

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

Discriminant: $D = b^2 - 4ac$.

- $D > 0 \Rightarrow$ Two real & distinct roots.
- $D = 0 \Rightarrow$ Two real & equal (coincident) roots ($x = -b/2a$).
- $D < 0 \Rightarrow$ Two complex conjugate roots.

Roots Under Special Conditions

For $ax^2 + bx + c = 0$ with $a \neq 0$:

- **b = 0** $\Rightarrow$ Roots are equal in magnitude, opposite in sign ($x = \pm\sqrt{(-c/a)}$).
- **c = 0** $\Rightarrow$ One root is 0, other is $-b/a$.
- **a = c** $\Rightarrow$ Roots are reciprocal of each other ($\alpha, 1/\alpha$).
- **a + b + c = 0** $\Rightarrow$ One root is 1, second root is c/a.
- **a − b + c = 0** [Added] $\Rightarrow$ One root is -1, second root is $-c/a$.
- **a & c opposite signs** ($ac < 0$) $\Rightarrow$ Roots are of opposite signs.

2. COEFFICIENT THEOREMS & CONJUGATE ROOTS

JEE Rules

Rationality of Roots ($a, b, c \in \mathbb{Q}$)

- If $a, b, c \in \mathbb{Q}$ and D is a **perfect square** of a rational number $\Rightarrow$ Roots are **rational**.
- If D is **not a perfect square** $\Rightarrow$ Roots are irrational and occur in **conjugate pairs**:

  $\alpha = p + \sqrt{q} \Rightarrow \beta = p - \sqrt{q}$ ($p, q \in \mathbb{Q}$, $\sqrt{q}$ is a surd)
- If $a = 1, b, c \in \mathbb{Z}$ and roots are rational $\Rightarrow$ Roots MUST be integers.

Complex Conjugate Roots ($a, b, c \in \mathbb{R}$)

- If coefficients a, b, c are real and one root is imaginary $p + iq$ (where $i = \sqrt{-1}$, $q \neq 0$), then the other root is its complex conjugate $p - iq$.
- **Identity Condition:** If $ax^2 + bx + c = 0$ is satisfied by more than 2 distinct numbers, it becomes an **Identity** in $x \Rightarrow a = b = c = 0$.

3. VIETA'S RELATIONS & SYMMETRIC FUNCTIONS OF ROOTS

Crucial for Main/Adv

Core Relations for Quadratic:

- Sum of roots: $S = \alpha + \beta = -b/a$
- Product of roots: $P = \alpha \cdot \beta = c/a$
- Difference of roots: $|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$

Equation Formation:

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \Rightarrow x^2 - Sx + P = 0$$

Factorized Form: $ax^2 + bx + c = a(x - \alpha)(x - \beta)$

Standard Symmetric Expressions (Rapid Computation Results):

Symmetric Combination	Formula in Terms of a, b, c
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$	$(b^2 - 2ac) / a^2$
$\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$	$-b\sqrt{(b^2 - 4ac)} / a^2$
$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$	$-b(b^2 - 3ac) / a^3$
$\alpha^3 - \beta^3 = (\alpha - \beta)(\alpha^2 + \alpha\beta + \beta^2)$	$[(\alpha+\beta)^2 - \alpha\beta] \cdot [\sqrt{D} / \lvert a\rvert]$
$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2$	$[(b^2 - 2ac) / a^2]^2 - 2c^2/a^2$
$\alpha^2 + \alpha\beta + \beta^2 = (\alpha + \beta)^2 - \alpha\beta$	$(b^2 - ac) / a^2$
$1/\alpha + 1/\beta = (\alpha + \beta) / \alpha\beta$	$-b / c$
$\alpha/\beta + \beta/\alpha = (\alpha^2 + \beta^2) / \alpha\beta$	$(b^2 - 2ac) / ac$

Symmetric Combination	Formula in Terms of a, b, c
$\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta)$	$-bc / a^2$
$(\alpha/\beta)^2 + (\beta/\alpha)^2 = (\alpha^4 + \beta^4) / \alpha^2\beta^2$	$[(b^2 - 2ac)^2 - 2a^2c^2] / a^2c^2$

4. TRANSFORMATION OF EQUATIONS (ROOTS SUBSTITUTION)

Direct Substitutions

If α, β are roots of $ax^2 + bx + c = 0$, to find equation whose roots are:

Target Roots	Substitution Rule	Transformed Equation
$-\alpha, -\beta$	Replace x with $-x$	$ax^2 - bx + c = 0$
$1/\alpha, 1/\beta$	Replace x with $1/x$	$cx^2 + bx + a = 0$
$k\alpha, k\beta$	Replace x with x/k	$ax^2 + kbx + k^2c = 0$
$\alpha/k, \beta/k$	Replace x with $k\,x$	$k^2ax^2 + kbx + c = 0$
$\alpha + k, \beta + k$	Replace x with $x - k$	$a(x - k)^2 + b(x - k) + c = 0$
$\alpha^n, \beta^n\ (n \in \mathbb{N})$	Replace x with $x^{1/n}$	$a(x^{1/n})^2 + b(x^{1/n}) + c = 0$
$\alpha^{1/n}, \beta^{1/n}$	Replace x with x^n	$a(x^n)^2 + b(x^n) + c = 0$

5. COMMON ROOTS OF TWO QUADRATIC EQUATIONS

High Priority JEE

Exactly One Common Root

Let equations be $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$.

Let common root be α. By Cramer's Rule:

$$\frac{\alpha^2}{b_1c_2 - b_2c_1} = \frac{\alpha}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

Common Root value: $\alpha = \dfrac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} = \dfrac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$

Condition for One Common Root:

$$(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

Both Roots Common

If both roots of $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ are common, the equations are proportional:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Shortcut Note: If one equation has imaginary/irrational roots (and coefficients are real/rational), then the **other root must also be common**. Thus, check if $D_1 < 0$; if so, apply the *both roots common* ratio condition directly!

6. QUADRATIC EXPRESSION & GRAPH CHARACTERISTICS

Graph & Range

Quadratic Expression: $f(x) = ax^2 + bx + c = a\left[\left(x + \dfrac{b}{2a}\right)^2 - \dfrac{D}{4a^2}\right]$

Represents a **Parabola** with vertical axis $x = -b/(2a)$ and Vertex $A(-b/2a, -D/4a)$.

Condition	Graph Shape	Range of f(x)	Extremum Value
a > 0	Opens Upward (∪)	$[\ -D / 4a\ , +\infty\ )$	Minimum value = $-D / 4a$ at $x = -b / 2a$
a < 0	Opens Downward (∩)	$(\ -\infty\ , -D / 4a\ ]$	Maximum value = $-D / 4a$ at $x = -b / 2a$

7. SIGN OF QUADRATIC EXPRESSION F(X) = AX² + BX + C

Inequalities

Always Positive

$f(x) > 0, \forall x \in \mathbb{R}$

a > 0 and D < 0

Parabola stays strictly above the x-axis.

Always Negative

$f(x) < 0, \forall x \in \mathbb{R}$

a < 0 and D < 0

Parabola stays strictly below the x-axis.

Non-Negative / Non-Positive

$f(x) \geq 0, \forall x \in \mathbb{R} \Rightarrow$ **a > 0 & D ≤ 0**

$f(x) \leq 0, \forall x \in \mathbb{R} \Rightarrow$ **a < 0 & D ≤ 0**

(Touches x-axis if $D = 0$).

Sign Distribution for D > 0 (Two Real Roots α < β)

- If **a > 0**: $f(x) > 0$ for $x \in (-\infty, \alpha) \cup (\beta, \infty)$ and $f(x) < 0$ for $x \in (\alpha, \beta)$.
- If **a < 0**: $f(x) < 0$ for $x \in (-\infty, \alpha) \cup (\beta, \infty)$ and $f(x) > 0$ for $x \in (\alpha, \beta)$.

8. LOCATION OF ROOTS (BOUND CONDITIONS)

Top Advanced Topic

Let $f(x) = ax^2 + bx + c = 0$ have real roots ($D \geq 0$) and $k, k_1, k_2 \in \mathbb{R}$:

Location Condition	Necessary & Sufficient Conditions
1. Both roots greater than k ($\alpha, \beta > k$)	(i) $D \geq 0$ (ii) $a \cdot f(k) > 0$ (iii) $-b / 2a > k$
2. Both roots smaller than k ($\alpha, \beta < k$)	(i) $D \geq 0$ (ii) $a \cdot f(k) > 0$ (iii) $-b / 2a < k$
3. k lies between the roots ($\alpha < k < \beta$)	(i) $a \cdot f(k) < 0$ *(D > 0 is automatically satisfied!)*
4. Both roots positive ($\alpha, \beta > 0$)	(i) $D \geq 0$ (ii) $\alpha + \beta = -b/a > 0$ (iii) $\alpha\beta = c/a > 0$
5. Both roots negative ($\alpha, \beta < 0$)	(i) $D \geq 0$ (ii) $\alpha + \beta = -b/a < 0$ (iii) $\alpha\beta = c/a > 0$
6. Roots of opposite signs	(i) $\alpha\beta = c/a < 0$ or $a \cdot f(0) < 0$
7. Both roots lie in interval (k_1, k_2)	(i) $D \geq 0$ (ii) $a \cdot f(k_1) > 0$ (iii) $a \cdot f(k_2) > 0$ (iv) $k_1 < -b/2a < k_2$
8. Exactly one root in (k_1, k_2)	(i) $f(k_1) \cdot f(k_2) < 0$ (if roots are non-endpoint)

9. HIGHER DEGREE POLYNOMIAL EQUATIONS (THEORY OF EQUATIONS)

Polynomials

General Polynomial: $P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \ldots + a_1 x + a_0 = 0$ ($a_n \neq 0$).

Let roots be $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n$.

Vieta's Formulas for Degree n:

- $\sum \alpha_1 = \alpha_1 + \alpha_2 + \ldots + \alpha_n = -\frac{a_{n-1}}{a_n}$
- $\sum \alpha_1\alpha_2 = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \ldots = +\frac{a_{n-2}}{a_n}$
- $\sum \alpha_1\alpha_2\alpha_3 = -\frac{a_{n-3}}{a_n}$
- Product of all roots: $\alpha_1\alpha_2\ldots\alpha_n = (-1)^n \frac{a_0}{a_n}$

Cubic Equation Relations ($ax^3 + bx^2 + cx + d = 0$):

- $\alpha + \beta + \gamma = -b / a$
- $\alpha\beta + \beta\gamma + \gamma\alpha = c / a$
- $\alpha\,\beta\,\gamma = -d / a$
- $x^3 - (\sum\alpha)x^2 + (\sum\alpha\beta)x - (\alpha\beta\gamma) = 0$

10. IMPORTANT POLYNOMIAL THEOREMS & ROOT PROPERTIES

JEE Advanced Special

Odd & Even Degree Theorems

- **Odd Degree Equation:** Every odd degree polynomial equation with real coefficients has **at least one real root** whose sign is opposite to that of its last term (when highest coefficient is +ve).
- **Even Degree Equation:** An even degree polynomial with $a_n > 0$ and last term negative ($a_0 < 0$) has **at least two real roots** (one positive, one negative).
- **All Positive Coefficients:** If an equation contains only *even powers* of x and all coefficients are positive, all roots are imaginary.

Descartes' Rule of Signs [Added]

- **Max Positive Real Roots:** Equal to the number of sign changes in the sequence of coefficients of $f(x)$.
- **Max Negative Real Roots:** Equal to the number of sign changes in $f(-x)$.
- **Remainder & Factor Theorem:** $f(x)$ divided by $(x - a)$ leaves remainder $f(a)$. If $f(a) = 0$, $(x - a)$ is a factor.

11. NEWTON'S SUMS THEOREM FOR POWERS OF ROOTS

High Weightage Shortcut

Newton's Theorem (Quadratic & Cubic) [Added]

Let α, β be roots of $ax^2 + bx + c = 0$. Define $S_n = \alpha^n + \beta^n$ (or $\alpha^n - \beta^n$).

Then for any integer $n \geq 2$:

$$a \cdot S_n + b \cdot S_{n-1} + c \cdot S_{n-2} = 0$$

Cubic Extension: If α, β, γ are roots of $ax^3 + bx^2 + cx + d = 0$ and $S_n = \alpha^n + \beta^n + \gamma^n$:

$$a \cdot S_n + b \cdot S_{n-1} + c \cdot S_{n-2} + d \cdot S_{n-3} = 0$$

12. RESOLUTION OF GENERAL 2D QUADRATIC EXPRESSION

Factorization

The general 2-variable quadratic expression:

$$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

can be resolved into **two linear factors** if and only if the determinant condition $\Delta = 0$ holds:

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

Or Matrix Determinant:

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

13. RANGE OF RATIONAL EXPRESSIONS Y = N(X) / D(X)

JEE Standard Method

To find range of $y = \dfrac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$ for $x \in \mathbb{R}$:

1. Cross multiply to convert into quadratic in x: $(a_1 - a_2y)x^2 + (b_1 - b_2y)x + (c_1 - c_2y) = 0$.
2. Since $x \in \mathbb{R}$, set Discriminant $D \geq 0$ on the quadratic in x.
3. Solve the resulting inequality in y to obtain the permissible range.
4. *Check boundary cases:* Check if coefficient of x^2 can be 0 (i.e. $y = a_1/a_2$).

JEE ADVANCED & MAIN FORMULA SHEET : CONIC SECTIONS

Complete Revision Guide: Circles | Parabola | Ellipse | Hyperbola

1. CIRCLES

Standard Equations

$$(x - h)^2 + (y - k)^2 = r^2$$

→ *Center: (h, k), Radius: r*

$$x^2 + y^2 = r^2$$

→ *Center at Origin (0, 0)*

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

→ *Center: (−g, −f), Radius: $r = \sqrt{(g^2 + f^2 - c)}$*

- **Real Circle:** $g^2 + f^2 - c > 0$
- **Point Circle:** $g^2 + f^2 - c = 0$
- **Imaginary Circle:** $g^2 + f^2 - c < 0$
- **Axis Intercepts:** x-int = $2\sqrt{(g^2 - c)}$, y-int = $2\sqrt{(f^2 - c)}$

Special Forms & Parametric

- **Diameter Form:** $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$
- **Parametric Form:** $x = h + r\cos\theta$, $y = k + r\sin\theta$
- **Touching Axes:** Radius r, Center (±r, ±r) ⇒ $(x \pm r)^2 + (y \pm r)^2 = r^2$

Position of a Point & Tangency

- **Point S_1:** ${x_1}^2 + {y_1}^2 + 2gx_1 + 2fy_1 + c >=< 0$ (Outside / On / Inside)
- **Length of Tangent:** $L = \sqrt{S_1}$ (Power of point = S_1)
- **Tangent Equation (T=0):** $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
- **Slope Form Tangent:** $y = mx \pm a\sqrt{(1 + m^2)}$ for $x^2 + y^2 = a^2$
- **Director Circle:** $x^2 + y^2 = 2a^2$ (Locus of ⊥ tangents)

Systems of Circles

- **Family passing through intersection of S=0 & L=0:** $S + \lambda L = 0$
- **Family passing through two circles S_1=0 & S_2=0:** $S_1 + \lambda S_2 = 0$ ($\lambda \neq -1$)
- **Common Chord / Radical Axis:** $S_1 - S_2 = 0$
- **Orthogonality Condition:** $2g_1g_2 + 2f_1f_2 = c_1 + c_2$ (or ${r_1}^2 + {r_2}^2 = d^2$)
- **Length of DCT:** $\sqrt{(d^2 - (r_1 - r_2)^2)}$
- **Length of TCT:** $\sqrt{(d^2 - (r_1 + r_2)^2)}$

2. PARABOLA

Property	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$
Vertex	(0, 0)	(0, 0)	(0, 0)
Focus	(a, 0)	(−a, 0)	(0, a)
Directrix	x = −a	x = a	y = −a
Axis	y = 0	y = 0	x = 0
Latus Rectum	4a	4a	4a

Parametric & Tangents

- **Parametric Coordinates:** $(at^2, 2at)$ for $y^2 = 4ax$
- **Focal Distance:** $SP = x_1 + a$
- **Point Tangent:** $yy_1 = 2a(x + x_1)$
- **Slope Form Tangent:** $y = mx + (a / m)$, Point of contact: $(a/m^2, 2a/m)$
- **Parametric Tangent:** $ty = x + at^2$
- **Intersection of Tangents at t_1, t_2:** $(at_1t_2, a(t_1 + t_2))$

Normals & Chords

- **Slope Form Normal:** $y = mx - 2am - am^3$ (Foot: $(am^2, -2am)$)
- **Parametric Normal:** $y + tx = 2at + at^3$
- **Chord joining t_1, t_2:** $y(t_1 + t_2) = 2x + 2at_1t_2$
- **Focal Chord Condition:** $t_1t_2 = -1$. Length = $a(t_1 - t_2)^2 = 4a\csc^2\alpha$
- **Subtends Right Angle at Vertex:** $t_1t_2 = -4$
- **Conormal Points:** Sum of ordinates = 0, sum of slopes = 0. Max 3 normals from a point.

3. ELLIPSE

$$x^2 / a^2 + y^2 / b^2 = 1 \qquad (a > b)$$

- **Eccentricity:** $b^2 = a^2(1 - e^2) \Rightarrow e = \sqrt{(1 - b^2/a^2)}$
- **Foci:** (±ae, 0) | **Directrices:** x = ± a/e
- **Vertices:** (±a, 0) | **Axes:** Major = 2a, Minor = 2b
- **Latus Rectum Length:** $2b^2 / a$
- **Parametric Form:** (a cos θ, b sin θ)
- **Focal Radii:** $SP = a - ex_1$, $S'P = a + ex_1$ ($SP + S'P = 2a$)
- **Auxiliary Circle:** $x^2 + y^2 = a^2$
- **Director Circle:** $x^2 + y^2 = a^2 + b^2$

Ellipse Tangents & Normals

- **Point Tangent:** $xx_1 / a^2 + yy_1 / b^2 = 1$
- **Slope Form Tangent:** $y = mx \pm \sqrt{(a^2m^2 + b^2)}$
- **Parametric Tangent:** $(x/a)\cos\theta + (y/b)\sin\theta = 1$
- **Point Normal:** $(a^2x / x_1) - (b^2y / y_1) = a^2 - b^2$
- **Slope Form Normal:** $y = mx \pm ((a^2 - b^2)m) / \sqrt{(a^2 + b^2m^2)}$

4. HYPERBOLA

$$x^2 / a^2 - y^2 / b^2 = 1$$

- **Eccentricity:** $b^2 = a^2(e^2 - 1) \Rightarrow e = \sqrt{(1 + b^2/a^2)}$
- **Foci:** (±ae, 0) | **Directrices:** x = ± a/e
- **Vertices:** (±a, 0) | **Transverse Axis = 2a, Conjugate Axis = 2b**
- **Latus Rectum Length:** $2b^2 / a$
- **Parametric Form:** (a sec θ, b tan θ)
- **Focal Distances:** $PS = ex_1 - a$, $P'S = ex_1 + a$ ($|PS - P'S| = 2a$)
- **Auxiliary Circle:** $x^2 + y^2 = a^2$
- **Director Circle:** $x^2 + y^2 = a^2 - b^2$ (Real if a > b)

Asymptotes & Rectangular Hyperbola

- **Asymptotes Equation:** $x^2 / a^2 - y^2 / b^2 = 0 \Rightarrow y = \pm(b/a)x$
- **Conjugate Hyperbola:** $-(x^2/a^2) + (y^2/b^2) = 1$ (Eccentricities: $1/e^2 + 1/(e')^2 = 1$)
- **Rectangular Hyperbola:** e = √2, Asymptotes ⊥ at 90°. Standard form: $xy = c^2$
- **Parametric ($xy = c^2$):** (ct, c/t) | **Tangent:** $x/t + yt = 2c$ | **Normal:** $xt^3 - yt = c(t^4 - 1)$

JEE MASTER FORMULA SHEET: TRIGONOMETRY & ITF

Comprehensive Revision Guide for JEE Main & Advanced (Extracted from Coaching Modules & Standard Resources)

1. TRIGONOMETRIC RATIOS & IDENTITIES

Fundamental Pythagorean Identities

```
sin²θ + cos²θ = 1
1 + tan²θ = sec²θ  (θ ≠ (2n+1)π/2)
1 + cot²θ = cosec²θ  (θ ≠ nπ)
```

Allied Angles & Transformation

Angles $n(\pi/2) \pm \theta$:

- Odd multiples of $\pi/2$: sin ↔ cos, tan ↔ cot, sec ↔ cosec (ASTC sign check).
- Even multiples of $\pi/2$: ratios retain their names.
- $\sin(-\theta) = -\sin\theta$, $\cos(-\theta) = \cos\theta$, $\tan(-\theta) = -\tan\theta$
- $\sin(n\pi \pm \theta) = (-1)^n \sin\theta$, $\cos(n\pi \pm \theta) = (-1)^n \cos\theta$

Compound Angle Formulae

```
sin(A ± B) = sinA cosB ± cosA sinB
cos(A ± B) = cosA cosB ∓ sinA sinB
tan(A ± B) = (tanA ± tanB) / (1 ∓ tanA tanB)
cot(A ± B) = (cotB cotA ∓ 1) / (cotB ± cotA)
```

Key results: $\sin(A+B)\sin(A-B) = \sin^2A - \sin^2B = \cos^2B - \cos^2A$; $\cos(A+B)\cos(A-B) = \cos^2A - \sin^2B$.

Factorization & Product-to-Sum

```
2 sinA cosB = sin(A+B) + sin(A-B)
2 cosA sinB = sin(A+B) - sin(A-B)
2 cosA cosB = cos(A+B) + cos(A-B)
2 sinA sinB = cos(A-B) - cos(A+B)

sinC + sinD = 2 sin((C+D)/2) cos((C-D)/2)
sinC - sinD = 2 cos((C+D)/2) sin((C-D)/2)
cosC + cosD = 2 cos((C+D)/2) cos((C-D)/2)
cosC - cosD = -2 sin((C+D)/2) sin((C-D)/2)
```

Multiple & Sub-Multiple Angles

```
sin 2θ = 2 sinθ cosθ = 2 tanθ / (1 + tan²θ)
cos 2θ = cos²θ - sin²θ = 2cos²θ - 1 = 1 - 2sin²θ
cos 2θ = (1 - tan²θ) / (1 + tan²θ),   tan 2θ = 2tanθ / (1 - tan²θ)
sin 3θ = 3 sinθ - 4 sin³θ
cos 3θ = 4 cos³θ - 3 cosθ
tan 3θ = (3 tanθ - tan³θ) / (1 - 3 tan²θ)
```

Half-angle: $\sin^2\theta = (1 - \cos 2\theta)/2$, $\cos^2\theta = (1 + \cos 2\theta)/2$.

Standard Angle Values (Extra Utility)

```
sin 15° = cos 75° = (√3 - 1) / (2√2)
cos 15° = sin 75° = (√3 + 1) / (2√2)
tan 15° = 2 - √3,   tan 75° = 2 + √3
sin 18° = (√5 - 1) / 4,   cos 36° = (√5 + 1) / 4
tan 22.5° = √2 - 1,   tan 67.5° = √2 + 1
```

Conditional Identities in ΔABC (A + B + C = π)

```
sin 2A + sin 2B + sin 2C = 4 sinA sinB sinC
cos 2A + cos 2B + cos 2C = -1 - 4 cosA cosB cosC
sin A + sin B + sin C = 4 cos(A/2) cos(B/2) cos(C/2)
cos A + cos B + cos C = 1 + 4 sin(A/2) sin(B/2) sin(C/2)
tan A + tan B + tan C = tan A tan B tan C
```

2. TRIGONOMETRIC EQUATIONS

Standard General Solutions

Equation	General Solution	Condition
$\sin\theta = 0$	$\theta = n\pi$	$n \in Z$
$\cos\theta = 0$	$\theta = (2n + 1)\pi / 2$	$n \in Z$
$\tan\theta = 0$	$\theta = n\pi$	$n \in Z$
$\sin\theta = \sin\alpha$	$\theta = n\pi + (-1)^n \alpha$	$\alpha \in [-\pi/2, \pi/2]$
$\cos\theta = \cos\alpha$	$\theta = 2n\pi \pm \alpha$	$\alpha \in [0, \pi]$
$\tan\theta = \tan\alpha$	$\theta = n\pi + \alpha$	$\alpha \in (-\pi/2, \pi/2)$
$\sin^2\theta = \sin^2\alpha$	$\theta = n\pi \pm \alpha$	$n \in Z$

Linear Equations: a cosθ + b sinθ = c

Solvable iff $|c| \le \sqrt{(a^2 + b^2)}$. Using auxiliary angle $\varphi = \tan^{-1}(b/a)$, reduces to $\sin(\theta + \varphi) = c / \sqrt{(a^2 + b^2)}$.

3. INVERSE TRIGONOMETRIC FUNCTIONS

Domains & Principal Value Branches

Function	Domain (D)	Principal Range
$\sin^{-1} x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$(-\infty, \infty)$	$(-\pi/2, \pi/2)$
$\cot^{-1} x$	$(-\infty, \infty)$	$(0, \pi)$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] \setminus \{\pi/2\}$
$\operatorname{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\pi/2, \pi/2] \setminus \{0\}$

Fundamental Properties & Identities

```
Property 1: sin(sin⁻¹ x) = x [-1 ≤ 1];   sin⁻¹(sin x) = x [-π/2, π/2]
cos(cos⁻¹ x) = x [-1 ≤ 1];   cos⁻¹(cos x) = x [0, π]
tan(tan⁻¹ x) = x [R];   tan⁻¹(tan x) = x (-π/2, π/2)
```

```
Property 2 (Negative Arguments):
sin⁻¹(-x) = -sin⁻¹ x,   tan⁻¹(-x) = -tan⁻¹ x,
cosec⁻¹(-x) = -cosec⁻¹ x
cos⁻¹(-x) = π - cos⁻¹ x,   cot⁻¹(-x) = π - cot⁻¹ x,   sec⁻¹(-x) = π - sec⁻¹ x
```

```
Property 3 (Complementary Angles):
sin⁻¹ x + cos⁻¹ x = π/2   (x ∈ [-1, 1])
tan⁻¹ x + cot⁻¹ x = π/2   (x ∈ R)
sec⁻¹ x + cosec⁻¹ x = π/2   (|x| ≥ 1)
```

```
Property 4 (Reciprocal Arguments):
cosec⁻¹ x = sin⁻¹(1/x),   sec⁻¹ x = cos⁻¹(1/x)  (|x| ≥ 1)
cot⁻¹ x = tan⁻¹(1/x) (x > 0);   π + tan⁻¹(1/x) (x < 0)
```

Sum & Difference Formulas for ITF

```
tan⁻¹ x + tan⁻¹ y = tan⁻¹((x + y) / (1 - xy))  [xy < 1]
= π + tan⁻¹((x + y) / (1 - xy))   [x > 0, y > 0, xy > 1]
= -π + tan⁻¹((x + y) / (1 - xy))   [x < 0, y < 0, xy > 1]

tan⁻¹ x - tan⁻¹ y = tan⁻¹((x - y) / (1 + xy))  [xy > -1]

sin⁻¹ x ± sin⁻¹ y = sin⁻¹[x√(1 - y²) ± y√(1 - x²)]
  (x,y ≥ 0, x² + y² ≤ 1)
```

Generalized sum: $\tan^{-1}x_1 + \ldots + \tan^{-1}x_n = \tan^{-1}((S_1 - S_3 + \ldots) / (1 - S_2 + S_4 - \ldots))$.

Multiple Angle Inverse Relations

```
2 tan⁻¹ x = tan⁻¹(2x / (1 - x²))   (-1 < x < 1)
= sin⁻¹(2x / (1 + x²))   (-1 ≤ x ≤ 1)
= cos⁻¹((1 - x²) / (1 + x²))   (0 ≤ x < ∞)

3 sin⁻¹ x = sin⁻¹(3x - 4x³)   [-1/2 ≤ x ≤ 1/2]
3 cos⁻¹ x = cos⁻¹(4x³ - 3x)   [1/2 ≤ x ≤ 1]
3 tan⁻¹ x = tan⁻¹((3x - x³) / (1 - 3x²))   [-1/√3
< x < 1/√3]
```

Substitutions: $\sqrt{(a^2 - x^2)} \to x = a\sin\theta$; $\sqrt{(a^2 + x^2)} \to x = a\tan\theta$; $\sqrt{(x^2 - a^2)} \to x = a\sec\theta$.

JEE MATHEMATICS FORMULA REVISION SHEET

Comprehensive Quick-Revision Booklet: Sets, Relations & Functions

1. SET THEORY & FUNDAMENTALS

Representations: Roster $\{1, 2, 3\}$ & Set-builder $\{x : P(x)\}$

Power Set P(A): Collection of all subsets. If $n(A) = m$, then $n(P(A)) = 2^m$. Number of proper subsets = $2^m - 1$.

Subset & Superset: $A \subset B$ means $\forall x \in A \Rightarrow x \in B$. Empty set φ is a subset of every set.

Intervals on R: Closed $[a, b] = \{x : a \le x \le b\}$; Open $(a, b) = \{x : a < x < b\}$.

Set Operations:

- **Union:** $A \cup B = \{x : x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x : x \in A \text{ and } x \in B\}$
- **Difference:** $A - B = \{x : x \in A \text{ and } x \notin B\} = A \cap B'$
- **Symmetric Diff:** $A \Delta B = (A - B) \cup (B - A)$
- **Complement:** $A' = U - A = \{x \in U : x \notin A\}$

Key Laws of Algebra of Sets:

- Idempotent: $A \cup A = A, A \cap A = A$
- Commutative: $A \cup B = B \cup A, A \cap B = B \cap A$
- Associative: $(A \cup B) \cup C = A \cup (B \cup C)$
- Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- **De Morgan's Laws:** $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$
- Involution: $(A')' = A$
- Complement Laws: $A \cup A' = U, A \cap A' = \varphi, \varphi' = U, U' = \varphi$

Cardinal Number Formulas (Venn Diagram Results):

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$
- $n(A - B) = n(A) - n(A \cap B)$
- $n(A' \cap B') = n(A \cup B)' = n(U) - n(A \cup B)$
- $n(A' \cup B') = n(A \cap B)' = n(U) - n(A \cap B)$
- Number of elements in exactly one of A or B = $n(A) + n(B) - 2n(A \cap B)$

2. RELATIONS & PROPERTIES

Definition: Let A, B be non-empty sets. A relation R from A to B is a subset of cartesian product $A \times B$. $(a, b) \in R \Rightarrow a \, R \, b$.

Domain & Range: $\text{Dom}(R) = \{a : (a, b) \in R\}$, $\text{Range}(R) = \{b : (a, b) \in R\}$. Codomain is set B.

Inverse Relation: $R^{-1} = \{(b, a) : (a, b) \in R\}$. $\text{Dom}(R^{-1}) = \text{Range}(R)$.

Composition: If $R \subset A \times B$ and $S \subset B \times C$, then $S \circ R \subset A \times C$ defined by $(a, c) \in S \circ R$ if $\exists b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. Note: $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

Types of Relations on a Set A:

- **Void (Empty) & Universal:** $R = \varphi$ and $R = A \times A$.
- **Identity:** $I_A = \{(a, a) : a \in A\}$ (every element related to itself only).
- **Reflexive:** $(a, a) \in R \ \forall a \in A$.
- **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R \ \forall a, b \in A$ ($R = R^{-1}$).
- **Transitive:** $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.
- **Equivalence:** Reflexive, Symmetric, and Transitive simultaneously.
- **Antisymmetric:** $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$.

Counting Formulas (Set A has *n* elements, Set B has *m* elements):

Relation Type	Formula / Count	Relation Type	Formula / Count
Total Relations from A to B	2^{mn}	Symmetric Relations on A	$2^{n(n+1)/2}$
Reflexive Relations on A	$2^{n(n-1)}$	Reflexive & Antisymmetric	$3^{n(n-1)/2}$
Identity Relation on A	1	Binary Operations on A	n^{n^2}

Equivalence Classes & Congruence: For an equivalence relation R, equivalence class of a is $[a] = \{x : (x, a) \in R\}$. Disjoint partition property: $[a] = [b] \leftrightarrow (a, b) \in R$; otherwise $[a] \cap [b] = \varphi$.
Congruence Modulo n: $a \equiv b \pmod n \leftrightarrow n$ divides $(a - b)$.

3. FUNCTIONS — DEFINITION & CLASSIFICATION

Definition: A function $f : A \to B$ is a rule assigning each element of A a unique element in B. $f(a) = b$.

Domain, Codomain, Range: $D_f = A$ (or maximal set where defined), Codomain = B, Range $f(A) = \{f(x) : x \in A\} \subset B$.

Algebra of Functions: For $f, g : X \to R$:
• $(f \pm g)(x) = f(x) \pm g(x)$ (Dom: $D_1 \cap D_2$)
• $(fg)(x) = f(x)g(x)$
• $(f/g)(x) = f(x)/g(x)$ (where $g(x) \neq 0$)

Types of Mappings (Injectivity & Surjectivity):

- **One-One (Injective):** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ (or strictly monotonic).
- **Many-One:** Not one-one. At least two distinct elements have same image.
- **Onto (Surjective):** Range = Codomain (i.e., $\forall y \in B$, $\exists x \in A$ such that $f(x) = y$).
- **Into:** Range is a proper subset of Codomain.
- **Bijective:** Both One-One and Onto (invertible).

Counting Number of Functions (Set A has m elements, Set B has n elements):

Function Type	Formula / Condition	Function Type	Formula / Condition
Total Functions from A to B	n^m	Constant Functions	n
One-One (Injective) Functions	${}^nP_m$ (if $n \geq m$) else 0	Bijective Functions	$n!$ (if $n = m$) else 0
Onto (Surjective) Functions	$\sum_{r=0}^{n} (-1)^r {}^nC_r (n - r)^m$ ($n \leq m$)	Many-One Functions	$n^m - {}^nP_m$ (if $n \geq m$)

4. STANDARD FUNCTIONS & SPECIAL GRAPHS

Modulus Function $f(x) = |x|$:

$|x| = x$ if $x \geq 0$; $-x$ if $x < 0$. $\sqrt{(x^2)} = |x|$.

Key Inequalities & Properties:

- $|x| \leq a \leftrightarrow -a \leq x \leq a$ (for $a > 0$)
- $|x| \geq a \leftrightarrow x \leq -a$ or $x \geq a$
- $|x + y| \leq |x| + |y|$ (Triangle Inequality)
- $|x + y| = |x| + |y| \leftrightarrow xy \geq 0$

Greatest Integer Function $f(x) = [x]$:

Greatest integer less than or equal to x. Range = Z.

- $[x] \leq x < [x] + 1$ and $x - 1 < [x] \leq x$
- $[x + n] = [x] + n$ ($n \in Z$)
- $[x] + [-x] = 0$ if $x \in Z$; -1 if $x \notin Z$

Fractional Part Function $f(x) = \{x\}$:

$\{x\} = x - [x]$. Range = $[0, 1)$. Periodic with period 1.

Signum Function $f(x) = sgn(x)$:

$sgn(x) = |x|/x = 1$ $(x > 0)$, 0 $(x = 0)$, -1 $(x < 0)$.

Logarithmic & Exponential Functions:

- $y = \log_a x$ ($a > 0, a \neq 1, x > 0$). Dom: $(0, \infty)$, Range: R.
- $\log_a(xy) = \log_a |x| + \log_a |y|$
- $\log_a(x/n) = n \log_a x$; $\log_{a^n} x^m = (m/n) \log_a x$
- $a^{\log_a x\}} = x$; $x^{\log_a y\}} = y^{\log_a x\}}$
- Base $a > 1$: increasing; Base $0 < a < 1$: decreasing.

5. EVEN & ODD FUNCTIONS

Even Function: $f(-x) = f(x)\ \forall\ x$. Symmetric about Y-axis. Derivative of odd function is even.

Odd Function: $f(-x) = -f(x)\ \forall\ x$. Symmetric about origin. Derivative of even function is odd.

Decomposition: Every function can be expressed uniquely as sum of even and odd parts:

$$f(x) = [f(x) + f(-x)]/2 + [f(x) - f(-x)]/2$$

Algebra: Even ± Even = Even; Odd ± Odd = Odd; Even × Even = Even; Odd × Odd = Even; Even × Odd = Odd. An even function is never one-one.

6. PERIODIC FUNCTIONS

Definition: A function $f(x)$ is periodic if $f(x + T) = f(x)\ \forall\ x$. Smallest positive real T is the **fundamental period**.

Standard Periods:

- $\sin x, \cos x, \sec x, \operatorname{cosec} x \rightarrow$ Period 2π ($|\sin x|$ has period π)
- $\tan x, \cot x \rightarrow$ Period π
- $\{x\} = x - [x] \rightarrow$ Period 1
- Constant function $f(x) = c \rightarrow$ Periodic with no fundamental period.

Property: If $f(x)$ has period T, then $f(ax + b)$ has period $T / |a|$. If $f(x)$ has period T_1 and $g(x)$ has period T_2, then $f(x) \pm g(x)$ has period = LCM of $\{T_1, T_2\}$.

7. INVERSE FUNCTIONS

Invertibility: A function $f : A \rightarrow B$ is invertible (non-singular) if and only if it is bijective (one-one onto).

Properties:

- Inverse of a bijection is unique and is also a bijection.
- $f(x) = y \leftrightarrow f^{-1}(y) = x$. Thus $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.
- Graphs of $y = f(x)$ and $y = f^{-1}(x)$ are mirror images of each other across the line $y = x$.
- **Composition Inverse:** $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

8. STANDARD FUNCTIONAL EQUATIONS

Relations satisfied by standard functions: [Added/JEE Standard]

Functional Equation	General Solution Form
$f(x + y) = f(x) + f(y)$	$f(x) = kx$ (Cauchy's Eq.)
$f(x + y) = f(x) \cdot f(y)$	$f(x) = a^{kx}$ or $f(x) = 0$
$f(xy) = f(x) + f(y)$	$f(x) = k \ln x\ (x > 0)$
$f(xy) = f(x) \cdot f(y)$	$f(x) = x^n\ (n \in R)$ or $f(x) = 0$
$f(x + y) + f(x - y) = 2f(x)f(y)$	$f(x) = \cos(kx)$ or $\cosh(kx)$

9. IMPORTANT JEE RANGE & GRAPH TRANSFORMATION SHORTCUTS

Range of Trigonometric Expressions:

• $a \cos x + b \sin x \in [-\sqrt{(a^2 + b^2)}, \sqrt{(a^2 + b^2)}]$

• Range of $a \sin^2 x + b \cos^2 x$ (for $a > b$): $[b, a]$

Graph Transformations ($y = f(x)$):

• $y = f(x \pm a) \rightarrow$ Shift left/right by a

• $y = -f(x) \rightarrow$ Reflection across X-axis

• $y = f(-x) \rightarrow$ Reflection across Y-axis

• $y = |f(x)| \rightarrow$ Flip lower part above X-axis

Special Limits / Inequalities:

• Range of $x + 1/x \in (-\infty, -2] \cup [2, \infty)$

• If $f(x)\, f(1/x) = 1$, then $f(x) = \pm x^n \pm 1$

• Number of binary operations on set of n elements = n^{n^2}

JEE MATHEMATICS MASTER FORMULA SHEET

Comprehensive Revision Booklet: Vector Algebra | Three Dimensional Geometry | Straight Lines (2D)

1. VECTOR ALGEBRA — FUNDAMENTALS & ALGEBRA

Position Vector: Point P(x, y, z) relative to origin O:
$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, Magnitude: $|\mathbf{r}| = \sqrt{(x^2 + y^2 + z^2)}$

Unit Vector: $\hat{\mathbf{a}} = \mathbf{a} / |\mathbf{a}| \Rightarrow \mathbf{a} = |\mathbf{a}|\hat{\mathbf{a}}$

Vector Joining Two Points: For $P_1(x_1, y_1, z_1)$ & $P_2(x_2, y_2, z_2)$:
$\mathbf{P_1P_2} = (x_2 - x_1)\hat{\mathbf{i}} + (y_2 - y_1)\hat{\mathbf{j}} + (z_2 - z_1)\hat{\mathbf{k}}$

Section Formula: Position vector dividing AB in ratio m:n:
Internal: $\mathbf{r} = (m\mathbf{b} + n\mathbf{a}) / (m + n)$
External: $\mathbf{r} = (m\mathbf{b} - n\mathbf{a}) / (m - n)$, Mid-point: $\mathbf{r} = (\mathbf{a} + \mathbf{b}) / 2$

Collinearity & Coplanarity:
- Vectors a and b are collinear if $\mathbf{b} = \lambda\mathbf{a}$.
- Three vectors a, b, c are coplanar if $\mathbf{r} = x\mathbf{a} + y\mathbf{b}$ or $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = 0$.

Triangle & Parallelogram Laws:
- Triangle Law: $\mathbf{AB} + \mathbf{BC} = \mathbf{AC} \Rightarrow \mathbf{AB} + \mathbf{BC} + \mathbf{CA} = 0$
- Parallelogram Law: Diagonal $\mathbf{OC} = \mathbf{a} + \mathbf{b}$
- Triangle Inequality: $|\mathbf{a} + \mathbf{b}| \le |\mathbf{a}| + |\mathbf{b}|$ | $|\mathbf{a} - \mathbf{b}| \ge ||\mathbf{a}| - |\mathbf{b}||$

2. SCALAR (DOT) & VECTOR (CROSS) PRODUCTS

Scalar Dot Product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$
- In component form: $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$
- Angle between vectors: $\cos\theta = (\mathbf{a} \cdot \mathbf{b}) / (|\mathbf{a}||\mathbf{b}|)$
- Orthogonality: $\mathbf{a} \perp \mathbf{b} \Leftrightarrow \mathbf{a} \cdot \mathbf{b} = 0$
- Projection of a on b: $(\mathbf{a} \cdot \mathbf{b}) / |\mathbf{b}|$; Vector projection: $((\mathbf{a} \cdot \mathbf{b}) / |\mathbf{b}|^2)\,\mathbf{b}$

Vector Cross Product: $\mathbf{a} \times \mathbf{b} = (|\mathbf{a}||\mathbf{b}| \sin\theta)\,\hat{\mathbf{n}}$
- Determinant form: $\mathbf{a} \times \mathbf{b} = \det([\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}; a_1, a_2, a_3; b_1, b_2, b_3])$
- Area of Triangle $\mathbf{ABC} = \frac{1}{2}|\mathbf{AB} \times \mathbf{AC}|$
- Area of Parallelogram (adjacent sides a, b) $= |\mathbf{a} \times \mathbf{b}|$
- Parallel condition: $\mathbf{a} \times \mathbf{b} = 0$

3. TRIPLE PRODUCTS & ADVANCED VECTOR RELATIONS

Scalar Triple Product (STP): $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \det([\mathbf{a}, \mathbf{b}, \mathbf{c}])$
- Cyclic property: $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = [\mathbf{b}\ \mathbf{c}\ \mathbf{a}] = [\mathbf{c}\ \mathbf{a}\ \mathbf{b}]$
- Volume of parallelepiped $= |[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]|$
- Volume of tetrahedron $= \frac{1}{6}|[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]|$
- Coplanarity condition: $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = 0$

Vector Triple Product (VTP): $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$
Lagrange's Identity: $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$
Reciprocal System: $\mathbf{a}' = (\mathbf{b} \times \mathbf{c}) / [\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$ (with $\mathbf{a} \cdot \mathbf{a}' = 1$)

4. THREE DIMENSIONAL GEOMETRY (3D) — BASICS & LINES

Distance & Section Formulas in 3D:
- Distance $\mathbf{PQ} = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2)}$
- Centroid: $((x_1+x_2+x_3)/3, (y_1+y_2+y_3)/3, (z_1+z_2+z_3)/3)$

Direction Cosines (DC's) & Ratios (DR's):
- $l = \cos\alpha, m = \cos\beta, n = \cos\gamma \Rightarrow l^2 + m^2 + n^2 = 1$
- If DR's are a, b, c: $l = \pm a / \sqrt{(a^2+b^2+c^2)}$
- Angle between two lines with DR's (a_1,b_1,c_1) & (a_2,b_2,c_2):
$\cos\theta = |(\sum a_1a_2) / (\sqrt{(\sum a_1^2)}\ \sqrt{(\sum a_2^2)})|$

Equations of a Line:
- Vector Form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$
- Cartesian Form: $(x - x_1)/a = (y - y_1)/b = (z - z_1)/c = r$
- Two-Point Form: $(x - x_1)/(x_2 - x_1) = (y - y_1)/(y_2 - y_1) = (z - z_1)/(z_2 - z_1)$

Shortest Distance Between Lines:
- Parallel lines ($\mathbf{r} = \mathbf{a_1} + \lambda\mathbf{b}$ & $\mathbf{r} = \mathbf{a_2} + \mu\mathbf{b}$): $\mathbf{d} = |\mathbf{b} \times (\mathbf{a_2} - \mathbf{a_1})| / |\mathbf{b}|$
- Skew lines ($\mathbf{r} = \mathbf{a_1} + \lambda\mathbf{b_1}$ & $\mathbf{r} = \mathbf{a_2} + \mu\mathbf{b_2}$):
$\mathbf{d} = |(\mathbf{b_1} \times \mathbf{b_2}) \cdot (\mathbf{a_2} - \mathbf{a_1})| / |\mathbf{b_1} \times \mathbf{b_2}|$

5. PLANES IN 3D & SPHERE

Equations of a Plane:
- General Form: $ax + by + cz + d = 0$
- Normal Form: $lx + my + nz = p$ or $\mathbf{r} \cdot \hat{\mathbf{n}} = p$
- Point-Normal Form: $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$
- Intercept Form: $x/a + y/b + z/c = 1$
- Passing through 3 points: $\det([\mathbf{r}-\mathbf{a}, \mathbf{b}-\mathbf{a}, \mathbf{c}-\mathbf{a}]) = 0$

Distance from Point to Plane: $d = |ax_1 + by_1 + cz_1 + d| / \sqrt{(a^2 + b^2 + c^2)}$

Angle Between Two Planes / Line & Plane:
- Planes: $\cos\theta = |(\sum a_1a_2) / (\sqrt{(\sum a_1^2)}\ \sqrt{(\sum a_2^2)})|$
- Line and Plane angle: $\sin\theta = |(al + bm + cn) / (\sqrt{(\sum a^2)}\ \sqrt{(\sum l^2)})|$

Family of Planes & Sphere:
- Plane through intersection of two planes: $(P_1) + \lambda(P_2) = 0$
- Sphere General Equation: $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$
Centre: $(-u, -v, -w)$, Radius: $\sqrt{(u^2 + v^2 + w^2 - d)}$

6. STRAIGHT LINES (2D) — FORMS & RELATIONS

Slope & Angle Between Lines:

• Slope $\mathbf{m = \tan\theta}$; For points $\mathbf{(x_1, y_1), (x_2, y_2)}$: $\mathbf{m = (y_2 - y_1)/(x_2 - x_1)}$

• Acute angle between lines with slopes m_1, m_2: $\mathbf{\tan\theta = |(m_2 - m_1) / (1 + m_1m_2)|}$

• Parallel: $\mathbf{m_1 = m_2}$ | Perpendicular: $\mathbf{m_1m_2 = -1}$

Standard Forms of Straight Lines:

• Slope-Intercept: $\mathbf{y = mx + c}$ | Point-Slope: $\mathbf{y - y_1 = m(x - x_1)}$

• Intercept Form: $\mathbf{x/a + y/b = 1}$ | Normal Form: $\mathbf{x\cos\alpha + y\sin\alpha = p}$

• Parametric Form: $\mathbf{(x - x_1)/\cos\theta = (y - y_1)/\sin\theta = r}$

• General Form: $\mathbf{Ax + By + C = 0}$ (Slope $\mathbf{m = -A/B}$)

Distance & Concurrency Formulas:

• Perpendicular distance from $\mathbf{(x_1, y_1)}$ to $\mathbf{Ax + By + C = 0}$:
$\mathbf{d = |Ax_1 + By_1 + C| / \sqrt{(A^2 + B^2)}}$

• Distance between parallel lines $\mathbf{Ax + By + C_1 = 0}$ & $\mathbf{C_2 = 0}$:
$\mathbf{d = |C_1 - C_2| / \sqrt{(A^2 + B^2)}}$

• Concurrency condition for 3 lines: $\mathbf{\det([a_i, b_i, c_i]) = 0}$

Pair of Straight Lines:

• Homogeneous ($\mathbf{ax^2 + 2hxy + by^2 = 0}$):
$\mathbf{m_1 + m_2 = -2h/b}$, $\mathbf{m_1m_2 = a/b}$, $\mathbf{\tan\theta = |2\sqrt{(h^2 - ab)}| / (a + b)}$

• General 2nd Degree ($\mathbf{ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0}$):
Condition for pair of lines: $\mathbf{abc + 2fgh - af^2 - bg^2 - ch^2 = 0}$

JEE RAPID REVISION FORMULA SHEET: MATRICES & DETERMINANTS

Comprehensive Core Relations, Properties, Adjoint/Inverse Shortcuts & Advanced Results

1. MATRICES: FUNDAMENTALS & TYPES

Order: Matrix $A = [a_{ij}]$ of size $m \times n$ has m rows and n columns. Total elements = mn.

Square Matrix: Number of rows equals columns ($m = n$). Principal diagonal elements: $a_{11}, a_{22}, ..., a_{nn}$.

Trace of Matrix: $tr(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + ... + a_{nn}$

Trace Properties:
- $tr(kA) = k\ tr(A)$
- $tr(A \pm B) = tr(A) \pm tr(B)$
- $tr(AB) = tr(BA)$ (when products are defined)
- $tr(I_n) = n$, $tr(O) = 0$

Special Square Matrices:
- **Diagonal:** $a_{ij} = 0$ for $i \neq j$.
- **Scalar:** Diagonal matrix with $a_{ii} = k$.
- **Identity (I):** Diagonal matrix with $a_{ii} = 1$.
- **Upper Triangular:** $a_{ij} = 0$ for $i > j$.
- **Lower Triangular:** $a_{ij} = 0$ for $i < j$.

2. ALGEBRA & SPECIAL SQUARE MATRICES

Equality: $A = B$ if same order and $a_{ij} = b_{ij}$ for all i, j.

Multiplication: $A_{m \times n} \cdot B_{n \times p} = C_{m \times p}$ where $C_{ij} = \sum_{r=1}^{n} a_{ir} b_{rj}$. *Not commutative ($AB \neq BA$ in general).*

Symmetric Matrix: $A^T = A$ ($a_{ij} = a_{ji}$). Max distinct entries in $n \times n = n(n+1)/2$.

Skew-Symmetric Matrix: $A^T = -A$ ($a_{ij} = -a_{ji}$). All diagonal elements are zero ($a_{ii} = 0$).

Decomposition: Any square matrix can be uniquely expressed as sum of symmetric and skew-symmetric matrices:
$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$

Orthogonal Matrix: $AA^T = I = A^TA \Rightarrow A^{-1} = A^T$, $|det(A)| = 1$.

Idempotent Matrix: $A^2 = A \Rightarrow A^n = A\ (\forall n \geq 2)$, $det(A) \in \{0, 1\}$.

Involutory Matrix: $A^2 = I \Rightarrow A^{-1} = A$.

Nilpotent Matrix: $A^m = O$ for least positive integer m (index). *Non-invertible.*

Periodic Matrix: $A^{k+1} = A$ (least k is period).

4. ADJOINT OF A MATRIX

Definition: $adj(A) = [C_{ji}]^T$ where C_{ij} is the cofactor of a_{ij}.

Fundamental Identity:
$A \cdot adj(A) = adj(A) \cdot A = |A|\ I_n$

Important Adjoint Results (Order n):
- $|adj(A)| = |A|^{n-1}$
- $adj(adj(A)) = |A|^{n-2}\ A$
- $|adj(adj(A))| = |A|^{(n-1)^2}$
- $adj(AB) = adj(B) \cdot adj(A)$
- $adj(kA) = k^{n-1}\ adj(A)$ (k is scalar)
- $adj(A^m) = (adj(A))^m$
- $adj(I_n) = I_n$, $adj(O) = O$

5. DETERMINANTS: FUNDAMENTALS & EXPANSION

Order 2 Expansion:

$|A| = |a_1\ b_1 / a_2\ b_2| = a_1b_2 - a_2b_1$

Order 3 Expansion: Sum of products of elements of any row/column with their corresponding cofactors:
$|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

Minors & Cofactors:
Minor M_{ij} is determinant after deleting i-th row and j-th column.
Cofactor $C_{ij} = (-1)^{i+j} M_{ij}$.

Cofactor Property:
$\sum_j a_{ij} C_{kj} = |A|$ if $i = k$, and 0 if $i \neq k$.

6. PROPERTIES OF DETERMINANTS

- $|A^T| = |A|$ (Row-column interchange leaves value unchanged).
- Interchanging any two rows/columns changes sign by -1.
- If any two rows/columns are identical or proportional, $|A| = 0$.
- $|kA| = k^n\ |A|$ (where A is order n).
- $|AB| = |A|\ |B|$, $|A^m| = |A|^m$.
- $|A^{-1}| = 1 / |A| = |A|^{-1}$.
- Determinant of skew-symmetric matrix of odd order is 0; of even order is a non-zero perfect square.
- Determinant of triangular/diagonal matrix = product of principal diagonal elements.

3. TRANSPOSE & INVERSE PROPERTIES

Transpose Laws:

- $(A^T)^T = A$, $(A \pm B)^T = A^T \pm B^T$
- $(kA)^T = k\,A^T$
- $(AB)^T = B^T A^T$ (Reversal Law)
- $(A_1A_2...A_n)^T = A_n^T...A_2^T A_1^T$

Inverse Matrix: $A^{-1} = \text{adj}(A) / |A|$, defined iff $|A| \neq 0$ (Non-singular).

Inverse Properties:

- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1} A^{-1}$ (Reversal Law)
- $(A^T)^{-1} = (A^{-1})^T$
- $(A^k)^{-1} = (A^{-1})^k = A^{-k}$ ($k \in N$)

JEE RAPID REVISION FORMULA SHEET: MATRICES & DETERMINANTS (CONT.)

Standard Determinants, Calculus, Coordinate Geometry & Linear Equations

7. STANDARD & SPECIAL DETERMINANTS

Vandermonde Determinants:

- $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$
- $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$

Circulant Determinant:

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$$

Symmetric Determinant:

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2$$

8. CALCULUS OF DETERMINANTS

Differentiation: If $\Delta(x) = |\, f_{ij}(x) \,|$ (3×3), then $\Delta'(x)$ is the sum of three determinants obtained by differentiating row 1, then row 2, then row 3 one at a time while keeping others constant.

Integration: $\int_a^b \Delta(x)\,dx$ involves integrating only the row/column containing functions of x (if other rows are constants).

10. SYSTEM OF LINEAR EQUATIONS

Consider system:

$a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$

$a_3x + b_3y + c_3z = d_3$

Matrix form: $AX = B$, where $A = [a_{ij}]$.

Condition	System Type	Solution State
$\|A\| \neq 0$	Consistent	Unique ($X = A^{-1}B$)
$\|A\| = 0$ & $\text{adj}(A)B \neq O$	Inconsistent	No Solution
$\|A\| = 0$ & $\text{adj}(A)B = O$	Consistent / Inconsistent	Infinitely Many or No Solution

11. CRAMER'S RULE (3 VARIABLES)

Let D be determinant of coefficient matrix A. Replace columns with constant vector $[d_1, d_2, d_3]^T$ to get D_1, D_2, D_3.

Solutions given by: $x = D_1/D$, $y = D_2/D$, $z = D_3/D$

Cases:

1. $D \neq 0$: Unique non-trivial solution.
2. $D = 0$ and at least one $D_i \neq 0$: Inconsistent (No solution).
3. $D = 0$ and $D_1 = D_2 = D_3 = 0$: Consistent with **Infinitely Many Solutions** (put one variable = k and solve).

12. HOMOGENEOUS SYSTEM ($AX = O$)

- $|A| \neq 0$: Only **Trivial Solution** ($x = y = z = 0$).
- $|A| = 0$: **Infinitely Many Solutions** (non-trivial solutions exist).

9. APPLICATIONS IN COORDINATE GEOMETRY

Area of Triangle: Vertices (x_1, y_1), (x_2, y_2), (x_3, y_3):

$$\text{Area} = \tfrac{1}{2} \left| x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right|$$

Collinear condition: Area = 0 (determinant = 0).

Equation of Line: Passing through (x_1, y_1) and (x_2, y_2):

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Concurrency of 3 Lines ($a_i x + b_i y + c_i = 0$):

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Pair of Straight Lines ($ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$):

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

13. USEFUL JEE SHORTCUT RESULTS [Added]

- If A and B are square matrices of order n, $|I + AB| = |I + BA|$.
- If A is non-singular, $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$.
- Characteristic Equation of 2×2 matrix A: $\lambda^2 - \text{tr}(A)\lambda + |A| = 0$ (Satisfied by A: $A^2 - \text{tr}(A)A + |A|I = 0$).
- Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation.